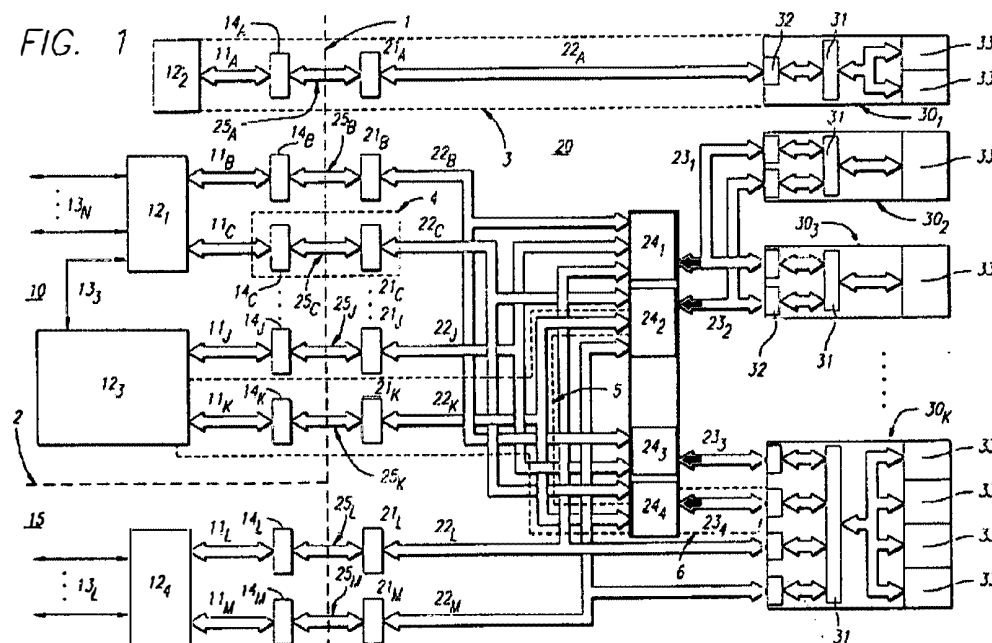


plurality of the channels to one or more of the memory clusters 30.” *Id.* at 11:52–54.

2. Claims 1 and 7

Independent claims 1 and 7 recite “a bi-directional data transfer fabric [having/with] two or more sets of data ports.” Petitioner argues that, if this phrase is construed to require two or more independent read/write paths to each of the volatile and non-volatile memory subsystems, such a feature was taught by Roy. Pet. 49–51. In particular, Petitioner relies upon Roy’s teaching of multiplexers 24_{1–4} (“a bi-directional data transfer fabric”), their interfaces to buses 23_{1–4} (“two or more sets of data ports”), some of which are coupled to memory cluster 30₃ (“a first set . . . coupled to”) and others of which are coupled to memory cluster 30_K (“a second set . . . coupled to”). Pet. 50–51. Petitioner also annotates Figure 1 of Roy to indicate multiplexers (red), interfaces (green), and memory clusters to which they are coupled (yellow):



Id. at 50.

With respect to why a person of ordinary skill in the art would have modified Best in view of Roy, Petitioner argues:

One of ordinary skill in the art would have been motivated to implement this architecture for all the reasons Roy describes, including allowing independent and simultaneous transactions, Ex. 1008, 7:37-40, and increased performance by providing a wide effective channel, *id.*, 7:45-49; Ex. 1003, ¶173. Roy also teaches that a multichannel architecture provides substantial flexibility. Ex. 1008, 9:30-42; Ex. 1003, ¶173.

Roy discloses that “nearly identical address and control information” can be applied to each channel such that “[s]ubsequent transfer[s] of data on each of these channels can be synchronized to provide an effectively wider channel.” Ex. 1008, 10:28-32. This provides particular motivation to combine with Best in light of Best’s disclosure that “multiple non-volatile storage dice and/or multiple volatile storage dice may be ... selected ... based on incoming address and/or control signals.” Ex. 1006, ¶15; Ex. 1003, ¶174.

Best suggests such a modification through his disclosure of overlapping and pipelined memory operations. Ex. 1006, ¶18. One of ordinary skill would understand that multiple channels allow for further overlapping or pipelining of operations, such as allowing Best to write data from volatile to non-volatile memory as part of the “Shadow Mode” operation while allowing the host to simultaneously write data to volatile memory, thus improving the operation and responsiveness of the system. Ex. 1003, ¶175.

Modifying Best to use a multichannel architecture such as Roy’s would have been an arrangement of old elements (Best’s hybrid memory, Roy’s multichannel architecture) with each performing the same function it had been known to perform and yielding no more than what one would expect from such an arrangement, *i.e.*, Best’s system with a multichannel architecture. Ex. 1003, ¶176. Multichannel architectures were known in the art, and using one in Best would have involved only routine skill to implement the functionality described by Roy. *Id.*

Such a modification would have therefore been obvious. *Id.*, ¶¶176, 240.

Pet. 56–53. On this record, we are persuaded that Petitioner has provided an articulated reasoning with some rational underpinning that would support the legal conclusion of obviousness. *See KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2017) (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)).

Additionally, Patent Owner relies upon the same argument as for Ground 1—i.e., that Best does not teach a “memory module”—and argues that the Petition fails to show that Roy teaches a “memory module.” Prelim. Resp. 35. We are not persuaded by that argument for the reasons discussed above.

3. Dependent claims 2–6 and 8–14

For dependent claims 2–6 and 8–14, Petitioner relies upon its explanations and supporting evidence from Ground 1. *See* Pet. 49–53. Patent Owner does not argue separately dependent claims 2–6 and 8–14. Based on the record before us, Petitioner has demonstrated a reasonable likelihood that it would prevail on its assertion that claims 2–6 and 8–14 would have been obvious over Best and Roy.

4. Conclusion

On this record, we are persuaded that Petitioner has established a reasonable likelihood that it would prevail in showing that claims 1–14 are unpatentable as obvious over the combination of Best and Roy.

*E. Claim 15: Obviousness over
Best, Mills, and Bonella, with or without Roy*

Petitioner argues that claim 15 is unpatentable under 35 U.S.C. § 103(a) as obvious over Best, Mills, and Bonella, with or without

Roy.² Pet. 57–68. In light of the arguments and evidence of record, we are persuaded that Petitioner has established a reasonable likelihood that the claim 15 is unpatentable as obvious over the combination of Best and Bonella, and over the combination of Best, Roy, and Bonella.

1. Bonella (Ex. 1013)

Bonella describes “A memory module including a volatile memory, a non-volatile memory, and a controller that provides address, data, and control interfaces to the memories and to a host system.” Ex. 1013, Abstract. Bonella teaches that, “[t]he memory module controller . . . is ‘Power State Aware.’” *Id.* ¶ 45. At “Power Level 4,” Bonella’s controller “reduces power by limiting the DRAM performance and the PCIe transaction performance.” *Id.* ¶ 48. Bonella teaches that “[r]eduction of power in the DRAM can be accomplished” by “reduc[ing] the frequency in which the DRAM is operating,” which “reduces power and, in general, produces no noticeable decrease in system performance.” *Id.* ¶ 49.

2. Claim 15

Claim 15 recites

operating the volatile memory subsystem at a first clock frequency when the memory module is in a first mode of operation in which data is communicated between the volatile memory subsystem and the memory controller;

Ex. 1001, 20:12–15. Petitioner relies upon Best’s teaching of a conventional DRAM, which one of ordinary skill in the art would have understood operates at a first clock frequency. Pet. 58–60.

² Although Petitioner does not include Mills (Ex. 1010) explicitly in its ground (Pet. 3), we include it here because Petitioner’s analysis relies upon it (*id.* at 61–64) for teaching part of a limitation.

Claim 15 further recites

operating the non-volatile memory subsystem at a second clock frequency when the memory module is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

Ex. 1001, 20:16–20. Petitioner concedes that Best does not explicitly disclose this limitation, but relies upon Mills’ teaching of a synchronous flash interface to argue that it would have been obvious to include this functionality in Best. Pet. 61 (citing Ex. 1010). Specifically, Petitioner argues that Mills teaches “a synchronous Flash interface where read and write operations are synchronized to the rising edge of a clock signal provided to the device and operating at a particular frequency.” *Id.* at 62.

With respect to why a person of ordinary skill in the art would have combined Best and Mills, Petitioner argues

it would have been obvious to one of ordinary skill in the art to employ a synchronous flash memory, such as disclosed in Mills, in the system of Best because to do so would have been merely an arrangement of old elements with each performing the same function it had been known to perform and yielding no more than what one would expect from such an arrangement, *i.e.*, the non-volatile storage of data. . . .

A skilled artisan would have been motivated to make such a combination because, as Mills explains, a synchronous flash interface “creates an average access time for sequential read accesses that is significantly less than the access time of an asynchronous flash device.” Ex. 1010, 17:6-9. In the context of Best, restoring data from the non-volatile flash memory would therefore have been faster by use of a synchronous flash memory, and reduced sequential read access times during other operations or uses of Best’s Flash memory, motivating one of ordinary skill in the art to use a synchronous interface generally. Ex. 1003, ¶299.

[O]ne of ordinary skill in the art would have therefore understood Best to suggest modification to work with any known Flash interface, including Mills' synchronous Flash interface. Ex. 1003, ¶300.

Pet. 62–64.

Patent Owner argues that Petitioner's reason to combine Mills and Best is insufficient because it "fails to show that Mills' interface provides any improvement for Best's write flushing to the alleged NVMS." This argument is not persuasive. Prelim. Resp. 36–40. Petitioner contends that there is an advantage when restoring, i.e., reading, data from the non-volatile memory, not when writing data to the non-volatile memory. Patent Owner does not explain why an advantage during reading would not be sufficient motivation. Patent Owner does not explain, for example, why a person of ordinary skill in the art would not make the proposed modification unless it also improved Best's write flushing.

Patent Owner also argues that the proposed modification adds complexity and consumes a large amount of power. *Id.* at 38–40 (citing Ex. 1010, 16:49–59). The portion of Mills on which Patent Owner relies, however, relates only to the embodiment of Figure 3. Petitioner is not relying solely upon Mills' teachings with respect to Figure 3. Petitioner is relying, for example, upon Mills' teachings with respect to Figure 6, which relies upon "a single flash memory chip" (Ex. 1010, 16:63) rather than the "at least two flash memory chips" in Figure 3. *See, e.g.*, Pet. 61 (citing 16:60–63). On this record, we are persuaded that Petitioner has provided an articulated reasoning with some rational underpinning that would support the legal conclusion of obviousness. *See KSR*, 550 U.S. at 418 (citing *Kahn*, 441 F.3d at 988).

Finally, claim 15 recites

operating the volatile memory subsystem at a third clock frequency when the memory module is in the second mode of operation, the third clock frequency being less than the first clock frequency.

Ex. 1001, 20:21–24. Petitioner concedes that Best does not explicitly disclose this limitation, but argues that “(1) reducing power during volatile to non-volatile flush operations prompted by a power loss was a well-known technique, and (2) one known way to reduce the power consumption of DRAM devices was to reduce their frequency of operation.” Pet. 64–65 (citing Ex. 1011; Ex. 1013). As for why a person of ordinary skill in the art would have modified Best in view of this knowledge, Petitioner states

One of ordinary skill in the art would have been motivated to reduce the power consumption during Best’s write flushing in response to a power loss. A skilled artisan would have been motivated to perform this power reduction technique for all the reasons that were known in the art: e.g., decreasing the risk of data loss due to insufficient backup power (Ex. 1011, ¶7) and enabling the use of a smaller-sized backup power source (Ex. 1012, 4:54-64). Reducing power consumption during write flushing in response to a power loss would also have been the arrangement of old elements, each performing the same function it had been known to perform, in a way that yields no more than one of ordinary skill in the art would expect from such an arrangement (reducing power consumption during a power loss event, as suggested by Long and Ashmore). Ex. 1003, ¶307

...

One of ordinary skill in the art would also have found it obvious to reduce power consumption during Best’s write flushing in response to a power loss using any known or conventional means, and would have also considered power consumption reduction techniques other than those of Ashmore and Long to obtain the same benefits, including those described in Bonella.

Pet. 66–67. On this record, we are persuaded that Petitioner has provided an articulated reasoning with some rational underpinning that would support the legal conclusion of obviousness. *See KSR*, 550 U.S. at 418 (citing *Kahn*, 441 F.3d at 988).

Patent Owner additionally argues that Petitioner fails to show evidence of the volatile memory subsystem operation at a “third clock frequency” during the “second mode.” Prelim. Resp. 40–43. Patent Owner argues that “[n]one of the references, however, teaches a reduction of the frequency of operation of DRAM devices prompted by the power loss” and “no one single reference, however, teaches all three requirements.” *Id.* at 41–42. This argument is not persuasive because Petitioner is relying upon the combination of the references. Nonobviousness cannot be established by attacking references individually where, as here, the ground of unpatentability is based upon the teachings of *a combination* of references. *In re Keller*, 642 F.2d 413, 426 (CCPA 1981). Rather, the test for obviousness is whether the combination of references, taken as a whole, would have suggested the patentee’s invention to a person of ordinary skill in the art. *In re Merck & Co., Inc.*, 800 F.2d 1091, 1097 (Fed. Cir. 1986).

Patent Owner also argues that Bonella teaches that “a reduction in the DRAM operation frequency would result in a write flush taking longer to complete” and “the total amount of power required for the write flush would not change.” Prelim. Resp. 42–43. At this stage of the proceeding, however, this attorney argument is unsupported by evidence and is, therefore, also unpersuasive.

3. Conclusion

On this record, we are persuaded that Petitioner has established a reasonable likelihood that it would prevail in showing that claim 15 is unpatentable as obvious over the combination of Best, Mills, and Bonella, with or without Roy.

F. Other Grounds

Petitioner argues that (1) claims 2 and 8 are unpatentable under 35 U.S.C. § 103 as obvious over Best and Tsunoda, with or without Roy (Pet. 53–54); (2) claims 5 and 12–14 are unpatentable under 35 U.S.C. § 103 as obvious over Best and Roohparvar, with or without Roy (*id.* at 55–57); and (3) claim 15 is unpatentable under 35 U.S.C. § 103 as obvious over Best, Bonella, and Ashmore, with or without Roy (*id.* at 68–69). The Board’s rules for AIA *inter partes* proceedings, including those pertaining to institution, are “construed to secure the just, speedy, and inexpensive resolution of every proceeding.” 37 C.F.R. § 42.1(b); *accord* 35 U.S.C. § 316(b) (regulations for AIA *inter partes* proceedings take into account “the efficient administration of the Office” and “the ability of the Office to timely complete [instituted] proceedings”). We institute an *inter partes* review of claims 1–15 based on the grounds discussed above. We exercise our discretion not to institute a review based on these other grounds. *See* 37 C.F.R. § 42.108(a) (“the Board may authorize the review to proceed . . . on all or some of the grounds of unpatentability asserted for each claim”); 35 U.S.C. § 314(a) (authorizing institution of an *inter partes* review under particular circumstances, but not requiring institution under any circumstances); *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1368 (Fed. Cir. 2016) (“[U]nder [37 C.F.R. § 42.108(a)], it is clear that the Board

may choose to institute some grounds and not institute others as part of its comprehensive institution decision.”).

III. CONCLUSION

For the foregoing reasons, we are persuaded that Petitioner has demonstrated that there is a reasonable likelihood that Petitioner would prevail in establishing the unpatentability of claims 1–15 of the ’831 patent. At this stage of the proceeding, we have not made a final determination with respect to the patentability of these challenged claims or to the construction of any claim term.

IV. ORDER

Accordingly, it is

ORDERED that pursuant to 35 U.S.C. § 314, an *inter partes* review is hereby instituted on the following grounds:

1. claims 1–14 under 35 U.S.C. § 102 as anticipated by Best;
2. claims 1–14 under 35 U.S.C. § 103(a) as obvious over Best;
3. claims 1–14 under 35 U.S.C. § 103(a) as obvious over Best and Roy;
4. claim 15 under 35 U.S.C. § 103(a) as obvious over Best, Mills, and Bonella, with or without Roy;

FURTHER ORDERED that no other grounds are authorized for this *inter partes* review other than those specifically identified above; and

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(d), and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial; the trial commences on the entry date of this Decision.

IPR2017-00692
Patent 8,874,831 B2

For PETITIONER:

Joseph A. Micallef
Samuel A. Dillon
SIDLEY AUSTIN LLP
jmicallef@sidley.com
samuel.dillon@sidley.com
Sidley-SKH-IPR@sidley.com

For PATENT OWNER:

Thomas J. Wimbiscus
Christopher C. Winslade
Gregory C. Schodde
Scott P. McBride
Ronald H. Spuhler
Wayne H. Bradley
McANDREWS, HELD & MALLOY, LTD.
twimbiscus@mcandrews-ip.com
cwinslade@mcandrews-ip.com
gschodde@mcandrews-ip.com
smcbride@mcandrews-ip.com
rspuhler@mcandrews-ip.com
wbradley@mcandrews-ip.com

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY TO PROSECUTE APPLICATIONS BEFORE THE USPTO

I hereby revoke all previous powers of attorney given in the application identified in the attached statement under 37 CFR 3.73(b).

I hereby appoint:



Practitioners associated with the Customer Number:

151145

OR



Practitioner(s) named below (if more than ten patent practitioners are to be named, then a customer number must be used):

Name	Registration Number	Name	Registration Number

as attorney(s) or agent(s) to represent the undersigned before the United States Patent and Trademark Office (USPTO) in connection with any and all patent applications assigned only to the undersigned according to the USPTO assignment records or assignment documents attached to this form in accordance with 37 CFR 3.73(b).

Please change the correspondence address for the application identified in the attached statement under 37 CFR 3.73(b) to:



The address associated with Customer Number:

151145

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

Assignee Name and Address:

Netlist, Inc.
175 Technology Drive, Suite 150
Irvine, CA 92618

A copy of this form, together with a statement under 37 CFR 3.73(b) (Form PTO/SB/96 or equivalent) is required to be filed in each application in which this form is used. The statement under 37 CFR 3.73(b) may be completed by one of the practitioners appointed in this form if the appointed practitioner is authorized to act on behalf of the assignee, and must identify the application in which this Power of Attorney is to be filed.

SIGNATURE of Assignee of Record

The individual whose signature and title is supplied below is authorized to act on behalf of the assignee

Signature		Date	12/5/17
Name	Noel B. Whitley	Telephone	949-679-0115
Title	VP, IP & Licensing		

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

STATEMENT UNDER 37 CFR 3.73(b)

Applicant/Patent Owner: Hyun Lee et al.

Application No./Patent No.: 13/559,476

Filed/Issue Date: 07-26-2012

Titled: FLASH-DRAM HYBRID MEMORY MODULE

Netlist, Inc., a Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)

the patent application/patent identified above, by virtue of either:

A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 029843, Frame 0824, or for which a copy therefore is attached.

OR

B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☐ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Khaled Shami/

Signature

December 8, 2017

Date

Khaled Shami, Reg. No. 38,745

Printed or Typed Name

Attorney for Assignee

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Electronic Acknowledgement Receipt

EFS ID:	31175292
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Casey Berger
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	08-DEC-2017
Filing Date:	26-JUL-2012
Time Stamp:	15:55:21
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00160010004_POA373.pdf	942088	yes	2
			b32c7c289974d43a421126e88f69d7eb2203ef69		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Power of Attorney	1	1
	Assignee showing of ownership per 37 CFR 3.73	2	2

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

Total Files Size (in bytes):	942088
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/559,476	07/26/2012	Hyun Lee	062453-010

46188
Nixon Peabody LLP
P.O. Box 26769
San Francisco, CA 94126

CONFIRMATION NO. 1046
POWER OF ATTORNEY NOTICE



Date Mailed: 12/14/2017

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 12/08/2017.

- The Power of Attorney to you in this application has been revoked by the assignee who has intervened as provided by 37 CFR 3.71. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/sleutchit/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/559,476	07/26/2012	Hyun Lee	062453-010

151145
Shami Messinger PLLC
1000 Potomac Street NW
Fifth Floor
Washington, DC 20007

CONFIRMATION NO. 1046
POA ACCEPTANCE LETTER



Date Mailed: 12/14/2017

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 12/08/2017.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/sleutchit/

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC., SK HYNIX AMERICA INC., and
SK HYNIX MEMORY SOLUTIONS INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2017-00692
Patent 8,874,831 B2

Before STEPHEN C. SIU, MATTHEW R. CLEMENTS, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

CLEMENTS, *Administrative Patent Judge*.

FINAL WRITTEN DECISION
Inter Partes Review
35 U.S.C. § 318(a) and 37 C.F.R. § 42.73

I. INTRODUCTION

In this *inter partes* review, instituted pursuant to 35 U.S.C. § 314, SK hynix Inc., SK hynix America Inc. and SK hynix memory solutions Inc. (“Petitioner”) challenges claims 1–15 (“the challenged claims”) of U.S. Patent No. 8,874,831 B2 (Ex. 1001, “the ’831 patent”), owned by Netlist, Inc. (“Patent Owner”). We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision is entered pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. For the reasons discussed below, Petitioner has shown by a preponderance of the evidence that the challenged claims are unpatentable. Petitioner’s Motion to Exclude is *dismissed*.

A. Procedural History

Petitioner filed a Petition requesting an *inter partes* review of claims 1–15 of the ’831 patent. Paper 1 (“Pet.”). Patent Owner filed a Preliminary Response. Paper 6. On July 21, 2017, we instituted *inter partes* review of (1) claims 1–14 of the ’831 patent as unpatentable under 35 U.S.C. § 102¹ as anticipated by Best;² (2) claims 1–14 under 35 U.S.C. § 103(a) as obvious over Best; (3) claims 1–14 under 35 U.S.C. § 103(a) as obvious over Best and Roy;³ and (4) claim 15 over Bowie under 35 U.S.C. § 103(a) as obvious

¹ The Leahy-Smith America Invents Act, Pub. L. No. 112–29, 125 Stat. 284 (2011) (“AIA”), amended 35 U.S.C. §§ 102 and 103. Because the ’831 patent has an effective filing date before the effective date of the applicable AIA amendments, we refer to the pre-AIA versions of 35 U.S.C. §§ 102 and 103.

² U.S. Patent Publication No. 2010/0110748 A1 (Ex. 1006, “Best”).

³ U.S. Patent No. 6,065,092 (Ex. 1008, “Roy”).

over Best, Mills,^{4,5} and Bonella,⁶ with or without Roy. Paper 7 (“Inst. Dec.”), 28.

Thereafter, Patent Owner filed a Patent Owner Response (Paper 12, “PO Resp.”), to which Petitioner filed a Reply (Paper 15, “Reply”). Petitioner filed a Motion to Exclude (Paper 17). Patent Owner filed an Opposition (Paper 20) to which Petitioner filed a Reply (Paper 22).

On April 24, 2018, we held a hearing and a transcript of the hearing is included in the record. Paper 24 (“Tr.”).

On May 3, 2018, following the Supreme Court’s decision in *SAS Inst., Inc. v. Iancu*, 138 S. Ct. 1348 (2018), we issued an Order (Paper 23) modifying our Institution Decision to include review of all challenged claims and all grounds presented in the Petition, including these grounds on which we had previously not instituted (Pet. 3, 28):

References	Basis	Claim(s) challenged
Best and Tsunoda, with or without Roy	§ 103	2 and 8
Best and Roohparvar, ⁷ with or without Roy	§ 103	5 and 12–14
Best, Mills, ⁸ Bonella, and Ashmore, with or without Roy	§ 103	15

⁴ U.S. Patent No. 6,026,465 (Ex. 1010, “Mills”).

⁵ Although Petitioner does not include Mills (Ex. 1010) explicitly in its ground (Pet. 3), we include it because Petitioner’s analysis relies upon it (*id.* at 61–64) for teaching part of a limitation.

⁶ U.S. Patent Publication No. 2007/0136523 A1 (Ex. 1013, “Bonella”).

⁷ U.S. Patent Publication No. 2005/0273548 A1 (Ex. 1019, “Roohparvar”).

⁸ Although Petitioner does not include Mills (Ex. 1010) explicitly in its

In our order, we also stated that, “If, after conferring, the parties wish to submit further briefing, the parties must, within one week of the date of this Order, request a conference call with the panel to seek authorization for such briefing.” Paper 23, 2. Neither party requested a conference call with the panel.

B. Related Proceedings

The parties indicate that the ’831 patent is the subject of several district court cases and related *inter partes* reviews. Pet. 2; Paper 4, 3.

C. The ’831 patent (Ex. 1001)

The ’831 patent, titled “Flash-Dram Hybrid Memory Module,” issued October 28, 2014, from U.S. Patent Application No. 13/559,476. Ex. 1001 at [54], [45], [21]. The ’831 patent generally relates to a memory module with a non-volatile memory, a volatile memory, and a data manager through which the volatile memory and non-volatile memory may exchange data, and a controller to receive read/write commands from a memory controller hub (“MCH”) and transfer data between any two or more of the MCH, volatile memory, and non-volatile memory. *Id.* at Abstract.

ground (Pet. 3), we include it because Petitioner’s analysis relies upon it (*id.* at 61–64) for teaching part of a limitation.

Figure 4A is reproduced below.

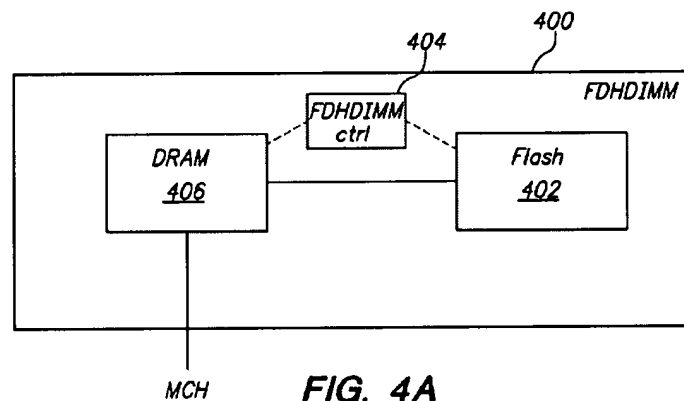


FIG. 4A

Figure 4A is a block diagram of a Flash-DRAM hybrid dynamic random access memory dual in-line memory module (DIMM). In this embodiment, volatile memory subsystem 406 (e.g. DRAM) is used as a data buffer such that data from Flash memory 402 is transferred to DRAM 406 at the Flash access speed, and buffered or collected into DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. *Id.* at 9:15–21. Similarly, when the MCH transfers data to DRAM 406, controller 404 manages the data transfer from DRAM 406 to Flash 402. *Id.* at 9:21–23.

Figure 5A is reproduced below.

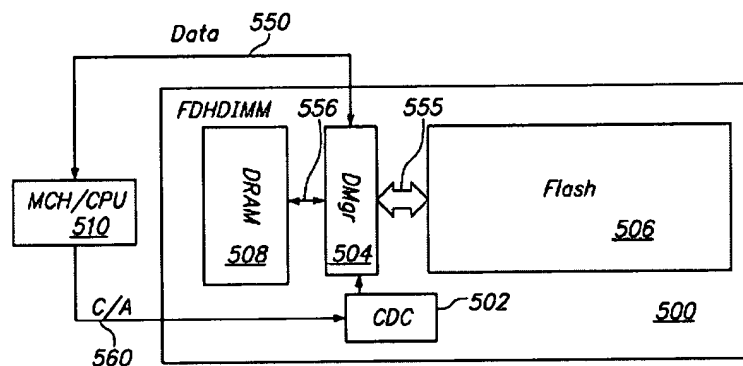


FIG. 5A

Figure 5A is a block diagram of memory module 500 in accordance with certain embodiments. Ex. 1001, 7:7–8. As shown in Figure 5, memory module 500 includes two on-module intermediary components: controller (CDC) 502 and data manager (DMgr) 504. *Id.* at 10:35–46. These components “manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.” *Id.* at 10:49–53. “In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510.” *Id.* at 10:54–56. “In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction.” *Id.* at 11:18–21.

Figure 6 is reproduced below.

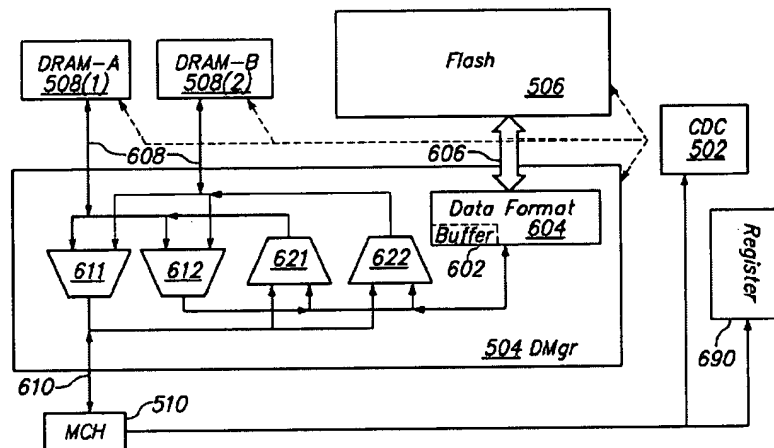


FIG. 6

Figure 6 is a block diagram showing some details of data manager 504. Ex. 1001, 7:11–12. “In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric.” *Id.* at 12:1–3. “For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM

508.” *Id.* at 12:3–5. “Multiplexers 611 and 612 provide controllable data paths from any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506.” *Id.* at 12:5–8. “Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B).” *Id.* at 12:8–11.

D. Illustrative Claim

Of the challenged claims, claims 1 and 7 are independent, claims 2–6 depend, directly or indirectly, from claim 1, and claims 8–15 depend, directly or indirectly, from claim 7. Independent claim 1 is illustrative of the challenged claims and is reproduced below:

1. A memory module couplable to a memory controller of a host system, comprising:
 - a non-volatile memory subsystem;
 - a data manager coupled to the non-volatile memory subsystem;
 - a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager; and
 - a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller, wherein:
 - at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments, each memory segment comprising at least one memory circuit, memory device, or memory die, and

the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports, a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystem, a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem, the two or more sets of data ports being operable by the data manager to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems, the data manager further including a data buffer for buffering data delivered to or from the non-volatile memory subsystem, and a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

Ex. 1001, 17:44–18:13.

E. Instituted Grounds of Unpatentability

Petitioner asserted that the challenged claims are unpatentable based on the following grounds (Pet. 3), and trial has been instituted on these grounds (*see supra* Section I.A):

Reference(s)	Basis	Claim(s) challenged
Best	§ 102	1–14
Best and Roy	§ 103	1–14
Best and Tsunoda, with or without Roy	§ 103	2 and 8
Best and Roohparvar, with or without Roy	§ 103	5 and 12–14
Best, Mills, ⁹ and Bonella, with or without Roy	§ 103	15
Best, Mills, ¹⁰ Bonella, and Ashmore, with or without Roy	§ 103	15

⁹ Petitioner does not include Mills (Ex. 1010) explicitly (Pet. 3), but the analysis relies upon it (*id.* at 61–64) for teaching part of a limitation.

¹⁰ *See* n.9, above.

II. ANALYSIS

A. Claim Construction

We interpret claims of an unexpired patent using the broadest reasonable construction in light of the specification of the patent in which they appear. *See* 37 C.F.R. § 42.100(b); *see Cuozzo Speed Techs., LLC v. Lee*, 136 S. Ct. 2131, 2142–46 (2016). Under the broadest reasonable construction standard, claim terms are given their ordinary and customary meaning, as would be understood by one of ordinary skill in the art in the context of the entire disclosure. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Any special definition for a claim term must be set forth in the specification with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994). We must be careful not to read a particular embodiment appearing in the written description into the claim if the claim language is broader than the embodiment. *See In re Van Geuns*, 988 F.2d 1181, 1184 (Fed. Cir. 1993). Only terms that are in controversy need to be construed, and then only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999).

1. “memory module”

Each independent claim recites a “memory module.” Petitioner did not propose a construction for “memory module” in the Petition. Patent Owner argued, in its Preliminary Response, that “memory module” should be construed to mean “a carrier that contains one or more chips.” Prelim. Resp. 21. In our Decision on Institution, we construed “memory module” to mean “a carrier that contains one or more memory chips,” where “carrier”

encompasses an integrated circuit package. Inst. Dec. 9–10. We observed that

The '831 patent does not define a “memory module.” The '831 patent depicts a memory module 500 in Figures 5A and 5B, and describes how “[i]n certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor” (Ex. 1001, 10:28–31), but the '831 patent also expressly contemplates other form factors (*see, e.g., id.* at 10:32–34 (“it is to be understood that in both structure and operation [memory module 500] may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B”), 10:46–49 (“While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well.”)).

Id.

In its Patent Owner Response, Patent Owner argues that “memory module” should be construed to mean “one or more memory segments on a printed circuit board” (“PCB”). PO Resp. 14–34. According to Patent Owner, “[e]very embodiment of a ‘memory module’ disclosed by the '831 patent comprises a PCB” and its proposed construction “parallels how the term ‘memory module’ is used by those in the industry and standards bodies.” *Id.* at 15. Specifically, Patent Owner annotates Figure 5A, arguing that it shows “a PCB (yellow)”:

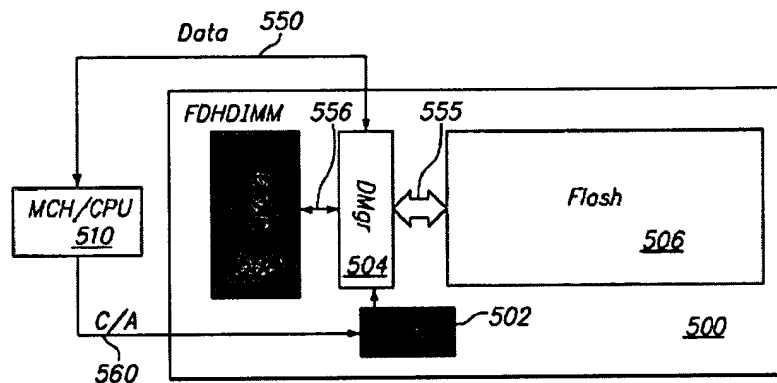


FIG. 5A

PO Resp. 16; *see also id.* at 20–23 (“Because the ’831 specification only discloses embodiments including memory segments on a PCB, the claimed ‘memory module’ should be construed to cover such embodiments.”). These arguments are not persuasive because the ’831 patent does *not* describe 500, or any other embodiment, as a PCB. Instead, the ’831 patent refers to 500 only as a “module.” Ex. 1001, 10:14–15.

Patent Owner also argues that the ’831 patent “repeatedly and consistently describes ‘memory modules’ as having a Dual In-Line Memory Module (‘DIMM’) form factor.” PO Resp. 17. As Petitioner points out, however, claims are not limited to the embodiments described in the Specification. Pet. Reply 3–4 (citing *ACUMED LLC v. Stryker Corp.*, 483 F.3d 800, 805 (Fed. Cir. 2007)). Moreover, as we noted above, the ’831 patent expressly contemplates other form factors for its “module” and, therefore, does not limit “module” to a DIMM form factor. *See, e.g.*, Ex. 1001, 10:32–49. The ’831 patent states, for example

These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for

integration known to those skilled in the art to achieve a specific design, application, or economic goal.

Id. at 10:36–43.

For the same reasons, we are unpersuaded by Patent Owner’s argument that boilerplate language cannot be used to broaden claims beyond what is disclosed in the Specification. PO Resp. 23–26.¹¹ The ’831 patent’s disclosure that “[w]hile the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well” (Ex. 1001, 10:46–49), is not mere boilerplate.

Patent Owner also relies upon disclosure in U.S. Patent Application No. 12/240,916 (“the ’916 application”), of which the ’831 patent is a continuation-in-part. PO Resp. 17–19; Ex. 1001 [63]. The ’916 patent explicitly describes an embodiment in which memory system 10 comprises PCB 20. Ex. 2018 ¶ 31. This argument is not persuasive, however, because when the application leading to the ’831 patent was filed, all references to “printed circuit board,” “PCB,” or even “board,” were deleted from the Specification and the more generic word “module” was used instead. That the applicant for the ’831 patent *removed* the words “PCB” and “printed circuit board” from the ’916 application and used only the word “module” instead suggests, if anything, that applicant intended “module” *not* to be limited to a PCB.

¹¹ Patent Owner’s citation to the “ID at 5–6” and the quote in parenthetical (Pet. 23–24) appears to be a citation to our Institution Decision in IPR2017-00560, not to our Institution Decision in this proceeding. Pet. Reply 10–11.

Patent Owner also relies upon a Final Written Decision in *Diablo Techs., Inc. v. Netlist, Inc.*, Case No. IPR2014-00882, concerning U.S. Patent 7,881,150 (“the ’150 patent”). PO Resp. 26–27; Ex. 2021. Patent Owner’s reliance on this decision is not persuasive, however, because the ’150 patent is not related to the ’831 patent and the Specification for the ’150 patent is different from the Specification of the ’831 patent. For example, the ’150 patent explicitly discloses a printed circuit board, whereas the ’831 patent does not.

Patent Owner also relies upon extrinsic evidence as support for its proposed construction. PO Resp. 27–31. There is no doubt that the DIMM form factor, including a PCB, was well-known. The issue, however, is whether the applicant for the ’831 patent intended the term “memory module” to require a PCB. In light of the intrinsic evidence that the applicant deleted the words “printed circuit board” and “PCB” from the Specification, explicitly contemplated non-DIMM form factors, and even contemplated a controller and data manager “integrated with other memory devices” (Ex. 1001, 10:39–40), Patent Owner’s extrinsic evidence does not persuade us that applicant intended “memory module” to require a PCB.

Patent Owner also argues that our preliminary determination that “carrier” encompasses an integrated circuit package is overly broad because (a) the ’831 patent “does not describe any IC package comprising a PCB” and (b) it lacks any support from the industry or standards bodies. PO Resp. 33–34. We disagree. As discussed, the ’831 patent explicitly contemplates integrating the controller and data manager “with other memory devices, for example in a three dimensional stack, or in any one of several other possible

expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal.” Ex. 1001, 10:36–43.

Finally, Patent Owner argues, with respect to Best, that our construction “requires two layers of IC packages, in which a first IC package (the Board’s ‘carrier’) contains one or more second IC packages (the Board’s memory chips).” PO Resp. 39. Petitioner counters that “[t]he Board’s construction does no such thing.” Pet. Reply 18–19. We agree with Petitioner. Patent Owner’s argument is predicated upon the erroneous assertion that “[e]ach memory chip in the Board’s ‘carrier’ requires a different IC package (*i.e.*, a second IC package) that encloses at least one memory die.” PO Resp. 39. Each memory chip does *not* require its own integrated circuit package. On the contrary, it may be, as Best discloses, a die enclosed with another die in a single IC package.

Having considered the arguments and evidence, we maintain our construction of “memory module” to mean “a carrier that contains one or more memory chips,” where “carrier” encompasses an integrated circuit package.

B. Level of Ordinary Skill in the Art

Petitioner contends that a hypothetical person of ordinary skill in the art, with respect to and at the time of the ’831 patent, “would be a person with a Bachelor’s degree in materials science, electrical engineering, computer engineering, computer science, or in a related field and at least one year of experience with the design or development of semiconductor non-volatile memory circuitry or systems.” Pet. 7; Ex. 1003 ¶¶ 55–56.

Patent Owner contends that such a person “would have been a person with a Bachelor’s degree in electrical and/or computer engineering and at

least five years of industry experience designing memory devices and controllers” but that “[a] Master of Science degree in electrical and/or computer engineering would substitute for two years of industry experience” and “[a] Doctorate degree in electrical and/or computer engineering would substitute for four years of industry experience.” PO Resp. 43; Ex. 2016 ¶ 32.

Patent Owner argues that a degree in material science would *not* be equivalent to a degree in electrical and/or computer engineering, and that Petitioner’s expert, Mr. Maltiel, who has only a degree in materials science, is not competent to testify to the understanding of a person of ordinary skill in the art. PO Resp. 43–44. Patent Owner did not, however, move to exclude the testimony of Mr. Maltiel. To the extent Patent Owner is arguing that we should accord Mr. Maltiel’s testimony little to no weight based on his qualifications, we decline to do so. After earning three degrees, Mr. Maltiel joined Intel in 1980 to work on the first commercial non-volatile EEPROM devices and, therefore, had 27 years of industry experience as of the earliest priority date to which the ’831 patent could be entitled. Ex. 1003 ¶¶ 3–9. Mr. Maltiel was recognized as a Senior Member of the IEEE in 2008 and is a named inventor on six United States patents covering various aspects of memory devices, including for combining non-volatile and DRAM memories. *Id.* ¶¶ 10–12. As a result, we are persuaded that Mr. Maltiel is qualified to testify as to the understanding of a person of ordinary skill in the art at the time of the ’831 patent.

Otherwise, we determine that no express finding on a specific corresponding level of technical education and experience is necessary. Here, the level of ordinary skill in the art is reflected by the prior art of

record. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001); *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995); *In re Oelrich*, 579 F.2d 86, 91 (CCPA 1978).

C. Whether to Give Weight to Mr. Maltiel's Testimony

Patent Owner argues that Mr. Maltiel's testimony should be given no weight because it is neither reliable nor credible. PO Resp. 41–42; *see also id.* at 62–64. In support of the argument, Patent Owner directs attention to portions of Dr. Maltiel's cross examination testimony where he allegedly (1) "contradicted himself on critical matters;" (2) "made a number of statements that are simply incorrect;" and (3) "concede[d] that he did not consider the complete intrinsic record, including the '916 patent application." *Id.*

Petitioner counters that Mr. Maltiel is qualified, reliable, and credible. Pet. Reply 24–25.

We have reviewed the arguments provided by Patent Owner and determine such arguments are insufficient to have Mr. Maltiel's declaration excluded in its entirety. Rather, it is within our discretion to assign the appropriate weight to be accorded evidence. *See* 37 C.F.R. § 42.65(a); *see also, e.g., Yorkey v. Diab*, 601 F.3d 1279, 1284 (Fed. Cir. 2010) (holding the Board has discretion to give more weight to one item of evidence over another "unless no reasonable trier of fact could have done so"); *In re Am. Acad. of Sci. Tech Ctr.*, 367 F.3d 1359, 1368 (Fed. Cir. 2004) ("[T]he Board is entitled to weigh the declarations and conclude that the lack of factual corroboration warrants discounting the opinions expressed in the declarations."); and *Velandar v. Garner*, 348 F.3d 1359, 1371 (Fed. Cir. 2003) ("In giving more weight to prior publications than to subsequent conclusory statements by experts, the Board acted well within [its]

discretion.”). Based on the record before us, we are not persuaded that we should give the entirety of Mr. Maltiel’s declaration no weight.

D. The Parties’ Post-Institution Arguments

In our Decision on Institution, we concluded that the arguments and evidence advanced by Petitioner demonstrated a reasonable likelihood that (1) claims 1–14 of the ’831 patent are unpatentable under 35 U.S.C. § 102 as anticipated by Best; (2) claims 1–14 under 35 U.S.C. § 103(a) are obvious over Best; (3) claims 1–14 under 35 U.S.C. § 103(a) are obvious over Best and Roy; and (4) claim 15 over Bowie under 35 U.S.C. § 103(a) is obvious over Best, Mills, and Bonella, with or without Roy. Inst. Dec. 28. We subsequently instituted on the remaining grounds asserted by Petitioner. Paper 23. We must now determine whether Petitioner has established by a preponderance of the evidence that claims 1–15 are unpatentable over the cited prior art. 35 U.S.C. § 316(e). We previously instructed Patent Owner that “any arguments for patentability not raised in the [Patent Owner Response] will be deemed waived.” Paper 3, 3; *see also* 37 C.F.R. § 42.23(a) (“Any material fact not specifically denied may be considered admitted.”); *In re Nuvasive, Inc.*, 842 F.3d 1376, 1379–1382 (Fed. Cir. 2016) (holding Patent Owner waived an argument addressed in Preliminary Response by not raising the same argument in the Patent Owner Response). Additionally, the Board’s Trial Practice Guide states that the Patent Owner Response “should identify all the involved claims that are believed to be patentable and state the basis for that belief.” Office Patent Trial Practice Guide, 77 Fed. Reg. 48,756, 48,766 (Aug. 14, 2012).

With a complete record before us, we note that we have reviewed arguments and evidence advanced by Petitioner to support its unpatentability

contentions where Patent Owner chose not to address certain limitations in its Patent Owner Response. In this regard, the record now contains persuasive, unrebutted arguments and evidence presented by Petitioner regarding the manner in which the asserted prior art teaches corresponding limitations of the claims against which that prior art is asserted. Based on the preponderance of the evidence before us, we conclude that the prior art identified by Petitioner teaches or suggests all uncontested limitations of the reviewed claims. The limitations that Patent Owner contests in the Patent Owner Response are addressed below.

E. Claims 1–14: Anticipation by Best

Petitioner argues that claims 1–14 are unpatentable under 35 U.S.C. § 102(e) as anticipated by Best. Pet. 20–49.

1. Principles of Law

To establish anticipation, “all of the elements and limitations of the claim must be shown in a single prior reference, arranged as in the claim.” *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). When evaluating a single prior art reference in the context of anticipation, the reference must be “considered together with the knowledge of one of ordinary skill in the pertinent art.” *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994) (citing *In re Samour*, 571 F.2d 559, 562 (CCPA 1978)). “[T]he dispositive question regarding anticipation[, therefore, i]s whether *one skilled in the art* would reasonably understand or infer from the [prior art reference’s] teaching’ that every claim element was disclosed in that single reference.” *Dayco Prods., Inc. v. Total Containment, Inc.*, 329 F.3d 1358, 1368 (Fed. Cir. 2003) (alterations in original) (quoting *In re Baxter Travenol Labs.*, 952 F.2d 388, 390 (Fed. Cir. 1991)). We analyze

this asserted ground based on anticipation with the principles stated above in mind.

2. Best Overview

Best is directed to a hybrid volatile and non-volatile memory device. Ex. 1006, Abstract. Specifically, Best discloses “[a] composite, hybrid memory device including a first storage die having an array of volatile storage cells and a second storage die having an array of non-volatile storage cells disposed within an integrated circuit package.” *Id.* “The hybrid memory device includes a shared interface circuit to receive memory access commands directed to the first storage die and the second storage die and to convey read and write data between an external data path and the first and second storage dice.” *Id.*

Figure 2 of Best is reproduced below.

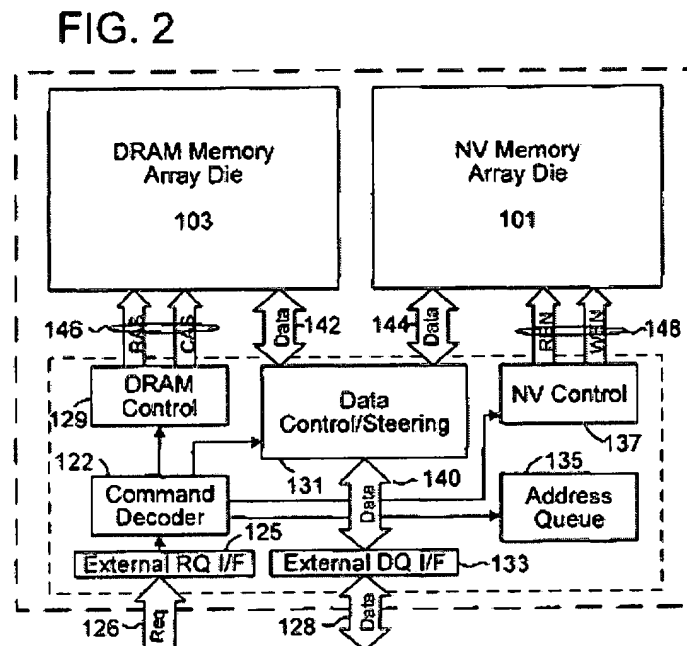


Figure 2 illustrates an embodiment of a hybrid, composite memory device with shared interface circuitry, volatile memory, and non-volatile memory.

Id. ¶ 17. “[T]he shared interface circuitry includes an external request interface 125, external data interface 133, command decoder 122, address queue 135, DRAM control circuit 129, Flash control circuit 137, and data control/steering circuit 131.” *Id.* “[I]ncoming control signals and addresses . . . are received in the external request interface 125 via control/address (CA) path 126, reformatted as necessary (e.g., deserialized to form a parallel command word and one or more address values) and then forwarded to the command decoder 122.” *Id.* “The command decoder 122 in turn forwards address to the address queue 135 and stores memory access commands.” *Id.* “[M]emory access operations may be automatically directed to either the volatile storage die or non-volatile storage die according to the memory address to be accessed.” *Id.* “[C]ommand decoder 122 outputs, from the head of the command queue, an enable signal and corresponding memory access control signals to the DRAM control circuit 129 and NV control circuit 137.” *Id.* ¶ 18. “[D]ata control/steering circuit 131 is used to control the transfer of data between a shared internal data bus and dedicated internal data buses associated with the volatile and non-volatile storage dice, respectively.” *Id.* ¶ 20.

Figure 3 of Best is reproduced below.

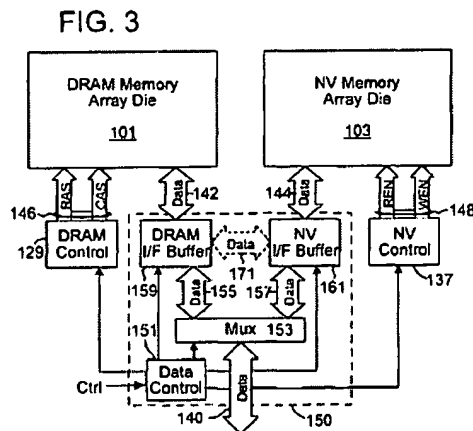


Figure 3 illustrates an embodiment of a data control/steering circuit 150 that may be used to implement the data control/steering circuit 131 of Figure 2.

Id. ¶ 21. “[D]ata control circuit 151 receives control signals from the command decoder that indicate the direction of data flow during a memory access operation (read or write) and whether the volatile or non-volatile storage die is the target of the memory access.” *Id.*

3. Petitioner’s Initial Positions

Petitioner contends that Best anticipates claims 1–14 of the ’831 patent. Pet. 20–49. We have reviewed the Petition, Patent Owner’s Response, and Petitioner’s Reply, as well as the relevant evidence discussed in those papers and other record papers, and are persuaded that the record establishes Petitioner’s contentions for claims 1–14, and we adopt Petitioner’s contentions discussed below as our own.

For example, Claim 1 recites “[a] memory module couplable to a memory controller of a host system.” Petitioner relies upon Best’s disclosure of “a ‘hybrid composite memory device having non-volatile and volatile memories implemented in distinct integrated circuit (IC) dice that are packaged together and accessed through a shared interface.’” Pet. 20 (quoting Ex. 1006 ¶ 12). In light of our construction of “memory module” to mean “a carrier that contains one or more memory chips,” where “carrier” encompasses an integrated circuit package, we are persuaded by Petitioner’s showing and find that Best’s integrated circuit package containing first and second storage die teaches the recited “memory module.”

Claim 1 further recites “a non-volatile memory subsystem.”

Petitioner relies upon Best’s disclosure of a Flash memory. Pet. 21. We are

persuaded by Petitioner's showing and find that Best's Flash memory teaches the recited "non-volatile memory subsystem."

Claim 1 further recites "a data manager coupled to the non-volatile memory subsystem." Petitioner relies upon Best's data control/steering circuit in combination with the external interface. Pet. 21–22 (citing Ex. 1006, Figure 3). We are persuaded by Petitioner's showing and find that Best's data control/steering circuit in combination with the external interface teaches the recited "data manager."

Claim 1 further recites "a volatile memory subsystem." Petitioner relies upon Best's disclosure of a DRAM. Pet. 22–23. We are persuaded by Petitioner's showing and find that Best's DRAM teaches this limitation.

Claim 1 further recites "a controller." Petitioner relies upon Best's command decoder 122, which receives "incoming control signals and addresses" and directs operation of the volatile and non-volatile memories by "output[ting] . . . an enable signal and corresponding memory access control signals to the DRAM control circuit . . . and NV control circuit," and transfers data between the memories and memory controller. Pet. 23–24 (citing Ex. 1006 ¶¶ 17–19, 21, 29). We are persuaded by Petitioner's showing and find that Best's command decoder 122 teaches the recited "controller."

Claim 1 further recites "at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments." Petitioner relies upon Best's disclosure that "the volatile and non-volatile memories [are] implemented by a DRAM die 103 and Flash memory die 101, respectively," each of which, according to Petitioner, is a memory segment. Pet. 24–25. We are persuaded by Petitioner's showing and find that Best's

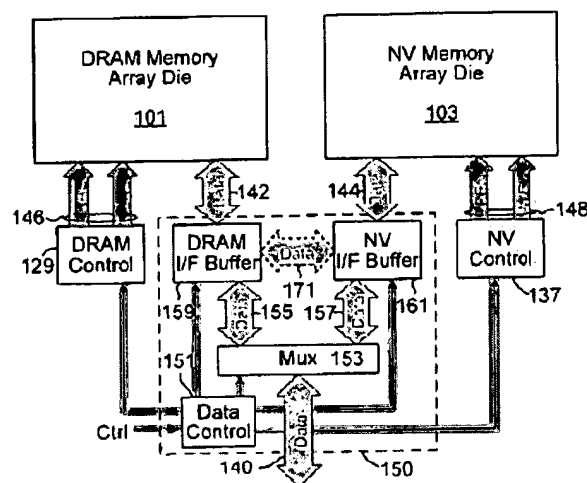
DRAM die 103 and Flash memory die 101 teach the recited “one or more memory segments.”

Claim 1 further recites

the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports, a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystem, a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem, the two or more sets of data ports being operable by the data manager to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

Petitioner relies upon Best’s disclosure of an interface to the primary volatile data path 142 between data control/steering circuit 150 and DRAM 101 (“a first set of data ports . . . coupled to the volatile memory subsystem”), and of an interface to primary non-volatile data path 144 between data control steering circuit 150 and NV memory 103 (“a second set of data ports . . . coupled to the non-volatile memory subsystem”). Pet. 25–26. Petitioner contends that each set of interconnections is “bi-directional” because data can flow in either direction, and illustrates the point with an annotated version of Figure 3, reproduced below:

FIG. 3



We are persuaded by Petitioner's showing and find that Best's interfaces to primary volatile data path 142 and to primary non-volatile data path 144, respectively, as illustrated in annotated Figure 3 above, teach the recited "two or more sets of data ports."

Finally, claim 1 recites

the data manager further including a data buffer for buffering data delivered to or from the non-volatile memory subsystem, and a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

Petitioner relies upon Best's disclosure of "non-volatile-storage-die interface buffer 161" as the recited "data buffer" (Pet. 27–28), and relies upon Best's disclosure of logic for serializing/deserializing within data steering/control circuit and the external data interface as the recited "data format module" (*id.* at 28–30). We are persuaded by Petitioner's showing and find that Best's non-volatile-storage-die interface buffer 161 teaches the recited "data buffer."

We also are persuaded that Petitioner's citations support its contentions with respect to claim 7, for which Petitioner relies upon largely the same disclosures in Best (Pet. 38–43).

Petitioner has provided a similar detailed analysis of claims 2–6 and 8–14. Pet. 30–37, 44–49. Notwithstanding Patent Owner's arguments, which we have considered and which we address below, we are persuaded by Petitioner's showing, which we adopt as our own findings and conclusions, that claims 1–14 are unpatentable under 35 U.S.C. § 102 as anticipated by Best.

*4. Patent Owner's
Assertions Concerning Best*

Patent Owner argues that Best does not disclose a “memory module” because it is implemented on a single integrated circuit package—i.e., a single chip. PO Resp. 34–41. Specifically, Patent Owner argues that (1) Petitioner’s expert testified that a “memory module” and an “IC package” are “totally different terms;” (2) Best’s dual-die chip teaches away from a memory module; (3) Best does not teach an IC package within another IC package; and (4) Best does not teach a PCB. *Id.*

Petitioner counters that (1) Mr. Maltiel testified specifically that Best’s specific disclosure of an IC package teaches a “memory module” as a person of ordinary skill in the art would have understood it; and (2) Patent Owner’s alleged “teaching away” is irrelevant in an anticipation analysis. Pet. Reply 17–18. We agree.

Moreover, Patent Owner’s remaining arguments are not persuasive because they are based upon Patent Owner’s proposed construction of “memory module,” which we do not adopt for the reasons discussed above. Because our broadest reasonable construction of “memory module” encompasses an integrated circuit package and because Best discloses an integrated circuit package with two die, we are persuaded that Best discloses this limitation.

Patent Owner does not separately argue claims 2–14.

5. Summary

For the foregoing reasons, we are persuaded that Petitioner has established, by a preponderance of the evidence, claims 1–14 of the ’831 patent are unpatentable under 35 U.S.C. § 102 as anticipated by Best.

F. Claim 1–14: Obviousness over Best

Although Petitioner challenges claims 1–14 as obvious over “Best in view of Roy” (Pet. 3), its other obviousness challenges are “with or without Roy” (*id.*). We, therefore, interpret the Petition to allege that claims 1–14 also are obvious over Best alone.

As discussed above, we are persuaded that Petitioner has established, by a preponderance of the evidence, that claims 1–14 are unpatentable as anticipated by Best. Patent Owner relies upon the same argument as for Ground 1—i.e., that Best does not teach a “memory module”—and argues that the Petition fails to show that Roy teaches a “memory module.” PO Resp. 45. We are not persuaded by that argument for the reasons discussed above.

Inasmuch as “anticipation is the epitome of obviousness” (*In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002)) and because there is no evidence of objective indicia of non-obviousness on the record, we determine that Petitioner has also established, by a preponderance of the evidence, that claims 1–14 would have been obvious over Best.

G. Claims 1–14: Obviousness over Best and Roy

Petitioner argues that claims 1–14 are unpatentable under 35 U.S.C. § 103(a) as obvious over the combination of Best and Roy. Pet. 49–53.

1. Principles of Law

A claim is unpatentable under § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.

KSR Int'l Co. v. Teleflex Inc., 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) when in evidence, objective indicia of non-obviousness (i.e., secondary considerations). *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). We analyze this asserted ground based on obviousness with the principles identified above in mind.

2. Roy Overview

Roy is directed generally to an “independent and cooperative multichannel memory architecture” that includes a plurality of independent channels between a master device and one or more memory clusters. Ex. 1008, Abstract. Figure 1 of Roy is reproduced below.

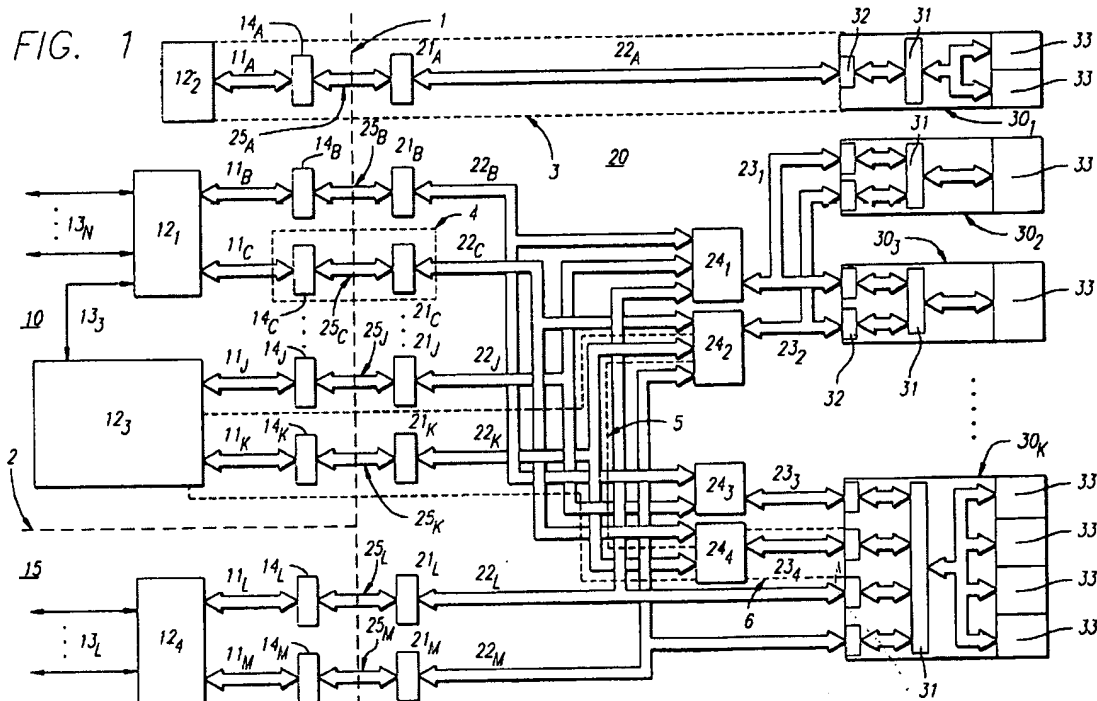


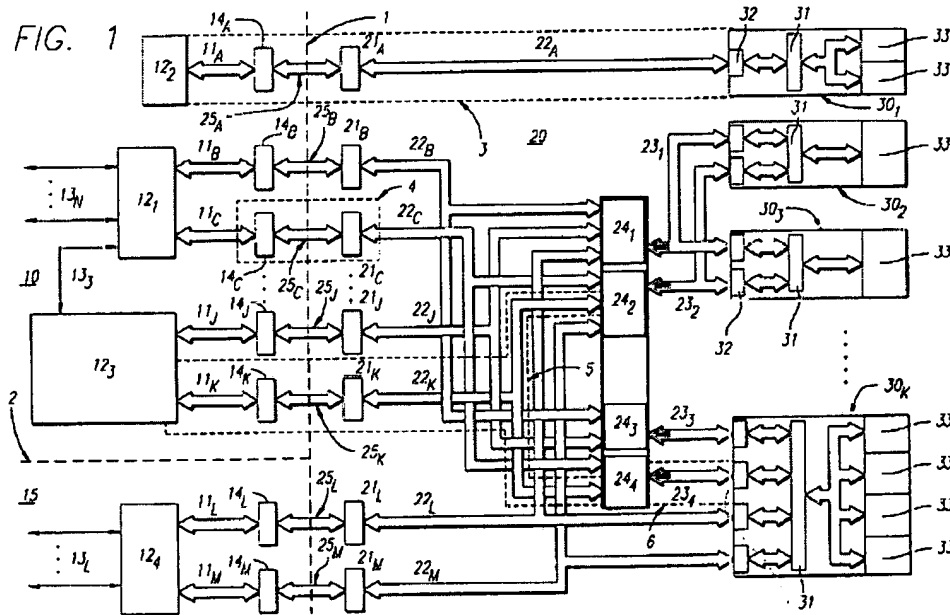
Figure 1 of Roy is a system level block diagram of a multichannel memory architecture disclosed in Roy. *Id.* at 7:59–60. As shown, “memory device 20 further includes multiplexer units 24_{1–4} which couple individual ones of a plurality of the channels to one or more of the memory clusters 30.” *Id.* at 11:52–54.

3. Analysis

Petitioner contends that, to the extent the claims require “two or more” independent read or write paths to the “volatile memory subsystem” and “non-volatile memory subsystem,” a combination of Best and Roy would have rendered obvious claims 1–14 of the ’831 patent. Pet. 49–53. We have reviewed the Petition, Patent Owner’s Response, and Petitioner’s Reply, as well as the relevant evidence discussed in those papers and other record papers, and are persuaded that the record establishes Petitioner’s contentions for claims 1–14, and we adopt Petitioner’s contentions discussed below as our own.

For example, independent claims 1 and 7 recite “a bi-directional data transfer fabric [having/with] two or more sets of data ports.” Petitioner argues that, if this phrase is construed to require two or more independent read/write paths to each of the volatile and non-volatile memory subsystems, such a feature was taught by Roy. Pet. 49–51. In particular, Petitioner relies upon Roy’s teaching of multiplexers 24_{1–4} (“a bi-directional data transfer fabric”), their interfaces to buses 23_{1–4} (“two or more sets of data ports”), some of which are coupled to memory cluster 30₃ (“a first set . . . coupled to”) and others of which are coupled to memory cluster 30_K (“a second set . . . coupled to”). Pet. 50–51. Petitioner also annotates Figure 1 of Roy to

indicate multiplexers (red), interfaces (green), and memory clusters to which they are coupled (yellow):



Id. at 50. An annotated version of Figure 1 of Roy is reproduced above.

We are persuaded by Petitioner’s showing and find that Roy’s multiplexers 24₁₋₄, buses 23₁₋₄, memory cluster 30₃, and memory cluster 30_K teach the recited “bi-directional data transfer fabric.”

With respect to why a person of ordinary skill in the art would have modified Best in view of Roy, Petitioner argues:

One of ordinary skill in the art would have been motivated to implement this architecture for all the reasons Roy describes, including allowing independent and simultaneous transactions, Ex. 1008, 7:37-40, and increased performance by providing a wide effective channel, *id.*, 7:45-49; Ex. 1003, ¶173. Roy also teaches that a multichannel architecture provides substantial flexibility. Ex. 1008, 9:30-42; Ex. 1003, ¶173.

Roy discloses that “nearly identical address and control information” can be applied to each channel such that “[s]ubsequent transfer[s] of data on each of these channels can be synchronized to provide an effectively wider channel.” Ex.

1008, 10:28-32. This provides particular motivation to combine with Best in light of Best’s disclosure that “multiple non-volatile storage dice and/or multiple volatile storage dice may be ... selected ... based on incoming address and/or control signals.” Ex. 1006, ¶15; Ex. 1003, ¶174.

Best suggests such a modification through his disclosure of overlapping and pipelined memory operations. Ex. 1006, ¶18. One of ordinary skill would understand that multiple channels allow for further overlapping or pipelining of operations, such as allowing Best to write data from volatile to non-volatile memory as part of the “Shadow Mode” operation while allowing the host to simultaneously write data to volatile memory, thus improving the operation and responsiveness of the system. Ex. 1003, ¶175.

Modifying Best to use a multichannel architecture such as Roy’s would have been an arrangement of old elements (Best’s hybrid memory, Roy’s multichannel architecture) with each performing the same function it had been known to perform and yielding no more than what one would expect from such an arrangement, *i.e.*, Best’s system with a multichannel architecture. Ex. 1003, ¶176. Multichannel architectures were known in the art, and using one in Best would have involved only routine skill to implement the functionality described by Roy. *Id.* Such a modification would have therefore been obvious. *Id.*, ¶¶176, 240.

Pet. 56–53. We determine that Petitioner has provided an articulated reasoning with some rational underpinning that would support the legal conclusion of obviousness. *See KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2017) (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)).

Patent Owner relies upon the same argument as for Ground 1—*i.e.*, that Best does not teach a “memory module”—and argues that the Petition fails to show that Roy teaches a “memory module.” PO Resp. 45. We are not persuaded by that argument for the reasons discussed above.

Having considered the arguments and evidence, we are persuaded by Petitioner's showing, which we adopt as our own findings and conclusions, that claims 1–14 are unpatentable under 35 U.S.C. § 103(a) as obvious over Best and Roy.

4. Summary

For the foregoing reasons, we are persuaded that Petitioner has established, by a preponderance of the evidence, that claims 1–14 of the '831 patent are unpatentable under 35 U.S.C. § 103(a) as obvious over Best and Roy.

H. Claim 15: Obviousness over Best, Mills, and Bonella, with or without Roy

Petitioner argues that claim 15 is unpatentable under 35 U.S.C. § 103(a) as obvious over Best, Mills, and Bonella, with or without Roy.¹² Pet. 57–69.

1. Bonella Overview

Bonella describes “A memory module including a volatile memory, a non-volatile memory, and a controller that provides address, data, and control interfaces to the memories and to a host system.” Ex. 1013, Abstract. Bonella teaches that, “[t]he memory module controller . . . is ‘Power State Aware.’” *Id.* ¶ 45. At “Power Level 4,” Bonella’s controller “reduces power by limiting the DRAM performance and the PCIe transaction performance.” *Id.* ¶ 48. Bonella teaches that “[r]eduction of power in the DRAM can be accomplished” by “reduc[ing] the frequency in

¹² Although Petitioner does not include Mills (Ex. 1010) explicitly in its ground (Pet. 3), we include it here because Petitioner’s analysis relies upon it (*id.* at 61–64) for teaching part of a limitation.

which the DRAM is operating,” which “reduces power and, in general, produces no noticeable decrease in system performance.” *Id.* ¶ 49.

2. Petitioner’s Initial Contentions

Petitioner contends that claim 15 of the ’831 patent is unpatentable as obvious over the combination of Best, Mills, and Bonella, and over the combination of Best, Mills, Roy, and Bonella. Pet. 57–68. We have reviewed the Petition, Patent Owner’s Response, and Petitioner’s Reply, as well as the relevant evidence discussed in those papers and other record papers, and are persuaded that the record establishes Petitioner’s contentions for claim 15, and we adopt Petitioner’s contentions discussed below as our own.

i. “first clock frequency”

For example, claim 15 recites

operating the volatile memory subsystem at a first clock frequency when the memory module is in a first mode of operation in which data is communicated between the volatile memory subsystem and the memory controller;

Ex. 1001, 20:12–15. Petitioner relies upon Best’s teaching of a conventional DRAM, which one of ordinary skill in the art would have understood operates at a first clock frequency. Pet. 58–60. We are persuaded by Petitioner’s showing and find that a person of ordinary skill in the art would have understood that Best’s DRAM operates at a first clock frequency.

ii. “second clock frequency”

Claim 15 further recites

operating the non-volatile memory subsystem at a second clock frequency when the memory module is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem;

Ex. 1001, 20:16–20. Petitioner concedes that Best does not explicitly disclose this limitation, but relies upon Mills’ teaching of a synchronous flash interface to argue that it would have been obvious to include this functionality in Best. Pet. 61 (citing Ex. 1010). Specifically, Petitioner argues that Mills teaches “a synchronous Flash interface where read and write operations are synchronized to the rising edge of a clock signal provided to the device and operating at a particular frequency.” *Id.* at 62. Petitioner also argues that “[a]s combined, Best’s Flash interface would conform to Mills’ synchronous Flash protocol and include a separate clock signal that controls read and write operations.” *Id.* at 63. We are persuaded by Petitioner’s showing and find that Mills teaches a synchronous Flash interface operating on a separate clock.

With respect to why a person of ordinary skill in the art would have combined Best and Mills, Petitioner argues

it would have been obvious to one of ordinary skill in the art to employ a synchronous flash memory, such as disclosed in Mills, in the system of Best because to do so would have been merely an arrangement of old elements with each performing the same function it had been known to perform and yielding no more than what one would expect from such an arrangement, *i.e.*, the non-volatile storage of data. . . .

A skilled artisan would have been motivated to make such a combination because, as Mills explains, a synchronous flash interface “creates an average access time for sequential read accesses that is significantly less than the access time of an asynchronous flash device.” Ex. 1010, 17:6-9. In the context of Best, restoring data from the non-volatile flash memory would therefore have been faster by use of a synchronous flash memory, and reduced sequential read access times during other operations or uses of Best’s Flash memory, motivating one of ordinary skill

in the art to use a synchronous interface generally. Ex. 1003, ¶ 299.

[O]ne of ordinary skill in the art would have therefore understood Best to suggest modification to work with any known Flash interface, including Mills' synchronous Flash interface. Ex. 1003, ¶300.

Pet. 62–64. We determine that Petitioner has provided an articulated reasoning with some rational underpinning that would support the legal conclusion of obviousness. *See KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2017) (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)).

iii. “third clock frequency”

Finally, claim 15 recites

operating the volatile memory subsystem at a third clock frequency when the memory module is in the second mode of operation, the third clock frequency being less than the first clock frequency.

Ex. 1001, 20:21–24. Petitioner concedes that Best does not explicitly disclose this limitation, but argues that Bonella teaches reducing DRAM operational frequency to reduce power. Pet. 66–67 (citing Ex. 1013 ¶ 50).

The cited portion of Bonella teaches

An improved power saving option is to slow the operating frequency of the device down and restrict the number of requests being serviced by the ExpressCard interface. By doing this a significant savings in power can be achieved. Voltage can be reduced to the DRAM, frequency is reduced to the DRAM, and the number of cycle requests to the DRAM are reduced making the device function in a very lower power state while maintaining a reasonable system performance level.

Ex. 1013 ¶ 50. We are persuaded by Petitioner's showing and find that Bonella teaches operating DRAM at a reduced frequency (i.e., the recited

“third clock frequency . . . less than the first clock frequency”) to conserve power (i.e., in the recited “second mode of operation”).

As for why a person of ordinary skill in the art would have modified Best in view of this knowledge, Petitioner states “(1) reducing power during volatile to non-volatile flush operations prompted by a power loss was a well-known technique, and (2) one known way to reduce the power consumption of DRAM devices was to reduce their frequency of operation.” Pet. 64–65 (citing Ex. 1011; Ex. 1013). As evidence that reducing power during volatile to non-volatile flush operations was well-known, Petitioner cites U.S. Patent Publication 2006/0212651 A1 (Ex. 1011, “Ashmore”), which “reduc[es] battery power consumption during a main power loss to reduce the likelihood of loss of user write-cached data in a write-caching mass storage controller” and U.S. Patent No. 7,421,552 (Ex. 1012, “Long”), which discloses that “if there is a loss of primary power 34, ... provid[ing] a significantly slower clock signal to ... the controller 40 while the controller 40 moves data from the volatile-memory storage cache 42 to the flash-based memory vault 44.” Pet. 65. As evidence that reducing power consumption of a DRAM by reducing its frequency of operation was well-known, Petitioner cites Bonella, which “explains that one way to reduce the power consumption of the memory module is to slow or reduce the operating frequency of the DRAM.” Pet. 66. Petitioner further states that

One of ordinary skill in the art would have been motivated to reduce the power consumption during Best’s write flushing in response to a power loss. A skilled artisan would have been motivated to perform this power reduction technique for all the reasons that were known in the art: e.g., decreasing the risk of data loss due to insufficient backup power (Ex. 1011, ¶7) and enabling the use of a smaller-sized backup power source (Ex.

1012, 4:54-64). Reducing power consumption during write flushing in response to a power loss would also have been the arrangement of old elements, each performing the same function it had been known to perform, in a way that yields no more than one of ordinary skill in the art would expect from such an arrangement (reducing power consumption during a power loss event, as suggested by Long and Ashmore). Ex. 1003, ¶307

...

One of ordinary skill in the art would also have found it obvious to reduce power consumption during Best's write flushing in response to a power loss using any known or conventional means, and would have also considered power consumption reduction techniques other than those of Ashmore and Long to obtain the same benefits, including those described in Bonella.

Pet. 66–67. On this record, we are persuaded that Petitioner has provided an articulated reasoning with some rational underpinning that would support the legal conclusion of obviousness. *See KSR*, 550 U.S. at 418 (citing *Kahn*, 441 F.3d at 988).

Notwithstanding Patent Owner's arguments, which we have considered and which we address below, we are persuaded by Petitioner's showing, which we adopt as our own findings and conclusions, that claim 15 is unpatentable under 35 U.S.C. § 103(a) as obvious over Best, Mills, and Bonella, and over the combination of Best, Mills, Roy, and Bonella.

3. Patent Owner's Assertions Concerning the References

Patent Owner relies upon the same argument as for Ground 1—i.e., that Best does not teach a “memory module”—and argues that the Petition fails to show that Roy teaches a “memory module.” PO Resp. 45. We are not persuaded by that argument for the reasons discussed above.

Patent Owner also argues that claim 15 is not obvious in view of Bonella's Power Level 4. PO Resp. 46–47. Specifically, Patent Owner argues that Bonella does not indicate “what frequency the DRAM is operating at during a power loss” because “there is no suggestion in Bonella about what frequency the DRAM runs at when powered by the UPS battery.” *Id.* at 46. Thus, concludes Patent Owner, Petitioner “fails to show evidence that operating in a ‘power saving’ mode prior to Bonella’s ‘power loss algorithm’ has any effect on the clock frequencies during the ‘power loss algorithm.’” *Id.* at 47.

Petitioner counters that it is relying upon Bonella's teaching to reduce power consumption by reducing DRAM operating frequency, not upon Bonella's Power Level 4 as a whole. Pet. Reply 21–22.

We agree with Petitioner. The Petition relies upon Bonella's teaching that reducing the operating frequency of a DRAM reduces power, and argues that a person of ordinary skill in the art would have known how, and been motivated, to apply that teaching to Best to reduce the power consumption of Best's DRAM during a power loss (i.e., in the “second mode of operation” recited in claim 15, when data is flushed from volatile memory subsystem to non-volatile memory subsystem) by operating it at a slower frequency (i.e., the “third clock frequency” recited in claim 15). Pet. 64–68. Although Bonella teaches frequency reduction in the context of Power Level 4, we are persuaded that a person of ordinary skill in the art would have known how to apply that teaching independent of the other aspects of Power Level 4 (e.g., limiting PCIe transaction performance). Pet. Reply 22.

Patent Owner also argues that claim 15 is not obvious in view of Petitioner's cited scenarios. PO Resp. 47–54. Specifically, Patent Owner

characterizes Petitioner's proposed combination as comprising "two scenarios [that] are independent of each other," one in which power is reduced to devices other than the volatile memory subsystem during a power loss, and another in which power is reduced to the volatile memory subsystem during normal operation (i.e., *not* during a power loss). *Id.* at 48. According to Patent Owner, Long teaches away from, and Ashmore does not teach, the "third clock frequency." PO Resp. 49–52.

These arguments are not persuasive because Petitioner relies upon Bonella, not upon Ashmore and Long, for teaching a "third clock frequency." Patent Owner also constructs a hypothetical system based on teachings of Ashmore, Long, and Bonella that is *not* relied upon by Petitioner and concludes that its hypothetical combination "does not reduce the DRAM clock frequency during power loss mode as Petitioner asserts." *Id.* at 52–53. This argument also is not persuasive because Petitioner is relying upon Long and Ashmore as evidence of what a person of ordinary skill in the art would have known and been motivated by (Pet. 65–68), not for teaching the "third clock frequency," much less for collateral teachings about the specific components in Long and Ashmore to which power is reduced during a power loss. Similarly, Petitioner relies upon Bonella's teaching that DRAM power consumption can be reduced by slowing the operating frequency down (Pet. 66–67 (Ex. 1013 ¶¶ 49–50)), not upon Bonella's teaching of a power-loss algorithm.

Patent Owner also argues that "Bonella does not teach that power (or frequency) is reduced during a power loss." *Id.* at 56. This argument is misplaced, however, because Petitioner is not relying upon Bonella to show power reducing frequency *during a power loss*. Petitioner is relying upon

Bonella's teaching that DRAM power consumption can be reduced by slowing the operating frequency down (Pet. 66–67 (Ex. 1013 ¶¶ 49–50)), and arguing that “[o]ne of ordinary skill in the art would have been motivated to reduce the power consumption during Best's write flushing in response to a power loss” (*id.* at 66).

Finally, Patent Owner argues a person of ordinary skill in the art would not have been motivated to combine Best with Bonella to “enable[e] the use of a smaller-sized backup power source,” as Petitioner contends (Pet. 66), because “[a person of ordinary skill in the art] would know that Petitioner's power reduction will not reduce the required battery capacity or increase the amount of data that can be transferred.” PO Resp. 60. Specifically, Patent Owner argues that “reducing the clock frequency . . . reduces power dissipation but simply spreads the task out over time, thereby consuming the same total energy.” *Id.* at 61. Thus, according to Patent Owner, “because Petitioner's power reduction technique extends the time required to back up the DRAM, energy requirements are not reduced.” *Id.* at 62.

Petitioner counters that Patent Owner's argument “ignores the energy consumption of an idle DRAM as it transfers data to a much-slower Flash memory.” Pet. Reply 22. Petitioner highlights Mr. Maltiel's testimony that “the transfer of data between DRAM and FLASH could not occur faster than the speed of the FLASH, which in the case of Best would be substantially slower than the DRAM.” Pet. Reply 23 (quoting Ex. 1003 ¶ 312). Patent Owner's declarant, Dr. Baker, testified that “accessing the memory in the big flash array takes considerably longer -- microseconds, as I testified -- than accessing memory in a DRAM, which could take, say, 10 to 50

nanoseconds, for example. So the access portion in a DRAM is considerably faster.” Ex. 1020, 15:14–19. Thus, according to Petitioner, reducing the operating frequency of the DRAM does *not* extend the time required to flush the DRAM to the Flash as long as the operating frequency of the DRAM is still faster than that of the Flash. *Id.* at 23–24.

We agree with Petitioner. Patent Owner’s argument assumes that the time to flush the DRAM to the Flash is directly proportional to the operating frequency of the DRAM. That process, however, is actually limited by the much slower operating frequency of the Flash memory. Ex. 1003 ¶ 312. We are persuaded by the arguments and evidence that a person of ordinary skill in the art would have known that reducing the operating frequency of the DRAM would enable the use of a smaller-sized backup power source because doing so reduces power consumption without extending the time required to flush the DRAM to the Flash.

4. Summary

For the foregoing reasons, we are persuaded that Petitioner has established, by a preponderance of the evidence, that claim 15 of the ’831 patent are unpatentable under 35 U.S.C. § 103(a) as obvious over Best, Mills, and Bonella, and over Best, Mills, Roy, and Bonella.

I. Other Grounds

Because claims 1–14 are unpatentable over Best, with or without Roy, and because claim 15 is unpatentable over Best, Mills, and Bonella, or Best, Mills, Roy, and Bonella, we need not reach Petitioner’s other grounds for unpatentability of these claims.

J. Petitioner's Motion to Exclude

Petitioner filed a Motion to Exclude (Paper 17). Patent Owner filed an Opposition (Paper 20, "Opp."), and Petitioner filed a Reply in support of its Motion (Paper 22). As movant, Petitioner has the burden of proof to establish that it is entitled to the requested relief. *See* 37 C.F.R. § 42.20(c).

We decline to assess the merits of Petitioner's Motion to Exclude. Even without excluding the identified evidence, we have concluded that Petitioner has demonstrated, by a preponderance of the evidence, that the challenged claims are unpatentable. Accordingly, Petitioner's Motion to Exclude is dismissed.

III. CONCLUSION

Petitioner has demonstrated, by a preponderance of the evidence, that (1) claims 1–14 of the '831 patent are unpatentable under 35 U.S.C. § 102 as anticipated by Best; (2) claims 1–14 are unpatentable under 35 U.S.C. § 103(a) as obvious over Best; (3) claims 1–14 are unpatentable under 35 U.S.C. § 103(a) as obvious over Best and Roy; and (4) claim 15 is unpatentable over Bowie under 35 U.S.C. § 103(a) as obvious over Best, Mills, and Bonella, with or without Roy.

IV. ORDER

Accordingly, it is

ORDERED that claims 1–15 of the '831 patent are held *unpatentable*;

FURTHER ORDERED that Petitioner's Motion to Exclude is *dismissed*; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2017-00692
Patent 8,874,831 B2

For PETITIONER:

Joseph A. Micallef
Samuel A. Dillon
SIDLEY AUSTIN LLP
jmicallef@sidley.com
samuel.dillon@sidley.com

For PATENT OWNER:

Thomas J. Wimbiscus
Christopher C. Winslade
Gregory C. Schodde
Scott P. McBride
Ronald H. Spuhler
Wayne H. Bradley
McANDREWS, HELD & MALLOY, LTD.
twimbiscus@mcandrews-ip.com
cwinslade@mcandrews-ip.com
gschodde@mcandrews-ip.com
smcbride@mcandrews-ip.com
rspuhler@mcandrews-ip.com
wbradley@mcandrews-ip.com



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY.DOCKET NO./TITLE	REQUEST ID
13/559,476	07/26/2012	Hyun Lee	0016.0010004	67267

Acknowledgement of Loss of Entitlement to Entity Status Discount

The entity status change request below filed through Private PAIR on 07/09/2018 has been accepted.

CERTIFICATIONS:

Change of Entity Status:

✕ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be notification of loss of entitlement to small or micro entity status, as applicable.

This portion must be completed by the signatory or signatories making the entity status change in accordance with 37 CFR 1.4(d)(4).

Signature:	/Khaled Shami/
Name:	Khaled Shami
Registration Number:	38745

(12) **INTER PARTES REVIEW CERTIFICATE** (1756th)

United States Patent
Lee et al.

(10) **Number:** **US 8,874,831 K1**
(45) **Certificate Issued:** **May 12, 2020**

(54) **FLASH-DRAM HYBRID MEMORY
MODULE**

(75) **Inventors: Hyun Lee; Chi-She Chen; Jeffrey C.
Solomon; Scott Milton; Jayesh
Bhakta**

(73) **Assignee: NETLIST, INC.**

Trial Number:

IPR2017-00692 filed Jan. 17, 2017

Inter Partes Review Certificate for:

Patent No.: **8,874,831**
Issued: **Oct. 28, 2014**
Appl. No.: **13/559,476**
Filed: **Jul. 26, 2012**

The results of IPR2017-00692 are reflected in this inter partes review certificate under 35 U.S.C. 318(b).

INTER PARTES REVIEW CERTIFICATE
U.S. Patent 8,874,831 K1
Trial No. IPR2017-00692
Certificate Issued May 12, 2020

1

2

AS A RESULT OF THE INTER PARTES
REVIEW PROCEEDING, IT HAS BEEN
DETERMINED THAT:

Claims **1-15** are cancelled.

5

* * * * *

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.040C1
		Application Number	
Title of Invention	NON-VOLATILE MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Chi-She		Chen		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Walnut	State/Province	CA	Country of Residence	US
Citizenship under 37 CFR 1.41(b)		TW			
Mailing Address of Applicant:					
Address 1	944 Crystal Water Lane				
Address 2					
City	Walnut	State/Province	CA		
Postal Code	91789	Country	US		
Applicant 2					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jeffrey	C.	Solomon		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Irvine	State/Province	CA	Country of Residence	US
Citizenship under 37 CFR 1.41(b)		US			
Mailing Address of Applicant:					
Address 1	6 Silver Fir				
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92604	Country	US		
Applicant 3					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Scott	H.	Milton		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Irvine	State/Province	CA	Country of Residence	US

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.040C1
		Application Number	
Title of Invention	NON-VOLATILE MEMORY MODULE		

Citizenship under 37 CFR 1.41(b)		US	
Mailing Address of Applicant:			
Address 1		49 Statehouse Place	
Address 2			
City	Irvine	State/Province	CA
Postal Code	92602	Country	US
Applicant 4			
Applicant Authority		☒ Inventor ☐ Legal Representative under 35 U.S.C. 117 ☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name
	Jayesh		Bhakta
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service			
City	Cerritos	State/Province	CA
Country of Residence		US	
Citizenship under 37 CFR 1.41(b)		US	
Mailing Address of Applicant:			
Address 1		12220 Rose Street	
Address 2			
City	Cerritos	State/Province	CA
Postal Code	90703	Country	US
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add			

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence Information of this application.	
Customer Number	20995
Email Address	efiling@kmob.com Add Email Remove Email

Application Information:

Title of the Invention	NON-VOLATILE MEMORY MODULE		
Attorney Docket Number	NETL.040C1	Small Entity Status Claimed ☐	
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	12	Suggested Figure for Publication (if any)	1

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.040C1
		Application Number	
Title of Invention	NON-VOLATILE MEMORY MODULE		

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☒	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	20995		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.			
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	12/131873	2008-06-02
Prior Application Status	Expired	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12/131873	non provisional of	60/941586	2007-06-01
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).			
Remove			
Application Number	Country ¹	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			

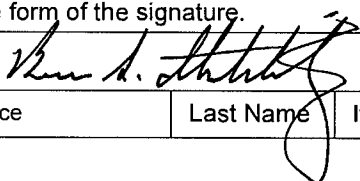
Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.040C1
		Application Number	
Title of Invention	NON-VOLATILE MEMORY MODULE		

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.			
Assignee 1			
If the Assignee is an Organization check here. ☒			
Organization Name	Netlist, Inc.		
Mailing Address Information:			
Address 1	51 Discovery		
Address 2	Suite 150		
City	Irvine	State/Province	CA
Country	US	Postal Code	92618
Phone Number		Fax Number	
Email Address			
Additional Assignee Data may be generated within this form by selecting the Add button.			

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature			Date (YYYY-MM-DD)	2008-09-29	
First Name	Bruce	Last Name	Itchkawitz	Registration Number	47677

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

NON-VOLATILE MEMORY MODULE**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] This application is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. Provisional Application No. 60/941,586, filed June 1, 2007. Each of these applications is incorporated in its entirety by reference herein.

BACKGROUND

[0002] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the computer system to provide volatile memory to the computer system.

[0003] Volatile memory generally maintains stored information only when it is powered. Batteries have been used to provide power to volatile memory during power failures or interruptions. However, batteries may require maintenance, may need to be replaced, are not environmentally friendly, and the status of batteries can be difficult to monitor.

[0004] Non-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory. In certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.

SUMMARY

[0005] In certain embodiments, a memory system coupled to a computer system is provided which includes a volatile memory subsystem, a non-volatile memory subsystem, and a controller operatively coupled to the non-volatile memory subsystem. The memory system can also include at least one circuit configured to selectively operatively decouple the controller from the volatile memory subsystem.

[0006] In some embodiments, a power module for providing a plurality of voltages to a memory system is described. The power module includes non-volatile and volatile memory, and the plurality of voltages include at least a first voltage and a second

voltage. The power module of certain embodiments includes an input providing a third voltage to the power module and a voltage conversion element configured to provide the second voltage to the memory system. The power module also includes a first power element configured to selectively provide a fourth voltage to the conversion element. The power module further includes a second power element configured to selectively provide a fifth voltage to the conversion element. The power module can be configured to selectively provide the first voltage to the memory system either from the conversion element or from the input.

[0007] The power module can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage is provided to the memory system from the input and the fourth voltage is provided to the conversion element from the first power element. In a second, state the fourth voltage is provided to the conversion element from the first power element and the first voltage is provided to the memory system from the conversion element. In a third state, the fifth voltage is provided to the conversion element from the second power element and the first voltage is provided to the memory system from the conversion element.

[0008] A method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems is provided in certain embodiments. The method includes, during a first condition, providing the first voltage to the memory system from an input power supply and providing the second voltage to the memory system from a first power subsystem. The method further includes detecting a second condition and, during the second condition, providing the first voltage and the second voltage to the memory system from the first power subsystem. The method also includes charging a second power subsystem and detecting a third condition. During the third condition, the method includes providing the first voltage and the second voltage to the memory system from the second power subsystem.

[0009] In certain embodiments, a method is provided for controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem. The method can include operating the volatile memory subsystem at a first frequency when the memory system is in a first mode of

operation in which data is communicated between the volatile memory subsystem and the host system. In certain embodiments, the method further includes operating the non-volatile memory subsystem at a second frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem. The method can also include operating the volatile memory subsystem at a third frequency when the memory system is in the second mode of operation, the third frequency less than the first frequency.

[0010] In certain embodiments, a method is provided for controlling a memory system operatively coupled to a host system. The memory system includes a volatile memory subsystem and a non-volatile memory subsystem. In certain embodiments, the method includes communicating data words between the volatile memory subsystem and the host system when the memory system is in a first mode of operation. The method can further include transferring data words from the volatile memory subsystem to the non-volatile memory subsystem when the memory system is in a second mode of operation. Transferring each data word can include storing a first portion of the data word in a buffer, storing a second portion of the data word in the buffer, and writing the entire data word from the buffer to the non-volatile memory subsystem.

[0011] A memory system operatively coupled to a host system is provided in certain embodiments. The memory system can include a volatile memory subsystem and a non-volatile memory subsystem comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system includes a controller operatively coupled to the volatile memory subsystem and operatively coupled to the non-volatile memory subsystem, the controller configured to allow data to be communicated between the volatile memory subsystem and the host system when the memory system is operating in a first state and to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem when the memory system is operating in a second state.

[0012] A method of controlling a memory system operatively coupled to a host system is provided in certain embodiments. The memory system includes a volatile memory subsystem and a non-volatile memory subsystem. The method can include communicating data between the volatile memory subsystem and the host system when the memory system is

in a first mode of operation. The method of certain embodiments further includes storing a first copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a first time when the memory system is in a second mode of operation. The method may further include restoring the first copy of data from the non-volatile memory subsystem to the volatile memory subsystem and erasing the first copy of data from the non-volatile memory subsystem. In certain embodiments, the method also includes storing a second copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a second time when the memory system is in the second mode of operation, wherein storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Figure 1 is a block diagram of an example memory system compatible with certain embodiments described herein.

[0014] Figure 2 is a block diagram of an example memory module with ECC (error-correcting code) having a volatile memory subsystem with nine volatile memory elements and a non-volatile memory subsystem with five non-volatile memory elements in accordance with certain embodiments described herein.

[0015] Figure 3 is a block diagram of an example memory module having a microcontroller unit and logic element integrated into a single device in accordance with certain embodiments described herein.

[0016] Figures 4A-4C schematically illustrate example embodiments of memory systems having volatile memory subsystems comprising registered dual in-line memory modules in accordance with certain embodiments described herein.

[0017] Figure 5 schematically illustrates an example power module of a memory system in accordance with certain embodiments described herein.

[0018] Figure 6 is a flowchart of an example method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems.

[0019] Figure 7 is a flowchart of an example method of controlling a memory system operatively coupled to a host system and which includes at least 100 percent more storage capacity in non-volatile memory than in volatile memory.

[0020] Figure 8 schematically illustrates an example clock distribution topology of a memory system in accordance with certain embodiments described herein.

[0021] Figure 9 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including operating a volatile memory subsystem at a reduced rate in a back-up mode.

[0022] Figure 10 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments of a volatile memory subsystem of a memory system to a controller of the memory system.

[0023] Figure 11 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including backing up and/or restoring a volatile memory subsystem in slices.

DETAILED DESCRIPTION

[0024] Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory, and a controller. The controller backs up the volatile memory using the non-volatile memory in the event of a trigger condition. Trigger conditions can include, for example, a power failure, power reduction, request by the host system, etc. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which does not comprise a battery and may include, for example, a capacitor or capacitor array.

[0025] In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example, one or more isolation devices may isolate the non-volatile memory and the controller from the volatile memory when the volatile memory is interacting with the host system and may allow communication between the volatile memory and the non-volatile memory when the data of the volatile memory is being restored or backed-up. This configuration generally

protects the operation of the volatile memory when isolated while providing backup and restore capability in the event of a trigger condition, such as a power failure.

[0026] In certain embodiments described herein, the memory system includes a power module which provides power to the various components of the memory system from different sources based on a state of the memory system in relation to a trigger condition (e.g., a power failure). The power module may switch the source of the power to the various components in order to efficiently provide power in the event of the power failure. For example, when no power failure is detected, the power module may provide power to certain components, such as the volatile memory, from system power while charging a secondary power source (e.g., a capacitor array). In the event of a power failure or other trigger condition, the power module may power the volatile memory elements using the previously charged secondary power source.

[0027] In certain embodiments, the power module transitions relatively smoothly from powering the volatile memory with system power to powering it with the secondary power source. For example, the memory system may power volatile memory with a third power source from the time the memory system detects that power failure is likely to occur until the time the memory system detects that the power failure has actually occurred.

[0028] In certain embodiments, the volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power. In some embodiments, during backup and/or restore operations, the volatile memory communicates with the non-volatile memory by writing and/or reading data words in bit-wise slices instead of by writing entire words at once. In certain embodiments, when each slice is being written to or read from the volatile memory the unused slice(s) of volatile memory is not active, which can reduce the power consumption of the system.

[0029] In yet other embodiments, the non-volatile memory can include at least 100 percent more storage capacity than the volatile memory. This configuration can allow the memory system to efficiently handle subsequent trigger conditions.

[0030] **Figure 1** is a block diagram of an example memory system 10 compatible with certain embodiments described herein. The memory system 10 can be coupled to a host

computer system and can include a volatile memory subsystem 30, a non-volatile memory subsystem 40, and a controller 62 operatively coupled to the non-volatile memory subsystem 40. In certain embodiments, the memory system 10 includes at least one circuit 52 configured to selectively operatively decouple the controller 62 from the volatile memory subsystem 30.

[0031] In certain embodiments, the memory system 10 comprises a memory module. The memory system 10 may comprise a printed-circuit board (PCB) 20. In certain embodiments, the memory system 10 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other volatile memory capacities are also compatible with certain embodiments described herein. In certain embodiments, the memory system 10 has a non-volatile memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, 16-GB, or 32-GB. Other non-volatile memory capacities are also compatible with certain embodiments described herein. In addition, memory systems 10 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 20 has an industry-standard form factor. For example, the PCB 20 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 20 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 20 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0032] In certain preferred embodiments, the memory system 10 is in electrical communication with the host system. In other embodiments, the memory system 10 may communicate with a host system using some other type of communication, such as, for example, optical communication. Examples of host systems include, but are not limited to, blade servers, 1U servers, personal computers (PCs), and other applications in which space is

constrained or limited. The memory system 10 can be in communication with a disk controller of a computer system, for example. The PCB 20 can comprise an interface 22 that is configured to be in electrical communication with the host system (not shown). For example, the interface 22 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 22 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 10 and the host system. For example, the interface 22 can comprise a standard 240-pin DDR2 edge connector.

[0033] The volatile memory subsystem 30 comprises a plurality of volatile memory elements 32 and the non-volatile memory subsystem 40 comprises a plurality of non-volatile memory elements 42. Certain embodiments described herein advantageously provide non-volatile storage via the non-volatile memory subsystem 40 in addition to high-performance (e.g., high speed) storage via the volatile memory subsystem 30. In certain embodiments, the first plurality of volatile memory elements 32 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 32 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of Figure 1, the first memory bank 30 comprises eight 64Mx8 DDR2 SDRAM elements 32. The volatile memory elements 32 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 32 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements 32 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0034] In certain embodiments, the second plurality of non-volatile memory elements 42 comprises one or more flash memory elements. Types of flash memory elements 42 compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, ONE-NAND flash, and multi-level cell (MLC). For example, in the block diagram of Figure 1, the second memory bank 40 comprises 512 MB of flash

memory organized as four 128Mbx8 NAND flash memory elements 42. In addition, non-volatile memory elements 42 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Non-volatile memory elements 42 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0035] **Figure 2** is a block diagram of an example memory module 10 with ECC (error-correcting code) having a volatile memory subsystem 30 with nine volatile memory elements 32 and a non-volatile memory subsystem 40 with five non-volatile memory elements 42 in accordance with certain embodiments described herein. The additional memory element 32 of the first memory bank 30 and the additional memory element 42 of the second memory bank 40 provide the ECC capability. In certain other embodiments, the volatile memory subsystem 30 comprises other numbers of volatile memory elements 32 (e.g., 2, 3, 4, 5, 6, 7, more than 9). In certain embodiments, the non-volatile memory subsystem 40 comprises other numbers of non-volatile memory elements 42 (e.g., 2, 3, more than 5).

[0036] Referring to Figure 1, in certain embodiments, the logic element 70 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 70 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 70 comprises an FPGA available from another vendor. The internal flash can improve the speed of the memory system 10 and save physical space. Other types of logic elements 70 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 70 is a custom device. In certain embodiments, the logic element 70 comprises various discrete electrical elements, while in certain other embodiments, the logic element 70 comprises one or more integrated circuits. **Figure 3** is a block diagram of an example memory module 10 having a microcontroller unit 60 and logic element 70 integrated into a single controller 62 in accordance with certain embodiments

described herein. In certain embodiments, the controller 62 includes one or more other components. For example, in one embodiment, an FPGA without an internal flash is used and the controller 62 includes a separate flash memory component which stores configuration information to program the FPGA.

[0037] In certain embodiments, the at least one circuit 52 comprises one or more switches coupled to the volatile memory subsystem 30, to the controller 62, and to the host computer (e.g., via the interface 22, as schematically illustrated by Figures 1-3). The one or more switches are responsive to signals (e.g., from the controller 62) to selectively operatively decouple the controller 62 from the volatile memory subsystem 30 and to selectively operatively couple the controller 62 to the volatile memory subsystem 30. In addition, in certain embodiments, the at least one circuit 52 selectively operatively couples and decouples the volatile memory subsystem 30 and the host system.

[0038] In certain embodiments, the volatile memory subsystem 30 can comprise a registered DIMM subsystem comprising one or more registers 160 and a plurality of DRAM elements 180, as schematically illustrated by **Figure 4A**. In certain such embodiments, the at least one circuit 52 can comprise one or more switches 172 coupled to the controller 62 (e.g., logic element 70) and to the volatile memory subsystem 30 which can be actuated to couple and decouple the controller 62 to and from the volatile memory subsystem 30, respectively. The memory system 10 further comprises one or more switches 170 coupled to the one or more registers 160 and to the plurality of DRAM elements 180 as schematically illustrated by **Figure 4A**. The one or more switches 170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 30 to the host system 150. In certain other embodiments, as schematically illustrated by **Figure 4B**, the one or more switches 174 are also coupled to the one or more registers 160 and to a power source 162 for the one or more registers 160. The one or more switches 174 can be selectively switched to turn power on or off to the one or more registers 160, thereby selectively operatively coupling the volatile memory subsystem 30 to the host system 150. As schematically illustrated by **Figure 4C**, in certain embodiments the at least one circuit 52 comprises a dynamic on-die termination (ODT) 176 circuit of the logic element 70. For example, the logic element 70 can comprise a dynamic ODT circuit 176 which selectively operatively couples and

decouples the logic element 70 to and from the volatile memory subsystem 30, respectively. In addition, and similar to the example embodiment of Figure 4A described above, the one or more switches 170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 30 to the host system 150.

[0039] Certain embodiments described herein utilize the non-volatile memory subsystem 40 as a flash “mirror” to provide backup of the volatile memory subsystem 30 in the event of certain system conditions. For example, the non-volatile memory subsystem 40 may backup the volatile memory subsystem 30 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the non-volatile memory subsystem 30 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 10 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

[0040] As schematically illustrated by Figures 1 and 2, in certain embodiments, the controller 62 may comprise a microcontroller unit (MCU) 60 and a logic element 70. In certain embodiments, the MCU 60 provides memory management for the non-volatile memory subsystem 40 and controls data transfer between the volatile memory subsystem 30 and the non-volatile memory subsystem 40. The MCU 60 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. As schematically illustrated by Figures 1 and 2, the logic element 70 of certain embodiments is in electrical communication with the non-volatile memory subsystem 40 and the MCU 60. The logic element 70 can provide signal level

translation between the volatile memory elements 32 (e.g., 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 42 (e.g., 3V TTL for NAND flash memory elements). In certain embodiments, the logic element 70 is also programmed to perform address/address translation between the volatile memory subsystem 30 and the non-volatile memory subsystem 40. In certain preferred embodiments, 1-NAND type flash are used for the non-volatile memory elements 42 because of their superior read speed and compact structure.

[0041] The memory system 10 of certain embodiments is configured to be operated in at least two states. The at least two states can comprise a first state in which the controller 62 and the non-volatile memory subsystem 40 are operatively decoupled (e.g., isolated) from the volatile memory subsystem 30 by the at least one circuit 52 and a second state in which the volatile memory subsystem 30 is operatively coupled to the controller 62 to allow data to be communicated between the volatile memory subsystem 30 and the non-volatile memory subsystem 40 via the controller 62. The memory system 10 may transition from the first state to the second state in response to a trigger condition, such as when the memory system 10 detects that there is a power interruption (e.g., power failure or reduction) or a system hang-up.

[0042] The memory system 10 may further comprise a voltage monitor 50. The voltage monitor circuit 50 monitors the voltage supplied by the host system via the interface 22. Upon detecting a low voltage condition (e.g., due to a power interruption to the host system), the voltage monitor circuit 50 may transmit a signal to the controller 62 indicative of the detected condition. The controller 62 of certain embodiments responds to the signal from the voltage monitor circuit 50 by transmitting a signal to the at least one circuit 52 to operatively couple the controller to the volatile memory system 30, such that the memory system 10 enters the second state. For example, the voltage monitor 50 may send a signal to the MCU 60 which responds by accessing the data on the volatile memory system 30 and by executing a write cycle on the non-volatile memory subsystem 40. During this write cycle, data is read from the volatile memory subsystem 30 and is transferred to the non-volatile memory subsystem 40 via the MCU 60. In certain embodiments, the voltage monitor circuit 50 is part of the controller 62 (e.g., part of the MCU 60) and the voltage monitor circuit 50

transmits a signal to the other portions of the controller 62 upon detecting a power threshold condition.

[0043] The isolation or operational decoupling of the volatile memory subsystem 30 from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 10 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 30. For example, in one embodiment during such periods of operation, the controller 62 and the non-volatile memory subsystem 40 do not add a significant capacitive load to the volatile memory system 30 when the memory system 10 is in the first state. In certain such embodiments, the capacitive load of the controller 62 and the non-volatile memory subsystem 40 do not significantly affect the signals propagating between the volatile memory subsystem 30 and the host system. This can be particularly advantageous in relatively high-speed memory systems where loading effects can be significant. In one preferred embodiment, the at least one circuit 52 comprises an FSA1208 Low-Power, Eight-Port, Hi-Speed Isolation Switch from Fairchild Semiconductor. In other embodiments, the at least one circuit 52 comprises other types of isolation devices.

[0044] Power may be supplied to the volatile memory subsystem 30 from a first power supply (e.g., a system power supply) when the memory system 10 is in the first state and from a second power supply 80 when the memory system 10 is in the second state. In certain embodiments, the memory system 10 is in the first state when no trigger condition (e.g., a power failure) is present and the memory system 10 enters the second state in response to a trigger condition. In certain embodiments, the memory system 10 has a third state in which the controller 62 is operatively decoupled from the volatile memory subsystem 30 and power is supplied to the volatile memory subsystem 30 from a third power supply (not shown). For example, in one embodiment the third power supply may provide power to the volatile memory subsystem 30 when the memory system 10 detects that a trigger condition is likely to occur but has not yet occurred.

[0045] In certain embodiments, the second power supply 80 does not comprise a battery. Because a battery is not used, the second power supply 80 of certain embodiments may be relatively easy to maintain, does not generally need to be replaced, and is relatively

environmentally friendly. In certain embodiments, as schematically illustrated by Figures 1-3, the second power supply 80 comprises a step-up transformer 82, a step-down transformer 84, and a capacitor bank 86 comprising one or more capacitors (e.g., double-layer capacitors). In one example embodiment, capacitors may take about three to four minutes to charge and about two minutes to discharge. In other embodiments, the one or more capacitors may take a longer time or a shorter time to charge and/or discharge. For example, in certain embodiments, the second power supply 80 is configured to power the volatile memory subsystem 30 for less than thirty minutes. In certain embodiments, the second power supply 80 may comprise a battery. For example, in certain embodiments, the second power supply 80 comprises a battery and one or more capacitors and is configured to power the volatile memory subsystem 30 for no more than thirty minutes.

[0046] In certain embodiments, the capacitor bank 86 of the second power supply 80 is charged by the first power supply while the memory system 10 is in the first state. As a result, the second power supply 80 is fully charged when the memory system 10 enters the second state. The memory system 10 and the second power supply 80 may be located on the same printed circuit board 20. In other embodiments, the second power supply 80 may not be on the same printed circuit board 20 and may be tethered to the printed circuit board 20, for example.

[0047] When operating in the first state, in certain embodiments, the step-up transformer 82 keeps the capacitor bank 86 charged at a peak value. In certain embodiments, the step-down transformer 84 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 32 and 3.0V to the non-volatile flash memory elements 42) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by Figures 1-3, the memory module 10 further comprises a switch 90 (e.g., FET switch) that switches power to the controller 62 the volatile memory subsystem 30, and the non-volatile memory subsystem 40 between the power from the second power supply 80 and the power from the first power supply (e.g., system power) received via the interface 22. For example, the switch 90 may switch from the first power supply to the second power supply 80 when the voltage monitor 50 detects a low voltage condition. The switch 90 of certain embodiments advantageously

ensures that the volatile memory elements 32 and non-volatile memory elements 42 are powered long enough for the data to be transferred from the volatile memory elements 32 and stored in the non-volatile memory elements 42. In certain embodiments, after the data transfer is complete, the switch 90 then switches back to the first power supply and the controller 62 transmits a signal to at least one circuit 50 to operatively decouple the controller 62 from the volatile memory subsystem 30, such that the memory system 10 re-enters the first state.

[0048] When the memory system 10 re-enters the first state, data may be transferred back from the non-volatile memory subsystem 40 to the volatile memory subsystem 30 via the controller 62. The host system can then resume accessing the volatile memory subsystem 30 of the memory module 10. In certain embodiments, after the memory system 10 enters or re-enters the first state (e.g., after power is restored), the host system accesses the volatile memory subsystem 30 rather than the non-volatile memory subsystem 40 because the volatile memory elements 32 have superior read/write characteristics. In certain embodiments, the transfer of data from the volatile memory bank 30 to the non-volatile memory bank 40, or from the non-volatile memory bank 40 to the volatile memory bank 30, takes less than one minute per GB.

[0049] In certain embodiments, the memory system 10 protects the operation of the volatile memory when communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure. In certain embodiments, the memory system 10 copies the entire contents of the volatile memory subsystem 30 into the non-volatile memory subsystem 40 on each backup operation. Moreover, in certain embodiments, the entire contents of the non-volatile memory subsystem 40 are copied back into the volatile memory subsystem 30 on each restore operation. In certain embodiments, the entire contents of the non-volatile memory subsystem 40 are accessed for each backup and/or restore operation, such that the non-volatile memory subsystem 40 (e.g., flash memory subsystem) is used generally uniformly across its memory space and wear-leveling is not performed by the memory system 10. In certain embodiments, avoiding wear-leveling can decrease cost and complexity of the memory system 10 and can improve the performance of the memory system 10. In certain other embodiments, the entire

contents of the volatile memory subsystem 30 are not copied into the non-volatile memory subsystem 40 on each backup operation, but only a partial copy is performed. In certain embodiments, other management capabilities such as bad-block management and error management for the flash memory elements of the non-volatile memory subsystem 40 are performed in the controller 62.

[0050] The memory system 10 generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem 30 which then writes the data to non-volatile storage which is not part of the memory system 10, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system 10 indicating that the data has been written to the hard disk or is otherwise secure. The memory system 10 of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system 10 will still be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system 10 may operated as a write-through cache or as some other type of cache.

[0051] **Figure 5** schematically illustrates an example power module 100 of the memory system 10 in accordance with certain embodiments described herein. The power module 100 provides power to the various components of the memory system 10 using different elements based on a state of the memory system 10 in relation to a trigger condition. In certain embodiments, the power module 100 comprises one or more of the components described above with respect to Figure 1. For example, in certain embodiments, the power module 100 includes the second power supply 80 and the switch 90.

[0052] The power module 100 provides a plurality of voltages to the memory system 10 comprising non-volatile and volatile memory subsystems 30, 40. The plurality of voltages comprises at least a first voltage 102 and a second voltage 104. The power module 100 comprises an input 106 providing a third voltage 108 to the power module 100 and a voltage conversion element 120 configured to provide the second voltage 104 to the memory system 10. The power module 100 further comprises a first power element 130 configured to

selectively provide a fourth voltage 110 to the conversion element 120. In certain embodiments, the first power element 130 comprises a pulse-width modulation power controller. For example, in one example embodiment, the first power element 130 is configured to receive a 1.8V input system voltage as the third voltage 108 and to output a modulated 5V output as the fourth voltage 110.

[0053] The power module 100 further comprises a second power element 140 can be configured to selectively provide a fifth voltage 112 to the conversion element 120. The power module 100 can be configured to selectively provide the first voltage 102 to the memory system 110 either from the conversion element 120 or from the input 106.

[0054] The power module 100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage 102 is provided to the memory system 10 from the input 106 and the fourth voltage 110 is provided to the conversion element 120 from the first power element 130. In a second state, the fourth voltage 110 is provided to the conversion element 120 from the first power element 130 and the first voltage 102 is provided to the memory system 10 from the conversion element 120. In the third state, the fifth voltage 112 is provided to the conversion element 120 from the second power element 140 and the first voltage 104 is provided to the memory system 10 from the conversion element 120.

[0055] In certain embodiments, the power module 100 transitions from the first state to the second state upon detecting that a trigger condition is likely to occur and transitions from the second state to the third state upon detecting that the trigger condition has occurred. For example, the power module 100 may transition to the second state when it detects that a power failure is about to occur and transitions to the third state when it detects that the power failure has occurred. In certain embodiments, providing the first voltage 102 in the second state from the first power element 130 rather than from the input 106 allows a smoother transition from the first state to the third state. For example, in certain embodiments, providing the first voltage 102 from the first power element 130 has capacitive and other smoothing effects. In addition, switching the point of power transition to be between the conversion element 120 and the first and second power elements 130, 140 (e.g.,

the sources of the pre-regulated fourth voltage 110 in the second state and the pre-regulated fifth voltage 112 in the third state) can smooth out potential voltage spikes.

[0056] In certain embodiments, the second power element 140 does not comprise a battery and may comprise one or more capacitors. For example, as schematically illustrated in Figure 4, the second power element 140 comprises a capacitor array 142, a buck-boost converter 144 which adjusts the voltage for charging the capacitor array and a voltage/current limiter 146 which limits the charge current to the capacitor array 142 and stops charging the capacitor array 142 when it has reached a certain charge voltage. In one example embodiment, the capacitor array 142 comprises two 50 farad capacitors capable of holding a total charge of 4.6V. For example, in one example embodiment, the buck-boost converter 144 receives a 1.8V system voltage (first voltage 108) and boosts the voltage to 4.3V which is outputted to the voltage current limiter 146. The voltage/current limiter 146 limits the current going to the capacitor array 142 to 1A and stops charging the array 142 when it is charged to 4.3V. Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the second power element 140 may include alternative embodiments. For example, different components and/or different value components may be used. For example, in other embodiments, a pure boost converter may be used instead of a buck-boost converter. In another embodiment, only one capacitor may be used instead of a capacitor array 142.

[0057] The conversion element 120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 120 may comprise a plurality of sub-blocks 122, 124, 126 as schematically illustrated by Figure 4, which can provide more voltages in addition to the second voltage 104 to the memory system 10. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 10. For example, in one embodiment, sub-block 122 comprises a buck converter, sub-block 124 comprises a dual buck converter, and sub-block 126 comprises a buck-boost converter as schematically illustrated by Figure 4. Various other components for the sub-blocks 122, 124, 126 of the conversion element 120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 120 receives

as input either the fourth voltage 110 from the first power element 130 or the fifth voltage 112 from the second power element 140, depending on the state of the power module 100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck-converter of sub-block 122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 32 (e.g., DRAM), the non-volatile memory elements 42 (e.g., flash), and the controller 62 (e.g., an FPGA) in one embodiment. The sub-block 124 can provide the second voltage 104 as well as another reduced voltage 105 to the memory system 10. In one example embodiment, the second voltage 104 is 2.5V and is used to power the at least one circuit 52 (e.g., isolation device) and the other reduced voltage 105 is 1.2V and is used to power the controller 62 (e.g., FPGA). The sub-block 126 can provide yet another voltage 107 to the memory system 10. For example, the voltage 107 may be 3.3V and may be used to power both the controller 62 and the at least one circuit 52.

[0058] Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the conversion element 120 may include alternative embodiments. For example, there may be more or less sub-blocks which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values. In one embodiment, the volatile memory elements 32 and non-volatile memory elements 42 are powered using independent voltages and are not both powered using the first voltage 102.

[0059] **Figure 6** is a flowchart of an example method 200 of providing a first voltage 102 and a second voltage 104 to a memory system 10 including volatile and non-volatile memory subsystems 30, 40. While the method 200 is described herein by reference to the memory system 10 schematically illustrated by Figures 1-4, other memory systems are also compatible with embodiments of the method 200. During a first condition, the method 200 at comprises providing the first voltage 102 to the memory system 10 from an input power supply 106 and providing the second voltage 104 to the memory system 10 from a first power subsystem in operational block 210. For example, in one embodiment, the first power subsystem comprises the first power element 130 and the voltage conversion element 120 described above with respect to Figure 4. In other embodiments, other first power subsystems are used.

[0060] The method 200 further comprises detecting a second condition in operational block 220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 200 comprises providing the first voltage 102 and the second voltage 104 to the memory system 10 from the first power subsystem in an operational block 230. For example, referring to Figure 4, a switch 148 can be toggled to provide the first voltage 102 from the conversion element 120 rather than from the input power supply.

[0061] The method 200 further comprises charging a second power subsystem in operational block 240. In certain embodiments, the second power subsystem comprises the second power element 140 or another power supply that does not comprise a battery. For example, in one embodiment, the second power subsystem comprises the second power element 140 and the voltage conversion element 120 described above with respect to Figure 4. In other embodiments, some other second power subsystem is used.

[0062] The method 200 further comprises detecting a third condition in an operational block 250 and during the third condition, providing the first voltage 102 and the second voltage 104 to the memory system 10 from the second power subsystem 140 in an operational block 250. In certain embodiments, detecting the third condition comprises detecting that the trigger condition has occurred. The trigger condition may comprise various conditions described herein. In various embodiments, for example, the trigger condition comprises a power reduction, power failure, or system hang-up. The operational blocks of the method 200 may be performed in different orders in various embodiments. For example, in certain embodiments, the second power subsystem 140 is charged before detecting the second condition.

[0063] In certain embodiments, the memory system 10 comprises a volatile memory subsystem 30 and a non-volatile memory subsystem 40 comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system 10 also comprises a controller 62 operatively coupled to the volatile memory subsystem 30 and operatively coupled to the non-volatile memory subsystem 40. The controller 62 can be configured to allow data to be communicated between the volatile memory subsystem 30 and the host system when the memory system 10 is operating in a first state and to allow data to

be communicated between the volatile memory subsystem 30 and the non-volatile memory subsystem 40 when the memory system 10 is operating in a second state.

[0064] Although the memory system 10 having extra storage capacity of the non-volatile memory subsystem 40 has been described with respect to certain embodiments, alternative configurations exist. For example, in certain embodiments, there may be more than 100 percent more storage capacity in the non-volatile memory subsystem 40 than in the volatile memory subsystem 30. In various embodiments, there may be at least 200, 300, or 400 percent more storage capacity in the non-volatile memory subsystem 40 than in the volatile memory subsystem 30. In other embodiments, the non-volatile memory subsystem 40 includes at least some other integer multiples of the storage capacity of the volatile memory subsystem 30. In some embodiments, the non-volatile memory subsystem 40 includes a non-integer multiple of the storage capacity of the volatile memory subsystem 30. In one embodiment, the non-volatile memory subsystem 40 includes less than 100 percent more storage capacity than does the volatile memory subsystem 30.

[0065] The extra storage capacity of the non-volatile memory subsystem 40 can be used to improve the backup capability of the memory system 10. In certain embodiments in which data can only be written to portions of the non-volatile memory subsystem 40 which do not contain data (e.g., portions which have been erased), the extra storage capacity of the non-volatile memory subsystem 40 allows the volatile memory subsystem 30 to be backed up in the event of a subsequent power failure or other trigger event. For example, the extra storage capacity of the non-volatile memory subsystem 40 may allow the memory system 10 to backup the volatile memory subsystem 30 efficiently in the event of multiple trigger conditions (e.g., power failures). In the event of a first power failure, for example, the data in the volatile memory system 30 is copied to a first, previously erased portion of the non-volatile memory subsystem 40 via the controller 62. Since the non-volatile memory subsystem 40 has more storage capacity than does the volatile memory subsystem 30, there is a second portion of the non-volatile memory subsystem 40 which does not have data from the volatile memory subsystem 30 copied to it and which remains free of data (e.g., erased). Once system power is restored, the controller 62 of the memory system 10 restores the data to the volatile memory subsystem 30 by copying the backed-up data from the non-volatile

memory subsystem 40 back to the volatile memory subsystem 30. After the data is restored, the memory system 10 erases the non-volatile memory subsystem 40. While the first portion of the non-volatile memory subsystem 40 is being erased, it may be temporarily un-accessible.

[0066] If a subsequent power failure occurs before the first portion of the non-volatile memory subsystem 40 is completely erased, the volatile memory subsystem 30 can be backed-up or stored again in the second portion of the non-volatile memory subsystem 40 as described herein. In certain embodiments, the extra storage capacity of the non-volatile memory subsystem 40 may allow the memory system 10 to operate more efficiently. For example, because of the extra storage capacity of the non-volatile memory subsystem 40, the memory system 10 can handle a higher frequency of trigger events that is not limited by the erase time of the non-volatile memory subsystem 40.

[0067] **Figure 7** is a flowchart of an example method 300 of controlling a memory system 10 operatively coupled to a host system and which includes a volatile memory subsystem 30 and a non-volatile memory subsystem 40. In certain embodiments, the non-volatile memory subsystem 40 comprises at least 100 percent more storage capacity than does the volatile memory subsystem 30 as described herein. While the method 300 is described herein by reference to the memory system 10 schematically illustrated by Figures 1-3, the method 300 can be practiced using other memory systems in accordance with certain embodiments described herein. In an operational block 310, the method 300 comprises communicating data between the volatile memory subsystem 30 and the host system when the memory system 10 is in a first mode of operation. The method 300 further comprises storing a first copy of data from the volatile memory subsystem 30 to the non-volatile memory subsystem 40 at a first time when the memory system 10 is in a second mode of operation in an operational block 320.

[0068] In an operational block 330, the method 300 comprises restoring the first copy of data from the non-volatile memory subsystem 40 to the volatile memory subsystem 30. The method 300 further comprises erasing the first copy of data from the non-volatile memory subsystem 40 in an operational block 340. The method further comprises storing a second copy of data from the volatile memory subsystem 30 to the non-volatile memory

subsystem 40 at a second time when the memory system 10 is in the second mode of operation in an operational block 350. Storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem 40.

[0069] In some embodiments, the memory system 10 enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the non-volatile memory subsystem 40. The method 300 can also include restoring the second copy of data from the non-volatile memory subsystem 40 to the volatile memory subsystem 30 in an operational block 360. The operational blocks of method 300 referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, the restoring the second copy of data to the volatile memory subsystem 30 at operational block 360 before the first copy of data is completely erased in the operational block 340.

[0070] **Figure 8** schematically illustrates an example clock distribution topology 400 of a memory system 10 in accordance with certain embodiments described herein. The clock distribution topology 400 generally illustrates the creation and routing of the clock signals provided to the various components of the memory system 10. A clock source 402 such as, for example, a 25 MHz oscillator, generates a clock signal. The clock source 402 may feed a clock generator 404 which provides a clock signal 406 to the controller 62, which may be an FPGA. In one embodiment, the clock generator 404 generates a 125 MHz clock signal 406. The controller 62 receives the clock signal 406 and uses it to clock the controller 62 master state control logic. For example, the master state control logic may control the general operation of an FPGA controller 62.

[0071] The clock signal 406 can also be input into a clock divider 410 which produces a frequency-divided version of the clock signal 406. In an example embodiment, the clock divider 410 is a divide by two clock divider and produces a 62.5 MHz clock signal in response to the 125 MHz clock signal 406. A non-volatile memory phase-locked loop (PLL) block 412 can be included (e.g., in the controller 62) which distributes a series of clock signals to the non-volatile memory subsystem 40 and to associated control logic. For example, a series of clock signals 414 can be sent from the controller 62 to the non-volatile memory subsystem 40. Another clock signal 416 can be used by the controller logic which is

dedicated to controlling the non-volatile memory subsystem 40. For example, the clock signal 416 may clock the portion of the controller 62 which is dedicated to generating address and/or control lines for the non-volatile memory subsystem 40. A feedback clock signal 418 is fed back into the non-volatile memory PLL block 412. In one embodiment, the PLL block 412 compares the feedback clock 418 to the reference clock 411 and varies the phase and frequency of its output until the reference 411 and feedback 418 clocks are phase and frequency matched.

[0072] A version of the clock signal 406 such as the backup clock signal 408 may be sent from the controller to the volatile memory subsystem 30. The clock signal 408 may be, for example, a differential version of the clock signal 406. As described herein, the backup clock signal 408 may be used to clock the volatile memory subsystem 30 when the memory system 10 is backing up the data from the volatile memory subsystem 30 into the non-volatile memory subsystem 40. In certain embodiments, the backup clock signal 408 may also be used to clock the volatile memory subsystem 30 when the memory system 10 is copying the backed-up data back into the volatile memory subsystem 30 from the non-volatile memory subsystem 40 (also referred to as restoring the volatile memory subsystem 30). The volatile memory subsystem 30 may normally be run at a higher frequency (e.g., DRAM running at 400MHz) than the non-volatile memory subsystem 40 (e.g., flash memory running at 62.5MHz) when communicating with the host system (e.g., when no trigger condition is present). However, in certain embodiments the volatile memory subsystem 30 may be operated at a reduced frequency (e.g., at twice the frequency of the non-volatile memory subsystem 40) without introducing significant delay into the system during backup operation and/or restore operations. Running the volatile memory subsystem 30 at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system 10.

[0073] In one embodiment, the backup clock 408 and the volatile memory system clock signal 420 are received by a multiplexer 422, as schematically illustrated by Figure 7. The multiplexer 422 can output either the volatile memory system clock signal 420 or the backup clock signal 408 depending on the backup state of the memory system 10. For example, when the memory system 10 is not performing a backup or restore operation and is

communicating with the host system (e.g., normal operation), the volatile memory system clock signal 420 may be provided by the multiplexer 422 to the volatile memory PLL block 424. When the memory system 10 is performing a backup (or restore) operation, the backup clock signal 408 may be provided.

[0074] The volatile memory PLL block 424 receives the volatile memory reference clock signal 423 from the multiplexer 422 and can generate a series of clock signals which are distributed to the volatile memory subsystem 30 and associated control logic. For example, in one embodiment, the PLL block 424 generates a series of clock signals 426 which clock the volatile memory elements 32. A clock signal 428 may be used to clock control logic associated with the volatile memory elements, such as one or more registers (e.g., the one or more registers of a registered DIMM). Another clock signal 430 may be sent to the controller 62. A feedback clock signal 432 is fed back into the volatile memory PLL block 424. In one embodiment, the PLL block 424 compares the feedback clock signal 432 to the reference clock signal 423 and varies the phase and frequency of its output until the reference clock signal 423 and the feedback clock signal 432 clocks are phase and frequency matched.

[0075] The clock signal 430 may be used by the controller 62 to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem 30. For example, control logic in the controller 62 may be used to control the volatile memory subsystem 30 during a backup or restore operation. The clock signal 430 may be used as a reference clock signal for the PLL block 434 which can generate one or more clocks 438 used by logic in the controller 62. For example, the PLL block 424 may generate one or more clock signals 438 used to drive logic circuitry associated with controlling the volatile memory subsystem 30. In certain embodiments, the PLL block 434 includes a feedback clock signal 436 and operates in a similar manner to other PLL blocks described herein.

[0076] The clock signal 430 may be used as a reference clock signal for the PLL block 440 which may generate one or more clock signals used by a sub-block 442 to generate one or more other clock signals 444. In one embodiment, for example, the volatile memory subsystem 30 comprises DDR2 SDRAM elements and the sub-block 442 generates one or

more DDR2 compatible clock signals 444. A feedback clock signal 446 is fed back into the PLL block 440. In certain embodiments, the PLL block 440 operates in a similar manner to other PLL blocks described herein.

[0077] While described with respect to the example embodiment of Figure 7, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem 30 operates on the volatile memory clock signal 420 and there is no backup clock signal 408. In some embodiments, the volatile memory subsystem 30 is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem 30 is operated at a reduced frequency during a restore operation and not during a backup operation.

[0078] **Figure 9** is a flowchart of an example method 500 of controlling a memory system 10 operatively coupled to a host system. Although described with respect to the memory system 10 described herein with respect to Figures 1-3 and 7, the method 500 is compatible with other memory systems. The memory system 10 may include a clock distribution topology 400 similar to the one described above with respect to Figure 7 or another clock distribution topology. The memory system 10 can include a volatile memory subsystem 30 and a non-volatile memory subsystem 40.

[0079] In an operational block 510, the method 500 comprises operating the volatile memory subsystem 40 at a first frequency when the memory system 10 is in a first mode of operation in which data is communicated between the volatile memory subsystem 30 and the host system. In an operational block 520, the method 500 comprises operating the non-volatile memory subsystem 40 at a second frequency when the memory system 10 is in a second mode of operation in which data is communicated between the volatile memory subsystem 30 and the non-volatile memory subsystem 40. The method 500 further comprises operating the volatile memory subsystem 30 at a third frequency in an operational block 530 when the memory system 10 is in the second mode of operation. In certain embodiments, the memory system 10 is not powered by a battery when it is in the second mode of operation.

The memory system 10 may switch from the first mode of operation to the second mode of operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

[0080] The third frequency can be less than the first frequency. For example, the third frequency can be approximately equal to the second frequency. In certain embodiments, the reduced frequency operation is an optional mode. In yet other embodiments, the first, second and/or third frequencies are configurable by a user or by the memory system 10.

[0081] **Figure 10** schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 630, 640 of a volatile memory subsystem 30 of a memory system 10 to a controller 62 of the memory system 10. While the example of Figure 10 shows a topology including two DRAM segments 630, 640 for the purposes of illustration, each address location of the volatile memory subsystem 30 comprises more than the two segments in certain embodiments. The data lines 632, 642 from the first DRAM segment 630 and the second DRAM segment 640 of the volatile memory subsystem 30 are coupled to switches 650, 652 which are coupled to the controller 62 (e.g., logic element 70) of the memory system 10. The chip select lines 634, 644 and the self-refresh lines 636, 646 (e.g., CKE signals) of the first and second DRAM segments 630, 640, respectively, are coupled to the controller 62. In certain embodiments, the controller 62 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 30. In certain embodiments, the buffer is a first-in, first out buffer (FIFO). In certain embodiments, data slices from each DRAM segment 630, 640 comprise a portion of the volatile memory subsystem data bus. In one embodiment, for example, the volatile memory subsystem 30 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 640 may comprise the remaining 32 bits

of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.

[0082] In certain embodiments, the switches 650, 652 can each be selectively switched to selectively operatively couple the data lines 632, 642, respectively from the first and second DRAM segments 630, 640 to the controller 62. The chip select lines 634, 644 enable the first and second DRAM segments 630, 640, respectively, of the volatile memory subsystem 30, and the self-refresh lines 636, 646 toggle the first and second DRAM segments 630, 640, respectively, from self-refresh mode to active mode. In certain embodiments, the first and second DRAM segments 630, 640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.

[0083] In certain embodiments, when the memory system 10 is backing up the volatile memory system 30, data slices from only one of the two DRAM segments 630, 640 at a time are sent to the controller 62. For example, when the first slice is being written to the controller 62 during a back-up, the controller 62 sends a signal via the CKe line 636 to the first DRAM segment 630 to put the first DRAM segment 630 in active mode. In certain embodiments, the data slice from the first DRAM segment 630 for multiple words (e.g., a block of words) is written to the controller 62 before writing the second data slice from the second DRAM segment 640 to the controller 62. While the first data slice is being written to the controller 62, the controller 62 also sends a signal via the CKe line 646 to put the second DRAM segment 640 in self-refresh mode. Once the first data slice for one word or for a block of words is written to the controller 62, the controller 62 puts the first DRAM segment 630 into self-refresh mode by sending a signal via the CKe line 636 to the first DRAM segment 640. The controller 62 also puts the second DRAM segment 640 into active mode by sending a signal via the CKe line 646 to the DRAM segment 640. The second slice for a word or for a block of words is written to the controller 62. In certain embodiments, when the first and second data slices are written to the buffer in the controller 62, the controller 62 combines the first and second data slices 630, 640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem

40. In certain embodiments, this process is called “slicing” the volatile memory subsystem 30.

[0084] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the non-volatile memory elements 42 write each backed-up data word to the controller 62 which writes a first slice of the data word to the volatile memory subsystem 30 and then a second slice of the data word to the volatile memory subsystem 30. In certain embodiments, slicing the volatile memory subsystem 30 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 30 during a backup operation.

[0085] **Figure 11** is a flowchart of an example method 600 of controlling a memory system 10 operatively coupled to a host system and which includes a volatile memory subsystem 30 and a non-volatile memory subsystem 40. Although described with respect to the memory system 10 described herein with respect to Figures 1-3 and 10, the method 600 is compatible with other memory systems. The method 600 comprises communicating data words between the volatile memory subsystem 40 and the host system when the memory system 10 is in a first mode of operation in an operational block 610. For example, the memory system 10 may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation or is not being powered by a secondary power supply.

[0086] In an operational block 620, the method further comprises transferring data words from the volatile memory subsystem 30 to the non-volatile memory subsystem 40 when the memory system 10 is in a second mode of operation. In certain embodiments, each data word comprises the data stored in a particular address of the memory system 10. The memory system 10 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 622, storing a second portion of the data word in the buffer in an operational block 624, and writing the entire data word from the buffer to the non-volatile memory subsystem 40 in an operational block 626.

[0087] In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or “slice”) may comprise 40 bits of the data word, and the second portion (or “slice”) may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 62. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 62 which comprises an FPGA. The method 600 may generally be referred to as “slicing” the volatile memory during a backup operation. In the example embodiment, the process of “slicing” the volatile memory during a backup includes bringing the 32-bit slice out of self refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 62 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).

[0088] In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.

[0089] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the non-volatile memory elements 40 write each backed-up data word to the controller 62 which writes a first portion of the data word to the volatile memory subsystem 30 and then a second portion of the data word to the volatile memory 30. In certain embodiments, slicing the volatile memory subsystem 30 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 30 during a backup operation.

[0090] The method 600 can advantageously provide significant power savings and can lead to other advantages. For example, in one embodiment where the volatile

memory subsystem 30 comprises DRAM elements, only the slice of the DRAM which is currently being accessed (e.g., written to the buffer) during a backup is configured in full-operational mode. The slice or slices that are not being accessed may be put in self-refresh mode. Because DRAM in self-refresh mode uses significantly less power than DRAM in full-operational mode, the method 600 can allow significant power savings. In certain embodiments, each slice of the DRAM includes a separate self-refresh enable (e.g., CKe) signal which allows each slice to be accessed independently.

[0091] In addition, the connection between the DRAM elements and the controller 62 may be as large as the largest slice instead of as large as the data bus. In the example embodiment, the connection between the controller 62 and the DRAM may be 40 bits instead of 72 bits. As a result, pins on the controller 62 may be used for other purposes or a smaller controller may be used due to the relatively low number of pin-outs used to connect to the volatile memory subsystem 30. In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 30 and the controller 62 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.

[0092] Various embodiments of the present invention have been described above. Although this invention has been described with reference to these specific embodiments, the descriptions are intended to be illustrative of the invention and are not intended to be limiting. Various modifications and applications may occur to those skilled in the art without departing from the true spirit and scope of the invention as defined in the appended claims.

WHAT IS CLAIMED IS:

1. A memory system coupled to a computer system, comprising:
a volatile memory subsystem;
a non-volatile memory subsystem;
a controller operatively coupled to the non-volatile memory subsystem;
at least one circuit configured to selectively operatively decouple the controller from the volatile memory subsystem.
2. The memory system of Claim 1, wherein the memory system is configured to be operated in at least two states, the at least two states comprising:
a first state in which the controller and the non-volatile memory subsystem are decoupled from the volatile memory subsystem by the at least one circuit; and
a second state in which the volatile memory subsystem is operatively coupled to the controller to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem via the controller.
3. The memory system of Claim 2, wherein power is supplied to the volatile memory subsystem from a first power supply when the memory system is in the first state and power is supplied to the volatile memory subsystem from a second power supply when the memory system is in the second state.
4. The memory system of Claim 3, wherein the second power supply does not comprise a battery.
5. The memory system of Claim 3, wherein the second power supply comprises one or more capacitors.
6. The memory system of Claim 5, wherein the second power supply comprises a battery and is configured to power the volatile memory subsystem for less than thirty minutes.
7. The memory system of Claim 5, wherein the one or more capacitors comprises one or more double layered capacitors.
8. The memory system of Claim 5, wherein the one or more capacitors are charged by the first power supply while the memory system is in the first state.

9. The memory system of Claim 3, further comprising a printed circuit board, wherein the second power supply and the memory system are located on the printed circuit board.

10. The memory system of Claim 3, wherein the at least two states further comprise a third state in which power is supplied to the volatile memory subsystem from a third power supply.

11. The memory system of Claim 2, wherein the controller and the non-volatile memory subsystem do not add a significant capacitive load to the volatile memory system when the memory system is in the first state.

12. The memory system of Claim 1, wherein the volatile memory subsystem comprises one or more DRAM memory elements.

13. The memory system of Claim 1, wherein the volatile memory subsystem comprises one or more SRAM memory elements.

14. The memory system of Claim 1, wherein the non-volatile memory subsystem comprises one or more flash memory elements.

15. The memory system of Claim 1, wherein the memory system is implemented on a memory module.

16. The memory system of Claim 1, wherein the at least one circuit comprise one or more switches coupled to the controller, to the volatile memory subsystem, and to the host computer.

17. The memory system of Claim 1, wherein the volatile memory subsystem comprises a registered DIMM comprising one or more registers and a plurality of DRAM elements, wherein the one or more registers are coupled to the computer system.

18. The memory system of Claim 17, wherein the at least one circuit comprise one or more switches coupled to the one or more registers and to the plurality of DRAM elements and configured to selectively operatively couple the volatile memory subsystem and to the computer system.

19. The memory system of Claim 17, wherein the one or more registers are coupled to the plurality of DRAM elements and the at least one circuit comprises one or more switches coupled to the one or more registers and to a power source for the one or more

registers, wherein the one or more switches are configured to selectively operatively couple the one or more registers to the power source for the one or more registers.

20. The memory system of Claim 1, wherein the non-volatile memory subsystem comprises at least 100 percent more storage capacity than does the volatile memory subsystem.

21. A power module for providing a plurality of voltages to a memory system comprising non-volatile and volatile memory, the plurality of voltages comprising at least a first voltage and a second voltage, the power module comprising:

- an input providing a third voltage to the power module;

- a voltage conversion element configured to provide the second voltage to the memory system;

- a first power element configured to selectively provide a fourth voltage to the conversion element;

- a second power element configured to selectively provide a fifth voltage to the conversion element, wherein the power module is configured to selectively provide the first voltage to the memory system either from the conversion element or from the input, wherein the power module is configured to be operated in at least three states comprising:

- a first state in which the first voltage is provided to the memory system from the input and the fourth voltage is provided to the conversion element from the first power element;

- a second state in which the fourth voltage is provided to the conversion element from the first power element and the first voltage is provided to the memory system from the conversion element;

- a third state in which the fifth voltage is provided to the conversion element from the second power element and the first voltage is provided to the memory system from the conversion element.

22. The power module of Claim 21, wherein the first power element comprises a pulse-width modulation power controller.

23. The power module of Claim 21, wherein the second power element does not comprise a battery.

24. The power module of Claim 21, wherein the second power element comprises one or more capacitors.

25. The power module of Claim 24, wherein the second power element comprises a battery and is configured to power the volatile memory for less than thirty minutes.

26. The power module of Claim 21, wherein the conversion element comprises one or more buck converters.

27. The power module of Claim 21, wherein the conversion element comprises one or more buck-boost converters.

28. A method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems, the method comprising:

 during a first condition, providing the first voltage to the memory system from an input power supply and providing the second voltage to the memory system from a first power subsystem;

 detecting a second condition;

 during the second condition, providing the first voltage and the second voltage to the memory system from the first power subsystem;

 charging a second power subsystem;

 detecting a third condition;

 during the third condition, providing the first voltage and the second voltage to the memory system from the second power subsystem.

29. The method of Claim 28, wherein detecting the second condition comprises detecting that a trigger condition is likely to occur.

30. The method of Claim 29, wherein detecting the third condition comprises detecting that the trigger condition has occurred.

31. The method of Claim 28, wherein the trigger condition comprises a power reduction.

32. The method of Claim 28, wherein the trigger condition comprises a power failure.

33. The method of Claim 28, wherein the trigger condition comprises a system hang-up.

34. The method of Claim 28, wherein the first power subsystem comprises a pulse-width modulation power controller and a voltage conversion subsystem.

35. The method of Claim 28, wherein the second power subsystem does not comprise a battery.

36. The method of Claim 35, wherein the second power subsystem comprises one or more capacitors and the voltage conversion subsystem.

37. A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third frequency when the memory system is in the second mode of operation, the third frequency less than the first frequency.

38. The method of Claim 37, wherein the third frequency is approximately equal to the second frequency.

39. The method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation.

40. The method of Claim 7, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

41. The method of Claim 40, wherein the trigger condition comprises a power failure condition.

42. The method of Claim 37, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

43. A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

- communicating data words between the volatile memory subsystem and the host system when the memory system is in a first mode of operation; and

- transferring data words from the volatile memory subsystem to the non-volatile memory subsystem when the memory system is in a second mode of operation, wherein transferring each data word comprises:

- storing a first portion of the data word in a buffer;

- storing a second portion of the data word in the buffer; and

- writing the entire data word from the buffer to the non-volatile memory subsystem.

44. The method of Claim 43, wherein the transferring each data word further comprises storing a third portion of each data word in the buffer.

45. A memory system operatively coupled to a host system, comprising:

- a volatile memory subsystem;

- a non-volatile memory subsystem comprising at least 100 percent more storage capacity than does the volatile memory subsystem; and

- a controller operatively coupled to the volatile memory subsystem and operatively coupled to the non-volatile memory subsystem, the controller configured to allow data to be communicated between the volatile memory subsystem and the host system when the memory system is operating in a first state and to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem when the memory system is operating in a second state.

46. The memory system of Claim 45, wherein power is supplied to the volatile memory subsystem from a first power supply when the memory system is in the first state.

47. The memory system of Claim 46, wherein power is supplied to the volatile memory subsystem from a second power supply when the memory system is in the second state.

48. The memory system of Claim 47, wherein the second power supply does not comprise a battery.

49. The memory system of Claim 45, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem, the non-volatile memory subsystem, and the controller are located on the printed circuit board.

50. A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

communicating data between the volatile memory subsystem and the host system when the memory system is in a first mode of operation;

storing a first copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a first time when the memory system is in a second mode of operation;

restoring the first copy of data from the non-volatile memory subsystem to the volatile memory subsystem;

erasing the first copy of data from the non-volatile memory subsystem; and

storing a second copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a second time when the memory system is in the second mode of operation, wherein storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem.

51. The method of Claim 50, further comprising restoring the second copy of data from the non-volatile memory subsystem to the volatile memory subsystem.

52. The method of Claim 50, wherein the memory system enters the second mode of operation in response to a power failure.

53. The method of Claim 50, wherein the memory system is not powered by a battery when it is in the second mode of operation.

54. The method of Claim 50, wherein the first copy of data and the second copy of data are stored in separate portions of the non-volatile memory subsystem.

NON-VOLATILE MEMORY MODULE

ABSTRACT OF THE DISCLOSURE

Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory and a controller which are configured such that the controller backs up the volatile memory using the non-volatile memory in the event of a trigger condition. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which is not a battery and may include, for example, a capacitor or capacitor array. The memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or the controller when the volatile memory is interacting with the host system.

5620397

Figure 1:

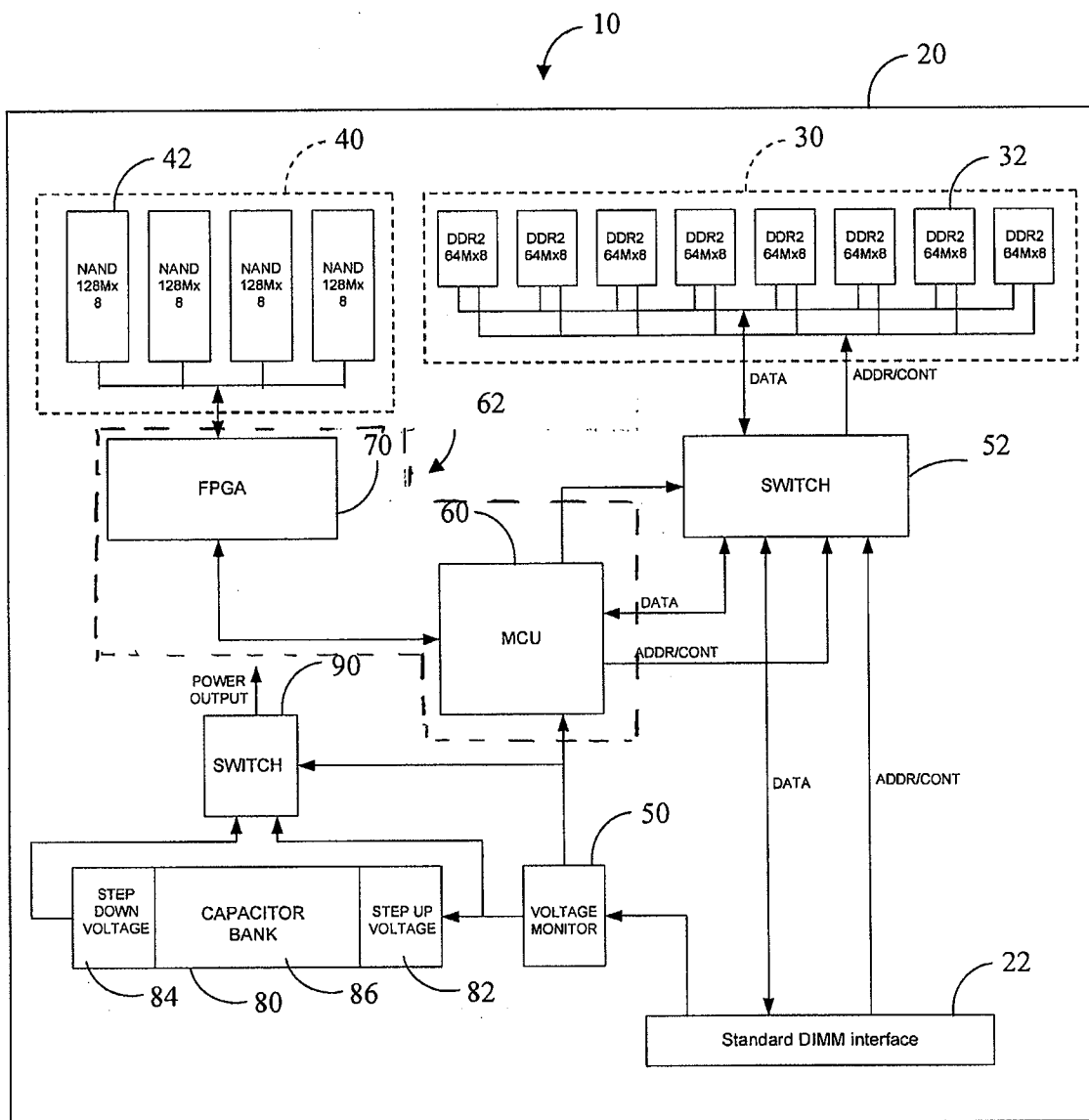


Figure 2:

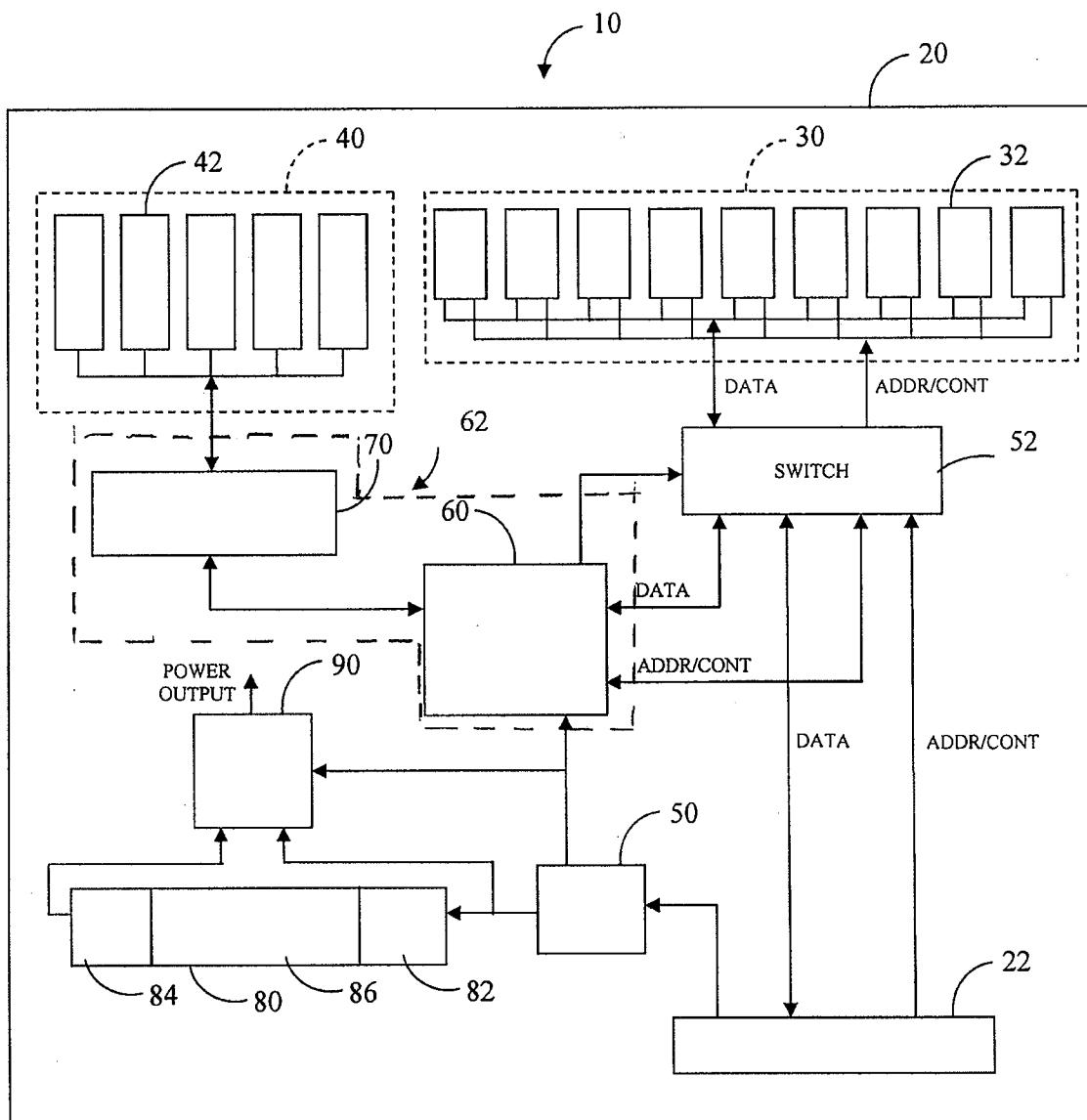


Figure 3:

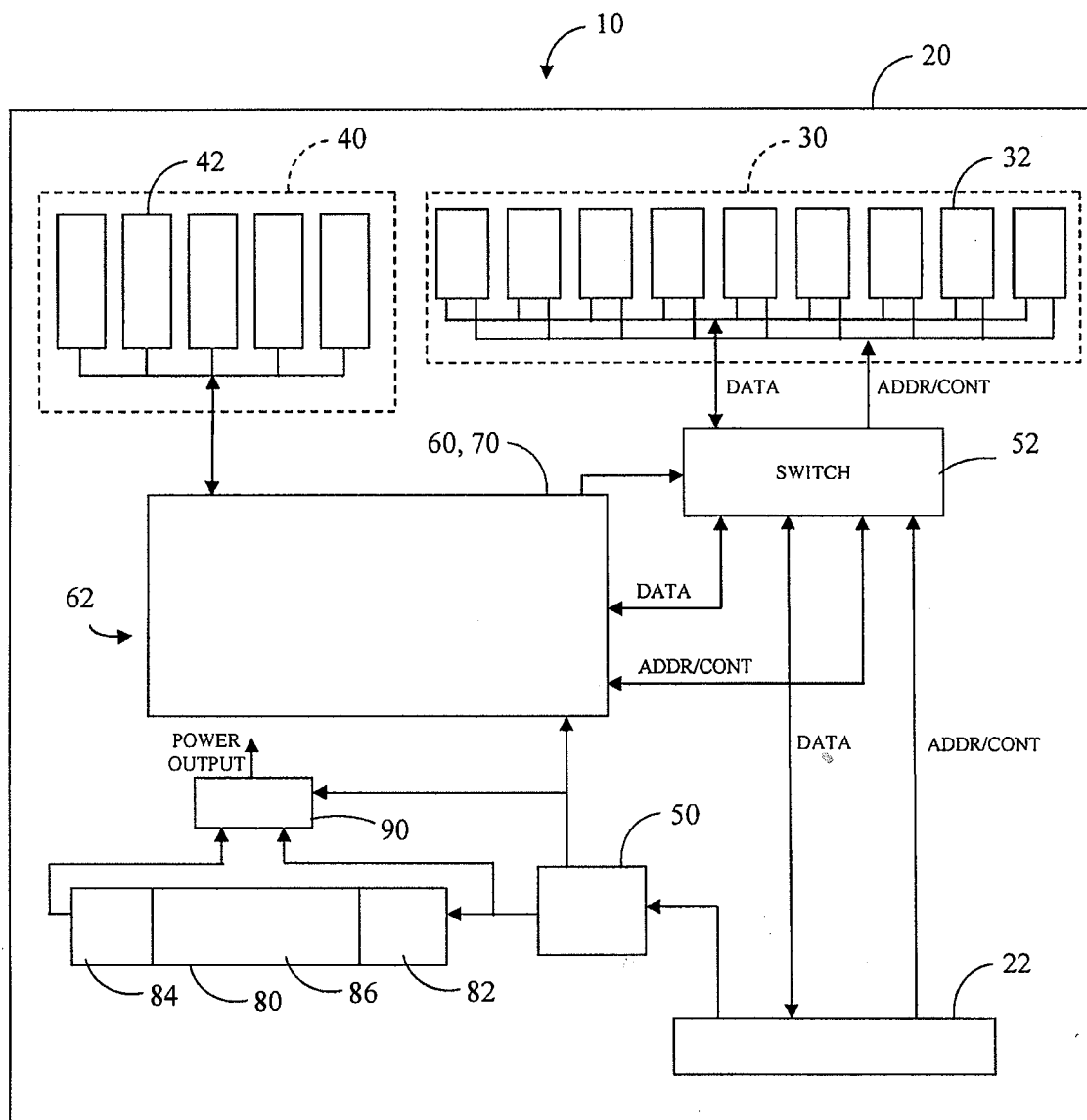


Figure 4A:

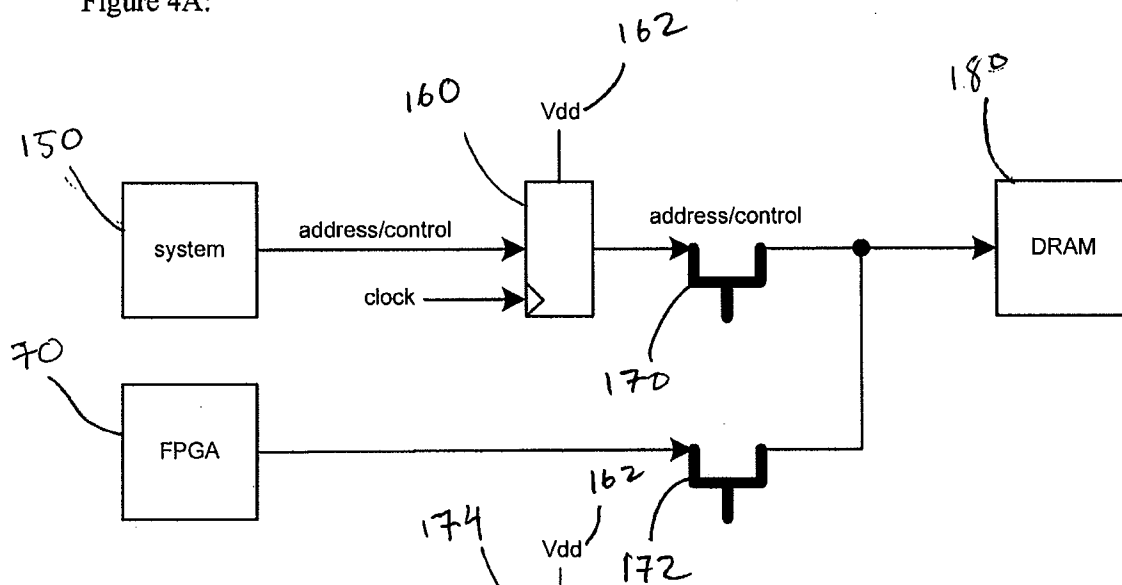


Figure 4B:

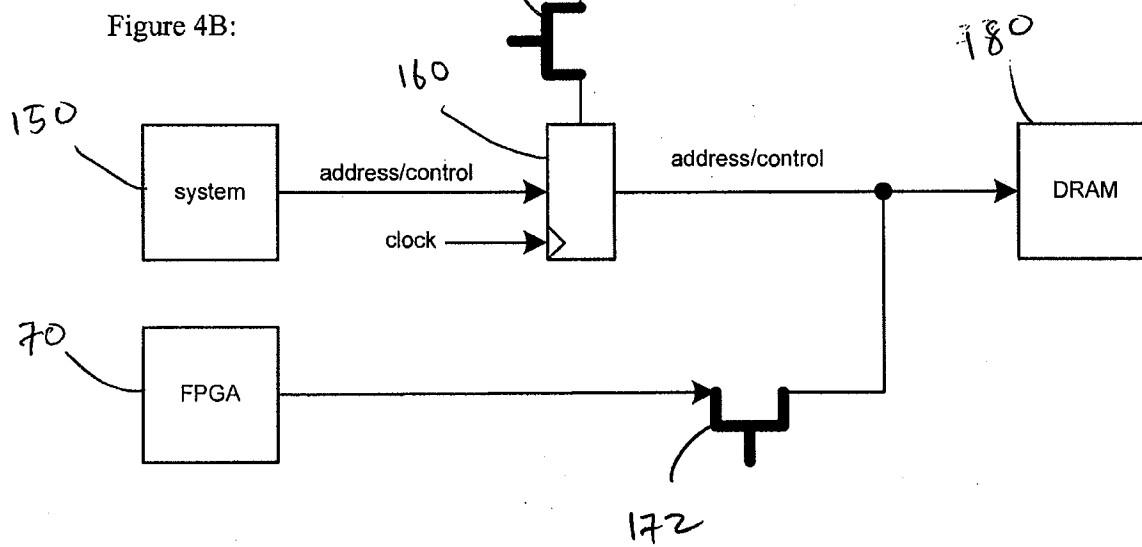


Figure 4C:

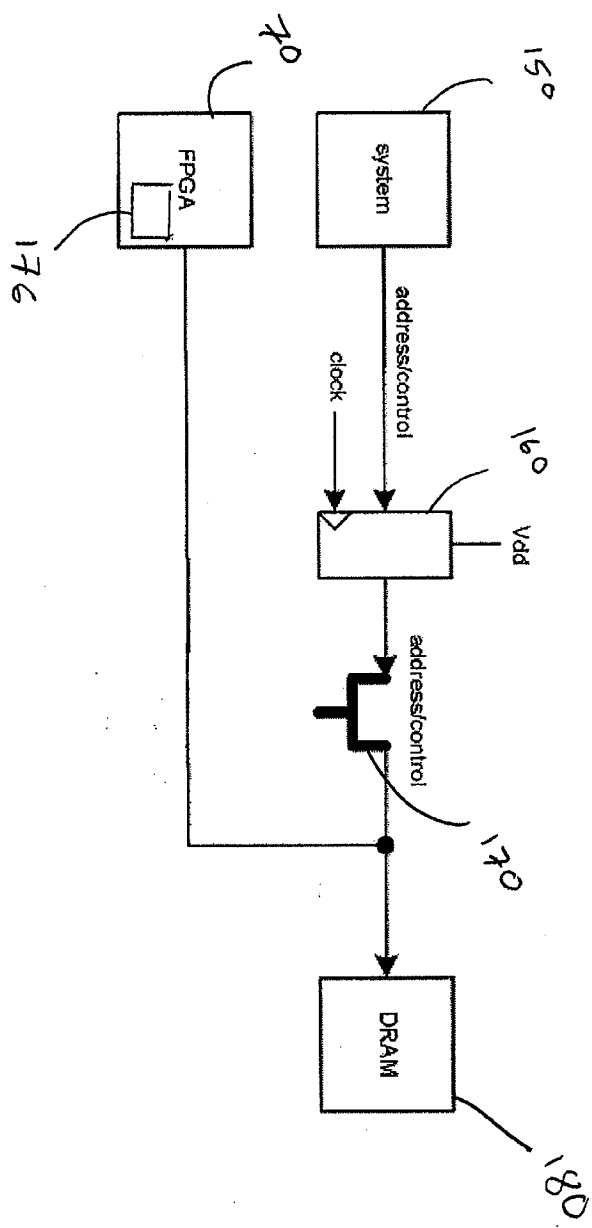


Figure 5

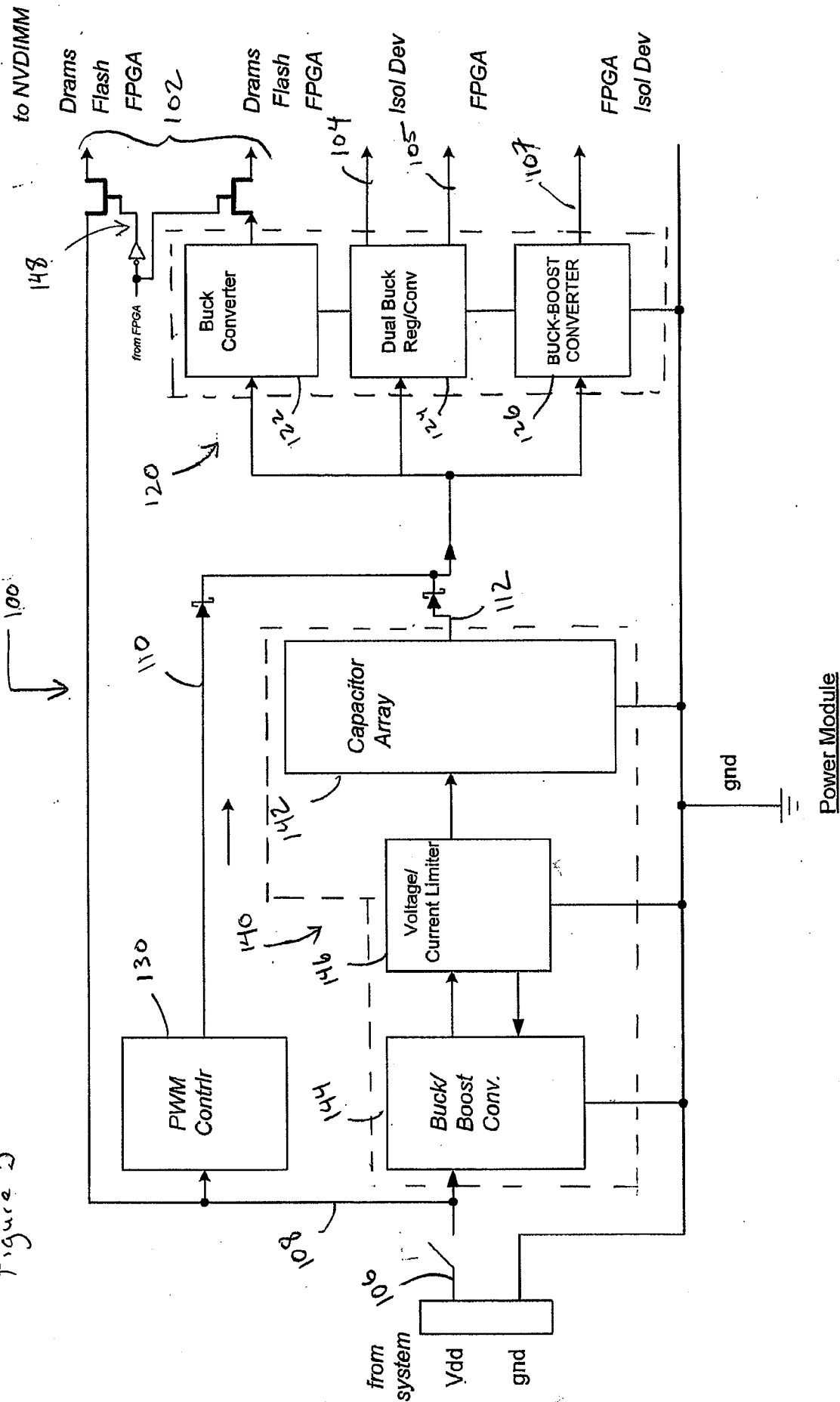


Figure 6

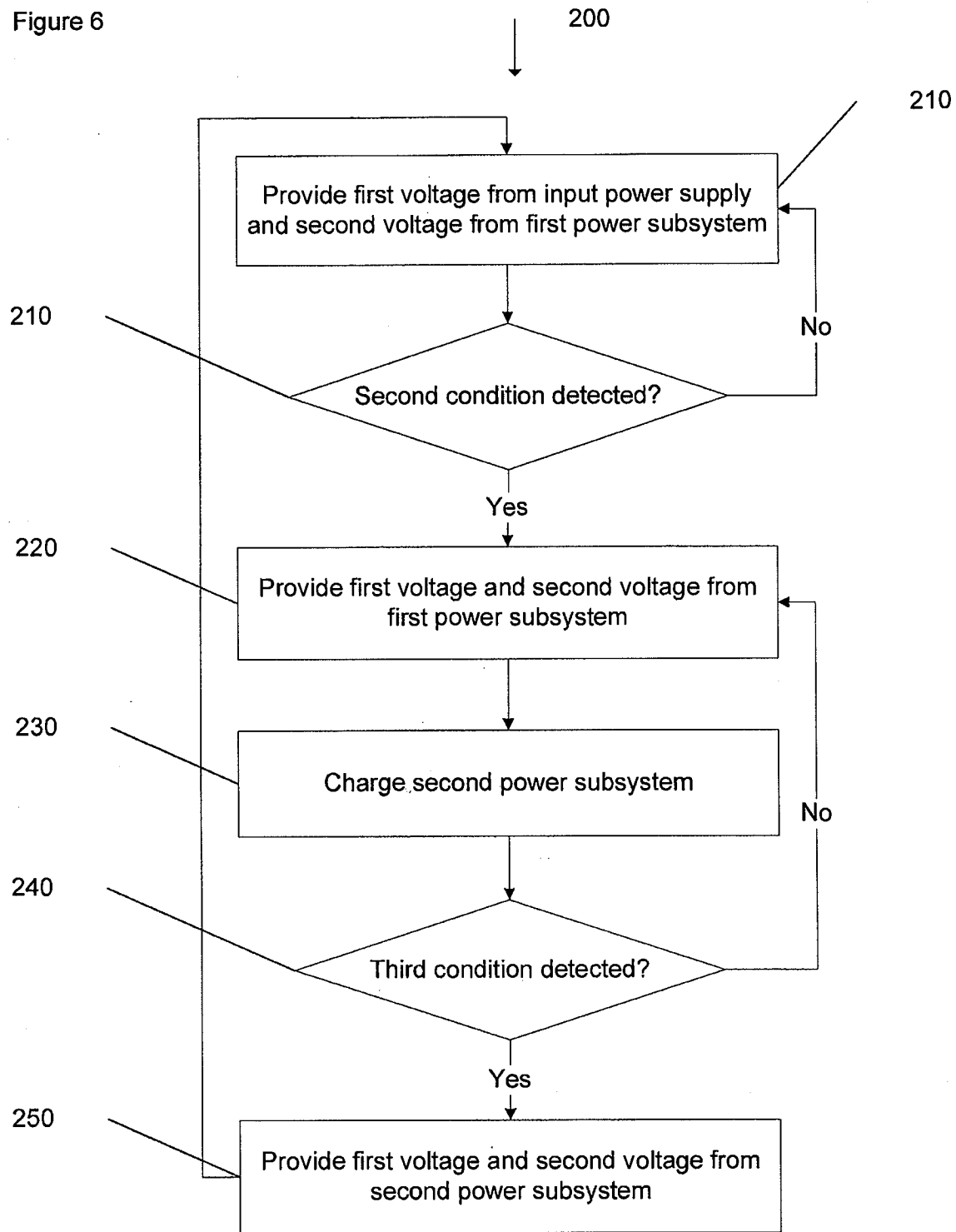


Figure 7

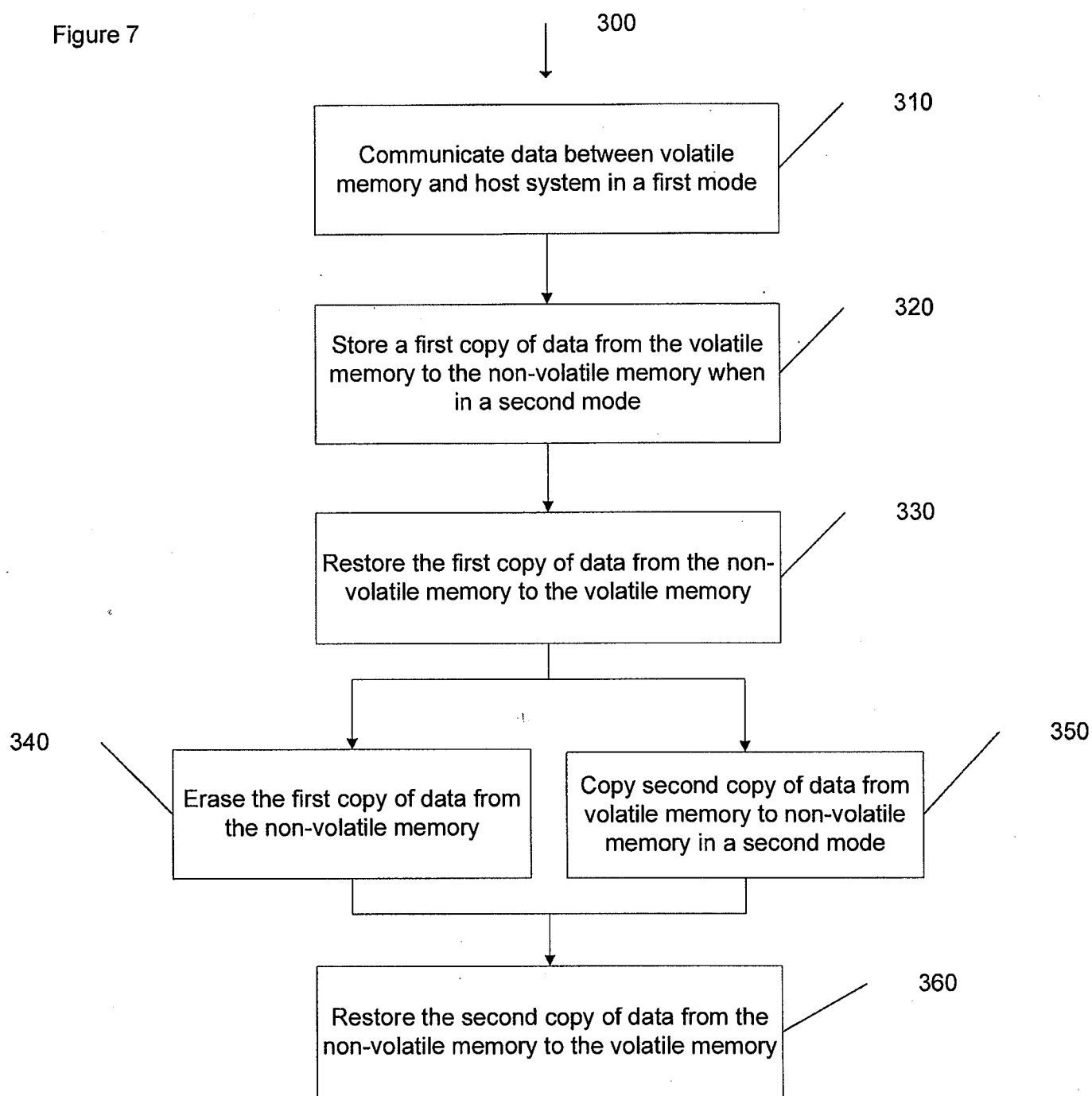


Figure 9

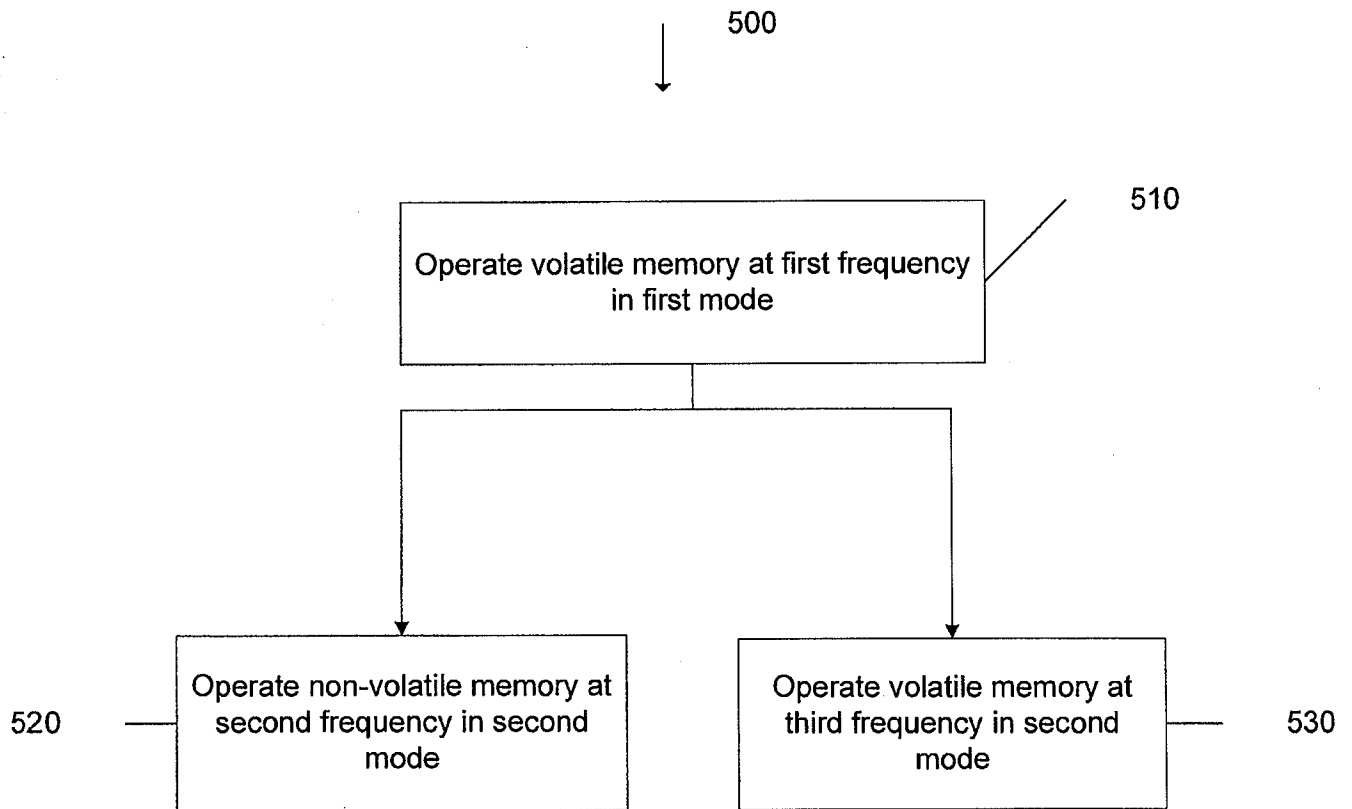


Figure 10

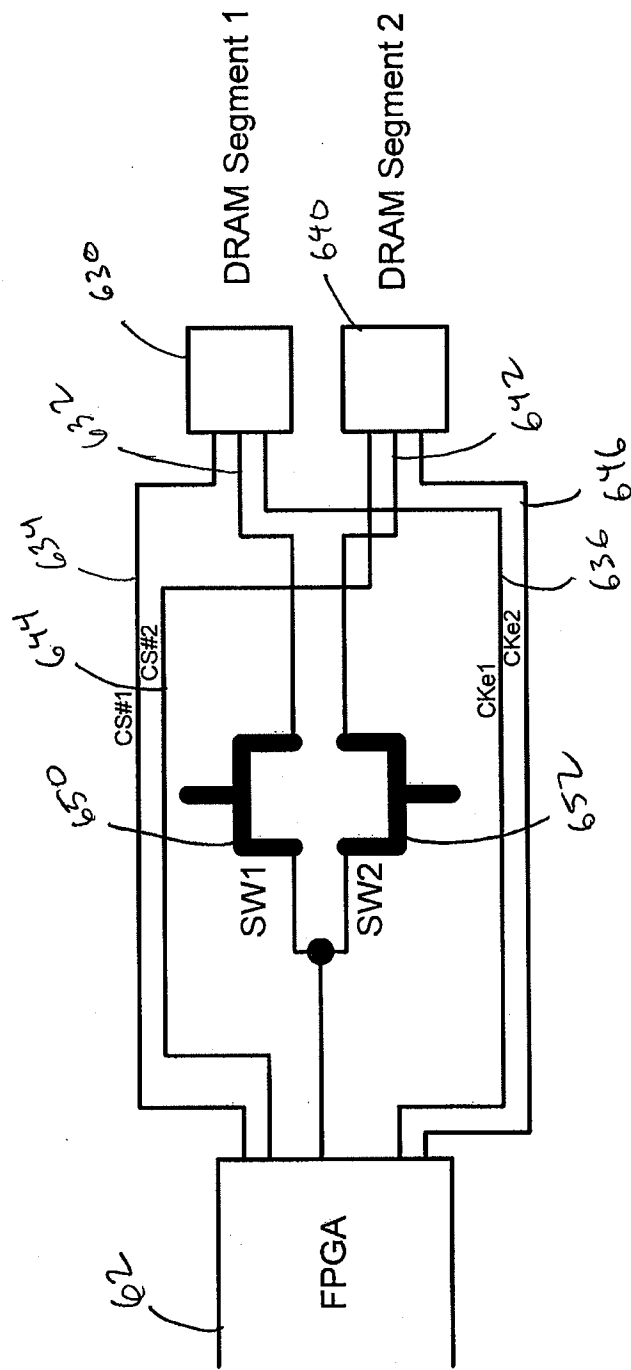
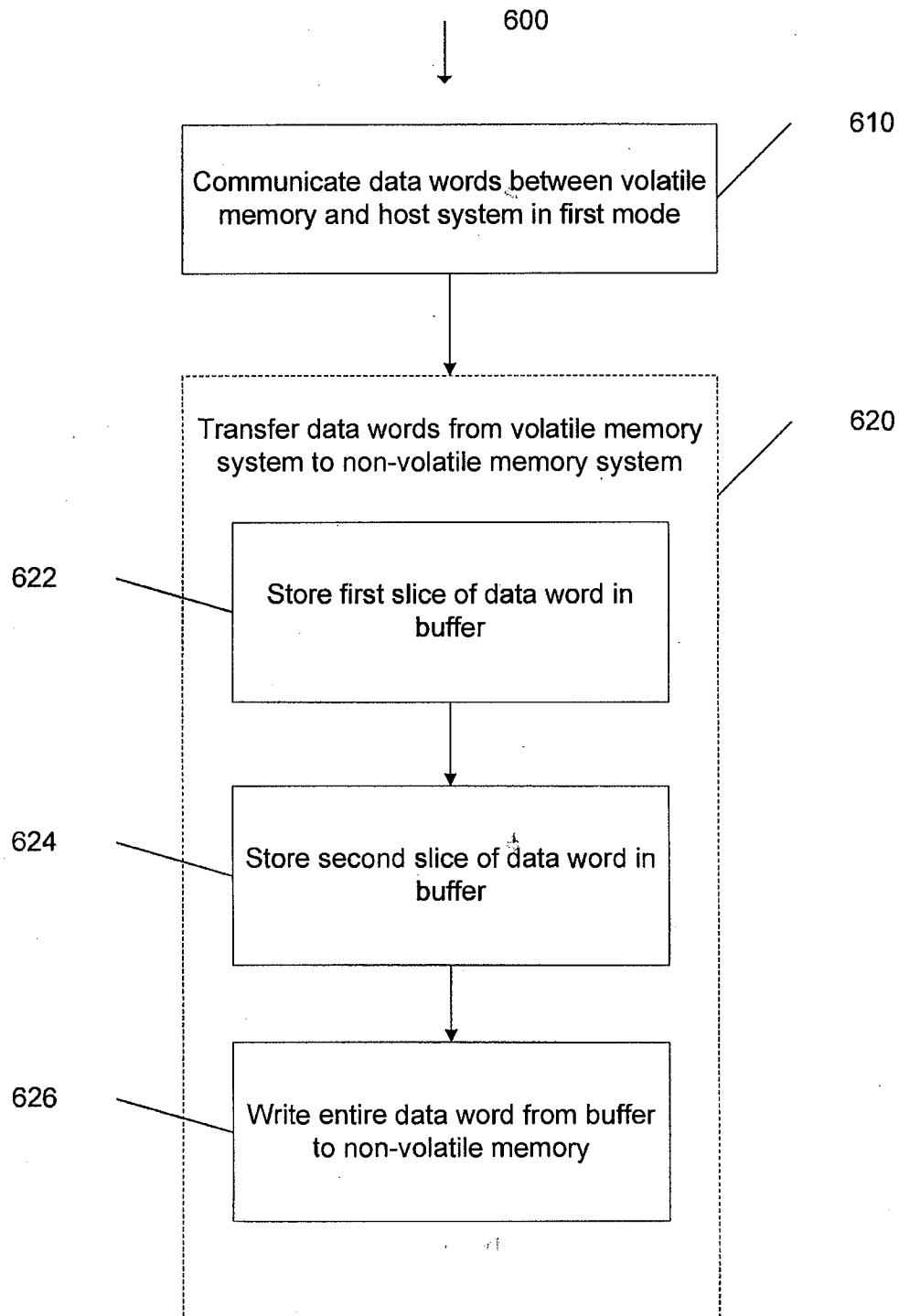


Figure 11



COPY

**DECLARATION FOR UTILITY OR DESIGN APPLICATION
UNDER 37 CFR 1.63**

Docket No.: NETL.040A

Page 1 of 2

Title: NON-VOLATILE MEMORY MODULE

Inventors: Chi-She Chen, Jeffrey C. Solomon; Scott Milton; Jayesh Bhakta

Please Direct All Correspondence to Customer Number 20995

This Declaration is directed to the invention that:

Was filed as Serial No. 12/131,873 filed on June 2, 2008

As a below named inventor:

I believe the inventor(s) named below to be the original and first inventor(s) of the subject matter which is described and claimed and for which a patent is sought;

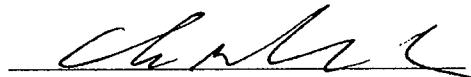
I have reviewed and understand the contents of the above-identified application, including the claims, and any amendment filed herewith or identified above;

I acknowledge the duty to disclose information which is material to patentability as defined in 37 CFR 1.56;

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful, false statements may jeopardize the validity of the application or any patent issued thereon.

Full name of first inventor: Chi-She Chen

Signature:

 Date: 09-15-08

Citizenship:

Taiwan

Mailing Address:

944 Crystal Water Lane, Walnut, CA 91789

Residence Address:

Same as above

(if different than above)

**DECLARATION FOR UTILITY OR DESIGN APPLICATION
UNDER 37 CFR 1.63**

Docket No.: NETL.040A

Page 2 of 2

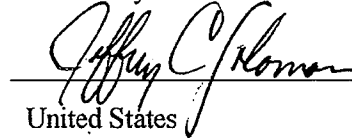
Title: NON-VOLATILE MEMORY MODULE

Inventors: Chi-She Chen, Jeffrey C. Solomon; Scott Milton; Jayesh Bhakta

Please Direct All Correspondence to Customer Number 20995

Full name of **second** inventor: Jeffery C. Solomon

Signature:



Date:

09-15-08

Citizenship:

United States

Mailing Address:

6 Silver Fir, Irvine, CA 92604

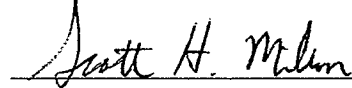
Residence Address:

Same as above

(if different than above)

Full name of **third** inventor: Scott Milton

Signature:



Date:

09/15/2008

Citizenship:

United States

Mailing Address:

49 Statehouse Place, Irvine, CA 92602

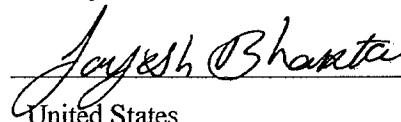
Residence Address:

Same as above

(if different than above)

Full name of **fourth** inventor: Jayesh Bhakta

Signature:



Date:

9-15-08

Citizenship:

United States

Mailing Address:

12220 Rose Street, Cerritos, CA 90703

Residence Address:

Same as above

(if different than above)

Send Correspondence To:

KNOBBE, MARTENS, OLSON & BEAR, LLP

Customer No. 20,995

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

NON-VOLATILE MEMORY MODULE

First Named Inventor/Applicant Name:

CHI-SHE CHEN

Filer:

Bruce S. Itchkawitz/Khylo Rhoden

Attorney Docket Number:

NETL.040C1

Filed as Large Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	310	310
Utility Search Fee	1111	1	510	510
Utility Examination Fee	1311	1	210	210
Pages:				
Claims:				
Claims in excess of 20	1202	34	50	1700
Independent claims in excess of 3	1201	4	210	840
Miscellaneous-Filing:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				3570

Electronic Acknowledgement Receipt

EFS ID:	4027882
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	CHI-SHE CHEN
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Chelsea Pearsall
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.040C1
Receipt Date:	29-SEP-2008
Filing Date:	
Time Stamp:	21:11:02
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 3570
RAM confirmation Number	5851
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NETL040C1.pdf	3115757 d6a44207cc2e372ce42cc20b95e514aa36eafe65	yes	58
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Application Data Sheet		1	4	
	Specification		5	35	
	Claims		36	43	
	Abstract		44	44	
	Drawings-only black and white line drawings		45	56	
	Oath or Declaration filed		57	58	
Warnings:					
Information:					
2	Fee Worksheet (PTO-06)	fee-info.pdf	37878 528a30fcd98f429c7514b64ce48d6f5f365481e6	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			3153635		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.					
<u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.					
<u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.					
<u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

09/29/08

Approved for use through 7/31/2006. OMB 0651-0032

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					12/240,916	
APPLICATION AS FILED – PART I						
(Column 1)		(Column 2)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY
FOR	NUMBER FILED	NUMBER EXTRA		RATE (\$)	FEE (\$)	RATE (\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A		N/A		N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A		N/A		N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A		N/A		N/A
TOTAL CLAIMS (37 CFR 1.16(j))	54	minus 20 =	*	X	25=	X
INDEPENDENT CLAIMS (37 CFR 1.16(h))	7	minus 3 =	*	X	105=	X
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR					
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				N/A		N/A
				TOTAL		TOTAL
						3570
* If the difference in column 1 is less than zero, enter "0" in column 2.						
APPLICATION AS AMENDED – PART II						
(Column 1)		(Column 2)		(Column 3)		SMALL ENTITY
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDI-TIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X
	Application Size Fee (37 CFR 1.16(s))					N/A
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
						TOTAL ADD'T FEE
(Column 1)		(Column 2)		(Column 3)		OR OTHER THAN SMALL ENTITY
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDI-TIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X
	Application Size Fee (37 CFR 1.16(s))					N/A
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
						TOTAL ADD'T FEE
(Column 1)		(Column 2)		(Column 3)		OR OTHER THAN SMALL ENTITY
AMENDMENT C	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDI-TIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X
	Application Size Fee (37 CFR 1.16(s))					N/A
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
						TOTAL ADD'T FEE
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/240,916	09/29/2008	Chi-She Chen	NETL.040C1

CONFIRMATION NO. 6240

FORMALITIES LETTER



OC000000032525402

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 10/15/2008

NOTICE TO FILE CORRECTED APPLICATION PAPERS

Filing Date Granted

An application number and filing date have been accorded to this application. The application is informal since it does not comply with the regulations for the reason(s) indicated below. Applicant is given TWO MONTHS from the date of this Notice within which to correct the informalities indicated below. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

The required item(s) identified below must be timely submitted to avoid abandonment:

- Replacement drawings in compliance with 37 CFR 1.84 and 37 CFR 1.121(d) are required. The drawings submitted are not acceptable because:
 - The drawings must be reasonably free from erasures and must be free from alterations, overwriting, interlineations, folds, and copy marks. See Figure(s) 8.
 - Numbers, letters, and reference characters on the drawings must measure at least 0.32 cm (1/8 inch) in height. See Figure(s) 8.
 - The drawings submitted to the Office are not electronically reproducible because portions of figures 8 are missing and/or blurry.

Applicant is cautioned that correction of the above items may cause the specification and drawings page count to exceed 100 pages. If the specification and drawings exceed 100 pages, applicant will need to submit the required application size fee.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/agizaw/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/240,916	09/29/2008	2189	3570	NETL.040C1	54	7

CONFIRMATION NO. 6240

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000032525401

Date Mailed: 10/15/2008

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Chi-She Chen, Walnut, CA;
Jeffery C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 12/131,873 06/02/2008
which claims benefit of 60/941,586 06/01/2007

Foreign Applications

If Required, Foreign Filing License Granted: 10/08/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/240,916**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

NON-VOLATILE MEMORY MODULE

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

Please Direct All Correspondence to Customer Number 20,995

RESPONSE TO FORMALITIES NOTICE

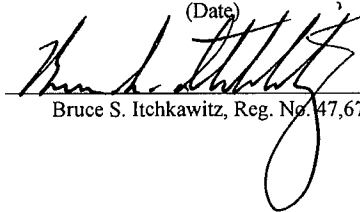
Applicant : Chi-She Chen et al.
App. No : 12/240,916
Filed : September 29, 2008
For : NON-VOLATILE MEMORY MODULE
Art Unit : 2189

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence, and any other attachment noted on the automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web server on:

December 15, 2008

(Date)


Bruce S. Itchkawitz, Reg. No. 47,677

**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

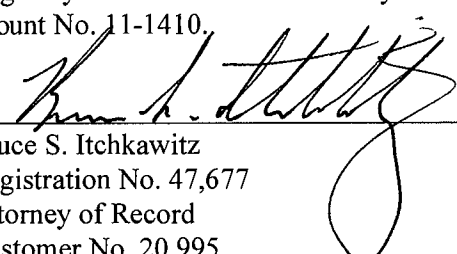
Dear Sir:

Enclosed for filing in the above-referenced application are the following:

- (X) 12 Sheets of Replacement Drawings.
- (X) A Supplemental Declaration in 2 pages.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

Please note that a copy of the declaration filed in U.S. Application No. 12/131,873, the parent of the above-referenced application, was previously filed in this case in accordance with 37 C.F.R. § 1.63(d)(1). However, the name of inventor Jeffrey C. Solomon was incorrectly spelled "Jeffery C. Solomon" in that previously filed declaration. The enclosed newly executed declaration includes the correct spelling.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

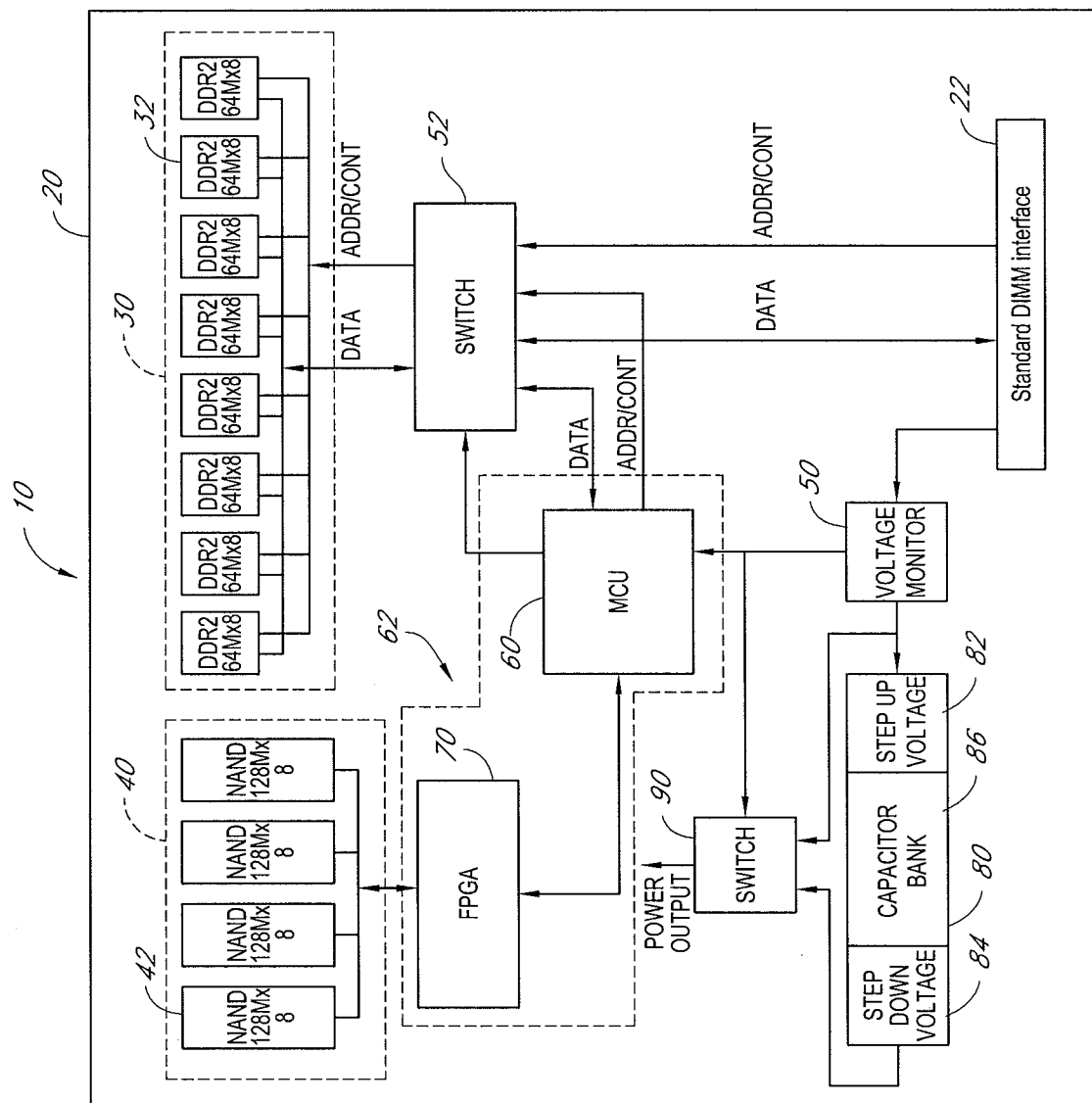


FIG. 1

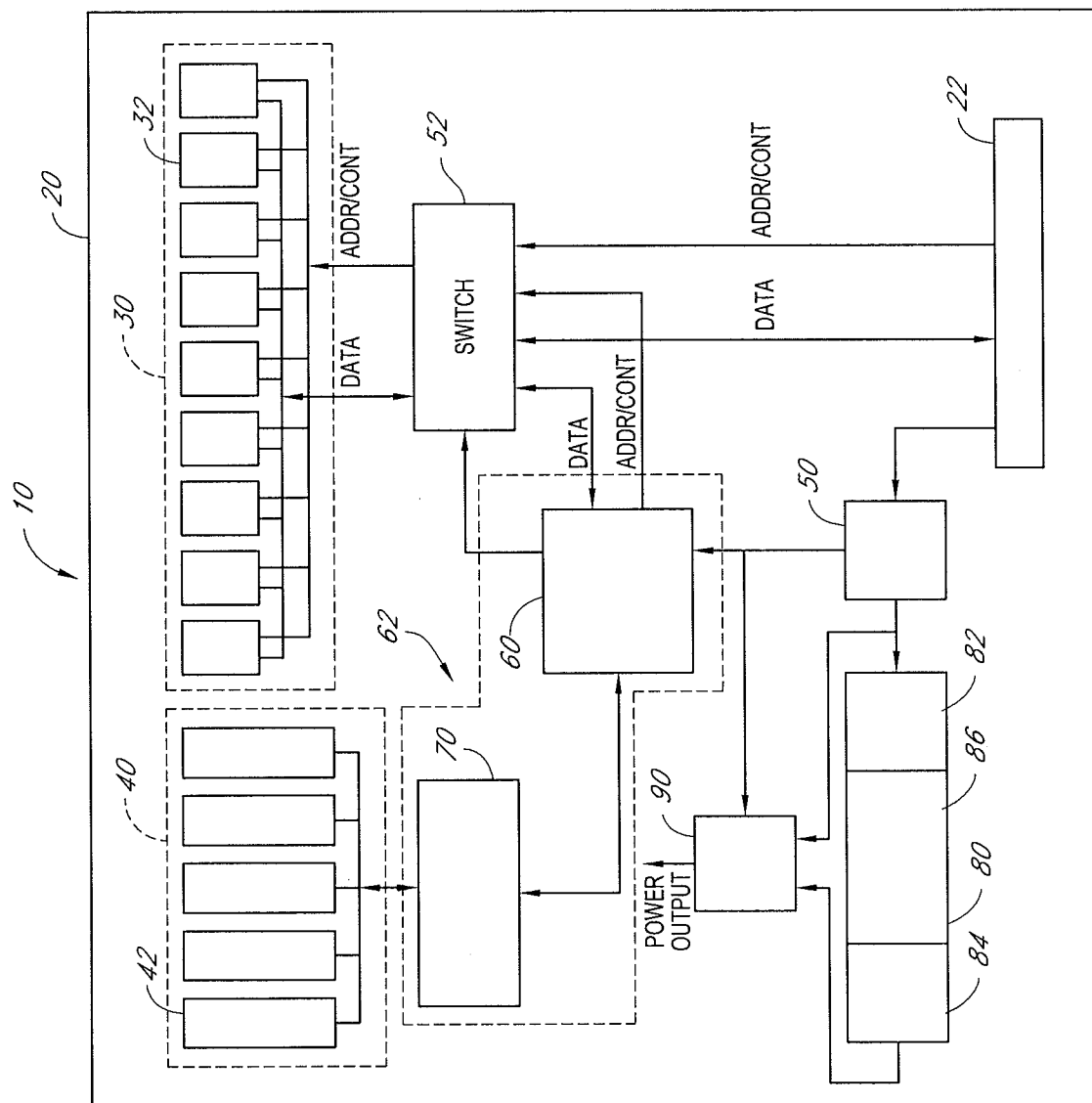


FIG. 2

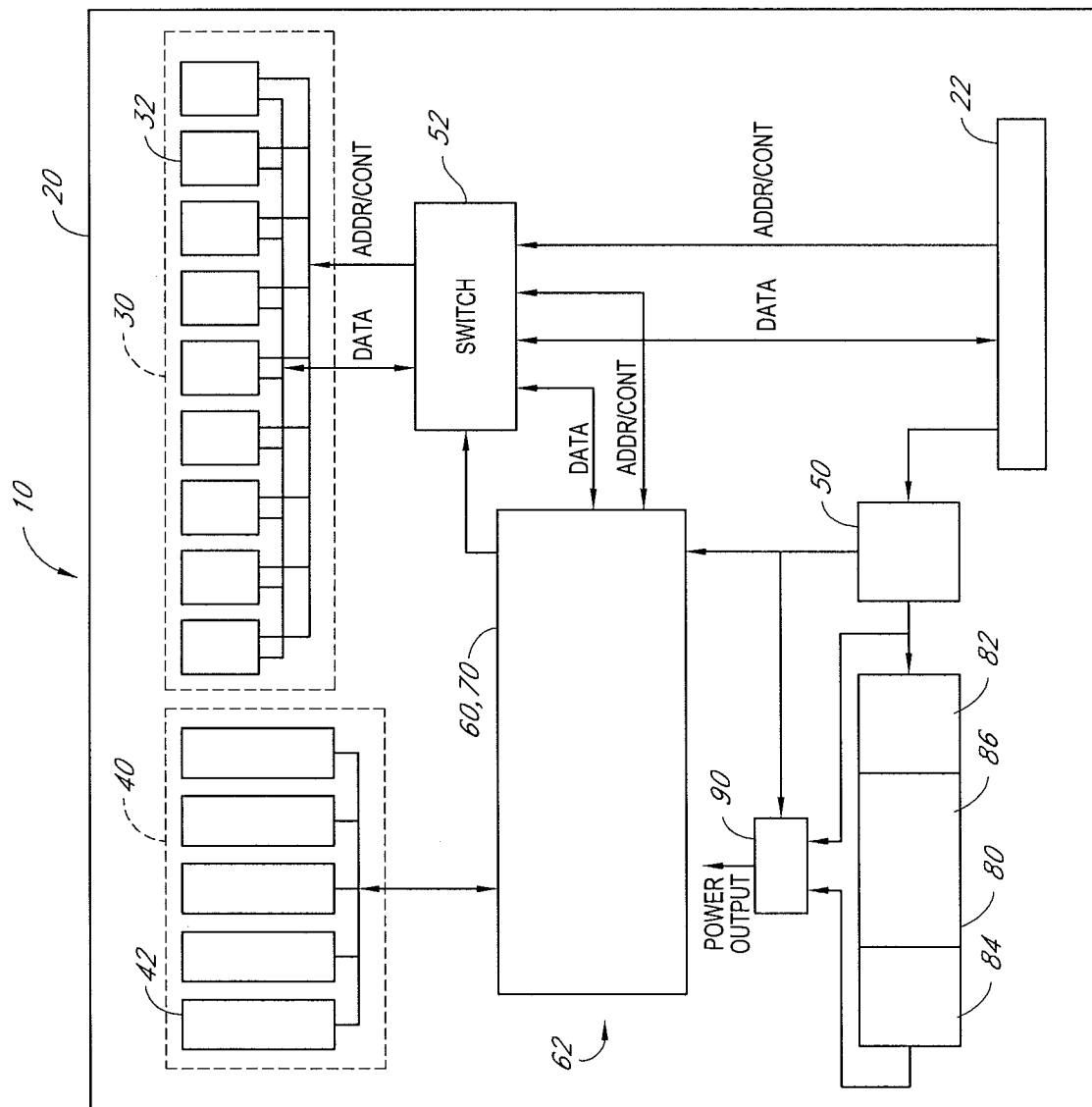
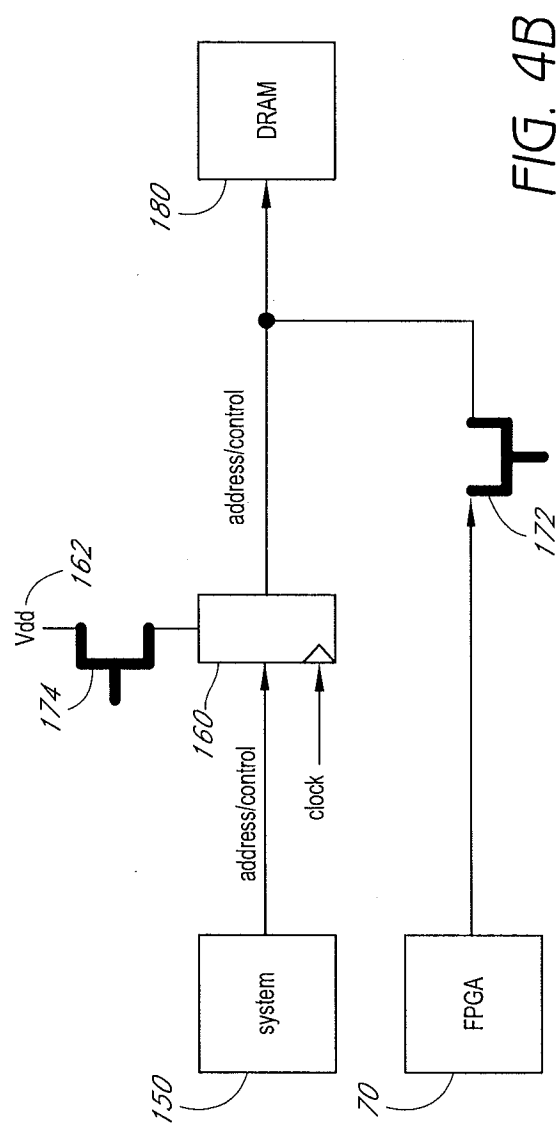
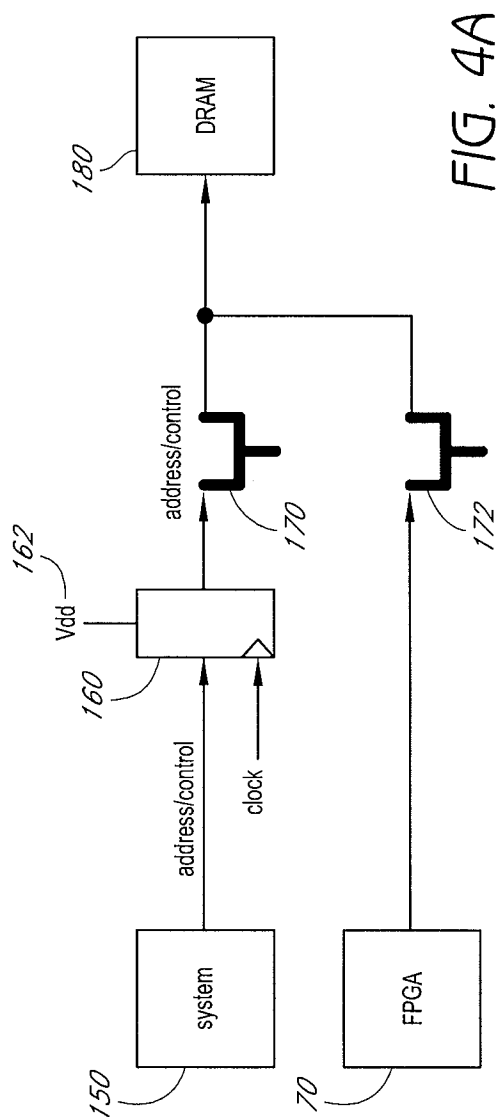


FIG. 3



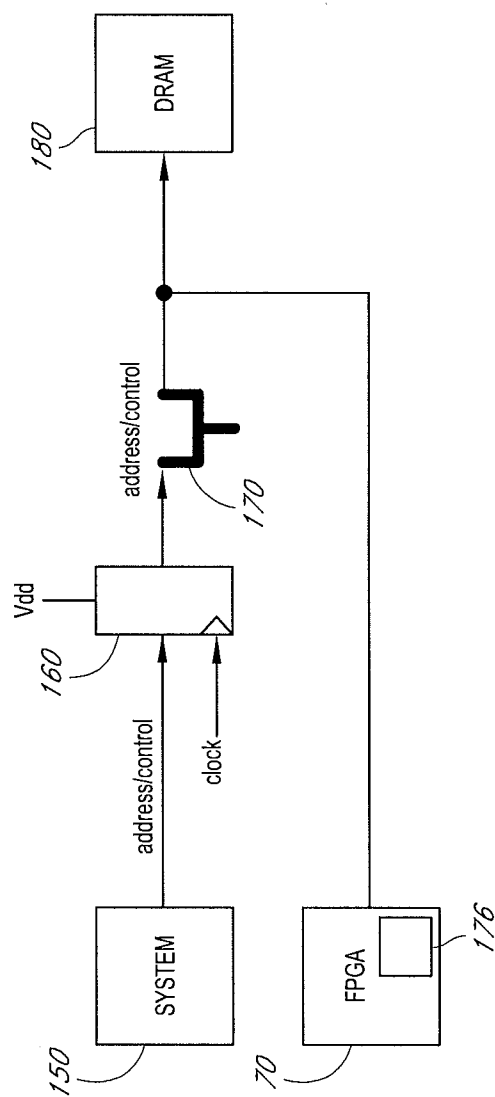


FIG. 4C

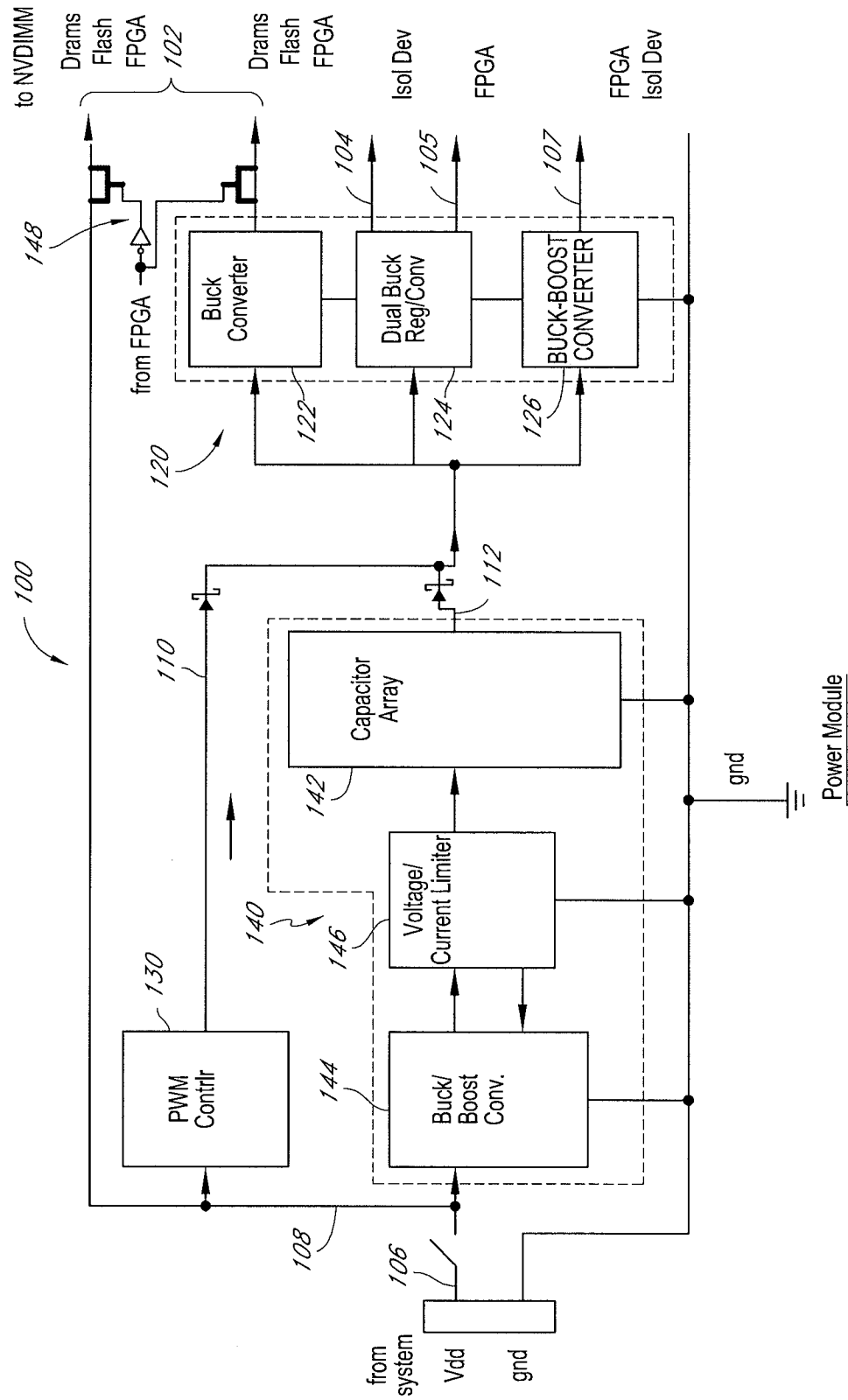


FIG. 5

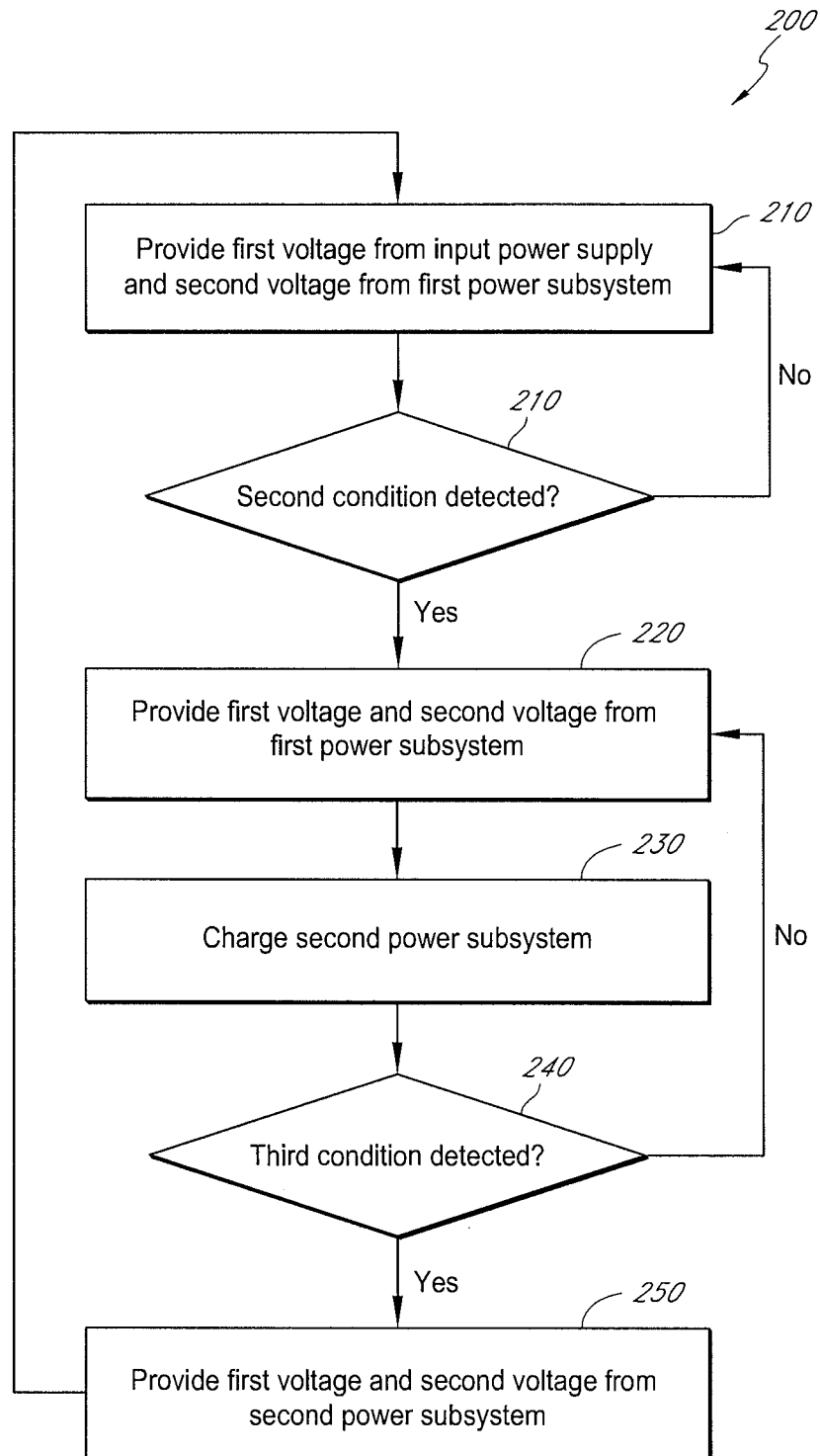


FIG. 6

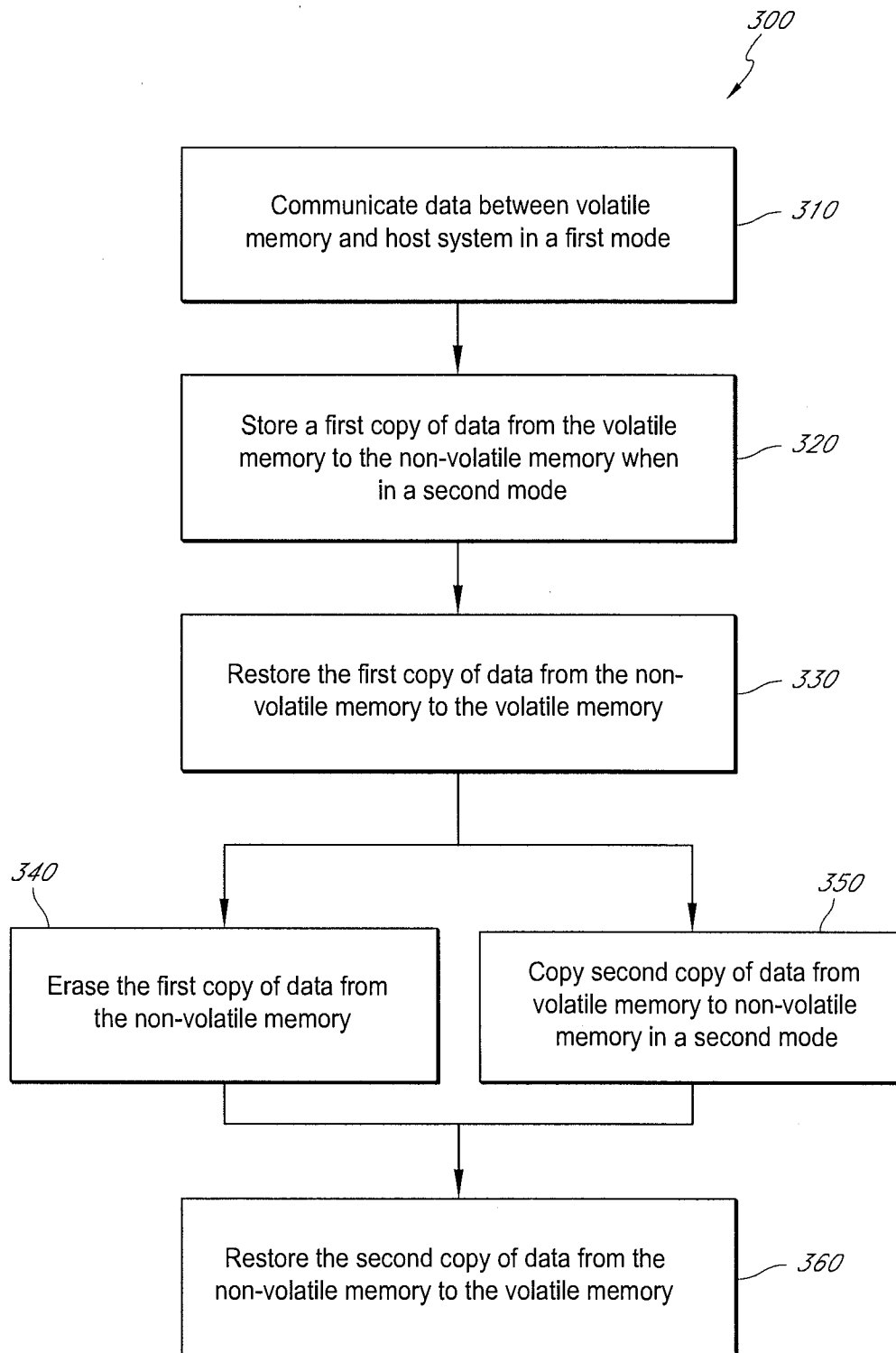
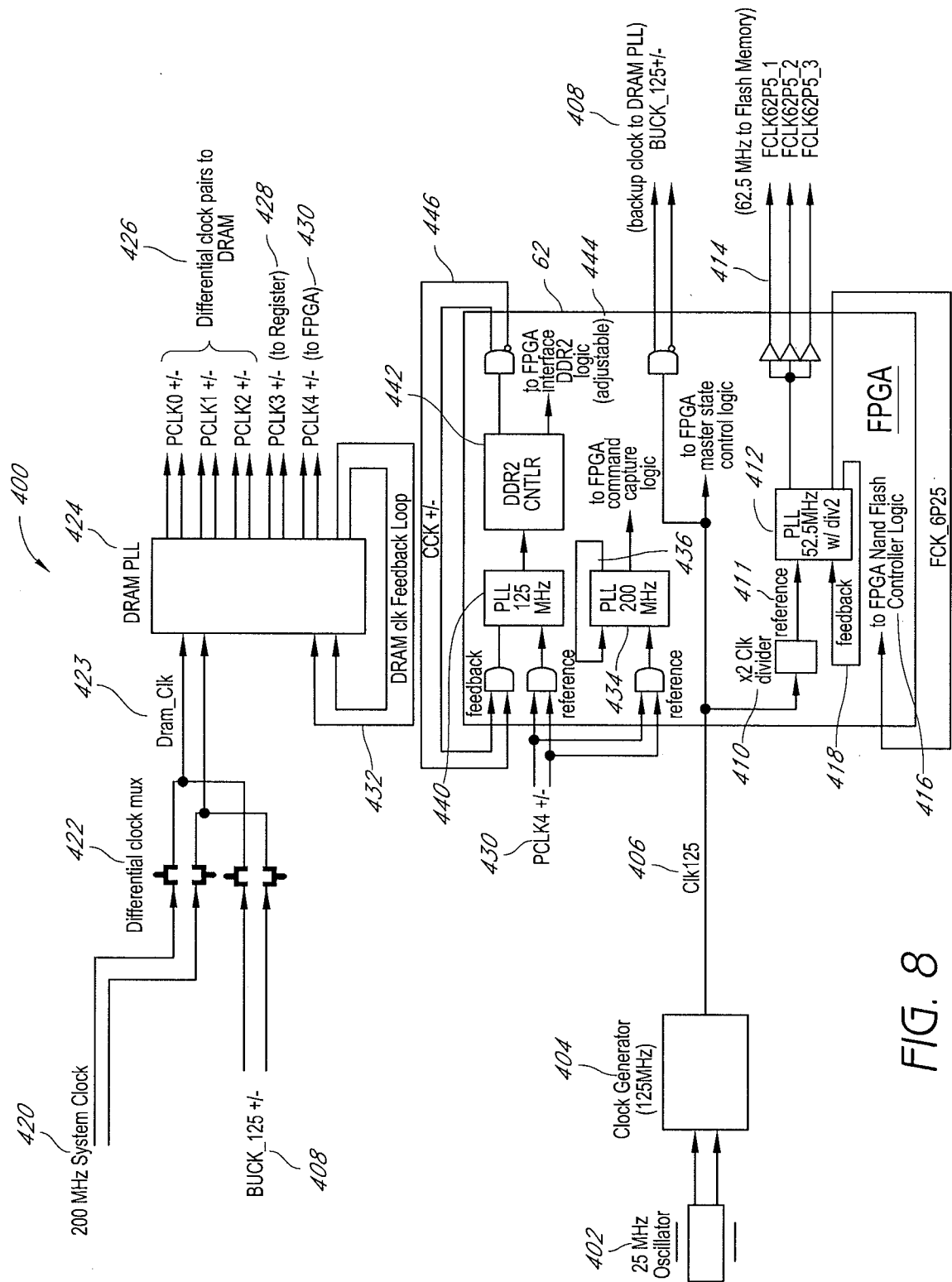


FIG. 7



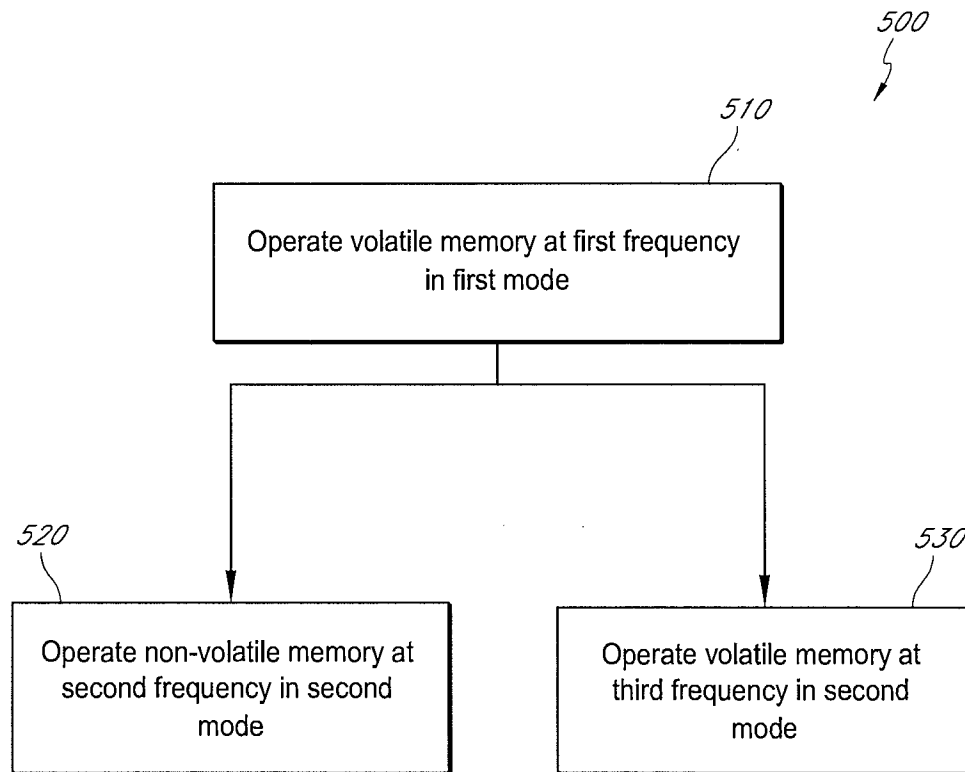


FIG. 9

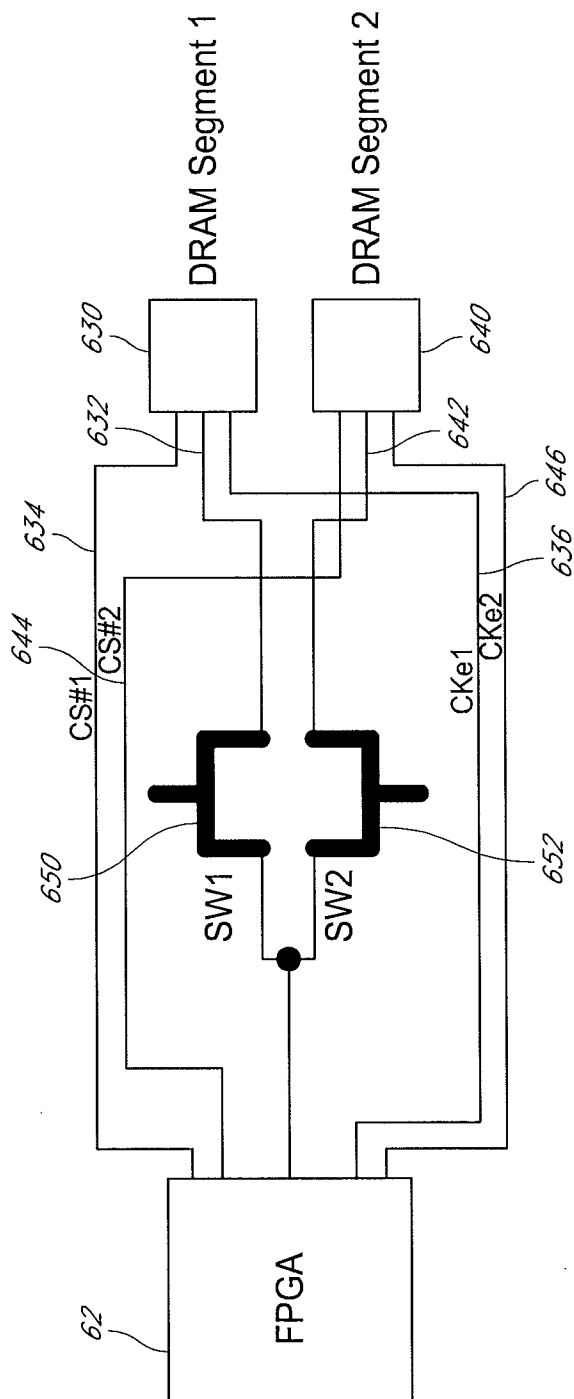


FIG. 10

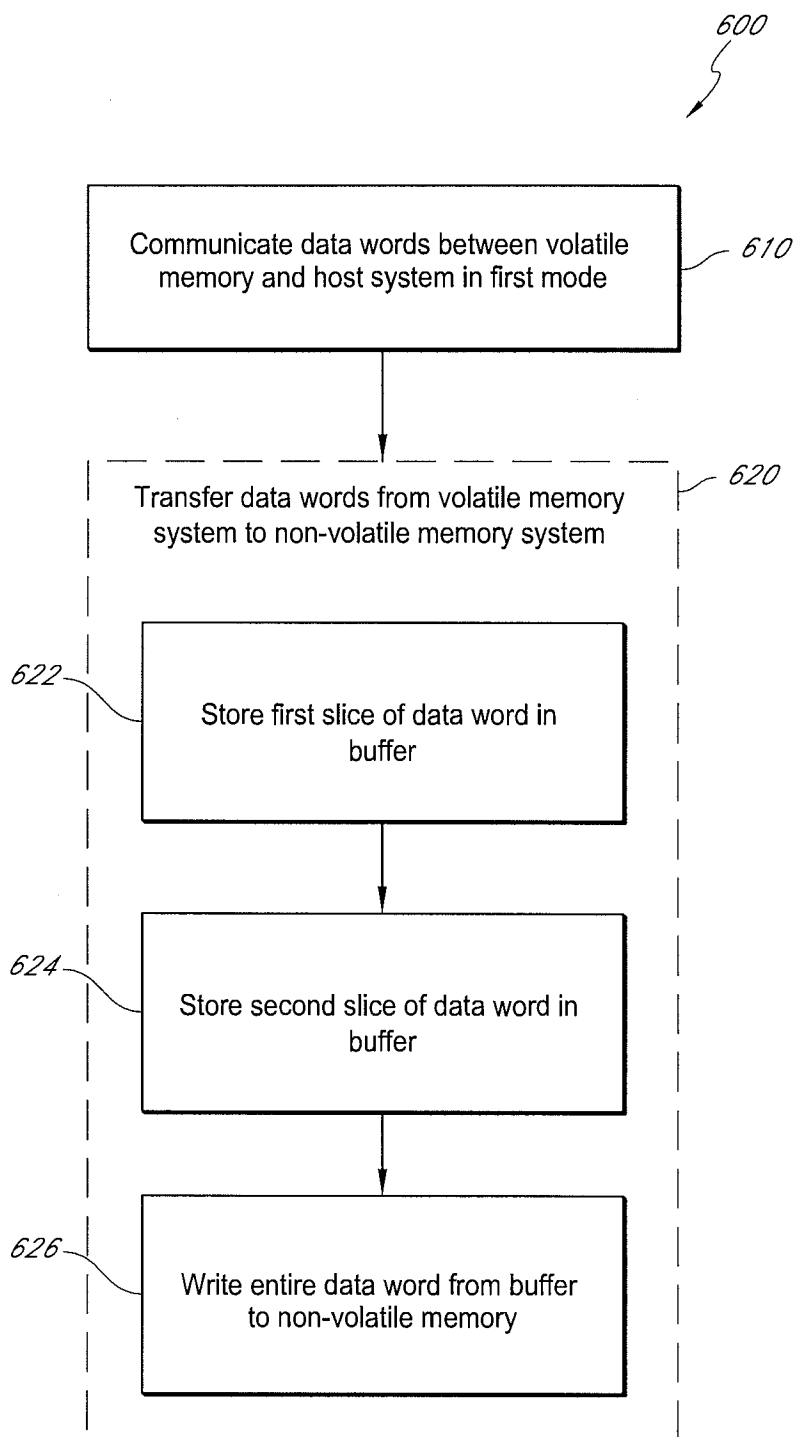


FIG. 11

**DECLARATION FOR UTILITY OR DESIGN APPLICATION
UNDER 37 CFR 1.63**

Docket No.: NETL.040C1

Page 1 of 2

Title: NON-VOLATILE MEMORY MODULE

Inventors: Chi-She Chen, Jeffrey C. Solomon; Scott Milton; Jayesh Bhakta

Please Direct All Correspondence to Customer Number 20995

This Declaration is directed to the invention that:

Was filed as Serial No. 12/240,916 filed on September 29, 2008

As a below named inventor:

I believe the inventors named below to be the original and first inventors of the subject matter which is described and claimed and for which a patent is sought;

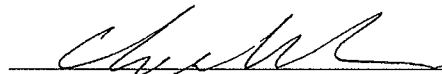
I have reviewed and understand the contents of the above-identified application, including the claims, and any amendment filed herewith or identified above;

I acknowledge the duty to disclose information which is material to patentability as defined in 37 CFR 1.56;

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful, false statements may jeopardize the validity of the application or any patent issued thereon.

Full name of **first** inventor: Chi-She Chen

Signature:



Date:

12-11-08

Citizenship:

Taiwan

Mailing Address:

944 Crystal Water Lane, Walnut, CA 91789

Residence Address:

Same as above

(if different than above)

**DECLARATION FOR UTILITY OR DESIGN APPLICATION
UNDER 37 CFR 1.63**

Docket No.: NETL.040C1

Page 2 of 2

Title: NON-VOLATILE MEMORY MODULE

Inventors: Chi-She Chen, Jeffrey C. Solomon; Scott Milton; Jayesh Bhakta

Please Direct All Correspondence to Customer Number 20995

Full name of **second** inventor: Jeffrey C. Solomon

Signature: _____

Date: 12-11-08

Citizenship: _____

United States

Mailing Address: _____

6 Silver Fir, Irvine, CA 92604

Residence Address: _____

Same as above

(if different than above)

Full name of **third** inventor: Scott Milton

Signature: _____

Date: 12/11/2008

Citizenship: _____

United States

Mailing Address: _____

49 Statehouse Place, Irvine, CA 92602

Residence Address: _____

Same as above

(if different than above)

Full name of **fourth** inventor: Jayesh Bhakta

Signature: _____

Date: 12-11-08

Citizenship: _____

United States

Mailing Address: _____

12220 Rose Street, Cerritos, CA 90703

Residence Address: _____

Same as above

(if different than above)

Send Correspondence To:

KNOBBE, MARTENS, OLSON & BEAR, LLP

Customer No. 20,995

Electronic Acknowledgement Receipt

EFS ID:	4458805
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Chelsea Pearsall
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.040C1
Receipt Date:	15-DEC-2008
Filing Date:	29-SEP-2008
Time Stamp:	19:53:50
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NETL040C1_MP_RESPONSE.pdf	388696 47b5f229aef5b2992864ba502fa0f296353e cba5	yes	15

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Applicant Response to Pre-Exam Formalities Notice	1	1
	Drawings-only black and white line drawings	2	13
	Oath or Declaration filed	14	15

Warnings:

Information:

Total Files Size (in bytes):	388696
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/240,916	09/29/2008	2189	3570	NETL.040C1	54	7

CONFIRMATION NO. 6240

UPDATED FILING RECEIPT



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 12/24/2008

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Chi-She Chen, Walnut, CA;
Jeffery C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications

If Required, Foreign Filing License Granted: 10/08/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/240,916**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

NON-VOLATILE MEMORY MODULE

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

**TRANSMITTAL LETTER
(General - Patent Pending)**

Docket No.
987-04-CON

In Re Application Of: **Chen et al.**

Application No.	Filing Date	Examiner	Customer No.	Group Art Unit	Confirmation No.
12/240,916	September 26,2008	Unknown	22145	2189	6240

Title: **NON-VOLATILE MEMORY MODULE**

COMMISSIONER FOR PATENTS:

Transmitted herewith is:

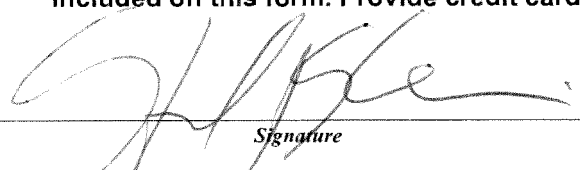
Statement under 37 CFR 3.73(b) (1 page)

Power of Attorney by Assignee (1 page)

in the above identified application.

- ☒ No additional fee is required.
- ☐ A check in the amount of _____ is attached.
- ☒ The Director is hereby authorized to charge and credit Deposit Account No. **11-1159** as described below.
- ☐ Charge the amount of _____
- ☒ Credit any overpayment.
- ☒ Charge any additional fee required.
- ☐ Payment by credit card. Form PTO-2038 is attached.

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.


Signature

Howard J. Klein (Reg. No. 28,727)
Klein, O'Neill & Singh, LLP (Customer No. 22145)
43 Corporate Park, Suite 204
Irvine, California 92606
Tel: 949-955-1920
Fax: 949-955-1921

Dated: **February 19, 2009**

I hereby certify that this correspondence is being deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to the "Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450" [37 CFR 1.8(a)] on _____

(Date)

Signature of Person Mailing Correspondence

Typed or Printed Name of Person Mailing Correspondence

CC:

STATEMENT UNDER 37 CFR 3.73(b)Applicant/Patent Owner: Chi-She Chen et al.Application No./Patent No.: 12/240,916Filed/Issue Date: September 29, 2008Entitled: **NON-VOLATILE MEMORY MODULE**NETLIST INC., a Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest; or
2. ☐ an assignee of less than the entire right, title and interest
(The extent (by percentage) of its ownership interest is _____ %)

in the patent application/patent identified above by virtue of either:

- A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel _____, Frame _____, or for which a copy thereof is attached.

OR

- B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

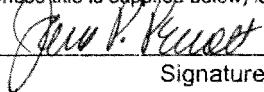
1. From: _____ To: _____
The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.
2. From: _____ To: _____
The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.
3. From: _____ To: _____
The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

- ☐ Additional documents in the chain of title are listed on a supplemental sheet.

- ☒ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.



Signature

James Perrott

Printed or Typed Name

SVP of Engineering and Marketing

Title

2/19/2009

Date

949-679-0152

Telephone number

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

ASSIGNMENT

WHEREAS, We, Chi-She Chen, a Taiwanese citizen, residing at 944 Crystal Water Lane, Walnut, CA 91789; Jeffrey C. Solomon, a United States citizen, residing at 16 Silver Fir, Irvine, CA 92604; Scott Milton, a United States citizen, residing at 49 Statehouse Place, Irvine, CA 92602; and Jayesh Bhakta, a United States citizen, residing at 12220 Rose Street, Cerritos, CA 90703, have invented certain new and useful improvements in a NON-VOLATILE MEMORY MODULE for which we have filed an application for Letters Patent in the United States, Application No. 12/131,873, Filed on June 2, 2008;

AND WHEREAS, Netlist, Inc. (hereinafter "ASSIGNEE"), a Delaware Corporation, with its principal place of business at 51 Discovery, Ste #150, Irvine, CA 92618, desires to acquire the entire right, title, and interest in and to said improvements and said Application:

NOW, THEREFORE, for good and valuable consideration, the receipt of which is hereby acknowledged, we, the inventors, do hereby acknowledge that we have sold, assigned, transferred and set over, and by these presents do hereby sell, assign, transfer and set over, unto said ASSIGNEE, its successors, legal representatives and assigns, the entire right, title, and interest throughout the world in, to and under said improvements, and said application including all provisional applications relating thereto (including but not limited to U.S. Provisional Application No. 60/941,586, filed June 1, 2007), and all divisions, renewals and continuations thereof, and all Letters Patent of the United States which may be granted thereon and all reissues and extensions thereof, and all rights of priority under International Conventions and applications for Letters Patent which may hereafter be filed for said improvements in any country or countries foreign to the United States, and all Letters Patent which may be granted for said improvements in any country or countries foreign to the United States and all extensions, renewals and reissues thereof; and we hereby authorize and request the Commissioner of Patents of the United States, and any Official of any country or countries foreign to the United States, whose duty it is to issue patents on applications as aforesaid, to issue all Letters Patent for said improvements to said ASSIGNEE, its successors, legal representatives and assigns, in accordance with the terms of this instrument.

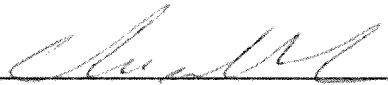
AND WE DO HEREBY sell, assign, transfer, and convey to ASSIGNEE, its successors, legal representatives, and assigns all claims for damages and all remedies arising out of any violation of the rights assigned hereby that may have accrued prior to the date of assignment to ASSIGNEE, or may accrue hereafter, including, but not limited to, the right to sue for, collect, and retain damages for past infringements of said Letters Patent before or after issuance.

AND WE HEREBY covenant and agree that we will communicate to said ASSIGNEE, its successors, legal representatives and assigns, any facts known to us respecting said improvements, and testify in any legal proceeding, sign all lawful papers, execute all divisional, continuing and reissue applications, make all rightful oaths and generally do everything possible to aid said ASSIGNEE, its successors, legal representatives and assigns, to obtain and enforce proper patent protection for said improvements in all countries.

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 2

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 24th day of October, 2008.


Chi-She Chen

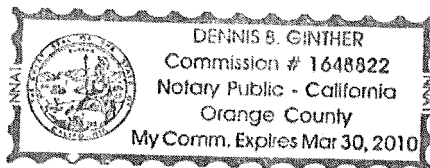
STATE OF California }
COUNTY OF Orange } ss.

On October, 24, 2008, before me, Dennis B. Ginter, notary public, personally appeared Chi-She Chen who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]

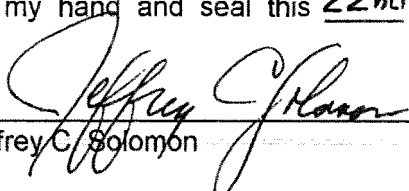



Notary Signature

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 3

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 22nd day of October, 2008.



Jeffrey C. Solomon

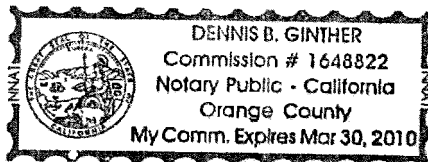
STATE OF California }
COUNTY OF Orange } ss.

On October 22, 2008, before me, Dennis B. Ginther, notary public, personally appeared Jeffrey C. Solomon who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]





Notary Signature

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 4

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 24 day of October, 2008.

Scott H. Milton
Scott Milton

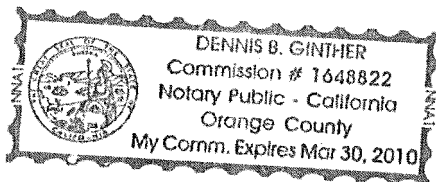
STATE OF California }
COUNTY OF Orange } ss.

On October 24, 2008, before me Dennis B. Ginter, notary public, personally appeared Scott Milton who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]



[Signature]
Notary Signature

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 22 day of October, 2008.

Jayesh Bhakta
Jayesh Bhakta

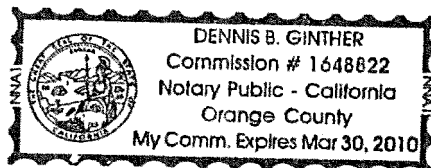
STATE OF California }
COUNTY OF Orange } ss.

On October 22, 2008, before me, Dennis B. Ginter, notary public, personally appeared Jayesh Bhakta who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]



[Signature]
Notary Signature

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re the Application of:	)	
Chen et al.	)	Group Art Unit: 2189
	)	
Serial No.: 12/240,916	)	Confirmation No.: 6240
	)	
Filed: September 29, 2008	)	Examiner: To be assigned
	)	
For: NON-VOLATILE MEMORY	)	
MODULE	)	

POWER OF ATTORNEY BY ASSIGNEE OF ENTIRE INTEREST
(REVOCATION OF PRIOR POWERS)

Commissioner for Patents
P. O. Box 1450
Alexandria, VA 22313-1450

Sir:

As representative of the assignee of an entire interest in the above-identified application, the undersigned hereby revokes all previous powers of attorney and appoints the practitioners associated with Customer Number **22145** as attorney for the assignee herein, to prosecute this application and to transact all business in the Patent and Trademark Office connected therewith.

Please direct all correspondence to:

Howard J. Klein (Reg. No.: 28,727)
Klein, O'Neill & Singh, LLP (Customer No. 22145)
43 Corporate Park, Suite 204
Irvine, California 92606
Telephone: (949) 955-1920
Fax: (949) 955-1921

Respectfully submitted,

Netlist Inc.

Dated: 2/19/2009

By: 

Name: James Perrott
Title: SVP of Engineering and Marketing

NETLIST INC.
51 Discovery, Suite 150
Irvine, California 92618

Electronic Acknowledgement Receipt

EFS ID:	4825579
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	20995
Filer:	Howard J. Klein/Carrie Cheung
Filer Authorized By:	Howard J. Klein
Attorney Docket Number:	NETL.040C1
Receipt Date:	19-FEB-2009
Filing Date:	29-SEP-2008
Time Stamp:	19:11:26
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		987-04-CON-- POAasFiled02192009.pdf	671314 2c6063649702d1d0d3fac6cc01cb3e03a769b4c7	yes	8

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Miscellaneous Incoming Letter	1	1
	Assignee showing of ownership per 37 CFR 3.73(b).	2	7
	Power of Attorney	8	8

Warnings:

Information:

Total Files Size (in bytes):	671314
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	12/240,916
	Filing Date	September 29, 2008
	First Named Inventor	Chi-She Chen et al.
	Title	NON-VOLATILE MEMORY MODULE
	Art Unit	2189
	Examiner Name	Unassigned
	Attorney Docket Number	NETL.040C1

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

22,145

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the:

☐ Applicant/Inventor.

OR

☒ Assignee of record of the entire interest. See 37 CFR 3.71.

Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on 10-28-2008

SIGNATURE of Applicant or Assignee of Record

Signature

Date

Name

Telephone

Title and Company

Netlist, Inc.

SVP, Marketing + Engineering

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.☒ *Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Acknowledgement Receipt

EFS ID:	5079583
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Brittany West
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.040C1
Receipt Date:	01-APR-2009
Filing Date:	29-SEP-2008
Time Stamp:	17:50:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	SIGNED_REVOCATION_NETL040C1.PDF	93520 d712cfa6b8e26d87ceac1e6f2bdd475447c9ab8e	no	1

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/240,916	09/29/2008	Chi-She Chen	987-04-CON

CONFIRMATION NO. 6240

POA ACCEPTANCE LETTER

22145
KLEIN, O'NEILL & SINGH, LLP
43 CORPORATE PARK
SUITE 204
IRVINE, CA 92606



Date Mailed: 04/07/2009

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 02/19/2009.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/sleutchit/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Substitute for form 1449A/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Use as many sheets as necessary)				<i>Complete if Known</i> <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%;">Application Number</td> <td>12/240,916</td> </tr> <tr> <td>Filing Date</td> <td>September 29, 2008</td> </tr> <tr> <td>First Named Inventor</td> <td>Chi-She Chen</td> </tr> <tr> <td>Art Unit</td> <td>2189</td> </tr> <tr> <td>Examiner Name</td> <td>Unknown</td> </tr> </table>		Application Number	12/240,916	Filing Date	September 29, 2008	First Named Inventor	Chi-She Chen	Art Unit	2189	Examiner Name	Unknown
Application Number	12/240,916														
Filing Date	September 29, 2008														
First Named Inventor	Chi-She Chen														
Art Unit	2189														
Examiner Name	Unknown														
Sheet	1	of	1	Attorney Docket No: 987-04-CON-H											

US PATENT DOCUMENTS					
Examiner Initial *	Cite No	USP Document Number	Publication Date	Name of Patentee or Applicant of cited Document	Filing Date If Appropriate
		4,420,821	12-13-1983	Hoffman	02-19-1982
		4,449,205	05-15-1984	Hoffman	02-19-1982
		5,519,663	05-21-1996	Harper, Jr. et al.	09-28-1994
		6,158,015	12-05-2000	Klein	03-30-1998
		6,336,176	01-01-2002	Li et al.	08-09-1999
		6,487,623	11-26-2002	Emerson et al.	04-30-1999
		6,658,507	12-02-2003	Chan	08-31-1998
		6,799,244	09-28-2004	Tanaka et al.	12-06-2002
		2002/0083368	06-27-2002	Abe et al.	12-20-2001
		2004/0190210	09-30-2004	Leete	03-26-2003

FOREIGN PATENT DOCUMENTS					
Examiner Initials*	Cite No	Foreign Document No	Publication Date	Name of Patentee or Applicant of cited Document	T ²

OTHER DOCUMENTS -- NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²

EXAMINER

DATE CONSIDERED

Electronic Acknowledgement Receipt

EFS ID:	5144773
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	22145
Filer:	Howard J. Klein/Monique Le
Filer Authorized By:	Howard J. Klein
Attorney Docket Number:	987-04-CON
Receipt Date:	13-APR-2009
Filing Date:	29-SEP-2008
Time Stamp:	17:32:58
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Corrected Filing Receipt	987-04-CON-- RequestFilingReceiptCorrectio ns04132009.pdf	377274 c25cd326ff0c67d7ab3a86a6bee89bd853b b9136	no	6

Warnings:

Information:

2	Information Disclosure Statement (IDS) Filed (SB/08)	987-04-CON-- IDSasFiled04132009.pdf	91382 34b3e6148abfa59d8bf9924c4752a25ca784852f	no	1
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
Total Files Size (in bytes):				468656	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s): Chen et al.
Assignee: Netlist, Inc.
Title: **NON-VOLATILE MEMORY MODULE**
Serial No.: 12/240,916 Filed: September 29, 2008
Examiner: Unknown Group Art Unit: 2189
Confirmation No.: 6240
Attorney Docket No.: 987-04-CON-HK

Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

REQUEST FOR FILING RECEIPT CORRECTIONS

Sir:

Attached hereto is a marked-up copy of the Official Filing Receipt in connection with the above-identified application.

THE FOLLOWING CORRECTION(S) IS/ARE RESPECTFULLY REQUESTED:

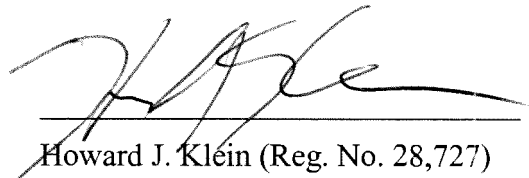
The second inventor's name should be **Jeffrey C. Solomon** instead of Jeffery C. Solomon.

It is respectfully requested that the U.S. Patent and Trademark Office forward/issue a new Filing Receipt with the correction indicated above. Support for the correction is readily apparent on the enclosed photocopy of the executed Declaration and Power of Attorney document.

If necessary, the Commissioner is hereby authorized in this, concurrent, and future replies, to charge payment or credit any overpayment to Deposit Account No. 11-1159 for any additional fee required under 37 C.F.R. §§ 1.16 or 1.17.

Respectfully submitted,

Date: April 13, 2009



Howard J. Klein (Reg. No. 28,727)
Klein, O'Neill & Singh, LLP (CN 22145)
43 Corporate Park, Suite 204
Irvine, California 92606
Tel: (949) 955-1920
Fax: (949) 955 1921



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/240,916	09/29/2008	2189	3570	NETL.040C1	54	7

CONFIRMATION NO. 6240

UPDATED FILING RECEIPT



0000000033714085

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 12/24/2008

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

This should be "Jeffrey" → Chi-She Chen, Walnut, CA;
Jeffery C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications

If Required, Foreign Filing License Granted: 10/08/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/240,916**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

NON-VOLATILE MEMORY MODULE

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

**DECLARATION FOR UTILITY OR DESIGN APPLICATION
UNDER 37 CFR 1.63**

Docket No.: NETL.040C1

Page 1 of 2

Title: NON-VOLATILE MEMORY MODULE

Inventors: Chi-She Chen, Jeffrey C. Solomon; Scott Milton; Jayesh Bhakta

Please Direct All Correspondence to Customer Number 20995

This Declaration is directed to the invention that:

Was filed as Serial No. 12/240,916 filed on September 29, 2008

As a below named inventor:

I believe the inventors named below to be the original and first inventors of the subject matter which is described and claimed and for which a patent is sought;

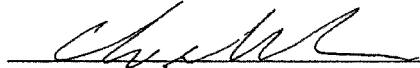
I have reviewed and understand the contents of the above-identified application, including the claims, and any amendment filed herewith or identified above;

I acknowledge the duty to disclose information which is material to patentability as defined in 37 CFR 1.56;

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful, false statements may jeopardize the validity of the application or any patent issued thereon.

Full name of first inventor: Chi-She Chen

Signature:



Date:

12-11-08

Citizenship:

Taiwan

Mailing Address:

944 Crystal Water Lane, Walnut, CA 91789

Residence Address:

Same as above

(if different than above)

**DECLARATION FOR UTILITY OR DESIGN APPLICATION
UNDER 37 CFR 1.63**

Docket No.: NETL040C1

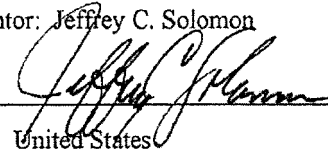
Page 2 of 2

Title: NON-VOLATILE MEMORY MODULE

Inventors: Chi-She Chen, Jeffrey C. Solomon; Scott Milton; Jayesh Bhakta

Please Direct All Correspondence to Customer Number 20995

Full name of **second** inventor: Jeffrey C. Solomon

Signature: 

Date: 12-11-08

Citizenship:

United States

Mailing Address:

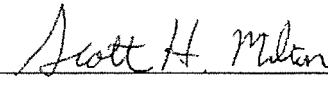
6 Silver Fir, Irvine, CA 92604

Residence Address:

Same as above

(if different than above)

Full name of **third** inventor: Scott Milton

Signature: 

Date: 12/11/2008

Citizenship:

United States

Mailing Address:

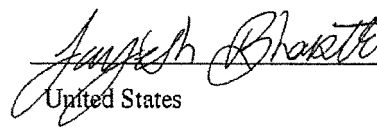
49 Statehouse Place, Irvine, CA 92602

Residence Address:

Same as above

(if different than above)

Full name of **fourth** inventor: Jayesh Bhakta

Signature: 

Date: 12-11-08

Citizenship:

United States

Mailing Address:

12220 Rose Street, Cerritos, CA 90703

Residence Address:

Same as above

(if different than above)

Send Correspondence To:

KNOBBE, MARTENS, OLSON & BEAR, LLP

Customer No. 20,995



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/240,916	09/29/2008	2189	3570	987-04-CON	54	7

CONFIRMATION NO. 6240

CORRECTED FILING RECEIPT



22145
KLEIN, O'NEILL & SINGH, LLP
43 CORPORATE PARK
SUITE 204
IRVINE, CA 92606

Date Mailed: 05/14/2009

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 22145

Domestic Priority data as claimed by applicant

This application is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications

If Required, Foreign Filing License Granted: 10/08/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/240,916**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

NON-VOLATILE MEMORY MODULE

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

Substitute for form 1449A/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				<i>Complete if Known</i>	
				Application Number	12/240,916
				Filing Date	September 29, 2008
				First Named Inventor	Chi-She Chen
				Art Unit	2189
				Examiner Name	Unknown
				IDS Filing Date	October 5, 2009
Sheet	1	of	1	Attorney Docket No: 987-04-CON-H	

US PATENT DOCUMENTS					
Examiner Initial *	Cite No	USP Document Number	Publication Date	Name of Patentee or Applicant of cited Document	Filing Date If Appropriate
		6,336,174	01-01-2002	Li et al.	08-09-1999
		7,409,590	08-05-2008	Moshayedi et al.	07-28-2006

FOREIGN PATENT DOCUMENTS					
Examiner Initials*	Cite No	Foreign Document No	Publication Date	Name of Patentee or Applicant of cited Document	T ²

OTHER DOCUMENTS -- NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²

EXAMINER

DATE CONSIDERED

Electronic Acknowledgement Receipt

EFS ID:	6201185
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	22145
Filer:	Howard J. Klein/Carrie Cheung
Filer Authorized By:	Howard J. Klein
Attorney Docket Number:	987-04-CON
Receipt Date:	05-OCT-2009
Filing Date:	29-SEP-2008
Time Stamp:	14:13:34
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Filed (SB/08)	987-04-CON-- IDSasFiled10052009.pdf	26559 8ccf00b1405571457647ea376a00d8a611715bc4	no	1

Warnings:

Information:

Total Files Size (in bytes):

26559

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

12/240,916

09/29/2008

Chi-She Chen

987-04-CON

6240

22145 7590 03/31/2011
KLEIN, O'NEILL & SINGH, LLP
18200 VON KARMAN AVENUE
SUITE 725
IRVINE, CA 92612

EXAMINER

ROJAS, MIDYS

ART UNIT

PAPER NUMBER

2185

MAIL DATE

DELIVERY MODE

03/31/2011

PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Office Action Summary	Application No. 12/240,916	Applicant(s) CHEN ET AL.	
	Examiner MIDYS ROJAS	Art Unit 2185	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 1 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 29 September 2008.
- 2a) ☐ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 4) ☒ Claim(s) 1-54 is/are pending in the application.
- 4a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 5) ☐ Claim(s) _____ is/are allowed.
- 6) ☐ Claim(s) _____ is/are rejected.
- 7) ☐ Claim(s) _____ is/are objected to.
- 8) ☒ Claim(s) 1-54 are subject to restriction and/or election requirement.

Application Papers

- 9) ☐ The specification is objected to by the Examiner.
- 10) ☒ The drawing(s) filed on 15 December 2008 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 11) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☐ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413) |
| 2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) | Paper No(s)/Mail Date. _____ |
| 3) ☐ Information Disclosure Statement(s) (PTO/SB/08) | 5) ☐ Notice of Informal Patent Application |
| Paper No(s)/Mail Date _____ | 6) ☐ Other: _____ |

DETAILED ACTION

Election/Restrictions

Restriction to one of the following inventions is required under 35 U.S.C. 121:

- I. Claims 1-20, and 43-49, drawn to a non-volatile and a volatile memory systems, classified in class 711, subclass 100 and 104.
- II. Claims 21-36, drawn to a power module for a volatile and non-volatile memory including a voltage conversion element, classified in class 327, subclass 51; class 365, subclass 242; and class 714, subclass 721.
- III. Claims 37-42 drawn to non-volatile and a volatile memory systems wherein the memory systems operate at different frequencies depending on the mode of operation classified in class 715, subclass 745, 789, 811.
- IV. Claims 50-54, drawn to restoring data, classified in class 707 subclass 679, 680, 681.

The inventions are distinct, each from the other because of the following reasons:

Inventions I and II are related as subcombinations disclosed as usable together in a single combination. The subcombinations are distinct if they do not overlap in scope and are not obvious variants, and if it is shown that at least one subcombination is separately usable. In the instant case, subcombination II has separate utility such as use in a system where a voltage conversion element is used for convert operational voltage in order to operation in different modes. See MPEP § 806.05(d).

Inventions I and III are related as subcombinations disclosed as usable together in a single combination. The subcombinations are distinct if they do not overlap in scope and are not obvious variants, and if it is shown that at least one subcombination is separately usable. In the instant case, subcombination III has separate utility such as use in a system where different modes of operation represent different operational frequencies. See MPEP § 806.05(d).

Inventions I and IV are related as subcombinations disclosed as usable together in a single combination. The subcombinations are distinct if they do not overlap in scope and are not obvious variants, and if it is shown that at least one subcombination is separately usable. In the instant case, subcombination IV has separate utility such as use in a system where data can be restored. See MPEP § 806.05(d).

Inventions II and III are related as subcombinations disclosed as usable together in a single combination. The subcombinations are distinct if they do not overlap in scope and are not obvious variants, and if it is shown that at least one subcombination is separately usable. In the instant case, subcombination III has separate utility such as use in a system where different modes of operation represent different operational frequencies. See MPEP § 806.05(d).

Inventions II and IV are related as subcombinations disclosed as usable together in a single combination. The subcombinations are distinct if they do not overlap in scope and are not obvious variants, and if it is shown that at least one subcombination is separately usable. In the instant case, subcombination IV has separate utility such as use in a system where data can be restored. See MPEP § 806.05(d).

Inventions III and IV are related as subcombinations disclosed as usable together in a single combination. The subcombinations are distinct if they do not overlap in scope and are not obvious variants, and if it is shown that at least one subcombination is separately usable. In the instant case, subcombination IV has separate utility such as use in a system where data can be restored. See MPEP § 806.05(d).

The examiner has required restriction between subcombinations usable together. Where applicant elects a subcombination and claims thereto are subsequently found allowable, any claim(s) depending from or otherwise requiring all the limitations of the allowable subcombination will be examined for patentability in accordance with 37 CFR 1.104. See MPEP § 821.04(a). Applicant is advised that if any claim presented in a continuation or divisional application is anticipated by, or includes all the limitations of, a claim that is allowable in the present application, such claim may be subject to provisional statutory and/or nonstatutory double patenting rejections over the claims of the instant application.

Restriction for examination purposes as indicated is proper because all these inventions listed in this action are independent or distinct for the reasons given above and there would be a serious search and/or examination burden if restriction were not required because at least the following reason(s) apply:

- (a) the inventions have acquired a separate status in the art in view of their different classification;
- (b) the inventions have acquired a separate status in the art due to their recognized divergent subject matter;

Art Unit: 2185

(c) the inventions require a different field of search (for example, searching different classes/subclasses or electronic resources, or employing different search queries);

Applicant is advised that the reply to this requirement to be complete must include (i) an election of a invention to be examined even though the requirement may be traversed (37 CFR 1.143) and (ii) identification of the claims encompassing the elected invention.

The election of an invention may be made with or without traverse. To reserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the restriction requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable upon the elected invention.

Should applicant traverse on the ground that the inventions are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing the inventions to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the inventions unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103(a) of the other invention.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MIDYS ROJAS whose telephone number is (571)272-4207. The examiner can normally be reached on M-TH 6:00am - 4:30pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Sanjiv Shah can be reached on (571) 272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Midys Rojas/
Examiner, Art Unit 2185

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s): Chen et al.
Assignee: Netlist, Inc.
Title: **NON-VOLATILE MEMORY MODULE**
Serial No.: 12/240,916 Filed: September 29, 2008
Examiner: Rojas, Midys Group Art Unit: 2185
Attorney Docket No.: 987-004.201 Confirmation No.: 6240

Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

PRELIMINARY AMENDMENT
AND REPLY TO ELECTION/RESTRICTION REQUIREMENT

Sir:

In response to the restriction/election requirement mailed March 31, 2011, please amend the application as shown on the pages below.

Amendments to the Claims are described on page 2 of this paper.

Remarks begin on page 9 of this paper.

A Request for a One-Month Extension of Time under 37 CFR 1.136(a) is submitted herewith, along with the fee prescribed by 37 CFR 1.17(a)(1).

Amendments to the Claims:

This listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Withdrawn) A memory system coupled to a computer system, comprising:
a volatile memory subsystem;
a non-volatile memory subsystem;
a controller operatively coupled to the non-volatile memory subsystem;
at least one circuit configured to selectively operatively decouple the controller from the volatile memory subsystem.
2. (Withdrawn) The memory system of Claim 1, wherein the memory system is configured to
be operated in at least two states, the at least two states comprising:
a first state in which the controller and the non-volatile memory subsystem are decoupled from the volatile memory subsystem by the at least one circuit; and
a second state in which the volatile memory subsystem is operatively coupled to the controller to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem via the controller.
3. (Withdrawn) The memory system of Claim 2, wherein power is supplied to the volatile memory subsystem from a first power supply when the memory system is in the first state and power is supplied to the volatile memory subsystem from a second power supply when the memory system is in the second state.
4. (Withdrawn) The memory system of Claim 3, wherein the second power supply does not comprise a battery.
5. (Withdrawn) The memory system of Claim 3, wherein the second power supply comprises one or more capacitors.
6. (Withdrawn) The memory system of Claim 5, wherein the second power supply comprises a battery and is configured to power the volatile memory subsystem for less

than thirty minutes.

7. (Withdrawn) The memory system of Claim 5, wherein the one or more capacitors comprises one or more double layered capacitors.

8. (Withdrawn) The memory system of Claim 5, wherein the one or more capacitors are charged by the first power supply while the memory system is in the first state.

9. (Withdrawn) The memory system of Claim 3, further comprising a printed circuit board, wherein the second power supply and the memory system are located on the printed circuit board.

10. (Withdrawn) The memory system of Claim 3, wherein the at least two states further comprise a third state in which power is supplied to the volatile memory subsystem from a third power supply.

11. (Withdrawn) The memory system of Claim 2, wherein the controller and the non-volatile memory subsystem do not add a significant capacitive load to the volatile memory system when the memory system is in the first state.

12. (Withdrawn) The memory system of Claim 1, wherein the volatile memory subsystem comprises one or more DRAM memory elements.

13. (Withdrawn) The memory system of Claim 1, wherein the volatile memory subsystem comprises one or more SRAM memory elements.

14. (Withdrawn) The memory system of Claim 1, wherein the non-volatile memory subsystem comprises one or more flash memory elements.

15. (Withdrawn) The memory system of Claim 1, wherein the memory system is implemented on a memory module.

16. (Withdrawn) The memory system of Claim 1, wherein the at least one circuit comprise one or more switches coupled to the controller, to the volatile memory subsystem, and to the host computer.

17. (Withdrawn) The memory system of Claim 1, wherein the volatile memory subsystem comprises a registered DIMM comprising one or more registers and a plurality of DRAM elements, wherein the one or more registers are coupled to the computer system.

18. (Withdrawn) The memory system of Claim 17, wherein the at least one circuit comprise one or more switches coupled to the one or more registers and to the plurality of DRAM elements and configured to selectively operatively couple the volatile memory subsystem and

to the computer system.

19. (Withdrawn) The memory system of Claim 17, wherein the one or more registers are coupled to the plurality of DRAM elements and the at least one circuit comprises one or more switches coupled to the one or more registers and to a power source for the one or more registers, wherein the one or more switches are configured to selectively operatively couple the one or more registers to the power source for the one or more registers.

20. (Withdrawn) The memory system of Claim 1, wherein the non-volatile memory subsystem comprises at least 100 percent more storage capacity than does the volatile memory subsystem.

21. (Withdrawn) A power module for providing a plurality of voltages to a memory system comprising non-volatile and volatile memory, the plurality of voltages comprising at least a first voltage and a second voltage, the power module comprising:

- an input providing a third voltage to the power module;

- a voltage conversion element configured to provide the second voltage to the memory system;

- a first power element configured to selectively provide a fourth voltage to the conversion element;

- a second power element configured to selectively provide a fifth voltage to the conversion element, wherein the power module is configured to selectively provide the first voltage to the memory system either from the conversion element or from the input, wherein the power module is configured to be operated in at least three states comprising:

- a first state in which the first voltage is provided to the memory system from the input and the fourth voltage is provided to the conversion element from the first power element;

- a second state in which the fourth voltage is provided to the conversion element from the first power element and the first voltage is provided to the memory system from the conversion element;

- a third state in which the fifth voltage is provided to the conversion element from the second power element and the first voltage is provided to the memory system from the conversion element.

22. (Withdrawn) The power module of Claim 21, wherein the first power element comprises a pulse-width modulation power controller.

23. (Withdrawn) The power module of Claim 21, wherein the second power element does not comprise a battery.

24. (Withdrawn) The power module of Claim 21, wherein the second power element comprises one or more capacitors.

25. (Withdrawn) The power module of Claim 24, wherein the second power element comprises a battery and is configured to power the volatile memory for less than thirty minutes.

26. (Withdrawn) The power module of Claim 21, wherein the conversion element comprises one or more buck converters.

27. (Withdrawn) The power module of Claim 21, wherein the conversion element comprises one or more buck-boost converters.

28. (Withdrawn) A method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems, the method comprising:

during a first condition, providing the first voltage to the memory system from an input power supply and providing the second voltage to the memory system from a first power subsystem;

detecting a second condition;

during the second condition, providing the first voltage and the second voltage to the memory system from the first power subsystem;

charging a second power subsystem;

detecting a third condition;

during the third condition, providing the first voltage and the second voltage to the memory system from the second power subsystem.

29. (Withdrawn) The method of Claim 28, wherein detecting the second condition comprises detecting that a trigger condition is likely to occur.

30. (Withdrawn) The method of Claim 29, wherein detecting the third condition comprises detecting that the trigger condition has occurred.

31. (Withdrawn) The method of Claim 28, wherein the trigger condition comprises a power reduction.

32. (Withdrawn) The method of Claim 28, wherein the trigger condition comprises a power failure.

33. (Withdrawn) The method of Claim 28, wherein the trigger condition comprises a system hang-up.

34. (Withdrawn) The method of Claim 28, wherein the first power subsystem comprises a pulse-width modulation power controller and a voltage conversion subsystem.

35. (Withdrawn) The method of Claim 28, wherein the second power subsystem does not comprise a battery.

36. (Withdrawn) The method of Claim 35, wherein the second power subsystem comprises one or more capacitors and the voltage conversion subsystem.

37. (Original) A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third frequency when the memory system is in the second mode of operation, the third frequency less than the first frequency.

38. (Original) The method of Claim 37, wherein the third frequency is approximately equal to the second frequency.

39. (Original) The method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation.

40. (Currently Amended) The method of Claim ~~[[7]]~~ 37, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

41. (Original) The method of Claim 40, wherein the trigger condition comprises a power failure condition.

42. (Original) The method of Claim 37, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

43. (Withdrawn) A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

- communicating data words between the volatile memory subsystem and the host system when the memory system is in a first mode of operation; and

- transferring data words from the volatile memory subsystem to the nonvolatile memory subsystem when the memory system is in a second mode of operation, wherein transferring each data word comprises:

- storing a first portion of the data word in a buffer;

- storing a second portion of the data word in the buffer;

- and writing the entire data word from the buffer to the non-volatile memory subsystem.

44. (Withdrawn) The method of Claim 43, wherein the transferring each data word further comprises storing a third portion of each data word in the buffer.

45. (Withdrawn) A memory system operatively coupled to a host system, comprising:

- a volatile memory subsystem;

- a non-volatile memory subsystem comprising at least 100 percent more storage capacity than does the volatile memory subsystem; and

- a controller operatively coupled to the volatile memory subsystem and operatively coupled to the non-volatile memory subsystem, the controller configured to allow data to be communicated between the volatile memory subsystem and the host system when the memory system is operating in a first state and to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem when the memory system is opeiating in a second state.

46. (Withdrawn) The memory system of Claim 45, wherein power is supplied to the volatile memory subsystem from a first power supply when the memory system is in the first state.

47. (Withdrawn) The memory system of Claim 46, wherein power is supplied to the volatile memory subsystem from a second power supply when the memory system is in the second state.

48. (Withdrawn) The memory system of Claim 47, wherein the second power supply does not comprise a battery.

49. (Withdrawn) The memory system of Claim 45, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem, the non-volatile memory subsystem, and the controller are located on the printed circuit board.

50. (Withdrawn) A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

communicating data between the volatile memory subsystem and the host system when the memory system is in a first mode of operation;

storing a first copy of data from the volatile memory subsystem to the nonvolatile memory subsystem at a first time when the memory system is in a second mode of operation;

restoring the first copy of data from the non-volatile memory subsystem to the volatile memory subsystem;

erasing the first copy of data from the non-volatile memory subsystem; and

storing a second copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a second time when the memory system is in the second mode of operation, wherein storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem.

51. (Withdrawn) The method of Claim 50, further comprising restoring the second copy of data from the non-volatile memory subsystem to the volatile memory subsystem.

52. (Withdrawn) The method of Claim 50, wherein the memory system enters the second mode of operation in response to a power failure.

53. (Withdrawn) The method of Claim 50, wherein the memory system is not powered by a battery when it is in the second mode of operation.

54. (Withdrawn) The method of Claim 50, wherein the first copy of data and the second copy of data are stored in separate portions of the non-volatile memory subsystem.

REMARKS

In response to the restriction/election requirement mailed March 31, 2011, applicant elects to prosecute Group III. At least the following claims read on the elected group: Claims 37-42.

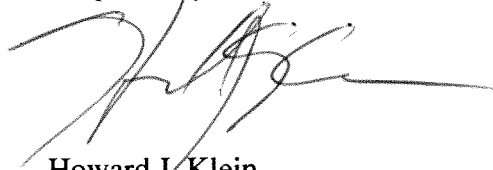
This election is made without traverse.

A Request for a One Month Extension of Time under 37 CFR 1.136(a) is submitted herewith, along with the fee prescribed by 37 CFR 1.17(a)(1).

Should there be any outstanding matters that need to be resolved in the present application, the Examiner is respectfully requested to contact the undersigned.

If necessary, the Commissioner is hereby authorized in this, concurrent, and future replies, to charge payment or credit any overpayment to Deposit Account No. 11-1159 for any additional fees required under 37 C.F.R. § 1.16 or under 37 C.F.R. § 1.17; particularly, extension of time fees.

Respectfully submitted,



Howard J. Klein
Registration No; 28,727

Date: May 20, 2011

Attorney's Email: hjklein@koslaw.com
Attorney's Direct Tel.: (949) 390-2730

Klein, O'Neill & Singh, LLP (Customer No.: 22145)
18200 Von Karman Avenue, Suit 725
Irvine, CA 92612
Tel: (949) 955-1920
Fax: (949) 955 1921

Electronic Patent Application Fee Transmittal

Application Number:	12240916			
Filing Date:	29-Sep-2008			
Title of Invention:	NON-VOLATILE MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Howard J. Klein/Carrie Cheung			
Attorney Docket Number:	987-004.201			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Extension - 1 month with \$0 paid	1251	1	130	130

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				130

Electronic Acknowledgement Receipt

EFS ID:	10141197
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	22145
Filer:	Howard J. Klein/Carrie Cheung
Filer Authorized By:	Howard J. Klein
Attorney Docket Number:	987-004.201
Receipt Date:	20-MAY-2011
Filing Date:	29-SEP-2008
Time Stamp:	19:46:08
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 130
RAM confirmation Number	6239
Deposit Account	111159
Authorized User	CHEUNG,CARRIE A.

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		987-004-201-- RRRof03312011asFiled0520201 1.pdf	710587 7e2b20deb8661f44c60460364d2bb04cd1115084	yes	11
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Extension of Time		1	2	
	Response to Election / Restriction Filed		3	3	
	Claims		4	10	
	Applicant Arguments/Remarks Made in an Amendment		11	11	

Warnings:**Information:**

2	Fee Worksheet (PTO-875)	fee-info.pdf	30306 eb01d48a823f3dcdff6025d3b6e2400d64fa4a034	no	2
---	-------------------------	--------------	--	----	---

Warnings:**Information:**

Total Files Size (in bytes):	740893
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

**Amendment Transmittal & Petition for Extension
of Time under 37 CFR 1.136(a)**

Docket Number

987-004.101

Address To
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

Title of Invention

NON-VOLATILE MEMORY MODULE

First Named Inventor	Chi-She Chen
Application No.	12/240,916
Filing Date	September 29, 2008
Examiner	Midys Rojas
Art Unit	2185

Transmitted herewith is an amendment in the above-identified application.

This is also a petition under the provisions of 37 CFR 1.136(a) to extend the period for filing a reply in the above identified application.

The requested extension and fee are as shown below (check time period desired).

Fee Calculation**Extension of Time Fee**

☒ One month (37 CFR 1.17(a)(1)) ☐ Two months (37 CFR 1.17(a)(2)) ☐ Three months (37 CFR 1.17(a)(3))
☐ Four months (37 CFR 1.17(a)(4)) ☐ Five months (37 CFR 1.17(a)(5))

Claims as Amended

For	#Filed	#Previously Paid For	#Extra	Rate	Fee
Total Claims	6	- 20 =		x 52 =	
Total Indep. Claims	1	- 3 =		x 220 =	
Multiple Dependent Claims (check if applicable) ☐					
Extension Fee (from above)					\$130
☐ Applicant claims small entity status. See 37 CFR 1.27.					
TOTAL					\$130

Method of Payment

☐ Deposit Account ☒ Credit Card ☐ Check ☐ Money Order ☐ Other: _____

Deposit Account Number 11-1159

For the above-identified deposit account, the Director is hereby authorized to: (check all that apply)

- ☐ Charge the fee(s) set forth above
☒ Charge any additional fee(s) or underpayments of fee(s) under 37 CFR 1.16 and 1.17
☐ Charge fee(s) indicated above, **except for the filing fee**
☒ Credit any overpayments
☒ If an additional extension of time is required, please consider this a petition therefor and charge any additional fees which may be required to the Deposit Account above.

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on form PTO-2038.

Amount Grand Total \$130

Amendment Transmittal & Petition for Extension of Time under 37 CFR 1.136(a)

Docket Number

987-004.101

Correspondence Address

Customer Number 22145

-OR-

Name

Address

City

State

Country

Postal Code

Phone Number

E-mail Address

Certificate of Mailing by Express Mail

I hereby certify that this Amendment and Petition for Extension of Time, accompanying documents, and fee are being deposited with the United States Postal Service "Express Mail Post Office to Addressee" service under 37 CFR 1.10 in an envelope addressed to Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450 on the date indicated below:

(Date of Mailing)

(Typed or Printed Name of Person Mailing Correspondence)

(Signature of Person Mailing Correspondence)

("Express Mail" Mailing Label Number)

Certificate of Mailing by First Class Mail

I hereby certify that this Amendment and Petition for Extension of Time, accompanying documents, and fee are being deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450 on the date indicated below:

(Date of Mailing)

(Name of Person Mailing Correspondence)

(Signature of Person Mailing Correspondence)

Certificate of Transmission

I hereby certify that this Amendment and Petition for Extension of Time, accompanying documents, and fee authorization are being facsimile transmitted to the United States Patent and Trademark Office on the date indicated below:

(Date of Transmission)

(Name of Person Transmitting Correspondence)

(Signature of Person Transmitting Correspondence)

Signature Instructions

Select the name of the person who will electronically sign the Amendment and Petition for Extension of Time from the drop-down box below.

If a practitioner is not present in the drop-down list, you must close this form and select 'Add Practitioner...' in the Form Manager's Utility menu.

Verify that the signatory information is correct and press the 'eSign' button to electronically sign the submission. If you prefer to sign the form manually, simply do not click the 'eSign' button; just print and manually sign.

Signatory Drop-Down Box

Howard J. Klein

Name

Howard J. Klein

Registration Number

28727

Signatory Capacity

Attorney for Applicant(s)

E-mail Address

eSign

Date Signed

5/20/2011

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/240,916		Filing Date 09/29/2008		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
			TOTAL			TOTAL				
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	05/20/2011	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 54	Minus	** 54	=	0		X \$52=	0	
	Independent (37 CFR 1.16(h))	* 7	Minus	*** 7	=	0		X \$220=	0	
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE			TOTAL ADD'L FEE	0	
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			X \$	=	
	Independent (37 CFR 1.16(h))	*	Minus	***	=			X \$	=	
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE			TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/PATRICIA LEWIS/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

12/240,916

09/29/2008

Chi-She Chen

987-004.201

6240

22145 7590 07/29/2011
KLEIN, O'NEILL & SINGH, LLP
18200 VON KARMAN AVENUE
SUITE 725
IRVINE, CA 92612

EXAMINER

ROJAS, MIDYS

ART UNIT

PAPER NUMBER

2185

MAIL DATE

DELIVERY MODE

07/29/2011

PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Office Action Summary	Application No. 12/240,916	Applicant(s) CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 20 May 2011.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 4) ☒ Claim(s) 37-42 is/are pending in the application.
- 4a) Of the above claim(s) 1-36 and 43-54 is/are withdrawn from consideration.
- 5) ☐ Claim(s) _____ is/are allowed.
- 6) ☒ Claim(s) 37-42 is/are rejected.
- 7) ☐ Claim(s) _____ is/are objected to.
- 8) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

Application Papers

- 9) ☐ The specification is objected to by the Examiner.
- 10) ☒ The drawing(s) filed on 29 September 2008 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
 Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
 Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 11) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).
- * See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| <p>1) ☒ Notice of References Cited (PTO-892)</p> <p>2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948)</p> <p>3) ☒ Information Disclosure Statement(s) (PTO/SB/08)
 Paper No(s)/Mail Date <u>4/13/2009; 10/5/2009</u>.</p> | <p>4) ☐ Interview Summary (PTO-413)
 Paper No(s)/Mail Date. _____.</p> <p>5) ☐ Notice of Informal Patent Application</p> <p>6) ☐ Other: _____.</p> |
|--|---|

DETAILED ACTION

Election/Restrictions

Applicant's election without traverse of claims 37-42 in the reply filed on 5/20/2011 is acknowledged.

Information Disclosure Statement

The information disclosure statements (IDS) submitted on 4/13/2009 and 10/5/2009 were considered by the examiner.

Drawings

The drawings received on 9/29/2008 have been accepted by the examiner.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 37-42 are rejected under 35 U.S.C. 103(a) as being unpatentable over Li et al. [US 6,336,174], Applicant cited art, in view of Oshikiri [US 2007/0192627].

Claim 37, Li et al. discloses a method of controlling [via controller 206, Fig. 2] a memory system [hardware assisted memory module 104, of Fig. 1 and 2] operatively coupled to a host system [host system 100 including CPU 102, Fig. 1] and which includes a volatile memory subsystem [volatile memory 202, Fig. 2] and a non-volatile memory subsystem [nonvolatile memory 204, Fig. 2], the method comprising:

operating the volatile memory subsystem when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system [volatile memory 202 switches between modes of operation by switching isolation devices on/off in response to trigger events, see col. 6, lines 48-57; during normal operation, when the isolation devices are on, the HAMM behaves like a conventional memory module, see Abstract, and stores digital information received from the data bus of the host, see col. 3, lines 5-9 and col. 5, line 55 – col. 6, line 3, thus representing the first mode of operation claimed];

operating the non-volatile memory subsystem when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem [when a trigger event occurs, such as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67].

Li et al. does not teach that in the first mode, the volatile memory operates at a first frequency while in the second mode, the non-volatile memory operates at a second frequency and the volatile memory operates at a third frequency that is less than the first frequency.

Oshikiri discloses a semiconductor memory 2 being used in an information processing apparatus 1 [Fig. 1] which switches between two operation modes [see par. 0029] wherein a first operation mode is detected when a first mode switching command

Art Unit: 2185

is received from the information processing apparatus [see par. 0031] and a second operation mode is detected when a second mode switching command is received from the information processing apparatus [see par. 0039]. In either operating mode, the information processing apparatus executes various processes in the semiconductor memory by using the CPU 11 and RAM 12 [par. 0021]. When the semiconductor memory operates in the second mode, it is operating at a lower level of security but at a higher processing speed, thus representing a higher operating frequency [par. 0040]. Therefore, when in the second mode, all elements accessing the memory operate at this higher operating frequency, thus representing the claimed first mode of operation [higher first frequency]. In Oshikiri's first mode, all elements operate at the lower operating speed, thus representing the claimed second mode of operation [lower second and third frequency]

It would have been obvious to one of ordinary skill in the art at the time the invention was made to combine the teachings of Li et al. with the operating seed of Oshikiri since providing for memory operations in different frequencies depending on the memory mode provided since different operating speeds have different power requirements.

Claim 38, Li et al. in view of Oshikiri discloses the method of Claim 37, wherein the third frequency is approximately equal to the second frequency [In Oshikiri's first mode, all elements operate at the lower operating speed, thus representing the claimed second mode of operation, lower second and third frequency, par. 0031 and 0040]

Claim 39, Li et al. in view of Oshikiri discloses the method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation [when the trigger condition that causes the switch of operation mode is a O/S hang-up, a backup battery is not required, see col. 7, lines 59-66 of Li et al.].

Claim 40, Li et al. in view of Oshikiri discloses the method of Claim 37, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition [detection of a trigger event, see Abstract of Li et al.].

Claim 41, Li et al. in view of Oshikiri discloses the method of Claim 40, wherein the trigger condition comprises a power failure condition [trigger event may be power failure, see Abstract of Li et al.].

Claim 42, Li et al. in view of Oshikiri discloses the method of Claim 37, wherein the memory system 104 further comprises a printed circuit board [as shown in Fig. 2 all elements are within one board] and the volatile memory subsystem 202 and the non-volatile memory subsystem 204 are located on the printed circuit board [see Fig. 2, and col. 2, line 58-col. 3, line 4].

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MIDYS ROJAS whose telephone number is (571)272-4207. The examiner can normally be reached on M-TH 6:00am - 4:30pm.

Art Unit: 2185

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Sanjiv Shah can be reached on (571) 272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Midys Rojas/
Primary Examiner, Art Unit 2185

MR

Notice of References Cited	Application/Control No. 12/240,916	Applicant(s)/Patent Under Reexamination CHEN ET AL.	
	Examiner MIDYS ROJAS	Art Unit 2185	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-2007/0192627	08-2007	Oshikiri, Takashi	713/191
	B	US-			
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	07/28/2011								
	1	N								
	2	N								
	3	N								
	4	N								
	5	N								
	6	N								
	7	N								
	8	N								
	9	N								
	10	N								
	11	N								
	12	N								
	13	N								
	14	N								
	15	N								
	16	N								
	17	N								
	18	N								
	19	N								
	20	N								
	21	N								
	22	N								
	23	N								
	24	N								
	25	N								
	26	N								
	27	N								
	28	N								
	29	N								
	30	N								
	31	N								
	32	N								
	33	N								
	34	N								
	35	N								
	36	N								

<i>Index of Claims</i> 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	07/28/2011							
	37	✓							
	38	✓							
	39	✓							
	40	✓							
	41	✓							
	42	✓							
	43	N							
	44	N							
	45	N							
	46	N							
	47	N							
	48	N							
	49	N							
	50	N							
	51	N							
	52	N							
	53	N							
	54	N							

Search Notes 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

SEARCHED			
Class	Subclass	Date	Examiner
711	160, 161, 162	7/28/2011	Mr
710	10	7/28/2011	Mr

SEARCH NOTES		
Search Notes	Date	Examiner
EAST search: limited search of 711/160, 161, 162; 710/10; and text searches	7/28/2011	Mr
PALM inventor name search		Mr

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner

	/MIDYS ROJAS/ Primary Examiner.Art Unit 2185
--	---

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	4347	((711/162) or (711/160) or (711/161) or (710/10)).CCLS.	USPAT; USOCR	OR	OFF	2011/07/28 19:20
L2	0	L1 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/28 19:21
L3	1	L1 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 19:21
S1	0	("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/27 15:21
S2	12	("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:21
S3	0	host same (volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:34
S4	1	(volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:34
S5	87	(memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:35
S6	56	"711"/\$.ccls. and (memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:35
S7	1013	memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:44
S8	128	"711"/\$.ccls. and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2011/07/27 15:44

			JPO			
S9	46	("4072852" "4815074" "4959774" "5283792" "5379431" "5799200").PN. OR ("6336174").URPN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/27 15:44
S10	8	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) with (frequency or speed or clock)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:08
S11	2	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:09
S12	0	memory same (host near3 frequency) same (non-volatile near3 frequency) same mode	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:11
S13	1	(memory with mode) same (host with frequency) same (non-volatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:12
S14	6	(memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:12
S15	13	"711"/\$.ccls. and (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:14
S16	40	"711"/\$.ccls. and (host with speed) same (nonvolatile with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:22
S17	16	"711"/\$.ccls. and (host with speed) same (backup with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:23
S18	50	(backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:29
S19	48	power and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:29
S20	14	(power near3 (loss or fail\$3)) and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:30
S21	8	(power near3 (loss or fail\$3)) and (copy\$3 or tranfer\$3) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:38
S22	8	"711"/\$.ccls. and (copy\$3 or tranfer\$3 or back-up or (back adj up)) with (high\$3 near3 (speed or frequency) near3 (non- volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:41
S23	88	"711"/\$.ccls. and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:44

EAST Search History (Interference)

<This search history is empty>

7/ 28/ 2011 7:22:23 PM

C:\ Users\ mrojas\ Documents\ EAST\ Workspaces\ 12240916.wsp

Substitute for form 1449A/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Use as many sheets as necessary)				<i>Complete if Known</i>	
				Application Number	12/240,916
				Filing Date	September 29, 2008
				First Named Inventor	Chi-She Chen
				Art Unit	2189
				Examiner Name	Unknown
Sheet	1	of	1	Attorney Docket No: 987-04-CON-H	

US PATENT DOCUMENTS					
Examiner Initial *	Cite No	USP Document Number	Publication Date	Name of Patentee or Applicant of cited Document	Filing Date If Appropriate
/M.R./		4,420,821	12-13-1983	Hoffman	02-19-1982
		4,449,205	05-15-1984	Hoffman	02-19-1982
		5,519,663	05-21-1996	Harper, Jr. et al.	09-28-1994
		6,158,015	12-05-2000	Klein	03-30-1998
		6,336,176	01-01-2002	Li et al.	08-09-1999
		6,487,623	11-26-2002	Emerson et al.	04-30-1999
		6,658,507	12-02-2003	Chan	08-31-1998
		6,799,244	09-28-2004	Tanaka et al.	12-06-2002
		2002/0083368	06-27-2002	Abe et al.	12-20-2001
		2004/0190210	09-30-2004	Leete	03-26-2003

FOREIGN PATENT DOCUMENTS					
Examiner Initials*	Cite No	Foreign Document No	Publication Date	Name of Patentee or Applicant of cited Document	T ²

OTHER DOCUMENTS -- NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²

EXAMINER

/Midys Rojas/

DATE CONSIDERED

07/27/2011

Substitute for form 1449A/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Use as many sheets as necessary)				<i>Complete if Known</i>	
				Application Number	12/240,916
				Filing Date	September 29, 2008
				First Named Inventor	Chi-She Chen
				Art Unit	2189
				Examiner Name	Unknown
				IDS Filing Date	October 5, 2009
Sheet	1	of	1	Attorney Docket No: 987-04-CON-H	

US PATENT DOCUMENTS					
Examiner Initial *	Cite No	USP Document Number	Publication Date	Name of Patentee or Applicant of cited Document	Filing Date If Appropriate
/M.R./		6,336,174	01-01-2002	Li et al.	08-09-1999
/M.R./		7,409,590	08-05-2008	Moshayedi et al.	07-28-2006

FOREIGN PATENT DOCUMENTS					
Examiner Initials*	Cite No	Foreign Document No	Publication Date	Name of Patentee or Applicant of cited Document	T ²

OTHER DOCUMENTS -- NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²

EXAMINER

/Midys Rojas/

DATE CONSIDERED 07/27/2011

Substitute Disclosure Statement Form (PTO-1449)

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional) ² Applicant is to place a check mark here if English language Translation is attached

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /M.R./

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	12/240,916
	Filing Date	2008-09-29
	First Named Inventor	Chi-She Chen
	Title	Non-volatile Memory Module
	Art Unit	2185
	Examiner Name	Midys Rojas
	Attorney Docket Number	062453-002

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46,188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the:

☐ Applicant/Inventor.

OR

☒ Assignee of record of the entire interest. See 37 CFR 3.71.

Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____

SIGNATURE of Applicant or Assignee of Record

Signature

Date

Name

Gail Sasaki

Telephone

Title and Company

Vice-president and Chief Financial Officer, Netlist, Inc.

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.☐ *Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

STATEMENT UNDER 37 CFR 3.73(b)Applicant/Patent Owner: Netlist, Inc.Application No./Patent No.: 12/240,916Filed/Issue Date: September 29, 2008

Titled:

NON-VOLATILE MEMORY MODULE

Netlist, Inc.

, a corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)

the patent application/patent identified above, by virtue of either:

- A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel _____, Frame _____, or for which a copy therefore is attached.

OR

- B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☐ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Khaled Shami/

Signature

September 7, 2011

Date

Khaled Shami

Printed or Typed Name

Attorney of Record

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 1

ASSIGNMENT

WHEREAS, We, Chi-She Chen, a Taiwanese citizen, residing at 944 Crystal Water Lane, Walnut, CA 91789; Jeffrey C. Solomon, a United States citizen, residing at 16 Silver Fir, Irvine, CA 92604; Scott Milton, a United States citizen, residing at 49 Statehouse Place, Irvine, CA 92602; and Jayesh Bhakta, a United States citizen, residing at 12220 Rose Street, Cerritos, CA 90703, have invented certain new and useful improvements in a NON-VOLATILE MEMORY MODULE for which we have filed an application for Letters Patent in the United States, Application No. 12/131,873, Filed on June 2, 2008;

AND WHEREAS, Netlist, Inc. (hereinafter "ASSIGNEE"), a Delaware Corporation, with its principal place of business at 51 Discovery, Ste #150, Irvine, CA 92618, desires to acquire the entire right, title, and interest in and to said improvements and said Application:

NOW, THEREFORE, for good and valuable consideration, the receipt of which is hereby acknowledged, we, the inventors, do hereby acknowledge that we have sold, assigned, transferred and set over, and by these presents do hereby sell, assign, transfer and set over, unto said ASSIGNEE, its successors, legal representatives and assigns, the entire right, title, and interest throughout the world in, to and under said improvements, and said application including all provisional applications relating thereto (including but not limited to U.S. Provisional Application No. 60/941,586, filed June 1, 2007), and all divisions, renewals and continuations thereof, and all Letters Patent of the United States which may be granted thereon and all reissues and extensions thereof, and all rights of priority under International Conventions and applications for Letters Patent which may hereafter be filed for said improvements in any country or countries foreign to the United States, and all Letters Patent which may be granted for said improvements in any country or countries foreign to the United States and all extensions, renewals and reissues thereof; and we hereby authorize and request the Commissioner of Patents of the United States, and any Official of any country or countries foreign to the United States, whose duty it is to issue patents on applications as aforesaid, to issue all Letters Patent for said improvements to said ASSIGNEE, its successors, legal representatives and assigns, in accordance with the terms of this instrument.

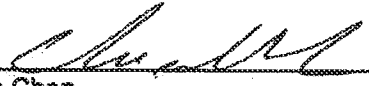
AND WE DO HEREBY sell, assign, transfer, and convey to ASSIGNEE, its successors, legal representatives, and assigns all claims for damages and all remedies arising out of any violation of the rights assigned hereby that may have accrued prior to the date of assignment to ASSIGNEE, or may accrue hereafter, including, but not limited to, the right to sue for, collect, and retain damages for past infringements of said Letters Patent before or after issuance.

AND WE HEREBY covenant and agree that we will communicate to said ASSIGNEE, its successors, legal representatives and assigns, any facts known to us respecting said improvements, and testify in any legal proceeding, sign all lawful papers, execute all divisional, continuing and reissue applications, make all rightful oaths and generally do everything possible to aid said ASSIGNEE, its successors, legal representatives and assigns, to obtain and enforce proper patent protection for said improvements in all countries.

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL040A
Page 2

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 24th day of October, 2008.


Chi-She Chen

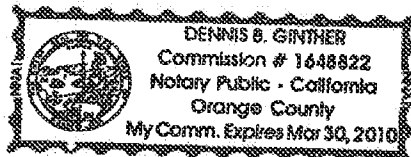
STATE OF California }
COUNTY OF Orange } ss.

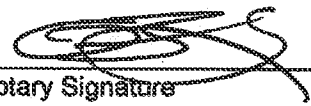
On October 24, 2008, before me, Dennis B. Ginter, notary public, personally appeared Chi-She Chen who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]




Notary Signature

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 3

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 22nd day of October, 2008.


Jeffrey C. Solomon

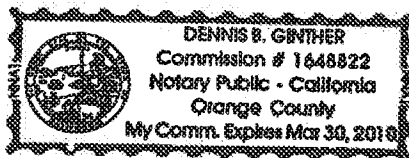
STATE OF California }
COUNTY OF Orange } ss.

On October 22, 2008, before me, Dennis B. Ginter, notary public, personally appeared Jeffrey C. Solomon who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]




Notary Signature

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL040A
Page 4

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 24 day of
October, 2008.

Scott H. Milton
Scott Milton

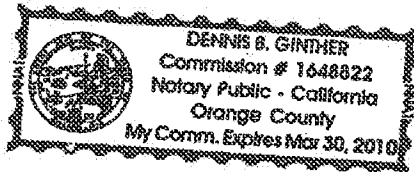
STATE OF California }
COUNTY OF Orange } ss.

On October 24, 2008, before me Dennis B. Ginter, notary public,
personally appeared Scott Milton who proved to me on the basis of satisfactory evidence to be
the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to
me that he executed the same in his authorized capacity(ies), and that by his signature(s) on
the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed
the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that
the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]



[Signature]
Notary Signature

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL040A
Page 5

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 21 day of October, 2008.

Jayesh Bhakta
Jayesh Bhakta

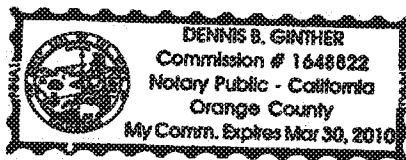
STATE OF California }
COUNTY OF Orange } ss.

On October 22, 2008, before me, Dennis B. Ginter, notary public, personally appeared Jayesh Bhakta who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]



[Signature]
Notary Signature

Electronic Acknowledgement Receipt

EFS ID:	10892151
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	22145
Filer:	Khaled Shami
Filer Authorized By:	
Attorney Docket Number:	987-004.201
Receipt Date:	07-SEP-2011
Filing Date:	29-SEP-2008
Time Stamp:	13:08:39
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	executed_POA.pdf	786534 f3fdb296722a812ecc166ffd91530418e76f028f	no	2

Warnings:

Information:

2	Assignee showing of ownership per 37 CFR 3.73(b).	3_73_wa_statement.pdf	1057450 b67aa7077f37cc10a89659f8dd85732f60a22746	no	7
Warnings:					
Information:					
Total Files Size (in bytes):				1843984	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/240,916	09/29/2008	Chi-She Chen	062453-002

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

CONFIRMATION NO. 6240
POA ACCEPTANCE LETTER



Date Mailed: 09/15/2011

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 09/07/2011.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/ddinh/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/240,916	09/29/2008	Chi-She Chen	987-004.201

22145
KLEIN, O'NEILL & SINGH, LLP
18200 VON KARMAN AVENUE
SUITE 725
IRVINE, CA 92612

CONFIRMATION NO. 6240
POWER OF ATTORNEY NOTICE



Date Mailed: 09/15/2011

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 09/07/2011.

- The Power of Attorney to you in this application has been revoked by the assignee who has intervened as provided by 37 CFR 3.71. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

/ddinh/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Chen et al.

CONFIRMATION NO.: 6240

SERIAL NO.: 12/240,916

FILING DATE: September 29, 2008

TITLE: NON-VOLATILE MEMORY MODULE

EXAMINER: ROJAS, Midys

ART UNIT: 2185

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Sir:

In response to the Office Action mailed July 29, 2011, please amend the subject application as indicated.

Amendments to the Specification begin on page **2**.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page **8**.

Remarks begin on page **14**.

In the Specification

Please amend paragraph [0039] as follows:

[0039] Certain embodiments described herein utilize the non-volatile memory subsystem 40 as a flash "mirror" to provide backup of the volatile memory subsystem 30 in the event of certain system conditions. For example, the non-volatile memory subsystem 40 may backup the volatile memory subsystem 30 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the non-volatile memory subsystem 30 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 10 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

Please amend paragraph [0047] as follows:

[0047] When operating in the first state, in certain embodiments, the step-up transformer 82 keeps the capacitor bank 86 charged at a peak value. In certain embodiments, the step-down transformer 84 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 32 and 3.0V to the non-volatile flash memory elements 42) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by Figures 1-3, the memory module 10 further comprises a switch 90 (e.g., FET switch) that

switches power provided to the controller 62, the volatile memory subsystem 30, and the non-volatile memory subsystem 40, between the power from the second power supply 80 and the power from the first power supply (e.g., system power) received via the interface 22. For example, the switch 90 may switch from the first power supply to the second power supply 80 when the voltage monitor 50 detects a low voltage condition. The switch 90 of certain embodiments advantageously ensures that the volatile memory elements 32 and non-volatile memory elements 42 are powered long enough for the data to be transferred from the volatile memory elements 32 and stored in the non-volatile memory elements 42. In certain embodiments, after the data transfer is complete, the switch 90 then switches back to the first power supply and the controller 62 transmits a signal to ~~a~~ at least one circuit 50 to operatively decouple the controller 62 from the volatile memory subsystem 30, such that the memory system 10 re-enters the first state.

Please amend paragraph [0050] as follows:

[0050] The memory system 10 generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem 30 which then writes the data to non-volatile storage which is not part of the memory system 10, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system 10 indicating that the data has been written to the hard disk or is otherwise secure. The memory system 10 of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system 10 will still be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system 10 may be operated as a write-through cache or as some other type of cache.

Please amend paragraph [0053] as follows:

[0053] The power module 100 further comprises a second power element 140 can be configured to selectively provide a fifth voltage 112 to the conversion element 120. The power module 100 can be configured to selectively provide the first voltage 102 to the memory system ~~110~~10 either from the conversion element 120 or from the input 106.

Please amend paragraph [0059] as follows:

[0059] Figure 6 is a flowchart of an example method 200 of providing a first voltage 102 and a second voltage 104 to a memory system 10 including volatile and nonvolatile memory subsystems 30, 40. While the method 200 is described herein by reference to the memory system 10 schematically illustrated by Figures 1-4, other memory systems are also compatible with embodiments of the method 200. During a first condition, the method 200 ~~at~~ comprises providing the first voltage 102 to the memory system 10 from an input power supply 106 and providing the second voltage 104 to the memory system 10 from a first power subsystem in operational block 210. For example, in one embodiment, the first power subsystem comprises the first power element 130 and the voltage conversion element 120 described above with respect to Figure 4. In other embodiments, other first power subsystems are used.

Please amend paragraph [0069] as follows:

[0069] In some embodiments, the memory system 10 enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the nonvolatile memory subsystem 40. The method 300 can also include restoring the second copy of data from the non-volatile memory subsystem 40 to the volatile memory subsystem 30 in an operational block 360. The operational blocks of method 300 referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, ~~the restoring the second copy of data~~ is restored to the volatile memory subsystem

30 at operational block 360 before the first copy of data is completely erased in the operational block 340.

Please amend paragraph [0073] as follows:

[0073] In one embodiment, the backup clock 408 and the volatile memory system clock signal 420 are received by a multiplexer 422, as schematically illustrated by Figure 78. The multiplexer 422 can output either the volatile memory system clock signal 420 or the backup clock signal 408 depending on the backup state of the memory system 10. For example, when the memory system 10 is not performing a backup or restore operation and is communicating with the host system (e.g., normal operation), the volatile memory system clock signal 420 may be provided by the multiplexer 422 to the volatile memory PLL block 424. When the memory system 10 is performing a backup (or restore) operation, the backup clock signal 408 may be provided.

Please amend paragraph [0075] as follows:

[0075] The clock signal 430 may be used by the controller 62 to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem 30. For example, control logic in the controller 62 may be used to control the volatile memory subsystem 30 during a backup or restore operation. The clock signal 430 may be used as a reference clock signal for the PLL block 434 which can generate one or more clocks 438 used by logic in the controller 62. For example, the PLL block 424434 may generate one or more clock signals 438 used to drive logic circuitry associated with controlling the volatile memory subsystem 30. In certain embodiments, the PLL block 434 includes a feedback clock signal 436 and operates in a similar manner to other PLL blocks described herein.

Please amend paragraph [0077] as follows:

[0077] While described with respect to the example embodiment of Figure 78, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem 30 operates on the volatile memory clock signal 420 and there is no backup clock signal 408. In some embodiments, the volatile memory subsystem 30 is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem 30 is operated at a reduced frequency during a restore operation and not during a backup operation.

Please amend paragraph [0078] as follows:

[0078] **Figure 9** is a flowchart of an example method 500 of controlling a memory system 10 operatively coupled to a host system. Although described with respect to the memory system 10 described herein, ~~with respect to Figures 1-3 and 7,~~ the method 500 is compatible with other memory systems. The memory system 10 may include a clock distribution topology 400 similar to the one described above with respect to Figure 78 or another clock distribution topology. The memory system 10 can include a volatile memory subsystem 30 and a non-volatile memory subsystem 40.

Please amend paragraph [0079] as follows:

[0079] In an operational block 510, the method 500 comprises operating the volatile memory subsystem 4030 at a first frequency when the memory system 10 is in a first mode of operation in which data is communicated between the volatile memory subsystem 30 and the host system. In an operational block 520, the method 500 comprises operating the non-volatile memory subsystem 40 at a second

frequency when the memory system 10 is in a second mode of operation in which data is communicated between the volatile memory subsystem 30 and the non-volatile memory subsystem 40. The method 500 further comprises operating the volatile memory subsystem 30 at a third frequency in an operational block 530 when the memory system 10 is in the second mode of operation. In certain embodiments, the memory system 10 is not powered by a battery when it is in the second mode of operation. The memory system 10 may switch from the first mode of operation to the second mode of operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

In the Claims

The following Listing of Claims replaces all prior versions in the application:

LISTING OF CLAIMS

1-36. (Canceled)

37. (Currently amended) A method ~~of~~for controlling a memory system operatively coupled to a host system ~~and which includes, the memory system including~~ a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third frequency when the memory system is in the second mode of operation, the third frequency being less than the first frequency.

38. (Original) The method of Claim 37, wherein the third frequency is approximately equal to the second frequency.

39. (Original) The method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation.

40. (Currently amended) The method of Claim ~~737~~37, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

41. (Original) The method of Claim 40, wherein the trigger condition comprises a power failure condition.
42. (Original) The method of Claim 37, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.
- 43-54. (Canceled)
55. (New) The method of claim 40, wherein the trigger condition is a request by the host system.
56. (New) The method of claim 40, wherein the trigger condition is a system hang-up.
57. (New) The method of claim 40, wherein the trigger condition is a power reduction.
58. (New) The method of claim 40, wherein the trigger condition is a drop below a voltage threshold.
59. (New) The method of claim 40, wherein the trigger condition is a rise above a voltage threshold.
60. (New) The method of claim 40, wherein the trigger condition is a drop below a first voltage threshold or a rise above a second voltage threshold.
61. (New) The method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.
62. (New) The method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.

63. (New) The method of claim 62, wherein the backup operation is conducted on a regular basis.
64. (New) The method of claim 63, wherein the backup operation is initiated in response to a trigger event.
65. (New) The method of claim 37, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.
66. (New) The method of claim 37, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.
67. (New) The method of claim 37, wherein one or more of the first, second or third frequencies is configurable by the memory system.
68. (New) The method of claim 37, wherein one or more of the first, second or third frequencies is configurable by a user.
69. (New) A memory system operatively coupled to a host system, the memory system comprising:
- a volatile memory subsystem operable at a first frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system; and
 - a non-volatile memory subsystem operable at a second frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem,
 - the volatile memory subsystem further being operable at a third frequency when the memory system is in the second mode of operation, the third frequency being less than the first frequency.

70. (New) The memory system of claim 69, further comprising:
a controller configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation and to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation.
71. (New) The memory system of claim 69, further comprising a plurality of power supplies and a switch configured to selectively deliver power from the plurality of power supplies to the volatile memory subsystem and the non-volatile memory subsystem as a function of the mode of operation.
72. (New) The memory system of claim 69, wherein the third frequency is approximately equal to the second frequency.
73. (New) The memory system of claim 69, wherein the memory system is not powered by a battery when it is in the second mode of operation.
74. (New) The memory system of claim 69, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.
75. (New) The memory system of claim 74, wherein the trigger condition comprises a power failure condition.
76. (New) The memory system of claim 69, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.
77. (New) The memory system of claim 74, wherein the trigger condition is a request by the host system.
78. (New) The memory system of claim 74, wherein the trigger condition is a system hang-up.
79. (New) The memory system of claim 74, wherein the trigger condition is a power reduction.

80. (New) The memory system of claim 74, wherein the trigger condition is drop below a voltage threshold.
81. (New) The memory system of claim 74, wherein the trigger condition is rise above a voltage threshold.
82. (New) The memory system of claim 74, wherein the trigger condition is a drop below a first voltage threshold or a rise above a second voltage threshold.
83. (New) The memory system of claim 69, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.
84. (New) The memory system of claim 69, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.
85. (New) The memory system of claim 84, wherein the backup operation is conducted on a regular basis.
86. (New) The memory system of claim 85, wherein the backup operation is initiated in response to a trigger event.
87. (New) The memory system of claim 69, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.
88. (New) The memory system of claim 69, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.

89. (New) The memory system of claim 69, wherein one or more of the first, second or third frequencies is configurable by the memory system.

90. (New) The memory system of claim 69, wherein one or more of the first, second or third frequencies is configurable by a user.

REMARKS

The Office Action mailed July 29, 2011, has been carefully considered. Reconsideration in view of the following remarks is respectfully requested.

Canceled Claims

Claims 1-36 and 43-54 have been canceled without prejudice or disclaimer of the subject matter contained therein.

Rejection(s) Under 35 U.S.C. § 103(a)

Claims 37-42 stand rejected under 35 U.S.C. § 103(a) as allegedly being unpatentable over U.S. pat. no. 6,336,174 to Li et al. (hereinafter, “Li”) in view of U.S. pat. pub. no. 2007/0192627 to Oshikiri (hereinafter, “Oshikiri”).

The Office action acknowledges Li’s failure to “teach that in the first mode, the volatile memory operates at a first frequency while in the second mode, the non-volatile memory operates at a second frequency and the volatile memory operates at a third frequency that is less than the first frequency.” The Office action asserts that this feature can be found in Oshikiri, and the combination of these two references to achieve the claimed invention would have been obvious to one of ordinary skill in the art at the time the invention was made.

The Office action correctly points out that Oshikiri operates in two different modes—a high security mode and a low security mode. These modes impose different computational burdens, and therefore result in different processing speeds. (See for example Oshikiri at col.1 lines 33-37, stating “When a command is encrypted, however, the processing speed disadvantageously decreases. That is because the encrypted command needs to be decrypted by using a predetermined algorithm and the load of this processing is heavy.”) Operation in the low security mode, being less resource-intensive,

naturally results in a higher processing speed (faster) than operation in the high security mode, which is more resource-intensive and therefore slower. Importantly, however, the different processing speeds do not equate to different memory subsystem operation frequency. In claim 37, memory subsystem operation frequencies, which can be thought of as their data or bit rates, can vary depending on the mode of operation. There is no indication in Oshikiri that the memory subsystem operation frequency is different in the two modes, and it is precisely because the memory subsystem operation frequencies are the same in Oshikiri that processing occurs more slowly for the computationally-intensive mode. If the memory subsystem operation frequencies were different, or were controllable or adjustable, then an increased memory subsystem operation frequency could be used to compensate for the greater computational burden, and a constant processing speed could be maintained. Therefore Oshikiri fails to remedy Li's admitted failure to disclose "that in the first mode, the volatile memory operates at a first frequency while in the second mode, the non-volatile memory operates at a second frequency and the volatile memory operates at a third frequency that is less than the first frequency." Accordingly, even if Li and Oshikiri were properly combinable, which is not conceded, the presently claimed invention would not result. The rejection of claim 37-42 under 35 U.S.C. § 103(a) based on the combination of Li and Oshikiri is therefore improper and its withdrawal is respectfully requested.

Newly-Added Claims

Claims 55-91 have been added to further particularly point out and distinctly claim the subject matter regarded as the invention. With respect to claim 69 *et seq*, these are apparatus counterparts to method claim 39 *et seq* and are patentable for at least the same reasons.

Conclusion

In view of the preceding discussion, Applicants respectfully urge that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present invention, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: October 18, 2011

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

NIXON PEABODY LLP
3000 EL CAMINO REAL
5TH FLOOR
PALO ALTO, CA
94306
TEL. (650) 320-7700
FAX. (650) 320-7701

Electronic Acknowledgement Receipt

EFS ID:	11212191
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-002
Receipt Date:	18-OCT-2011
Filing Date:	29-SEP-2008
Time Stamp:	16:34:22
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453_002_Resp_to_OA.pdf	149948 76fe5026b04352c91a2788735572a99546eae43c	yes	16

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1
	Specification	2	7
	Claims	8	13
	Applicant Arguments/Remarks Made in an Amendment	14	16

Warnings:

Information:

Total Files Size (in bytes):	149948
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/240,916		Filing Date 09/29/2008		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	10/18/2011	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 42	Minus	** 54	= 0	X \$	=	OR	X \$60=	0
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 7	= 0	X \$	=	OR	X \$250=	0
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
TOTAL ADD'L FEE						OR TOTAL ADD'L FEE				
0										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	10/18/2011	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=	OR	X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=	OR	X \$	=
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
TOTAL ADD'L FEE						OR TOTAL ADD'L FEE				
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/ANDREA FREEMAN/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/240,916	09/29/2008	Chi-She Chen	062453-002	6240
46188	7590	02/01/2012	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ROJAS, MIDYS	
			ART UNIT	PAPER NUMBER
			2185	
			NOTIFICATION DATE	DELIVERY MODE
			02/01/2012	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentsv@nixonpeabody.com
ocastanon@nixonpeabody.com

Office Action Summary	Application No. 12/240,916	Applicant(s) CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 18 October 2011.
- 2a) ☒ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 4) ☒ Claim(s) 37-90 is/are pending in the application.
- 4a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 5) ☐ Claim(s) _____ is/are allowed.
- 6) ☒ Claim(s) 37-54, 61-76, 83-90 is/are rejected.
- 7) ☐ Claim(s) _____ is/are objected to.
- 8) ☒ Claim(s) 55-60 and 77-82 are subject to restriction and/or election requirement.

Application Papers

- 9) ☐ The specification is objected to by the Examiner.
- 10) ☒ The drawing(s) filed on 29 September 2008 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 11) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|--|
| <p>1) ☐ Notice of References Cited (PTO-892)</p> <p>2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948)</p> <p>3) ☐ Information Disclosure Statement(s) (PTO/SB/08)
Paper No(s)/Mail Date _____.</p> | <p>4) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____.</p> <p>5) ☐ Notice of Informal Patent Application</p> <p>6) ☐ Other: _____.</p> |
|---|--|

DETAILED ACTION***Response to Arguments***

Applicant's arguments filed 10/18/2011 have been fully considered but they are not persuasive.

Applicant argues that Oshikiri does not teach the claimed different operating frequencies because although it does disclose different processing speeds when operating in different modes, the memories still operate at the same frequency during both modes. The examiner disagrees. The claimed subject matter requires the operation frequency of the memory subsystem to be different in different operation modes. The memory subsystem includes more elements than just a memory. Since Oshikiri discloses that the memory system operates at a high processing speed in one mode and at a low processing speed in another mode, this teaching correlates to the changes in speeds and operating frequencies of the entire memory subsystem (as claimed) when operating in different modes. It does not represent the operating speed of just the memory. The claim is not directed to the operating speed of a memory, but instead it is drawn to the operating speed of a memory subsystem.

Election/Restrictions

Newly submitted claims 55-60 and 77-82 are directed to an invention that is independent or distinct from the invention originally claimed for the following reasons:

Art Unit: 2185

These claims are drawn to different embodiments for the claimed trigger condition. However, the originally presented invention states that the trigger condition is a power failure [Claim 41].

Since applicant has received an action on the merits for the originally presented invention, this invention has been constructively elected by original presentation for prosecution on the merits. Accordingly, claims 55-60 and 77-82 are withdrawn from consideration as being directed to a non-elected invention. See 37 CFR 1.142(b) and MPEP § 821.03.

Claim Rejections - 35 USC § 112

The following is a quotation of the second paragraph of 35 U.S.C. 112:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 63 and 85 are rejected under 35 U.S.C. 112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which applicant regards as the invention. It is not clear what time interval of operation is defined by "the backup operation is conducted on a regular basis". The term "regular basis" does not clearly define when the backup operations occur.

Claim Rejections - 35 USC § 103

Art Unit: 2185

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 37-42, 61-62, 64-76, 83-84, and 86-90 are rejected under 35 U.S.C. 103(a) as being unpatentable over Li et al. [US 6,336,174], Applicant cited art, in view of Oshikiri [US 2007/0192627].

Claim 37, Li et al. discloses a method of controlling [via controller 206, Fig. 2] a memory system [hardware assisted memory module 104, of Fig. 1 and 2] operatively coupled to a host system [host system 100 including CPU 102, Fig. 1] and which includes a volatile memory subsystem [volatile memory 202, Fig. 2] and a non-volatile memory subsystem [nonvolatile memory 204, Fig. 2], the method comprising:

operating the volatile memory subsystem when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system [volatile memory 202 switches between modes of operation by switching isolation devices on/off in response to trigger

Art Unit: 2185

events, see col. 6, lines 48-57; during normal operation, when the isolation devices are on, the HAMM behaves like a conventional memory module, see Abstract, and stores digital information received from the data bus of the host, see col. 3, lines 5-9 and col. 5, line 55 – col. 6, line 3, thus representing the first mode of operation claimed];

operating the non-volatile memory subsystem when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem [when a trigger event occurs, such as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67].

Li et al. does not teach that in the first mode, the volatile memory operates at a first frequency while in the second mode, the non-volatile memory operates at a second frequency and the volatile memory operates at a third frequency that is less than the first frequency.

Oshikiri discloses a semiconductor memory 2 being used in an information processing apparatus 1 [Fig. 1] which switches between two operation modes [see par. 0029] wherein a first operation mode is detected when a first mode switching command is received from the information processing apparatus [see par. 0031] and a second operation mode is detected when a second mode switching command is received from the information processing apparatus [see

Art Unit: 2185

par. 0039]. In either operating mode, the information processing apparatus executes various processes in the semiconductor memory by using the CPU 11 and RAM 12 [par. 0021]. When the semiconductor memory operates in the second mode, it is operating at a lower level of security but at a higher processing speed, thus representing a higher operating frequency [par. 0040]. Therefore, when in the second mode, all elements accessing the memory operate at this higher operating frequency, thus representing the claimed first mode of operation [higher first frequency]. In Oshikiri's first mode, all elements operate at the lower operating speed, thus representing the claimed second mode of operation [lower second and third frequency]

It would have been obvious to one of ordinary skill in the art at the time the invention was made to combine the teachings of Li et al. with the operating seed of Oshikiri since providing for memory operations in different frequencies depending on the memory mode provided since different operating speeds have different power requirements.

Claim 38, Li et al. in view of Oshikiri discloses the method of Claim 37, wherein the third frequency is approximately equal to the second frequency [In Oshikiri's first mode, all elements operate at the lower operating speed, thus representing the claimed second mode of operation, lower second and third frequency, par. 0031 and 0040]

Claim 39, Li et al. in view of Oshikiri discloses the method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation [when the trigger condition that causes the switch of operation

Art Unit: 2185

mode is a O/S hang-up, a backup battery is not required, see col. 7, lines 59-66 of Li et al.].

Claim 40, Li et al. in view of Oshikiri discloses the method of Claim 37, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition [detection of a trigger event, see Abstract of Li et al.].

Claim 41, Li et al. in view of Oshikiri discloses the method of Claim 40, wherein the trigger condition comprises a power failure condition [trigger event may be power failure, see Abstract of Li et al.].

Claim 42, Li et al. in view of Oshikiri discloses the method of Claim 37, wherein the memory system 104 further comprises a printed circuit board [as shown in Fig. 2 all elements are within one board] and the volatile memory subsystem 202 and the non-volatile memory subsystem 204 are located on the printed circuit board [see Fig. 2, and col. 2, line 58-col. 3, line 4].

Claim 61, Li et al. in view of Oshikiri discloses the method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation [volatile memory 202 switches between modes of operation by switching isolation devices on/off in response to trigger events, see col. 6, lines 48-57; during normal operation, when the isolation devices are on, the HAMM behaves like a conventional memory module, see Abstract, and stores digital information received from the data bus of the host, see col. 3, lines 5-9

Art Unit: 2185

and col. 5, line 55 – col. 6, line 3, thus representing the first mode of operation claimed, Li et al.].

Claim 62, Li et al. in view of Oshikiri discloses the method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation [when a trigger event occurs, such as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67, Li et al.].

Claim 64, Li et al. in view of Oshikiri discloses the method of claim 63, wherein the backup operation is initiated in response to a trigger event [when a trigger event occurs, such as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67, Li et al.].

Claim 65, Li et al. in view of Oshikiri discloses the method of claim 37, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem [when a trigger event occurs, such as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host

Art Unit: 2185

system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67, Li et al.].

Claim 66, Li et al. in view of Oshikiri discloses the method of claim 37, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem [Restoring from the HAMM, see Abstract of Li et al.].

Claim 67, Li et al. in view of Oshikiri discloses the method of claim 37, wherein one or more of the first, second or third frequencies is configurable by the memory system [frequency changing commands are received from the information processing apparatus, Oshikiri par. 0029, 0031, 0039; therefore, they are being configured by the information processing apparatus. In this case, the information processing apparatus is part of a memory system].

Claim 68, Li et al. in view of Oshikiri discloses the method of claim 37, wherein one or more of the first, second or third frequencies is configurable by a user [frequency changing commands are received from the information processing apparatus, Oshikiri par. 0029, 0031, 0039; therefore, they are being configured by the information processing apparatus. In this case, the information processing apparatus is a user of an operation mode switching command generated by the CPU in order to perform this mode switch].

Claim 69 is rejected using the same rationale as that of Claim 37.

Claim 70, Li et al. in view of Oshikiri discloses the memory system of claim 69, further comprising: a controller configured to decouple the non-volatile

Art Unit: 2185

memory subsystem from the volatile memory subsystem in the first mode of operation and to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation [trigger event causes the HAMM to isolate is self, therefore, decoupling itself. When no trigger has occurred, the HAMM is coupled, see Abstract of Li et al].

Claim 71, Li et al. in view of Oshikiri discloses the memory system of claim 69, further comprising a plurality of power supplies and a switch configured to selectively deliver power from the plurality of power supplies to the volatile memory subsystem and the non-volatile memory subsystem as a function of the mode of operation [trigger even may be a power failure, Li et al. Abstract].

Claim 72 is rejected using the same rationale as that of Claim 38.

Claim 73 is rejected using the same rationale as that of Claim 39.

Claim 74 is rejected using the same rationale as that of Claim 40.

Claim 75 is rejected using the same rationale as that of Claim 41.

Claim 76 is rejected using the same rationale as that of Claim 42.

Claim 83 is rejected using the same rationale as that of Claim 61.

Claim 84 is rejected using the same rationale as that of Claim 62.

Claim 86 is rejected using the same rationale as that of Claim 64.

Claim 87 is rejected using the same rationale as that of Claim 65.

Claim 88 is rejected using the same rationale as that of Claim 66.

Claim 89 is rejected using the same rationale as that of Claim 62.

Claim 90 is rejected using the same rationale as that of Claim 68.

Conclusion

THIS ACTION IS MADE FINAL. Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MIDYS ROJAS whose telephone number is (571)272-4207. The examiner can normally be reached on M-TH 6:00am - 4:30pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Sanjiv Shah can be reached on (571) 272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information

Art Unit: 2185

for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Midys Rojas/

Primary Examiner, Art Unit 2185

MR

1/3/2012

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Chen et al.

CONFIRMATION NO.: 6240

SERIAL NO.: 12/240,916

FILING DATE: September 29, 2008

TITLE: NON-VOLATILE MEMORY MODULE

EXAMINER: ROJAS, Midys

ART UNIT: 2185

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Sir:

In response to the final Office Action mailed February 1, 2012, please amend the subject application as indicated.

Amendments to the Specification begin on page **2**.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page **4**.

Remarks begin on page **9**.

In the Specification

Please amend paragraph [0039] as follows:

[0039] Certain embodiments described herein utilize the non-volatile memory subsystem 40 as a flash "mirror" to provide backup of the volatile memory subsystem 30 in the event of certain system conditions. For example, the non-volatile memory subsystem 40 may backup the volatile memory subsystem 30 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the non-volatile memory subsystem ~~30~~40 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 10 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

Please amend paragraph [0047] as follows:

[0047] When operating in the first state, in certain embodiments, the step-up transformer 82 keeps the capacitor bank 86 charged at a peak value. In certain embodiments, the step-down transformer 84 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 32 and 3.0V to the non-volatile flash memory elements 42) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by Figures 1-3, the memory module 10 further comprises a switch 90 (e.g., FET switch) that

switches power provided to the controller 62, the volatile memory subsystem 30, and the non-volatile memory subsystem 40, between the power from the second power supply 80 and the power from the first power supply (e.g., system power) received via the interface 22. For example, the switch 90 may switch from the first power supply to the second power supply 80 when the voltage monitor 50 detects a low voltage condition. The switch 90 of certain embodiments advantageously ensures that the volatile memory elements 32 and non-volatile memory elements 42 are powered long enough for the data to be transferred from the volatile memory elements 32 and stored in the non-volatile memory elements 42. In certain embodiments, after the data transfer is complete, the switch 90 then switches back to the first power supply and the controller 62 transmits a signal to the at least one circuit ~~50~~52 to operatively decouple the controller 62 from the volatile memory subsystem 30, such that the memory system 10 re-enters the first state.

In the Claims

The following Listing of Claims replaces all prior versions in the application:

LISTING OF CLAIMS

1-36. (Canceled)

37. (Currently amended) A method for controlling a memory system operatively coupled to a host system, the memory system including a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

38. (Currently amended) The method of Claim 37, wherein the third clock frequency is approximately equal to the second clock frequency.

39. (Original) The method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation.

40. (Previously presented) The method of Claim 37, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

41. (Original) The method of Claim 40, wherein the trigger condition comprises a power failure condition.

42. (Original) The method of Claim 37, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

43-60. (Canceled)

61. (Previously presented) The method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.

62. (Previously presented) The method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.

63. (Currently amended) The method of claim 62, wherein the backup operation is conducted ~~on a~~ regular basis at repeating time intervals.

64. (Previously presented) The method of claim 63, wherein the backup operation is initiated in response to a trigger event.

65. (Previously presented) The method of claim 37, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.

66. (Previously presented) The method of claim 37, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.

67. (Currently amended) The method of claim 37, wherein one or more of the first, second or third clock frequencies is configurable by the memory system.

68. (Currently amended) The method of claim 37, wherein one or more of the first, second or third clock frequencies is configurable by a user.

69. (Currently amended) A memory system operatively coupled to a host system, the memory system comprising:

a volatile memory subsystem operable at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system; and

a non-volatile memory subsystem operable at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem,

the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.

70. (Previously presented) The memory system of claim 69, further comprising:
a controller configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation and to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation.

71. (Previously presented) The memory system of claim 69, further comprising a plurality of power supplies and a switch configured to selectively deliver power from the plurality of power supplies to the volatile memory subsystem and the non-volatile memory subsystem as a function of the mode of operation.

72. (Currently amended) The memory system of claim 69, wherein the third clock frequency is approximately equal to the second clock frequency.

73. (Previously presented) The memory system of claim 69, wherein the memory system is not powered by a battery when it is in the second mode of operation.

74. (Previously presented) The memory system of claim 69, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

75. (Previously presented) The memory system of claim 74, wherein the trigger condition comprises a power failure condition.

76. (Previously presented) The memory system of claim 69, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

77-82. (Canceled)

83. (Previously presented) The memory system of claim 69, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.

84. (Previously presented) The memory system of claim 69, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.

85. (Currently amended) The memory system of claim 84, wherein the backup operation is conducted ~~on a regular basis~~ at repeating time intervals.

86. (Previously presented) The memory system of claim 85, wherein the backup operation is initiated in response to a trigger event.
87. (Previously presented) The memory system of claim 69, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.
88. (Previously presented) The memory system of claim 69, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.
89. (Currently amended) The memory system of claim 69, wherein one or more of the first, second or third clock frequencies is configurable by the memory system.
90. (Currently amended) The memory system of claim 69, wherein one or more of the first, second or third clock frequencies is configurable by a user.

REMARKS

The final Office Action mailed February 1, 2012, has been carefully considered. Reconsideration in view of the following remarks is respectfully requested.

Interview Record

Applicants gratefully acknowledge the courtesy and consideration extended to Applicants' undersigned representative, Khaled Shami, during the telephone interview with Examiner Mydis Rohas on March 1, 2012.

During the interview, Mr. Shami explained the distinction between the present invention and U.S. Pat. Pub. no. 2007/0192627 to Oshikiri (hereinafter, "Oshikiri"), and proposed modifying "frequency" with "clock," as reflected in the presently amended claims. Examiner Rohas agreed that such amendment would overcome the prior art rejections of the claims, and should therefore be submitted in the present after-final response.

Specification

The specification has been amended to correct minor typographical errors, changing the numerals "30" to "40" and "50" to "52" where appropriate. No new matter has been introduced.

Election/Restriction

Claims 55-60 and 77-82 were alleged to be drawn to different (non-elected) embodiments and are canceled without prejudice or disclaimer of their subject matter.

Rejection Under 35 U.S.C. § 112, Second Paragraph

Claims 63 and 65 have been amended to clarify that the backups may be performed at repeating time intervals. This language is used instead of “regular basis” even though the term “regular basis” is expressly used in paragraph [0039], line 8, and is in fact intended to mean repeating time intervals.

Rejection(s) Under 35 U.S.C. § 103(a)

Claims 37-42, 61-62, 64-76, 83-84, and 89-90 stand rejected under 35 U.S.C. § 103(a) as allegedly being unpatentable over U.S. pat. no. 6,336,174 to Li et al. (hereinafter, “Li”) in view of Oshikiri (hereinafter, “Oshikiri”).

All of the independent claims—namely, claims 1 and 69—have been amended to modify “frequency” with “clock” and now expressly recite operation or operability “at a first *clock* frequency,” “a second *clock* frequency,” “a third *clock* frequency” (emphasis added), and so on. Such modification, it was agreed in the Examiner interview, overcomes the prior art rejection, and these claims and their dependency should now be passed to allowance.

Conclusion

In view of the preceding discussion, Applicants respectfully urge that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present invention, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: March 2, 2012

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

NIXON PEABODY LLP
P.O. BOX 60610
PALO ALTO, CA 94306
TEL. (650) 320-7700
FAX. (650) 320-7701

Electronic Acknowledgement Receipt

EFS ID:	12216001
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-002
Receipt Date:	02-MAR-2012
Filing Date:	29-SEP-2008
Time Stamp:	17:11:45
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453_002_Resp_to_FOA.pdf	158906 ff811dc4dc919a96424b86e629968ac86b34 ffd5	yes	11

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment After Final	1	1
	Specification	2	2
	Claims	3	8
	Claims	9	11

Warnings:

Information:

Total Files Size (in bytes):	158906
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/240,916		Filing Date 09/29/2008		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)		RATE (\$)	FEE (\$)			
☒ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	310			
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=	OR	X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.										
			TOTAL			TOTAL	310			
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	03/02/2012	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)		RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 30	Minus	** 54	= 0	X \$	=	OR	X \$60=	0
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 7	= 0	X \$	=	OR	X \$250=	0
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	0
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)		RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=	OR	X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=	OR	X \$	=
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/TAMMY MCBETH BROWN/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/240,916	09/29/2008	Chi-She Chen	062453-002	6240
46188	7590	03/13/2012	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ROJAS, MIDYS	
			ART UNIT	PAPER NUMBER
			2185	
			NOTIFICATION DATE	DELIVERY MODE
			03/13/2012	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentsv@nixonpeabody.com
ocastanon@nixonpeabody.com

Advisory Action Before the Filing of an Appeal Brief	Application No. 12/240,916	Applicant(s) CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

--The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

THE REPLY FILED 02 March 2012 FAILS TO PLACE THIS APPLICATION IN CONDITION FOR ALLOWANCE.

1. ☒ The reply was filed after a final rejection, but prior to or on the same day as filing a Notice of Appeal. To avoid abandonment of this application, applicant must timely file one of the following replies: (1) an amendment, affidavit, or other evidence, which places the application in condition for allowance; (2) a Notice of Appeal (with appeal fee) in compliance with 37 CFR 41.31; or (3) a Request for Continued Examination (RCE) in compliance with 37 CFR 1.114. The reply must be filed within one of the following time periods:

- a) ☐ The period for reply expires _____ months from the mailing date of the final rejection.
b) ☒ The period for reply expires on: (1) the mailing date of this Advisory Action, or (2) the date set forth in the final rejection, whichever is later. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of the final rejection.

Examiner Note: If box 1 is checked, check either box (a) or (b). ONLY CHECK BOX (b) WHEN THE FIRST REPLY WAS FILED WITHIN TWO MONTHS OF THE FINAL REJECTION. See MPEP 706.07(f).

Extensions of time may be obtained under 37 CFR 1.136(a). The date on which the petition under 37 CFR 1.136(a) and the appropriate extension fee have been filed is the date for purposes of determining the period of extension and the corresponding amount of the fee. The appropriate extension fee under 37 CFR 1.17(a) is calculated from: (1) the expiration date of the shortened statutory period for reply originally set in the final Office action; or (2) as set forth in (b) above, if checked. Any reply received by the Office later than three months after the mailing date of the final rejection, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

NOTICE OF APPEAL

2. ☐ The Notice of Appeal was filed on _____. A brief in compliance with 37 CFR 41.37 must be filed within two months of the date of filing the Notice of Appeal (37 CFR 41.37(a)), or any extension thereof (37 CFR 41.37(e)), to avoid dismissal of the appeal. Since a Notice of Appeal has been filed, any reply must be filed within the time period set forth in 37 CFR 41.37(a).

AMENDMENTS

3. ☒ The proposed amendment(s) filed after a final rejection, but prior to the date of filing a brief, will not be entered because
(a) ☒ They raise new issues that would require further consideration and/or search (see NOTE below);
(b) ☐ They raise the issue of new matter (see NOTE below);
(c) ☐ They are not deemed to place the application in better form for appeal by materially reducing or simplifying the issues for appeal; and/or
(d) ☐ They present additional claims without canceling a corresponding number of finally rejected claims.

NOTE: _____. (See 37 CFR 1.116 and 41.33(a)).

4. ☐ The amendments are not in compliance with 37 CFR 1.121. See attached Notice of Non-Compliant Amendment (PTOL-324).
5. ☐ Applicant's reply has overcome the following rejection(s): _____.
6. ☐ Newly proposed or amended claim(s) _____ would be allowable if submitted in a separate, timely filed amendment canceling the non-allowable claim(s).
7. ☒ For purposes of appeal, the proposed amendment(s): a) ☒ will not be entered, or b) ☐ will be entered and an explanation of how the new or amended claims would be rejected is provided below or appended.
The status of the claim(s) is (or will be) as follows:
Claim(s) allowed: _____.
Claim(s) objected to: _____.
Claim(s) rejected: _____.
Claim(s) withdrawn from consideration: _____.

AFFIDAVIT OR OTHER EVIDENCE

8. ☐ The affidavit or other evidence filed after a final action, but before or on the date of filing a Notice of Appeal will not be entered because applicant failed to provide a showing of good and sufficient reasons why the affidavit or other evidence is necessary and was not earlier presented. See 37 CFR 1.116(e).
9. ☐ The affidavit or other evidence filed after the date of filing a Notice of Appeal, but prior to the date of filing a brief, will not be entered because the affidavit or other evidence failed to overcome all rejections under appeal and/or appellant fails to provide a showing a good and sufficient reasons why it is necessary and was not earlier presented. See 37 CFR 41.33(d)(1).
10. ☐ The affidavit or other evidence is entered. An explanation of the status of the claims after entry is below or attached.

REQUEST FOR RECONSIDERATION/OTHER

11. ☒ The request for reconsideration has been considered but does NOT place the application in condition for allowance because:
Applicants remarks are based on the claims as amended. The claim amendments are not being entered because they change the scope of the claims and raise new issues that require further search and consideration. Furthermore, the claim amendments do not place the claims in condition for allowance. The examiner is aware of Prior Art that teaches nonvolatile memory which operates by changing the clock frequency being used to read and write data. .
12. ☐ Note the attached Information *Disclosure Statement*(s). (PTO/SB/08) Paper No(s). _____.
13. ☐ Other: _____.

/Midys Rojas/
Primary Examiner, Art Unit 2185



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/240,916	09/29/2008	Chi-She Chen	062453-002	6240
46188	7590	03/13/2012	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ROJAS, MIDYS	
			ART UNIT	PAPER NUMBER
			2185	
			NOTIFICATION DATE	DELIVERY MODE
			03/13/2012	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentsv@nixonpeabody.com
ocastanon@nixonpeabody.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 12/240,916	Applicant(s) CHEN ET AL.	
	Examiner MIDYS ROJAS	Art Unit 2185	

All participants (applicant, applicant's representative, PTO personnel):

(1) MIDYS ROJAS. (3) ____.

(2) Khaled Shami [38,745]. (4) ____.

Date of Interview: 01 March 2012.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: ____.

Issues Discussed ☐101 ☐112 ☐102 ☒103 ☐Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1 and 69.

Identification of prior art discussed: Li et al. [6,336,174] and Oshikiri [2007/0192627].

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Applicant presented amendments to overcome the previously presented rejection. The examiner agreed that the amendments should overcome the previously presented rejection. Further search and considaration will be required..

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

/Midys Rojas/ Primary Examiner, Art Unit 2185	
--	--

Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiner's responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicant's correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicant's record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiner's version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, "Interview Record OK" on the paper recording the substance of the interview along with the date and the examiner's initials.

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	12240916	Filing Date	2008-09-29	Docket Number (if applicable)	062453-002	Art Unit	2185
First Named Inventor	Chi-She Chen			Examiner Name	Midys Rojas		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☒ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☒ Other Response to Final Office Action filed March 2, 2012

☐ Enclosed

☐ Amendment/Reply

☐ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
Deposit Account No 503557

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

☐ Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2012-03-20
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	12240916			
Filing Date:	29-Sep-2008			
Title of Invention:	NON-VOLATILE MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Pamela Wilson			
Attorney Docket Number:	062453-002			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Request for continued examination	1801	1	930	930
Total in USD (\$)				930

Electronic Acknowledgement Receipt

EFS ID:	12351500
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-002
Receipt Date:	20-MAR-2012
Filing Date:	29-SEP-2008
Time Stamp:	18:34:54
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$930
RAM confirmation Number	5753
Deposit Account	503557
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	062453_002_RCE.pdf	114286 df181dfab020ad0f393025e018222e8e17c094ef	no	3

Warnings:

This is not a USPTO supplied RCE SB30 form.

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30466 a288ac34c852c555b2ca41b8c460828dd3ba06377	no	2
---	----------------------	--------------	--	----	---

Warnings:**Information:**

Total Files Size (in bytes):			144752
------------------------------	--	--	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/240,916		Filing Date 09/29/2008		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	03/20/2012	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 30	Minus	** 54	= 0	X \$	=	OR	X \$60=	0
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 7	= 0	X \$	=	OR	X \$250=	0
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
						TOTAL ADD'L FEE	OR	TOTAL ADD'L FEE	0	
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=	OR	X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=	OR	X \$	=
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
						TOTAL ADD'L FEE	OR	TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/DEBRA R. WYATT/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/240,916	09/29/2008	Chi-She Chen	062453-002	6240
46188	7590	04/03/2012	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ROJAS, MIDYS	
			ART UNIT	PAPER NUMBER
			2185	
			NOTIFICATION DATE	DELIVERY MODE
			04/03/2012	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentsv@nixonpeabody.com
ocastanon@nixonpeabody.com

Office Action Summary	Application No. 12/240,916	Applicant(s) CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 20 March 2012.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 4) ☒ Claim(s) 37-54, 61-76 and 83-90 is/are pending in the application.
- 4a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 5) ☐ Claim(s) _____ is/are allowed.
- 6) ☒ Claim(s) 37-54, 61-76 and 83-90 is/are rejected.
- 7) ☐ Claim(s) _____ is/are objected to.
- 8) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

Application Papers

- 9) ☐ The specification is objected to by the Examiner.
- 10) ☒ The drawing(s) filed on 29 September 2008 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 11) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) | 5) ☐ Notice of Informal Patent Application |
| 3) ☐ Information Disclosure Statement(s) (PTO/SB/08)
Paper No(s)/Mail Date _____ | 6) ☐ Other: _____ |

DETAILED ACTION

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection, on 3/20/2012. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 3/02/2012 has been entered.

Response to Arguments

Applicant's arguments with respect to claims 37-42, 61-62, 64-76, 83-84, and 86-90 have been considered but are moot because the arguments do not apply to the references being used in the current rejection.

Claim Rejections - 35 USC § 112

The rejection of claims 63 and 85 under 35 U.S.C. 112, second paragraph, as being indefinite for failing to particularly point out and distinctly claiming the subject matter which applicant regards as the invention is being withdrawn in view of Applicant's amendment to these claims.

The following is a quotation of the second paragraph of 35 U.S.C. 112:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 38 and 72 are rejected under 35 U.S.C. 112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which applicant regards as the invention. It is unclear how the third clock frequency can be **approximately** equal to the second clock frequency. It is not clear what makes these two values approximately equal or how two values can be approximately equal. Following mathematical logic, it is held that two values can only be equal or not equal. Clarification is required.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 37-42, 61-62, 64-76, 83-84, and 86-90 are rejected under 35 U.S.C. 103(a) as being unpatentable over Li et al. [US 6,336,174], Applicant cited art, in view of Cope [US 2008/0195806].

Claim 37, Li et al. discloses a method of controlling [via controller 206, Fig. 2] a memory system [hardware assisted memory module 104, of Fig. 1 and 2] operatively coupled to a host system [host system 100 including CPU 102, Fig. 1] and which includes a volatile memory subsystem [volatile memory 202, Fig. 2] and a non-volatile memory subsystem [nonvolatile memory 204, Fig. 2], the method comprising:

operating the volatile memory subsystem when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system [volatile memory 202 switches between modes of operation by switching isolation devices on/off in response to trigger events, see col. 6, lines 48-57; during normal operation, when the isolation devices are on, the HAMM behaves like a conventional memory module, see Abstract, and stores digital information received from the data bus of the host, see col. 3, lines 5-9 and col. 5, line 55 – col. 6, line 3, thus representing the first mode of operation claimed];

operating the non-volatile memory subsystem when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem [when a trigger event occurs, such as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67].

Li et al. does not teach that in the first mode, the volatile memory operates at a first clock frequency while in the second mode, the non-volatile memory operates at a second clock frequency and the volatile memory operates at a third clock frequency that is less than the first clock frequency.

Cope discloses a memory [volatile memory device such a DRAM device, par. 0039] that is part of a memory system wherein after a pin grant transition 410, an adjust clock signal 412 transitions from a clock stable signal [which represents the claimed first

Art Unit: 2185

mode of operation] to a clock adjustment signal 414 [representing the second mode of operation] which is generated by the first memory controller to signal the volatile memory to discontinue operation at a normal clock frequency [first clock frequency] and to begin operation at a new clock frequency [second or third clock frequencies, see par. 0039]. It would have been obvious to one of ordinary skill in the art at the time the invention was made to combine the teachings of Li et al. with the changing clock frequencies of Cope since providing for memory operations at different clock frequencies depending on the memory being executed enables the memory system to operate at optimal power consumption depending on the operation [see par. 0046 of Cope].

Claim 38, Li et al. in view of Cope discloses the method of Claim 37, wherein the third frequency is approximately equal to the second frequency [In Cope's first mode, all elements operate at the first clock frequency; the second mode represents a change of frequency clock for operation, therefore, the operations of the memories must be synchronized to this clock, par. 0039]

Claim 39, Li et al. in view of Cope discloses the method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation [when the trigger condition that causes the switch of operation mode is a O/S hang-up, a backup battery is not required, see col. 7, lines 59-66 of Li et al.].

Claim 40, Li et al. in view of Cope discloses the method of Claim 37, wherein the memory system switches from the first mode of operation to the second mode of

operation in response to a trigger condition [detection of a trigger event, see Abstract of Li et al.].

Claim 41, Li et al. in view of Cope discloses the method of Claim 40, wherein the trigger condition comprises a power failure condition [trigger event may be power failure, see Abstract of Li et al.].

Claim 42, Li et al. in view of Cope discloses the method of Claim 37, wherein the memory system 104 further comprises a printed circuit board [as shown in Fig. 2 all elements are within one board] and the volatile memory subsystem 202 and the non-volatile memory subsystem 204 are located on the printed circuit board [see Fig. 2, and col. 2, line 58-col. 3, line 4].

Claim 61, Li et al. in view of Cope discloses the method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation [volatile memory 202 switches between modes of operation by switching isolation devices on/off in response to trigger events, see col. 6, lines 48-57; during normal operation, when the isolation devices are on, the HAMM behaves like a conventional memory module, see Abstract, and stores digital information received from the data bus of the host, see col. 3, lines 5-9 and col. 5, line 55 – col. 6, line 3, thus representing the first mode of operation claimed, Li et al.].

Claim 62, Li et al. in view of Cope discloses the method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation [when a trigger event occurs, such

Art Unit: 2185

as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67, Li et al.].

Claim 64, Li et al. in view of Cope discloses the method of claim 63, wherein the backup operation is initiated in response to a trigger event [when a trigger event occurs, such as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67, Li et al.].

Claim 65, Li et al. in view of Cope discloses the method of claim 37, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem [when a trigger event occurs, such as a system fault or a power failure, the HAMM isolates the volatile memory from the system memory bus by turning off isolation devices and isolating the HAMM from the host system. Then to store operation begins which includes copying data, address by address, from the volatile memory to the nonvolatile memory, see col. 9, lines 49-67, Li et al.].

Claim 66, Li et al. in view of Cope discloses the method of claim 37, wherein the second mode of operation comprises a restore operation in which data is communicated

Art Unit: 2185

from the non-volatile memory subsystem to the volatile memory subsystem [Restoring from the HAMM, see Abstract of Li et al.].

Claim 67, Li et al. in view of Cope discloses the method of claim 37, wherein one or more of the first, second or third frequencies is configurable by the memory system [clock adjustment signal 414 are generated by the memory controller; therefore, they are being configured by the information processing apparatus. In this case, the information processing apparatus is part of a memory system].

Claim 68, Li et al. in view of Cope discloses the method of claim 37, wherein one or more of the first, second or third frequencies is configurable by a user [clock adjustment signal 414 are generated by the memory controller; therefore, they are being configured by the information processing apparatus. In this case, the information processing apparatus is a user of an operation mode switching command generated by the CPU in order to perform this mode switch].

Claim 69 is rejected using the same rationale as that of Claim 37.

Claim 70, Li et al. in view of Cope discloses the memory system of claim 69, further comprising: a controller configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation and to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation [trigger event causes the HAMM to isolate is self, therefore, decoupling itself. When no trigger has occurred, the HAMM is coupled, see Abstract of Li et al].

Claim 71, Li et al. in view of Cope discloses the memory system of claim 69, further comprising a plurality of power supplies and a switch configured to selectively deliver power from the plurality of power supplies to the volatile memory subsystem and the non-volatile memory subsystem as a function of the mode of operation [trigger even may be a power failure, Li et al. Abstract].

Claim 72 is rejected using the same rationale as that of Claim 38.

Claim 73 is rejected using the same rationale as that of Claim 39.

Claim 74 is rejected using the same rationale as that of Claim 40.

Claim 75 is rejected using the same rationale as that of Claim 41.

Claim 76 is rejected using the same rationale as that of Claim 42.

Claim 83 is rejected using the same rationale as that of Claim 61.

Claim 84 is rejected using the same rationale as that of Claim 62.

Claim 86 is rejected using the same rationale as that of Claim 64.

Claim 87 is rejected using the same rationale as that of Claim 65.

Claim 88 is rejected using the same rationale as that of Claim 66.

Claim 89 is rejected using the same rationale as that of Claim 62.

Claim 90 is rejected using the same rationale as that of Claim 68.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MIDYS ROJAS whose telephone number is (571)272-4207. The examiner can normally be reached on M-TH 6:00am - 4:30pm.

Art Unit: 2185

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Sanjiv Shah can be reached on (571) 272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Midys Rojas/

Primary Examiner, Art Unit 2185

MR

3/25/2012

Notice of References Cited	Application/Control No. 12/240,916	Applicant(s)/Patent Under Reexamination CHEN ET AL.	
	Examiner MIDYS ROJAS	Art Unit 2185	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-2008/0195806	08-2008	Cope, Bryan	711/111
	B	US-			
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	0	("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/27 15:21
S2	12	("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:21
S3	0	host same (volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:34
S4	1	(volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:34
S5	87	(memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:35
S6	56	"711"/\$.ccls. and (memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:35
S7	1013	memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:44
S8	128	"711"/\$.ccls. and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:44
S9	46	("4072852" "4815074" "4959774" "5283792" "5379431" "5799200").PN. OR ("6336174").URPN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/27 15:44
S10	8	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) with (frequency or speed or clock)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:08
S11	2	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:09

S12	0	memory same (host near3 frequency) same (non-volatile near3 frequency) same mode	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:11
S13	1	(memory with mode) same (host with frequency) same (non-volatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:12
S14	6	(memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:12
S15	13	"711"/\$.ccls. and (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:14
S16	40	"711"/\$.ccls. and (host with speed) same (nonvolatile with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:22
S17	16	"711"/\$.ccls. and (host with speed) same (backup with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:23
S18	50	(backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:29
S19	48	power and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:29
S20	14	(power near3 (loss or fail\$3)) and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:30
S21	8	(power near3 (loss or fail\$3)) and (copy\$3 or tranfer\$3) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:38
S22	8	"711"/\$.ccls. and (copy\$3 or tranfer\$3 or back-up or (back adj up)) with (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:41
S23	88	"711"/\$.ccls. and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:44
S24	4347	((711/162) or (711/160) or (711/161) or (710/10)).CCLS.	USPAT; USOCR	OR	OFF	2011/07/28 19:20
S25	0	S24 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/28 19:21
S26	1	S24 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 19:21
S27	8	"711"/\$.ccls. and ((non-volatile or nonvolatile) near3 memory) with ((vary\$3 or alternat\$3 or alter\$3 or fluctuat\$3 or chang\$3) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/03/08 13:52
S28	1	"711"/\$.ccls. and ((non-volatile or nonvolatile) near3 memory) with (dynamic near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2012/03/08 13:56

			JPO			
--	--	--	-----	--	--	--

EAST Search History (Interference)

< This search history is empty >

3/ 26/ 2012 12:43:27 AM

C:\ Users\ mrojas\ Documents\ EAST\ Workspaces\ 12240916.wsp

Search Notes 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

SEARCHED			
Class	Subclass	Date	Examiner
711	160, 161, 162	3/25/2012	MR
710	10	3/25/2012	MR

SEARCH NOTES		
Search Notes	Date	Examiner
EAST Updated search: limited search of 711/160, 161, 162; 710/10; and text searches	3/25/2012	MR
PALM inventor name search	3/25/2012	MR

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner

	/MIDYS ROJAS/ Primary Examiner.Art Unit 2185
--	---

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Chen et al.

CONFIRMATION NO.: 6240

SERIAL NO.: 12/240,916

FILING DATE: September 29, 2008

TITLE: NON-VOLATILE MEMORY MODULE

EXAMINER: ROJAS, Midys

ART UNIT: 2185

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Sir:

In response to the Office Action mailed April 3, 2012, please amend the subject application as indicated.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 7.

In the Claims

The following Listing of Claims replaces all prior versions in the application:

LISTING OF CLAIMS

1-36. (Canceled)

37. (Previously presented) A method for controlling a memory system operatively coupled to a host system, the memory system including a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

38. (Currently amended) The method of Claim 37, wherein the third clock frequency is ~~approximately~~ substantially equal to the second clock frequency.

39. (Original) The method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation.

40. (Previously presented) The method of Claim 37, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

41. (Original) The method of Claim 40, wherein the trigger condition comprises a power failure condition.

42. (Original) The method of Claim 37, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

43-60. (Canceled)

61. (Previously presented) The method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.

62. (Previously presented) The method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.

63. (Previously presented) The method of claim 62, wherein the backup operation is conducted at repeating time intervals.

64. (Previously presented) The method of claim 63, wherein the backup operation is initiated in response to a trigger event.

65. (Previously presented) The method of claim 37, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.

66. (Previously presented) The method of claim 37, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.

67. (Previously presented) The method of claim 37, wherein one or more of the first, second or third clock frequencies is configurable by the memory system.

68. (Previously presented) The method of claim 37, wherein one or more of the first, second or third clock frequencies is configurable by a user.

69. (Previously presented) A memory system operatively coupled to a host system, the memory system comprising:

a volatile memory subsystem operable at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system; and

a non-volatile memory subsystem operable at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem,

the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.

70. (Previously presented) The memory system of claim 69, further comprising:

a controller configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation and to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation.

71. (Previously presented) The memory system of claim 69, further comprising a plurality of power supplies and a switch configured to selectively deliver power from the plurality of power supplies to the volatile memory subsystem and the non-volatile memory subsystem as a function of the mode of operation.

72. (Currently amended) The memory system of claim 69, wherein the third clock frequency is ~~approximately~~ substantially equal to the second clock frequency.

73. (Previously presented) The memory system of claim 69, wherein the memory system is not powered by a battery when it is in the second mode of operation.

74. (Previously presented) The memory system of claim 69, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

75. (Previously presented) The memory system of claim 74, wherein the trigger condition comprises a power failure condition.

76. (Previously presented) The memory system of claim 69, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

77-82. (Canceled)

83. (Previously presented) The memory system of claim 69, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.

84. (Previously presented) The memory system of claim 69, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.

85. (Previously presented) The memory system of claim 84, wherein the backup operation is conducted at repeating time intervals.

86. (Previously presented) The memory system of claim 85, wherein the backup operation is initiated in response to a trigger event.
87. (Previously presented) The memory system of claim 69, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.
88. (Previously presented) The memory system of claim 69, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.
89. (Previously presented) The memory system of claim 69, wherein one or more of the first, second or third clock frequencies is configurable by the memory system.
90. (Previously presented) The memory system of claim 69, wherein one or more of the first, second or third clock frequencies is configurable by a user.

REMARKS

The Office Action mailed April 3, 2012, has been carefully considered. Reconsideration in view of the following remarks is respectfully requested.

Rejection Under 35 U.S.C. § 112, Second Paragraph

Claims 63 and 85 stand rejected under 35 U.S.C. § 112, second paragraph, as allegedly being indefinite for failing to particularly point out and distinctly claim the subject matter which Applicants regard as the invention. The term “approximately” was objected to and has been replaced with the term “substantially.” This term is used to account for the fact that while in theory two clock frequencies can be identical, in practice there will always be a difference, even if it is infinitesimal. This is a common approach in patent practice and has been deemed acceptable in the case law. See for instance *Andrew Corp. v. Gabriel Electronics*, 847 F.2d 819, 6 USPQ2d 2010 (Fed. Cir. 1988), discussed in MPEP 2173.05(b), (“The court held that the limitation ‘which produces substantially equal E and H plane illumination patterns’ was definite because one of ordinary skill in the art would know what was meant by ‘substantially equal.’”)

Rejection(s) Under 35 U.S.C. § 103(a)

Claims 37-42, 61-62, 64-76, 83-84, and 89-90 stand rejected under 35 U.S.C. § 103(a) as allegedly being unpatentable over U.S. pat. no. 6,336,174 to *Li et al.* (hereinafter, “*Li*”) in view of U.S. pat. pub. no. 2008/0195806 to *Cope* (hereinafter, “*Cope*”).

The independent claims—namely, claims 37 and 69—recite:

- operation of the volatile memory subsystem at *a first clock frequency in a first mode* (in which data is communicated between the volatile memory subsystem and the host system),

- operation of the non-volatile memory subsystem at *a second clock frequency in a second mode* (in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem), and
- operation of the volatile memory subsystem at *a third clock frequency in the second mode* (the third clock frequency is less than the first clock frequency).

According to the above, the volatile memory subsystem operates at a lower clock frequency in the second mode (data communicated between volatile and non-volatile memory subsystems) than in the first mode (data communicated between volatile memory subsystem and host).

The Office action acknowledges that *Li* “does not teach that in the first mode, the volatile memory operates at a first clock frequency while in the second mode, the non-volatile memory operates at a second clock frequency and the volatile memory operates at a third clock frequency that is less than the first clock frequency.” The Office action cites *Cope* to cure this defect, stating:

Cope discloses a memory [volatile memory device such a DRAM device, par. 0039] that is part of a memory system wherein after a pin grant transition 410, an adjust clock signal 412 transitions from a clock stable signal [which represents the claimed first mode of operation] to a clock adjustment signal 414 [representing the second mode of operation] which is generated by the first memory controller to signal the volatile memory to discontinue operation at a normal clock frequency [first clock frequency] and to begin operation at a new clock frequency [second or third clock frequencies, see par. 0039]. It would have been obvious to one of ordinary skill in the art at the time the invention was made to combine the teachings of *Li et al.* with the changing clock frequencies of *Cope* since providing for memory operations at different clock frequencies depending on the memory being executed enables the memory system to operate at optimal power consumption depending on the operation [see par. 0046 of *Cope*].

A close examination of *Cope* reveals that during memory operation, **the two memories are synchronized with one another.** Depending on which of the two memories has access to the shared contact 262, the memories operate together at either a first clock frequency or a second clock frequency. **Importantly, they never operate at different clock frequencies.**

This is evident from paragraph [0021], which states:

In a particular embodiment, the external clock circuit can generate a common clock signal at a first clock frequency **for both** of the first memory controller 204 and the second memory controller 238 when the first memory controller 204 performs memory operations and can generate a common clock signal at a second clock frequency **for both** of the first memory controller 204 and the second memory controller 238 when the second memory controller 238 performs memory operations. In a specific embodiment, by adjusting the common clock signal frequency **for both** the first memory controller 204 and the second memory controller 238, lower power consumption can be achieved than by operating both memory controllers 204 and 238 at a constant clock frequency. In a specific embodiment, the second clock frequency can be predetermined to reduce number of cycles performed by the second memory controller 238 over a time period to reduce power consumption.

(Emphasis added)

It is also evident from paragraph [0026], which describes what happens after the hand off of control of the shared contact 262 from the default second memory (volatile memory 258) to the first memory (non-volatile memory 230). Paragraph [0026] details synchronization of the clock-frequencies so that they are both the same after the hand-off occurs, and states:

In addition, the logic 206 can send a clock adjust signal to the second memory controller 238 indicating that the external clock circuit has changed the clock frequency. The second memory controller 238 can **synchronize** to the new clock frequency and return a clock adjust complete signal to the logic 206 **when the synchronization is complete**.

(Emphasis added)

In some instances, the shared frequency is lowered in order to reduce power consumption, but as in all cases, both memories are then run at the reduced frequency. This is evident from paragraph [0021], which further states:

In a specific embodiment, by adjusting the common clock signal frequency for both the first memory controller 204 and the second memory controller 238, lower power consumption can be achieved than by operating both memory controllers 204 and 238 at a constant clock frequency. In a specific embodiment, the second clock frequency can be predetermined to reduce number of cycles performed by the second memory controller 238 over a time period to reduce power consumption.

(Emphasis added)

The above clearly indicates that the frequencies of the *Li* memories are tied together such that the two memories both operate at a first frequency **or** a second frequency. Therefore *Cope* cannot be combined with *Li* to achieve the different clock frequencies of operation for two modes.

FIG. 4 of *Cope* is reproduced below, with labels added for time segments A – F for ease of discussion. The clock frequencies for the first (non-volatile) and second (volatile) memories during these time segments are clearly described in paragraphs [0037] – [0042] in *Cope*. These clock frequencies are listed below, with “transition” indicating an indeterminate clock frequency as clearly described by *Cope* in paragraph [0041]: “[T]he adjust clock signal transition 420 can be generated by the first memory controller to signal to the second memory controller that a clock signal has stabilized after transitioning to a different or new clock frequency.”

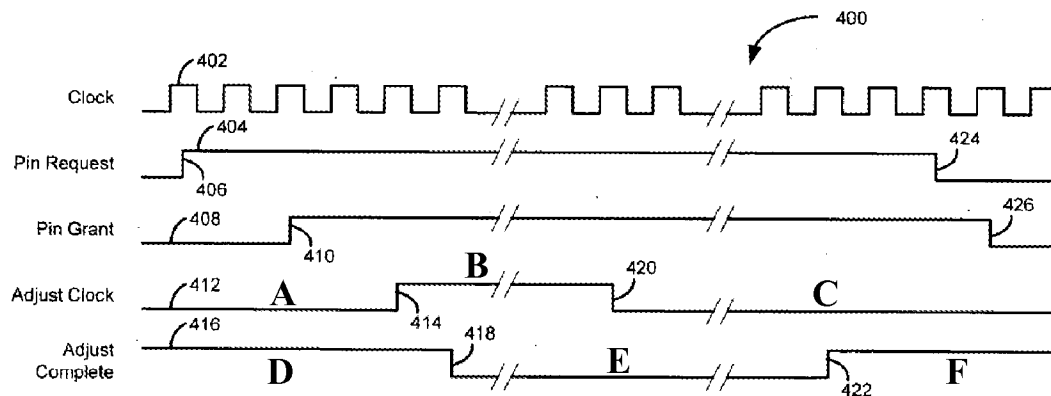


FIG. 4

- Time Segment A: First memory (FLASH) at 1st frequency
- Time Segment B: Transition
- Time Segment C: First memory (FLASH) at 2nd frequency
- Time Segment D: Second memory (DRAM) at 1st frequency
- Time Segment E: Transition
- Time Segment F: Second memory (DRAM) at 2nd frequency

Cope describes a 1st memory controller (FLASH controller) and a 2nd memory controller (DRAM controller) sharing a contact or data pin 262 (see [0012]-[0015] and that shared data pin (bus 106, 108, and 110 of Fig. 1 or bus 208, 210, and 212). *Cope* further describes that only ONE of the FLASH controller and DRAM controllers can use the shared data pin at a time, and that the DRAM controller is the default memory controller for controlling the shared contact (see paragraphs [0022], [0030], and [0045]).

In order for the FLASH controller to utilize the shared data bus in *Cope*, the FLASH controller must request the use of the shared data bus from the DRAM controller via a PIN REQUEST signal 404 at transition 406. The DRAM controller can release the shared data bus to the FLASH controller by asserting the PIN GRANT signal 408 at transition 410 (see [0037]-[0038]).

Cope describes changing the clock frequency for **both** the DRAM controller and FLASH controller (see [0021]). Once the DRAM controller releases the shared data bus to the FLASH controller, as indicated by the “pin grant transition 410” see [0038], the FLASH controller can assert the ADJUST CLOCK signal 412 at transition 414 to “to signal [the] DRAM device, to discontinue operation at a normal clock frequency and to begin operation at a different or new clock frequency” (see [0039]). It should be noted that, according to *Cope*:

- a. when the ADJUST SIGNAL 412 is LOW (not asserted or logic 0, see Fig. 4, Time Segment A and Time Segment C) then it is indicative of a clock signal that is stable, regardless of what the frequency of the clock signal is.
- b. when the ADJUST SIGNAL 412 is HIGH (asserted or logic 1, see Fig. 4, Time Segment B) then it is indicative of:
 - i. a clock signal that is NOT stable and with indeterminate frequency, see [0041] and as noted above,
 - ii. at transition 414, the DRAM device has already released the shared bus and therefore cannot possibly be operable to communicate data and note that *Cope* further describes additional steps that must be performed before

the DRAM device actually may starts operating using the new clock frequency, as clearly described in paragraph [0042] the ADJUST COMPLETE signal at “transition 422 is generated by the second memory controller to signal to the first memory controller that synchronization to the new clock frequency is complete”, and

iii *Cope* further describes in paragraph [0040] that “In response to the adjust clock transition 414, an adjust complete signal 416 transitions to a low state at transition 418, indicating that synchronization to a clock signal is being performed.” Or “...synchronization to a different or new clock frequency has not been completed.”

c. It is very clear that *Cope* describes that:

- i. during Time Segment A (see reproduced FIG. 4 above), the clock is stable at a first frequency,
- ii. during Time Segment B, the clock is NOT stable, at an indeterminate frequency, and
- iii. during Time Segment C, the clock is stable at a second frequency.

Therefore, it is not possible to have the ADJUST SIGNAL 412 while LOW to represent the claimed first mode of operation, because *Cope* uses the LOW level of ADJUST SIGNAL 412 to simply indicate that a “clock signal has stabilized after transitioning to a different or new clock frequency” **regardless of what that frequency is.** In fact, at ADJUST SIGNAL 412 LOW at Time Segment A, the clock is at a first frequency, while at ADJUST SIGNAL 412 LOW at Time Segment C, the clock is at a second frequency. In contrast, the claimed first mode of operation requires “operating the volatile memory subsystem at a first frequency.” The DRAM device cannot be operating at two different frequencies, as alleged in the Office action, when the memory system is in a first mode.

Similarly, in the claimed second mode of operation, the “non-volatile memory subsystem operates at a second frequency” and “the volatile memory subsystem [operates] at a third frequency...” The Office action alleges that ADJUST CLOCK 412 transitioning at 414 to HIGH represents the claimed second mode of operation. It is very clear that *Cope* describes that:

- a. during Time Segment B, when ADJUST CLOCK 412 transitions at 414 to HIGH, it is indicative of a clock that is NOT stable and thus the clock's frequency cannot be determined, see above, and
- b. during Time Segment B, *Cope*'s DRAM devices:
 - i. are inoperative to communicate data on the shared data bus since "after a pin grant transition 410" the DRAM controller has released the shared data bus to be used exclusively by the FLASH controller,
 - ii. cannot be operative since the clock frequency is unknown or indeterminate, and thus a catastrophic failure would occur if data access to the DRAM were to be gained somehow while the frequency "is not stable", and
 - iii. inoperable to communicate data since *Cope* clearly describes that the "adjust clock signal transition 414 can cause the volatile memory device to enter a self-refresh mode" and it is well known that if a DRAM device enters a self-refresh mode, which is akin to a power down mode, the DRAM uses an on-chip timer to generate internal refresh cycles as necessary, and the system clock may even be stopped during this time. Thus, no data read or write operation can be performed during DRAM self-refresh mode.

Cope cannot be used to describe two modes of operation, where a DRAM in a first mode operates at a first clock frequency and in a second mode operates at another frequency. This is because *Cope*'s DRAM devices are operable only in a first Mode at a first frequency, and are inoperable to communicate data in a second mode where the clock is beginning to change or has changed to another frequency.

In *Cope*, after the FLASH controller asserts ADJUST SIGNAL 412 to a HIGH level, the DRAM controller asserts ADJUST COMPLETE signal 416 to LOW at transition 418 "to signal to the first (FLASH) memory controller that synchronization to a different or new clock frequency has not been completed" (see [0040]). Once the clock has stabilized, the FLASH controller returns ADJUST SIGNAL 412 to a LOW level, at 420, and after an unknown time period (indicated by the breaks in signals and clock in Fig. 4) the DRAM controller asserts ADJUST COMPLETE signal 416 to HIGH at

transition 422 “to signal to the FLASH memory controller that synchronization to the new clock frequency is complete” (see [0042]).

Further, as noted above, when the ADJUST CLOCK pin is asserted at 414 (to begin the alleged second mode of operation), the volatile memory is in fact being taken offline, as indicated by the pin grant transition 410, because *Cope* describes that only one of the first and second memories is allowed access to the shared data pin 262 at a time. In this mode, therefore, the DRAM is non-operational. At best, it is directed to enter a self-refresh mode, but it is effectively disconnected from the system. The combination of *Cope* with *Li* therefore would not teach or suggest operating the volatile memory at a first clock frequency in a first mode and at a lower clock frequency in a second mode, wherein in the first mode the volatile memory is communicating with the host, and in the second mode it is communicating with the non-volatile memory.

It should also be noted that the transition 414 and the asserted ADJUST CLOCK during Time Segment B thereafter do not demarcate a stable second clock frequency—the new clock frequency is not stable for possible usage by the DRAM until transition 422, when the ADJUST COMPLETE pin is returned to high by the DRAM at transition 422 and during Time Segment F thereafter.

Conclusion

In view of the preceding discussion, Applicants respectfully urge that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present invention, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: May 24, 2012

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

NIXON PEABODY LLP
P.O. BOX 60610
PALO ALTO, CA 94306
TEL. (650) 320-7700
FAX. (650) 320-7701

Electronic Acknowledgement Receipt

EFS ID:	12855993
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-002
Receipt Date:	24-MAY-2012
Filing Date:	29-SEP-2008
Time Stamp:	13:14:07
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453_002_Resp_to_OA.pdf	236279 298f4c5d348d21e5683e1f782bcb796e34f4e5df	yes	15

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1
	Claims	2	6
	Applicant Arguments/Remarks Made in an Amendment	7	15

Warnings:

Information:

Total Files Size (in bytes):	236279
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/240,916		Filing Date 09/29/2008		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☒ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	310			
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL		TOTAL	310		
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	05/24/2012	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 30	Minus	** 54	= 0	X \$	=		X \$60=	0
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 7	= 0	X \$	=		X \$250=	0
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	0
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=		X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=		X \$	=
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/TER JACKSON WOODRUFF/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Chen et al.

CONFIRMATION NO.: 6240

SERIAL NO.: 12/240,916

FILING DATE: September 29, 2008

TITLE: NON-VOLATILE MEMORY MODULE

EXAMINER: ROJAS, Midys

ART UNIT: 2185

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

SUPPLEMENTAL AMENDMENT AND/OR REPLY TO OFFICE ACTION

Sir:

In the Amendment and/or Reply to Office Action mailed May 24, 2012, two minor errors in the Remarks section were noted and are corrected hereby. On page 7, line 3 of the Amendment, "Claims 63 and 85," should read "Claims 38 and 72," and on page 10, lines 9-10, "*Li*" should be changed to "*Cope*," such that lines 9-10 of page 10 should read: "The above clearly indicates that the frequencies of the *Cope* memories are tied together such that the two memories both operate at a first frequency **or** a second frequency." In all other respects the following remarks and changes continue to apply.

In response to the Office Action mailed April 3, 2012, please amend the subject application as indicated.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 7.

In the Claims

The following Listing of Claims replaces all prior versions in the application:

LISTING OF CLAIMS

1-36. (Canceled)

37. (Previously presented) A method for controlling a memory system operatively coupled to a host system, the memory system including a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

38. (Currently amended) The method of Claim 37, wherein the third clock frequency is ~~approximately~~ substantially equal to the second clock frequency.

39. (Original) The method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation.

40. (Previously presented) The method of Claim 37, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

41. (Original) The method of Claim 40, wherein the trigger condition comprises a power failure condition.

42. (Original) The method of Claim 37, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

43-60. (Canceled)

61. (Previously presented) The method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.

62. (Previously presented) The method of claim 37, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.

63. (Previously presented) The method of claim 62, wherein the backup operation is conducted at repeating time intervals.

64. (Previously presented) The method of claim 63, wherein the backup operation is initiated in response to a trigger event.

65. (Previously presented) The method of claim 37, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.

66. (Previously presented) The method of claim 37, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.

67. (Previously presented) The method of claim 37, wherein one or more of the first, second or third clock frequencies is configurable by the memory system.
68. (Previously presented) The method of claim 37, wherein one or more of the first, second or third clock frequencies is configurable by a user.
69. (Previously presented) A memory system operatively coupled to a host system, the memory system comprising:
- a volatile memory subsystem operable at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system; and
 - a non-volatile memory subsystem operable at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem,
 - the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.
70. (Previously presented) The memory system of claim 69, further comprising:
- a controller configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation and to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation.
71. (Previously presented) The memory system of claim 69, further comprising a plurality of power supplies and a switch configured to selectively deliver power from the plurality of power supplies to the volatile memory subsystem and the non-volatile memory subsystem as a function of the mode of operation.

72. (Currently amended) The memory system of claim 69, wherein the third clock frequency is ~~approximately~~ substantially equal to the second clock frequency.

73. (Previously presented) The memory system of claim 69, wherein the memory system is not powered by a battery when it is in the second mode of operation.

74. (Previously presented) The memory system of claim 69, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

75. (Previously presented) The memory system of claim 74, wherein the trigger condition comprises a power failure condition.

76. (Previously presented) The memory system of claim 69, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

77-82. (Canceled)

83. (Previously presented) The memory system of claim 69, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.

84. (Previously presented) The memory system of claim 69, wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.

85. (Previously presented) The memory system of claim 84, wherein the backup operation is conducted at repeating time intervals.

86. (Previously presented) The memory system of claim 85, wherein the backup operation is initiated in response to a trigger event.
87. (Previously presented) The memory system of claim 69, wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.
88. (Previously presented) The memory system of claim 69, wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.
89. (Previously presented) The memory system of claim 69, wherein one or more of the first, second or third clock frequencies is configurable by the memory system.
90. (Previously presented) The memory system of claim 69, wherein one or more of the first, second or third clock frequencies is configurable by a user.

REMARKS

The Office Action mailed April 3, 2012, has been carefully considered. Reconsideration in view of the following remarks is respectfully requested.

Rejection Under 35 U.S.C. § 112, Second Paragraph

Claims ~~63-38~~ and ~~85-72~~ stand rejected under 35 U.S.C. § 112, second paragraph, as allegedly being indefinite for failing to particularly point out and distinctly claim the subject matter which Applicants regard as the invention. The term “approximately” was objected to and has been replaced with the term “substantially.” This term is used to account for the fact that while in theory two clock frequencies can be identical, in practice there will always be a difference, even if it is infinitesimal. This is a common approach in patent practice and has been deemed acceptable in the case law. See for instance *Andrew Corp. v. Gabriel Electronics*, 847 F.2d 819, 6 USPQ2d 2010 (Fed. Cir. 1988), discussed in MPEP 2173.05(b), (“The court held that the limitation ‘which produces substantially equal E and H plane illumination patterns’ was definite because one of ordinary skill in the art would know what was meant by ‘substantially equal.’”)

Rejection(s) Under 35 U.S.C. § 103(a)

Claims 37-42, 61-62, 64-76, 83-84, and 89-90 stand rejected under 35 U.S.C. § 103(a) as allegedly being unpatentable over U.S. pat. no. 6,336,174 to *Li et al.* (hereinafter, “*Li*”) in view of U.S. pat. pub. no. 2008/0195806 to *Cope* (hereinafter, “*Cope*”).

The independent claims—namely, claims 37 and 69—recite:

- operation of the volatile memory subsystem at *a first clock frequency in a first mode* (in which data is communicated between the volatile memory subsystem and the host system),
- operation of the non-volatile memory subsystem at *a second clock frequency in a second mode* (in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem), and
- operation of the volatile memory subsystem at *a third clock frequency in the second mode* (the third clock frequency is less than the first clock frequency).

According to the above, the volatile memory subsystem operates at a lower clock frequency in the second mode (data communicated between volatile and non-volatile memory subsystems) than in the first mode (data communicated between volatile memory subsystem and host).

The Office action acknowledges that *Li* “does not teach that in the first mode, the volatile memory operates at a first clock frequency while in the second mode, the non-volatile memory operates at a second clock frequency and the volatile memory operates at a third clock frequency that is less than the first clock frequency.” The Office action cites *Cope* to cure this defect, stating:

Cope discloses a memory [volatile memory device such a DRAM device, par. 0039] that is part of a memory system wherein after a pin grant transition 410, an adjust clock signal 412 transitions from a clock stable signal [which represents the claimed first mode of operation] to a clock adjustment signal 414 [representing the second mode of operation] which is generated by the first memory controller to signal the volatile memory to discontinue operation at a normal clock frequency [first clock frequency] and to begin operation at a new clock frequency [second or third clock frequencies, see par. 0039]. It would have been obvious to one of ordinary skill in the art at the time the invention was made to combine the teachings of *Li et al.* with the changing clock frequencies of *Cope* since providing for memory operations at different clock frequencies depending on the memory being executed enables the memory system to operate at optimal power consumption depending on the operation [see par. 0046 of *Cope*].

A close examination of *Cope* reveals that during memory operation, **the two memories are synchronized with one another**. Depending on which of the two memories has access to the shared contact 262, the memories operate together at either a first clock frequency or a second clock frequency. **Importantly, they never operate at different clock frequencies.**

This is evident from paragraph [0021], which states:

In a particular embodiment, the external clock circuit can generate a common clock signal at a first clock frequency **for both** of the first memory controller 204 and the second memory controller 238 when the first memory controller 204 performs memory operations and can generate a common clock signal at a second clock frequency **for both** of the first memory controller 204 and the second memory controller 238 when the second memory controller 238 performs memory operations. In a specific embodiment, by adjusting the common clock signal frequency **for both** the first memory controller 204 and the second memory controller 238, lower power consumption can be achieved than by operating both memory controllers 204 and 238 at a constant clock frequency. In a specific embodiment, the second clock frequency can be predetermined to reduce number of cycles performed by the second memory controller 238 over a time period to reduce power consumption.

(Emphasis added)

It is also evident from paragraph [0026], which describes what happens after the hand off of control of the shared contact 262 from the default second memory (volatile memory 258) to the first memory (non-volatile memory 230). Paragraph [0026] details synchronization of the clock-frequencies so that they are both the same after the hand-off occurs, and states:

In addition, the logic 206 can send a clock adjust signal to the second memory controller 238 indicating that the external clock circuit has changed the clock frequency. The second memory controller 238 can **synchronize** to the new clock frequency and return a clock adjust complete signal to the logic 206 **when the synchronization is complete**.

(Emphasis added)

In some instances, the shared frequency is lowered in order to reduce power consumption, but as in all cases, both memories are then run at the reduced frequency. This is evident from paragraph [0021], which further states:

In a specific embodiment, by adjusting the common clock signal frequency for both the first memory controller 204 and the second memory controller 238, lower power consumption can be achieved than by operating both memory controllers 204 and 238 at a constant clock frequency. In a specific embodiment, the second clock frequency can be predetermined to reduce number of cycles performed by the second memory controller 238 over a time period to reduce power consumption.

(Emphasis added)

The above clearly indicates that the frequencies of the ~~*Li*~~ *Cope* memories are tied together such that the two memories both operate at a first frequency **or** a second frequency. Therefore *Cope* cannot be combined with *Li* to achieve the different clock frequencies of operation for two modes.

FIG. 4 of *Cope* is reproduced below, with labels added for time segments A – F for ease of discussion. The clock frequencies for the first (non-volatile) and second (volatile) memories during these time segments are clearly described in paragraphs [0037] – [0042] in *Cope*. These clock frequencies are listed below, with “transition” indicating an indeterminate clock frequency as clearly described by *Cope* in paragraph [0041]: “[T]he adjust clock signal transition 420 can be generated by the first memory controller to signal to the second memory controller that a clock signal has stabilized after transitioning to a different or new clock frequency.”

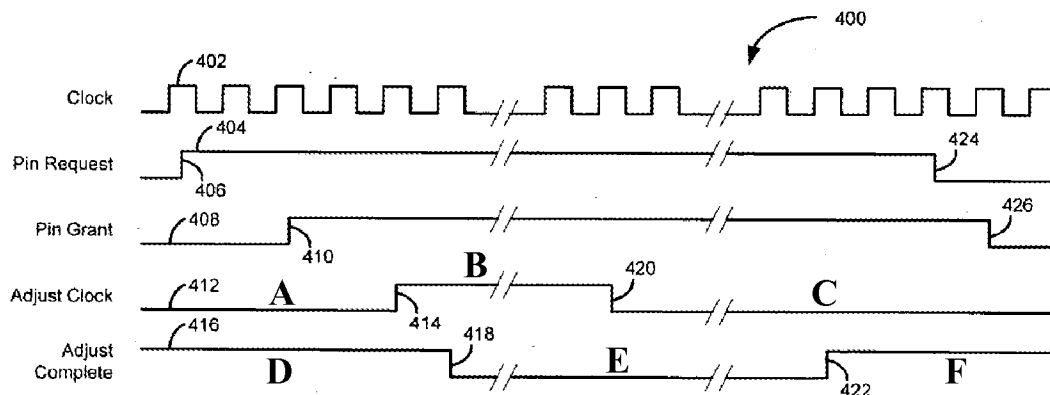


FIG. 4

- Time Segment A: First memory (FLASH) at 1st frequency
- Time Segment B: Transition
- Time Segment C: First memory (FLASH) at 2nd frequency
- Time Segment D: Second memory (DRAM) at 1st frequency
- Time Segment E: Transition
- Time Segment F: Second memory (DRAM) at 2nd frequency

Cope describes a 1st memory controller (FLASH controller) and a 2nd memory controller (DRAM controller) sharing a contact or data pin 262 (see [0012]-[0015] and that shared data pin (bus 106, 108, and 110 of Fig. 1 or bus 208, 210, and 212). *Cope* further describes that only ONE of the FLASH controller and DRAM controllers can use the shared data pin at a time, and that the DRAM controller is the default memory controller for controlling the shared contact (see paragraphs [0022], [0030], and [0045]).

In order for the FLASH controller to utilize the shared data bus in *Cope*, the FLASH controller must request the use of the shared data bus from the DRAM controller via a PIN REQUEST signal 404 at transition 406. The DRAM controller can release the shared data bus to the FLASH controller by asserting the PIN GRANT signal 408 at transition 410 (see [0037]-[0038]).

Cope describes changing the clock frequency for **both** the DRAM controller and FLASH controller (see [0021]). Once the DRAM controller releases the shared data bus to the FLASH controller, as indicated by the “pin grant transition 410” see [0038], the FLASH controller can

assert the ADJUST CLOCK signal 412 at transition 414 to “to signal [the] DRAM device, to discontinue operation at a normal clock frequency and to begin operation at a different or new clock frequency” (see [0039]). It should be noted that, according to *Cope*:

- a. when the ADJUST SIGNAL 412 is LOW (not asserted or logic 0, see Fig. 4, Time Segment A and Time Segment C) then it is indicative of a clock signal that is stable, regardless of what the frequency of the clock signal is.
- b. when the ADJUST SIGNAL 412 is HIGH (asserted or logic 1, see Fig. 4, Time Segment B) then it is indicative of:
 - i. a clock signal that is NOT stable and with indeterminate frequency, see [0041] and as noted above,
 - ii. at transition 414, the DRAM device has already released the shared bus and therefore cannot possibly be operable to communicate data and note that *Cope* further describes additional steps that must be performed before the DRAM device actually may starts operating using the new clock frequency, as clearly described in paragraph [0042] the ADJUST COMPLETE signal at “transition 422 is generated by the second memory controller to signal to the first memory controller that synchronization to the new clock frequency is complete”, and
 - iii. *Cope* further describes in paragraph [0040] that “In response to the adjust clock transition 414, an adjust complete signal 416 transitions to a low state at transition 418, indicating that synchronization to a clock signal is being performed.” Or “...synchronization to a different or new clock frequency has not been completed.”
- c. It is very clear that *Cope* describes that:
 - i. during Time Segment A (see reproduced FIG. 4 above), the clock is stable at a first frequency,
 - ii. during Time Segment B, the clock is NOT stable, at an indeterminate frequency, and
 - iii. during Time Segment C, the clock is stable at a second frequency.

Therefore, it is not possible to have the ADJUST SIGNAL 412 while LOW to represent the claimed first mode of operation, because *Cope* uses the LOW level of ADJUST SIGNAL 412 to simply indicate that a “clock signal has stabilized after transitioning to a different or new clock frequency” **regardless of what that frequency is**. In fact, at ADJUST SIGNAL 412 LOW at Time Segment A, the clock is at a first frequency, while at ADJUST SIGNAL 412 LOW at Time Segment C, the clock is at a second frequency. In contrast, the claimed first mode of operation requires “operating the volatile memory subsystem at a first frequency.” The DRAM device cannot be operating at two different frequencies, as alleged in the Office action, when the memory system is in a first mode.

Similarly, in the claimed second mode of operation, the “non-volatile memory subsystem operates at a second frequency” and “the volatile memory subsystem [operates] at a third frequency...” The Office action alleges that ADJUST CLOCK 412 transitioning at 414 to HIGH represents the claimed second mode of operation. It is very clear that *Cope* describes that:

- a. during Time Segment B, when ADJUST CLOCK 412 transitions at 414 to HIGH, it is indicative of a clock that is NOT stable and thus the clock’s frequency cannot be determined, see above, and
- b. during Time Segment B, *Cope*’s DRAM devices:
 - i. are inoperative to communicate data on the shared data bus since “after a pin grant transition 410” the DRAM controller has released the shared data bus to be used exclusively by the FLASH controller,
 - ii. cannot be operative since the clock frequency is unknown or indeterminate, and thus a catastrophic failure would occur if data access to the DRAM were to be gained somehow while the frequency “is not stable”, and
 - iii. inoperable to communicate data since *Cope* clearly describes that the “adjust clock signal transition 414 can cause the volatile memory device to enter a self-refresh mode” and it is well known

that if a DRAM device enters a self-refresh mode, which is akin to a power down mode, the DRAM uses an on-chip timer to generate internal refresh cycles as necessary, and the system clock may even be stopped during this time. Thus, no data read or write operation can be performed during DRAM self-refresh mode.

Cope cannot be used to describe two modes of operation, where a DRAM in a first mode operates at a first clock frequency and in a second mode operates at another frequency. This is because *Cope*'s DRAM devices are operable only in a first Mode at a first frequency, and are inoperable to communicate data in a second mode where the clock is beginning to change or has changed to another frequency.

In *Cope*, after the FLASH controller asserts ADJUST SIGNAL 412 to a HIGH level, the DRAM controller asserts ADJUST COMPLETE signal 416 to LOW at transition 418 "to signal to the first (FLASH) memory controller that synchronization to a different or new clock frequency has not been completed" (see [0040]). Once the clock has stabilized, the FLASH controller returns ADJUST SIGNAL 412 to a LOW level, at 420, and after an unknown time period (indicated by the breaks in signals and clock in Fig. 4) the DRAM controller asserts ADJUST COMPLETE signal 416 to HIGH at transition 422 "to signal to the FLASH memory controller that synchronization to the new clock frequency is complete" (see [0042]).

Further, as noted above, when the ADJUST CLOCK pin is asserted at 414 (to begin the alleged second mode of operation), the volatile memory is in fact being taken offline, as indicated by the pin grant transition 410, because *Cope* describes that only one of the first and second memories is allowed access to the shared data pin 262 at a time. In this mode, therefore, the DRAM is non-operational. At best, it is directed to enter a self-refresh mode, but it is effectively disconnected from the system. The combination of *Cope* with *Li* therefore would not teach or suggest operating the volatile memory at a first clock frequency in a first mode and at a lower clock frequency in a second mode, wherein in the first mode the volatile memory is communicating with the host, and in the second mode it is communicating with the non-volatile memory.

It should also be noted that the transition 414 and the asserted ADJUST CLOCK during Time Segment B thereafter do not demarcate a stable second clock frequency—the new clock frequency is not stable for possible usage by the DRAM until transition 422, when the ADJUST COMPLETE pin is returned to high by the DRAM at transition 422 and during Time Segment F thereafter.

Conclusion

In view of the preceding discussion, Applicants respectfully urge that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present invention, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: June 14, 2012

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

NIXON PEABODY LLP
P.O. BOX 60610
PALO ALTO, CA 94306
TEL. (650) 320-7700
FAX. (650) 320-7701

Electronic Acknowledgement Receipt

EFS ID:	13013716
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-002
Receipt Date:	14-JUN-2012
Filing Date:	29-SEP-2008
Time Stamp:	13:32:43
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453_002_Supplemental_A mdt.pdf	237985 780a911a5630338b6ac2b958e08c8298341606af	yes	16

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Supplemental Response or Supplemental Amendment	1	1
	Claims	2	6
	Applicant Arguments/Remarks Made in an Amendment	7	16

Warnings:

Information:

Total Files Size (in bytes):	237985
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/240,916		Filing Date 09/29/2008		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)		SMALL ENTITY ☐ OR			OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☒ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	54 minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	7 minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL				
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)		SMALL ENTITY OR			OTHER THAN SMALL ENTITY		
AMENDMENT	06/14/2012	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)		RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 30	Minus	** 54	=	X \$	=		X \$	=
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 7	=	X \$	=		X \$	=
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE			TOTAL ADD'L FEE	
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)		RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=		X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=		X \$	=
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE			TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/JESSICA GAYNOR/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

RESCISSION OF PREVIOUS NONPUBLICATION REQUEST (35 U.S.C. 122(b)(2)(B)(ii)) AND, IF APPLICABLE, NOTICE OF FOREIGN FILING (35 U.S.C. 122(b)(2)(B)(iii))	Application Number	12240916
	Filing Date	2008-09-29
	First Named Inventor	Chi-She CHEN
	Title	Non-Volatile Memory Module
	Atty Docket Number	062453-002
	Art Unit	2185
	Examiner	Midys ROJAS

Send completed form to:
Mail Stop PG Pub
 Commissioner for Patents
 P.O. Box 1450
 Alexandria, VA 22313-1450
 FAX: (571) 273-8300

A request that the above-identified application not be published under 35 U.S.C. 122(b) (nonpublication request) was included with the above-identified application on filing pursuant to 35 U.S.C. 122(b)(2)(B)(i).
 I hereby **rescind** the previous nonpublication request.

If a notice of foreign or international filing is or will be required by 35 U.S.C. 122(b)(2)(B)(iii) and 37 CFR 1.213(c), I hereby provide such notice. This notice is being provided no later than forty-five (45) days after the date of such foreign or international filing.

If a notice of subsequent foreign or international filing required by 35 U.S.C. 122(b)(2)(B)(iii) and 37 CFR 1.213(c) was not filed within forty-five (45) days after the date of filing of the foreign or international application, the application is ABANDONED, and a petition to revive under 37 CFR 1.137(b) is required. See 37 CFR 1.137(f).

/Khaled Shami/

August 22, 2012

Signature

Date

Khaled Shami

38745

Typed or printed name

Registration Number, if applicable

650-320-7700

Telephone Number

This request must be signed in compliance with 37 CFR 1.33(b).

If information or assistance is needed in completing this form, please contact the Pre-Grant Publication Division at (703)605-4283 or by e-mail at PGPub@USPTO.gov.

CERTIFICATE OF MAILING OR TRANSMISSION

I hereby certify that this correspondence is being deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to: Mail Stop PG Pub, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450, or facsimile transmitted to the U.S. Patent and Trademark Office on the date shown below.

Signature

Name (Print/Type)

Date

This collection of information is required by 37 CFR 1.213(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 6 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Mail Stop PG Pub, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	13558325
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-002
Receipt Date:	22-AUG-2012
Filing Date:	29-SEP-2008
Time Stamp:	12:29:03
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Rescind Nonpublication Request for Pre Grant Pub	062453_002_Recission.pdf	259216 3eba2863f7701ba99a9092d6fffe5637163c7e0	no	2

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/240,916	09/29/2008	Chi-She Chen	062453-002

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

CONFIRMATION NO. 6240
NEW OR REVISED PPD NOTICE



NOTICE OF NEW OR REVISED PROJECTED PUBLICATION DATE

The above-identified application has a new or revised projected publication date. The current projected publication date for this application is 12/06/2012. If this is a new projected publication date (there was no previous projected publication date), the application has been cleared by Licensing & Review or a secrecy order has been rescinded and the application is now in the publication queue.

If this is a revised projected publication date (one that is different from a previously communicated projected publication date), the publication date has been revised due to processing delays in the USPTO or the abandonment and subsequent revival of an application. The application is anticipated to be published on a date that is more than six weeks different from the originally-projected publication date.

More detailed publication information is available through the private side of Patent Application Information Retrieval (PAIR) System. The direct link to access PAIR is currently <http://pair.uspto.gov>. Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Questions relating to this Notice should be directed to the Office of Data Management, Application Assistance Unit at (571) 272-4000, or (571) 272-4200, or 1-888-786-0101.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/240,916	09/29/2008	Chi-She Chen	062453-002

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

CONFIRMATION NO. 6240
NONPUBLICATION RESCISSION
LETTER



OC000000056230596

Date Mailed: 08/31/2012

**Communication Regarding Rescission Of
Nonpublication Request and/or Notice of Foreign Filing**

Applicant's rescission of the previously-filed nonpublication request and/or notice of foreign filing is acknowledged. The paper has been reflected in the Patent and Trademark Office's (USPTO's) computer records so that the earliest possible projected publication date can be assigned.

The projected publication date is 12/06/2012.

If applicant rescinded the nonpublication request before or on the date of "foreign filing,"¹ then no notice of foreign filing is required.

If applicant foreign filed the application after filing the above application and before filing the rescission, and the rescission did not also include a notice of foreign filing, then a notice of foreign filing (not merely a rescission) is required to be filed within 45 days of the date of foreign filing. See 35 U.S.C. § 122(b)(2)(B)(iii), and Clarification of the United States Patent and Trademark Office's Interpretation of the Provisions of 35 U.S.C. § 122(b)(2)(B)(ii)-(iv), 1272 Off. Gaz. Pat. Office 22 (July 1, 2003).

If a notice of foreign filing is required and is not filed within 45 days of the date of foreign filing, then the application becomes abandoned pursuant to 35 U.S.C. § 122(b)(2)(B)(iii). In this situation, applicant should either file a petition to revive or notify the Office that the application is abandoned. See 37 CFR 1.137(f). Any such petition to revive will be forwarded to the Office of Petitions for a decision. Note that the filing of the petition will not operate to stay any period of reply that may be running against the application.

Questions regarding petitions to revive should be directed to the Office of Petitions at (571) 272-3282.

¹ Note, for purpose of this notice, that "foreign filing" means "filing an application directed to the same invention in another country, or under a multilateral international agreement, that requires publication of applications 18 months after filing".

/hsarwari/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

46188 7590 09/17/2012
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

EXAMINER

ROJAS, MIDYS

ART UNIT

PAPER NUMBER

2185

DATE MAILED: 09/17/2012

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

12/240,916

09/29/2008

Chi-She Chen

062453-002

6240

TITLE OF INVENTION: NON-VOLATILE MEMORY MODULE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$300	\$0	\$2040	12/17/2012

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

46188 7590 09/17/2012
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

12/240,916 09/29/2008 Chi-She Chen 062453-002 6240

TITLE OF INVENTION: NON-VOLATILE MEMORY MODULE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
-------------	--------------	---------------	---------------------	----------------------	------------------	----------

nonprovisional NO \$1740 \$300 \$0 \$2040 12/17/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
----------	----------	----------------

ROJAS, MIDYS 2185 711-104000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/240,916	09/29/2008	Chi-She Chen	062453-002	6240
46188	7590	09/17/2012	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ROJAS, MIDYS	
			ART UNIT	PAPER NUMBER
			2185	
DATE MAILED: 09/17/2012				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 466 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 466 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No.	Applicant(s)	
	12/240,916	CHEN ET AL.	
	Examiner	Art Unit	
	MIDYS ROJAS	2185	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 6/14/2012.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 37-42,61-76 and 83-90.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
 - a) ☐ All b) ☐ Some* c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
 6. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
 - (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
 - 1) ☐ hereto or 2) ☐ to Paper No./Mail Date ____.
 - (b) ☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
7. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| <ol style="list-style-type: none"> 1. ☐ Notice of References Cited (PTO-892) 2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) 3. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____ 4. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | <ol style="list-style-type: none"> 5. ☐ Notice of Informal Patent Application 6. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. 7. ☐ Examiner's Amendment/Comment 8. ☒ Examiner's Statement of Reasons for Allowance 9. ☐ Other ____. |
|--|--|

/Midys Rojas/
Primary Examiner, Art Unit 2185

REASONS FOR ALLOWANCE

Claims 37-42, 61-76, and 83-90 are allowed.

The following is an examiner's statement of reasons for allowance:

The Prior Art of Record fails to teach and/or suggest in the claimed combination controlling a memory system coupled to a host, the memory system including a volatile memory subsystem and a non-volatile memory subsystem, the method comprising: operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system; operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Art Unit: 2185

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MIDYS ROJAS whose telephone number is (571)272-4207. The examiner can normally be reached on M-TH 6:00am - 4:30pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Sanjiv Shah can be reached on (571) 272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Midys Rojas/
Primary Examiner, Art Unit 2185

9/10/2012

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	0	("("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L2	12	("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L3	0	host same (volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L4	1	(volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L5	92	(memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L6	60	"711"/\$.ccls. and (memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L7	1121	memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L8	151	"711"/\$.ccls. and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L9	74	("4072852" "4815074" "4959774" "5283792" "5379431" "5799200").PN. OR ("6336174").URPN.	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L10	10	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) with (frequency or speed or clock)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L11	2	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28

L12	0	memory same (host near3 frequency) same (non-volatile near3 frequency) same mode	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L13	2	(memory with mode) same (host with frequency) same (non-volatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L14	6	(memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L15	20	"711"/\$.ccls. and (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L16	49	"711"/\$.ccls. and (host with speed) same (nonvolatile with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L17	29	"711"/\$.ccls. and (host with speed) same (backup with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L18	50	(backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L19	48	power and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L20	14	(power near3 (loss or fail\$3)) and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L21	8	(power near3 (loss or fail\$3)) and (copy\$3 or tranfer\$3) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L22	8	"711"/\$.ccls. and (copy\$3 or tranfer\$3 or back-up or (back adj up)) with (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L23	94	"711"/\$.ccls. and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L24	5081	((711/162) or (711/160) or (711/161) or (710/10)).CCLS.	USPAT; USOCR	OR	OFF	2012/09/10 13:28
L25	2	L24 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L26	1	L24 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L27	9	"711"/\$.ccls. and ((non-volatile or nonvolatile) near3 memory) with ((vary\$3 or alternat\$3 or alter\$3 or fluctuat\$3 or chang\$3) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L28	2	"711"/\$.ccls. and ((non-volatile or nonvolatile) near3 memory) with (dynamic near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2012/09/10 13:28

			JPO			
L40	5839	((711/162) or (711/160) or (711/161) or (710/10) or (711/104)).CCLS.	USPAT; USOCR	OR	OFF	2012/09/10 13:40
L41	3	L40 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:40
L42	5	L40 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:41
L43	1	L40 and (host with speed) same (nonvolatile with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:41
L44	0	L40 and ((non-volatile or nonvolatile) near3 memory) with (dynamic near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:41
L45	2	L40 and (memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:41
L46	2	L40 and (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:42
L47	8	L40 and (host with speed) same (backup with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:42
L48	3	L40 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:42

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L29	8101	((711/162) or (711/160) or (711/161) or (710/10)).CCLS.	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:28
L30	2	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:35
L31	524	(memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:36
L32	6	(memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:36
L33	1029	memory with mode near3 frequency	US-PGPUB;	OR	OFF	2012/09/10 13:37

			USPAT; UPAD			
L34	0	L29 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:37
L35	2	L29 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency) (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:37
L36	524	L29 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency) (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:37
L37	2	L29 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:39
L38	0	L29 and (memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:39
L39	3	L29 and memory with mode near3 frequency	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:39
L49	9362	((711/162) or (711/160) or (711/161) or (710/10) or (711/104)).OCLS.	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:42
L50	0	L49 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:42
L51	2	L49 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency) (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L52	524	L49 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency) (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L53	7	L49 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L54	0	L49 and (memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L55	8	L49 and memory with mode near3 frequency	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L56	0	netlist.as. with (memory same frequency)	US-PGPUB; USPAT;	OR	OFF	2012/09/10 14:10

			UPAD			
L57	42	netlist.as.	US- PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 14:10
L58	134096	netlist.as. (frequency with mode)	US- PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 14:11
L59	1	netlist.as. and (frequency with mode)	US- PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 14:11

9/ 10/ 2012 2:12:25 PM
C:\ Users\ mrojas\ Documents\ EAST\ Workspaces\ 12240916.wsp

Search Notes 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

SEARCHED			
Class	Subclass	Date	Examiner
711	104, 160, 161, 162	9/10/2012	MR
710	10	9/10/2012	MR

SEARCH NOTES		
Search Notes	Date	Examiner
EAST Updated search: limited search of 711/160, 161, 162; 710/10; and text searches	3/25/2012	MR
PALM inventor name search	3/25/2012	MR
EAST Updated search: limited search of 711/104, 160, 161, 162; 710/10; and text search	9/10/2012	MR
PALM Inventor Name search	9/10/2012	MR
EAST Interference search	9/10/2012	MR
EAST Assignee search	9/10/2012	MR

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner
711	104, 160, 161, 162		
710	10		

	/MIDYS ROJAS/ Primary Examiner.Art Unit 2185
--	---



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
 United States Patent and Trademark Office
 Address: COMMISSIONER FOR PATENTS
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
 www.uspto.gov

BIB DATA SHEET

CONFIRMATION NO. 6240

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.		
12/240,916	09/29/2008	711	2185	062453-002		
RULE						
APPLICANTS Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;						
** CONTINUING DATA ***** This application is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007						
** FOREIGN APPLICATIONS *****						
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 10/08/2008						
Foreign Priority claimed	☐ Yes ☒ No	☐ Met after Allowance	STATE OR COUNTRY	SHEETS DRAWINGS	TOTAL CLAIMS	INDEPENDENT CLAIMS
35 USC 119(a-d) conditions met	☐ Yes ☒ No	Initials	CA	12	54	7
Verified and Acknowledged	/MIDYS ROJAS/ Examiner's Signature					
ADDRESS						
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306 UNITED STATES						
TITLE						
NON-VOLATILE MEMORY MODULE						
FILING FEE RECEIVED 3570	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	0	("("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L2	12	("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L3	0	host same (volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L4	1	(volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L5	92	(memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L6	60	"711"/\$.ccls. and (memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L7	1121	memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L8	151	"711"/\$.ccls. and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L9	74	("4072852" "4815074" "4959774" "5283792" "5379431" "5799200").PN. OR ("6336174").URPN.	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L10	10	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) with (frequency or speed or clock)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L11	2	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28

L12	0	memory same (host near3 frequency) same (non-volatile near3 frequency) same mode	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L13	2	(memory with mode) same (host with frequency) same (non-volatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L14	6	(memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L15	20	"711"/\$.ccls. and (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L16	49	"711"/\$.ccls. and (host with speed) same (nonvolatile with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L17	29	"711"/\$.ccls. and (host with speed) same (backup with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L18	50	(backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L19	48	power and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L20	14	(power near3 (loss or fail\$3)) and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L21	8	(power near3 (loss or fail\$3)) and (copy\$3 or tranfer\$3) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L22	8	"711"/\$.ccls. and (copy\$3 or tranfer\$3 or back-up or (back adj up)) with (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L23	94	"711"/\$.ccls. and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L24	5081	((711/162) or (711/160) or (711/161) or (710/10)).CCLS.	USPAT; USOCR	OR	OFF	2012/09/10 13:28
L25	2	L24 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L26	1	L24 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:28
L27	9	"711"/\$.ccls. and ((non-volatile or nonvolatile) near3 memory) with ((vary\$3 or alternat\$3 or alter\$3 or fluctuat\$3 or chang\$3) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:28
L28	2	"711"/\$.ccls. and ((non-volatile or nonvolatile) near3 memory) with (dynamic near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2012/09/10 13:28

			JPO			
L40	5839	((711/162) or (711/160) or (711/161) or (710/10) or (711/104)).CCLS.	USPAT; USOCR	OR	OFF	2012/09/10 13:40
L41	3	L40 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:40
L42	5	L40 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:41
L43	1	L40 and (host with speed) same (nonvolatile with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:41
L44	0	L40 and ((non-volatile or nonvolatile) near3 memory) with (dynamic near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:41
L45	2	L40 and (memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:41
L46	2	L40 and (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:42
L47	8	L40 and (host with speed) same (backup with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2012/09/10 13:42
L48	3	L40 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/09/10 13:42
S1	0	("(20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/27 15:21
S2	12	("20020083368" "20040190210" "4420821" "4449205" "5519663" "6158015" "6336174" "6336176" "6487623" "6658507" "6799244" "7409590").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:21
S3	0	host same (volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:34
S4	1	(volatile with non-volatile with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:34
S5	87	(memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2011/07/27 15:35

			JPO			
S6	56	"711"/\$.ccls. and (memory with (back-up or copy) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:35
S7	1013	memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:44
S8	128	"711"/\$.ccls. and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/27 15:44
S9	46	("4072852" "4815074" "4959774" "5283792" "5379431" "5799200").PN. OR ("6336174").URPN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/27 15:44
S10	8	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) with (frequency or speed or clock)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:08
S11	2	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:09
S12	0	memory same (host near3 frequency) same (non-volatile near3 frequency) same mode	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:11
S13	1	(memory with mode) same (host with frequency) same (non-volatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:12
S14	6	(memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:12
S15	13	"711"/\$.ccls. and (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:14
S16	40	"711"/\$.ccls. and (host with speed) same (nonvolatile with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:22
S17	16	"711"/\$.ccls. and (host with speed) same (backup with speed)	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:23
S18	50	(backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:29
S19	48	power and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:29
S20	14	(power near3 (loss or fail\$3)) and (backup or back-up) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:30
S21	8	(power near3 (loss or fail\$3)) and (copy\$3 or tranfer\$3) same (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:38
S22	8	"711"/\$.ccls. and (copy\$3 or tranfer\$3 or	US-PGPUB;	OR	OFF	2011/07/28

		back-up or (back adj up)) with (high\$3 near3 (speed or frequency) near3 (non-volatile or nonvolatile))	USPAT; USOCR			16:41
S23	88	"711"/\$.cls. and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 16:44
S24	4347	((711/162) or (711/160) or (711/161) or (710/10)).CCLS.	USPAT; USOCR	OR	OFF	2011/07/28 19:20
S25	0	S24 and memory with mode near3 frequency	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2011/07/28 19:21
S26	1	S24 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; USOCR	OR	OFF	2011/07/28 19:21
S27	8	"711"/\$.cls. and ((non-volatile or nonvolatile) near3 memory) with ((vary\$3 or alternat\$3 or alter\$3 or fluctuat\$3 or chang\$3) near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/03/08 13:52
S28	1	"711"/\$.cls. and ((non-volatile or nonvolatile) near3 memory) with (dynamic near3 frequency)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO	OR	OFF	2012/03/08 13:56

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L29	8101	((711/162) or (711/160) or (711/161) or (710/10)).CCLS.	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:28
L30	2	(power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:35
L31	524	(memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:36
L32	6	(memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:36
L33	1029	memory with mode near3 frequency	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:37
L34	0	L29 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:37
L35	2	L29 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency) (power near3 (loss or fail\$3)) with (memory near3 (back-up or back	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:37

		adj up)) same (frequency)				
L36	524	L29 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency) (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:37
L37	2	L29 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:39
L38	0	L29 and (memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:39
L39	3	L29 and memory with mode near3 frequency	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:39
L49	9362	((711/162) or (711/160) or (711/161) or (710/10) or (711/104)).CCLS.	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:42
L50	0	L49 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:42
L51	2	L49 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency) (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L52	524	L49 and (power near3 (loss or fail\$3)) with (memory near3 (back-up or back adj up)) same (frequency) (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L53	7	L49 and (memory near3 mode) with (operat\$3 near3 (speed or frequency))	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L54	0	L49 and (memory with mode) same (host with frequency) same (nonvolatile with frequency)	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43
L55	8	L49 and memory with mode near3 frequency	US-PGPUB; USPAT; UPAD	OR	OFF	2012/09/10 13:43

<i>Index of Claims</i> 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	07/28/2011	09/10/2012						
	1	N	-						
	2	N	-						
	3	N	-						
	4	N	-						
	5	N	-						
	6	N	-						
	7	N	-						
	8	N	-						
	9	N	-						
	10	N	-						
	11	N	-						
	12	N	-						
	13	N	-						
	14	N	-						
	15	N	-						
	16	N	-						
	17	N	-						
	18	N	-						
	19	N	-						
	20	N	-						
	21	N	-						
	22	N	-						
	23	N	-						
	24	N	-						
	25	N	-						
	26	N	-						
	27	N	-						
	28	N	-						
	29	N	-						
	30	N	-						
	31	N	-						
	32	N	-						
	33	N	-						
	34	N	-						
	35	N	-						
	36	N	-						

<i>Index of Claims</i> 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	07/28/2011	09/10/2012						
	37	✓	=						
	38	✓	=						
	39	✓	=						
	40	✓	=						
	41	✓	=						
	42	✓	=						
	43	N	-						
	44	N	-						
	45	N	-						
	46	N	-						
	47	N	-						
	48	N	-						
	49	N	-						
	50	N	-						
	51	N	-						
	52	N	-						
	53	N	-						
	54	N	-						
	55		-						
	56		-						
	57		-						
	58		-						
	59		-						
	60		-						
	61		=						
	62		=						
	63		=						
	64		=						
	65		=						
	66		=						
	67		=						
	68		=						
	69		=						
	70		=						
	71		=						
	72		=						

<i>Index of Claims</i> 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	07/28/2011	09/10/2012						
	73		=						
	74		=						
	75		=						
	76		=						
	77		-						
	78		-						
	79		-						
	80		-						
	81		-						
	82		-						
	83		=						
	84		=						
	85		=						
	86		=						
	87		=						
	88		=						
	89		=						
	90		=						

<i>Issue Classification</i> 	Application/Control No. 12240916	Applicant(s)/Patent Under Reexamination CHEN ET AL.
	Examiner MIDYS ROJAS	Art Unit 2185

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/MIDYS ROJAS/ Primary Examiner.Art Unit 2185	09/10/2012	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	37	9

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

46188 7590 09/17/2012
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/240,916	09/29/2008	Chi-She Chen	062453-002	6240

TITLE OF INVENTION: NON-VOLATILE MEMORY MODULE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$300	\$0	\$2040	12/17/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
ROJAS, MIDYS	2185	711-104000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list

(1) the names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 **NIXON PEABODY LLP**

2 **Khaled Shami**

3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

NETLIST, INC.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

IRVINE, CALIFORNIA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
☒ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number **503557** (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27.

☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature 

Date **9/26/2012**

Typed or printed name **Khaled Shami**

Registration No. **38745**

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Electronic Patent Application Fee Transmittal

Application Number:	12240916			
Filing Date:	29-Sep-2008			
Title of Invention:	NON-VOLATILE MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Pamela Wilson			
Attorney Docket Number:	062453-002			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl issue fee	1501	1	1740	1740
Publ. Fee- early, voluntary, or normal	1504	1	300	300

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				2040

Electronic Acknowledgement Receipt

EFS ID:	13846688
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-002
Receipt Date:	26-SEP-2012
Filing Date:	29-SEP-2008
Time Stamp:	17:42:03
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 2040
RAM confirmation Number	4774
Deposit Account	503557
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Post Allowance Communication - Incoming	062453_002_comm_on_allowance.pdf	23385	no	2
			26dcff079c0b3cdd9dc0aaa217498b5a143381d2		

Warnings:**Information:**

2	Issue Fee Payment (PTO-85B)	062453_002_fee_transmittal.pdf	126328	no	1
			dce87fca09fc35d3659dbfb8e36cd6062841077d		

Warnings:**Information:**

3	Fee Worksheet (SB06)	fee-info.pdf	31956	no	2
			174e29407854eb6593e2a97ff467482113f20518		

Warnings:**Information:**

Total Files Size (in bytes):			181669
------------------------------	--	--	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Chi-She Chen
SERIAL NO.: 12/240,916 CONFIRMATION No.: 6240
FILING DATE: September 29, 2008
TITLE: NON-VOLATILE MEMORY MODULE
EXAMINER: Rojas, Midys
ART UNIT: 2185

**Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

APPLICANT'S COMMENTS ON REASONS FOR ALLOWANCE

These comments are responsive to the Notice of Allowance, mailed on September 17, 2012.

Applicant gratefully acknowledges the indication of allowance of Claims 37-42, 61-76 and 83-90. Applicant respectfully urges that additional and/or alternative reasons for allowance may exist apart from those advanced by the Examiner and the Applicant, and these reasons may each be independently sufficient to establish the patentability of each of the allowed claims.

Applicant respectfully reserves the right to introduce, articulate, or otherwise comment on any such additional reasons for allowance as may be appropriate in any future proceedings concerning the one or more claimed embodiments.

Please charge any additional required fee or credit any overpayment to our Deposit
Account number 50-3557.

Respectfully submitted,

NIXON PEABODY LLP

Dated: 9/26/12


Khaled Shami
Reg. No. 38,745

Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306
Tel. (650) 320-7700
Fax. (650) 320-7701

Substitute for form 1449A/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Use as many sheets as necessary)				<i>Complete if Known</i>	
				Application Number	12/240,916
				Filing Date	September 29, 2008
				First Named Inventor	Chi-She Chen
				Art Unit	2189
				Examiner Name	Unknown
Sheet	1	of	1	Attorney Docket No: 987-04-CON-H	

US PATENT DOCUMENTS

Examiner Initial *	Cite No	USP Document Number	Publication Date	Name of Patentee or Applicant of cited Document	Filing Date If Appropriate
/M.R./		4,420,821	12-13-1983	Hoffman	02-19-1982
		4,449,205	05-15-1984	Hoffman	02-19-1982
		5,519,663	05-21-1996	Harper, Jr. et al.	09-28-1994
		6,158,015	12-05-2000	Klein	03-30-1998
		6,336,176	01-01-2002	Li et al. Leyda, et al.	08-09-1999
		6,487,623	11-26-2002	Emerson et al.	04-30-1999
		6,658,507	12-02-2003	Chan	08-31-1998
		6,799,244	09-28-2004	Tanaka et al.	12-06-2002
		2002/0083368	06-27-2002	Abe et al.	12-20-2001
		2004/0190210	09-30-2004	Leete	03-26-2003

FOREIGN PATENT DOCUMENTS

Examiner Initials*	Cite No	Foreign Document No	Publication Date	Name of Patentee or Applicant of cited Document	T ²

OTHER DOCUMENTS -- NON PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²

EXAMINER

/Midys Rojas/

DATE CONSIDERED

07/27/2011



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/240,916	10/30/2012	8301833	062453-002	6240

46188 7590 10/10/2012
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 638 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

TO: Mail Stop 8 Director of the U.S. Patent & Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	---

In Compliance with 35 § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been
filed in the U.S. District Court NORTHERN CALIFORNIA on the following ☒ Patents or ☐ Trademarks:

DOCKET NO. CV 13-03901 DMR	DATE FILED 8/23/2013	U.S. DISTRICT COURT Northern District of California, 1301 Clay St. #400-S, Oakland, CA 94612
PLAINTIFF DIABLO TECHNOLOGIES INC		DEFENDANT NETLIST INC
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,001,434		***SEE ATTACHED COMPLAINT***
2 8,301,833		
3 8,359,501		
4 8,516,185		
5 8,516,187		

In the above—entitled case, the following patent(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading		
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK	
1			
2			
3			
4			
5			

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK Richard W. Wieking	(BY) DEPUTY CLERK Clara Pierce	DATE August 27, 2013
---------------------------------	---------------------------------------	-----------------------------

Copy 1—Upon initiation of action, mail this copy to Commissioner Copy 3—Upon termination of action, mail this copy to Commissioner
Copy 2—Upon filing document adding patent(s), mail this copy to Commissioner Copy 4—Case file copy

1 WILLIAM F. ABRAMS
(CA Bar. No. 88805)
2 wabrams@kslaw.com
SANJEET K. DUTTA
3 (CA Bar No. 203463)
sducta@kslaw.com
4 KING & SPALDING LLP
333 Twin Dolphin Drive, Suite 400
5 Redwood Shores, CA 94065
Telephone: (650) 590-0700
6 Facsimile: (650) 590-1900

7 Attorneys for Plaintiff
Diablo Technologies, Inc.
8

9 UNITED STATES DISTRICT COURT
10 NORTHERN DISTRICT OF CALIFORNIA
11 SAN FRANCISCO DIVISION

DMR

12 DIABLO TECHNOLOGIES, INC.,

13 Plaintiff,

14 v.

15 NETLIST, INC.,

16 Defendant.

CV Case No. 13 3901

COMPLAINT FOR DECLARATORY
JUDGMENT

DEMAND FOR JURY TRIAL

ORIGINAL
FILED

2013 AUG 23 A 9:23

RICHARD W. WILSON
CLERK, U.S. DISTRICT COURT
NORTHERN DISTRICT OF CALIFORNIA

1 For its Complaint against Defendant NETLIST, INC., ("NETLIST"), Plaintiff Diablo
2 Technologies, Inc. ("DIABLO") alleges as follows:

3 **FACTUAL BACKGROUND**

4 **DIABLO**

5 1. DIABLO is a corporation incorporated under the laws of Canada, with its
6 principal place of business at 80 Aberdeen Street, Suite 401, Ottawa, Ontario, K1S 5R5 Canada.

7 2. DIABLO, founded in 2003, delivers product solutions that enhance the
8 performance and capability of memory system designs. DIABLO developed the Memory
9 Channel Storage™ (MCST™) architecture for enterprise servers and storage systems.

10 3. DIABLO has been awarded "Most Innovative Flash Memory Enterprise Business
11 Application" in the Flash Memory Summit Best of Show Awards for enabling a new class of
12 ultra-low latency (ULL) SSDs and application accelerators. This innovative new architecture
13 radically transforms memory and storage for enterprise rack, blade and storage servers.

14
15 **NETLIST'S THREAT OF SUIT AGAINST DIABLO**

16 4. NETLIST purports to be incorporated under the laws of Delaware, with its
17 principal place of business at 51 Discovery, Suite 150, Irvine, California, 92618.

18 5. On August 22, 2013, NETLIST sent DIABLO a letter stating that "Netlist has
19 concluded that Smart Modular Technologies, Inc., Smart Storage Systems, Inc., Smart
20 Worldwide Holdings, Inc. and Diablo Technologies, Inc. (collectively "Defendants") are
21 directly and /or indirectly infringing the following United States patents at least through the
22 making, using, offering for sale and/or selling of the recently announced ULLtraDIMM product:

23 U.S. Patent No. 8,001,434, Memory Board With Self-Testing Capability

24 U.S. Patent No. 8,301,833, Non-Volatile Memory Module

25 U.S. Patent No. 8,359,501, Memory Board With Self-Testing Capability

26 U.S. Patent No. 8,516,185, System and Method Utilizing Distributed Byte-Wise Buffers
27 on a Memory Module

28 U.S. Patent No. 8,516,187, Data Transfer Scheme for Non-Volatile Memory Module"

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28

JURISDICTION AND VENUE

6. This Court has subject matter jurisdiction over this action pursuant to 28 U.S.C. §§ 1331, 1338, 1367, 2201, and 2202.

7. This Court has personal jurisdiction over DIABLO. Among other things, DIABLO transacts business in this jurisdiction.

8. This Court has personal jurisdiction over NETLIST. Among other things, NETLIST transacts business in this jurisdiction and is a plaintiff in this District at least in the pending action *Netlist, Inc. v. Google Inc.* 4-09-cv-05718 SBA (CAND).

9. Venue is proper in this District pursuant to 28 U.S.C. § 1391.

INTRADISTRICT ASSIGNMENT

10. For purposes of intradistrict assignment pursuant to Local Rule 3-2(c), this Intellectual Property Action is to be assigned on a district-wide basis.

COUNT I

(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,001,434)

11. DIABLO incorporates by reference, as if fully set forth herein, the allegations of the preceding paragraphs of this Complaint.

12. Upon information and belief, NETLIST is the owner of the '434 patent.

13. NETLIST is asserting the '434 patent against "ULLtraDIMM."

14. DIABLO, contends that it has not infringed and does not infringe, any valid and enforceable claim of the '434 patent.

15. An actual and justiciable controversy has thus arisen between DIABLO and NETLIST concerning the alleged infringement of the '434 patent.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123
124
125
126
127
128
129
130
131
132
133
134
135
136
137
138
139
140
141
142
143
144
145
146
147
148
149
150
151
152
153
154
155
156
157
158
159
160
161
162
163
164
165
166
167
168
169
170
171
172
173
174
175
176
177
178
179
180
181
182
183
184
185
186
187
188
189
190
191
192
193
194
195
196
197
198
199
200
201
202
203
204
205
206
207
208
209
210
211
212
213
214
215
216
217
218
219
220
221
222
223
224
225
226
227
228
229
230
231
232
233
234
235
236
237
238
239
240
241
242
243
244
245
246
247
248
249
250
251
252
253
254
255
256
257
258
259
260
261
262
263
264
265
266
267
268
269
270
271
272
273
274
275
276
277
278
279
280
281
282
283
284
285
286
287
288
289
290
291
292
293
294
295
296
297
298
299
300
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
327
328
329
330
331
332
333
334
335
336
337
338
339
340
341
342
343
344
345
346
347
348
349
350
351
352
353
354
355
356
357
358
359
360
361
362
363
364
365
366
367
368
369
370
371
372
373
374
375
376
377
378
379
380
381
382
383
384
385
386
387
388
389
390
391
392
393
394
395
396
397
398
399
400
401
402
403
404
405
406
407
408
409
410
411
412
413
414
415
416
417
418
419
420
421
422
423
424
425
426
427
428
429
430
431
432
433
434
435
436
437
438
439
440
441
442
443
444
445
446
447
448
449
450
451
452
453
454
455
456
457
458
459
460
461
462
463
464
465
466
467
468
469
470
471
472
473
474
475
476
477
478
479
480
481
482
483
484
485
486
487
488
489
490
491
492
493
494
495
496
497
498
499
500
501
502
503
504
505
506
507
508
509
510
511
512
513
514
515
516
517
518
519
520
521
522
523
524
525
526
527
528
529
530
531
532
533
534
535
536
537
538
539
540
541
542
543
544
545
546
547
548
549
550
551
552
553
554
555
556
557
558
559
560
561
562
563
564
565
566
567
568
569
570
571
572
573
574
575
576
577
578
579
580
581
582
583
584
585
586
587
588
589
590
591
592
593
594
595
596
597
598
599
600
601
602
603
604
605
606
607
608
609
610
611
612
613
614
615
616
617
618
619
620
621
622
623
624
625
626
627
628
629
630
631
632
633
634
635
636
637
638
639
640
641
642
643
644
645
646
647
648
649
650
651
652
653
654
655
656
657
658
659
660
661
662
663
664
665
666
667
668
669
670
671
672
673
674
675
676
677
678
679
680
681
682
683
684
685
686
687
688
689
690
691
692
693
694
695
696
697
698
699
700
701
702
703
704
705
706
707
708
709
710
711
712
713
714
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736
737
738
739
740
741
742
743
744
745
746
747
748
749
750
751
752
753
754
755
756
757
758
759
760
761
762
763
764
765
766
767
768
769
770
771
772
773
774
775
776
777
778
779
780
781
782
783
784
785
786
787
788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809
810
811
812
813
814
815
816
817
818
819
820
821
822
823
824
825
826
827
828
829
830
831
832
833
834
835
836
837
838
839
840
84

2

3
4

5

6
78
9

11

12

13
14

14

10

11
13

1
2

2

2

2
2

2

22

1 28. An actual and justiciable controversy has thus arisen between DIABLO and
2 NETLIST concerning the validity of the '833 patent.

3
4 **COUNT V**

5 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,359,501)**

6 29. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
7 the preceding paragraphs of this Complaint.

8 30. Upon information and belief, NETLIST is the owner of the '434 patent.

9 31. NETLIST is asserting the '501 patent against "ULLtraDIMM".

10 32. DIABLO contends that it has not infringed and does not infringe, any valid and
11 enforceable claim of the '501 patent.

12 33. An actual and justiciable controversy has thus arisen between DIABLO and
13 NETLIST concerning the alleged infringement of the '501 patent.

14
15 **COUNT VI**

16 **(Declaratory Judgment of Invalidity of U.S. Patent No. 8,359,501)**

17 34. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
18 the preceding paragraphs of this Complaint.

19 35. Upon information and belief, NETLIST contends that the '501 patent is valid.

20 36. DIABLO contends that the claims of the '501 patent are invalid under 35 U.S.C.
21 §§ 101, 102, 103, and/or 112.

22 37. An actual and justiciable controversy has thus arisen between DIABLO and
23 NETLIST concerning the validity of the '501 patent.

24
25 **COUNT VII**

26 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,516,185)**

27 38. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
28 the preceding paragraphs of this Complaint.

1 39. Upon information and belief, NETLIST is the owner of the '185 patent.

2 40. NETLIST is asserting the '185 patent against "ULLtraDIMM".

3 41. DIABLO, contends that it has not infringed and does not infringe, any valid and
4 enforceable claim of the '185 patent.

5 42. An actual and justiciable controversy has thus arisen between DIABLO and
6 NETLIST concerning the alleged infringement of the '185 patent.

7

8

COUNT VIII

9 **(Declaratory Judgment of Invalidity of U.S. Patent No. 8,516,185)**

10 43. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
11 the preceding paragraphs of this Complaint.

12 44. Upon information and belief, NETLIST contends that the '185 patent is valid.

13 45. DIABLO contends that the claims of the '185 patent are invalid under 35 U.S.C.
14 §§ 101, 102, 103, and/or 112.

15 46. An actual and justiciable controversy has thus arisen between DIABLO and
16 NETLIST concerning the validity of the '185 patent.

17

18

COUNT IX

19 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,516,187)**

20 47. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
21 the preceding paragraphs of this Complaint.

22 48. Upon information and belief, NETLIST is the owner of the '187 patent.

23 49. NETLIST is asserting the '187 patent against "ULLtraDIMM".

24 50. DIABLO, contends that it has not infringed and does not infringe, any valid and
25 enforceable claim of the '187 patent.

26 51. An actual and justiciable controversy has thus arisen between DIABLO and
27 NETLIST concerning the alleged infringement of the '187 patent.

28

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12
- 13
- 14
- 15
- 16
- 17
- 18
- 19
- 20
- 21
- 22
- 23
- 24
- 25
- 26
- 27
- 28
- 29
- 30
- 31
- 32
- 33
- 34
- 35
- 36
- 37
- 38
- 39
- 40
- 41
- 42
- 43
- 44
- 45
- 46
- 47
- 48
- 49
- 50
- 51
- 52
- 53
- 54
- 55
- 56
- 57
- 58
- 59
- 60
- 61
- 62
- 63
- 64
- 65
- 66
- 67
- 68
- 69
- 70
- 71
- 72
- 73
- 74
- 75
- 76
- 77
- 78
- 79
- 80
- 81
- 82
- 83
- 84
- 85
- 86
- 87
- 88
- 89
- 90
- 91
- 92
- 93
- 94
- 95
- 96
- 97
- 98
- 99
- 100

2

3
4

5

6
7

8
9

11

12

14

15

16

17
18

19

20
2122
23

1 Respectfully submitted this 23rd day of August 2013.

2 DIABLO TECHNOLOGIES, INC.,

3 By: Sanjeet Dutta

4 WILLIAM F. ABRAMS

5 (CA Bar. No. 88805)

6 wabrams@kslaw.com

7 SANJEET K. DUTTA

8 (CA Bar No. 203463)

9 sdutta@kslaw.com

10 KING & SPALDING LLP

11 333 Twin Dolphin Drive, Suite 400

12 Redwood Shores, CA 94065

13 Telephone: (650) 590-0700

14 Facsimile: (650) 590-1900

15 Attorneys for Plaintiff

16 DIABLO TECHNOLOGIES, INC.

TO: Mail Stop 8
 Director of the U.S. Patent & Trademark Office
 P.O. Box 1450
 Alexandria, VA 22313-1450

**REPORT ON THE
 FILING OR DETERMINATION OF AN
 ACTION REGARDING A PATENT OR
 TRADEMARK**

In Compliance with 35 § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court Northern District of California on the following:
 (X) Patents or () Trademarks

DOCKET NO:
 13-cv-05889-YGR

DATE FILED:
 October 7, 2014

UNITED STATES DISTRICT COURT
 Ronald Dellums Federal Building
 1301 Clay Street
 Oakland, CA 94612

PLAINTIFF:
 Netlist, Inc

DEFENDANT:
 Smart Modular Technologies, Inc

PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1. <u>8,001,434</u>		See attached THIRD AMENDED COMPLAINT FOR PATENT INFRINGEMENT
2. <u>8,301,833</u>		
3. <u>8,359,501</u>		
4. <u>8,516,185</u>		
5. <u>8,516,187</u>		

In the above-entitled case, the following patent(s) have been included.

DATE INCLUDED INCLUDED BY:
 () Amendment () Answer () Cross Bill () Other Pleading

PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1. <u>7,881,150</u>		
2. <u>8,081,536</u>		
3.		
4.		
5.		

In the above-entitled case, the following decision has been rendered or judgment issued:

DECISION/JUDGEMENT:

Richard W. Wieking, Clerk


 (by) Deputy Clerk, Clara Pierce

Copy 1 – Upon initiation of action, mail this copy to Commissioner
 Copy 2 – Upon filing document adding patent(s) mail this copy to Commissioner
 Copy 3 – Upon termination of action, mail this copy to the Commissioner
 Copy 4 – Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SANDISK CORPORATION,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-00994
Patent 8,301,833 B1

Before: LINDA M. GAUDETTE, BRYAN F. MOORE, and
GEORGIANNA W. BRADEN, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

DECISION

Denying Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

Sandisk Corporation, Inc. ("Petitioner") filed a Petition, on June 20, 2014, requesting an *inter partes* review of claims 1–30 of US Patent No. 8,301,833 B1 (Ex. 1001, "the '833 patent"). Paper 1 ("Pet."). Netlist, Inc.

(“Patent Owner”) filed a Preliminary Response on October 2, 2014. Paper 7 (“Prelim. Resp.”).

We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may be authorized only if “the information presented in the petition . . . and any [preliminary] response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a). Pursuant to 35 U.S.C. § 314, the Board does not find a reasonable likelihood that Petitioner would prevail with respect to at least one claim of the ’833 patent and, thus, does not authorize an *inter partes* review to be instituted as to those claims.

A. Related Proceedings

Petitioner indicates that the ’833 patent is involved in the following co-pending actions: *Netlist, Inc. v. Smart Modular Technologies, Inc.*, U.S. District Court for the Northern District of California, Civil Action No. 3:13-CV-05889-YGR; *Diablo Technologies, Inc. v. Netlist, Inc.*, U.S. District Court for the Northern District of California, Civil Action No. 4:13-CV-03901-YGR; and *Smart Modular Technologies, Inc. v. Netlist, Inc.*, U.S. District Court for the Northern District of California, Civil Action No. 4:13-CV-03916-YGR.). Pet. 59.

B. The ’833 Patent

The invention in the ’833 patent relates to a specific configuration of hybrid memory systems that addresses non-volatile memory backup, while running the volatile memory subsystem at lower power, and therefore, at lower clock speeds. Ex. 1001, col. 16, ll. 29–34. Specifically, the alleged invention of the ’833 patent includes circuitry for providing a regular high-

speed clock frequency (first clock frequency) during communications between the host and the volatile memory subsystem, and a slower clock frequency during communications between the volatile memory subsystem (using third clock frequency) and the non-volatile memory subsystem (using second clock frequency). *Id.* at col. 21, ll. 5–21. Further, the second and third clock frequencies may be substantially equal. *Id.* at col. 21, ll. 23–24.

C. Illustrative Claim

Of the challenged claims, 1 and 5 are independent claims. Claim 1 is illustrative of the claimed subject matter of the '833 patent, and is reproduced below:

1. A method for controlling a memory system operatively coupled to a host system, the memory system including a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:
 - operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;
 - operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem; and
 - operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

1. Prior Art Relied Upon

Petitioner relies upon the following prior art references:

Fukuzo ("Fukuzo," Ex. 1013)	US 2006/0294295 A1	June 24, 2005
Panabaker ("Panabaker," Ex. 1014)	US 7,716,411 B2	June 7, 2006
Li ("Li," Ex. 1015)	US 6,336,174 B1	August 9, 1999
Spiers ("Spiers," Ex. 1016)	US 2006/0080515 A1	October 12, 2004
Hansen ("Hansen," Ex. 1017)	US 2005/0132250 A1	December 16, 2003
Sun ("Sun," Ex. 1018)	US 7,102,391 B1	July 29, 2004
Komatsuzaki ("Komatsuzaki," Ex. 1019)	US 6,944,042 B2	December 31, 2002

2. The Asserted Grounds

Petitioner asserts that the challenged claims are unpatentable based on the following grounds:

Reference[s]	Basis	Claims challenged
Fukuzo	§ 102	1, 2, 6, 8, 11, 12, 15, 18, 22, 24, 27, and 28
Panabaker	§ 102	1-6, 8, 11-13, 15, 17-22, 24, and 27-29
Fukuzo and Li	§ 103	3 and 19
Fukuzo, Li, and Spiers	§ 103	3 and 19
Fukuzo and Hansen	§ 103	7 and 23
Fukuzo, Li, and Hansen	§ 103	7, 9, 10, 23, 25, and 26
Fukuzo and Sun	§ 103	14 and 30
Fukuzo, Li,	§ 103	14 and 30

Reference[s]	Basis	Claims challenged
and Sun		
Fukuzo and Komatsuzaki	§ 103	16
Fukuzo, Li, and Komatsuzaki	§ 103	16
Panabaker and Li	§ 103	1–6, 8, 11, 12, 15, 17–22, 24, 27, and 28
Panabaker and Spiers	§ 103	3 and 19
Panabaker, Li, and Spiers	§ 103	3 and 19
Panabaker and Hansen	§ 103	7, 9, 23, and 25
Panabaker, Li, and Hansen	§ 103	7, 9, 10, 23, 25, and 26
Panabaker and Fukuzo	§ 103	13 and 29
Panabaker, Li, and Fukuzo	§ 103	13 and 29
Panabaker and Sun	§ 103	14 and 30
Panabaker, Li, and Sun	§ 103	14 and 30
Panabaker and Komatsuzaki	§ 103	16
Panabaker, Li, and Komatsuzaki	§ 103	16

II. ANALYSIS

A. Claim Construction

In an *inter partes* review, claim terms in an unexpired patent are given their broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b). Under the broadest reasonable construction standard, claim terms are given their ordinary and customary meaning, as would be understood by one of ordinary skill in the art in the context of the entire disclosure. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Any special definition for a claim term must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

Petitioner does not present any claim constructions, rather, Petitioner states that “each claim should be construed in accordance with its plain and ordinary meaning under the required broadest reasonable interpretation.” Pet. 9–10. Patent Owner provides a construction for one term only—“clock frequency.” Patent Owner argues “the claimed ‘clock frequency’ should be construed literally as clock frequency.” Prelim Resp. 18. Because the claims refer specifically to “clock” frequency, on this record, and for purposes of this Decision, we determine that the broadest reasonable interpretation of “clock frequency” requires identification of a clock running at a particular frequency.

B. Claims 1, 2, 6, 8, 11, 12, 15, 18, 22, 24, 27, and 28—Anticipated by Fukuzo (Ex. 1013)

Petitioner argues that claims 1, 2, 6, 8, 11, 12, 15, 18, 22, 24, 27, and 28 are anticipated by Fukuzo under 35 U.S.C. § 102(a) and (e). Pet. 9–17. Fukuzo discloses an SDRAM memory chip device that comprises a non-

volatile memory controller operating a nonvolatile memory and a FIFO memory. Ex. 1013, Abstract, ¶ 27. Fukuzo's SDRAM memory chip device is used to store data to its internal SDRAM memory array (volatile memory) and to external FLASH (nonvolatile memory) using at least two additional pins as compared with conventional SDRAM standard. *Id.* Two further pins reflecting the flash memory status provide appropriate issuance of load or store signals by the host. *Id.* Fukuzo teaches using foreground and background operations such that read/write operations to volatile memory can occur simultaneously with read/write operations to non-volatile memory.

Id.

Below we discuss independent claims 1 and 15, from which all other dependent claims challenged in this ground depend. Claim 1 recites "operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency." Claim 15 recites "the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency." Figure 3 of Fukuzo is

reproduced below.

FIG 3

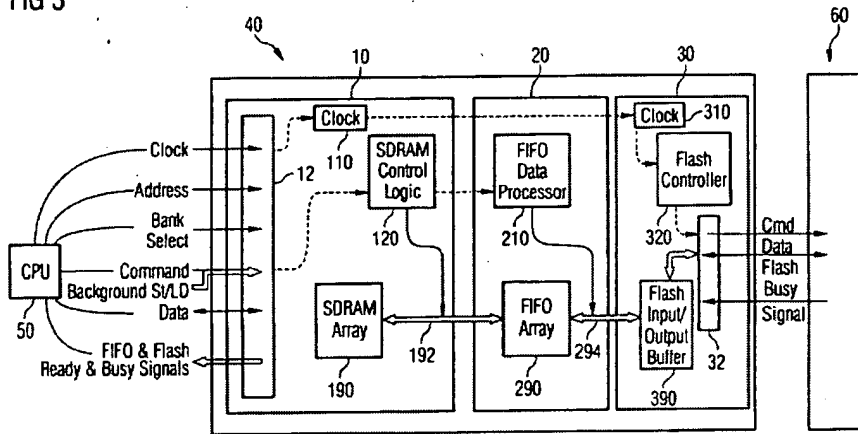


Figure 3, above, depicts a schematic block diagram of a memory chip device according to the invention of the Fukuzo device.

Petitioner asserts that Fukuzo's disclosure of SDRAM memory chip device 40 ("SDRAM Chip 40") is the claimed volatile memory subsystem and Fukuzo's disclosure of clock generator 110, which operates at 110 MHz and is valid for blocks 10 and 20 of the SDRAM chip 40, is the claimed first clock frequency. Pet. 14 (citing Ex. 1013 ¶¶ 84, 87-8). Further, Petitioner asserts that Fukuzo's disclosure of flash clock 310 which drives flash controller 320 and flash input/output buffer 390, which is described as operating at 20 MHz and is valid for block 30 of SDRAM chip 40, meets the limitation to a third clock frequency. Pet. 14 (citing Ex. 1013 ¶¶ 84, 87-88). We are not persuaded by Petitioner's argument or cited disclosure.

Petitioner reads the entire SDRAM chip 40 on the volatile memory subsystem. However, SDRAM chip 40 has at least two clocks running at different frequencies. Petitioner does not sufficiently explain how the chip reads on operating at a first frequency in one mode and a second frequency

in a second mode when the chip appears to be running both frequencies all the time or at least at the first frequency all the time. *See* Prelim. Resp. 21. It appears what Petitioner is actually doing is using blocks 10 and 20 to be the volatile memory subsystem at one time and block 30 to be the volatile memory subsystem at other times.

Anticipation requires the presence in a single prior art reference disclosure of each and every element of the claimed invention, arranged as in the claim. *Lindemann Maschinenfabrik GmbH v. American Hoist & Derrick Co.*, 730 F.2d 1452, 1458 (Fed. Cir. 1984). Nonetheless, Petitioner reads inconsistently the claim limitation on two distinct structures, SDRAM core section 10 and flash controller section 30 of the SDRAM chip 40, described in the reference as being driven by two different clocks.

Petitioner presents no persuasive or credible explanation or disclosure to support this way of reading Fukuzo on the claim language. Alternately, if the Petitioner reads the volatile memory subsystem to be chip 40 as a whole, then it does not appear the chip “operates” at the frequency of flash clock 310, but rather at the frequency of the system clock 110. *See* Prelim. Resp. 21. Nonetheless, Petitioner does not explain its reading of two different clocks which drive different sets of components as corresponding to the clock for the single volatile memory subsystem of the claim.

The claim language must be read in light of the specification as it would be interpreted by one of ordinary skill in the art. *In re Sneed*, 710 F.2d 1544, 1548, (Fed. Cir. 1983). While it would be unreasonable to ignore any interpretive guidance afforded by the specification, *see In re Morris*, 127 F.3d 1048, 1054–55, (Fed. Cir. 1997), it is improper to read limitations from the specification into the claims. *Id.* For example, the specification of the

'833 patent discloses only one bank of volatile memory devices 32. Ex. 1001, Fig. 1. The Specification of the '833 patent discloses only one clock input into the volatile memory subsystem and does not suggest that there are subsections of the volatile memory subsystem that are operating at different clock frequencies. *Id.* at col. 15, l. 41–col. 17, l. 38.

Additionally, the specification suggests “[r]unning the volatile memory subsystem 30 at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system 10.” *Id.* at col. 16, ll. 34–37. Petitioner’s reading of two separate sections of memory on the volatile memory subsystem is not consistent with the goal of reducing overall power consumption because the memory that Petitioner points to as running at a reduced frequency (flash/input output buffer 390) is not the same memory that is accessed by the host system (SDRAM array 190). Thus, on the record before us, Petitioner has not shown sufficiently that Fukuzo discloses the limitations to “operating the volatile memory subsystem at a third clock frequency . . . the third clock frequency being less than the first clock frequency” and “the volatile memory subsystem further being operable at a third clock frequency . . . the third clock frequency being less than the clock first frequency,” as recited in independent claims 1 and 15.

Thus, upon review of Petitioner’s analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood that it would prevail with respect to claims 1 and 15, or claims 2, 6, 8, 11, 12, 18, 22, 24, 27, and 28 that depend ultimately from claims 1 and 15, on the ground that these claims are anticipated by Fukuzo.

C. Claims 1–6, 8, 11–13, 15, 17–22, 24, and 27–29 — Anticipated by Panabaker (Ex. 1014)

Petitioner argues that claims 1–6, 8, 11–13, 15, 17–22, 24, and 27–29 are anticipated by Panabaker under 35 U.S.C. § 102(e). Pet. 9–17.

Panabaker discloses a hybrid memory device comprising a memory controller, a volatile memory (e.g. SDRAM) and nonvolatile memory (e.g. flash memory). *See e.g.*, Ex. 1014, Abstract, Fig. 3A, col. 7, ll. 8–11.

Panabaker’s memory controller operates such that the memory device has only a single memory interface with respect to voltage and access protocols defined for one type of memory. *Id.* Panabaker teaches that the memory controller allows access to both SDRAM (volatile memory) and non-volatile memory, with the non-volatile memory overlaid in one or more designated blocks of the volatile memory address space (or vice-versa). *Id.* In Panabaker, a host is coupled to the controller of the hybrid memory device, where the controller either allows a direct access to the volatile memory or uses a buffer for speed matching between the host and the nonvolatile memory. *Id.*

Below we discuss independent claims 1 and 15, from which all other dependent claims challenged in this ground depend. Claim 1 recites “operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.” Claim 15 recites “the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.”

Petitioner asserts that Panabaker’s disclosure of SDRAM buffer set 210 as the volatile memory subsystem “is at least as fast as a given SDRAM

chip and its protocol requires . . . “ meets the limitation to a first clock frequency. Pet. 19 (citing Ex. 1014, col. 4, ll. 33–47). Further, Petitioner asserts that Panabaker’s disclosure of speed matching buffer set as volatile memory subsystem meets the limitation to a first clock frequency. Pet. 20 (citing Ex. 1013 ¶¶ 84, 87–88). We are not persuaded by this cited disclosure.

Petitioner apparently reads the controller 308A and the buffer 310 and the SDRAM 304 on the volatile memory subsystem in the first mode. However, in the second mode, Petitioner is unclear whether it reads the volatile memory subsystem on the controller 308A or the speed matching buffer 310. Reading the controller as the volatile memory subsystem is inconsistent with specification which discloses a controller 62 that is separate from the volatile and non-volatile memory subsystems. Ex. 1001, Fig. 1. The controller 62 provides the clock signal to the volatile and nonvolatile memory subsystems. Ex. 1001, Fig. 6. As Patent Owner points out, Petitioner does not sufficiently explain what it considers the volatile memory subsystem and does not point to any clock which provides a first or third frequency to that subsystem. *See* Prelim. Resp. 28–29.

Petitioner does not specify that Panabaker teaches that the speed matching buffer operates at a third clock frequency. For example, Panabaker teaches that a firmware protocol is used for speed matching. *See e.g.*, Ex. 1014, 6:32–36 (“As can be readily appreciated, because SDRAM is presently one or more orders of magnitude faster than flash, the protocol includes a way for the controller 308A to signal to the firmware 330 when a flash read or write request is busy and when the request is ready.”). Petitioner does not explain how this scheme amounts to operating the speed

matching buffer at a third clock frequency. We find that one of ordinary skill at the time of the invention reasonably would not consider Panabaker as disclosing a volatile memory subsystem that operates at the first and third clock frequencies. Thus, on the record before us, Petitioner has not shown sufficiently that Panabaker discloses these limitations.

Therefore, upon review of Petitioner's analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood that it would prevail with respect to claims 1 and 15, and claims 2–6, 8, 11–13, 17–22, 24, and 27–29 that depend ultimately from claims 1 and 15, on the ground that these claims are anticipated by Panabaker.

D. Obviousness Grounds

A patent claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) where in evidence, so-called secondary considerations. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

Based on the arguments and evidence presented by Petitioner, we are unpersuaded that it has established a reasonable likelihood that it would prevail in showing the unpatentability of any of the challenged claim under

35 U.S.C. § 103 as obvious over a combination relying on either Fukuzo or Panabaker.

As discussed above in the context of the anticipation analysis with regard to Fukuzo and Panabaker, Petitioner has not shown sufficiently that Fukuzo or Panabaker discloses a volatile memory subsystem that operates at a third frequency, as is required by each of independent claims 1 and 15. Additionally, Petitioner does not provide any further explanation why Fukuzo or Panabaker disclose a volatile memory subsystem that operates at a third frequency under an obviousness analysis. Each of Petitioner's obviousness grounds assumes that either Fukuzo or Panabaker discloses that limitation. Pet. 7–9. Petitioner bears the burden of proof of showing that limitation for each independent claim. Nonetheless, as shown above, none of the challenges relied on to meet claims 1 and 15 has shown sufficiently the required “third frequency.”

Thus, upon review of Petitioner's analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood that it would prevail with respect to: the ground that claims 3 and 19 are unpatentable over Fukuzo and Li; the ground that claims 3 and 19 are unpatentable over Fukuzo, Li, and Spiers; the ground that claims 7 and 23 are unpatentable over Fukuzo and Hansen; the ground that claims 7, 9, 10, 23, 25, and 26 are unpatentable over Fukuzo, Li, and Hansen; the ground that claims 14 and 30 are unpatentable over Fukuzo and Sun; the ground that claims 14 and 30 are unpatentable over Fukuzo, Li, and Sun; the ground that claim 16 is unpatentable over Fukuzo and Komatsuzaki; the ground that claim 16 is unpatentable over Fukuzo, Li, and Komatsuzaki; the ground that claims 1–6, 8, 11, 12, 15, 17–22, 24, 27, and 28 are unpatentable over

Panabaker and Li; the ground that claims 3 and 19 are unpatentable over Panabaker and Spiers; the ground that claims 3 and 19 are unpatentable over Panabaker, Li, and Spiers; the ground that claims 7, 9, 23, and 25 are unpatentable over Panabaker and Hansen; the ground that claims 7, 9, 10, 23, 25, and 26 are unpatentable over Panabaker, Li, and Hansen; the ground that claims 13 and 29 are unpatentable over Panabaker and Fukuzo; the ground that claims 13 and 29 are unpatentable over Panabaker, Li, and Fukuzo; the ground that claims 14 and 30 are unpatentable over Panabaker and Sun; the ground that claims 14 and 30 are unpatentable over Panabaker, Li, and Sun; the ground that claim 16 is unpatentable over Panabaker and Komatsuzaki; the ground that claim 16 is unpatentable over Panabaker, Li, and Komatsuzaki.

III. CONCLUSION

The information presented does not show that there is a reasonable likelihood that Petitioner would prevail at trial with respect to at least one claim of the '833 patent, based on any ground presented in the petition. On this record, we deny the petition for *inter partes* review of claims 1–30.

IV. ORDER

Accordingly, it is

ORDERED that that the petition is *denied* as to all challenged claims, and no trial is instituted.

IPR2014-00994
Patent 8,301,833 B1

PETITIONER:

David B. Cochran
dcochran@jonesday.com

Joseph M. Sauer
jmsauer@jonesday.com

Joshua R. Nightingale
jrnightingale@jonesday.com

PATENT OWNER:

Thomas J. Wimbiscus
twimbiscus@mcandrews-ip.com

Gregory C. Schodde
gschodde@mcandrews-ip.com

Scott P. McBride
smcbride@mcandrews-ip.com

Ronald H. Spuhler
rspuhler@mcandrews-ip.com

Wayne Bradley
wbradley@mcandrews-ip.com

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01370
Patent 8,301,833 B1

Before: LINDA M. GAUDETTE, BRYAN F. MOORE, and
GEORGIANNA W. BRADEN, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

DECISION

Denying Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

Smart Modular Technologies Inc. (“Petitioner”) filed a Corrected Petition requesting an *inter partes* review of claims 1–30 of US Patent No. 8,301,833 B1 (Ex. 1009, “the ’833 patent”). Paper 8 (“Pet.”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response. Paper 11 (“Prelim. Resp.”).

We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may be authorized only if “the information presented in the petition . . . and any [preliminary] response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a). Pursuant to 35 U.S.C. § 314, the Board does not find a reasonable likelihood that Petitioner would prevail with respect to at least one claim of the ’833 patent and, thus, does not authorize an *inter partes* review to be instituted as to those claims.

A. Related Proceedings

Petitioner recites the District Court proceedings related to this *inter partes* review. Pet. 2–3. This *inter partes* review challenges the same patent at issue in the decision entered in IPR2014-00994 in which we denied institution. IPR2014-00994 (Paper 8).

B. The ’833 Patent

The invention in the ’833 patent relates to a specific configuration of hybrid memory systems that addresses non-volatile memory backup while running the volatile memory subsystem at lower power, and, therefore, at lower clock speeds. Ex. 1001, col. 16, ll. 29–34. Specifically, the alleged invention of the ’833 patent includes circuitry for providing a regular high-speed clock frequency (first clock frequency) during communications between the host and the volatile memory subsystem, and a slower clock frequency during communications between the volatile memory subsystem (using a third clock frequency) and the non-volatile memory subsystem (using a second clock frequency). *Id.* at col. 21, ll. 5–21. Further, the second and third clock frequencies may be substantially equal. *Id.* at col. 21,

ll. 23–24.

C. Illustrative Claim

Of the challenged claims, 1 and 5 are independent claims. Claim 1 is illustrative of the claimed subject matter of the '833 patent, and is reproduced below:

1. A method for controlling a memory system operatively coupled to a host system, the memory system including a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem; and

operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

D. Prior Art Relied Upon

Petitioner relies upon the following prior art references:

Reference	Patent Number	Exhibit Number
Fukuzo '295Pub	US 2006/0294295 A1	Ex. 1012
Leete '210Pub	US 2004/0190210 A1	Ex. 1013
Ichikawa '142	US 7,600,142 B2	Ex. 1014
Long '552	US 7,421,552 B2	Ex. 1015
Tsunoda '618	US 7,062,618 B2	Ex. 1016

E. The Asserted Grounds

Petitioner asserts that the challenged claims are unpatentable based on the following grounds:

Reference[s]	Basis	Claims Challenged
Fukuzo '295Pub	§ 102	1, 2, 4, 6–13, 15, 16, 18, 20, and 22–29 ¹
Fukuzo '295Pub and Leete '210Pub	§ 103	3, 5, 14, 17, 19, 21 and 30
Ichikawa '142	§ 102	1, 2, 7, 8, 11–13, 15, 18, 23, 24 and 27–29
Ichikawa '142 and Leete '210Pub	§ 103	3–6, 9, 10, 14, 16, 17, 19–22, 25, 26, and 30
Long '552	§ 102	1, 2, 4, 5, 7, 12, 13, 15, 18, 20, 21, 23, 28, and 29 ²

¹ We note the challenged claims are listed at page 6 of the Petition. Also, analysis is provided starting at page 26 of the Petition. Although Claim 16 is not listed, analysis provided at page 29 of the Petition.

² We note the Petition isn't consistent. The challenged claims are listed as 1, 2, 4, 5, 12, 13, 15, 18, 20, 21, 28, and 29. Pet. 7. Nonetheless, analysis

Reference[s]	Basis	Claims Challenged
Long '552 and Leete '210Pub	§ 103	3, 6–11, 14, 16, 17, 19, 22–27, and 30
Tsunoda '618	§ 102	1, 2, 4, 5, 12, 13, 15, 16, 18, 20, 21, 28, and 29
Tsunoda '618 and Leete '210Pub	§ 103	3, 6–11, 14, 16, 17, 19, 22–27, and 30

II. ANALYSIS

A. Claim Construction

In an *inter partes* review, claim terms in an unexpired patent are given their broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b). Under the broadest reasonable construction standard, claim terms are given their ordinary and customary meaning, as would be understood by one of ordinary skill in the art in the context of the entire disclosure. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Any special definition for a claim term must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

Petitioner and Patent Owner propose constructions for several terms. Pet. 19–24; Prelim Resp. 15–21. We determine that none of the terms cited by the parties require explicit construction for the purpose of this Decision.

provided for claims 1, 2, 4, 5, 7, 12, 13, 15, 18, 20, 21, 23, 28, and 29. *See* Pet. 43–44.

B. Claims 1, 2, 4, 6–13, 15, 16, 18, 20, and 22–29—Anticipation by Fukuzo '295Pub (Ex. 1013)

Petitioner argues that claims 1, 2, 4, 6–13, 15, 16, 18, 20, and 22–29 are anticipated by Fukuzo '295Pub under 35 U.S.C. § 102(b). Pet. 24–30. Fukuzo '295Pub discloses an SDRAM memory chip device that comprises a non-volatile memory controller operating a nonvolatile memory and a FIFO memory. Ex. 1013, Abstract, ¶ 27. Fukuzo '295Pub's SDRAM memory chip device is used to store data to its internal SDRAM memory array (volatile memory) and to external FLASH (nonvolatile memory) using at least two additional pins as compared with conventional SDRAM standard. *Id.* Two further pins reflecting the flash memory status provide appropriate issuance of load or store signals by the host. *Id.* Fukuzo '295Pub teaches using foreground and background operations such that read/write operations to volatile memory can occur simultaneously with read/write operations to non-volatile memory. *Id.*

Below we discuss independent claims 1 and 15, from which all other dependent claims challenged in this ground depend. Claim 1 recites “operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.” Claim 15 recites “the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.” Figure 3 of Fukuzo

'295Pub is reproduced below.

FIG 3

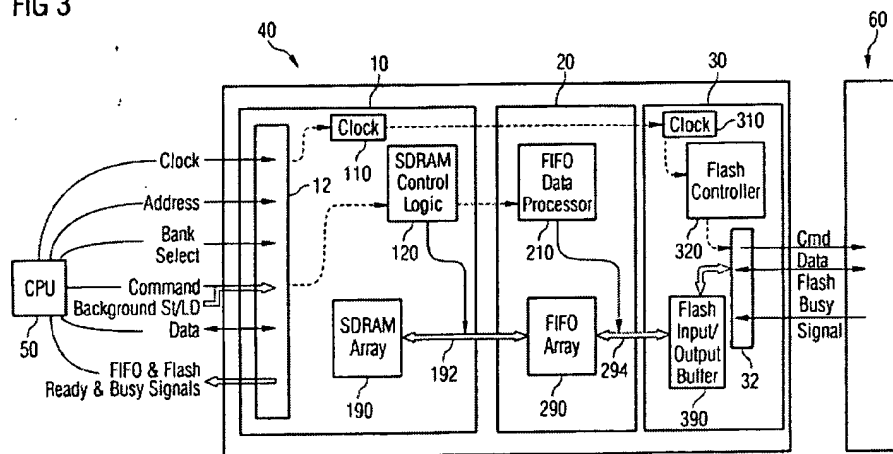


Figure 3, above, depicts a schematic block diagram of a memory chip device according to the invention of the Fukuzo '295Pub device.

Petitioner asserts that Fukuzo '295Pub's disclosure of FIFO Buffer 20 is the claimed volatile memory subsystem and Fukuzo '295Pub's disclosure that FIFO Buffer 20 operates at 130 MHz is the claimed first clock frequency. Pet. 25 (citing Ex. 1013 ¶¶ 84, 127). Further, Petitioner asserts that one of ordinary skill in the art would understand that FIFO Buffer 20 receives the same clock as the flash memory system (clocked at 20Mz) though FIFO timing generator 211, thus FIFO Buffer 20 meets the limitation to a third clock frequency. Pet. 14 (citing Ex. 1013 ¶¶ 84, 87–88). Petitioner's expert repeats this assertion without further explanation. Ex. 1010 ¶ 40. We are not persuaded by Petitioner's argument or cited disclosure.

Petitioner states, without explanation, that "the FIFO SDRAM buffer receives the same clock of the flash memory system, via the FIFO timing generator." Pet. 25. However, the cited disclosure, other than for a line in

Figure 3 between FIFO Buffer 20 and FIFO timing generator 211, does not disclose explicitly that the FIFO timing generator 211 provides the flash clock signal to FIFO Buffer 20. At best, the cited disclosure states that there is “data transfer between the SDRAM FIFO memory array 290 and the flash memory input/output buffer 390 (second data transfer bus 294).” Ex. 1001 ¶ 103. Petitioner’s declarant, Dr. Nader Baghezadeh, does not provide any further explanation. Ex. 1010 ¶ 40. Petitioner does not explain sufficiently how or why FIFO timing generator 211 provides a clock signal that runs FIFO Buffer 20 at 20 MHz. *See* Prelim. Resp. 27.

Anticipation requires the disclosure in a single prior art reference of each and every element of the claimed invention, arranged as in the claim. *Lindemann Maschinenfabrik GmbH v. American Hoist & Derrick Co.*, 730 F.2d 1452, 1458 (Fed. Cir. 1984). We are not persuaded Petitioner has shown sufficiently that Fukuzo ’295Pub discloses the limitations to “operating the volatile memory subsystem at a third clock frequency . . . the third clock frequency being less than the first clock frequency” and “the volatile memory subsystem further being operable at a third clock frequency . . . the third clock frequency being less than the clock first frequency,” as recited in independent claims 1 and 15.

Thus, upon review of Petitioner’s analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood it would prevail with respect to claims 1 and 15, or claims 2, 4, 6–13, 16, 18, 20, and 22–29 that depend ultimately from claims 1 and 15, on the ground that these claims are anticipated by Fukuzo ’295Pub.

C. Claims 3, 5, 14, 17, 19, 21 and 30—Obviousness over Fukuzo '295Pub and Leete '210Pub (Ex.1013)

Petitioner argues that claims 3, 5, 14, 17, 19, 21 and 30 would have been obvious over Fukuzo '295Pub and Leete '210Pub under 35 U.S.C. § 103(a). Pet. 30–32. Claims 3, 5, 14, 17, 19, 21 and 30 depend from independent claims 1 and 15. As discussed above, Fukuzo '295Pub fails to teach or suggest all of the elements of independent claims 1 and 15. Petitioner does not assert that Leete '210Pub overcomes the aforementioned deficiency in Fukuzo '295Pub. Thus, upon review of Petitioner's analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood it would prevail with respect to claims 3, 5, 14, 17, 19, 21 and 30, on the ground that these claims would have been obvious over Fukuzo '295Pub and Leete '210Pub.

D. Claims 1, 2, 7, 8, 11–13, 15, 18, 23, 24 and 27–29— Anticipated by Ichikawa '142 (Ex. 1014)

Petitioner argues that claims 1, 2, 7, 8, 11–13, 15, 18, 23, 24 and 27–29 are anticipated by Ichikawa '142 under 35 U.S.C. § 102(b). Pet. 32–35. Ichikawa '142 discloses a system including volatile memory RAM 13 and nonvolatile memory system combining serial input-output (SIO) 14 with external flash memory 30. Ex. 1014, 1:12–16, 1:23–25, 1:27–28, Fig 2. Below we discuss independent claims 1 and 15, from which all other dependent claims challenged in this ground depend.

Claim 1 recites “operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.” Claim 15 recites “the volatile memory subsystem further being

operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.”

Petitioner asserts that Ichikawa '142's disclosure of RAM 13 as the volatile memory subsystem discloses a first clock frequency because Ichikawa '142 “discloses that RAM 13 is clocked at 5 MHz during read and write operations with the CPU 11” and Ishikawa's disclosure of RAM 13 as the volatile memory subsystem discloses a second clock frequency because Ichikawa '142 “discloses that RAM [13] functions at the 1 MHz clock during operations with SIO and external flash memory.” Pet. 33 (citing Ex. 1014, 4:29–41, 55–61, 2:38–40, 3:50–56). The cited portions of Ichikawa '142 recite: “the 5-MHZ clock signal CKH is supplied to the CPU 11” (Ex. 1014, 4:37–38); “[t]he CPU 11 carries out prescribed computation and control processing . . . in synchronization with a clock signal CLK” (*id.* at 3:50–53); “[n]ormally, the central processing unit operates on a first clock signal” (*id.* at 2:38–39); and

Operating on the 1-MHZ clock signal, the CPU 11 reads one byte of the data stored in the RAM 13 (step 44), supplies the data to the serial input-output interface 14 (step 45), and gives a serial transfer command (step 46). Also operating on the 1-MHZ clock signal, the serial input-output interface 14 converts the data received from the CPU 11 to serial data and transfers the data to the external memory device 30 (step 47).

id. at 4:55–61. We are not persuaded by this cited disclosure.

The cited disclosure states that CPU 11 is operating on either a 5 MHZ clock signal or a 1 MHz clock signal. There is an underlying assumption in Petitioner's argument that RAM 13 is clocked at the same speed as the CPU. Petitioner does not provide sufficient support for this

assumption. *See* Prelim. Resp. 44–45. Petitioner relies on its declarant, Dr. Bagherzadeh, for the proposition that “a POSITA would understand that the external flash memory and the RAM are communicating at the same frequency of 1 MHz to be able to transfer data between the components,” (Pet. 33–34 (citing Ex. 1010 (Bagherzadeh Decl.) ¶¶ 86, 87)), but Dr. Bagherzadeh provides no evidentiary support for this conclusory statement. We note that the ’833 patent shows explicitly that controller 62 provides a specific clock signal to the volatile and nonvolatile memory subsystems. Ex. 1001, 17:1–14, Fig. 7, 8. We are not persuaded Petitioner has shown sufficiently that Ichikawa ’142 discloses these limitations.

Therefore, upon review of Petitioner’s analysis and supporting evidence, we determine Petitioner has not demonstrated that there is a reasonable likelihood it would prevail with respect to claims 1 and 15, and claims 2, 7, 8, 11–13, 18, 23, 24 and 27–29 that depend ultimately from claims 1 and 15, on the ground that these claims are anticipated by Ichikawa ’142.

E. Claims 3–6, 9, 10, 14, 16, 17, 19–22, 25, 26, and 30—Obviousness over Ichikawa ’142 and Leete ’210Pub

Petitioner argues that claims 3–6, 9, 10, 14, 16, 17, 19–22, 25, 26, and 30 would have been obvious over Ichikawa ’142 and Leete ’210Pub under 35 U.S.C. § 103(a). Pet. 35–41. Claims 3–6, 9, 10, 14, 16, 17, 19–22, 25, 26, and 30 depend from independent claims 1 and 15. As discussed above, Ichikawa ’142 fails to teach or suggest all of the elements of independent claims 1 and 15. Petitioner does not assert that Leete ’210Pub overcomes the aforementioned deficiency in Ichikawa ’142. Thus, upon review of Petitioner’s analysis and supporting evidence, we determine that Petitioner has not demonstrated there is a reasonable likelihood it would prevail with

respect to claims 3–6, 9, 10, 14, 16, 17, 19–22, 25, 26, and 30, on the ground that these claims would have been obvious over Ichikawa '142 and Leete '210Pub.

*F. Claims 1, 2, 4, 5, 7, 12, 13, 15, 18, 20, 21, 23, 28, and 29—
Anticipation by Long '552 (Ex. 1015)*

Petitioner argues that claims 1, 2, 4, 5, 7, 12, 13, 15, 18, 20, 21, 23, 28, and 29 are anticipated by Long '552 under 35 U.S.C. § 102(b). Pet. 41–44. Long '552 discloses a system including volatile memory cache 42 and nonvolatile memory vault 44. Ex. 1015, 3:47–50. Further, Long '552 discloses that the device is configured to operate at a fast clock speed, supplied with 50 to 100 watts of power when communicating normally with the host. *Id.*, 4:46–53 and 3:50–59). Long '552 discloses that the device is configured to operate at a significantly slower clock speed, supplied with 30 watts of power when communicating with the non-volatile memory 44. *Id.* 4:54–61.

Below we discuss independent claims 1 and 15, from which all other dependent claims challenged in this ground depend. Claim 1 recites “operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.” Claim 15 recites “the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.”

Petitioner asserts that Long '552's disclosure of controller 40³ as the volatile memory subsystem meets the limitation to a first clock frequency

³ We note that the Petition says “RAM 13,” but that's apparently a

because Long '552 discloses that the processing circuitry of controller 40 is clocked at a relatively fast clock signal supported by 50-100 watts of power during read and write operations with the host. Pet. 42. (citing Ex. 1015, 4:46–53; 3:50–59). The cited portion of Long '552 recites:

The clock generator circuit 46 is configured to provide a relatively-fast clock signal (or multiple clock signals) to the processing circuitry of the controller 40 during normal operation when the controller 40 is performing data storage operations on behalf of the set of hosts 22.

Ex. 1014, 4:46–51.

Petitioner asserts that Long '552 teaches a third clock frequency because “the flash memory vault is clocked at a significantly slower clock supported by less than 30 watts of power, which is applicable during read and write operations with the storage cache 42” and “one of ordinary skill would understand that since the available power is significantly less than during the host communication processes, the speed is significantly less for all the components in operation.” Pet. 42. (citing Ex. 1015, 4:54–61; Ex. 1010 ¶¶ 134–135). We are not persuaded by this cited disclosure.

The citation provided does not state that storage cache 42 or flash-based memory vault 44 is clocked at a relatively fast clock signal. Rather, the cited disclosure notes that the processing circuitry of the controller is running at a relatively high clock speed and then is run on a significantly slower clock speed at a later time. This disclosure is silent as to whether

typographical error because Long doesn't teach a RAM 13, and the cited section is about clock generator circuit 46 driving circuitry of controller 40 at “a relatively-fast clock signal (or multiple clock signals).”

storage cache 42 or flash-based memory vault 44 is running at the relatively high clock speed along with the controller. *See* Prelim. Resp. 48–49. Thus, it cannot be determined whether storage cache 42 or flash-based memory vault 44 is forced to run at a significantly lower clock speed when the available power is reduced. Petitioner does not provide objective evidence sufficient to support a finding that storage cache 42 must be necessarily clocked at the same speed as the controller. Additionally, the conclusory statements of Petitioner’s declarant that a “person of ordinary skill in the art would know from education and experience that the speed is proportional to the maximum power available to the components” and “since the available power is significantly less than during the host communication processes, the speed is significantly less for all the components in operation,” do not explain adequately how the this claim limitation is met. Ex. 1010 ¶ 134; *see also, e.g., In re Am. Acad. of Sci. Tech. Ctr.*, 367 F.3d 1359, 1368 (Fed. Cir. 2004) (“[T]he Board is entitled to weigh the declarations and conclude that the lack of factual corroboration warrants discounting the opinions expressed in the declarations.” (citations omitted)); *Velandar v. Garner*, 348 F.3d 1359, 1371 (Fed. Cir. 2003) (“In giving more weight to prior publications than to subsequent conclusory statements by experts, the Board acted well within [its] discretion.”). For example, these statements in the Petition and Bagherzadeh declaration do not make clear whether clock speed being proportional to available power is always the case or whether anything in the Long reference teaches these alleged facts. Thus, we are not persuaded Petitioner has shown sufficiently that Long ’552 discloses these limitations.

Therefore, upon review of Petitioner’s analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a

reasonable likelihood that it would prevail with respect to claims 1 and 15, and claims 2, 4, 5, 7, 12, 13, 18, 20, 21, 23, 28, and 29 that depend ultimately from claims 1 and 15, on the ground that these claims are anticipated by Long '552.

G. Claims 3, 6–11, 14, 16, 17, 19, 22–27, and 30—Obviousness over Long '552 and Leete '210Pub

Petitioner argues that claims 3, 6–11, 14, 16, 17, 19, 22–27, and 30 would have been obvious over Long '552 and Leete '210Pub under 35 U.S.C. § 103(a). Pet. 44–50. Claims 3, 6–11, 14, 16, 17, 19, 22–27, and 30 depend from independent claims 1 and 15. As discussed above, Long '552 fails to teach or suggest all of the elements of independent claims 1 and 15. Petitioner does not assert that Leete '210Pub overcomes the aforementioned deficiency in Long '552. Thus, upon review of Petitioner's analysis and supporting evidence, we determine Petitioner has not demonstrated that there is a reasonable likelihood it would prevail with respect to claims 3, 6–11, 14, 16, 17, 19, 22–27, and 30, on the ground that these claims would have been obvious over Long '552 and Leete '210Pub.

H. Claims 1, 2, 4, 5, 12, 13, 15, 16, 18, 20, 21, 28, and 29—Anticipation by Tsunoda '618 (Ex. 1016)

Petitioner argues that claims 1, 2, 4, 5, 12, 13, 15, 16, 18, 20, 21, 28, and 29 are anticipated by Tsunoda '618 under 35 U.S.C. § 102(b). Pet. 50–54. Tsunoda '618 discloses a system including SDRAM volatile memory cache 4010 and flash non-volatile memory 4020. Ex. 1016, 4:2–6. Further, Tsunoda '618 discloses that the device is configured to operate the volatile memory 42 at a high SDRAM speed, when communicating normally with the host (*id.*, 13:51–53, 11:51–54, and 12:27–31). Tsunoda '618 discloses that the device is configured to operate at a speed different from the high

speed, when communicating with the non-volatile memory 44. *Id.* 12:21–25, 13:55–58, and 11:51–54.

Below we discuss independent claims 1 and 15, from which all other dependent claims challenged in this ground depend. Claim 1 recites “operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.” Claim 15 recites “the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.”

Petitioner asserts that Tsunoda ’618’s disclosure of speed matching buffer 108 within the SDRAM discloses a first clock frequency. Pet. 51–52 (citing Ex. 1016, 11:51–54). We are not persuaded by this cited disclosure.

Petitioner apparently reads buffer 108 and SDRAM 103 on the volatile memory subsystem. However, speed matching buffer 108 is a part of memory control unit 104. Ex. 1016, Fig. 1. Reading the controller as the volatile memory subsystem is inconsistent with Tsunoda ’618’s disclosure that controller 62 is *separate from* the volatile and non-volatile memory subsystems. Ex. 1001, Fig. 1. Controller 62 provides the clock signal to the volatile and nonvolatile memory subsystems. Ex. 1001, Fig. 6. As Patent Owner points out, Petitioner does not explain sufficiently why the controller is a part of the volatile memory subsystem. *See* Prelim. Resp. 56–58. We find that one of ordinary skill at the time of the invention would not have considered reasonably Tsunoda ’618 as disclosing a volatile memory subsystem that operates at the first and third clock frequencies. Thus, , we

are not persuaded Petitioner has shown sufficiently that Tsunoda '618 discloses these limitations.

Therefore, upon review of Petitioner's analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood that it would prevail with respect to claims 1 and 15, and claims 2, 4, 5, 12, 13, 16, 18, 20, 21, 28, and 29 that depend ultimately from claims 1 and 15, on the ground that these claims are anticipated by Tsunoda '618.

I. Claims 3, 6–11, 14, 17, 19, 22–27, and 30—Obviousness over Tsunoda '618 and Leete '210Pub

Petitioner argues that claims 3, 6–11, 14, 17, 19, 22–27, and 30 would have been obvious over Tsunoda '618 and Leete '210Pub under 35 U.S.C. § 103(a). Pet. 54–59. Claims 3, 6–11, 14, 17, 19, 22–27, and 30 depend from independent claims 1 and 15. As discussed above, Tsunoda '618 fails to teach or suggest all of the elements of independent claims 1 and 15. Petitioner does not assert that Leete '210Pub overcomes the aforementioned deficiency in Tsunoda '618. Thus, upon review of Petitioner's analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood that it would prevail with respect to claims 3, 6–11, 14, 17, 19, 22–27, and 30, on the ground that these claims would have been obvious over Tsunoda '618 and Leete '210Pub.

III. CONCLUSION

The information presented does not show that there is a reasonable likelihood that Petitioner would prevail at trial with respect to at least one claim of the '833 patent, based on any ground presented in the Petition. We deny the Petition for *inter partes* review of claims 1–30.

IV. ORDER

Accordingly, it is

ORDERED that that the Petition is *denied* as to all challenged claims,
and no trial is instituted.

IPR2014-01370
Patent 8,301,833 B1

PETITIONER:

Sanjiva Reddy
sreddy@kslaw.com

Michael Heafey
mheafey@kslaw.com

PATENT OWNER:

Thomas J. Wimbiscus
twimbiscus@mcandrews-ip.com

Gregory C. Schodde
gschodde@mcandrews-ip.com

Scott P. McBride
smcbride@mcandrews-ip.com

Ronald H. Spuhler
rspuhler@mcandrews-ip.com

Wayne Bradley
wbradley@mcandrews-ip.com

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX
MEMORY SOLUTIONS INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2017-00649
Patent 8,301,833 B1

Before BRYAN F. MOORE, GEORGIANNA W. BRADEN, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

DECISION

Denying Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

SK hynix Inc., SK hynix America Inc. and SK hynix memory
solutions Inc. (“Petitioner”) requests *inter partes* review of claims 1–30 of

U.S. Patent No. 8,301,833 B2 (“the ‘833 Patent,” Ex. 1001) pursuant to 35 U.S.C. §§ 311 *et seq.* Paper 1 (“Pet.”). Netlist, Inc. (“Patent Owner”) filed a preliminary response. Paper 6 (“Prelim. Resp.”). Institution of an *inter partes* review is authorized by statute when “the information presented in the petition . . . and any response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a); *see* 37 C.F.R. § 42.108. Upon consideration of the Petition and Preliminary Response, we conclude the information presented shows there is not a reasonable likelihood that Petitioner would prevail in establishing the unpatentability of claims 1–30 of the ‘833 Patent.

A. Related Matters

Petitioner recites the District Court proceedings related to this *inter partes* review. Pet. 2. The Board has twice declined to institute an *inter partes* review of claims 1–30 of the ‘883 Patent. *Sandisk Corporation v. Netlist, Inc.*, Case IPR2014-00994 (PTAB December 16, 2014) (Paper 8) (rehearing denied, Paper 10); *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01370 (PTAB March 13, 2015) (Paper 13)).

Considering the particular circumstances of this case, we address the merits of the Petition and do not exercise our discretion under 35 U.S.C. § 325(d) (indicating “if another proceeding or matter involving the patent is before the Office, the Director may determine the manner in which the post-grant review or other processing or matter may proceed . . . and may take into account whether, and reject the petition or request because, the same or substantially the same prior art or arguments previously were presented to the Office”) and/or 35 U.S.C. § 314(a) (authorizing institution of an *inter*

partes review under particular circumstances, but not requiring institution under any circumstances). *See* 37 C.F.R. § 42.108(a) (“the Board may authorize the review to proceed”) (emphasis added); *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1367 (Fed. Cir. 2016) (explaining that under § 314(a), “the PTO is permitted, but never compelled, to institute an IPR proceeding”). Petitioner was not a party to any of the prior proceedings. In addition, this Petition raises new issues, including asserting obviousness in view of references not at issue in the previous proceedings. Pet. 3.

B. The '833 Patent

The invention in the '833 patent relates to a specific configuration of hybrid memory systems that addresses non-volatile memory backup while running the volatile memory subsystem at lower power, and, therefore, at lower clock speeds. Ex. 1001, 16:29–34. Specifically, the alleged invention of the '833 patent includes circuitry for providing a regular high-speed clock frequency (first clock frequency) during communications between the host and the volatile memory subsystem, and a slower clock frequency during communications between the volatile memory subsystem (using a third clock frequency) and the non-volatile memory subsystem (using a second clock frequency). *Id.* at 21:5–21. Furthermore, the second and third clock frequencies may be substantially equal. *Id.* at 21:23–24.

C. Illustrative Claim

Independent claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A method for controlling a memory system operatively coupled to a host system, the memory system including a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem; and

operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

Ex. 1001, 21:6–22.

D. Asserted Grounds of Unpatentability

Petitioner asserts that claims 1–30 are unpatentable based on the following grounds:

References	Basis	Claim(s) challenged
Bonella ¹ and Mills ²	§ 103	1–30
Bonella, Mills, and Ashmore ³	§ 103	1–30
Bonella, Mills, Ashmore and Larson ⁴	§ 103	7 and 23
Bonella, Mills, Ashmore and Windows 2000 ⁵	§ 103	8–10, 24–26

¹ US Publication No. 2007/0136523 A1, filed December 8, 2006 (“Bonella,” Ex. 1005). Claims priority to US Provisional No. 11/635,926 filed December 8, 2005.

² US Patent No. 6,026,465, issued February 15, 2000 (“Mills,” Ex. 1007).

³ US Publication No. 2006/0212651 A1, published September 21, 2006 (“Ashmore,” Ex. 1008).

⁴ US Patent No. 6,571,244 B1, issued May 27, 2003 (“Larson,” Ex. 1019).

⁵ MICROSOFT WINDOWS 2000 PROFESSIONAL RESOURCE KIT, lists Feb. 2, 2000 date of publication (“Windows 2000,” Ex. 1021).

Bonella, Mills, Ashmore and Klein ⁶	§ 103	16
Bonella, Mills, Ashmore and Maeda ⁷	§ 103	17

Pct. 3, 14–59.

II. DISCUSSION

A. Claim Construction

In an *inter partes* review, we construe claim terms in an unexpired patent according to their broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b). Consistent with the broadest reasonable construction, claim terms are presumed to have their ordinary and customary meaning as understood by a person of ordinary skill in the art in the context of the entire patent disclosure. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007).

At this juncture of the proceeding, we determine that it is not necessary to provide an express interpretation of any term of the claims.

B. Asserted Obviousness over Bonella and Mills

Petitioner contends claims 1–30 are unpatentable under 35 U.S.C. § 103(a) as obvious over Bonella and Mills. Pet. 14–48. Relying on the testimony of Ron Maltiel, Petitioner explains how Bonella and Mills allegedly describe all of the claim limitations. *Id.* (citing Ex. 1003).

⁶ US Patent No. 6,721,860 B2, issued April 3, 2004 (“Klein,” Ex. 1009).

⁷ US Publication No. 2005/0249011 A1, published November 10, 2005 (“Maeda,” Ex. 1013).

1. Bonella (Ex. 1005)

Bonella is directed to a plug-and-play end-user add-in memory module for computers and consumer electronic devices. Ex. 1005 ¶ 2. Bonella discloses a memory module including a volatile memory, a non-volatile memory, and a controller that provides address, data, and control interfaces to the memories and to a host system. *Id.* ¶ 6.

Bonella teaches a “Power Level 5” state that allows for full function, full performance operation. *Id.* ¶ 47. Bonella also teaches a “Power Level 4” state that reduces the power consumption of the memory module by limiting the DRAM performance. *Id.* ¶ 48.

Bonella also teaches that the DRAM write buffer is occasionally backed up to the internal FLASH memory so as to ensure data integrity in case of a power loss. *Id.* ¶ 96. This write buffer flushing can be triggered by a power loss event, which then causes Bonella’s “Power loss algorithm” to be executed:

When the memory module controller detects a power loss event, the data that is flagged as critical is flushed to the FLASH, a flag is set and the memory module then shuts down. At new power on the normal power on sequence is followed and data restored to the DRAM.

Id. ¶ 101.

During the execution of the power loss algorithm, Bonella’s memory relies on backup power such as, for example, Uninterruptible Power Supply (UPS) capacitors. *Id.* ¶ 29.

2. Mills (Ex. 1007)

Mills describes several interfaces for a FLASH memory device, one of which is a synchronous FLASH interface, i.e. “a synchronous flash

interface (SFI) flash memory integrated circuit 600 that incorporates a complete synchronous flash interface in a single flash memory chip.” Ex. 1007, 16:60–63. The synchronous FLASH interface includes a clock input such that all the external operations of the device are synchronized to the rising edge of the clock. *Id.* at 17:10–25.

Mills teaches that this synchronous operation is used for both read operations and write operations: “When SFI is enabled, interlace control 670 and [bank] select logic 674 operate to interlace read (and write) operations between flash bank A 610 and a flash bank B 620” *Id.* at 17:33–39. Because “the device is interleaved internally,” it “creates an average access time for sequential read accesses that is significantly less than the access time of an asynchronous flash device.” *Id.* at 17:1–9.

3. Analysis

Claims 1 and 15 recite generally three limitations (excluding the preamble). We determine that Petitioner has not shown a reasonable likelihood that Bonella teaches the third limitation of claims 1 and 15, i.e. “operating the volatile memory subsystem [operable] at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.” Below, we summarize Petitioner’s position as to the other two limitations of claims 1 and 15 to provide context for the discussion of the third limitation.

In a first limitation, claims 1 and 15 recite “operating the volatile memory subsystem [operable] at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system.” Petitioner argues that Bonella discloses this feature. Pet. 15–19. Petitioner contends

“[o]ne of ordinary skill in the art would understand that Bonella’s DRAM [volatile memory subsystem] is operating at the particular frequency of the clock signal (“CLK”).” *Id.* at 16 (citing Ex. 1005 ¶ 52; Ex. 1003 ¶ 92).

Petitioner also contends, Bonella teaches its memory module has multiple power states, including Power Level 5 [first mode] which “allows for full function, full performance operation” with “no preset restrictions placed on the DRAM” and when “a skilled artisan would understand . . . ‘data is communicated between the volatile memory subsystem and the host system’ during a read from and write to the DRAM write buffer.” Pet. 17–18 (quoting Ex. 1005 ¶ 6, 8–9, 47, Abstract; Ex. 1003 ¶ 98).

In a second limitation, claims 1 and 15 recite “operating the non-volatile memory subsystem [operable] at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem.” Petitioner argues that the combination of Bonella and Mills teach this feature. Pet. 19–24.

Petitioner admits “Bonella does not explicitly disclose “*operating the non-volatile memory subsystem*” (the FLASH memory) “*at a second clock frequency*” during this second mode” Pet. 21. According to Petitioner, Mills discloses a synchronous Flash interface where read and write operations are synchronized to the rising edge of a clock signal provided to the device and operating at a particular frequency (i.e., “*operating [a] non-volatile memory subsystem at a second clock frequency*”).” Pet. 21–22 (citing Ex. 1007, 16:60–63, 17:10–25; Ex. 1003 ¶ 106–107). As explained below, we determine Bonella does not teach operating the volatile memory subsystem [operable] at a third clock frequency when the memory system is

in the second mode of operation, the third clock frequency being less than the first clock frequency, as recited in claims 1 and 15, therefore, we do not decide whether Petitioner has combined Bonella and Mills properly. Below we discuss Petitioner's contentions regarding Bonella that are relevant to the discussion of the third claim limitation.

Petitioner contends Bonella teaches a FLASH memory and associated controller and in the alleged "first mode of operation" Bonella used the DRAM buffer as a write buffer and periodically flushes the write data to the FLASH memory. *Id.* at 20 (citing 1005 ¶ 96; Fig. 1, Ex 1003 ¶¶ 94–97). On the other hand, Petitioner further contends that during a power loss event, the memory module flushes critical data to the FLASH memory—this is Petitioner's alleged "second mode of operation." *Id.* (citing 1005 ¶ 101, Ex 1003 ¶¶ 103).

In a third limitation, claims 1 and 15 recite "operating the volatile memory subsystem [operable] at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency." Petitioner argues that Bonella discloses this feature. Pet. 23–30. Petitioner contends Bonella discloses modes of operation supported by the system, including a power saving mode [Power Level 4] which saves power in part by reducing the frequency at which the DRAM is operating. Pet. 24–25 (citing Ex 1003 ¶ 116). Table 1, showing the power modes according to Bonella, is reproduced below.

TABLE 1

Partial List of ACPI Power States			
ACPI Power State	Power Source	Power Level (0-5)	Memory Module operational Condition
D0	Line/Battery	5	No operational restrictions, full power
D0	Battery	4	PCIe/DRAM performance reduction, no FLASH restrictions
D0	Battery	3	PCIe/DRAM DRAM wr buffer only, FLASH restrictions
D1	Line/Battery	2	DRAM in standby, FLASH idle
D1	Battery	1	DRAM in standby, FLASH standby
D1	Battery	0	DRAM off, FLASH standby

Table 1 (Ex. 1005, Table 1), above, shows Power Level 4 which operates at a reduced frequency that Petitioner contends is the claimed “third frequency.” Pet. 24–25. According to Petitioner, “A skilled artisan would understand a DRAM operating at a ‘reduced’ frequency in such a power saving mode [Power Level 4] to be operating at a clock frequency lower than the clock frequency at which the DRAM would operate in a normal operating mode, such as Power Level 5 of Bonella.” *Id.* at 25 (citing Ex. 1003 ¶ 118).

Specifically, Petitioner asserts

Bonella’s power flush algorithm (i.e., the “second mode”) would still be triggered during Bonella’s “power saving” mode of operation (i.e., when the DRAM was operating at a “third clock frequency”) if a power loss event occurred. Bonella teaches that the memory module’s backup power source is located within the memory module, Ex. 1005, ¶29, and that the external power source can be a “Line” or “Battery,” even under less than “full power” operation, *id.*, ¶46. One of ordinary skill in the art would understand that Bonella’s memory module, when operating under a “power saving” mode that reduced the DRAM operating frequency (i.e., a “third clock frequency”), could still lose its external power source and be forced to switch to the power loss

algorithm (i.e., the “second mode”). Ex. 1003, ¶119. Bonella thus teaches to one of ordinary skill in the art at least one situation where the ‘*volatile memory subsystem*’ operates ‘*at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.*’

Pet. 25–26.

Patent Owner asserts that “Bonella does not teach that the alleged VMS [volatile memory system] operates at the alleged ‘third clock frequency’ during the alleged ‘second mode of operation.’” Prelim. Resp. 26. Petitioner asserts the “second mode of operation” is the power loss operation described in paragraph 101 of Bonella (Pet. 20), however, Patent Owner disagrees. *Id.* at 25. Contrary to Petitioner’s position, Patent Owner correctly explains, as a complement to that assertion, that the power source during that alleged “second mode of operation” is the battery uninterruptible power supply (“UPS”) and/or the capacitor (“CAP”) UPS because line and battery power from the alleged “host system” has been lost. *Id.* (citing Ex. 1005 ¶ 33).

Petitioner asserts that the DRAM (the alleged volatile memory system) operates at the alleged “*third clock frequency*” during Power Level 4. Pet. 20. As quoted above, Petitioner asserts “when operating under a ‘power saving’ mode that reduced the DRAM operating frequency (*i.e.*, a “*third clock frequency*”), could still lose its external power source and be forced to switch to the power loss algorithm (*i.e.*, the “*second mode*”).” *Id.* Petitioner does not point to any disclosure in Bonella, however, that indicates what frequency the DRAM is operating at during a power loss. Rather, as Patent Owner explains, the battery associated with Power Level 5 (alleged “first mode of operation”) and Power Level 4 (alleged “second

frequency”) is a not the UPS battery so there is no suggestion in Bonella about what frequency the DRAM runs at when powered by the UPS battery. Prelim Resp. 25–26. Thus, we agree with Patent Owner’s assertions that “the Petition fails to show evidence that operating in a ‘power saving’ mode prior to Bonella’s ‘power loss algorithm’ has any effect on the operating frequencies during the ‘power loss algorithm.’” Prelim. Resp. 29–30.

Petitioner does not provide sufficient objective evidence sufficient to support a finding that the DRAM operates at the frequency of Power Level 4 when a power failure occurs during Power Level 4. Additionally, the conclusory statements of Petitioner’s declarant, which are essentially identical to the statements in the Petition, do not explain adequately how the limitation to a “third frequency” is met. Ex. 1003 ¶ 116–120; *see also, e.g., In re Am. Acad. of Sci. Tech. Ctr.*, 367 F.3d 1359, 1368 (Fed. Cir. 2004) (“[T]he Board is entitled to weigh the declarations and conclude that the lack of factual corroboration warrants discounting the opinions expressed in the declarations.” (citations omitted)); *Velandier v. Garner*, 348 F.3d 1359, 1371 (Fed. Cir. 2003) (“In giving more weight to prior publications than to subsequent conclusory statements by experts, the Board acted well within [its] discretion.”). For example, statements in the Petition and Maltiel’s declaration do not make clear where Bonella states that during a power loss the DRAM runs at the frequency of the power lever state immediately preceding the power loss, or whether anything in the Bonella reference teaches that alleged fact.

Petitioner also asserts that

[u]sing Bonella’s power reduction technique during the power loss algorithm would also have been the arrangement of old elements (Bonella’s power reduction mode and Bonella’s power

loss algorithm), each performing the same function it had been known to perform, in a way that yields no more than one of ordinary skill in the art would expect from such an arrangement (reducing power consumption during a power loss event, as suggested by Long and Ashmore).

Pet. 29 (citing Ex. 1003 ¶ 126). Petitioner further asserts “[o]ne of ordinary skill in the art would know that Bonella’s backup power supply is not unlimited and therefore would have been motivated to conserve power during such emergency backup operations in order to ensure backup of all unsaved data, or at least as much as possible.” *Id.* at 27 (citing Ex. 1003 ¶ 122). Petitioner has not shown, however, how or why Bonella’s power reduction modes are applicable during a power loss event, given that during a power loss event power is supplied by the UPS, and not the external Line/battery that supplies power under normal operations.

Petitioner asserts that Ashmore supports that contention that running at the frequency associated with Power Level 4 during a power loss would have been obvious because Ashmore “provides a method for reducing battery power consumption during a main power loss to reduce the likelihood of loss of user write-cached data in a write-caching mass storage controller.” Pet. 28 (quoting Ex. 1008 ¶ 9) (emphasis added). Specifically, Petitioner asserts Ashmore “only provide[s] battery power to the critical memory banks, but not to the non-critical memory banks, in order to reduce the amount of battery power consumed during the main power outage.” *Id.* (quoting Ex. 1008 ¶ 9). We are not persuaded by this cited disclosure.

The Ashmore citation relied upon does not state that the operating frequency of the memory bank is adjusted. Petitioner does not point to a motivation to adjust the frequency of the memory banks specifically, simply a general motivation to reduce battery power during power loss. Thus, we

are not persuaded Ashmore supports that contention that running at the frequency associated with Power Level 4 during a power loss would have been obvious.

Petitioner also asserts that Long supports that contention that running at the frequency associated with Power Level 4 during a power loss would have been obvious because Long “‘provide[s] a significantly slower clock signal to the processing circuitry’ *of a memory* ‘while the controller 40 moves data from the volatile-memory storage cache 42 to the flash-based memory vault 44.’” Pet. 28 (quoting Ex. 1011, 4:54–64) (emphasis added). We are not persuaded by this cited disclosure.

The Long citation relied upon does not state that storage cache 42 or FLASH-based memory vault 44 is clocked at a relatively fast clock signal. Petitioner leaves out the part of the quote that states that it is specifically the “processing circuitry *of the controller 40*” is run at a lower clock speed. That is, the cited disclosure notes that the processing circuitry of the controller is running at a relatively high clock speed and then is run on a significantly slower clock speed at a later time. This disclosure is silent as to whether storage cache 42 or FLASH-based memory vault 44 is running at the relatively high clock speed along with the controller. Thus, it cannot be determined whether storage cache 42 or FLASH-based memory vault 44 is forced to run at a significantly lower clock speed when the available power is reduced. Thus, we are not persuaded Long supports that contention that running at the frequency associated with Power Level 4 during a power loss would have been obvious.

Accordingly, for the reasons stated above, Petitioner has not shown there is a reasonable likelihood of prevailing in establishing that claims 1

and 15, and claims 2–14 and 16–30 that depend therefrom, would have been obvious over Bonella and Mills.

C. Asserted Obviousness over Bonella, Mills, and Ashmore

Petitioner contends claims 1–30 are unpatentable under 35 U.S.C. § 103(a) as obvious over Bonella, Mills, and Ashmore. Pet. 48–49. As explained above, we determine that Petitioner has not shown a reasonable likelihood that Bonella and Mills renders obvious the “third frequency” limitation of claims 1 and 15, even considering the disclosure of Ashmore. Thus, Petitioner has not shown there is a reasonable likelihood of prevailing in establishing the unpatentability of claims 1–30 of the ‘833 Patent as obvious over Bonella, Mills, and Ashmore.

D. Remaining Asserted Obviousness Grounds

As discussed above in the context of the obviousness analysis with regard to Bonella, Mills, and Ashmore, Petitioner has not shown sufficiently that Bonella, Mills, and/or Ashmore teaches “a volatile memory subsystem” that operates at a “third frequency,” as is required by each of independent claims 1 and 15. Each of Petitioner’s remaining obviousness grounds assumes that either Bonella or Ashmore discloses that limitation. Pet. 50–59. Petitioner bears the burden of proof of showing that limitation for each independent claim. Nonetheless, as shown above, none of the challenges Petitioner relied on to meet claims 1 and 15 shows sufficiently the required “third frequency.”

Thus, upon review of Petitioner’s analysis and supporting evidence and Patent Owner’s response and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood that it

would prevail with respect to: (1) the ground that claims 7 and 23 are unpatentable over Bonella, Mills, with or without Ashmore, and Larson; (2) the ground that claims 8–10 and 24–26 are unpatentable over Bonella, Mills, with or without Ashmore, and Windows 2000; (3) the ground that claim 16 is unpatentable over Bonella, Mills, with or without Ashmore, and Klein; and (4) the ground that claim 17 is unpatentable over Bonella, Mills, with or without Ashmore, and Maeda.

III. CONCLUSION

The information presented does not show that there is a reasonable likelihood that Petitioner would prevail at trial with respect to at least one claim of the '833 patent, based on any ground presented in the Petition. On this record, we deny the Petition for *inter partes* review of claims 1–30.

IV. ORDER

Accordingly, it is

ORDERED that that the Petition is *denied* as to all challenged claims, and no trial is instituted.

IPR2017-00649
Patent 8,301,833 B1

PETITIONER:

Joseph A. Micallef
Samuel A. Dillon
SIDLEY AUSTIN LLP
jmicallef@sidley.com
samuel.dillon@sidley.com
Sidley-SKH-IPR@sidley.com

PATENT OWNER:

Thomas J. Wimbiscus
Christopher C. Winslade
Scott P. McBride
Ronald H. Spuhler
Wayne H. Bradley
MCANDREWS, HELD & MALLOY, LTD.
twimbiscus@mcandrews-ip.com
cwinslade@mcandrews-ip.com
smcbride@mcandrews-ip.com
rspuhler@mcandrews-ip.com
wbradley@mcandrews-ip.com

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

STATEMENT UNDER 37 CFR 3.73(b)

Applicant/Patent Owner: Chi-She Chen et al.

Application No./Patent No.: 8,301,833 Filed/Issue Date: 10-30-2012

Titled: NON-VOLATILE MEMORY MODULE

Netlist, Inc., a Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)

the patent application/patent identified above, by virtue of either:

A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 028504, Frame 0323, or for which a copy therefore is attached.

OR

B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☐ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Khaled Shami/

Signature

July 10, 2018

Date

Khaled Shami, Reg. No. 38,745

Printed or Typed Name

Attorney for Assignee

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY TO PROSECUTE APPLICATIONS BEFORE THE USPTO

I hereby revoke all previous powers of attorney given in the application identified in the attached statement under 37 CFR 3.73(b).

I hereby appoint:



Practitioners associated with the Customer Number:

151145

OR



Practitioner(s) named below (if more than ten patent practitioners are to be named, then a customer number must be used):

Name	Registration Number	Name	Registration Number

as attorney(s) or agent(s) to represent the undersigned before the United States Patent and Trademark Office (USPTO) in connection with any and all patent applications assigned only to the undersigned according to the USPTO assignment records or assignment documents attached to this form in accordance with 37 CFR 3.73(b).

Please change the correspondence address for the application identified in the attached statement under 37 CFR 3.73(b) to:



The address associated with Customer Number:

151145

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

Assignee Name and Address:

Netlist, Inc.
175 Technology Drive, Suite 150
Irvine, CA 92618

A copy of this form, together with a statement under 37 CFR 3.73(b) (Form PTO/SB/96 or equivalent) is required to be filed in each application in which this form is used. The statement under 37 CFR 3.73(b) may be completed by one of the practitioners appointed in this form if the appointed practitioner is authorized to act on behalf of the assignee, and must identify the application in which this Power of Attorney is to be filed.

SIGNATURE of Assignee of Record

The individual whose signature and title is supplied below is authorized to act on behalf of the assignee

Signature		Date	12/5/17
Name	Noel B. Whitley	Telephone	949-679-0115
Title	VP, IP & Licensing		

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Acknowledgement Receipt

EFS ID:	33131184
Application Number:	12240916
International Application Number:	
Confirmation Number:	6240
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	46188
Filer:	Khaled Shami/Casey Berger
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-002
Receipt Date:	10-JUL-2018
Filing Date:	29-SEP-2008
Time Stamp:	12:11:34
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	00160010002_POA.pdf	941908	no	2
			938e52dd4bb70fc441d95d4ead37c66e97f12c2		

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

Total Files Size (in bytes):

941908

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/240,916	09/29/2008	Chi-She Chen	062453-002

46188
Nixon Peabody LLP
P.O. Box 26769
San Francisco, CA 94126

CONFIRMATION NO. 6240 POWER OF ATTORNEY NOTICE



OC000000101000575

Date Mailed: 07/17/2018

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 07/10/2018.

- The Power of Attorney to you in this application has been revoked by the assignee who has intervened as provided by 37 CFR 3.71. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/nrhayden/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/240,916	09/29/2008	Chi-She Chen	

151145
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

CONFIRMATION NO. 6240
POA ACCEPTANCE LETTER



Date Mailed: 07/17/2018

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 07/10/2018.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/nrhayden/

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	--

In Compliance with 35 U.S.C. (290 and/or 15 U.S.C. (1116 you are hereby advised that a court action has been
 filed in the U.S. District Court **Western District of Texas** on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. (292.);

DOCKET NO. 6:21-cv-00430	DATE FILED 4/28/2021	U.S. DISTRICT COURT Western District of Texas
PLAINTIFF Netlist, Inc.		DEFENDANT Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,301,833	10/30/2012	Netlist, Inc.
2		
3		
4		
5		

In the above entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY	☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
2		
3		
4		
5		

In the above entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
-------	-------------------	------

Copy 1' Upon initiation of action, mail this copy to Director Copy 3' Upon termination of action, mail this copy to Director
 Copy 2' Upon filing document adding patent(s), mail this copy to Director Copy 4' Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

MICRON TECHNOLOGY, INC.; MICRON SEMICONDUCTOR
PRODUCTS, INC.; and MICRON TECHNOLOGY TEXAS LLC,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00418
Patent 8,301,833 B1

Before GEORGIANNA W. BRADEN, SHEILA F. McSHANE, and
KARA L. SZPONDOWSKI, *Administrative Patent Judges*.

SZPONDOWSKI, *Administrative Patent Judge*.

DECISION
Granting Institution of *Inter Partes* Review
35 U.S.C. § 314

I. INTRODUCTION

Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC (collectively “Petitioner”) filed a Petition (Paper 2, “Pet.”) to institute an *inter partes* review of claims 1, 3–17, and 19–30 of U.S. Patent 8,301,833 B1, issued on October 30, 2012 (Ex. 1001, “the ’833 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 11, “Prelim. Resp.”). We have jurisdiction under 35 U.S.C. § 314.

Institution of an *inter partes* review is authorized when “the information presented in the petition . . . and any response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a). Based on the current record, and for the reasons explained below, we determine that Petitioner has established a reasonable likelihood that it would prevail with respect to at least one challenged claim. Accordingly, we institute an *inter partes* review.

II. BACKGROUND

A. Real Parties in Interest

Petitioner identifies itself as the real party in interest. Pet. 64. Patent Owner also identifies itself as the real party in interest. Paper 3 (“Patent Owner’s Mandatory Disclosure”), 1.

B. Related Matters

The parties advise that the ’833 patent is the subject of *Netlist, Inc. v. Micron Technology, Inc., et al.*, 6:21-cv-00430 (W.D. Tex.). Pet. 64; Paper 3, 2. Petitioner also advises that the ’833 patent has been the subject of the following three *inter partes* review proceedings: IPR2014-00994,

IPR2014-001370, and IPR2017-00649. Pet. 64. Additionally, Petitioner advises that “a nearly identical claim to the ’833 Patent’s claim 15 was invalidated as obvious in IPR2017-00692.” *Id.* at 65.

C. The ’833 Patent (Ex. 1001)

The ’833 patent is titled “Non-Volatile Memory Module” and is generally directed to “a memory system which can communicate with a host system such as a disk controller of a computer system.” Ex. 1001, codes (54), (57).

The ’833 patent states that “[n]on-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory,” so, “[i]n certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.” *Id.* at 1:28–31. The invention in the ’833 patent relates to a configuration of hybrid memory systems that “can include volatile and non-volatile memory and a controller which are configured such that the controller backs up the volatile memory using the non-volatile memory in the event of a trigger condition,” such as a power failure or power reduction. *Id.* at code (57), 3:65–67, 17:66–18:1. “[T]he volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power.” *Id.* at 4:41–44.

Figure 9, which depicts a flowchart of an example method of a volatile memory subsystem operating at a reduced rate in back-up mode, is reproduced below:

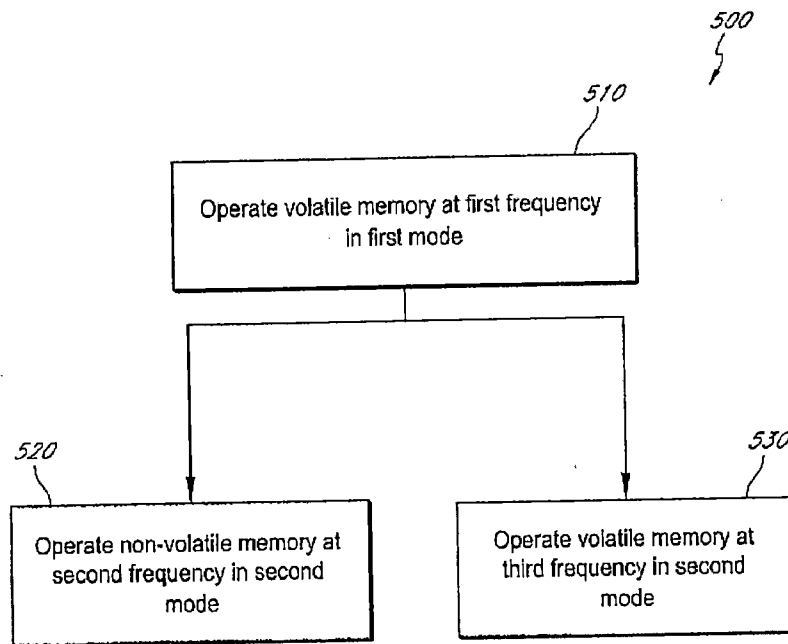


FIG. 9

Id. at 3:45–48. Figure 9 depicts block 510, “[o]perate volatile memory at first frequency in first mode,” block 520, “[o]perate non-volatile memory at second frequency in second mode,” and block 530, “[o]perate volatile memory at third frequency in second mode.” The memory system “may switch from the first mode of operation to the second mode of operation in response to a trigger condition,” such as “a power failure condition.” *Id.* at 17:64–8:1.” The second mode of operation may include, for example, backup and/or restore operations. *Id.* at 18:1–7. The ’833 patent also describes that “[t]he third frequency can be less than the first frequency,” and “can be approximately equal to the second frequency.” *Id.* at 8:8–10.

D. Illustrative Claims

Among the challenged claims, claims 1 and 15 are independent.

Independent claim 15 is representative and is reproduced below.

15. A memory system operatively coupled to a host system, the memory system comprising:

a volatile memory subsystem operable at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

and a non-volatile memory subsystem operable at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem,

the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.

Ex. 1001, 21:61–22:17.

E. Prior Art and Asserted Challenges to Patentability

Petitioner asserts that claims 1, 3–17, and 19–30 are unpatentable on the following challenges (Pet. 2):

Claims Challenged	35 U.S.C. § ¹	Basis/References
1, 15	§ 103(a)	Collateral Estoppel

¹ Because the '833 patent issued from a patent application that was filed before March 16, 2013, patentability is governed by the version of 35 U.S.C. § 103 preceding the Leahy-Smith America Invents Act ("AIA"), Pub L. No. 112–29, 125 Stat. 284 (2011).

Claims Challenged	35 U.S.C. § ¹	Basis/References
1, 3–17, 19–30	§ 103(a)	Best, ² Bonella, ³ Mills ⁴

In support of its proposed challenges, Petitioner relies on the Declaration of Ron Maltiel. *See* Ex. 1003.

III. ANALYSIS

A. Discretion to Deny Institution Under § 35 U.S.C. 314(a)

Patent Owner asserts that the Petition should be denied under 35 U.S.C. § 314(a), as a “follow on” petition, by application of the factors set forth in *General Plastic Indus. Co. v. Canon Kabushiki Kaisha*, IPR2016-01357, Paper 19 at 16 (PTAB Sept. 6, 2017) (precedential as to § II.B.4.i). Prelim. Resp. 9–21. Under the circumstances of this case, we decline to exercise discretion to deny the Petition.

1. Background

The '833 patent has been challenged in three prior *inter partes* review proceedings, each of which was denied institution on the merits. Prelim. Resp. 10. These three proceedings are:

In IPR2014-00994, SanDisk Corp. challenged all claims of the '833 patent, asserting twenty-two challenges based upon seven references, none of which are asserted in this case. *Id.*; IPR2014-00994, Paper 1. The Board denied institution and denied SanDisk's request for rehearing. Prelim. Resp. 10; IPR2014-00994, Papers 8, 10.

² US 2010/0110748 A1, published May 6, 2010 (Ex. 1006).

³ US 2007/0136523 A1, published June 14, 2007 (Ex. 1008).

⁴ US 6,026,465, issued Feb. 15, 2000 (Ex. 1009).

In IPR2014-01370, SMART Modular Technologies, Inc. challenged all claims of the '833 patent, asserting eight challenges based upon five references. Prelim. Resp. 10; IPR2014-01370, Paper 1. None of the references are asserted in this case, although Petitioner here relies on the Long⁵ reference, which was asserted in IPR2014-01370, in support of the knowledge of one of ordinary skill in the art and of its rationale to combine Best and Bonella. *See* Pet. 35–37; IPR2014-01370, Paper 1 at 7. The Board denied institution. Prelim. Resp. 10; IPR2014-01370, Paper 13.

In IPR2017-00649, Hynix challenged all claims of the '833 patent, asserting six challenges based on seven references (“Hynix IPR”). Prelim. Resp. 10; IPR2017-00649, Paper 1. In particular, one challenge asserted that all claims were unpatentable under 35 U.S.C. § 103 over Bonella and Mills. IPR2017-00649, Paper 1 at 3. Petitioner here also relies upon Ashmore⁶ and Long, both of which were asserted in IPR2017-00649, in support of the knowledge of one of ordinary skill in the art and of its rationale to combine Best and Bonella. Pet. 35–37; IPR2017-00649, Paper 1 at 3. Hynix also relied upon a declaration from Ron Maltiel, the same declarant Petitioner relies on in this proceeding. IPR2017-00649, Ex. 1003. The Board denied institution and denied Hynix’s request for rehearing. Prelim. Resp. 11; IPR2017-00649, Papers 7, 9.

In addition, in IPR2017-00692, Hynix challenged all claims of U.S. Patent No. 8,874,831 (“the '831 patent”), which is a continuation-in-part of

⁵ US 7,421,552, issued Sep. 2, 2008 (Ex. 1011).

⁶ US 2006/0212651 A1, published Sep. 21, 2006 (Ex. 1010).

the '833 patent, asserting six challenges based on seven⁷ references, including Best, Bonella, Mills, and Ashmore (“Hynix ’831 IPR”). Prelim. Resp. 11; IPR2017-00692, Paper 1. Hynix also relied upon a declaration from Ron Maltiel, the same declarant Petitioner relies on in this proceeding. IPR2017-00692, Ex. 1003. In a final written decision, the Board found all challenged claims unpatentable. IPR2017-00692, Paper 25. In particular, the Board found challenged dependent claim 15 unpatentable over Best, Mills, and Bonella, with or without Roy. IPR2017-00692, Paper 25 at 31–40. Patent Owner did not appeal that judgment or request rehearing. *See* Pet. 23–24.

2. *Analysis*

General Plastic is directed to multiple petitions serially filed by the same petitioner against the same patent. In *Valve Corp. v. Elec. Scripting Prods., Inc.*, IPR2019-00062, -00063, -00084, Paper 11 (PTAB Apr. 2, 2019) (designated precedential May 7, 2019) (“*Valve*”), the Board extended application of *General Plastic* factors to the situation where multiple petitions for review of the same patent are filed by different petitioners. *Valve*, Paper 11 at 2 (“when different petitioners challenge the same patent, we consider any relationship between those petitioners when weighing the *General Plastic* factors.”).

⁷ In IPR2017-00692, Hynix did not explicitly identify Mills as part of the challenge in the Petition (Paper 1 at 3), but in the Institution Decision, the Board stated that “Although Petitioner does not include Mills . . . explicitly in its ground . . . , we include it here because Petitioner’s analysis relies upon it . . . for teaching part of a limitation.” *See* IPR2017-00692, Paper 7 at n.2. Thereafter, both parties treated the ground as though Mills were included. Paper 12 at 45 (Patent Owner Response); Paper 15 at 19 (Petitioner’s Reply).

The *General Plastic* factors are as follows:

1. whether the same petitioner previously filed a petition directed to the same claims of the same patent;
2. whether at the time of filing of the first petition the petitioner knew of the prior art asserted in the second petition or should have known of it;
3. whether at the time of filing of the second petition the petitioner already received the patent owner's preliminary response to the first petition or received the Board's decision on whether to institute review in the first petition;
4. the length of time that elapsed between the time the petitioner learned of the prior art asserted in the second petition and the filing of the second petition;
5. whether the petitioner provides adequate explanation for the time elapsed between the filings of multiple petitions directed to the same claims of the same patent;
6. the finite resources of the Board; and
7. the requirement under 35 U.S.C. § 316(a)(11) to issue a final determination not later than 1 year after the date on which the Director notices institution of review.

IPR2016-01357, Paper 19 at 9–10.

The parties agree that factors 2, 4, and 7 are neutral. *See* Pet. 64; Prelim. Resp. 17, 19–20. Furthermore, it is undisputed that Petitioner here is different from the petitioners in the prior IPRs described above. Pet. 63; Prelim. Resp. 12. And although the Petition here does not challenge two of the thirty claims previously challenged in the three prior IPRs, we find that,

due to the extent of the overlap, Petitioner has effectively filed a petition directed to the same claims of the '833 patent.

The crux of Patent Owner's arguments as to the remaining factors is that "Petitioners clearly used prior IPR petitions against the '833 Patent and its continuation-in-part, the '831 Patent, as a roadmap to formulate the instant Petition." Prelim. Resp. 13. According to Patent Owner, Petitioner (1) used the same declarant as in the Hynix IPR and the Hynix '831 IPR and copied large portions of the prior declarations; and (2) used the Hynix IPR and Hynix '831 IPR as a roadmap and copied verbatim portions of Hynix's arguments. *Id.* at 12–16. According to Patent Owner, these actions created a relationship between Petitioner and the prior petitioners. *Id.* Patent Owner asserts that "Petitioners' tactics are precisely the sort of serial attacks that favor denial, as they implicate the efficiency concerns underpinning *General Plastic* and promote a manifest unfairness by allowing endless attacks on valid patents with perpetually refined arguments." *Id.* at 21.

Under the circumstances in this case, we disagree with Patent Owner's arguments. When different petitioners challenge the same patent, the relationship between those petitioners may be considered when weighing the *General Plastic* factors. *Valve*, Paper 11 at 9. Here, there is no evidence in the record that Petitioner has any significant relationship with the petitioners of the three prior IPRs challenging the '833 patent—for example, that Petitioner is a co-defendant with any of the other three petitioners or was accused of infringing the '833 patent based on the same product. *See Valve*, Paper 11 at 10. For this reason, we find that factor 1 does not weigh in favor of exercising our discretion to deny institution.

It is, however, undisputed that the full record of the prior IPR proceedings against the '833 patent and the Hynix '831 IPR were available when the Petition was filed. Accordingly, the timing raises the potential for abuse because Petitioner had ample opportunity to study the arguments raised by Patent Owner, and the Board's findings and conclusions of law regarding the claims of the '833 patent, as well as the claims of the related '831 patent. However, here, there is no indication in the record that Petitioner specifically waited to file the Petition in a strategic move to gain any sort of advantage. That is, it does not appear that the Petition was filed with any consideration of the timing of the prior IPR proceedings, because Petitioner did not file the Petition until over three years after the prior IPRs concluded, and only after being sued for infringement of the '833 patent by Patent Owner. *See* Pet. 64. That Petitioner apparently may have reviewed and considered strategies undertaken by prior petitioners on similar issues does not, in and of itself, constitute unfairness to Patent Owner or create a relationship with any of the prior petitioners. This is unlike a situation where a petitioner purposely waited in order to learn from prior petitioners' mistakes or success. Although Patent Owner contends that institution was denied in other non-precedential Board decisions alleged to be similar (*see* Prelim. Resp. 12–16), we find that argument unavailing because “[e]ach case is decided on the basis of its own facts, and the Board’s consideration of the factors . . . may vary from case to case.” *General Plastic*, Paper 19 at 21; *see also id.* at 18. For the foregoing reasons, we are not persuaded that any of factors 1, 5, or 6 weigh in favor of exercising our discretion to deny institution.

Accordingly, upon review of the evidence and the *General Plastic* factors, we are not persuaded that we should exercise our discretion to deny institution under § 314.

B. Discretion to Deny Under 35 U.S.C. § 325(d)

Patent Owner asserts that we should exercise our discretion under 35 U.S.C. § 325(d) to deny institution because the Petition presents the same art and arguments previously analyzed in the Hynix IPR and Petitioner fails to show that the Board erred. Prelim. Resp. 21–32. We do not agree with Patent Owner for the reasons discussed below.

1. Standards of Law

In evaluating the exercise of discretion to deny institution under Section 325(d), the Board uses the following two-part framework: (1) determining whether the same or substantially the same art previously was presented to the Office or whether the same or substantially the same arguments previously were presented to the Office; and (2) if either condition of the first part of the framework is satisfied, determining whether the petitioner has demonstrated that the Office erred in a manner material to the patentability of challenged claims. *Advanced Bionics, LLC v. Med-El Elektromedizinische Geräte GmbH*, IPR2019-01469, Paper 6 at 8 (PTAB Feb. 13, 2020) (precedential).

In applying the two-part framework, we consider several non-exclusive factors, including:

- (a) the similarities and material differences between the asserted art and the prior art involved during examination;
- (b) the cumulative nature of the asserted art and the prior art evaluated during examination;

(c) the extent to which the asserted art was evaluated during examination, including whether the prior art was the basis for rejection;

(d) the extent of the overlap between the arguments made during examination and the manner in which petitioner relies on the prior art or patent owner distinguishes the prior art;

(e) whether petitioner has pointed out sufficiently how the examiner erred in its evaluation of the asserted prior art; and

(f) the extent to which additional evidence and facts presented in the petition warrant reconsideration of the prior art or arguments.

Becton, Dickinson & Co. v. B. Braun Melsungen AG, IPR2017-01586, Paper 8 at 17–18 (PTAB Dec. 15, 2017) (precedential as to Section III.C.5, first paragraph) (“*Becton, Dickinson*”). *Becton, Dickinson* factors (a), (b), and (d) relate to whether the art or arguments presented in the Petition are the same or substantially the same as those previously presented to the Office. *Advanced Bionics*, Paper 6 at 10. Factors (c), (e), and (f) “relate to whether the petitioner has demonstrated a material error by the Office” in its prior consideration of that art or arguments. *Id.* Only if the same or substantially the same art or arguments were previously presented to the Office do we consider whether petitioner has demonstrated a material error by the Office. *Id.* “At bottom, this framework reflects a commitment to defer to previous Office evaluations of the evidence of record unless material error is shown.” *Id.* at 9.

2. *Whether the Same or Substantially the Same Art or Arguments
Were Previously Presented to the Office*

As discussed above, the Hynix IPR involved a challenge to all claims of the '833 patent, and in particular, a challenge to all claims based upon the combination of Bonella and Mills. IPR2017-00649, Paper 7 at 4. Petitioner here challenges claims 1, 3–17, and 19–30 based upon the combination of Best, Bonella, and Mills. Pet. 2. Petitioner contends that this petition involves a different prior art combination than in the Hynix IPR. Pet. 7–8. There is no dispute that Best was not part of the Hynix IPR. Thus, a different combination of references is presented here.

Patent Owner focuses its argument on *Becton, Dickinson's* factor (b), arguing that Best is merely cumulative of Bonella. Prelim. Resp. 24–25. Patent Owner argues that for the preamble of claim 1, “Best teaches no more than what a reasonable examiner would consider to be taught by Bonella” and “Petitioners fail to obscure the substantial overlap between the arguments made in the Hynix 833 IPR—which the Board rejected—and those made here.” *Id.* at 26. Turning to the body of claim 1, for the first clause, Patent Owner argues that “Petitioner relies on the same functionality in Best as purportedly disclosing the claimed ‘first mode of operation’ as Hynix relied on with Bonella in the Hynix 833 IPR.” *Id.* at 27. Patent Owner contends that “[t]here was no need for Petitioners to rely on Best as opposed to Bonella in the instant Petition—other than to manufacture a superficial difference between this Petition and the Hynix 833 IPR Petition.” *Id.* at 28. For the second and third clauses, Patent Owner again argues that Petitioner merely swapped Best for Bonella. *Id.* at 28–31.

We do not agree with Patent Owner's arguments that Best is merely cumulative of Bonella. Initially, we note that Patent Owner's arguments are difficult to reconcile, given that the Board denied institution in the Hynix IPR over the combination of Bonella and Mills but found a facially similar claim in the '831 patent unpatentable in a final written decision over the combination of *Best*, Bonella, and Mills in IPR2017-00692. IPR2017-00649, Paper 7 at 5–15; IPR2017-00692, Paper 25 at 31–40. This would seem to indicate that Best *is not* merely cumulative of Bonella.

Moreover, in the Hynix IPR, petitioner had not shown, for purposes of institution, that Bonella alone taught the limitations of the third clause in claim 1. IPR2017-00649, Paper 7 at 5–15. Specifically, the Board found that petitioner “does not point to any disclosure in Bonella, however, that indicates what frequency the DRAM is operating at during a power loss.” *Id.* at 11; *see also id.* at 12 (“Petitioner does not provide sufficient objective evidence sufficient to support a finding that the DRAM operates at the frequency of Power Level 4 when a power failure occurs during Power Level 4”). Here, Petitioner relies on a combination of Best and Bonella to teach the third clause's limitations in claim 1 (and claim 15), where Bonella is only relied upon to teach the “third clock frequency” portion of the limitation, as opposed to the entire limitation, as was asserted in the Hynix IPR. *See* Pet. 34–38. We find this sufficient to determine that the art and arguments presented in the Hynix IPR are not substantially similar.

We conclude that the art raised in the Petition is not the same or substantially the same as that presented to the Office during the Hynix IPR; nor do we consider the arguments for unpatentability to be substantially

similar. As such, we decline to exercise our discretion under 35 U.S.C. § 325(d) to deny institution.

C. Legal Standards

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations. *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted); *see KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re NuVasive, Inc.*, 842 F.3d 1376,

1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’” (citation omitted)).

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3)); *see also Intelligent Bio-Sys.*, 821 F.3d at 1369. Therefore, to prevail in an *inter partes* review, Petitioner must explain how the proposed combinations of prior art would have rendered the challenged claims unpatentable. At this preliminary stage, we determine whether the information presented in the Petition shows there is a reasonable likelihood that Petitioner would prevail in establishing that at least one of the challenged claims would have been obvious over the proposed combinations of prior art.

D. Level of Ordinary Skill in the Art

Petitioner asserts a person of ordinary skill in the art would have been a person with a bachelor’s degree in materials science, electrical engineering, computer engineering, computer science, or in a related field and at least one year of experience with the design or development of semiconductor non-volatile memory circuitry or systems.

Pet. 4 (citing Ex. 1003 ¶¶ 48–52). Patent Owner does not dispute Petitioner’s proposed level of skill in the art. *See generally* Prelim. Resp.

We find Petitioner’s proposal is consistent with the level of ordinary skill in the art reflected by the prior art of record, and, therefore, adopt Petitioner’s proposed level of ordinary skill in the art for purposes of this Decision. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

E. Claim Construction

Petitioner provides the parties' proposed constructions from the related district court litigation, but asserts that "[t]hese construction disputes from the related litigations do not affect the outcome of this Petition with respect to any claim." Pet. 9 (citing Exs. 1012, 1013). Petitioner also asserts that the term "clock frequency" was construed in IPR2014-00994, Paper 8 at 6 as "identification of a clock running at a particular frequency," which Petitioner asserts is consistent with the '833 patent's Specification. *Id.* (citing Ex. 1001, 17:25–18:13; Ex. 1003 ¶ 136). Petitioner states that it has applied this construction of "clock frequency" in its analysis. *Id.*

Patent Owner does not present any arguments on claim construction. *See generally* Prelim. Resp. At this stage in the proceeding, only terms that are in controversy need to be construed, and then only to the extent necessary to determine whether to institute *inter partes* review. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))). After reviewing the parties' arguments and evidence, we determine that we do not need to expressly construe any term for purposes of this Decision.

F. Collateral Estoppel

"Collateral estoppel protects a party from having to litigate issues that have been fully and fairly tried in a previous action and adversely resolved against a party-opponent." *Ohio Willow Wood Co. v. Alps South, LLC*, 735 F.3d 1333, 1342 (Fed. Cir. 2013) (citing *Pharmacia & Upjohn Co. v. Mylan Pharm., Inc.*, 170 F.3d 1373, 1379 (Fed. Cir. 1999)). "It is well established that collateral estoppel, also known as issue preclusion, applies in the

administrative context.” *MaxLinear, Inc. v. CF CRESPE LLC*, 880 F.3d 1373, 1376 (Fed. Cir. 2018) (citations omitted). The application of collateral estoppel is not limited to patent claims that are identical. *Ohio Willow Wood*, 735 F.3d at 1342. “Rather, it is the identity of the issues that were litigated that determines whether collateral estoppel should apply If the differences between the unadjudicated patent claims and the adjudicated patent claims do not materially alter the question of invalidity, collateral estoppel applies.” *Id.*

Collateral estoppel precludes a party from relitigating an issue if: (1) the prior action presents an identical issue; (2) the prior action actually litigated and adjudged that issue; (3) the judgment in that prior action necessarily required determination of the identical issue; and (4) the prior action featured full representation of the estopped party. *VirnetX Inc. v. Apple, Inc.*, 909 F.3d 1375, 1377 (Fed. Cir. 2018) (quoting *Stephen Slesinger, Inc. v. Disney Enters., Inc.*, 702 F.3d 640, 644 (Fed. Cir. 2012)).

1. Collateral Estoppel As a Separate Challenge

Petitioner asserts “Unpatentable as obvious under the doctrine of collateral estoppel” as Challenge 1 to patentability. *See* Pet. 2. At this point, we agree with Patent Owner that this is not *per se* an appropriate challenge for an *inter partes* review petition. Prelim. Resp. 33. “A petitioner in an *inter partes* review may request to cancel as unpatentable 1 or more claims of a patent *only on a ground that could be raised under section 102 or 103 and only on the basis of prior art consisting of patents or printed publications.*” 35 U.S.C. § 311(b) (emphasis added).

Nonetheless, although we make no determinations here, collateral estoppel may or may not be a consideration in evaluating issues relating to

findings under Sections 102 and 103⁸. Accordingly, in order to provide some guidance to the parties, we discuss the parties' collateral estoppel arguments.

2. *Petitioner's Contentions*

Petitioner asserts that collateral estoppel bars Patent Owner from relitigating the validity of claims 1 and 15, based upon the final written decision in the Hynix '831 IPR, finding claim 15 of the '831 patent unpatentable over Best, Mills, and Bonella. Pet. 18.

First, Petitioner contends that “the substantive features of challenged claims 1 and 15 of the '833 Patent are materially identical to the substantive features of unpatentable claim 15 of the '831 Patent,” and provides a side by side chart showing the comparison. *Id.* at 20–21. Petitioner asserts that “[t]he patent claims need not be verbatim identical, but ‘substantially similar’ so that the issues of validity are materially the same.” *Id.* at 21. Petitioner argues that there is only a “very slight difference in a few words” between the claims. *Id.* at 22. Petitioner also argues that “the prior art references, and supporting arguments” provided here are “also identical to the prior art references, and supporting arguments” evaluated in the Hynix '831 IPR. *Id.*

Second, Petitioner contends that the ultimate issue of validity of claim 15 of the '831 Patent was actually litigated and adjudged in the Hynix '831 IPR, “as well as the sub-issues pertaining to the same prior art, materially similar specific claim features, and materially similar arguments.” *Id.* at 23.

⁸ As discussed above, in the Hynix '831 IPR, all challenged claims were found unpatentable.

Third, Petitioner contends that “the invalidity holding of claim 15 of the ’831 Patent undoubtedly ‘necessarily required determination of the identical issue’ of the validity of claim 15 of the ’831 Patent because the validity of claim 15 of the ’831 Patent was the only issue litigated in the” Hynix ’831 IPR with respect to that claim. *Id.* at 23. Petitioner also asserts that the finality of the final written decision is undisputed, as Patent Owner did not appeal and the time for appeal has expired. *Id.* at 23–24.

Fourth, Petitioner contends that Patent Owner was fully represented by counsel in the Hynix ’831 IPR and ‘has not argued or suggested that it was not adequately represented.’” *Id.* at 24.

3. *Patent Owner’s Contentions*

Patent Owner contends that the elements of collateral estoppel have not been met. Prelim. Resp. 37. Specifically, Patent Owner argues that “[d]ifferent claim construction standards apply,” the ’833 and ’831 patents “have different claim limitations and vastly different written descriptions, and different priority date issues are in play with respect to the asserted references.” *Id.* Patent Owner further argues that because a different claim construction must be applied, “identical issues of claim construction were not actually litigated in the prior proceedings.” *Id.* at 42. Patent Owner also argues that “[w]hether Best is actually entitled to the 2007 priority date was not adjudicated in the [Hynix ’831 IPR] because the ’831 Patent’s earliest priority date (September 29, 2008) rendered the point moot, so Best’s status as prior art was not actually confirmed.” *Id.*

Patent Owner also contends that, even if Petitioner establishes the conditions for collateral estoppel, exceptions apply. Prelim. Resp. 43–47. First, Patent Owner argues that it “did not have an adequate opportunity or

incentive to obtain a full and fair adjudication in the initial action.” *Id.* at 43. According to Patent Owner, it did not have adequate incentive in the Hynix ’831 IPR because it had never alleged any party infringed the ’831 patent, so it was only responding to a petition attacking the patent. *Id.* at 44. Patent Owner also argues that it did not have the opportunity to antedate claim 15 of the ’831 patent. *Id.* Patent Owner asserts that now, it “is asserting the ’833 Patent against defendants in multiple lawsuits, including against Petitioners.” *Id.* at 45. Second, Patent Owner argues that there are prior determinations inconsistent with the judgment sought to be applied for issue preclusion. *Id.* at 45–46. For example, Patent Owner points out that the Board denied institution in the Hynix IPR. *Id.* at 46–47.

4. *Analysis*

At this stage of the proceeding, and because we determine that Petitioner has sufficiently established a reasonable likelihood of success on the merits of its prior art ground for purposes of institution, we determine that whether collateral estoppel applies to Patent Owner’s arguments, and to which issues, is best resolved on a full record at trial.

In the Hynix ’831 IPR, the Board determined that claim 15 of the ’831 patent was obvious over Best, Bonella, and Mills. IPR2017-00692, Paper 25 at 31–40. An essential part of that decision was that all of the elements of claim 15 were found in the combination of references. *Id.* Thus, the finding that the elements of claim 15 were taught in the prior art, and that Hynix had provided sufficient rationale for the combination of references, was an essential part of the ruling. *Id.* As Petitioner points out (Pet. 23–24), Patent Owner did not appeal that decision, rendering the Board’s decision final.

Patent Owner was also represented by counsel in that proceeding and has not argued or suggested that it was not adequately represented.

We agree with Petitioner that the claims 1 and 15 of the '833 patent share some of the same claim limitations addressed by the Board in the Hynix '831 IPR and, on their face, appear quite similar. However, as Patent Owner points out, the language recited in claims 1 and 15 of the '833 patent is not identical to the language recited in claim 15 of the '831 patent.

Although Petitioner provides a side by side comparison of the claims, Petitioner does not further address the differences, i.e., “module” versus “system,” “controlling” versus “managing,” or “operatively coupled to a host system” versus “memory controller,” but simply states that there is “a very slight difference in few words.” *See* Pet. 22. At this stage, neither party has substantively addressed these differences, how such differences may affect the scope of the claims, and whether such differences “materially alter” the question of invalidity in terms of a collateral estoppel analysis. We encourage the parties to address these issues during trial.

Moreover, we agree with Patent Owner that a different claim construction standard applies. However, at this stage of the proceeding, neither party has proposed any terms for construction, nor has Patent Owner explained how any proposed construction would affect this proceeding differently than in the Hynix '831 IPR. We note that in the Hynix '831 IPR, the parties only construed the term “memory module,” which is not recited in claims 1 or 15 of the '833 patent. IPR2017-00692, Paper 25 at 9–14. We encourage the parties to further develop the record on this issue during trial.

Finally, at this stage of the proceeding, we do not know what arguments Patent Owner will present in its Patent Owner Response, and, as a

result, it is premature to determine whether such hypothetical arguments are collaterally estopped. For example, many of Patent Owner's arguments against collateral estoppel relate to claim construction, the difference in the written description of the respective '831 and '833 patents, the difference in the claim limitations of the respective patents, and differences in priority date issues. *See* Prelim. Resp. 37–41. However, at this stage of the proceeding, Patent Owner has not presented arguments pertaining to any of these issues. Accordingly, we encourage the parties to specifically identify and discuss which issues may or may not be collaterally estopped during trial.

We, likewise, determine that Patent Owner's arguments pertaining to whether an exception may apply is best decided on a full record at trial. However, we note that we disagree with Patent Owner's assertion that the decisions in the Hynix IPR and the Hynix '831 IPR are inconsistent. As has been discussed above, the Hynix IPR involved only the Bonella and Mills references, whereas the Hynix '831 IPR added Best to the combination.

G. Challenge 2: Alleged Obviousness Over Best, Bonella, and Mills

Petitioner contends claims 1, 3–17, and 19–23 would have been obvious over the combination of Best, Bonella, and Mills. Pet. 24–60. For the reasons that follow, we are persuaded that the evidence, including Mr. Maltiel's testimony, sufficiently supports Petitioner's arguments and, therefore, establishes a reasonable likelihood of prevailing with respect to this challenge at this stage of the proceeding.

1. Best (Ex. 1006)

Best is titled "Hybrid Volatile and Non-Volatile Memory Device" and is generally directed to a "composite, hybrid memory device including a first

storage die having an array of volatile storage cells and a second storage die having an array of non-volatile storage cells disposed within an integrated circuit package.” Ex. 1006, codes (54), (57). “The hybrid memory device includes a shared interface circuit to receive memory access commands directed to the first storage die and the second storage die and to convey read and write data between an external data path and the first and second storage dice.” *Id.* at code (57).

Figure 2 of Best, which illustrates an embodiment of a hybrid, composite memory device, is reproduced below.

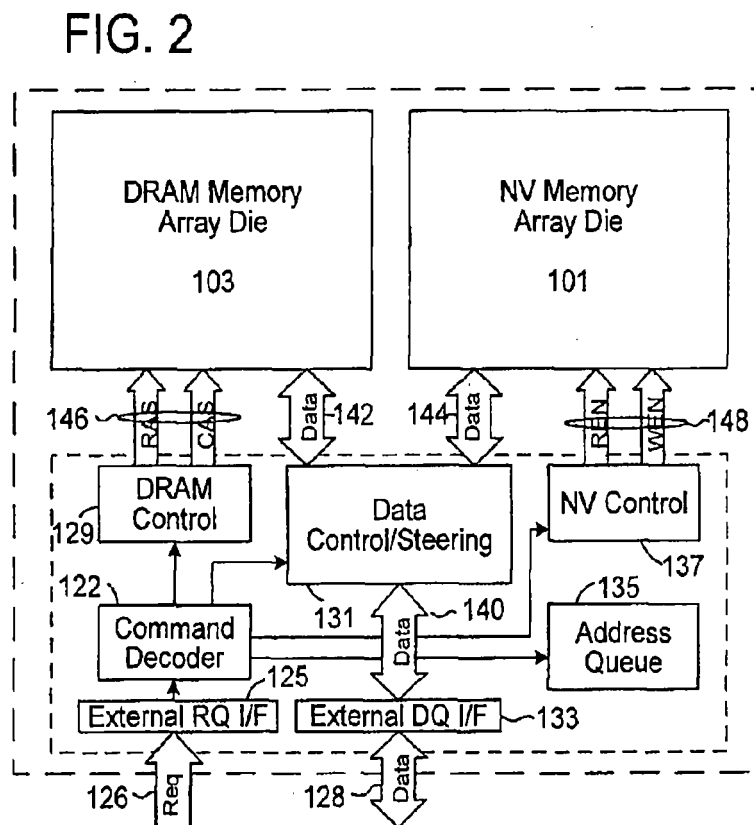


Figure 2, depicted above, illustrates an embodiment of a hybrid, composite memory device with shared interface circuitry, volatile memory

implemented with DRAM die 103, and non-volatile memory implemented with Flash memory die 101. *Id.* ¶ 17. “[T]he shared interface circuitry includes an external request interface 125, external data interface 133, command decoder 122, address queue 135, DRAM control circuit 129, Flash control circuit 137, and data control/steering circuit 131.” *Id.* “[I]ncoming control signals and addresses . . . are received in the external request interface 125 via control/address (CA) path 126, reformatted as necessary (e.g., deserialized to form a parallel command word and one or more address values) and then forwarded to the command decoder 122.” *Id.* “The command decoder 122 in turn forwards address to the address queue 135 and stores memory access commands.” *Id.* “[M]emory access operations may be automatically directed to either the volatile storage die or non-volatile storage die according to the memory address to be accessed.” *Id.* “[C]ommand decoder 122 outputs, from the head of the command queue, an enable signal and corresponding memory access control signals to the DRAM control circuit 129 and NV control circuit 137.” *Id.* ¶ 18. “[D]ata control/steering circuit 131 is used to control the transfer of data between a shared internal data bus and dedicated internal data buses associated with the volatile and non-volatile storage dice, respectively.” *Id.* ¶ 20.

Figure 3 of Best, which illustrates an embodiment of a data control/steering circuit, is reproduced below.

FIG. 3

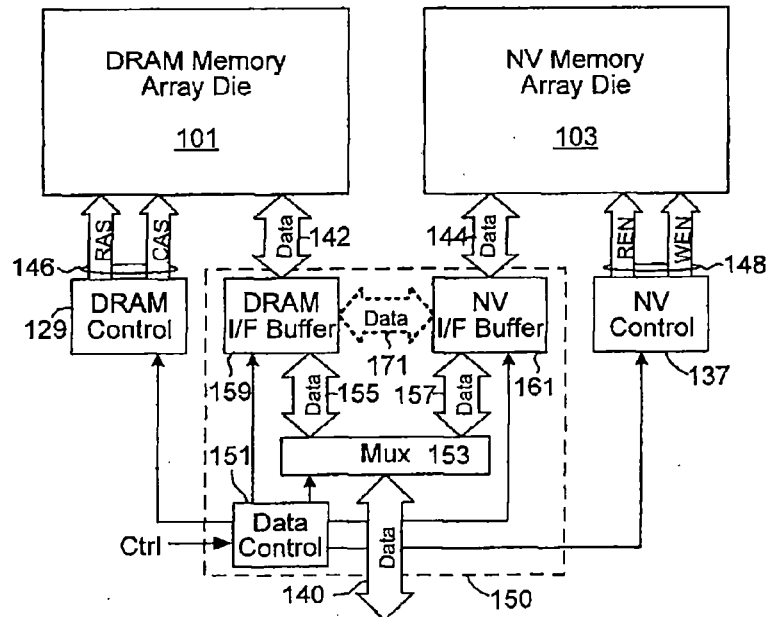


Figure 3, depicted above, illustrates an embodiment of a data control/steering circuit 150 that may be used to implement the data control/steering circuit 131 of Figure 2. *Id.* ¶ 21. “[D]ata control circuit 151 receives control signals from the command decoder that indicate the direction of data flow during a memory access operation (read or write) and whether the volatile or non-volatile storage die is the target of the memory access.” *Id.*

2. *Bonella (Ex. 1008)*

Bonella is titled “Advanced Dynamic Disk Memory Module Special Operations” and is generally directed to “[a] memory module including a volatile memory, a non-volatile memory, and a controller that provides address, data, and control interfaces to the memories and to a host system.” *Ex. 1008*, codes (54), (57). *Bonella* states that “[r]eduction of power in the

DRAM can be accomplished” by “reduc[ing] the frequency in which the DRAM is operating,” which “reduces power and, in general, produces no noticeable decrease in system performance.” *Id.* ¶ 49; *see also id.* ¶ 50.

3. *Mills (Ex. 1009)*

Mills is titled “Flash Memory Including a Mode Register for Indicating Synchronous or Asynchronous Mode of Operation” and is generally directed to a “flash memory chip that can be switched into four different read modes.” Ex. 1009, codes (54), (57). Mills describes several interfaces for a FLASH memory device, one of which is a synchronous FLASH interface, i.e. “a synchronous flash interface (SFI) flash memory integrated circuit 600 that incorporates a complete synchronous flash interface in a single flash memory chip.” *Id.* at 16:60–63. The synchronous FLASH interface includes a clock input such that all the external operations of the device are synchronized to the rising edge of the clock. *Id.* at 17:10–25. Mills teaches that this synchronous operation is used for both read operations and write operations: “When SFI is enabled, interlace control 670 and [bank] select logic 674 operate to interlace read (and write) operations between flash bank A 610 and a flash bank B 620” *Id.* at 17:33–39. Because “the device is interleaved internally,” it “creates an average access time for sequential read accesses that is significantly less than the access time of an asynchronous flash device.” *Id.* at 17:1–9.

4. *Analysis of Independent Claims 1 and 15*

Petitioner contends that Best, Bonella, and Mills, alone or in combination, teach each limitation of independent claims 1 and 15. Pet. 24–38. In support, Petitioner identifies certain passages and figures in the references and explains their significance with respect to the corresponding

claim limitation. *Id.* Petitioner also provides reasons, supported by the testimony of Mr. Maltiel, why it would have been obvious to one of ordinary skill in the art to combine Best, Bonella, and Mills. *Id.* Petitioner analyzes claims 1 and 15 together. *Id.* Claims 1 and 15 are substantially similar, although claim 1 is a method claim and claim 15 is a system claim. *See, e.g.,* Ex. 1001, 21:6–22, 21:61–22:17. We select claim 15 as representative.

a) *Preamble: A memory system operatively coupled to a host system, the memory system comprising:*

Petitioner relies on Best to teach the preamble. Pet. 25–26.

Specifically, Petitioner contends that Best’s memory device (as shown in Fig. 2) is “operatively coupled to a host system” because it “includes an interface that ‘receive[s] commands from a controller device.’” *Id.* at 25 (citing Ex. 1006 ¶ 14; Ex. 1003 ¶ 142). According to Petitioner, “Best discloses techniques for using the shared interface to control access by the controller device to the non-volatile and/or volatile memories.” *Id.* at 25–26 (citing Ex. 1006 ¶¶ 12–17, Fig. 2; Ex. 1003 ¶¶ 140–142).

Patent Owner does not specifically respond to these arguments. *See generally* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has sufficiently shown that Best teaches the recited elements for purposes of institution.⁹

b) *A volatile memory subsystem operable at a first clock frequency when the memory system is in a first mode of*

⁹ Neither party argues whether the preamble limits claim 1 or claim 15. Although we find that the evidence supports that the prior art teaches the preamble, we make no determination at this stage of the proceeding whether the preamble of claim 1 and claim 15 is limiting.

operation in which data is communicated between the volatile memory subsystem and the host system;

Petitioner relies on Best to teach this limitation. Pet. 26–29.

Specifically, Petitioner contends that Best’s DRAM teaches “a volatile memory system.” Pet. 26 (citing Ex. 1006 ¶¶ 17, 21; Ex. 1003 ¶ 145).

Petitioner further contends that Best’s volatile memory (DRAM) die can be synchronous or asynchronous, and that Best “explains that a conventional synchronous DRAM interface relies on ‘clock and clock-enable signals.’” *Id.* at 26–27 (citing Ex. 1006 ¶ 14). According to Petitioner, a person of ordinary skill in the art would understand this “to mean that a synchronous DRAM is ‘operable at a first clock frequency’ because it is operated synchronously by a clock signal.” *Id.* at 27 (citing Ex. 1003 ¶¶ 146–147). Petitioner contends that “[t]his corresponds to the usual operation of DRAM memory, as would have been well-known by a” person of ordinary skill in the art.” *Id.*

Petitioner further contends that Best teaches “a first mode of operation in which data is communicated between the volatile memory subsystem and the host system” because “Best discloses a ‘Shadow Operation’^[10] embodiment where a portion of the DRAM is used as a write buffer for the non-volatile memory.” *Id.* (citing Ex. 1006 ¶ 24; Ex. 1003 ¶ 148). In this embodiment, “[d]ata is stored in a fast access volatile storage die,” and later

¹⁰ Best states: “In memory shadowing, memory accesses are directed first to the non-volatile storage device to enable relatively high-speed, non-degrading memory accesses, with the data eventually being written-back (retired or recorded) within the non-volatile storage in response to a triggering event referred to herein as a write-back trigger. Applications of memory shadowing include non-volatile DRAM, and rapid-access non-volatile memory.” Ex. 1006 ¶ 24.

“a write-back trigger is detected within the shared interface circuitry,” such as a “power-loss” event. *Id.* at 30 (citing Best ¶¶ 25–26). According to Petitioner, “[o]nce the power-loss event is detected, ‘internal data transfer operations are performed to transfer data . . . from the DRAM to . . . the NV memory.’” *Id.* (citing Ex. 1006 ¶¶ 25–26; Ex. 1003 ¶ 148). Petitioner, therefore, asserts that Best “describes at least two modes of operation: (1) the operation of the system when no write-back trigger is detected (*‘a first mode of operation’*), and (2) the operation of the system when a power-loss event write-back trigger is detected such as a power-loss (*‘a second mode of operation’*).” *Id.* at 30 (citing Ex. 1006 ¶¶ 25–26; Ex. 1003 ¶¶ 148–150).

According to Petitioner, Best’s “*first mode of operation*” “is one in which ‘*data is communicated between the volatile memory subsystem and the host system*’ because initially ‘data is stored in a fast-access volatile storage die’ and only later after a write-back trigger is written to non-volatile memory.” *Id.* (citing Ex. 1006 ¶ 25; Ex. 1003 ¶¶ 148–150).

Petitioner further contends that it would have been obvious to operate Best’s DRAM at any of the frequencies described by the DDR DRAM and DDR2 DRAM standards because it “would have been merely the use of a known structure (a standard DRAM module and clock frequency) for its known use (operating as a DRAM module) to achieve a predictable result (operating a DRAM module at a standard clock frequency in Best’s system).” *Id.* at 28–29 (citing Ex. 1003 ¶¶ 151–152).

Patent Owner does not specifically respond to these arguments. *See generally* Prelim. Resp. Nonetheless, the burden remains on Petitioner to

demonstrate unpatentability. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015).

Based on our review and consideration of the current record, we determine that Petitioner has sufficiently shown that Best teaches this limitation for purposes of institution.

- c) *And a non-volatile memory subsystem operable at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem,*

(1) Petitioner's Contentions

Petitioner relies on the combination of Best and Mills to teach this limitation. Pet. 29–34. Specifically, Petitioner contends that Best's Flash memory teaches “a non-volatile memory system.” Pet. 30 (citing Ex. 1006 ¶¶ 13, 17; Ex. 1003 ¶ 155). Petitioner relies on Best's “Shadow Operation” embodiment, discussed above, to teach “a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem.” *Id.* (citing Best ¶¶ 24–26; Ex. 1003 ¶ 156). According to Petitioner, Best's “‘second mode of operation’ (i.e., the operation of the system when a power-loss write-back trigger is detected) involves ‘a non-volatile memory subsystem operable’ (Best's ‘NV’ memory) such that ‘data is communicated between the volatile memory subsystem and the non-volatile memory subsystem,’ e.g. when ‘internal data transfer operations are performed to transfer data . . . from the DRAM to . . . the NV memory.’” *Id.* at 31 (citing Ex. 1006 ¶¶ 25–26; Ex. 1003 ¶¶ 156–158).

Petitioner contends that “Best does not explicitly disclose ‘operating the non-volatile memory subsystem’ (the Flash memory) ‘at a second clock

frequency’ during this second mode, but it would have been obvious to include that functionality in the system of Best.” *Id.* (citing Ex. 1003 ¶ 159). Petitioner, therefore, relies on a Flash implementation in Mills, which Petitioner asserts describes a synchronous flash interface that causes the Flash memory to “*opera[te] at a second clock frequency.*” *Id.* at 31–32 (citing Ex. 1009, 16:60–63, 17:10–25; Ex. 1003 ¶ 160). According to Petitioner, “Mills teaches that this synchronous operation is used for both read operations and write operations.” *Id.* at 32.

Petitioner asserts that Best and Mills are analogous art to the ’833 patent. *Id.* Petitioner further asserts that the combination (i.e., employing a synchronous flash memory as described in Mills in the system of Best) would have been obvious to a person of ordinary skill in the art because it “would have been merely an arrangement of old elements with each performing the same function it had been known to perform and yielding no more than what one would expect from such an arrangement, i.e. the non-volatile storage of data.” *Id.* at 32–33 (citing Ex. 1003 ¶ 163). Petitioner contends that, as combined, Best’s Flash memory would operate “at a second clock frequency.” *Id.* at 33 (citing Ex. 1003 ¶ 164). Petitioner contends that one of ordinary skill in the art would have been motivated to make the combination because “restoring data from the non-volatile flash memory would therefore have been faster by use of a synchronous flash memory, and reduced sequential read access times during other operations or uses of Best’s Flash memory.” *Id.* at 33–34 (citing Ex. 1003 ¶ 165). Petitioner also asserts that a person of ordinary skill in the art would have understood Best to work with any known Flash interface, including Mills’ synchronous Flash interface. *Id.* at 34 (citing Ex. 1003 ¶ 166).

(2) Patent Owner's Contentions

Patent Owner argues that a person of ordinary skill in the art would not be motivated to combine Best with Mills. Prelim. Resp. 47–59.

First, Patent Owner contends that Petitioner's argument "is inconsistent with Best's actual teachings and inventive purpose." Prelim. Resp. 49. Specifically, Patent Owner argues that "the inventive purpose of Best is to be flexible and accommodate different types of memory devices," so "it is apparent that Petitioners 'relied on the ['833 Patent] itself as [their] roadmap.'" *Id.*

Second, Patent Owner argues that Petitioner's rationale is contradictory, in that Petitioner relies on Best's data transfer from DRAM to Flash to teach the claimed "second mode of operation," but then asserts that a person of ordinary skill in the art would have been motivated to combine Mills with Best because Mills provides faster transfer of data from Flash to DRAM. Prelim. Resp. 50–51.

Third, Patent Owner argues that "Best already discloses that it can be used with either synchronous or asynchronous non-volatile and volatile memories," so combining Best with Mills is redundant. Prelim. Resp. 51–52.

Fourth, Patent Owner argues that Petitioner's rationale employs impermissible hindsight reasoning because Petitioner's have not provided an explanation why a person of ordinary skill in the art, "would have been motivated to modify Best specifically with the Mills synchronous flash interface." Prelim. Resp. 52–53.

Fifth, Patent Owner contends that Petitioner "fails to consider Mills as a whole." Prelim. Resp. 53.

Sixth, Patent Owner contends that Petitioner fails to sufficiently explain why a person of ordinary skill in the art would have been motivated to combine Mills with Best, “particularly since . . . Best already teaches that a synchronous non-volatile memory interface may be used, and that his invention allows for data to be ‘quickly restored’ from flash to DRAM following a power failure event.” Prelim. Resp. 55–56.

Seventh, Patent Owner contends that “it is Mills’ unique interleaved interface design that purportedly leads to reduced access times, and Mills does not teach that access times are increased simply by using a synchronous flash device,” so a person of ordinary skill in the art would not have been motivated to modify Best with Mills. Prelim. Resp. 56–57.

Eighth, Patent Owner contends that Petitioner has improperly offered a generic motivation to combine the references. Prelim. Resp. 57.

(3) *Analysis*¹¹

At this stage of the proceeding, and for purposes of institution, we find that Petitioner has sufficiently shown that the combination of references teaches this limitation, and Petitioner has provided a sufficient rationale for combining the references.

¹¹ We note that Patent Owner did not challenge the rationale for the combination of Best and Mills in the Hynix ’831 IPR. IPR2017-00692, Paper 12 at 45–64 (Patent Owner Response). As discussed above in Section III.F., Patent Owner may be estopped from making such arguments at trial. Nevertheless, given that we find that Petitioner has provided sufficient rationale for the combination for purposes of institution, we need not rely on or decide Petitioner’s collateral estoppel argument at this stage of the proceeding.

Patent Owner's arguments do not address the combination as relied upon by Petitioner. Petitioner contends that "Best does not *explicitly* disclose '*operating the non-volatile memory subsystem*' (the Flash memory) '*at a second clock frequency*' during the second mode." Pet. 31 (first emphases added). Petitioner, therefore, relies on Mills to *explicitly* teach the operation of the (non-volatile) Flash memory "at a second clock frequency." *Id.* at 31–32 (citing Ex. 1009, 17:10–25, which states "A clock input is a part of the interface . . . All the external operations of the device are synchronized to the rising edge clock.")). Petitioner's rationale for the combination is based upon the assertion that synchronous flash memory interfaces that were operated by clock signals were well known in the art, and offers Mills' synchronous flash interface as one such prior art flash implementation. *See* Pet. 31–32.

As Patent Owner acknowledges, Best teaches that it can "accommodate[s] different types of memory devices," and discloses that it may be used with synchronous memory devices. *See* Prelim. Resp. 49–92; Ex. 1006 ¶ 14. Based upon this disclosure, we are not persuaded that Petitioner's rationale for the combination employs hindsight reasoning. Any judgment on obviousness is necessarily a reconstruction based upon hindsight reasoning, but so long as it takes into account only knowledge that was within the level of ordinary skill at the time the claimed invention was made and does not include knowledge gleaned only from the challenged patent's disclosure, such a reconstruction is proper. *In re McLaughlin*, 443 F.2d 1392, 1313–14 (CCPA 1971). For purposes of this Decision, we determine that is the case here. Patent Owner has not identified any knowledge relied upon by Petitioner that was gleaned only from the

disclosure of the '833 patent and that was not otherwise taught by the prior art reference (i.e., Best) or within the level of ordinary skill in the art at the time the invention was made (i.e., synchronous flash memory interfaces that were operated by clock signals were well known in the art). Moreover, we find Petitioner's argument that the combination would have been the result of "merely an arrangement of old elements with each performing the same function it had been known to perform and yielding no more than what one would expect from such an arrangement, i.e., the non-volatile storage of data" to be sufficient for purposes of institution, and consistent with the disclosure relied upon in Best and Mills. *See* Pet. 32–33.

Thus, on the current record, we determine Petitioner's arguments, as discussed above, provide sufficient reasoning with rational underpinning for combining the teachings of Best and Mills.

d) The volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the clock first frequency.

Petitioner relies on the combination of Best and Bonella to teach this limitation. Pet. 34–38.

Petitioner asserts that Best does not explicitly disclose this limitation, but it would have been obvious for two reasons: "(1) reducing power during volatile to non-volatile flush operations prompted by a power loss was a well-known technique, and (2) one known way to reduce the power consumption of DRAM devices was to reduce their frequency of operation." *Id.* at 35 (citing Ex. 1003 ¶ 170).

Petitioner contends that the first reason and its advantages were well known in the art, and cites to Ashmore and Long in support of this assertion.

Id. at 35–36 (citing Ex. 1010 ¶ 9; Ex. 1011, 4:54–64; Ex. 1003 ¶¶ 171–172). Petitioner contends that a person of ordinary skill in the art “would have been motivated to reduce the power consumption during Best’s write flushing in response to a power loss” (1) for all the reasons that were known in the art; and (2) because it would have been the arrangement of old elements, each performing the same function it had been known to perform, in a way that yields no more than what would be expected from such an arrangement. *Id.* at 36 (citing Ex. 1010 ¶ 7; Ex. 1011, 4:54–64; Ex. 1003 ¶ 173).

Petitioner relies on Bonella in support of the second reason, asserting that Bonella “explains that one way to reduce the power consumption of the memory module is to slow or reduce the operating frequency of the DRAM.” Pet. 37 (citing Ex. 1008 ¶¶ 49–50; Ex. 1003 ¶ 174). Petitioner contends that Best and Bonella are analogous art to the ’833 patent because both are from the field of memory systems. *Id.* Petitioner contends that a person of ordinary skill in the art “would also have found it obvious to reduce power consumption during Best’s write flushing in response to a power loss using any known or conventional means, and would have also considered power consumption reduction techniques other than those of Ashmore and Long to obtain the same benefits, including those described in Bonella.” *Id.* According to Petitioner, a person of ordinary skill in the art “would have found it obvious to perform Bonella’s power reduction technique (DRAM operational frequency reduction) during the performance of Bests’ write flushing in response to a power loss, in view of the specific motivations evidenced by Ashmore and Long.” *Id.* (citing Ex. 1003 ¶ 176). Petitioner also argues that Bonella itself provides motivation because “it

results in ‘major power savings.’” *Id.* (citing Ex. 1008 ¶ 50). Petitioner contends that “[u]sing Bonella’s power reduction technique during Best’s write flushing would also have been the arrangement of old elements, each performing the same function it had been known to perform, in a way that yields no more” would be expected from such an arrangement. *Id.* at 38.

Patent Owner does not specifically respond to these arguments. *See generally* Prelim. Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

Based on our review and consideration of the current record, for purposes of institution, we determine that Petitioner has sufficiently shown that the combination of Best and Bonella teach this limitation and Petitioner has provided a sufficient rationale for combining the references.

e) Conclusion for Independent Claims 1 and 15

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Best, Bonella, and Mills teaches the limitations in claims 1 and 15, and provided sufficient rationale for combining the references for purposes of institution. We determine, on the current record and for purposes of this Decision, that the information presented in the Petition demonstrates a reasonable likelihood that claims 1 and 15 are unpatentable under 35 U.S.C. § 103(a) over Best, Bonella, and Mills.

5. Dependent Claims 3–14, 16, 17, and 19–30

Petitioner contends that dependent claims 3–14, 16, 17, and 19–30, which depend directly or indirectly from independent claims 1 and 15, would have been obvious over Best, Bonella, and Mills, and provides

explanation as to how the prior art teaches each claim limitation. Pet. 38–60.

Patent Owner does not specifically respond to these arguments. *See* Prelim. Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

Based on our review and consideration of the current record, Best, Bonella, and Mills teach the limitations in claims 3–14, 16, 17, and 19–30 for purposes of institution. We determine, on the current record and for purposes of this Decision, that the information presented in the Petition demonstrates a reasonable likelihood that claims 3–14, 16, 17, and 19–30 are unpatentable under 35 U.S.C. § 103(a) over Best, Bonella, and Mills.

6. *Summary of Challenge 2*

We find that Petitioner has demonstrated a reasonable likelihood that claims 1, 2–17, and 19–30 are unpatentable under 35 U.S.C. § 103(a) over Best, Bonella, and Mills.

IV. CONCLUSION

After considering the evidence and arguments presented in the Petition and Preliminary Response, we determine Petitioner has established a reasonable likelihood of prevailing on its assertion that at least one claim of the '833 patent is unpatentable. Accordingly, we institute an *inter partes* review on all the challenged claims and challenges presented in the Petition. *See PGS Geophysical AS v. Iancu*, 891 F.3d 1354, 1360 (Fed. Cir. 2018) (interpreting the statute to require “a simple yes-or-no institution choice respecting a petition, embracing all challenges included in the petition”). At this stage of the proceeding, we have not made a final determination as to the patentability of these challenged claims.

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that pursuant to 35 U.S.C. § 314, *inter partes* review is instituted as to the challenged claims of the '833 patent and all challenges of unpatentability presented in the Petition; and

FURTHER ORDERED that *inter partes* review is commenced on the entry date of this Order, and pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial.

IPR2022-00418
Patent 8,301,833 B1

PETITIONER:

Juan C. Yaquian
WINSTON & STRAWN LLP
JYaquian@winston.com

PATENT OWNER:

Rex Hwang
Michael D. Ricketts
Ryan A. Hargrave
SKIERMONT DERBY LLP
rhwang@skiermontderby.com
mricketts@skiermontderby.com
rhargrave@skiermontderby.com

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

MICRON TECHNOLOGY, INC.; MICRON SEMICONDUCTOR
PRODUCTS, INC.; and MICRON TECHNOLOGY TEXAS LLC,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00418
Patent 8,301,833 B1

Before GEORGIANNA W. BRADEN, SHEILA F. McSHANE, and
KARA L. SZPONDOWSKI, *Administrative Patent Judges*.

SZPONDOWSKI, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

We instituted *inter partes* review of claims 1, 3–17, and 19–30 of U.S. Patent 8,301,833 B1, issued on October 30, 2012 (Ex. 1001, “the ’833 patent”) in response to a Petition (Paper 2, “Pet.”) filed by Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC (collectively “Petitioner”). Paper 14 (“Inst. Dec.”). During the trial, Netlist, Inc. (“Patent Owner”) filed a Response to the Petition (Paper 19, “PO Resp.”), Petitioner filed a Reply (Paper 22, “Reply”), and Patent Owner filed a Sur-reply (Paper 23, “Sur-reply”).

An oral hearing was held on June 7, 2023, and a copy of the transcript was entered into the record. Paper 29 (“Tr.”).

We have jurisdiction under 35 U.S.C. § 6. This Decision is a Final Written Decision under 35 U.S.C. § 318(a) as to the patentability of the claims on which we instituted trial. Based on the complete record, Petitioner has shown by a preponderance of the evidence that claims 1, 3–17, and 19–30 of the ’833 patent are unpatentable.

II. BACKGROUND

A. *Real Parties-in-Interest*

Petitioner identifies itself as the real party-in-interest. Pet. 64. Patent Owner also identifies itself as the real party-in-interest. Paper 3 (“Patent Owner’s Mandatory Disclosure”), 1.

B. *Related Matters*

The parties advise that the ’833 patent is the subject of *Netlist, Inc. v. Micron Technology, Inc., et al.*, 6:21-cv-00430 (W.D. Tex.). Pet. 64; Paper 3, 2. Petitioner also advises that the ’833 patent has been the subject of the following three *inter partes* review proceedings: IPR2014-00994,

IPR2014-01370, and IPR2017-00649. Pet. 64. Additionally, Petitioner advises that “a nearly identical claim to the ’833 Patent’s claim 15 was invalidated as obvious in IPR2017-00692.” *Id.* at 65.

C. The ’833 Patent (Ex. 1001)

The ’833 patent is titled “Non-Volatile Memory Module” and is generally directed to “a memory system which can communicate with a host system such as a disk controller of a computer system.” Ex. 1001, codes (54), (57).

The ’833 patent states that “[n]on-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory,” so, “[i]n certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.” *Id.* at 1:28–31. The invention in the ’833 patent relates to a configuration of hybrid memory systems that “can include volatile and non-volatile memory and a controller which are configured such that the controller backs up the volatile memory using the non-volatile memory in the event of a trigger condition,” such as a power failure or power reduction. *Id.* at code (57), 3:65–67, 17:66–18:1. “[T]he volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power.” *Id.* at 4:41–44.

Figure 9, which depicts a flowchart of an example method of a volatile memory subsystem operating at a reduced rate in back-up mode, is reproduced below:

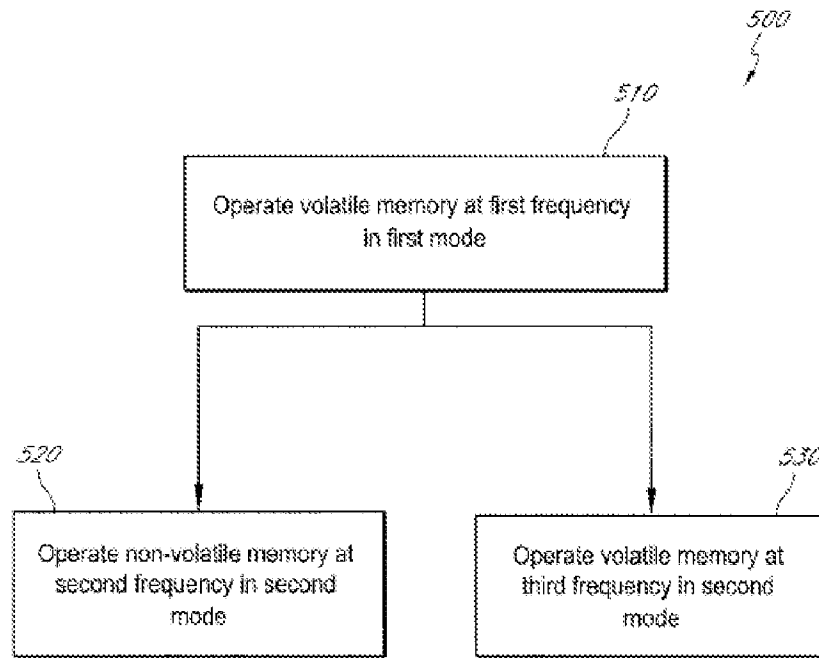


FIG. 9

Id. at 3:45–48. Figure 9 depicts block 510, “[o]perate volatile memory at first frequency in first mode,” block 520, “[o]perate non-volatile memory at second frequency in second mode,” and block 530, “[o]perate volatile memory at third frequency in second mode.” The memory system “may switch from the first mode of operation to the second mode of operation in response to a trigger condition,” such as “a power failure condition.” *Id.* at 17:64–18:1. The second mode of operation may include, for example, backup and/or restore operations. *Id.* at 18:1–7. The ’833 patent also describes that “[t]he third frequency can be less than the first frequency,” and “can be approximately equal to the second frequency.” *Id.* at 18:8–10.

D. Illustrative Claims

Among the challenged claims, claims 1 and 15 are independent.

Independent claim 15 is representative and is reproduced below.

15. A memory system operatively coupled to a host system, the memory system comprising:

a volatile memory subsystem operable at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

and a non-volatile memory subsystem operable at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem,

the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

Ex. 1001, 21:61–22:11.

E. Prior Art and Asserted Challenge to Patentability

Petitioner asserts that claims 1, 3–17, and 19–30 are unpatentable on the following challenge (Pet. 2):¹

¹ Petitioner also asserts, as a separate challenge, that Patent Owner is collaterally estopped from re-litigating the issue of validity of claims 1 and 15 of the '833 patent based upon the Final Written Decision in IPR2017-00692, where claim 15 of U.S. Patent No. 8,874,831, which is a continuation-in-part of the '833 patent, was determined invalid over the combination of Best, Bonella, and Mills. Pet. 18–24; IPR2017-00692, Paper 25 at 31–40. Petitioner asserts that claim 15 of the '831 patent is substantively identical to claims 1 and 15 of the '833 patent. Pet. 18. As we

Claim(s) Challenged	35 U.S.C. § ²	Reference(s)/Basis
1, 3–17, 19–30	§ 103(a)	Best, ³ Bonella, ⁴ Mills ⁵

In support of its proposed challenge, Petitioner relies on the Declaration of Ron Maltiel. *See* Ex. 1003. Patent Owner relies on the Declaration of Dr. Steven Przybylski. Ex. 2004. Deposition transcripts have been submitted for both witnesses. Exs. 1025 (deposition of Dr. Przybylski), 2005 (deposition of Mr. Maltiel).

III. ANALYSIS

A. Legal Standards

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the

stated in our Institution Decision, collateral estoppel is not an appropriate *separate* ground for *inter partes* review. Inst. Dec. 19 (citing 35 U.S.C. § 311(b), “A petitioner in an inter partes review may request to cancel as unpatentable 1 or more claims of a patent *only on a ground that could be raised under section 102 or 103 and only on the basis of prior art consisting of patents or printed publications.*”). Additionally, even if had been presented appropriately (i.e., as an argument prohibiting Patent Owner from contesting the unpatentability of claims 1 and 15 over Best, Bonella, and Mills rather than as a separate ground), because we determine that Petitioner has established that claims 1 and 15 are unpatentable based upon the combination of Best, Bonella, and Mills, it is not necessary to consider Petitioner’s arguments regarding collateral estoppel.

² Because the ’833 patent issued from a patent application that was filed before March 16, 2013, patentability is governed by the version of 35 U.S.C. § 103 preceding the Leahy-Smith America Invents Act (“AIA”), Pub L. No. 112–29, 125 Stat. 284 (2011).

³ US 2010/0110748 A1, published May 6, 2010 (Ex. 1006).

⁴ US 2007/0136523 A1, published June 14, 2007 (Ex. 1008).

⁵ US 6,026,465, issued Feb. 15, 2000 (Ex. 1009).

invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations. *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted); *see KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re NuVasive, Inc.*, 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’”).

B. Level of Ordinary Skill in the Art

Petitioner asserts a person of ordinary skill in the art “would have been a person with a bachelor’s degree in materials science, electrical engineering, computer engineering, computer science, or in a related field

and at least one year of experience with the design or development of semiconductor non-volatile memory circuitry or systems.” Pet. 4 (citing Ex. 1003 ¶¶ 48–52).

Patent Owner “disputes Petitioners’ definition of the level of a person of ordinary skill in the art . . . [but] resolution of such dispute is not necessary for the Board to determine the validity of the Challenged Claims.” PO Resp. 13. Dr. Przybylski testifies that he does not take exception to the education and years of experience identified by Mr. Maltiel, however, he believes that “an individual with that level of experience would have less knowledge and familiarity than Mr. Maltiel presumes, as described herein.” Ex. 2004 ¶ 42. We do not discern any further explanation of that statement in the testimony nor does Patent Owner propose any alternative qualifications of a person of ordinary skill in the art.

In our Institution Decision, we preliminarily adopted Petitioner’s unopposed⁶ proposed level of ordinary skill in the art. Inst. Dec. 17–18. Nothing in the full record persuades us that our preliminary finding as to the level of ordinary skill in the art was incorrect. Accordingly, we maintain our adoption of Petitioner’s proposed level of ordinary skill in the art, as consistent with the level of ordinary skill in the art reflected by the prior art of record. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

C. Claim Construction

Petitioner provides the parties’ proposed constructions from the related district court litigation, but asserts that “[t]hese construction disputes from the related litigations do not affect the outcome of this Petition with

⁶ Patent Owner did not challenge the proposed level of ordinary skill in the art in the Preliminary Response. *See generally* Paper 11.

respect to any claim.” Pet. 9 (citing Exs. 1012, 1013). Petitioner also asserts that the term “clock frequency” was construed in IPR2014-00994, Paper 8 at 6 as “identification of a clock running at a particular frequency,” which Petitioner contends is consistent with the ’833 patent’s Specification. *Id.* (citing Ex. 1001, 17:25–18:13; Ex. 1003 ¶ 136). Petitioner states that it has applied this construction of “clock frequency” in its analysis. *Id.*

Patent Owner asserts that “a first mode of operation” and “a second mode of operation,” as recited in independent claims 1 and 15, require construction. PO Resp. 15–27. Patent Owner contends that Petitioner improperly interprets “a first mode of operation” as “the operation of the system when no write-back trigger is detected,” and “a second mode of operation” as “the operation of the system when a power-loss event write-back trigger is detected.” *Id.* at 15.

1. *“a first mode of operation”*

a) *The Parties’ Contentions*

Patent Owner contends that “a first mode of operation” should be construed as “a mode of operation wherein no backup or restore operations are performed, wherein the non-volatile memory subsystem does not communicate with the volatile memory subsystem or the host system.” PO Resp. 16.

According to Patent Owner, the ’833 patent describes the first mode of operation as having “three key features.” PO Resp. 16 (citing Ex. 2004 ¶¶ 45–46).

First, Patent Owner contends that during a first mode of operation, “data is communicated between the volatile memory subsystem and the host

system.” PO Resp. 16 (citing Ex. 1001, 2:18–22, 2:35–37, 2:63–66, 15:6–10, Figs. 6, 7; Ex. 2004 ¶ 47).

Second, Patent Owner contends that during the first mode of operation, “memory system 10 is not performing a backup or restore operation,” and cites to disclosure from the ’833 patent allegedly supporting this assertion. PO Resp. 16–17 (citing Ex. 1001, 16:43–48, 9:15–17, 19:41–44). Patent Owner asserts, a person of ordinary skill in the art “would understand that no backup or restore operations are occurring during the ‘first mode of operation.’” *Id.* (citing Ex. 2004 ¶ 48).

Third, Patent Owner contends that during the first mode of operation, “the non-volatile memory subsystem does not communicate with the volatile memory subsystem or the host.” PO Resp. 17 (citing Ex. 2004 ¶ 49). Patent Owner argues that “the specification expressly touts the importance of isolating the non-volatile memory subsystem during the first mode of operation as advantageous.” *Id.* (citing Ex. 1001, 4:5–9, 8:57–62; Ex. 2004 ¶¶ 49–50). Patent Owner argues “[b]ecause of the importance of isolating the non-volatile memory subsystem during the first mode of operation, the ’833 Patent discloses multiple examples of ‘isolation devices’ used to isolate the data bus and address busses of the host system and the volatile memory system when the volatile memory is interacting with the host system.” *Id.* at 18. In support, Patent Owner cites to the embodiments disclosed in Figures 3, 4A, and 4B, and their associated circuit 52, which is described as a “switch” or “isolation device.” *Id.* at 18–21.

In Reply, Petitioner contends that Patent Owner’s proposed construction is improper and seeks “to fundamentally modify the claim language.” Reply 1–2. Petitioner contends that this term is clear on its face

because the claim language recites the requirements for the “first mode of operation,” and, therefore, this term need not be construed. *Id.* at 2–4.

Petitioner also argues that some dependent claims already recite the limitations Patent Owner proposes for the independent claims. *Id.* at 4–5.

Petitioner further argues that Dr. Przybylski’s testimony is unreliable because, among other things, it cites to portions of the Specification that refer to embodiments or to portions of the Specification that do not use the term “first mode of operation” and omit embodiment language, such as “for example,” from the cited portions of the Specification. *Id.* at 5–6.

In Sur-reply, Patent Owner argues that Petitioner’s application of Best to the “first mode of operation” defines the term with respect to backup or restore operations. Sur-reply 2. Patent Owner contends that the Specification and both experts support its proposed construction. *Id.* at 3–5. Regarding Petitioner’s claim differentiation argument, Patent Owner argues that Petitioner ignores claim scope differences between the independent and dependent claims. *Id.* at 3–4. For example, Patent Owner argues that “decouple from,” as recited in claim 16, conveys a narrower claim scope than “does not communicate with.” *Id.* at 3. Patent Owner also reiterates that the ’833 patent “consistently and repeatedly teaches the importance of not allowing the non-volatile memory to communicate in the first mode in order to preserve the memory system’s integrity and ensure its operation is not adversely affected.” *Id.* at 6; *see also id.* at 7–8.

b) Analysis

We construe each claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent,” the same

standard used to construe the claim in a civil action. 37 C.F.R. § 42.100(b) (2019). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–1317 (Fed. Cir. 2005) (en banc)). “[T]he claims themselves provide substantial guidance as to the meaning of particular claim terms,” but “do not stand alone.” *Phillips*, 415 F.3d at 1314–1315. They are part of “‘a fully integrated written instrument,’ consisting principally of a specification that concludes with the claims,” and must therefore “be read in view of the specification.” *Id.* at 1315 (quoting *Markman v. Westview Instruments, Inc.*, 52 F.3d 967, 978–79 (Fed. Cir. 1995)). There is a “heavy presumption,” however, that a claim term carries its ordinary and customary meaning. *CCS Fitness, Inc. v. Brunswick Corp.*, 288 F.3d 1359, 1366 (Fed. Cir. 2002) (citation omitted). “[T]he ordinary and customary meaning of a claim term is the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention.” *Phillips*, 415 F.3d at 1313.

Independent claim 15 recites, in part, “the memory system is in *a first mode of operation* in which data is communicated between the volatile memory subsystem and the host system.” Ex. 1001, 21:64–22:2 (emphasis added). We agree with Petitioner that the plain language of the claim clearly recites the requirements for the “first mode of operation,” i.e., data is communicated between the volatile memory subsystem and the host system. *See* Reply 3. This language replicates the first of the “three key features”

that Patent Owner contends is part of the “first mode of operation.” PO Resp. 16.

Patent Owner’s proposed construction seeks to limit “a first mode of operation” to particular embodiments disclosed in the Specification, which are themselves exemplary embodiments. Patent Owner supports its contentions that “no backup or restore operations are performed” in the “first mode of operation” with citations from the ’833 patent, but each of Patent Owner’s citations explicitly refers to examples and “certain embodiments.”⁷ See Ex. 1001, 16:43–48 (“*For example*, when the memory system 10 is not performing a backup or restore operation and is communicating with the host system (e.g., normal operation), the volatile memory system clock signal 420 may be provided by the multiplexer 422 to the volatile memory PLL block 424.”) (emphasis added), 9:15–17 (“*In certain embodiments*, the memory system 10 is in the first state when no trigger condition . . . is present”) (emphasis added), 19:41–44 (“*For example*, the memory system 10 may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation”) (emphasis added).

Patent Owner similarly provides citations to exemplary embodiments in support of its contention that the “non-volatile memory subsystem does not communicate with the volatile memory subsystem or the host system” during a “first mode of operation.” Patent Owner asserts that “the specification expressly touts the importance of isolating the non-volatile memory subsystem during the first mode of operation as advantageous,” (PO

⁷ In its citations, Patent Owner omits the language from the Specification referring to “for example” or “certain embodiments.”

Resp. 17) but Patent Owner’s citations to the ’833 patent again are directed to “certain embodiments.” *See* Ex. 1001, 4:5–9 (“*In certain embodiments*, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example”) (emphasis added); 8:22–23, 8:57–62 (“The memory system 10 *of certain embodiments* is configured to be operated in at least two states . . . The isolation or operational decoupling of the volatile memory subsystem 30 from the on-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 10 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 30”) (emphasis added).

Patent Owner’s reliance on switch 52 in the Figure 3, 4A, and 4B embodiments, and supporting citations is similarly unavailing. *See, e.g.*, Ex. 1001, 4:61–64 (“*In certain embodiments*, the memory system 10 includes at least one circuit 52 configured to selectively operatively decouple the controller 62 from the volatile memory subsystem 30”) (emphasis added); 6:63–7:4 (“*In certain embodiments*, the at least one circuit 52 comprises one or more switches coupled to the volatile memory subsystem 30, to the controller 62, and to the host computer (e.g., via the interface 22)”) (emphasis added); 7:4–6 (“In addition, *in certain embodiments* the at least one circuit 52 selectively operatively couples and decouples the volatile memory subsystem 30 and the host system”) (emphasis added); 7:11–16 (“*In certain such embodiments*, the at least one circuit 52 can comprise one or more switches 172 coupled to the controller 62 (e.g., logic element 70) and to the volatile memory subsystem 30”) (emphasis added).

We are not persuaded that the proper interpretation of “a first mode of operation” requires that we read in the unclaimed features of these embodiments. Claims are not limited to specific embodiments described in the specification. *See Phillips*, 415 F.3d at 1323.

Though understanding the claim language may be aided by explanations contained in the written description, it is important not to import into a claim limitations that are not a part of the claim. For example, a particular embodiment appearing in the written description may not be read into a claim when the claim language is broader than the embodiment.

SuperGuide Corp. v. DirecTV Enterprises, Inc., 358 F.3d 870, 875 (Fed Cir. 2004) (citing *Electro Med. Sys. S.A. v. Cooper Life Sci., Inc.*, 34 F.3d 1048, 1054 (Fed. Cir. 1994)); *see also Thorner v. Sony Computer Ent. Am. LLC*, 669 F.3d 1362, 1366 (Fed. Cir. 2012) (“It is likewise not enough that the only embodiments, or all of the embodiments, contain a particular limitation. We do not read limitations from the specification into claims; we do not redefine words.”). The mere description of a preferred embodiment is not a limiting definition for a claim term. *See E-Pass Techs., Inc. v. 3Com Corp.*, 343 F.3d 1364, 1369 (Fed. Cir. 2003); *Liebel-Flarsheim Co. v. Medrad, Inc.*, 358 F.3d 898, 908–09 (Fed. Cir. 2004).

Patent Owner’s contention that the ’833 patent includes “numerous and consistent teachings describing the importance of the structures used to perform the claimed functions” is not supported by the ’833 patent. Although the specification’s emphasis of a particular feature’s importance to the invention can indicate “what the patentee had claimed and disclaimed,” *see SafeTCare Mfg., Inc. v. Tele-Made, Inc.*, 497 F.3d 1262, 1269–70 (Fed. Cir. 2007), Patent Owner has not shown that here. The

Federal Circuit has acknowledged “the difficulty in drawing the ‘fine line between construing the claims in light of the specification and improperly importing a limitation from the specification into the claims.’” *Continental Circuits LLC v. Intel Corp.*, 915 F.3d 788, 797 (Fed. Cir. 2019) (citing *Retractable Techs., Inc. v. Becton, Dickinson & Co.*, 653 F.3d 1296, 1305 (Fed. Cir. 2011)). To avoid improperly importing limitations into the claims, “it is important to keep in mind that the purposes of the specification are to teach and enable those of skill in the art to make and use the invention and to provide a best mode for doing so.” *Phillips*, 415 F.3d at 1323.

The written description includes embodiments with a switch that may operate to isolate the non-volatile memory subsystem from the volatile memory subsystem or the host system, but does not indicate that this is an important or essential part of the invention, or that it is important or essential in distinguishing the prior art. Patent Owner focuses on the following two statements from the Specification: (1) “In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system”; and (2) “The isolation or operational decoupling of the volatile memory subsystem 30 from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 10 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 30.” *See* PO Resp. 17 (citing Ex. 1001, 4:5–9, 8:57–62); Sur-reply 6. However, as discussed above, these statements are limited to “certain embodiments” and are not statements clearly limiting or expressly requiring that the claimed “first

mode of operation” include this functionality, or setting forth its importance. *See, e.g., Continental Circo.*, 915 F.3d at 797; *Hill-Rom Services, Inc. v. Stryker Corp.*, 755 F.3d 1367, 1372–1373 (Fed. Cir. 2014). Indeed, the ’833 patent expressly states that “[a]lthough this invention has been described with reference to these specific embodiments, the descriptions are intended to be illustrative of the invention and are not intended to be limiting.” Ex. 1001, 20:64–67. Nothing in the language of the Specification suggests that the “first mode of operation” should be limited to operations when “no backup or restore operations are performed” and “the non-volatile memory subsystem does not communicate with the volatile memory subsystem or the host system.” Thus, there is no reason to depart from the plain and ordinary meaning of a “first mode of operation,” as recited in the claim language.

Moreover, we find Patent Owner’s assertions that Mr. Maltiel agrees with its interpretation of this term to be disingenuous. *See* Sur-reply 4–6. Patent Owner cites to testimony on pages 58 and 59 of Mr. Maltiel’s deposition in support of its assertions that Mr. Maltiel testified that the non-volatile memory does not communicate with the volatile memory in the first mode (*id.* at 4–5), but when Mr. Maltiel’s testimony is read in context, it is clear that Mr. Maltiel was being questioned about certain passages in the ’833 patent, not about the interpretation of a “first mode of operation” as recited in the claims. *See* Ex. 2005, 53:3–54:20 (asking Mr. Maltiel to read aloud the sentences beginning at Ex. 1001, 2:18, and 2:34); *see also id.* at 59:5–8 (“Q. And isn’t it true that *neither of those passages* discuss the communication of data between the volatile memory and the nonvolatile memory in the first mode of operation?”) (emphasis added); *see* Tr. 38:9–16.

As discussed above, there is no dispute that during a first mode of operation “data is communicated between the volatile memory subsystem and the host system.” PO Resp. 16; Ex. 2005, 56:20–57:1. This is precisely what the claim language recites. For the reasons explained above, we decline to import limitations into the claim, as Patent Owner urges. *See, e.g., SuperGuide Corp.*, 358 F.3d at 875 (“Though understanding the claim language may be aided by explanations contained in the written description, it is important not to import into a claim limitations that are not a part of the claim.”)

Accordingly, we agree with Petitioner that the term “a first mode of operation” does not require construction, and its meaning is evident from the claim language, i.e., “a first mode of operation in which data is communicated between the volatile memory subsystem and the host system.”

2. “*a second mode of operation*”

a) *The Parties’ Contentions*

Patent Owner contends that “a second mode of operation” should be construed as “a backup or restore mode of operation wherein the host system does not communicate with the volatile memory subsystem.” PO Resp. 23.

According to Patent Owner, the ’833 patent “makes clear that the second mode of operation is a mode in which backup or restore operations are performed.” PO Resp. 23 (citing Ex. 1001, 7:41–48, 10:33–58, 18:1–7; Ex. 2004 ¶ 61). Patent Owner also contends that Mr. Maltiel agrees with this interpretation. *Id.* (citing Ex. 2005, 68:5–8). As with the “first mode of operation,” Patent Owner similarly relies on circuit 52 (switch) and switch 174 as shown in Figures 3, 4A and 4B, respectively, which show

“that the host system may be isolated from the volatile memory subsystem using isolation devices, such as switches.” *Id.* at 24–26 (citing Ex. 1001, 7:20–22, 7:26–29, 8:22–35, 10:19–23, Figs. 3, 4A, 4B; Ex. 2004 ¶¶ 62–64, 66). Similarly, Patent Owner contends that Mr. Maltiel agrees that “in the second mode of operation, when backup or restore operations are performed, the volatile memory does not communicate with the host.” *Id.* at 26 (citing Ex. 2005, 68:13–19).

In Reply, Petitioner contends that Patent Owner’s proposed construction is improper and seeks “to fundamentally modify the claim language.” Reply 1–2. Petitioner contends that this term is clear on its face because the claim language clearly recites the requirements for the “second mode of operation,” and, therefore, this term need not be construed. *Id.* at 2–4. Petitioner also argues that some dependent claims already recite the limitations Patent Owner proposes for the independent claims. *Id.* at 4–5. Petitioner also argues that Dr. Przybylski’s testimony is unreliable because, among other things, it cites to portions of the Specification that refer to embodiments or do not use the term “second mode of operation” and omits embodiment language, such as “for example,” from the cited portions of the Specification. *Id.* at 5–6.

In Sur-reply, Patent Owner argues that Petitioner’s application of Best to the “second mode of operation” defines the term with respect to backup or restore operations. Sur-reply 2. Patent Owner contends that the Specification and both experts support its proposed construction. *Id.* at 3–5. Regarding Petitioner’s claim differentiation argument, Patent Owner argues that Petitioner ignores claim scope differences between the independent and dependent claims. *Id.* at 3–4. For example, Patent Owner argues that the

language in dependent claim 28, which recites “wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem,” conveys a narrower claim scope than the language in claim 15. *Id.* Patent Owner also reiterates that the ’833 patent “consistently and repeatedly teaches the importance of not allowing the non-volatile memory to communicate in the first mode in order to preserve the memory system’s integrity and ensure its operation is not adversely affected.” *Id.* at 6; *see also id.* at 7–8.

b) Analysis

Independent claim 15 recites, in part, “the memory system is in *a second mode of operation* in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem.” Ex. 1001, 22:4–7 (emphasis added). We agree with Petitioner that the plain language of the claim clearly recites the requirements for the second mode of operation, i.e., data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem. Reply 3.

We do not adopt Patent Owner’s proposed construction for the same reasons as discussed above for “a first mode of operations.” For example, we find that Patent Owner’s proposed construction seeks to limit “a second mode of operation” to particular embodiments disclosed in the Specification, which are themselves exemplary embodiments. *See* Ex. 1001, 7:41–48 (“*Certain embodiments* described herein utilize the non-volatile memory subsystem 40 as a flash ‘mirror’ to provide backup of the volatile memory subsystem 30 . . .”) (emphasis added); 10:33–58 (*In certain embodiments*, the memory system 10 protects the operation of the volatile memory when

communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure”) (emphasis added); 18:1–7 (“*In certain embodiments*, the second mode of operation includes both backup and restore operations as described herein. *In other embodiments*, the second mode of operation includes backup operations but not restore operations. *In yet other embodiments*, the second mode of operation includes restore operations but not backup operations”) (emphasis added). We note that the ’833 patent supports that the “second mode of operation” could be a backup and restore operation, but not that it is *limited to* a backup and restore operation, as Patent Owner argues.

Patent Owner’s reliance on switch 52 in the Figure 3, 4A, and 4B embodiments, and supporting citations, to support that the host system may be isolated from the volatile memory subsystem, are similarly unavailing, for the same reasons set forth above. *See, e.g.*, Ex. 1001, 7:20–29 (referring to “*certain embodiments*”), 8:22–35 (“The memory subsystem 10 of *certain embodiments* is configured to be operated in at least two states. . . a second state in which the volatile memory subsystem 30 is operatively coupled to the controller 62 to allow data to be communicated between the volatile memory subsystem 30 and the non-volatile memory subsystem 40 via the controller 62”) (emphasis added).

Moreover, we again find Patent Owner’s assertions that Mr. Maltiel agrees with its interpretation of this term to be disingenuous. *See* PO Resp. 23, 26; Sur-reply 4–6. Patent Owner cites to testimony on page 68 of Mr. Maltiel’s deposition in support of its assertions that Mr. Maltiel agrees that backup or restore operations are performed in the second mode and testified that the non-volatile memory does not communicate with the host in

the second mode (PO Resp. 23, 26; Sur-reply 4–5), but when Mr. Maltiel’s testimony is read in context, it is clear that Mr. Maltiel was being questioned about certain passages in the ’833 patent, not about the interpretation of a “second mode of operation” as recited in the claims. *See* Ex. 2005, 63:5–21 (asking Mr. Maltiel to read the paragraph beginning at Ex. 1001, 6:63).

There appears to be no dispute that during a second mode of operation “data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem,” as recited in the claim language. For the reasons explained above, we decline to import limitations into the claim, as Patent Owner urges. *See, e.g., SuperGuide Corp.* 58 F.3d at 875. Accordingly, we agree with Petitioner that the term “a second mode of operation” does not require construction, and its meaning is evident from the claim language, i.e., “a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem.”

D. Alleged Obviousness Over Best, Bonella, and Mills

Petitioner contends claims 1, 3–17, and 19–30 would have been obvious over the combination of Best, Bonella, and Mills. Pet. 24–60. After reviewing the entire record developed at trial, we determine that Petitioner has shown by a preponderance of the evidence that claims 1, 3–17, and 19–30 are unpatentable over Best, Bonella, and Mills.

1. Best (Ex. 1006)

Best is titled “Hybrid Volatile and Non-Volatile Memory Device” and is generally directed to a “composite, hybrid memory device including a first storage die having an array of volatile storage cells and a second storage die having an array of non-volatile storage cells disposed within an integrated

circuit package.” Ex. 1006, codes (54), (57). “The hybrid memory device includes a shared interface circuit to receive memory access commands directed to the first storage die and the second storage die and to convey read and write data between an external data path and the first and second storage dice.” *Id.* at code (57).

Figure 2 of Best, which illustrates an embodiment of a hybrid, composite memory device, is reproduced below.

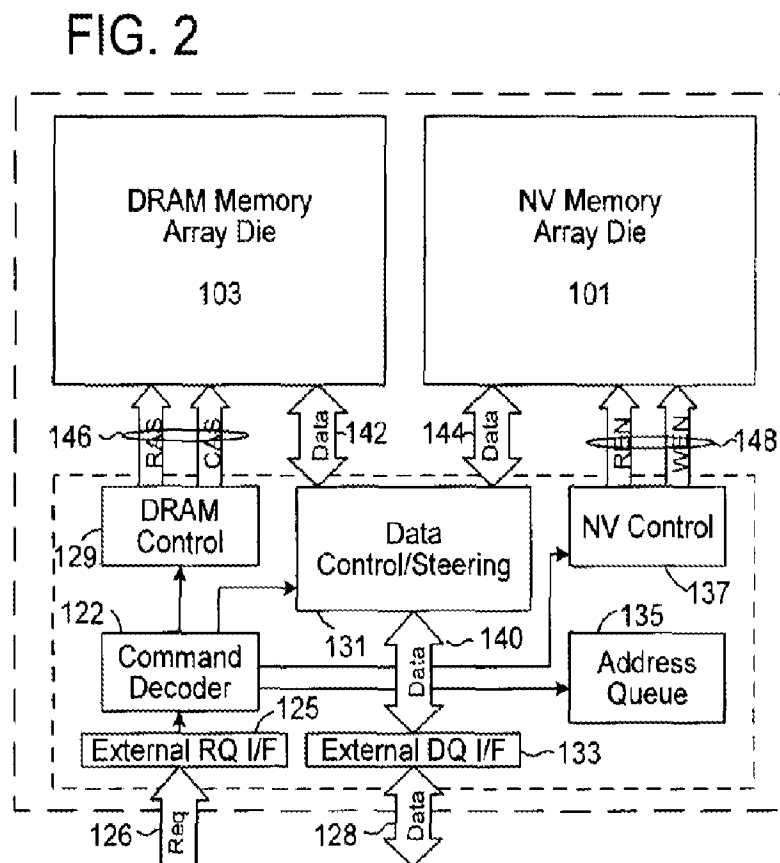


Figure 2, depicted above, illustrates an embodiment of a hybrid, composite memory device with shared interface circuitry, volatile memory implemented with DRAM die 103, and non-volatile memory implemented with Flash memory die 101. *Id.* ¶ 17. “[T]he shared interface circuitry

includes an external request interface 125, external data interface 133, command decoder 122, address queue 135, DRAM control circuit 129, Flash control circuit 137, and data control/steering circuit 131.” *Id.* “[I]ncoming control signals and addresses . . . are received in the external request interface 125 via control/address (CA) path 126, reformatted as necessary (e.g., deserialized to form a parallel command word and one or more address values) and then forwarded to the command decoder 122.” *Id.* “The command decoder 122 in turn forwards address to the address queue 135 and stores memory access commands.” *Id.* “[M]emory access operations may be automatically directed to either the volatile storage die or non-volatile storage die according to the memory address to be accessed.” *Id.* “[C]ommand decoder 122 outputs, from the head of the command queue, an enable signal and corresponding memory access control signals to the DRAM control circuit 129 and NV control circuit 137.” *Id.* ¶ 18. “[D]ata control/steering circuit 131 is used to control the transfer of data between a shared internal data bus and dedicated internal data buses associated with the volatile and non-volatile storage dice, respectively.” *Id.* ¶ 20.

Figure 3 of Best, which illustrates an embodiment of a data control/steering circuit, is reproduced below.

FIG. 3

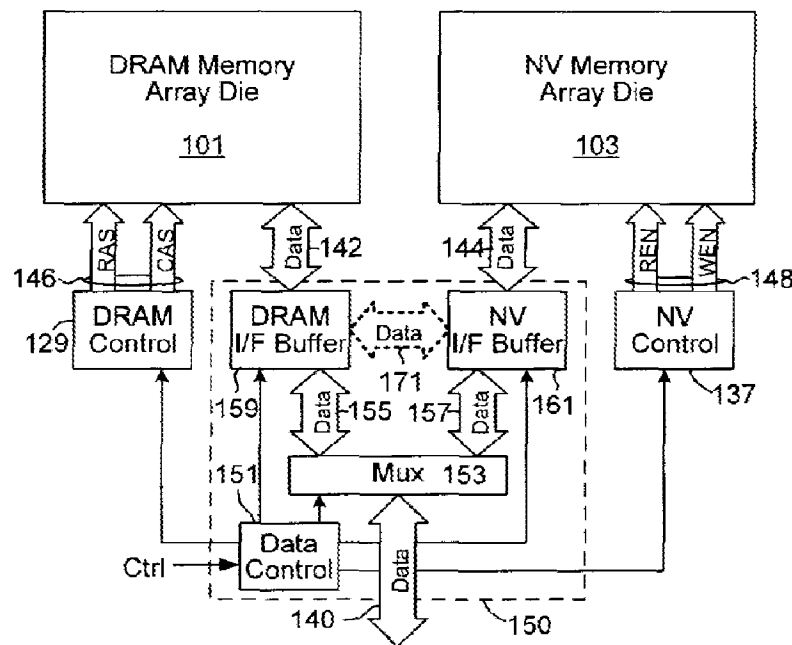


Figure 3, depicted above, illustrates an embodiment of a data control/steering circuit 150 that may be used to implement the data control/steering circuit 131 of Figure 2. *Id.* ¶ 21. “[D]ata control circuit 151 receives control signals from the command decoder that indicate the direction of data flow during a memory access operation (read or write) and whether the volatile or non-volatile storage die is the target of the memory access.” *Id.*

2. *Bonella* (Ex. 1008)

Bonella is titled “Advanced Dynamic Disk Memory Module Special Operations” and is generally directed to “[a] memory module including a volatile memory, a non-volatile memory, and a controller that provides address, data, and control interfaces to the memories and to a host system.” Ex. 1008, codes (54), (57). *Bonella* states that “[r]eduction of power in the

DRAM can be accomplished” by “reduc[ing] the frequency in which the DRAM is operating,” which “reduces power and, in general, produces no noticeable decrease in system performance.” *Id.* ¶ 49; *see id.* ¶ 50.

3. *Mills (Ex. 1009)*

Mills is titled “Flash Memory Including a Mode Register for Indicating Synchronous or Asynchronous Mode of Operation” and is generally directed to a “flash memory chip that can be switched into four different read modes.” Ex. 1009, codes (54), (57). Mills describes several interfaces for a FLASH memory device, one of which is a synchronous FLASH interface, i.e. “a synchronous flash interface (SFI) flash memory integrated circuit 600 that incorporates a complete synchronous flash interface in a single flash memory chip.” *Id.* at 16:60–63. The synchronous FLASH interface includes a clock input such that all the external operations of the device are synchronized to the rising edge of the clock. *Id.* at 17:10–25. Mills teaches that this synchronous operation is used for both read operations and write operations: “When SFI is enabled, interlace control 670 and [bank] select logic 674 operate to interlace read (and write) operations between flash bank A 610 and a flash bank B 620” *Id.* at 17:33–39. Because “the device is interleaved internally,” it “creates an average access time for sequential read accesses that is significantly less than the access time of an asynchronous flash device.” *Id.* at 17:1–9.

4. *Analysis of Independent Claims 1 and 15*

Petitioner contends that Best, Bonella, and Mills, alone or in combination, teach each limitation of independent claims 1 and 15. Pet. 24–38. In support, Petitioner identifies certain passages and figures in the references and explains their significance with respect to the corresponding

claim limitation. *Id.* Petitioner also provides reasons, supported by the testimony of Mr. Maltiel, why it would have been obvious to one of ordinary skill in the art to combine Best, Bonella, and Mills. *Id.* Petitioner analyzes claims 1 and 15 together.⁸ *Id.* Claims 1 and 15 are substantially similar, although claim 1 is a method claim and claim 15 is a system claim. *See, e.g.,* Ex. 1001, 21:6–22, 21:61–22:11. We select claim 15 as representative.

a) *“A memory system operatively coupled to a host system, the memory system comprising:”*

Petitioner relies on Best to teach the preamble. Pet. 25–26. Specifically, Petitioner contends that Best’s memory device (as shown in Fig. 2) is “operatively coupled to a host system” because it “includes an interface that ‘receive[s] commands from a controller device.’” *Id.* at 25 (citing Ex. 1006 ¶ 14; Ex. 1003 ¶ 142). According to Petitioner, “Best discloses techniques for using the shared interface to control access by the controller device to the non-volatile and/or volatile memories.” *Id.* at 25–26 (citing Ex. 1006 ¶¶ 12–17, Fig. 2; Ex. 1003 ¶¶ 140–142).

Patent Owner does not specifically respond to these arguments. *See generally* PO Resp.

After considering the evidence and arguments of the complete record, and for the reasons set forth by Petitioner, we determine that Petitioner has shown, by a preponderance of the evidence, that Best teaches the preamble.⁹

b) *“a volatile memory subsystem operable at a first clock frequency when the memory system is in a first mode of*

⁸ Patent Owner also addresses claims 1 and 15 together. PO Resp. 35 n.3.

⁹ Neither party argues whether the preamble limits claim 1 or claim 15. Because we find that the evidence supports that the prior art teaches the preamble, we need not make a determination whether the preamble of claim 1 and claim 15 is limiting.

operation in which data is communicated between the volatile memory subsystem and the host system;”

Petitioner relies on Best to teach this limitation. Pet. 26–29.

Specifically, Petitioner contends that Best’s DRAM teaches “a volatile memory subsystem.” *Id.* at 26 (citing Ex. 1006 ¶¶ 17, 21; Ex. 1003 ¶ 145).

Petitioner further contends that Best’s volatile memory (DRAM) die can be synchronous or asynchronous, and that Best “explains that a conventional synchronous DRAM interface relies on ‘clock and clock-enable signals.’” *Id.* at 26–27 (citing Ex. 1006 ¶ 14). According to Petitioner, a person of ordinary skill in the art would understand this “to mean that a synchronous DRAM is ‘operable at a first clock frequency’ because it is operated synchronously by a clock signal.” *Id.* at 27 (citing Ex. 1003 ¶¶ 146–147). Petitioner contends that “[t]his corresponds to the usual operation of DRAM memory, as would have been well-known by a” person of ordinary skill in the art. *Id.*

Petitioner further contends that Best teaches “a first mode of operation in which data is communicated between the volatile memory subsystem and the host system” because “Best discloses a ‘Shadow Operation’¹⁰ embodiment where a portion of the DRAM is used as a write buffer for the non-volatile memory.” Pet. 27 (citing Ex. 1006 ¶ 24; Ex. 1003 ¶ 148). In this embodiment, “[d]ata is stored in a fast access volatile storage die,” and

¹⁰ Best states: “In memory shadowing, memory accesses are directed first to the non-volatile storage device to enable relatively high-speed, non-degrading memory accesses, with the data eventually being written-back (retired or recorded) within the non-volatile storage in response to a triggering event referred to herein as a write-back trigger. Applications of memory shadowing include non-volatile DRAM, and rapid-access non-volatile memory.” Ex. 1006 ¶ 24.

later “a write-back trigger is detected within the shared interface circuitry,” such as a “power-loss” event. *Id.* (citing Ex. 1006 ¶¶ 25–26). According to Petitioner, “[o]nce the power-loss event is detected, ‘internal data transfer operations are performed to transfer data . . . from the DRAM to . . . the NV memory.’” *Id.* (citing Ex. 1006 ¶¶ 25–26; Ex. 1003 ¶ 148) (second two alterations in original). Petitioner, therefore, asserts that Best “describes at least two modes of operation: (1) the operation of the system when no write-back trigger is detected (*‘a first mode of operation’*), and (2) the operation of the system when a power-loss event write-back trigger is detected such as a power-loss (*‘a second mode of operation’*).” *Id.* at 28 (citing Ex. 1006 ¶¶ 25–26; Ex. 1003 ¶¶ 148–150).

According to Petitioner, Best’s “*first mode of operation*” “is one in which ‘data is communicated between the volatile memory subsystem and the host system’ because initially ‘data is stored in a fast-access volatile storage die’ and only later after a write-back trigger is written to non-volatile memory.” Pet. 28 (citing Ex. 1006 ¶ 25; Ex. 1003 ¶¶ 148–150).

Petitioner further contends that it would have been obvious to operate Best’s DRAM at any of the frequencies described by the DDR DRAM and DDR2 DRAM standards because it “would have been merely the use of a known structure (a standard DRAM module and clock frequency) for its known use (operating as a DRAM module) to achieve a predictable result (operating a DRAM module at a standard clock frequency in Best’s system).” *Id.* at 28–29 (citing Ex. 1003 ¶¶ 151–152).

Patent Owner argues that Best’s “Shadow Operation” does not disclose “*a first mode of operation.*” PO Resp. 35–36. Specifically, Patent Owner argues that during Best’s normal operation, when no write-back

trigger is detected, “the host communicates with **both** the volatile memory and the non-volatile memories.” *Id.* at 36 (citing Ex. 2004 ¶ 83). According to Patent Owner, Best “repeatedly and consistently emphasizes the host’s ability to communicate with both the volatile and non-volatile memories, even during ‘Shadow Operation.’” *Id.* at 37 (citing Ex. 2004 ¶ 86).

We agree with Petitioner (Reply 17) that Patent Owner’s arguments are based upon its proposed claim construction for a “first mode of operation.” As discussed above, we decline to adopt Patent Owner’s proposed construction that interprets the “first mode of operation” to *require* that the “the non-volatile memory subsystem does not communicate with the volatile memory subsystem or the host system.” Rather, the claim simply recites that “in a first mode of operation . . . data is communicated between the volatile memory subsystem and the host system.” Patent Owner does not appear to dispute that in Best, data is communicated between the volatile memory subsystem and the host system. *See* PO Resp. 36 (“Best teaches that during normal operation, when no write-back trigger is detected, the host communicates with both the volatile memory and the non-volatile memories.”).

Therefore, after considering the evidence and arguments of the complete record, and for the reasons set forth by Petitioner, we determine that Petitioner has shown, by a preponderance of the evidence, that Best teaches this limitation.

- c) “*and a non-volatile memory subsystem operable at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem,*”

Petitioner relies on the combination of Best and Mills to teach this limitation. Pet. 29–34. Specifically, Petitioner contends that Best’s Flash memory teaches “a non-volatile memory subsystem.” *Id.* at 30 (citing Ex. 1006 ¶¶ 13, 17; Ex. 1003 ¶ 155). Petitioner relies on Best’s “Shadow Operation” embodiment, discussed above, to teach “a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem.” *Id.* (citing Ex. 1006 ¶¶ 24–26; Ex. 1003 ¶ 156). According to Petitioner, Best’s “‘*second mode of operation*’ (i.e., the operation of the system when a power-loss write-back trigger is detected) involves ‘*a non-volatile memory subsystem operable*’ (Best’s ‘NV’ memory) such that ‘*data is communicated between the volatile memory subsystem and the non-volatile memory subsystem,*’ e.g. when ‘internal data transfer operations are performed to transfer data . . . from the DRAM to . . . the NV memory.’” *Id.* at 31 (citing Ex. 1006 ¶¶ 25–26; Ex. 1003 ¶¶ 156–158).

Petitioner contends that “Best does not explicitly disclose ‘operating the non-volatile memory subsystem’ (the Flash memory) ‘at a second clock frequency’ during this second mode, but it would have been obvious to include that functionality in the system of Best.” Pet. 31 (citing Ex. 1003 ¶ 159). Petitioner, therefore, relies on a Flash implementation in Mills, which Petitioner asserts describes a synchronous flash interface that causes the Flash memory to “*opera[te] at a second clock frequency.*” *Id.* at 31–32

(citing Ex. 1009, 16:60–63, 17:10–25; Ex. 1003 ¶ 160) (alteration in original). According to Petitioner, “Mills teaches that this synchronous operation is used for both read operations and write operations.” *Id.* at 32.

Petitioner asserts that Best and Mills are analogous art to the ’833 patent. Pet. 32. Petitioner further asserts that the combination (i.e., employing a synchronous flash memory as described in Mills in the system of Best) would have been obvious to a person of ordinary skill in the art because it “would have been merely an arrangement of old elements with each performing the same function it had been known to perform and yielding no more than what one would expect from such an arrangement, i.e. the non-volatile storage of data.” *Id.* at 32–33 (citing Ex. 1003 ¶ 163). Petitioner contends that, as combined, Best’s Flash memory would operate “at a second clock frequency.” *Id.* at 33 (citing Ex. 1003 ¶ 164). Petitioner contends that one of ordinary skill in the art would have been motivated to make the combination because “restoring data from the non-volatile flash memory would therefore have been faster by use of a synchronous flash memory, and reduced sequential read access times during other operations or uses of Best’s Flash memory.” *Id.* at 33–34 (citing Ex. 1003 ¶ 165). Petitioner also asserts that a person of ordinary skill in the art would have understood Best to work with any known Flash interface, including Mills’ synchronous Flash interface. *Id.* at 34 (citing Ex. 1003 ¶ 166).

Patent Owner argues that Best’s “Shadow Operation” does not disclose “*a second mode of operation.*” PO Resp. 38–40. Specifically, Patent Owner contends that “Best does not teach that the host system does not communicate with the volatile memory subsystem as part of the ‘Shadow Operation’ embodiment when a power-loss event write-back

trigger is detected.” *Id.* at 38; *see id.* at 40–41. According to Patent Owner, “Best teaches away from isolating the host system from the volatile memory subsystem when a power-loss event write-back trigger is detected by emphasizing that the volatile and non-volatile memories can be accessed simultaneously even during ‘Shadow Operation.’” *Id.* at 38–39 (citing Ex. 1006 ¶¶ 18, 29; Ex. 2004 ¶ 91)).

We agree with Petitioner (Reply 17) that Patent Owner’s arguments are based upon its proposed claim construction for a “second mode of operation.” As discussed above, we decline to adopt Patent Owner’s proposed construction that interprets the “second mode of operation” to require that the “the host system does not communicate with the volatile memory subsystem.” Rather, the claim recites “a second mode of operation in which data is communicated between the volatile memory subsystem and the nonvolatile memory subsystem.” We agree with Petitioner that Best teaches this limitation. *See, e.g.*, Ex. 1006 ¶ 25 (“Some time after one or more write operations have been performed, a write-back trigger is detected within the shared interface circuitry and, in response, one or more internal data transfer operations are performed *to transfer data* from the locations indicated by the write-back table to be dirty . . . *from the DRAM to corresponding locations (e.g., 205) within the NV memory*”) (emphasis added).

Therefore, after considering the evidence and arguments of the complete record, and for the reasons set forth by Petitioner, we determine that Petitioner has shown, by a preponderance of the evidence, that the combination of Best and Mills teaches this limitation.

- d) *“the volatile memory subsystem further being operable at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.”*

(1) The Parties’ Contentions

Petitioner relies on the combination of Best and Bonella to teach this limitation. Pet. 34–38. Petitioner asserts that Best does not explicitly disclose this limitation, but it would have been obvious for two reasons: “(1) reducing power during volatile to non-volatile flush operations prompted by a power loss was a well-known technique, and (2) one known way to reduce the power consumption of DRAM devices was to reduce their frequency of operation.” *Id.* at 35 (citing Ex. 1003 ¶ 170). According to Petitioner, it would have been obvious to operate Best’s DRAM “*at a third clock frequency*” that is “*less than the [first clock] frequency*” when performing Best’s flush operation (i.e., “*when the memory system is in the second mode of operation.*”). *Id.*

Petitioner contends that the first reason and its advantages were well known in the art, and cites to Ashmore¹¹ and Long¹² in support of this assertion. Pet. 35–36 (citing Ex. 1010 ¶ 9; Ex. 1011, 4:54–64; Ex. 1003 ¶¶ 171–172). Petitioner contends that a person of ordinary skill in the art “would have been motivated to reduce the power consumption during Best’s write flushing in response to a power loss” (1) for all the reasons that were known in the art, such as decreasing the risk of data loss due to insufficient backup power and enabling the use of a smaller-sized backup power source; and (2) because it would have been the arrangement of old elements, each

¹¹ US 2006/0212651 A1, published Sep. 21, 2006 (Ex. 1010).

¹² US 7,421,552 B2, issued Sep. 2, 2008 (Ex. 1011).

performing the same function it had been known to perform, in a way that yields no more than what would be expected from such an arrangement, i.e., reducing power consumption during a power loss event, as suggested by Long and Ashmore. *Id.* at 36 (citing Ex. 1010 ¶ 7; Ex. 1011, 4:54–64; Ex. 1003 ¶ 173).

Petitioner relies on Bonella in support of the second reason, asserting that Bonella “explains that one way to reduce the power consumption of the memory module is to slow or reduce the operating frequency of the DRAM.” Pet. 37 (citing Ex. 1008 ¶¶ 49–50; Ex. 1003 ¶ 174). Petitioner contends that Best and Bonella are analogous art to the ’833 patent because both are from the field of memory systems. *Id.* Petitioner contends that a person of ordinary skill in the art “would also have found it obvious to reduce power consumption during Best’s write flushing in response to a power loss using any known or conventional means, and would have also considered power consumption reduction techniques other than those of Ashmore and Long to obtain the same benefits, including those described in Bonella.” *Id.* According to Petitioner, a person of ordinary skill in the art “would have found it obvious to perform Bonella’s power reduction technique (DRAM operational frequency reduction) during the performance of Best’s write flushing in response to a power loss, in view of the specific motivations evidenced by Ashmore and Long.” *Id.* (citing Ex. 1003 ¶ 176). Petitioner also argues that Bonella itself provides motivation because “it results in ‘major power savings.’” *Id.* (citing Ex. 1008 ¶ 50). Petitioner contends that “[u]sing Bonella’s power reduction technique during Best’s write flushing would also have been the arrangement of old elements, each performing the same function it had been known to perform, in a way that

yields no more” than would be expected from such an arrangement. *Id.* at 37–38.

Patent Owner argues that a person of ordinary skill in the art would not have been motivated to combine Bonella with Best. PO Resp. 45. Specifically, Patent Owner argues that “[r]educing the DRAM frequency, as taught by Bonella, reduces DRAM performance” and “Best emphasizes the need to maintain high-speed, non-degraded DRAM performance, including during write-back operations.” *Id.* at 49; *id.* at 45–46 (citing Ex. 2004 ¶ 109; Ex. 1006 ¶ 12); *see id.* at 46 (citing Ex. 1006 ¶ 18; Ex. 2004 ¶ 110). Patent Owner argues that “Best teaches away from reducing DRAM performance during write-back operation.” *Id.* at 45 (citing Ex. 2004 ¶ 109).

Patent Owner argues that “Best neither teaches nor suggests . . . that the hybrid memory device behaves differently during a write-back operation depending on the write-back trigger.” PO Resp. 47 (citing Ex. 2004 ¶ 112). Patent Owner argues that “all of Best’s potential write-back triggers would occur during normal operation during which, Best teaches, the performance of the hybrid memory device should not be degraded.” *Id.* (citing Ex. 2004 ¶ 112). Patent Owner also argues that “Best teaches away from reducing the DRAM’s clock frequency, and thus its performance, during write-back operations, because the host may access the DRAM during such operations.” *Id.* at 48 (citing Ex. 2004 ¶ 113).

Patent Owner argues that “Best’s write-back operations triggered by a power-loss event are the same as write-back operations initiated by other triggers.” PO Resp. 49 (citing Ex. 1006 ¶ 26; Ex. 2004 ¶ 117). Therefore, according to Patent Owner, the “same teaching away applies to write-back operations that are triggered by a power-loss event.” *Id.* at 48 (citing

Ex. 2004 ¶ 114). Patent Owner contends that “Best’s power-loss trigger indicates that power will be unavailable soon, not that power is already unavailable.” *Id.* (citing Ex. 1006 ¶ 26). Patent Owner argues that “other operations within the host system would need to be performed prior to the loss of power . . . [and a person of ordinary skill in the art] would understand that these common shutdown tasks, which would be performed by the host system, should not be delayed by artificially reducing the frequency of—and, thus, the performance of—Best’s DRAM.” *Id.* (citing Ex. 2004 ¶ 114).

In Reply, Petitioner argues that Patent Owner “fails to show that Best ever discloses or teaches that the DRAM must maintain high-speed and high-performance communication with the host even after it has lost power or is powering down.” Reply 19. Petitioner contends that Best does not support Patent Owner’s assertions that write-back operations can be triggered by “a power-loss trigger [that] indicates that power will be unavailable soon, not that power is already unavailable.” *Id.* (alteration in original). Rather, Petitioner contends that Best “discloses that the trigger that initiates the write-back flush operations is the actual detection of ‘a power-loss or power-down signal or event,’ . . . not the near-future unavailability of power, as [Patent Owner] suggests.” *Id.* at 20. Petitioner further argues that “[t]here is no disclosure or teaching [in Best] that the DRAM must maintain, or is even capable of maintaining, high-speed and high-performance read/write communications with the host during a power-loss event when ‘dirty data’ is being written to Flash.” *Id.*

In Sur-reply, Patent Owner cites to paragraph 26 of Best, and argues that a person of ordinary skill in the art “would understand that Best teaches performing write-backs as part of a power down procedure, prior to the

removal of power to the system.” Sur-reply 22 (emphasis omitted). According to Patent Owner, a person of ordinary skill in the art “would understand that the ability to perform any module operation after the removal or failure of system power would require an alternate power source such as a battery or capacitor,” and Best “does not disclose a secondary power source.” *Id.* Patent Owner repeats its contention that “Best does not teach that his device performs differently depending on the write-back trigger source.” *Id.* at 23.

(2) *Analysis*

On this record, we are persuaded that Petitioner has provided an articulated reasoning with rational underpinning for the combination of Best and Bonella. *See KSR*, 550 U.S. at 418. We are not persuaded that Best’s description of “relatively high-speed, non-degrading memory accesses” teaches away from the proposed combination with Bonella. *See* Ex. 1006 ¶ 24; *see also id.* ¶ 12 (“in one application of the hybrid memory device, a DRAM memory is provided for relatively high-speed memory access and degradation-free performance”); *id.* ¶ 18 (“accesses may be overlapped at least in part to enable multiple accesses to the DRAM to be carried out while awaiting a memory read or programming operation (memory write) within the slower non-volatile die”). In an exemplary description of the memory shadowing operation, Best states that “Initially . . . data is stored in a fast-access volatile storage die.” *Id.* ¶ 25. Best goes on to describe that “some time after one or more write operations have been performed, a write-back trigger is detected within the shared interface circuitry and, in response, one or more internal data transfer operations are performed to transfer data from the locations indicated by the write-back

table to be dirty . . . from the DRAM to the corresponding locations . . . within the NV memory.” *Id.* Best states that the write-back trigger may be “detecting a power-loss or power-down signal or event (i.e., performing all necessary write-backs—an operation referred to herein as flushing the write-back table—as part of power down).” *Id.* ¶ 26. According to Patent Owner, Best teaches away from reducing DRAM performance (i.e., by reducing DRAM frequency) during this “power-loss trigger [that] indicates that power will be unavailable soon, not that power is already unavailable.” *See* PO Resp. 49.

We are not persuaded that the foregoing description of a “power-loss” or “power-down signal” or event exclusively refers to an indication that “power will be unavailable soon,” (PO Resp. 49) as opposed to the actual detection of a power-loss. *See* Reply 19–20. Even taking into account Best’s description of “relatively high-speed, non-degrading memory accesses,” we are not persuaded that this teaches away from the combination of Best and Bonella. As the Federal Circuit has explained, “[a] reference may be said to teach away when a person of ordinary skill, upon reading the reference, would be discouraged from following the path set out in the reference, or would be led in a direction divergent from the path that was taken by the applicant.” *Galderma Labs. LP v. Tolmar Inc.*, 737 F.3d 731, 738 (Fed. Cir. 2013). It is not sufficient for “teaching away” if the reference “merely expresses a general preference for an alternative invention but does not criticize, discredit, or otherwise discourage investigation into the invention claimed.” *Id.* We agree with Petitioner that there is “no disclosure or teaching [in Best] that the DRAM must maintain, or is even capable of maintaining, high-speed and high-performance read/write communications

with the host during a power-loss event when ‘dirty data’ is being written to Flash.” *See* Reply 20. We are persuaded that a person of ordinary skill in the art would be motivated to use Bonella’s power reduction technique (DRAM operational frequency reduction) during the performance of Best’s write flushing for the reasons set forth by Petitioner.

After considering the evidence and arguments of the complete record, and for the reasons set forth by Petitioner, we determine that Petitioner has shown, by a preponderance of the evidence, that the combination of Best and Bonella teaches this limitation and Petitioner has provided a sufficient reason for combining the references.

e) Conclusion for Independent Claims 1 and 15

Based on our review and consideration, we determine that Petitioner has demonstrated that the combination of Best, Bonella, and Mills teaches the limitations in claims 1 and 15, and provided sufficient reasoning for combining the references. We determine that the information presented in the Petition demonstrates, by a preponderance of the evidence, that claims 1 and 15 are unpatentable under 35 U.S.C. § 103(a) over Best, Bonella, and Mills.

5. Dependent Claim 16

Claim 16 depends from claim 15 and recites “a controller configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation and to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation.” Ex. 1001, 22:12–17.

Petitioner contends that Best discloses a “*controller*” in the form of a shared interface circuit. Pet. 39 (citing Ex. 1006 ¶ 12; Ex. 1003 ¶ 181); *see also id.* (citing Ex. 1006 ¶¶ 17, 21, Figs. 2–3).

Petitioner contends that Best’s shared interface circuitry is “*configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation*” because when no write-back trigger is detected (i.e., the *first mode of operation*), the Flash memory is decoupled from the DRAM. Pet. 39–40 (Ex. 1006 ¶¶ 17, 21, 25–26; Ex. 1003 ¶¶ 182–183). In particular, Petitioner contends that “access to only one of the DRAM or Flash memory is enabled, and when access to one is enabled access to the other is not enabled.” *Id.* at 40 (citing Ex. 1006 ¶¶ 17, 21, 22, 25; Ex. 1003 ¶ 183). In particular, Petitioner identifies that the data control circuit 151 may switch multiplexer 153 to convey data from the shared internal path 140 to secondary data path 155 (volatile) or secondary data path 157 (non-volatile). *Id.* at 40 (citing Ex. 1006 ¶ 21); *see* Ex. 1006, Figs. 2, 3.

Petitioner further contends that Best’s shared interface circuitry is configured “*to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation*” because Best discloses that during a power-loss event write-back trigger event (i.e., a second mode of operation), “the Flash memory is coupled to the DRAM so that ‘the data read out of the volatile storage die may be transferred from the volatile-die interface buffer 159 to the non-volatile-die interface buffer 161 via the inter-die data path 171.’” Pet. 41 (citing Ex. 1006 ¶ 21); *see* Ex. 1006, Fig. 3. Petitioner contends that “Best therefore discloses that at least the following elements of the shared interface circuitry couple the Flash memory to the

DRAM: volatile-die interface buffer 159, non-volatile-die interface buffer 161, and inter-die data path 171.” Pet. 41–42 (citing Ex. 1006 ¶ 21; Ex. 1003 ¶ 187).

Patent Owner argues that Petitioner has not shown that Best’s shared interface circuitry is “*configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation*” because Petitioner relies on data paths between the host system and the Flash or DRAM, which do not “ha[ve] anything to do with decoupling Best’s volatile memory from the non-volatile memory (or coupling, for that matter).” PO Resp. 43. Patent Owner cites to testimony from Mr. Maltiel that it asserts supports the conclusion that Petitioner’s position has nothing to do with coupling or decoupling the non-volatile and volatile memories from each other. *Id.* at 44 (citing Ex. 2005, 174:8–12; 144:13–17; Ex. 2004 ¶¶ 104, 105).

In Reply, Petitioner contends that showing that the non-volatile and volatile memories are coupled in a second mode of operation via inter-die data path 171 renders obvious that they are decoupled during the first mode of operation. Reply 18–19 (citing Ex. 1003 ¶¶ 185, 188).

In Sur-reply, Patent Owner argues that Petitioner did not argue that inter-die data path 171 is disabled while data paths 155 and 157 are enabled, as Petitioner contends in the Reply, so this new argument should be rejected as untimely. Sur-reply 21. Patent Owner argues that “Best teaches that data may flow via inter-die data path 171 from non-volatile memory to volatile memory, and vice versa,” but “Best does not disclose that inter-die data path is ever disabled.” *Id.* (citing Ex. 1006 ¶ 21).

We disagree that Petitioner’s evidence does not support that Best’s shared interface circuitry is “*configured to decouple the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation.*” Figure 3 of Best is reproduced below.

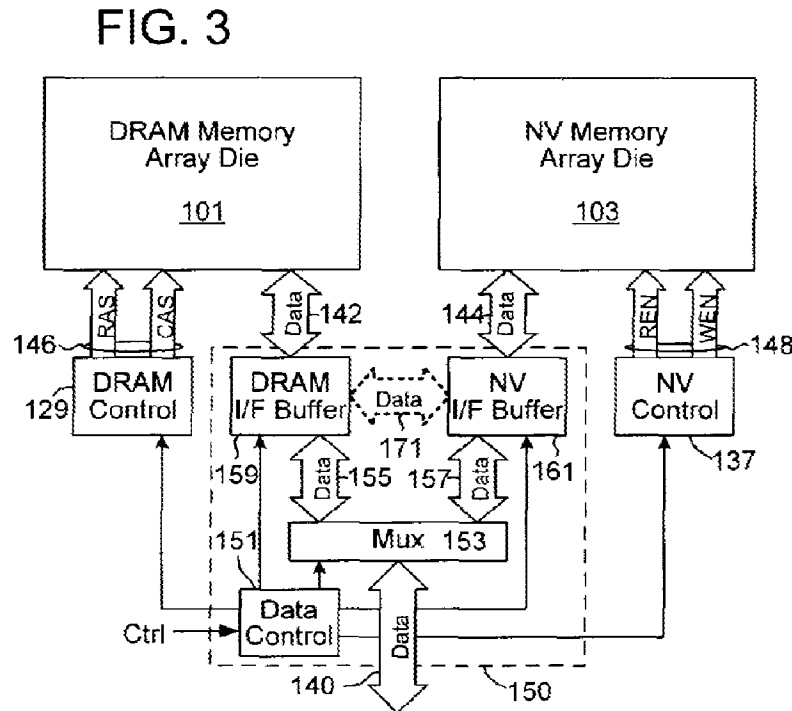


Figure 3, depicted above, illustrates an embodiment of a data control/steering circuit 150 that includes control circuit 151, multiplexer 153, secondary volatile and non-volatile data paths 155 and 157, volatile-storage-die interface buffer 159, non-volatile-storage-die interface buffer 161, and inter-die data path 171. *Id.* ¶ 21.

Best describes that:

In a memory write operation, *data received via shared internal data path 140 is passed, via multiplexer 153, to one of the secondary data paths 155, 157 according to the memory access target (e.g., indicated by memory-select signal MSel as described in reference to FIG. 4).* If the volatile storage die 103

is the target of the memory write, the data control circuit 151 *switches the multiplexer 153* to convey data from the shared internal path 140 to secondary data path 155, and signals the DRAM interface buffer 159 to enable the data to be buffered (or queued) therein for eventual transfer to the DRAM die via data path 142.

Id. (emphasis added). We agree with Petitioner that this disclosure, describing use of a multiplexer 153 to convey data to one of paths 155 or 157 depending on the memory access target (i.e., volatile storage die or non-volatile storage die), teaches “*decoupl[ing] the non-volatile memory subsystem from the volatile memory subsystem in the first mode of operation.*” That is, “during Best’s first mode of operation, the non-volatile and volatile memory subsystems are ***decoupled*** because the only memory paths that are enabled originate from the host to either volatile or non-volatile memory.” See Reply 18; see Pet. 40–41. Patent Owner’s argument that Petitioner’s cited evidence “has nothing to do with coupling or decoupling the non-volatile and volatile memories from each other” (Sur-reply 21) is not persuasive, given Best’s disclosure above.

We also disagree that Petitioner’s evidence does not support that Best teaches that its shared interface circuitry is configured “*to couple the non-volatile memory subsystem to the volatile memory subsystem in the second mode of operation.*” This is supported by Best’s disclosure of an alternative embodiment during the memory shadow operation, where data may be transferred from the volatile-die interface buffer 159 to the non-volatile-die interface buffer 161 and vice versa *via the inter-die data path 171*, depicted

above in Figure 3.¹³ Ex. 1006 ¶ 21. We agree with Petitioner that “Best’s disclosure of *coupling* the non-volatile and volatile memory subsystems during the second mode of operation renders obvious that the memory systems are *decoupled* during the first mode of operation.” See Reply 18–19; *see also* Pet. 41–42. For example, there does not appear to be, and Patent Owner does not identify, any path by which the volatile and non-volatile memories are coupled in Figure 3, aside from inter-die data path 171, which is described in context of the memory shadowing operation described above.

Accordingly, we determine that the information presented in the Petition demonstrates, by a preponderance of the evidence, that claim 16 is unpatentable under 35 U.S.C. § 103(a) over Best, Bonella, and Mills.

6. Dependent Claims 3–14, 17, and 19–30

Petitioner contends that dependent claims 3–14, 17, and 19–30, which depend directly or indirectly from independent claims 1 and 15, would have been obvious over Best, Bonella, and Mills, and provides explanation as to how the prior art teaches each claim limitation. Pet. 38–60.

For example, claim 17 depends from claim 15 and recites “a plurality of power supplies and a switch configured to selectively deliver power from the plurality of power supplies to the volatile memory subsystem and the non-volatile memory subsystem as a function of the mode of operation.” Ex. 1001, 22:18–22. Petitioner contends that these features were “well-known in the art” and are disclosed by Bonella’s standard and backup power

¹³ Petitioner explicitly referred to inter-die data path 171 twice on page 40 of the Petition, so Patent Owner’s contention that Petitioner did not identify inter-die data path 171 in the Petition is not persuasive.

sources. Pet. 42–43 (citing Ex. 1008 ¶¶ 29–31, 33, 101; Ex. 1003 ¶ 192). Petitioner contends that it would have been obvious to a person of ordinary skill in the art “to include Bonella’s power supplies and associated selective power switching functionality/circuitry, including the power transistors, in Best’s hybrid memory devices because to do so would have been merely an arrangement of old elements . . . each performing the same function it had been known to perform . . . in a way that yields no more than a [person of ordinary skill in the art] would expect from such an arrangement.” *Id.* at 43–44 (citing Ex. 1006 ¶¶ 25, 26; Ex. 1008 ¶¶ 29–31, 33, 101; Ex. 1003 ¶ 194).

Claims 3 and 19 depend from claims 1 and 15, respectively, and recite “wherein the memory system is not powered by a battery when it is in the second mode of operation.” Ex. 1001, 21:25–27, 22:25–27. Petitioner contends that Bonella discloses that a power loss event is powered by either a battery or capacitors, so “when Bonella’s capacitors power the memory module for emergency shutdown operations . . . the memory module ‘is not powered by a battery.’” Pet. 44–45 (citing Ex. 1008 ¶ 29; Ex. 1003 ¶ 198). According to Petitioner, “as combined, Best’s system would use Bonella’s power techniques/capacitors to power Best’s hybrid memory device system when Best’s hybrid memory device system is also operating in a state similar to Bonella’s ‘emergency shutdown operations,’ such as Best’s operation in response to ‘power-loss’/‘power-down’ events.” *Id.* at 45–46 (citing Ex. 1006 ¶¶ 25–26; Ex. 1008 ¶¶ 29–31, 33, 101; Ex. 1003 ¶ 199).

Claims 4 and 20 depend from claims 1 and 15, respectively, and recite “wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.” Ex. 1001, 21:28–30, 22:28–30. Petitioner relies on Best to teach this limitation,

specifically, Best’s detection of the power-loss event write-back trigger.
Pet. 46–47 (citing Ex. 1006 ¶¶ 25–26; Ex. 1003 ¶ 202).

Claims 5 and 21 depend from claims 4 and 20, respectively, and recite “wherein the trigger condition comprises a power failure condition.” Ex. 1001, 21:31–32, 22:32–33. Petitioner relies on Best’s write-back trigger detecting a power-loss or power-down signal or event to teach this limitation. Pet. 47 (citing Ex. 1006 ¶ 26).

Claims 6 and 22 depend from claims 1 and 15, respectively, and recite “wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.” Ex. 1001, 21:33–36, 22:33–36. Petitioner relies on Figure 1A of Best, and Best’s disclosure that “the DRAM and Flash are typically implemented in separate integrated circuit devices having distinct control and data interfaces and disposed in distinct regions of an integrated circuit board.” Pet. 47–48 (citing Ex. 1006 ¶¶ 2, 13, Fig. 1A).

Claims 7 and 23 depend from claims 1 and 15, respectively, and recite “wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is intermediate data from a host computer system computation.” Ex. 1001, 21:37–40, 22:37–40. Petitioner relies on Mills to teach this limitation, specifically Mills’ saving of intermediate results generated during the course of execution of the program. Pet. 49 (citing Ex. 1009, 3:21–26; Ex. 1003 ¶ 207). Petitioner contends that, “as combined, Best’s system would operate as usual such that when ‘a write-back trigger is detected,’ such as a power-loss event, shadowing operations are performed, e.g., ‘internal data transfer operations are performed to

transfer data . . . from the DRAM to . . . the NV memory,’ where the data being transferred in Best’s system includes the ‘intermediate results’ disclosed by Mills.” *Id.* at 50 (alterations in original). According to Petitioner a person of ordinary skill in the art would be motivated to make such a combination “to prevent the loss of all data forever, especially because prevention of data loss is a goal of both Best and Mills.” *Id.* at 51 (citing Ex. 1006 ¶¶ 25–26; Ex. 1009, 3:21–26; Ex. 1003 ¶ 212).

Claims 8 and 24 depend from claims 1 and 15, respectively, and recite “wherein data communicated between the volatile memory subsystem and the non-volatile memory subsystem is backup data from a backup operation.” Ex. 1001, 21:41–44, 22:41–44. Petitioner relies on Best’s memory shadowing operations to teach this limitation. Pet. 52 (citing Ex. 1006 ¶¶ 24–25; Ex. 1003 ¶¶ 215, 216, 218, 219).

Claims 9 and 25 depend from claims 8 and 24 and recite “wherein the backup operation is conducted at repeating time intervals.” Ex. 1001, 21:44–45, 22:45–46. Petitioner relies on Best to disclose this limitation, for example, that Best teaches “[t]he write-back trigger itself may include . . . periodically performing write-back.” Pet. 52–53 (citing Ex. 1006 ¶¶ 25, 26; Ex. 1003 ¶ 220) (alteration in original).

Claims 10 and 26 depend from claims 9 and 25, respectively, and recite “wherein the backup operation is initiated in response to a trigger event.” Ex. 1001, 21:46–47, 22:47–48. Petitioner relies on Best to teach this limitation, for example, that Best’s “write-back operations” occur “in response” to when a “write-back trigger” is detected. Pet. 53 (citing Ex. 1006 ¶¶ 25–26; Ex. 1003 ¶ 221).

Claims 11 and 27 depend from claims 1 and 15, respectively, and recite “wherein the second mode of operation comprises a backup operation in which data is communicated from the volatile memory subsystem to the non-volatile memory subsystem.” Ex. 1001, 21:48–51, 22:50–53. Petitioner relies on Best’s memory shadow operation to teach this limitation. Pet. 54 (citing Ex. 1006 ¶¶ 25, 26; Ex. 1003 ¶ 223).

Claims 12 and 28 depend from claims 1 and 15, respectively, and recite “wherein the second mode of operation comprises a restore operation in which data is communicated from the non-volatile memory subsystem to the volatile memory subsystem.” Ex. 1001, 21:52–55, 22:54–57. Petitioner relies on Best to teach this limitation, for example, that Best discloses that communication between the volatile memory and non-volatile memory includes the performance of restore operations in which data is “transferred . . . from the non-volatile die 101 to the volatile storage die 103 . . . to restore the state of the volatile storage die 103 to a pre-power-down or pre-sleep condition.” Pet. 55 (citing Ex. 1006 ¶ 21; Ex. 1003 ¶ 226).

Claims 13 and 29 depend from claims 1 and 15, respectively, and recite “wherein the one or more of the first, second, or third clock frequencies is configurable by the memory system.” Ex. 1001, 21:56–58, 22:58–60. Petitioner relies on Bonella to teach this limitation, for example, that Bonella teaches that the memory module has power management features that are “user configured and/or system configured” and that at least the primary operating frequency of the DRAM is configurable by the memory system based on the user configuration. Pet. 56 (citing Ex. 1008 ¶¶ 36, 45, 48–50; Ex. 1003 ¶¶ 230, 231). Petitioner contends that a person of ordinary skill in the art would have been motivated to modify Best to

obtain the benefit of configurability and power savings. *Id.* at 57 (citing Ex. 1003 ¶ 234).

Claims 14 and 30 depend from claims 1 and 15, respectively, and recite “wherein one or more of the first, second or third clock frequencies is configurable by a user.” Ex. 1001, 21:59–60, 22:60–62. Petitioner relies on Bonella to teach this limitation, for many of the same reasons set forth for claims 13 and 29. Pet. 58–60.

Patent Owner does not specifically respond to Petitioner’s contentions, but instead relies on the arguments presented for claim 15. PO Resp. 45.

Based on our review and consideration of the record, we determine that Petitioner has demonstrated that the combination of Best, Bonella, and Mills teach the limitations in claims 3–14, 17, and 19–30 and provided sufficient reasoning for combining the references. We determine that the information presented in the Petition demonstrates, by a preponderance of the evidence, that claims 3–14, 17, and 19–30 are unpatentable under 35 U.S.C. § 103(a) over Best, Bonella, and Mills.

7. *Summary*

We find that Petitioner has demonstrated by a preponderance of the evidence that claims 1, 3–17, and 19–30 are unpatentable under 35 U.S.C. § 103(a) over Best, Bonella, and Mills.

IV. CONCLUSION

For the foregoing reasons, we are persuaded that Petitioner established by a preponderance of the evidence that claims 1, 3–17, and 19–30 of the '833 patent are unpatentable.¹⁴

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1, 3– 17, 19– 30	103(a)	Best, Bonella, Mills	1, 3–17, 19– 30	

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1, 3–17, and 19–30 of the '833 patent have been shown to be unpatentable under 35 U.S.C. § 103(a); and

FURTHER ORDERED that, because this is a final written decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

¹⁴ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

IPR2022-00418
Patent 8,301,833 B1

For PETITIONER:

Juan Yaquian
WINSTON & STRAWN LLP
jyaquian@winston.com

For PATENT OWNER:

Rex Hwang
Michael Ricketts
Ryan Hargrave
SKIERMONT DERBY LLP
rhwang@skiermontderby.com
mricketts@skiermontderby.com
rhargrave@skiermontderby.com

(12) **INTER PARTES REVIEW CERTIFICATE** (3313th)

United States Patent
Chen et al.

(10) **Number:** **US 8,301,833 K1**
(45) **Certificate Issued:** **Nov. 13, 2023**

(54) **NON-VOLATILE MEMORY MODULE**

(75) **Inventors:** **Chi-She Chen; Jeffrey C. Solomon;**
Scott Milton; Jayesh Bhakta

(73) **Assignee:** **NETLIST, INC.**

Trial Number:

IPR2022-00418 filed Jan. 14, 2022

Inter Partes Review Certificate for:

Patent No.: **8,301,833**
Issued: **Oct. 30, 2012**
Appl. No.: **12/240,916**
Filed: **Sep. 29, 2008**

The results of IPR2022-00418 are reflected in this inter partes review certificate under 35 U.S.C. 318(b).

INTER PARTES REVIEW CERTIFICATE
U.S. Patent 8,301,833 K1
Trial No. IPR2022-00418
Certificate Issued Nov. 13, 2023

1

2

AS A RESULT OF THE INTER PARTES
REVIEW PROCEEDING, IT HAS BEEN
DETERMINED THAT:

Claims 1, 3-17 and 19-30 are cancelled.

5

* * * * *



United States Patent and Trademark Office

Office of the Chief Financial Officer

Document Code:WFEE

User :Dorethea Allen

Sale Accounting Date:04/24/2024

Sale Item Reference Number

12240916

Effective Date

04/22/2024

Document Number	Fee Code	Fee Code Description	Amount Paid	Payment Method
I20244N043185428	1553	MAINTENANCE FEE DUE AT 11.5 YEARS	\$7,700.00	Wire

UTILITY APPLICATION

Attorney Docket No.: NETL.040A
First Named Inventor: to be determined
Title: NON-VOLATILE MEMORY MODULE

Direct all correspondence to Customer No.: 20995

Date: June 2, 2008

Page 1

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

The following enclosures are transmitted herewith to be filed in the patent application of:

Inventor(s):

1. to be determined

APPLICATION:

(X) Specification in 40 pages.

(X) Drawings in 12 sheets.

CONTINUITY INFORMATION:

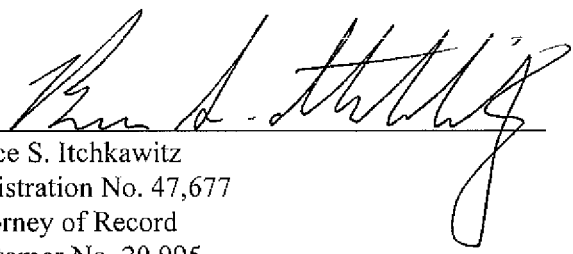
Application	Relationship	Parent App. No.	Filing Date	Status
This Application	Non-Provisional of	60/941,586	6/01/2007	closed

Reference to prior domestic applications is made in the:

(X) Specification.

OTHER APPLICATION PARTS:**FILING FEES:**

(X) The total fees will be paid at a later date.


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

5458952
060208

NON-VOLATILE MEMORY MODULE**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] The present application claims the benefit of priority from U.S. Provisional Application No. 60/941,586, filed June 1, 2007, which is incorporated in its entirety by reference herein.

BACKGROUND

[0002] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the computer system to provide volatile memory to the computer system.

[0003] Volatile memory generally maintains stored information only when it is powered. Batteries have been used to provide power to volatile memory during power failures or interruptions. However, batteries may require maintenance, may need to be replaced, are not environmentally friendly, and the status of batteries can be difficult to monitor.

[0004] Non-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory. In certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.

SUMMARY

[0005] In certain embodiments, a memory system coupled to a computer system is provided which includes a volatile memory subsystem, a non-volatile memory subsystem, and a controller operatively coupled to the non-volatile memory subsystem. The memory system can also include at least one circuit configured to selectively operatively decouple the controller from the volatile memory subsystem.

[0006] In some embodiments, a power module for providing a plurality of voltages to a memory system is described. The power module includes non-volatile and volatile memory, and the plurality of voltages include at least a first voltage and a second voltage. The power module of certain embodiments includes an input providing a third

voltage to the power module and a voltage conversion element configured to provide the second voltage to the memory system. The power module also includes a first power element configured to selectively provide a fourth voltage to the conversion element. The power module further includes a second power element configured to selectively provide a fifth voltage to the conversion element. The power module can be configured to selectively provide the first voltage to the memory system either from the conversion element or from the input.

[0007] The power module can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage is provided to the memory system from the input and the fourth voltage is provided to the conversion element from the first power element. In a second, state the fourth voltage is provided to the conversion element from the first power element and the first voltage is provided to the memory system from the conversion element. In a third state, the fifth voltage is provided to the conversion element from the second power element and the first voltage is provided to the memory system from the conversion element.

[0008] A method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems is provided in certain embodiments. The method includes, during a first condition, providing the first voltage to the memory system from an input power supply and providing the second voltage to the memory system from a first power subsystem. The method further includes detecting a second condition and, during the second condition, providing the first voltage and the second voltage to the memory system from the first power subsystem. The method also includes charging a second power subsystem and detecting a third condition. During the third condition, the method includes providing the first voltage and the second voltage to the memory system from the second power subsystem.

[0009] In certain embodiments, a method is provided for controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem. The method can include operating the volatile memory subsystem at a first frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the

host system. In certain embodiments, the method further includes operating the non-volatile memory subsystem at a second frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem. The method can also include operating the volatile memory subsystem at a third frequency when the memory system is in the second mode of operation, the third frequency less than the first frequency.

[0010] In certain embodiments, a method is provided for controlling a memory system operatively coupled to a host system. The memory system includes a volatile memory subsystem and a non-volatile memory subsystem. In certain embodiments, the method includes communicating data words between the volatile memory subsystem and the host system when the memory system is in a first mode of operation. The method can further include transferring data words from the volatile memory subsystem to the non-volatile memory subsystem when the memory system is in a second mode of operation. Transferring each data word can include storing a first portion of the data word in a buffer, storing a second portion of the data word in the buffer, and writing the entire data word from the buffer to the non-volatile memory subsystem.

[0011] A memory system operatively coupled to a host system is provided in certain embodiments. The memory system can include a volatile memory subsystem and a non-volatile memory subsystem comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system includes a controller operatively coupled to the volatile memory subsystem and operatively coupled to the non-volatile memory subsystem, the controller configured to allow data to be communicated between the volatile memory subsystem and the host system when the memory system is operating in a first state and to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem when the memory system is operating in a second state.

[0012] A method of controlling a memory system operatively coupled to a host system is provided in certain embodiments. The memory system includes a volatile memory subsystem and a non-volatile memory subsystem. The method can include communicating data between the volatile memory subsystem and the host system when the memory system is in a first mode of operation. The method of certain embodiments further includes storing a

first copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a first time when the memory system is in a second mode of operation. The method may further include restoring the first copy of data from the non-volatile memory subsystem to the volatile memory subsystem and erasing the first copy of data from the non-volatile memory subsystem. In certain embodiments, the method also includes storing a second copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a second time when the memory system is in the second mode of operation, wherein storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Figure 1 is a block diagram of an example memory system compatible with certain embodiments described herein.

[0014] Figure 2 is a block diagram of an example memory module with ECC (error-correcting code) having a volatile memory subsystem with nine volatile memory elements and a non-volatile memory subsystem with five non-volatile memory elements in accordance with certain embodiments described herein.

[0015] Figure 3 is a block diagram of an example memory module having a microcontroller unit and logic element integrated into a single device in accordance with certain embodiments described herein.

[0016] Figures 4A-4C schematically illustrate example embodiments of memory systems having volatile memory subsystems comprising registered dual in-line memory modules in accordance with certain embodiments described herein.

[0017] Figure 5 schematically illustrates an example power module of a memory system in accordance with certain embodiments described herein.

[0018] Figure 6 is a flowchart of an example method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems.

[0019] Figure 7 is a flowchart of an example method of controlling a memory system operatively coupled to a host system and which includes at least 100 percent more storage capacity in non-volatile memory than in volatile memory.

[0020] Figure 8 schematically illustrates an example clock distribution topology of a memory system in accordance with certain embodiments described herein.

[0021] Figure 9 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including operating a volatile memory subsystem at a reduced rate in a back-up mode.

[0022] Figure 10 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments of a volatile memory subsystem of a memory system to a controller of the memory system.

[0023] Figure 11 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including backing up and/or restoring a volatile memory subsystem in slices.

DETAILED DESCRIPTION

[0024] Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory, and a controller. The controller backs up the volatile memory using the non-volatile memory in the event of a trigger condition. Trigger conditions can include, for example, a power failure, power reduction, request by the host system, etc. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which does not comprise a battery and may include, for example, a capacitor or capacitor array.

[0025] In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example, one or more isolation devices may isolate the non-volatile memory and the controller from the volatile memory when the volatile memory is interacting with the host system and may allow communication between the volatile memory and the non-volatile memory when the data of the volatile memory is being restored or backed-up. This configuration generally protects the operation of the volatile memory when isolated while providing backup and restore capability in the event of a trigger condition, such as a power failure.

[0026] In certain embodiments described herein, the memory system includes a power module which provides power to the various components of the memory system from different sources based on a state of the memory system in relation to a trigger condition (e.g., a power failure). The power module may switch the source of the power to the various components in order to efficiently provide power in the event of the power failure. For example, when no power failure is detected, the power module may provide power to certain components, such as the volatile memory, from system power while charging a secondary power source (e.g., a capacitor array). In the event of a power failure or other trigger condition, the power module may power the volatile memory elements using the previously charged secondary power source.

[0027] In certain embodiments, the power module transitions relatively smoothly from powering the volatile memory with system power to powering it with the secondary power source. For example, the memory system may power volatile memory with a third power source from the time the memory system detects that power failure is likely to occur until the time the memory system detects that the power failure has actually occurred.

[0028] In certain embodiments, the volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power. In some embodiments, during backup and/or restore operations, the volatile memory communicates with the non-volatile memory by writing and/or reading data words in bit-wise slices instead of by writing entire words at once. In certain embodiments, when each slice is being written to or read from the volatile memory the unused slice(s) of volatile memory is not active, which can reduce the power consumption of the system.

[0029] In yet other embodiments, the non-volatile memory can include at least 100 percent more storage capacity than the volatile memory. This configuration can allow the memory system to efficiently handle subsequent trigger conditions.

[0030] **Figure 1** is a block diagram of an example memory system 10 compatible with certain embodiments described herein. The memory system 10 can be coupled to a host computer system and can include a volatile memory subsystem 30, a non-volatile memory subsystem 40, and a controller 62 operatively coupled to the non-volatile memory subsystem

40. In certain embodiments, the memory system 10 includes at least one circuit 52 configured to selectively operatively decouple the controller 62 from the volatile memory subsystem 30.

[0031] In certain embodiments, the memory system 10 comprises a memory module. The memory system 10 may comprise a printed-circuit board (PCB) 20. In certain embodiments, the memory system 10 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other volatile memory capacities are also compatible with certain embodiments described herein. In certain embodiments, the memory system 10 has a non-volatile memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, 16-GB, or 32-GB. Other non-volatile memory capacities are also compatible with certain embodiments described herein. In addition, memory systems 10 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 20 has an industry-standard form factor. For example, the PCB 20 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 20 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 20 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0032] In certain preferred embodiments, the memory system 10 is in electrical communication with the host system. In other embodiments, the memory system 10 may communicate with a host system using some other type of communication, such as, for example, optical communication. Examples of host systems include, but are not limited to, blade servers, 1U servers, personal computers (PCs), and other applications in which space is constrained or limited. The memory system 10 can be in communication with a disk controller of a computer system, for example. The PCB 20 can comprise an interface 22 that

is configured to be in electrical communication with the host system (not shown). For example, the interface 22 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 22 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 10 and the host system. For example, the interface 22 can comprise a standard 240-pin DDR2 edge connector.

[0033] The volatile memory subsystem 30 comprises a plurality of volatile memory elements 32 and the non-volatile memory subsystem 40 comprises a plurality of non-volatile memory elements 42. Certain embodiments described herein advantageously provide non-volatile storage via the non-volatile memory subsystem 40 in addition to high-performance (e.g., high speed) storage via the volatile memory subsystem 30. In certain embodiments, the first plurality of volatile memory elements 32 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 32 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of Figure 1, the first memory bank 30 comprises eight 64Mx8 DDR2 SDRAM elements 32. The volatile memory elements 32 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 32 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements 32 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0034] In certain embodiments, the second plurality of non-volatile memory elements 42 comprises one or more flash memory elements. Types of flash memory elements 42 compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, ONE-NAND flash, and multi-level cell (MLC). For example, in the block diagram of Figure 1, the second memory bank 40 comprises 512 MB of flash memory organized as four 128Mbx8 NAND flash memory elements 42. In addition, non-volatile memory elements 42 having bit widths of 4, 8, 16, 32, as well as other bit widths, are

compatible with certain embodiments described herein. Non-volatile memory elements 42 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0035] **Figure 2** is a block diagram of an example memory module 10 with ECC (error-correcting code) having a volatile memory subsystem 30 with nine volatile memory elements 32 and a non-volatile memory subsystem 40 with five non-volatile memory elements 42 in accordance with certain embodiments described herein. The additional memory element 32 of the first memory bank 30 and the additional memory element 42 of the second memory bank 40 provide the ECC capability. In certain other embodiments, the volatile memory subsystem 30 comprises other numbers of volatile memory elements 32 (e.g., 2, 3, 4, 5, 6, 7, more than 9). In certain embodiments, the non-volatile memory subsystem 40 comprises other numbers of non-volatile memory elements 42 (e.g., 2, 3, more than 5).

[0036] Referring to Figure 1, in certain embodiments, the logic element 70 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 70 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 70 comprises an FPGA available from another vendor. The internal flash can improve the speed of the memory system 10 and save physical space. Other types of logic elements 70 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 70 is a custom device. In certain embodiments, the logic element 70 comprises various discrete electrical elements, while in certain other embodiments, the logic element 70 comprises one or more integrated circuits. **Figure 3** is a block diagram of an example memory module 10 having a microcontroller unit 60 and logic element 70 integrated into a single controller 62 in accordance with certain embodiments described herein. In certain embodiments, the controller 62 includes one or more other components. For example, in one embodiment, an FPGA without an internal flash is used

and the controller 62 includes a separate flash memory component which stores configuration information to program the FPGA.

[0037] In certain embodiments, the at least one circuit 52 comprises one or more switches coupled to the volatile memory subsystem 30, to the controller 62, and to the host computer (e.g., via the interface 22, as schematically illustrated by Figures 1-3). The one or more switches are responsive to signals (e.g., from the controller 62) to selectively operatively decouple the controller 62 from the volatile memory subsystem 30 and to selectively operatively couple the controller 62 to the volatile memory subsystem 30. In addition, in certain embodiments, the at least one circuit 52 selectively operatively couples and decouples the volatile memory subsystem 30 and the host system.

[0038] In certain embodiments, the volatile memory subsystem 30 can comprise a registered DIMM subsystem comprising one or more registers 160 and a plurality of DRAM elements 180, as schematically illustrated by **Figure 4A**. In certain such embodiments, the at least one circuit 52 can comprise one or more switches 172 coupled to the controller 62 (e.g., logic element 70) and to the volatile memory subsystem 30 which can be actuated to couple and decouple the controller 62 to and from the volatile memory subsystem 30, respectively. The memory system 10 further comprises one or more switches 170 coupled to the one or more registers 160 and to the plurality of DRAM elements 180 as schematically illustrated by Figure 4A. The one or more switches 170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 30 to the host system 150. In certain other embodiments, as schematically illustrated by **Figure 4B**, the one or more switches 174 are also coupled to the one or more registers 160 and to a power source 162 for the one or more registers 160. The one or more switches 174 can be selectively switched to turn power on or off to the one or more registers 160, thereby selectively operatively coupling the volatile memory subsystem 30 to the host system 150. As schematically illustrated by **Figure 4C**, in certain embodiments the at least one circuit 52 comprises a dynamic on-die termination (ODT) 176 circuit of the logic element 70. For example, the logic element 70 can comprise a dynamic ODT circuit 176 which selectively operatively couples and decouples the logic element 70 to and from the volatile memory subsystem 30, respectively. In addition, and similar to the example embodiment of Figure 4A described above, the one or

more switches 170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 30 to the host system 150.

[0039] Certain embodiments described herein utilize the non-volatile memory subsystem 40 as a flash “mirror” to provide backup of the volatile memory subsystem 30 in the event of certain system conditions. For example, the non-volatile memory subsystem 40 may backup the volatile memory subsystem 30 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the non-volatile memory subsystem 30 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 10 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

[0040] As schematically illustrated by Figures 1 and 2, in certain embodiments, the controller 62 may comprise a microcontroller unit (MCU) 60 and a logic element 70. In certain embodiments, the MCU 60 provides memory management for the non-volatile memory subsystem 40 and controls data transfer between the volatile memory subsystem 30 and the non-volatile memory subsystem 40. The MCU 60 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. As schematically illustrated by Figures 1 and 2, the logic element 70 of certain embodiments is in electrical communication with the non-volatile memory subsystem 40 and the MCU 60. The logic element 70 can provide signal level translation between the volatile memory elements 32 (e.g., 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 42 (e.g., 3V TTL for NAND flash memory

elements). In certain embodiments, the logic element 70 is also programmed to perform address/address translation between the volatile memory subsystem 30 and the non-volatile memory subsystem 40. In certain preferred embodiments, 1-NAND type flash are used for the non-volatile memory elements 42 because of their superior read speed and compact structure.

[0041] The memory system 10 of certain embodiments is configured to be operated in at least two states. The at least two states can comprise a first state in which the controller 62 and the non-volatile memory subsystem 40 are operatively decoupled (e.g., isolated) from the volatile memory subsystem 30 by the at least one circuit 52 and a second state in which the volatile memory subsystem 30 is operatively coupled to the controller 62 to allow data to be communicated between the volatile memory subsystem 30 and the non-volatile memory subsystem 40 via the controller 62. The memory system 10 may transition from the first state to the second state in response to a trigger condition, such as when the memory system 10 detects that there is a power interruption (e.g., power failure or reduction) or a system hang-up.

[0042] The memory system 10 may further comprise a voltage monitor 50. The voltage monitor circuit 50 monitors the voltage supplied by the host system via the interface 22. Upon detecting a low voltage condition (e.g., due to a power interruption to the host system), the voltage monitor circuit 50 may transmit a signal to the controller 62 indicative of the detected condition. The controller 62 of certain embodiments responds to the signal from the voltage monitor circuit 50 by transmitting a signal to the at least one circuit 52 to operatively couple the controller to the volatile memory system 30, such that the memory system 10 enters the second state. For example, the voltage monitor 50 may send a signal to the MCU 60 which responds by accessing the data on the volatile memory system 30 and by executing a write cycle on the non-volatile memory subsystem 40. During this write cycle, data is read from the volatile memory subsystem 30 and is transferred to the non-volatile memory subsystem 40 via the MCU 60. In certain embodiments, the voltage monitor circuit 50 is part of the controller 62 (e.g., part of the MCU 60) and the voltage monitor circuit 50 transmits a signal to the other portions of the controller 62 upon detecting a power threshold condition.

[0043] The isolation or operational decoupling of the volatile memory subsystem 30 from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 10 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 30. For example, in one embodiment during such periods of operation, the controller 62 and the non-volatile memory subsystem 40 do not add a significant capacitive load to the volatile memory system 30 when the memory system 10 is in the first state. In certain such embodiments, the capacitive load of the controller 62 and the non-volatile memory subsystem 40 do not significantly affect the signals propagating between the volatile memory subsystem 30 and the host system. This can be particularly advantageous in relatively high-speed memory systems where loading effects can be significant. In one preferred embodiment, the at least one circuit 52 comprises an FSA1208 Low-Power, Eight-Port, Hi-Speed Isolation Switch from Fairchild Semiconductor. In other embodiments, the at least one circuit 52 comprises other types of isolation devices.

[0044] Power may be supplied to the volatile memory subsystem 30 from a first power supply (e.g., a system power supply) when the memory system 10 is in the first state and from a second power supply 80 when the memory system 10 is in the second state. In certain embodiments, the memory system 10 is in the first state when no trigger condition (e.g., a power failure) is present and the memory system 10 enters the second state in response to a trigger condition. In certain embodiments, the memory system 10 has a third state in which the controller 62 is operatively decoupled from the volatile memory subsystem 30 and power is supplied to the volatile memory subsystem 30 from a third power supply (not shown). For example, in one embodiment the third power supply may provide power to the volatile memory subsystem 30 when the memory system 10 detects that a trigger condition is likely to occur but has not yet occurred.

[0045] In certain embodiments, the second power supply 80 does not comprise a battery. Because a battery is not used, the second power supply 80 of certain embodiments may be relatively easy to maintain, does not generally need to be replaced, and is relatively environmentally friendly. In certain embodiments, as schematically illustrated by Figures 1-3, the second power supply 80 comprises a step-up transformer 82, a step-down transformer

84, and a capacitor bank 86 comprising one or more capacitors (e.g., double-layer capacitors). In one example embodiment, capacitors may take about three to four minutes to charge and about two minutes to discharge. In other embodiments, the one or more capacitors may take a longer time or a shorter time to charge and/or discharge. For example, in certain embodiments, the second power supply 80 is configured to power the volatile memory subsystem 30 for less than thirty minutes. In certain embodiments, the second power supply 80 may comprise a battery. For example, in certain embodiments, the second power supply 80 comprises a battery and one or more capacitors and is configured to power the volatile memory subsystem 30 for no more than thirty minutes.

[0046] In certain embodiments, the capacitor bank 86 of the second power supply 80 is charged by the first power supply while the memory system 10 is in the first state. As a result, the second power supply 80 is fully charged when the memory system 10 enters the second state. The memory system 10 and the second power supply 80 may be located on the same printed circuit board 20. In other embodiments, the second power supply 80 may not be on the same printed circuit board 20 and may be tethered to the printed circuit board 20, for example.

[0047] When operating in the first state, in certain embodiments, the step-up transformer 82 keeps the capacitor bank 86 charged at a peak value. In certain embodiments, the step-down transformer 84 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 32 and 3.0V to the non-volatile flash memory elements 42) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by Figures 1-3, the memory module 10 further comprises a switch 90 (e.g., FET switch) that switches power to the controller 62 the volatile memory subsystem 30, and the non-volatile memory subsystem 40 between the power from the second power supply 80 and the power from the first power supply (e.g., system power) received via the interface 22. For example, the switch 90 may switch from the first power supply to the second power supply 80 when the voltage monitor 50 detects a low voltage condition. The switch 90 of certain embodiments advantageously ensures that the volatile memory elements 32 and non-volatile memory elements 42 are powered long enough for the data to be transferred from the volatile memory elements 32 and

stored in the non-volatile memory elements 42. In certain embodiments, after the data transfer is complete, the switch 90 then switches back to the first power supply and the controller 62 transmits a signal to at least one circuit 50 to operatively decouple the controller 62 from the volatile memory subsystem 30, such that the memory system 10 re-enters the first state.

[0048] When the memory system 10 re-enters the first state, data may be transferred back from the non-volatile memory subsystem 40 to the volatile memory subsystem 30 via the controller 62. The host system can then resume accessing the volatile memory subsystem 30 of the memory module 10. In certain embodiments, after the memory system 10 enters or re-enters the first state (e.g., after power is restored), the host system accesses the volatile memory subsystem 30 rather than the non-volatile memory subsystem 40 because the volatile memory elements 32 have superior read/write characteristics. In certain embodiments, the transfer of data from the volatile memory bank 30 to the non-volatile memory bank 40, or from the non-volatile memory bank 40 to the volatile memory bank 30, takes less than one minute per GB.

[0049] In certain embodiments, the memory system 10 protects the operation of the volatile memory when communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure. In certain embodiments, the memory system 10 copies the entire contents of the volatile memory subsystem 30 into the non-volatile memory subsystem 40 on each backup operation. Moreover, in certain embodiments, the entire contents of the non-volatile memory subsystem 40 are copied back into the volatile memory subsystem 30 on each restore operation. In certain embodiments, the entire contents of the non-volatile memory subsystem 40 are accessed for each backup and/or restore operation, such that the non-volatile memory subsystem 40 (e.g., flash memory subsystem) is used generally uniformly across its memory space and wear-leveling is not performed by the memory system 10. In certain embodiments, avoiding wear-leveling can decrease cost and complexity of the memory system 10 and can improve the performance of the memory system 10. In certain other embodiments, the entire contents of the volatile memory subsystem 30 are not copied into the non-volatile memory subsystem 40 on each backup operation, but only a partial copy is performed. In certain

embodiments, other management capabilities such as bad-block management and error management for the flash memory elements of the non-volatile memory subsystem 40 are performed in the controller 62.

[0050] The memory system 10 generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem 30 which then writes the data to non-volatile storage which is not part of the memory system 10, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system 10 indicating that the data has been written to the hard disk or is otherwise secure. The memory system 10 of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system 10 will still be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system 10 may operated as a write-through cache or as some other type of cache.

[0051] **Figure 5** schematically illustrates an example power module 100 of the memory system 10 in accordance with certain embodiments described herein. The power module 100 provides power to the various components of the memory system 10 using different elements based on a state of the memory system 10 in relation to a trigger condition. In certain embodiments, the power module 100 comprises one or more of the components described above with respect to Figure 1. For example, in certain embodiments, the power module 100 includes the second power supply 80 and the switch 90.

[0052] The power module 100 provides a plurality of voltages to the memory system 10 comprising non-volatile and volatile memory subsystems 30, 40. The plurality of voltages comprises at least a first voltage 102 and a second voltage 104. The power module 100 comprises an input 106 providing a third voltage 108 to the power module 100 and a voltage conversion element 120 configured to provide the second voltage 104 to the memory system 10. The power module 100 further comprises a first power element 130 configured to selectively provide a fourth voltage 110 to the conversion element 120. In certain embodiments, the first power element 130 comprises a pulse-width modulation power

controller. For example, in one example embodiment, the first power element 130 is configured to receive a 1.8V input system voltage as the third voltage 108 and to output a modulated 5V output as the fourth voltage 110.

[0053] The power module 100 further comprises a second power element 140 can be configured to selectively provide a fifth voltage 112 to the conversion element 120. The power module 100 can be configured to selectively provide the first voltage 102 to the memory system 110 either from the conversion element 120 or from the input 106.

[0054] The power module 100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage 102 is provided to the memory system 10 from the input 106 and the fourth voltage 110 is provided to the conversion element 120 from the first power element 130. In a second state, the fourth voltage 110 is provided to the conversion element 120 from the first power element 130 and the first voltage 102 is provided to the memory system 10 from the conversion element 120. In the third state, the fifth voltage 112 is provided to the conversion element 120 from the second power element 140 and the first voltage 104 is provided to the memory system 10 from the conversion element 120.

[0055] In certain embodiments, the power module 100 transitions from the first state to the second state upon detecting that a trigger condition is likely to occur and transitions from the second state to the third state upon detecting that the trigger condition has occurred. For example, the power module 100 may transition to the second state when it detects that a power failure is about to occur and transitions to the third state when it detects that the power failure has occurred. In certain embodiments, providing the first voltage 102 in the second state from the first power element 130 rather than from the input 106 allows a smoother transition from the first state to the third state. For example, in certain embodiments, providing the first voltage 102 from the first power element 130 has capacitive and other smoothing effects. In addition, switching the point of power transition to be between the conversion element 120 and the first and second power elements 130, 140 (e.g., the sources of the pre-regulated fourth voltage 110 in the second state and the pre-regulated fifth voltage 112 in the third state) can smooth out potential voltage spikes.

[0056] In certain embodiments, the second power element 140 does not comprise a battery and may comprise one or more capacitors. For example, as schematically illustrated in Figure 4, the second power element 140 comprises a capacitor array 142, a buck-boost converter 144 which adjusts the voltage for charging the capacitor array and a voltage/current limiter 146 which limits the charge current to the capacitor array 142 and stops charging the capacitor array 142 when it has reached a certain charge voltage. In one example embodiment, the capacitor array 142 comprises two 50 farad capacitors capable of holding a total charge of 4.6V. For example, in one example embodiment, the buck-boost converter 144 receives a 1.8V system voltage (first voltage 108) and boosts the voltage to 4.3V which is outputted to the voltage current limiter 146. The voltage/current limiter 146 limits the current going to the capacitor array 142 to 1A and stops charging the array 142 when it is charged to 4.3V. Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the second power element 140 may include alternative embodiments. For example, different components and/or different value components may be used. For example, in other embodiments, a pure boost converter may be used instead of a buck-boost converter. In another embodiment, only one capacitor may be used instead of a capacitor array 142.

[0057] The conversion element 120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 120 may comprise a plurality of sub-blocks 122, 124, 126 as schematically illustrated by Figure 4, which can provide more voltages in addition to the second voltage 104 to the memory system 10. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 10. For example, in one embodiment, sub-block 122 comprises a buck converter, sub-block 124 comprises a dual buck converter, and sub-block 126 comprises a buck-boost converter as schematically illustrated by Figure 4. Various other components for the sub-blocks 122, 124, 126 of the conversion element 120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 120 receives as input either the fourth voltage 110 from the first power element 130 or the fifth voltage 112 from the second power element 140, depending on the state of the power module 100,

and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck-converter of sub-block 122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 32 (e.g., DRAM), the non-volatile memory elements 42 (e.g., flash), and the controller 62 (e.g., an FPGA) in one embodiment. The sub-block 124 can provide the second voltage 104 as well as another reduced voltage 105 to the memory system 10. In one example embodiment, the second voltage 104 is 2.5V and is used to power the at least one circuit 52 (e.g., isolation device) and the other reduced voltage 105 is 1.2V and is used to power the controller 62 (e.g., FPGA). The sub-block 126 can provide yet another voltage 107 to the memory system 10. For example, the voltage 107 may be 3.3V and may be used to power both the controller 62 and the at least one circuit 52.

[0058] Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the conversion element 120 may include alternative embodiments. For example, there may be more or less sub-blocks which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values. In one embodiment, the volatile memory elements 32 and non-volatile memory elements 42 are powered using independent voltages and are not both powered using the first voltage 102.

[0059] **Figure 6** is a flowchart of an example method 200 of providing a first voltage 102 and a second voltage 104 to a memory system 10 including volatile and non-volatile memory subsystems 30, 40. While the method 200 is described herein by reference to the memory system 10 schematically illustrated by Figures 1-4, other memory systems are also compatible with embodiments of the method 200. During a first condition, the method 200 comprises providing the first voltage 102 to the memory system 10 from an input power supply 106 and providing the second voltage 104 to the memory system 10 from a first power subsystem in operational block 210. For example, in one embodiment, the first power subsystem comprises the first power element 130 and the voltage conversion element 120 described above with respect to Figure 4. In other embodiments, other first power subsystems are used.

[0060] The method 200 further comprises detecting a second condition in operational block 220. In certain embodiments, detecting the second condition comprises

detecting that a trigger condition is likely to occur. During the second condition, the method 200 comprises providing the first voltage 102 and the second voltage 104 to the memory system 10 from the first power subsystem in an operational block 230. For example, referring to Figure 4, a switch 148 can be toggled to provide the first voltage 102 from the conversion element 120 rather than from the input power supply.

[0061] The method 200 further comprises charging a second power subsystem in operational block 240. In certain embodiments, the second power subsystem comprises the second power element 140 or another power supply that does not comprise a battery. For example, in one embodiment, the second power subsystem comprises the second power element 140 and the voltage conversion element 120 described above with respect to Figure 4. In other embodiments, some other second power subsystem is used.

[0062] The method 200 further comprises detecting a third condition in an operational block 250 and during the third condition, providing the first voltage 102 and the second voltage 104 to the memory system 10 from the second power subsystem 140 in an operational block 250. In certain embodiments, detecting the third condition comprises detecting that the trigger condition has occurred. The trigger condition may comprise various conditions described herein. In various embodiments, for example, the trigger condition comprises a power reduction, power failure, or system hang-up. The operational blocks of the method 200 may be performed in different orders in various embodiments. For example, in certain embodiments, the second power subsystem 140 is charged before detecting the second condition.

[0063] In certain embodiments, the memory system 10 comprises a volatile memory subsystem 30 and a non-volatile memory subsystem 40 comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system 10 also comprises a controller 62 operatively coupled to the volatile memory subsystem 30 and operatively coupled to the non-volatile memory subsystem 40. The controller 62 can be configured to allow data to be communicated between the volatile memory subsystem 30 and the host system when the memory system 10 is operating in a first state and to allow data to be communicated between the volatile memory subsystem 30 and the non-volatile memory subsystem 40 when the memory system 10 is operating in a second state.

[0064] Although the memory system 10 having extra storage capacity of the non-volatile memory subsystem 40 has been described with respect to certain embodiments, alternative configurations exist. For example, in certain embodiments, there may be more than 100 percent more storage capacity in the non-volatile memory subsystem 40 than in the volatile memory subsystem 30. In various embodiments, there may be at least 200, 300, or 400 percent more storage capacity in the non-volatile memory subsystem 40 than in the volatile memory subsystem 30. In other embodiments, the non-volatile memory subsystem 40 includes at least some other integer multiples of the storage capacity of the volatile memory subsystem 30. In some embodiments, the non-volatile memory subsystem 40 includes a non-integer multiple of the storage capacity of the volatile memory subsystem 30. In one embodiment, the non-volatile memory subsystem 40 includes less than 100 percent more storage capacity than does the volatile memory subsystem 30.

[0065] The extra storage capacity of the non-volatile memory subsystem 40 can be used to improve the backup capability of the memory system 10. In certain embodiments in which data can only be written to portions of the non-volatile memory subsystem 40 which do not contain data (e.g., portions which have been erased), the extra storage capacity of the non-volatile memory subsystem 40 allows the volatile memory subsystem 30 to be backed up in the event of a subsequent power failure or other trigger event. For example, the extra storage capacity of the non-volatile memory subsystem 40 may allow the memory system 10 to backup the volatile memory subsystem 30 efficiently in the event of multiple trigger conditions (e.g., power failures). In the event of a first power failure, for example, the data in the volatile memory system 30 is copied to a first, previously erased portion of the non-volatile memory subsystem 40 via the controller 62. Since the non-volatile memory subsystem 40 has more storage capacity than does the volatile memory subsystem 30, there is a second portion of the non-volatile memory subsystem 40 which does not have data from the volatile memory subsystem 30 copied to it and which remains free of data (e.g., erased). Once system power is restored, the controller 62 of the memory system 10 restores the data to the volatile memory subsystem 30 by copying the backed-up data from the non-volatile memory subsystem 40 back to the volatile memory subsystem 30. After the data is restored, the memory system 10 erases the non-volatile memory subsystem 40. While the first portion

of the non-volatile memory subsystem 40 is being erased, it may be temporarily un-accessible.

[0066] If a subsequent power failure occurs before the first portion of the non-volatile memory subsystem 40 is completely erased, the volatile memory subsystem 30 can be backed-up or stored again in the second portion of the non-volatile memory subsystem 40 as described herein. In certain embodiments, the extra storage capacity of the non-volatile memory subsystem 40 may allow the memory system 10 to operate more efficiently. For example, because of the extra storage capacity of the non-volatile memory subsystem 40, the memory system 10 can handle a higher frequency of trigger events that is not limited by the erase time of the non-volatile memory subsystem 40.

[0067] **Figure 7** is a flowchart of an example method 300 of controlling a memory system 10 operatively coupled to a host system and which includes a volatile memory subsystem 30 and a non-volatile memory subsystem 40. In certain embodiments, the non-volatile memory subsystem 40 comprises at least 100 percent more storage capacity than does the volatile memory subsystem 30 as described herein. While the method 300 is described herein by reference to the memory system 10 schematically illustrated by Figures 1-3, the method 300 can be practiced using other memory systems in accordance with certain embodiments described herein. In an operational block 310, the method 300 comprises communicating data between the volatile memory subsystem 30 and the host system when the memory system 10 is in a first mode of operation. The method 300 further comprises storing a first copy of data from the volatile memory subsystem 30 to the non-volatile memory subsystem 40 at a first time when the memory system 10 is in a second mode of operation in an operational block 320.

[0068] In an operational block 330, the method 300 comprises restoring the first copy of data from the non-volatile memory subsystem 40 to the volatile memory subsystem 30. The method 300 further comprises erasing the first copy of data from the non-volatile memory subsystem 40 in an operational block 340. The method further comprises storing a second copy of data from the volatile memory subsystem 30 to the non-volatile memory subsystem 40 at a second time when the memory system 10 is in the second mode of

operation in an operational block 350. Storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem 40.

[0069] In some embodiments, the memory system 10 enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the non-volatile memory subsystem 40. The method 300 can also include restoring the second copy of data from the non-volatile memory subsystem 40 to the volatile memory subsystem 30 in an operational block 360. The operational blocks of method 300 referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, the restoring the second copy of data to the volatile memory subsystem 30 at operational block 360 before the first copy of data is completely erased in the operational block 340.

[0070] **Figure 8** schematically illustrates an example clock distribution topology 400 of a memory system 10 in accordance with certain embodiments described herein. The clock distribution topology 400 generally illustrates the creation and routing of the clock signals provided to the various components of the memory system 10. A clock source 402 such as, for example, a 25 MHz oscillator, generates a clock signal. The clock source 402 may feed a clock generator 404 which provides a clock signal 406 to the controller 62, which may be an FPGA. In one embodiment, the clock generator 404 generates a 125 MHz clock signal 406. The controller 62 receives the clock signal 406 and uses it to clock the controller 62 master state control logic. For example, the master state control logic may control the general operation of an FPGA controller 62.

[0071] The clock signal 406 can also be input into a clock divider 410 which produces a frequency-divided version of the clock signal 406. In an example embodiment, the clock divider 410 is a divide by two clock divider and produces a 62.5 MHz clock signal in response to the 125 MHz clock signal 406. A non-volatile memory phase-locked loop (PLL) block 412 can be included (e.g., in the controller 62) which distributes a series of clock signals to the non-volatile memory subsystem 40 and to associated control logic. For example, a series of clock signals 414 can be sent from the controller 62 to the non-volatile memory subsystem 40. Another clock signal 416 can be used by the controller logic which is dedicated to controlling the non-volatile memory subsystem 40. For example, the clock

signal 416 may clock the portion of the controller 62 which is dedicated to generating address and/or control lines for the non-volatile memory subsystem 40. A feedback clock signal 418 is fed back into the non-volatile memory PLL block 412. In one embodiment, the PLL block 412 compares the feedback clock 418 to the reference clock 411 and varies the phase and frequency of its output until the reference 411 and feedback 418 clocks are phase and frequency matched.

[0072] A version of the clock signal 406 such as the backup clock signal 408 may be sent from the controller to the volatile memory subsystem 30. The clock signal 408 may be, for example, a differential version of the clock signal 406. As described herein, the backup clock signal 408 may be used to clock the volatile memory subsystem 30 when the memory system 10 is backing up the data from the volatile memory subsystem 30 into the non-volatile memory subsystem 40. In certain embodiments, the backup clock signal 408 may also be used to clock the volatile memory subsystem 30 when the memory system 10 is copying the backed-up data back into the volatile memory subsystem 30 from the non-volatile memory subsystem 40 (also referred to as restoring the volatile memory subsystem 30). The volatile memory subsystem 30 may normally be run at a higher frequency (e.g., DRAM running at 400MHz) than the non-volatile memory subsystem 40 (e.g., flash memory running at 62.5MHz) when communicating with the host system (e.g., when no trigger condition is present). However, in certain embodiments the volatile memory subsystem 30 may be operated at a reduced frequency (e.g., at twice the frequency of the non-volatile memory subsystem 40) without introducing significant delay into the system during backup operation and/or restore operations. Running the volatile memory subsystem 30 at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system 10.

[0073] In one embodiment, the backup clock 408 and the volatile memory system clock signal 420 are received by a multiplexer 422, as schematically illustrated by Figure 7. The multiplexer 422 can output either the volatile memory system clock signal 420 or the backup clock signal 408 depending on the backup state of the memory system 10. For example, when the memory system 10 is not performing a backup or restore operation and is communicating with the host system (e.g., normal operation), the volatile memory system

clock signal 420 may be provided by the multiplexer 422 to the volatile memory PLL block 424. When the memory system 10 is performing a backup (or restore) operation, the backup clock signal 408 may be provided.

[0074] The volatile memory PLL block 424 receives the volatile memory reference clock signal 423 from the multiplexer 422 and can generate a series of clock signals which are distributed to the volatile memory subsystem 30 and associated control logic. For example, in one embodiment, the PLL block 424 generates a series of clock signals 426 which clock the volatile memory elements 32. A clock signal 428 may be used to clock control logic associated with the volatile memory elements, such as one or more registers (e.g., the one or more registers of a registered DIMM). Another clock signal 430 may be sent to the controller 62. A feedback clock signal 432 is fed back into the volatile memory PLL block 424. In one embodiment, the PLL block 424 compares the feedback clock signal 432 to the reference clock signal 423 and varies the phase and frequency of its output until the reference clock signal 423 and the feedback clock signal 432 clocks are phase and frequency matched.

[0075] The clock signal 430 may be used by the controller 62 to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem 30. For example, control logic in the controller 62 may be used to control the volatile memory subsystem 30 during a backup or restore operation. The clock signal 430 may be used as a reference clock signal for the PLL block 434 which can generate one or more clocks 438 used by logic in the controller 62. For example, the PLL block 424 may generate one or more clock signals 438 used to drive logic circuitry associated with controlling the volatile memory subsystem 30. In certain embodiments, the PLL block 434 includes a feedback clock signal 436 and operates in a similar manner to other PLL blocks described herein.

[0076] The clock signal 430 may be used as a reference clock signal for the PLL block 440 which may generate one or more clock signals used by a sub-block 442 to generate one or more other clock signals 444. In one embodiment, for example, the volatile memory subsystem 30 comprises DDR2 SDRAM elements and the sub-block 442 generates one or more DDR2 compatible clock signals 444. A feedback clock signal 446 is fed back into the

PLL block 440. In certain embodiments, the PLL block 440 operates in a similar manner to other PLL blocks described herein.

[0077] While described with respect to the example embodiment of Figure 7, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem 30 operates on the volatile memory clock signal 420 and there is no backup clock signal 408. In some embodiments, the volatile memory subsystem 30 is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem 30 is operated at a reduced frequency during a restore operation and not during a backup operation.

[0078] **Figure 9** is a flowchart of an example method 500 of controlling a memory system 10 operatively coupled to a host system. Although described with respect to the memory system 10 described herein with respect to Figures 1-3 and 7, the method 500 is compatible with other memory systems. The memory system 10 may include a clock distribution topology 400 similar to the one described above with respect to Figure 7 or another clock distribution topology. The memory system 10 can include a volatile memory subsystem 30 and a non-volatile memory subsystem 40.

[0079] In an operational block 510, the method 500 comprises operating the volatile memory subsystem 40 at a first frequency when the memory system 10 is in a first mode of operation in which data is communicated between the volatile memory subsystem 30 and the host system. In an operational block 520, the method 500 comprises operating the non-volatile memory subsystem 40 at a second frequency when the memory system 10 is in a second mode of operation in which data is communicated between the volatile memory subsystem 30 and the non-volatile memory subsystem 40. The method 500 further comprises operating the volatile memory subsystem 30 at a third frequency in an operational block 530 when the memory system 10 is in the second mode of operation. In certain embodiments, the memory system 10 is not powered by a battery when it is in the second mode of operation. The memory system 10 may switch from the first mode of operation to the second mode of

operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

[0080] The third frequency can be less than the first frequency. For example, the third frequency can be approximately equal to the second frequency. In certain embodiments, the reduced frequency operation is an optional mode. In yet other embodiments, the first, second and/or third frequencies are configurable by a user or by the memory system 10.

[0081] **Figure 10** schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 630, 640 of a volatile memory subsystem 30 of a memory system 10 to a controller 62 of the memory system 10. While the example of Figure 10 shows a topology including two DRAM segments 630, 640 for the purposes of illustration, each address location of the volatile memory subsystem 30 comprises more than the two segments in certain embodiments. The data lines 632, 642 from the first DRAM segment 630 and the second DRAM segment 640 of the volatile memory subsystem 30 are coupled to switches 650, 652 which are coupled to the controller 62 (e.g., logic element 70) of the memory system 10. The chip select lines 634, 644 and the self-refresh lines 636, 646 (e.g., CKE signals) of the first and second DRAM segments 630, 640, respectively, are coupled to the controller 62. In certain embodiments, the controller 62 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 30. In certain embodiments, the buffer is a first-in, first out buffer (FIFO). In certain embodiments, data slices from each DRAM segment 630, 640 comprise a portion of the volatile memory subsystem data bus. In one embodiment, for example, the volatile memory subsystem 30 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 640 may comprise the remaining 32 bits

of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.

[0082] In certain embodiments, the switches 650, 652 can each be selectively switched to selectively operatively couple the data lines 632, 642, respectively from the first and second DRAM segments 630, 640 to the controller 62. The chip select lines 634, 644 enable the first and second DRAM segments 630, 640, respectively, of the volatile memory subsystem 30, and the self-refresh lines 636, 646 toggle the first and second DRAM segments 630, 640, respectively, from self-refresh mode to active mode. In certain embodiments, the first and second DRAM segments 630, 640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.

[0083] In certain embodiments, when the memory system 10 is backing up the volatile memory system 30, data slices from only one of the two DRAM segments 630, 640 at a time are sent to the controller 62. For example, when the first slice is being written to the controller 62 during a back-up, the controller 62 sends a signal via the CKE line 636 to the first DRAM segment 630 to put the first DRAM segment 630 in active mode. In certain embodiments, the data slice from the first DRAM segment 630 for multiple words (e.g., a block of words) is written to the controller 62 before writing the second data slice from the second DRAM segment 640 to the controller 62. While the first data slice is being written to the controller 62, the controller 62 also sends a signal via the CKE line 646 to put the second DRAM segment 640 in self-refresh mode. Once the first data slice for one word or for a block of words is written to the controller 62, the controller 62 puts the first DRAM segment 630 into self-refresh mode by sending a signal via the CKE line 636 to the first DRAM segment 640. The controller 62 also puts the second DRAM segment 640 into active mode by sending a signal via the CKE line 646 to the DRAM segment 640. The second slice for a word or for a block of words is written to the controller 62. In certain embodiments, when the first and second data slices are written to the buffer in the controller 62, the controller 62 combines the first and second data slices 630, 640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem

40. In certain embodiments, this process is called “slicing” the volatile memory subsystem 30.

[0084] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the non-volatile memory elements 42 write each backed-up data word to the controller 62 which writes a first slice of the data word to the volatile memory subsystem 30 and then a second slice of the data word to the volatile memory subsystem 30. In certain embodiments, slicing the volatile memory subsystem 30 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 30 during a backup operation.

[0085] **Figure 11** is a flowchart of an example method 600 of controlling a memory system 10 operatively coupled to a host system and which includes a volatile memory subsystem 30 and a non-volatile memory subsystem 40. Although described with respect to the memory system 10 described herein with respect to Figures 1-3 and 10, the method 600 is compatible with other memory systems. The method 600 comprises communicating data words between the volatile memory subsystem 40 and the host system when the memory system 10 is in a first mode of operation in an operational block 610. For example, the memory system 10 may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation or is not being powered by a secondary power supply.

[0086] In an operational block 620, the method further comprises transferring data words from the volatile memory subsystem 30 to the non-volatile memory subsystem 40 when the memory system 10 is in a second mode of operation. In certain embodiments, each data word comprises the data stored in a particular address of the memory system 10. The memory system 10 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 622, storing a second portion of the data word in the buffer in an operational block 624, and writing the entire data word from the buffer to the non-volatile memory subsystem 40 in an operational block 626.

[0087] In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or “slice”) may comprise 40 bits of the data word, and the second portion (or “slice”) may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 62. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 62 which comprises an FPGA. The method 600 may generally be referred to as “slicing” the volatile memory during a backup operation. In the example embodiment, the process of “slicing” the volatile memory during a backup includes bringing the 32-bit slice out of self refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 62 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).

[0088] In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.

[0089] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the non-volatile memory elements 40 write each backed-up data word to the controller 62 which writes a first portion of the data word to the volatile memory subsystem 30 and then a second portion of the data word to the volatile memory 30. In certain embodiments, slicing the volatile memory subsystem 30 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 30 during a backup operation.

[0090] The method 600 can advantageously provide significant power savings and can lead to other advantages. For example, in one embodiment where the volatile

memory subsystem 30 comprises DRAM elements, only the slice of the DRAM which is currently being accessed (e.g., written to the buffer) during a backup is configured in full-operational mode. The slice or slices that are not being accessed may be put in self-refresh mode. Because DRAM in self-refresh mode uses significantly less power than DRAM in full-operational mode, the method 600 can allow significant power savings. In certain embodiments, each slice of the DRAM includes a separate self-refresh enable (e.g., CKe) signal which allows each slice to be accessed independently.

[0091] In addition, the connection between the DRAM elements and the controller 62 may be as large as the largest slice instead of as large as the data bus. In the example embodiment, the connection between the controller 62 and the DRAM may be 40 bits instead of 72 bits. As a result, pins on the controller 62 may be used for other purposes or a smaller controller may be used due to the relatively low number of pin-outs used to connect to the volatile memory subsystem 30. In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 30 and the controller 62 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.

[0092] Various embodiments of the present invention have been described above. Although this invention has been described with reference to these specific embodiments, the descriptions are intended to be illustrative of the invention and are not intended to be limiting. Various modifications and applications may occur to those skilled in the art without departing from the true spirit and scope of the invention as defined in the appended claims.

WHAT IS CLAIMED IS:

1. A memory system coupled to a computer system, comprising:
a volatile memory subsystem;
a non-volatile memory subsystem;
a controller operatively coupled to the non-volatile memory subsystem;
at least one circuit configured to selectively operatively decouple the controller from the volatile memory subsystem.
2. The memory system of Claim 1, wherein the memory system is configured to be operated in at least two states, the at least two states comprising:
a first state in which the controller and the non-volatile memory subsystem are decoupled from the volatile memory subsystem by the at least one circuit; and
a second state in which the volatile memory subsystem is operatively coupled to the controller to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem via the controller.
3. The memory system of Claim 2, wherein power is supplied to the volatile memory subsystem from a first power supply when the memory system is in the first state and power is supplied to the volatile memory subsystem from a second power supply when the memory system is in the second state.
4. The memory system of Claim 3, wherein the second power supply does not comprise a battery.
5. The memory system of Claim 3, wherein the second power supply comprises one or more capacitors.
6. The memory system of Claim 5, wherein the second power supply comprises a battery and is configured to power the volatile memory subsystem for less than thirty minutes.
7. The memory system of Claim 5, wherein the one or more capacitors comprises one or more double layered capacitors.
8. The memory system of Claim 5, wherein the one or more capacitors are charged by the first power supply while the memory system is in the first state.

9. The memory system of Claim 3, further comprising a printed circuit board, wherein the second power supply and the memory system are located on the printed circuit board.

10. The memory system of Claim 3, wherein the at least two states further comprise a third state in which power is supplied to the volatile memory subsystem from a third power supply.

11. The memory system of Claim 2, wherein the controller and the non-volatile memory subsystem do not add a significant capacitive load to the volatile memory system when the memory system is in the first state.

12. The memory system of Claim 1, wherein the volatile memory subsystem comprises one or more DRAM memory elements.

13. The memory system of Claim 1, wherein the volatile memory subsystem comprises one or more SRAM memory elements.

14. The memory system of Claim 1, wherein the non-volatile memory subsystem comprises one or more flash memory elements.

15. The memory system of Claim 1, wherein the memory system is implemented on a memory module.

16. The memory system of Claim 1, wherein the at least one circuit comprise one or more switches coupled to the controller, to the volatile memory subsystem, and to the host computer.

17. The memory system of Claim 1, wherein the volatile memory subsystem comprises a registered DIMM comprising one or more registers and a plurality of DRAM elements, wherein the one or more registers are coupled to the computer system.

18. The memory system of Claim 17, wherein the at least one circuit comprise one or more switches coupled to the one or more registers and to the plurality of DRAM elements and configured to selectively operatively couple the volatile memory subsystem and to the computer system.

19. The memory system of Claim 17, wherein the one or more registers are coupled to the plurality of DRAM elements and the at least one circuit comprises one or more switches coupled to the one or more registers and to a power source for the one or more

registers, wherein the one or more switches are configured to selectively operatively couple the one or more registers to the power source for the one or more registers.

20. The memory system of Claim 1, wherein the non-volatile memory subsystem comprises at least 100 percent more storage capacity than does the volatile memory subsystem.

21. A power module for providing a plurality of voltages to a memory system comprising non-volatile and volatile memory, the plurality of voltages comprising at least a first voltage and a second voltage, the power module comprising:

- an input providing a third voltage to the power module;

- a voltage conversion element configured to provide the second voltage to the memory system;

- a first power element configured to selectively provide a fourth voltage to the conversion element;

- a second power element configured to selectively provide a fifth voltage to the conversion element, wherein the power module is configured to selectively provide the first voltage to the memory system either from the conversion element or from the input, wherein the power module is configured to be operated in at least three states comprising:

- a first state in which the first voltage is provided to the memory system from the input and the fourth voltage is provided to the conversion element from the first power element;

- a second state in which the fourth voltage is provided to the conversion element from the first power element and the first voltage is provided to the memory system from the conversion element;

- a third state in which the fifth voltage is provided to the conversion element from the second power element and the first voltage is provided to the memory system from the conversion element.

22. The power module of Claim 21, wherein the first power element comprises a pulse-width modulation power controller.

23. The power module of Claim 21, wherein the second power element does not comprise a battery.

24. The power module of Claim 21, wherein the second power element comprises one or more capacitors.

25. The power module of Claim 24, wherein the second power element comprises a battery and is configured to power the volatile memory for less than thirty minutes.

26. The power module of Claim 21, wherein the conversion element comprises one or more buck converters.

27. The power module of Claim 21, wherein the conversion element comprises one or more buck-boost converters.

28. A method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems, the method comprising:

 during a first condition, providing the first voltage to the memory system from an input power supply and providing the second voltage to the memory system from a first power subsystem;

 detecting a second condition;

 during the second condition, providing the first voltage and the second voltage to the memory system from the first power subsystem;

 charging a second power subsystem;

 detecting a third condition;

 during the third condition, providing the first voltage and the second voltage to the memory system from the second power subsystem.

29. The method of Claim 28, wherein detecting the second condition comprises detecting that a trigger condition is likely to occur.

30. The method of Claim 29, wherein detecting the third condition comprises detecting that the trigger condition has occurred.

31. The method of Claim 28, wherein the trigger condition comprises a power reduction.

32. The method of Claim 28, wherein the trigger condition comprises a power failure.

33. The method of Claim 28, wherein the trigger condition comprises a system hang-up.

34. The method of Claim 28, wherein the first power subsystem comprises a pulse-width modulation power controller and a voltage conversion subsystem.

35. The method of Claim 28, wherein the second power subsystem does not comprise a battery.

36. The method of Claim 35, wherein the second power subsystem comprises one or more capacitors and the voltage conversion subsystem.

37. A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

operating the volatile memory subsystem at a first frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third frequency when the memory system is in the second mode of operation, the third frequency less than the first frequency.

38. The method of Claim 37, wherein the third frequency is approximately equal to the second frequency.

39. The method of Claim 37, wherein the memory system is not powered by a battery when it is in the second mode of operation.

40. The method of Claim 7, wherein the memory system switches from the first mode of operation to the second mode of operation in response to a trigger condition.

41. The method of Claim 40, wherein the trigger condition comprises a power failure condition.

42. The method of Claim 37, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem and the non-volatile memory subsystem are located on the printed circuit board.

43. A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

- communicating data words between the volatile memory subsystem and the host system when the memory system is in a first mode of operation; and

- transferring data words from the volatile memory subsystem to the non-volatile memory subsystem when the memory system is in a second mode of operation, wherein transferring each data word comprises:

- storing a first portion of the data word in a buffer;

- storing a second portion of the data word in the buffer; and

- writing the entire data word from the buffer to the non-volatile memory subsystem.

44. The method of Claim 43, wherein the transferring each data word further comprises storing a third portion of each data word in the buffer.

45. A memory system operatively coupled to a host system, comprising:

- a volatile memory subsystem;

- a non-volatile memory subsystem comprising at least 100 percent more storage capacity than does the volatile memory subsystem; and

- a controller operatively coupled to the volatile memory subsystem and operatively coupled to the non-volatile memory subsystem, the controller configured to allow data to be communicated between the volatile memory subsystem and the host system when the memory system is operating in a first state and to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem when the memory system is operating in a second state.

46. The memory system of Claim 45, wherein power is supplied to the volatile memory subsystem from a first power supply when the memory system is in the first state.

47. The memory system of Claim 46, wherein power is supplied to the volatile memory subsystem from a second power supply when the memory system is in the second state.

48. The memory system of Claim 47, wherein the second power supply does not comprise a battery.

49. The memory system of Claim 45, wherein the memory system further comprises a printed circuit board and the volatile memory subsystem, the non-volatile memory subsystem, and the controller are located on the printed circuit board.

50. A method of controlling a memory system operatively coupled to a host system and which includes a volatile memory subsystem and a non-volatile memory subsystem, the method comprising:

communicating data between the volatile memory subsystem and the host system when the memory system is in a first mode of operation;

storing a first copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a first time when the memory system is in a second mode of operation;

restoring the first copy of data from the non-volatile memory subsystem to the volatile memory subsystem;

erasing the first copy of data from the non-volatile memory subsystem; and

storing a second copy of data from the volatile memory subsystem to the non-volatile memory subsystem at a second time when the memory system is in the second mode of operation, wherein storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem.

51. The method of Claim 50, further comprising restoring the second copy of data from the non-volatile memory subsystem to the volatile memory subsystem.

52. The method of Claim 50, wherein the memory system enters the second mode of operation in response to a power failure.

53. The method of Claim 50, wherein the memory system is not powered by a battery when it is in the second mode of operation.

54. The method of Claim 50, wherein the first copy of data and the second copy of data are stored in separate portions of the non-volatile memory subsystem.

NON-VOLATILE MEMORY MODULE

ABSTRACT OF THE DISCLOSURE

Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory and a controller which are configured such that the controller backs up the volatile memory using the non-volatile memory in the event of a trigger condition. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which is not a battery and may include, for example, a capacitor or capacitor array. The memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or the controller when the volatile memory is interacting with the host system.

5430569
052708

Electronic Acknowledgement Receipt

EFS ID:	3392259
Application Number:	12131873
International Application Number:	
Confirmation Number:	6645
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	to be determined to be determined
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Segovia Phan
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.040A
Receipt Date:	02-JUN-2008
Filing Date:	
Time Stamp:	20:45:04
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Drawings-only black and white line drawings	drawings.PDF	269349 828146be2f520b280ab963b8d9952d8a6b77aadd	no	12

Warnings:

Information:

2	Transmittal of New Application	tran.pdf	21093 95ce43136e8111ef4b7f4d107713f5d2538b07b2	no	1
Warnings:					
Information:					
3		spec.pdf	2156614 c21b75c6bc2ff6efe33d7bde740779f9dc989257c	yes	40
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	31	
	Claims		32	39	
	Abstract		40	40	
Warnings:					
Information:					
Total Files Size (in bytes):			2447056		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Figure 1:

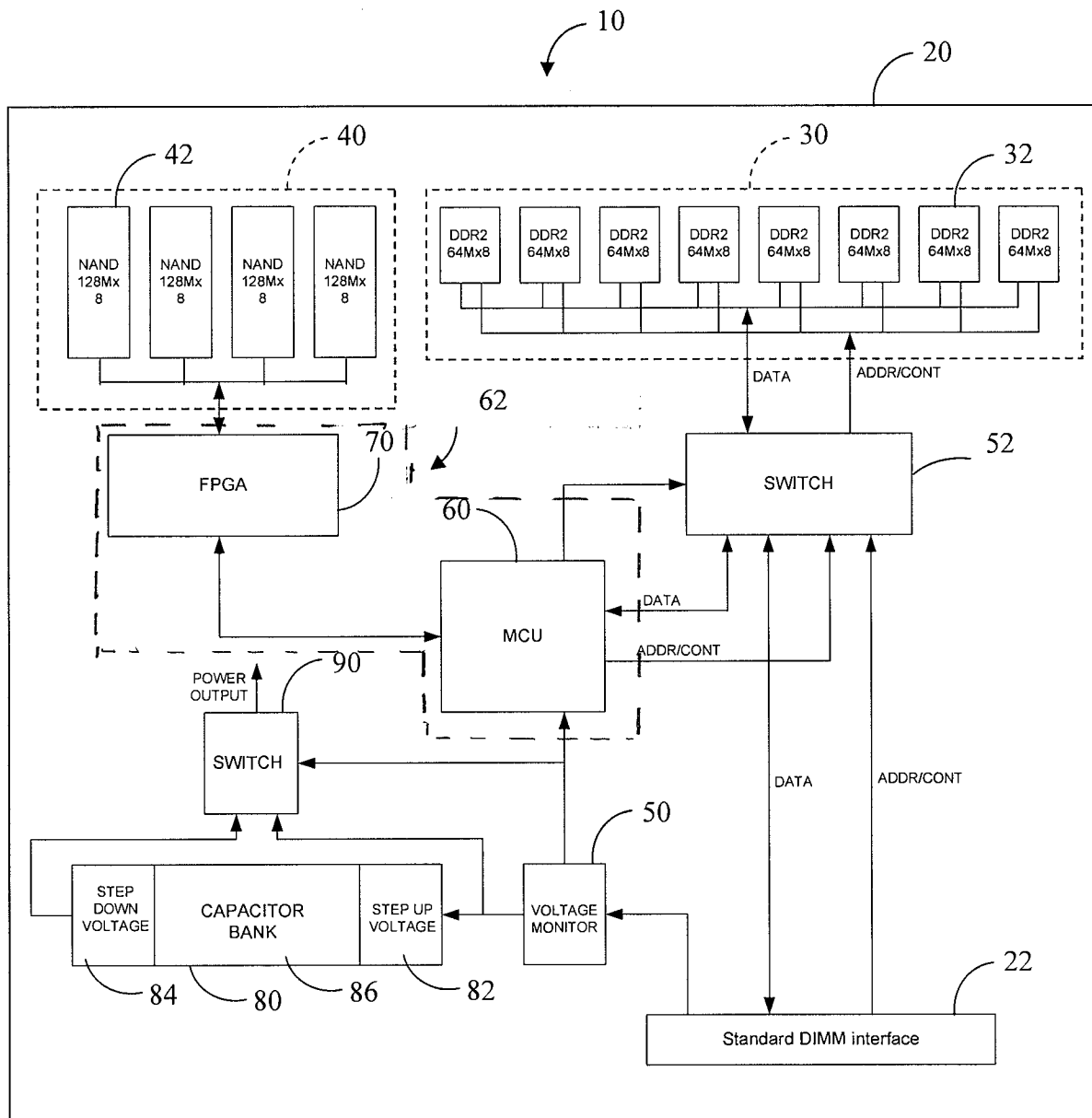


Figure 2:

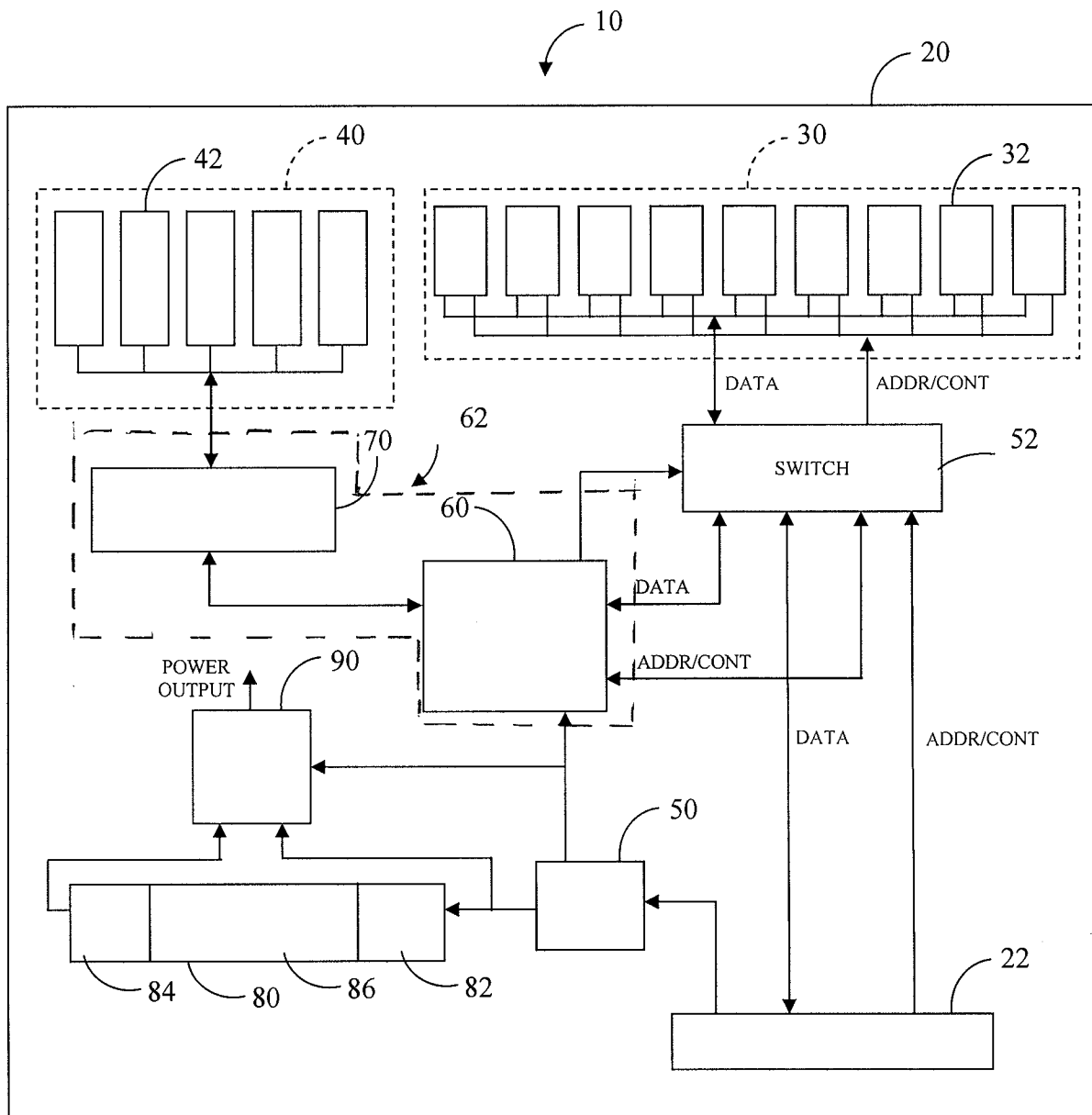


Figure 3:

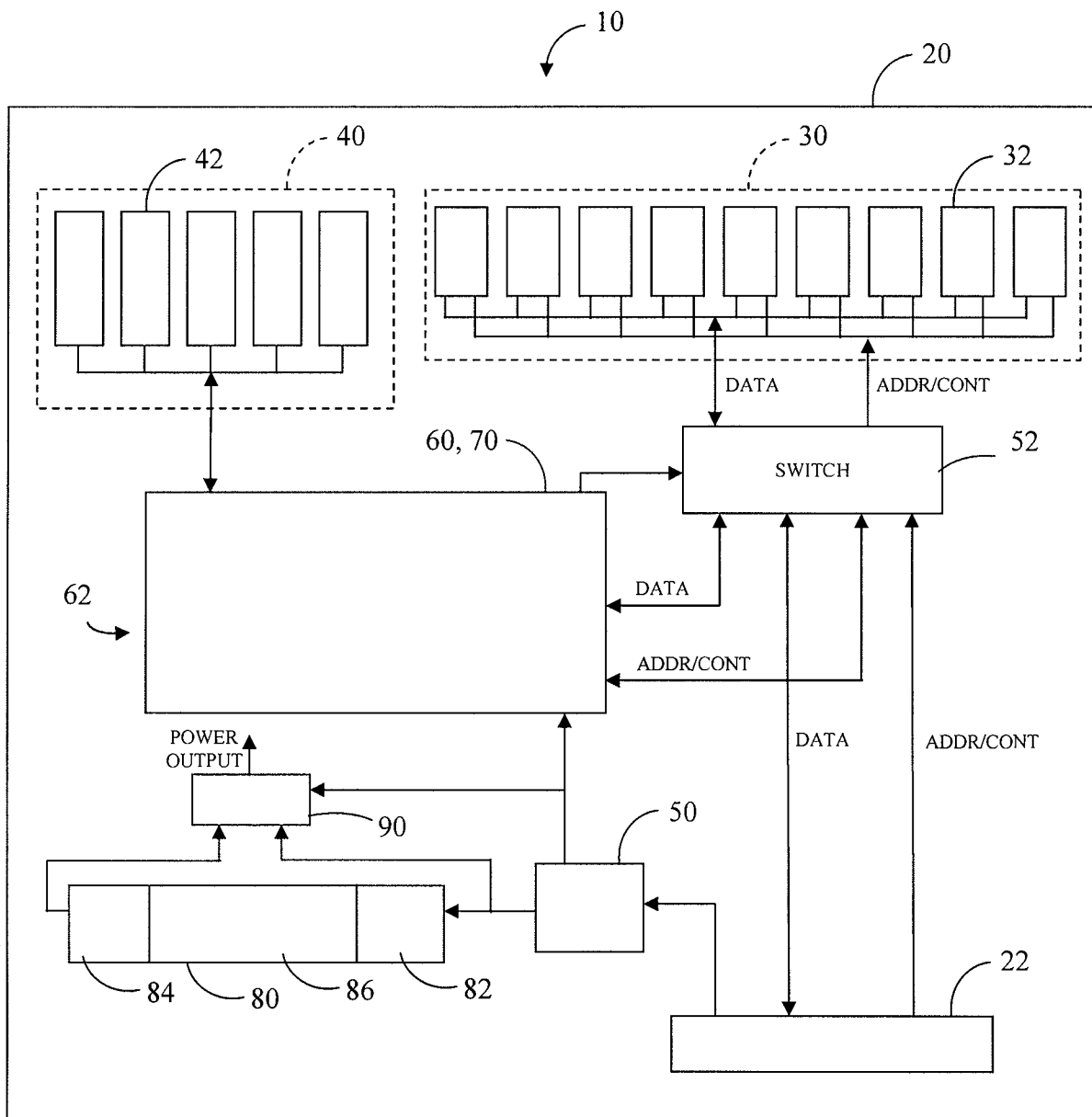


Figure 4A:

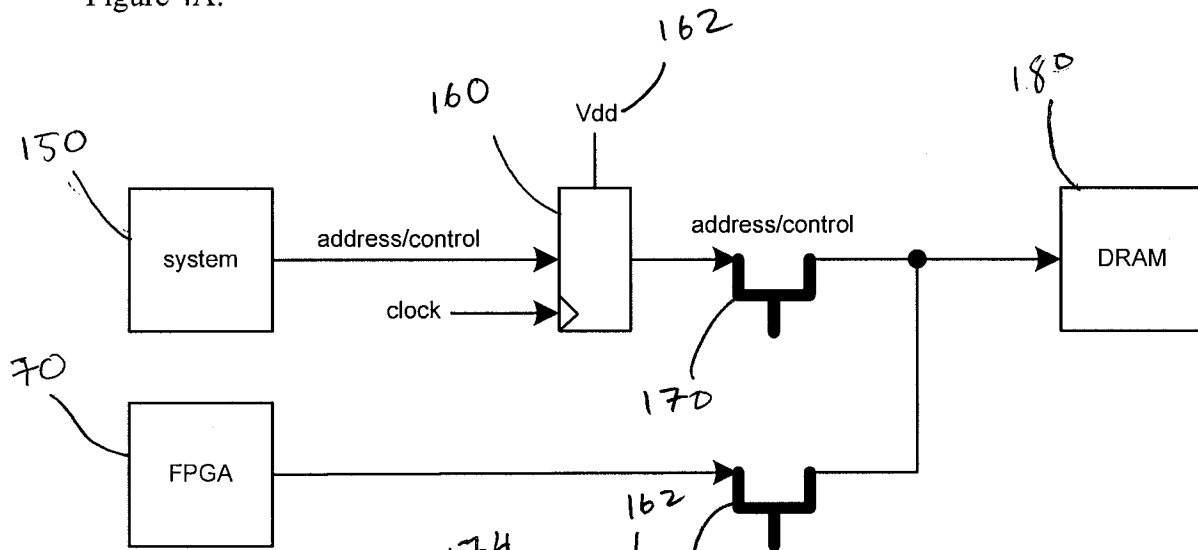


Figure 4B:

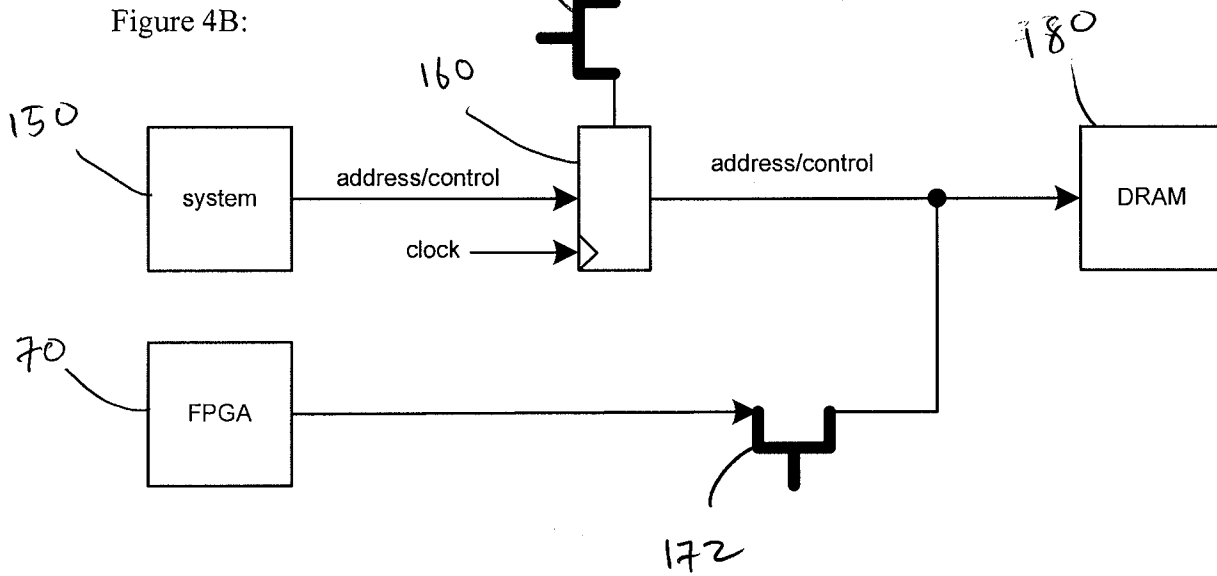


Figure 4C:

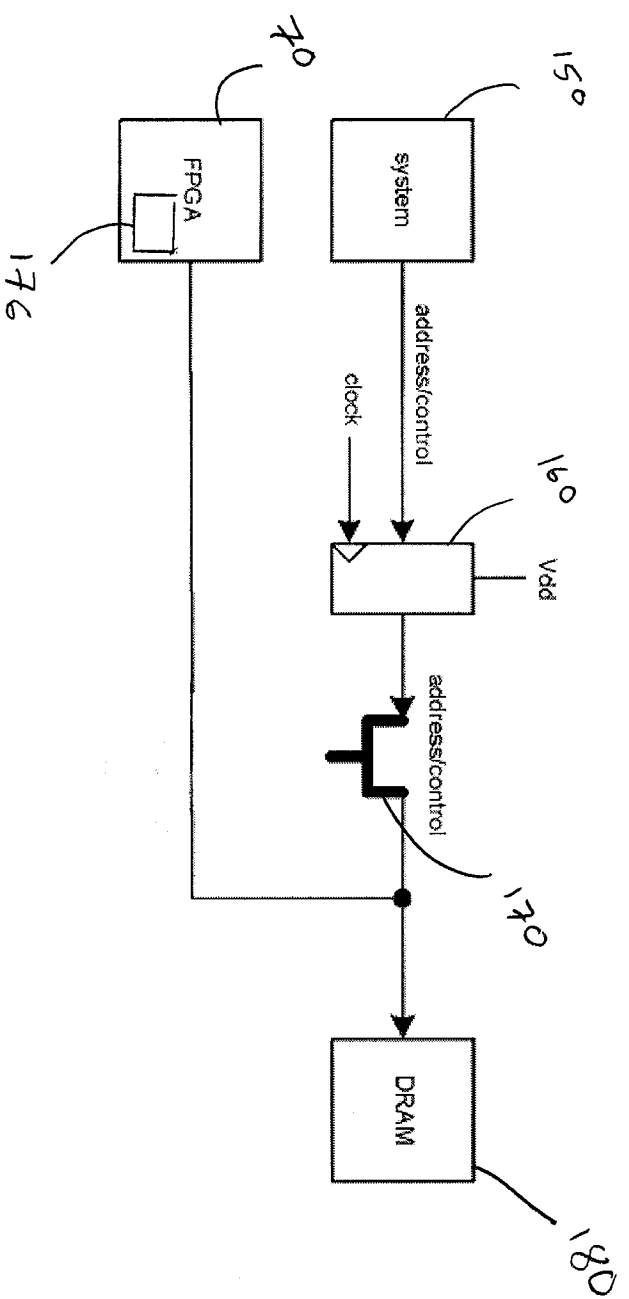


Figure 5

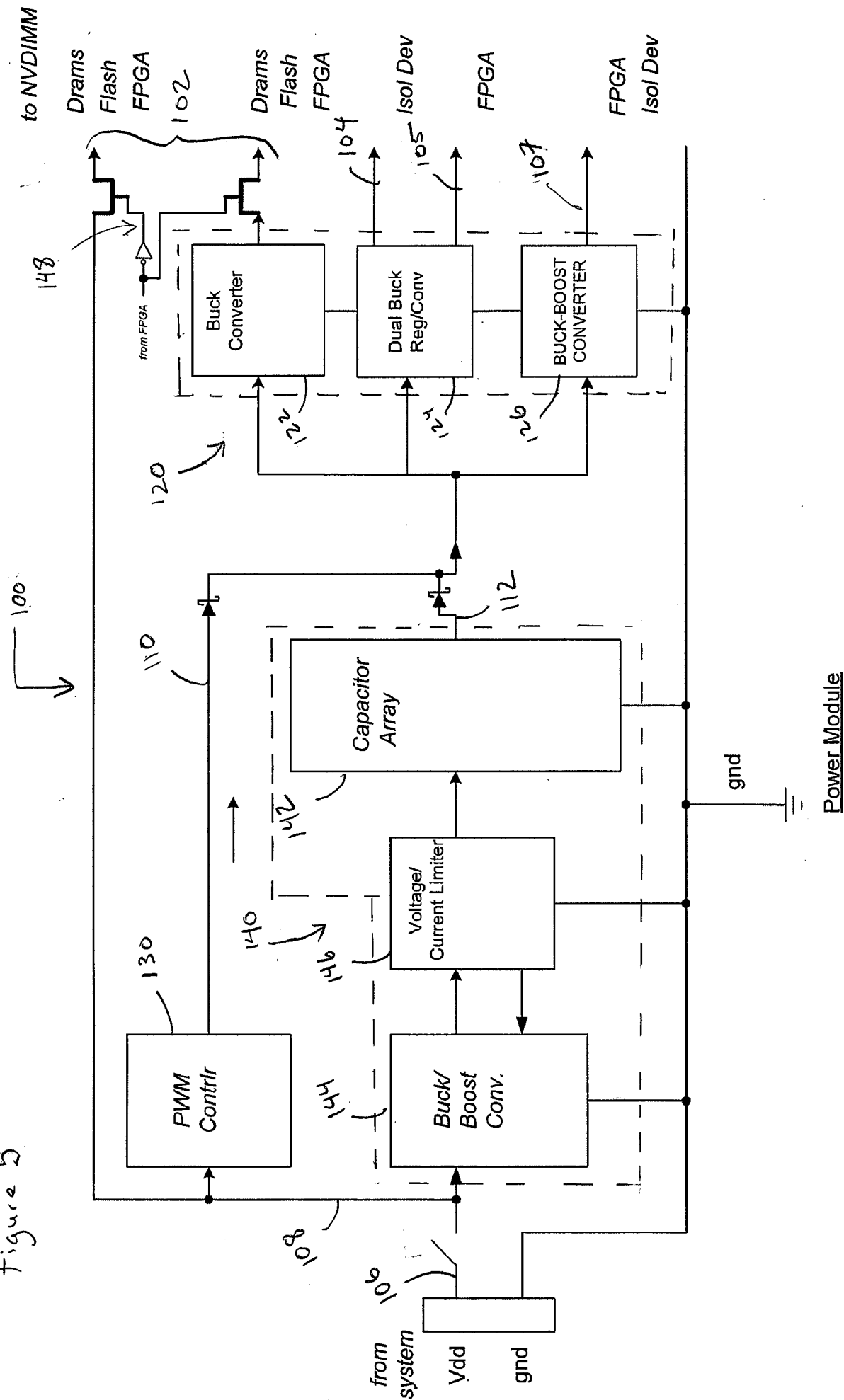


Figure 6

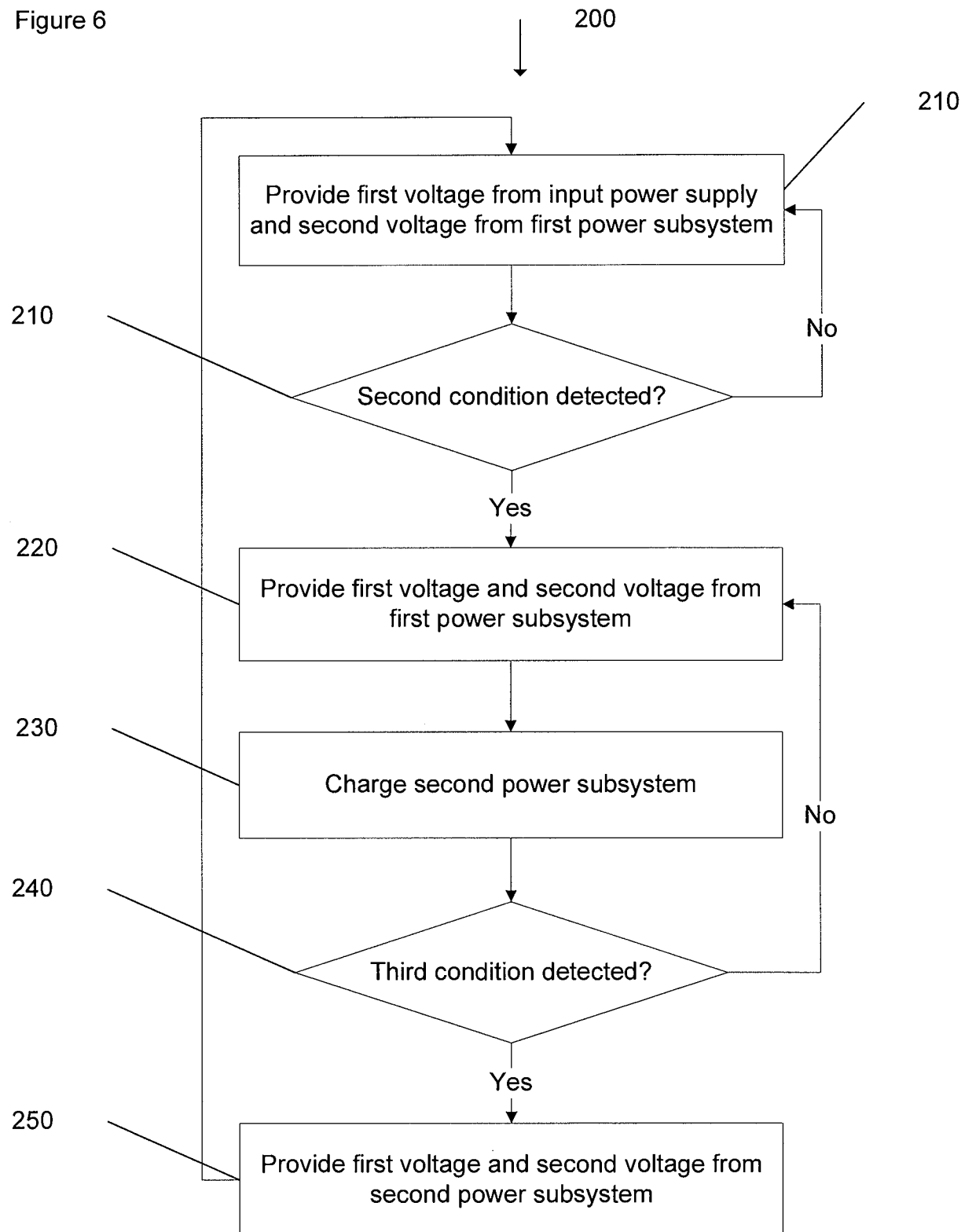


Figure 7

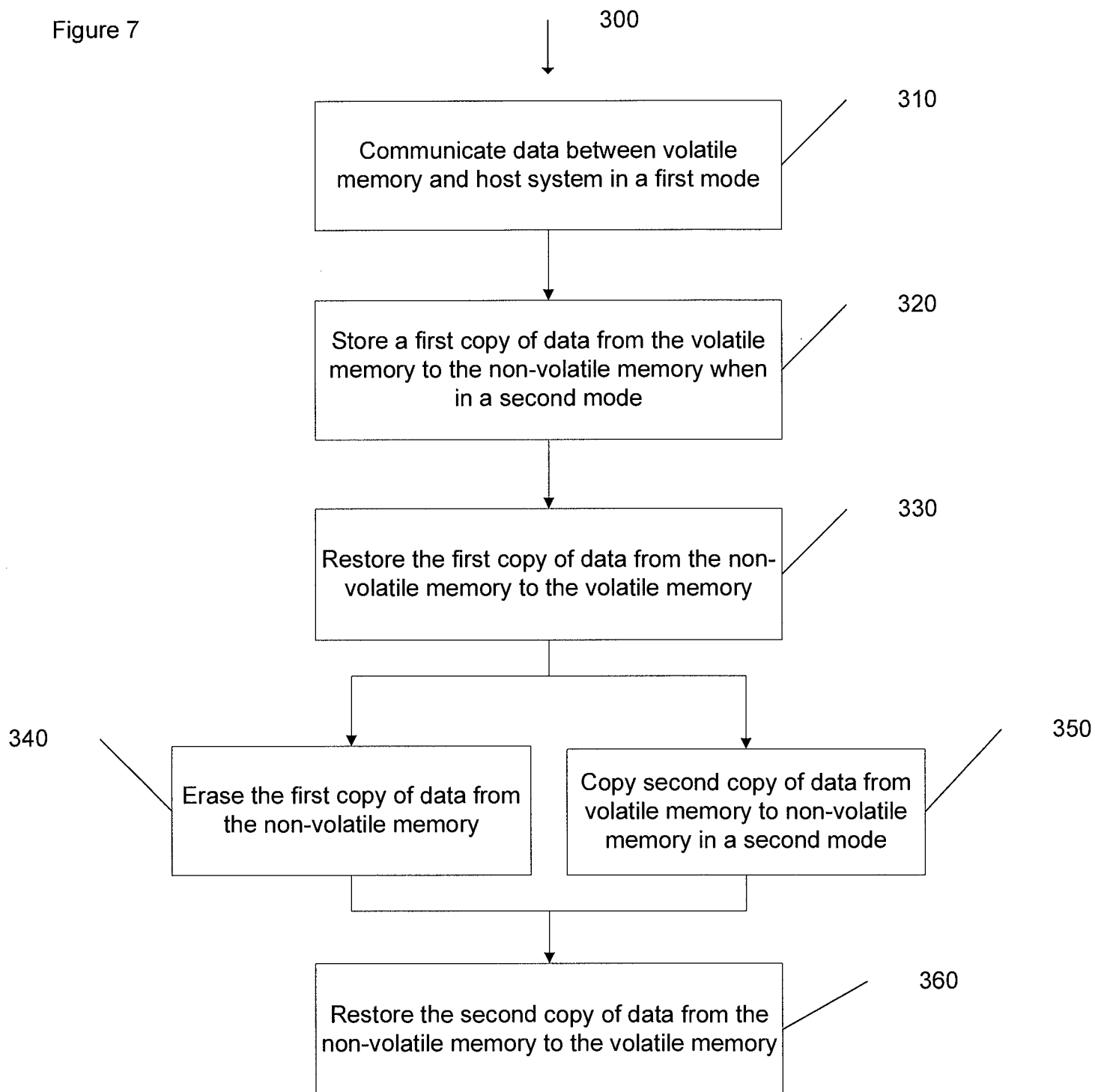


Figure 8

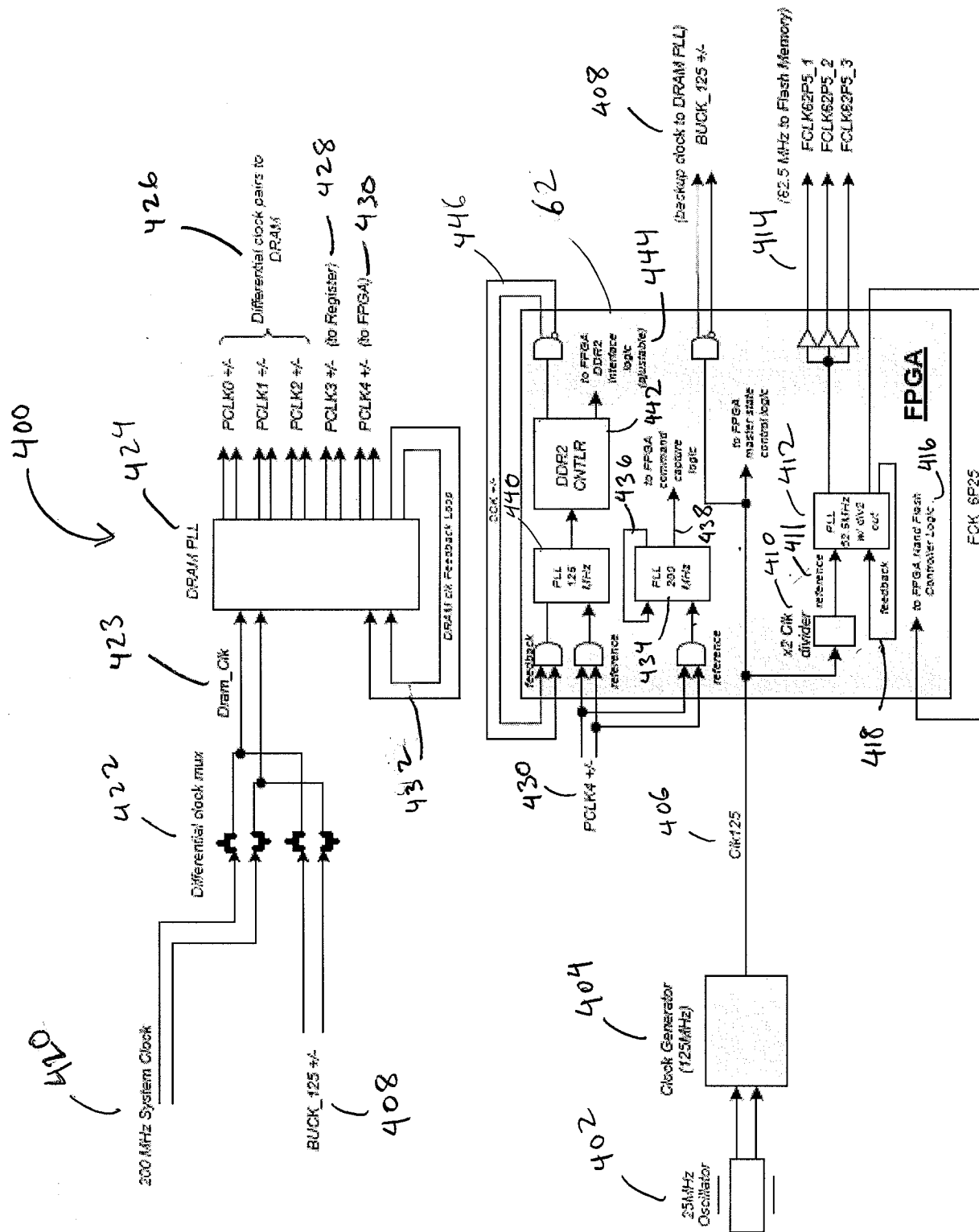


Figure 9

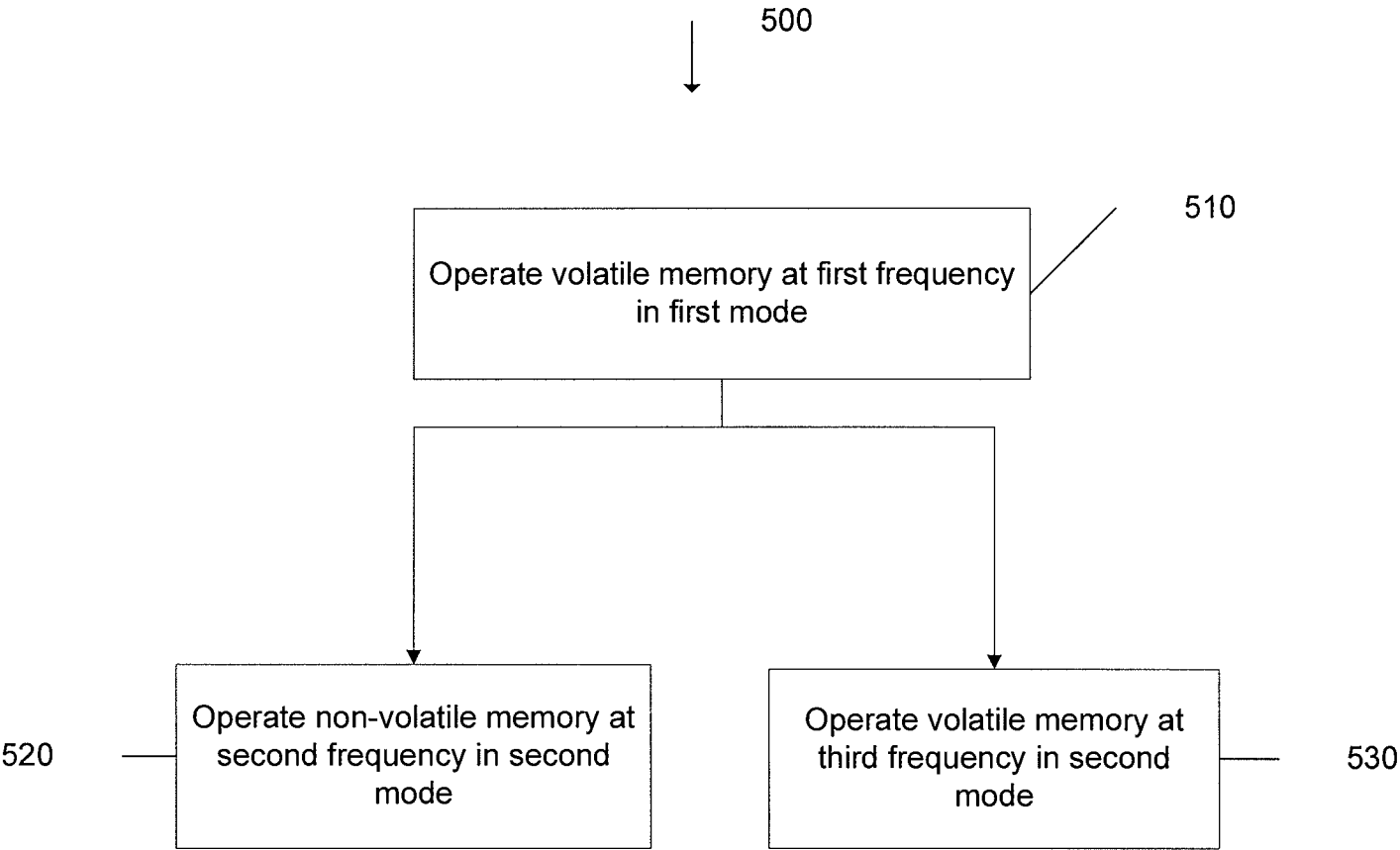


Figure 10

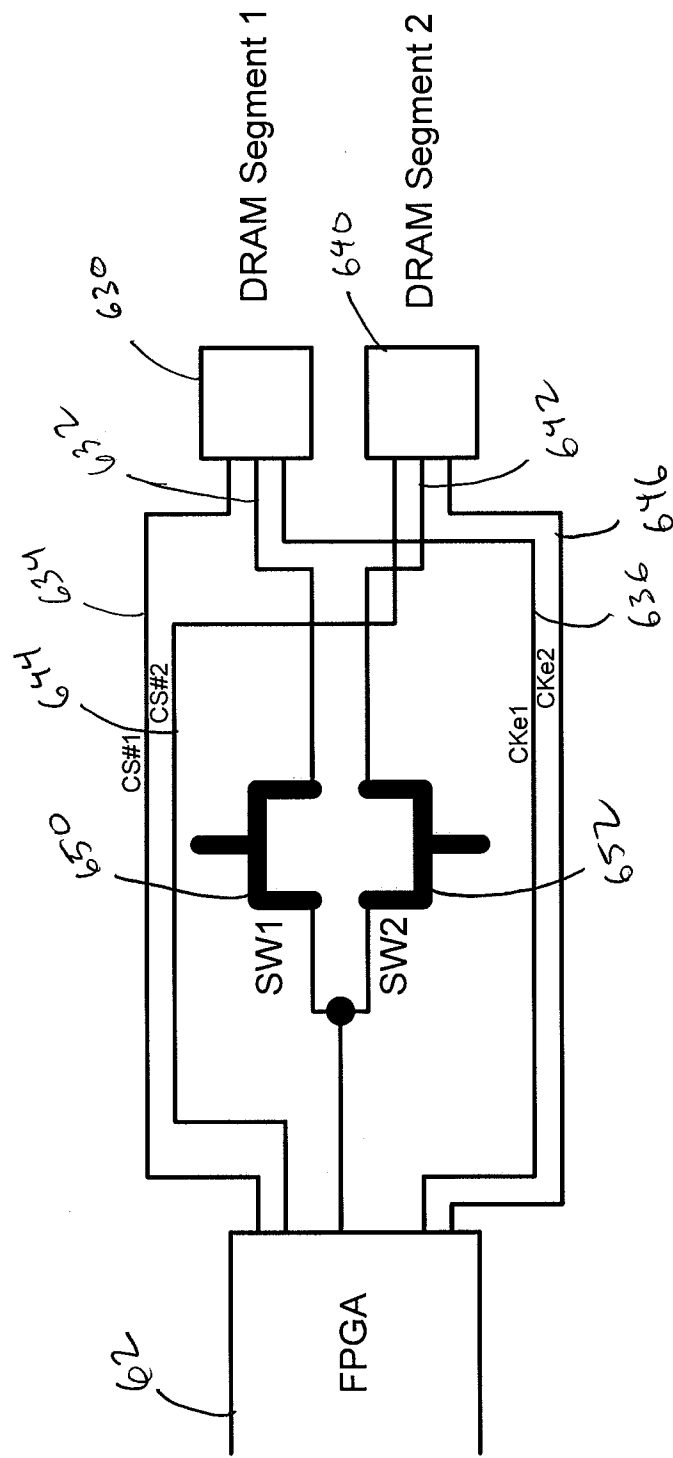
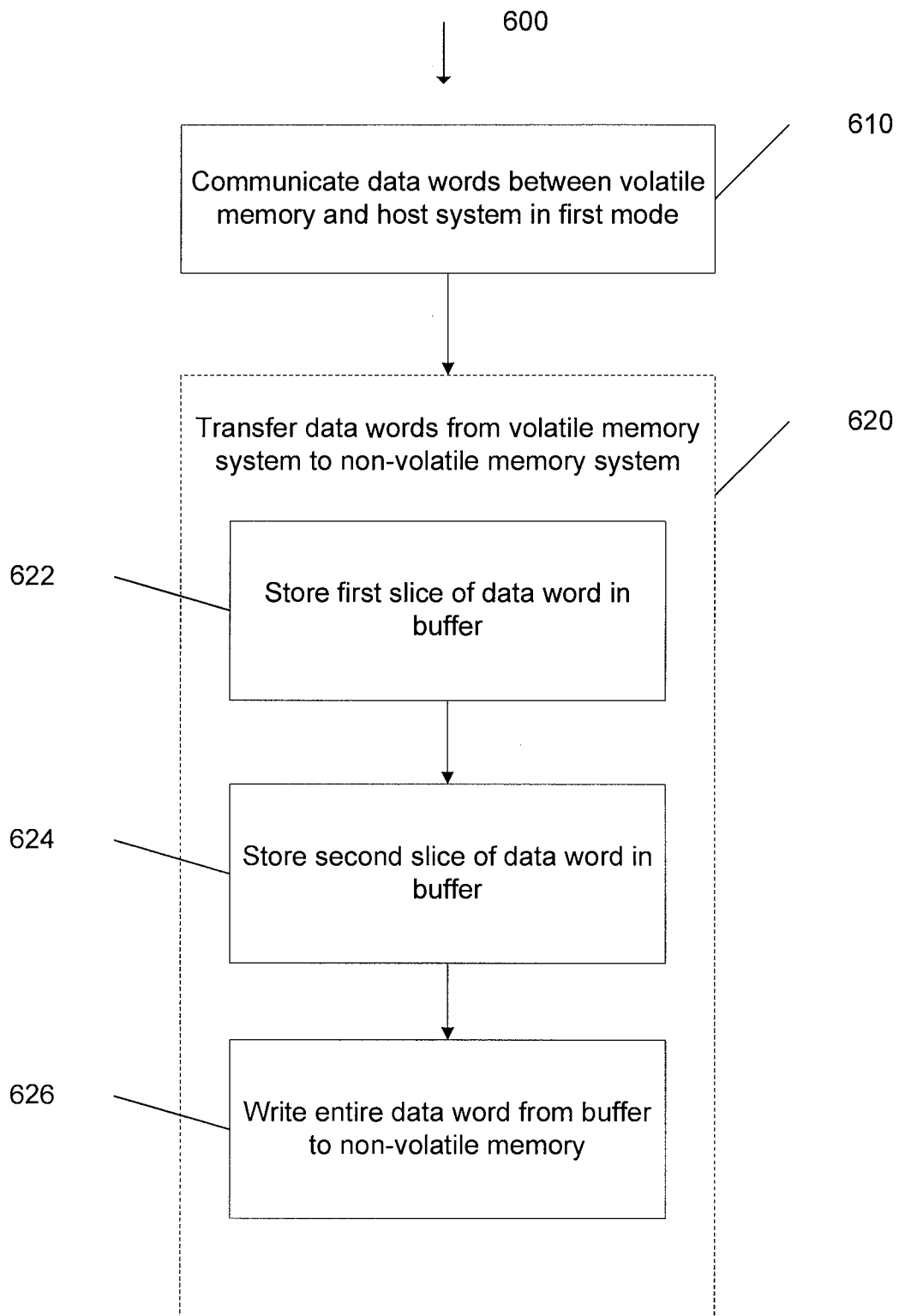


Figure 11



Filing Date: 06/02/08

Approved for use through 7/31/2006. OMB 0651-0032

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875						Application or Docket Number 12/131,873	
APPLICATION AS FILED – PART I							
(Column 1)		(Column 2)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA		RATE (\$)		RATE (\$)	
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A		N/A		N/A	310
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A		N/A		N/A	510
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A		N/A		N/A	210
TOTAL CLAIMS (37 CFR 1.16(i))	54	34		X\$ 25		X\$50	1700
INDEPENDENT CLAIMS (37 CFR 1.16(h))	7	4		X\$105		X\$210	840
APPLICATION SIZE FEE (37 CFR 1.16(s))							
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							
* If the difference in column 1 is less than zero, enter "0" in column 2.				TOTAL	0	TOTAL	3570
APPLICATION AS AMENDED – PART II							
(Column 1)		(Column 2)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDI-TIONAL FEE (\$)	RATE (\$)	ADDI-TIONAL FEE (\$)
Total (37 CFR 1.16(i))	*	Minus **	=	X =		X =	
Independent (37 CFR 1.16(h))	*	Minus ***	=	X =		X =	
Application Size Fee (37 CFR 1.16(s))							
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
				TOTAL ADD'T FEE		TOTAL ADD'T FEE	
(Column 1)		(Column 2)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDI-TIONAL FEE (\$)	RATE (\$)	ADDI-TIONAL FEE (\$)
Total (37 CFR 1.16(i))	*	Minus **	=	X =		X =	
Independent (37 CFR 1.16(h))	*	Minus ***	=	X =		X =	
Application Size Fee (37 CFR 1.16(s))							
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
				TOTAL ADD'T FEE		TOTAL ADD'T FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.							

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/131,873	06/02/2008	2189	0.00	NETL.040A	54	7

CONFIRMATION NO. 6645

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000030435239

Date Mailed: 06/16/2008

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 60/941,586 06/01/2007

Foreign Applications

If Required, Foreign Filing License Granted: 06/11/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/131,873**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No

Title

NON-VOLATILE MEMORY MODULE

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent

in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/131,873	06/02/2008		NETL.040A

CONFIRMATION NO. 6645

FORMALITIES LETTER



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 06/16/2008

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
Applicant must submit \$310 to complete the basic filing fee for a non-small entity. If appropriate, applicant may make a written assertion of entitlement to small entity status and pay the small entity filing fee (37 CFR 1.27).
- The oath or declaration is missing.
A properly signed oath or declaration in compliance with 37 CFR 1.63, identifying the application by the above Application Number and Filing Date, is required.
Note: If a petition under 37 CFR 1.47 is being filed, an oath or declaration in compliance with 37 CFR 1.63 signed by all available joint inventors, or if no inventor is available by a party with sufficient proprietary interest, is required.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- Additional claim fees of **\$2540** as a non-small entity, including any required multiple dependent claim fee, are required. Applicant must submit the additional claim fees or cancel the additional claims for which fees are due.
- To avoid abandonment, a surcharge (for late submission of filing fee, search fee, examination fee or oath or declaration) as set forth in 37 CFR 1.16(f) of **\$130** for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is **\$3700** for a non-small entity

- **\$310** Statutory basic filing fee.
- **\$130** Surcharge.
- The application search fee has not been paid. Applicant must submit **\$510** to complete the search fee.

- The application examination fee has not been paid. Applicant must submit **\$210** to complete the examination fee for a non-small entity.
- Total additional claim fee(s) for this application is **\$2540**
 - **\$840** for **4** independent claims over 3.
 - **\$1700** for **34** total claims over 20.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/mkanno/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Please Direct All Correspondence to Customer Number 20995

RESPONSE TO FORMALITIES NOTICE

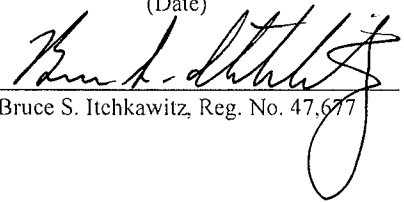
Applicant : Jeffrey C. Solomon et al.
App. No : 12/131,873
Filed : June 2, 2008
For : NON-VOLATILE MEMORY MODULE
Art Unit : 2189
Conf. No. : 6645

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence, and any other attachment noted on the automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web server on:

September 16, 2008

(Date)


Bruce S. Itchkawitz, Reg. No. 47,677

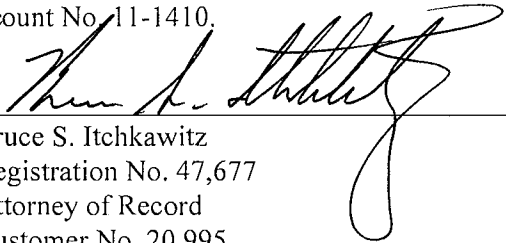
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without a Declaration and filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following.

- (X) A Declaration in 2 pages.
- (X) A Notice to File Missing Parts dated June 16, 2008.
- (X) Basic filing and surcharge fees will be paid via EFS Web. Other remaining fees as described on the attached Notice to File Missing Parts will not be paid at this time. Extension of time is requested by payment of any extension fee.
- (X) A Preliminary Amendment in 5 pages.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/131,873	06/02/2008		NETL.040A

CONFIRMATION NO. 6645

FORMALITIES LETTER



0000000030435240

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 06/16/2008

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
Applicant must submit \$310 to complete the basic filing fee for a non-small entity. If appropriate, applicant may make a written assertion of entitlement to small entity status and pay the small entity filing fee (37 CFR 1.27).
- The oath or declaration is missing.
A properly signed oath or declaration in compliance with 37 CFR 1.63, identifying the application by the above Application Number and Filing Date, is required.
Note: If a petition under 37 CFR 1.47 is being filed, an oath or declaration in compliance with 37 CFR 1.63 signed by all available joint inventors, or if no inventor is available by a party with sufficient proprietary interest, is required.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- Additional claim fees of **\$2540** as a non-small entity, including any required multiple dependent claim fee, are required. Applicant must submit the additional claim fees or cancel the additional claims for which fees are due.
- To avoid abandonment, a surcharge (for late submission of filing fee, search fee, examination fee or oath or declaration) as set forth in 37 CFR 1.16(f) of **\$130** for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is **\$3700** for a non-small entity

- **\$310** Statutory basic filing fee.
- **\$130** Surcharge.
- The application search fee has not been paid. Applicant must submit **\$510** to complete the search fee.

- The application examination fee has not been paid. Applicant must submit **\$210** to complete the examination fee for a non-small entity.
- Total additional claim fee(s) for this application is **\$2540**
 - **\$840** for **4** independent claims over 3.
 - **\$1700** for **34** total claims over 20.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.

<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/mkanno/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

**DECLARATION FOR UTILITY OR DESIGN APPLICATION
UNDER 37 CFR 1.63**

Docket No.: NETL.040A

Page 1 of 2

Title: NON-VOLATILE MEMORY MODULE

Inventors: Chi-She Chen, Jeffrey C. Solomon; Scott Milton; Jayesh Bhakta

Please Direct All Correspondence to Customer Number 20995

This Declaration is directed to the invention that:

Was filed as Serial No. 12/131,873 filed on June 2, 2008

As a below named inventor:

I believe the inventor(s) named below to be the original and first inventor(s) of the subject matter which is described and claimed and for which a patent is sought;

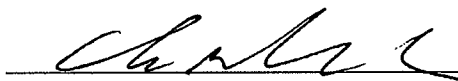
I have reviewed and understand the contents of the above-identified application, including the claims, and any amendment filed herewith or identified above;

I acknowledge the duty to disclose information which is material to patentability as defined in 37 CFR 1.56;

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful, false statements may jeopardize the validity of the application or any patent issued thereon.

Full name of **first** inventor: Chi-She Chen

Signature:

 Date: 09-15-08

Citizenship:

Taiwan

Mailing Address:

944 Crystal Water Lane, Walnut, CA 91789

Residence Address:

Same as above

(if different than above)

**DECLARATION FOR UTILITY OR DESIGN APPLICATION
UNDER 37 CFR 1.63**

Docket No.: NETL.040A

Page 2 of 2

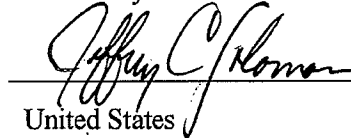
Title: NON-VOLATILE MEMORY MODULE

Inventors: Chi-She Chen, Jeffrey C. Solomon; Scott Milton; Jayesh Bhakta

Please Direct All Correspondence to Customer Number 20995

Full name of **second** inventor: Jeffery C. Solomon

Signature:



Date:

09-15-08

Citizenship:

United States

Mailing Address:

6 Silver Fir, Irvine, CA 92604

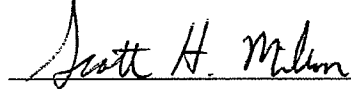
Residence Address:

Same as above

(if different than above)

Full name of **third** inventor: Scott Milton

Signature:



Date:

09/15/2008

Citizenship:

United States

Mailing Address:

49 Statehouse Place, Irvine, CA 92602

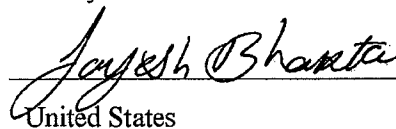
Residence Address:

Same as above

(if different than above)

Full name of **fourth** inventor: Jayesh Bhakta

Signature:



Date:

9-15-08

Citizenship:

United States

Mailing Address:

12220 Rose Street, Cerritos, CA 90703

Residence Address:

Same as above

(if different than above)

Send Correspondence To:

KNOBBE, MARTENS, OLSON & BEAR, LLP

Customer No. 20,995

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

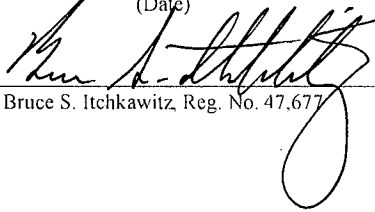
Applicant : Jeffrey C. Solomon et al.
App. No : 12/131,873
Filed : June 2, 2008
For : NON-VOLATILE MEMORY
MODULE
Examiner : Unknown
Art Unit : 2189
Conf # : 6645

CERTIFICATE OF EFS WEB
TRANSMISSION

I hereby certify that this correspondence, and any other attachment noted on the automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web server on:

September 16, 2008

(Date)


Bruce S. Itchkawitz, Reg. No. 47,677

PRELIMINARY AMENDMENT**Mail Stop Amendment**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Prior to examination on the merits, please amend the above-captioned application as described herein.

Amendments to the Claims: Begin on Page 2 of this Preliminary Amendment.

Remarks: Begin on page 5 of this paper.

AMENDMENTS TO THE CLAIMS

Please cancel Claims 21-54 without prejudice, as indicated below.

A complete listing of all claims is presented below with insertions underlined (e.g., insertion), and deletions struckthrough or in double brackets (e.g., ~~deletion~~ or [[deletion]]):

1. (Original) A memory system coupled to a computer system, comprising:
a volatile memory subsystem;
a non-volatile memory subsystem;
a controller operatively coupled to the non-volatile memory subsystem;
at least one circuit configured to selectively operatively decouple the controller from the volatile memory subsystem.
2. (Original) The memory system of Claim 1, wherein the memory system is configured to be operated in at least two states, the at least two states comprising:
a first state in which the controller and the non-volatile memory subsystem are decoupled from the volatile memory subsystem by the at least one circuit; and
a second state in which the volatile memory subsystem is operatively coupled to the controller to allow data to be communicated between the volatile memory subsystem and the non-volatile memory subsystem via the controller.
3. (Original) The memory system of Claim 2, wherein power is supplied to the volatile memory subsystem from a first power supply when the memory system is in the first state and power is supplied to the volatile memory subsystem from a second power supply when the memory system is in the second state.
4. (Original) The memory system of Claim 3, wherein the second power supply does not comprise a battery.
5. (Original) The memory system of Claim 3, wherein the second power supply comprises one or more capacitors.
6. (Original) The memory system of Claim 5, wherein the second power supply comprises a battery and is configured to power the volatile memory subsystem for less than thirty minutes.
7. (Original) The memory system of Claim 5, wherein the one or more capacitors comprises one or more double layered capacitors.

8. (Original) The memory system of Claim 5, wherein the one or more capacitors are charged by the first power supply while the memory system is in the first state.

9. (Original) The memory system of Claim 3, further comprising a printed circuit board, wherein the second power supply and the memory system are located on the printed circuit board.

10. (Original) The memory system of Claim 3, wherein the at least two states further comprise a third state in which power is supplied to the volatile memory subsystem from a third power supply.

11. (Original) The memory system of Claim 2, wherein the controller and the non-volatile memory subsystem do not add a significant capacitive load to the volatile memory system when the memory system is in the first state.

12. (Original) The memory system of Claim 1, wherein the volatile memory subsystem comprises one or more DRAM memory elements.

13. (Original) The memory system of Claim 1, wherein the volatile memory subsystem comprises one or more SRAM memory elements.

14. (Original) The memory system of Claim 1, wherein the non-volatile memory subsystem comprises one or more flash memory elements.

15. (Original) The memory system of Claim 1, wherein the memory system is implemented on a memory module.

16. (Original) The memory system of Claim 1, wherein the at least one circuit comprise one or more switches coupled to the controller, to the volatile memory subsystem, and to the host computer.

17. (Original) The memory system of Claim 1, wherein the volatile memory subsystem comprises a registered DIMM comprising one or more registers and a plurality of DRAM elements, wherein the one or more registers are coupled to the computer system.

18. (Original) The memory system of Claim 17, wherein the at least one circuit comprise one or more switches coupled to the one or more registers and to the plurality of DRAM elements and configured to selectively operatively couple the volatile memory subsystem and to the computer system.

Application No.: 12/131,873
Filing Date: June 2, 2008

19. (Original) The memory system of Claim 17, wherein the one or more registers are coupled to the plurality of DRAM elements and the at least one circuit comprises one or more switches coupled to the one or more registers and to a power source for the one or more registers, wherein the one or more switches are configured to selectively operatively couple the one or more registers to the power source for the one or more registers.

20. (Original) The memory system of Claim 1, wherein the non-volatile memory subsystem comprises at least 100 percent more storage capacity than does the volatile memory subsystem.

21.-54. (Cancelled)

Application No.: 12/131,873
Filing Date: June 2, 2008

REMARKS

Original Claims 1-20 remain pending and original Claims 21-54 have been cancelled without prejudice. Applicants believe that none of the foregoing amendments have added new matter to the application.

All of the claims currently pending in the present application are believed to be allowable over the prior art, and such action is earnestly solicited. If, however, any matters remain which could be resolved by Examiner's Amendment, the Examiner is cordially invited to contact the undersigned by telephone so that any such matters can be promptly resolved.

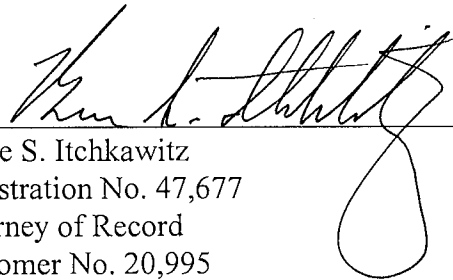
Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: September 16, 2008

By: _____

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

A handwritten signature in black ink, appearing to read 'Bruce S. Itchkawitz', is written over a horizontal line. The signature is stylized with a large loop at the end.

5933718

Electronic Patent Application Fee Transmittal

Application Number:	12131873			
Filing Date:	02-Jun-2008			
Title of Invention:	NON-VOLATILE MEMORY MODULE			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz/ALEXIS SWAN			
Attorney Docket Number:	NETL.040A			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	310	310
Pages:				
Claims:				
Miscellaneous-Filing:				
Late filing fee for oath or declaration	1051	1	130	130
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				440

Electronic Acknowledgement Receipt

EFS ID:	3953462
Application Number:	12131873
International Application Number:	
Confirmation Number:	6645
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Alexandra Benitez
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.040A
Receipt Date:	16-SEP-2008
Filing Date:	02-JUN-2008
Time Stamp:	19:01:00
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 440
RAM confirmation Number	13659
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NETL040A_MPARTS.pdf	411431 7c4ea10deaefcfab736e80390c17f981809ef497	yes	10
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Applicant Response to Pre-Exam Formalities Notice		1	3	
	Oath or Declaration filed		4	5	
	Preliminary Amendment		6	10	
Warnings:					
Information:					
2	Fee Worksheet (PTO-06)	fee-info.pdf	31782 a4817ac094a56ff4ab5396dcd1280629affff79c	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			443213		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/131,873	06/02/2008	2189	440	NETL.040A	20	1

CONFIRMATION NO. 6645

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000032226843

Date Mailed: 09/30/2008

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Chi-She Chen, Walnut, CA;
Jeffery C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 60/941,586 06/01/2007

Foreign Applications

If Required, Foreign Filing License Granted: 06/11/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/131,873**

Projected Publication Date: Incomplete Reply

Non-Publication Request: No

Early Publication Request: No

Title

NON-VOLATILE MEMORY MODULE

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/131,873	06/02/2008	Chi-She Chen	NETL.040A

CONFIRMATION NO. 6645

FORMALITIES LETTER



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 09/30/2008

NOTICE OF INCOMPLETE REPLY (NONPROVISIONAL)

Filing Date Granted

The U.S. Patent and Trademark Office has received your reply on 09/16/2008 to the Notice to File Missing Parts (Notice) mailed 06/16/2008 and it has been entered into the nonprovisional application. The reply, however, does not include the following items required in the Notice.

The period of reply remains as set forth in the Notice. You may, however, obtain EXTENSIONS OF TIME under the provisions of 37 CFR 1.136 (a) accompanied by the appropriate fee (37 CFR 1.17(a)).

A complete reply must be timely filed to prevent ABANDONMENT of the above-identified application. Replies should be mailed to: Mail Stop Missing Parts, Commissioner for Patents, P.O. Box 1450, Alexandria VA 22313-1450.

Items Required to Avoid Abandonment:

The required items noted below SHOULD be filed along with any items required above. The filing date of this nonprovisional application will be the date of receipt of the items required above.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is **\$720** for a non-small entity

- The application search fee has not been paid. Applicant must submit **\$510** to complete the search fee.
- The application examination fee has not been paid. Applicant must submit **\$210** to complete the examination fee for a non-small entity.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/eulanday/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

**REVOCATION
AND
GENERAL POWER OF ATTORNEY**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

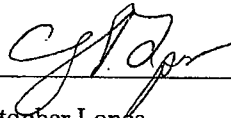
The undersigned is an empowered representative of the Assignee and hereby appoints the registrants of Knobbe, Martens, Olson & Bear, LLP, **Customer No. 20,995**, as attorneys and agents to represent the Assignee before the United States Patent and Trademark Office (USPTO) in connection with any and all patent applications assigned to the Assignee according to the USPTO assignment records or assignment documents supplied with an accompanying Statement Under 37 CFR § 3.73(b). This appointment is to be to the exclusion of the inventor(s) and his attorney(s) in accordance with the provisions of 37 CFR § 3.71.

Submission of this paper in connection with any matter of the below named assignee, together with a statement under 37 CFR 3.73(b), shall serve to revoke any previous powers of attorney in that matter.

A Statement Under 37 CFR § 3.73(b), signed by a registrant of Knobbe, Martens, Olson & Bear, LLP, is attached setting forth a full chain of title for the subject application owned by the Assignee named below.

Please recognize or change the correspondence address for the application identified in the attached Statement to **Customer No. 20,995**.

By: _____



Date: _____

6/7/07

Name: Christopher Lopes

Title: Vice President

Assignee: NETLIST, INC.
475 Goddard, Suite 100
Address: Irvine, CA 92618

ASSIGNMENT

WHEREAS, We, Chi-She Chen, a Taiwanese citizen, residing at 944 Crystal Water Lane, Walnut, CA 91789; Jeffrey C. Solomon, a United States citizen, residing at 16 Silver Fir, Irvine, CA 92604; Scott Milton, a United States citizen, residing at 49 Statehouse Place, Irvine, CA 92602; and Jayesh Bhakta, a United States citizen, residing at 12220 Rose Street, Cerritos, CA 90703, have invented certain new and useful improvements in a NON-VOLATILE MEMORY MODULE for which we have filed an application for Letters Patent in the United States, Application No. 12/131,873, Filed on June 2, 2008;

AND WHEREAS, Netlist, Inc. (hereinafter "ASSIGNEE"), a Delaware Corporation, with its principal place of business at 51 Discovery, Ste #150, Irvine, CA 92618, desires to acquire the entire right, title, and interest in and to said improvements and said Application:

NOW, THEREFORE, for good and valuable consideration, the receipt of which is hereby acknowledged, we, the inventors, do hereby acknowledge that we have sold, assigned, transferred and set over, and by these presents do hereby sell, assign, transfer and set over, unto said ASSIGNEE, its successors, legal representatives and assigns, the entire right, title, and interest throughout the world in, to and under said improvements, and said application including all provisional applications relating thereto (including but not limited to U.S. Provisional Application No. 60/941,586, filed June 1, 2007), and all divisions, renewals and continuations thereof, and all Letters Patent of the United States which may be granted thereon and all reissues and extensions thereof, and all rights of priority under International Conventions and applications for Letters Patent which may hereafter be filed for said improvements in any country or countries foreign to the United States, and all Letters Patent which may be granted for said improvements in any country or countries foreign to the United States and all extensions, renewals and reissues thereof; and we hereby authorize and request the Commissioner of Patents of the United States, and any Official of any country or countries foreign to the United States, whose duty it is to issue patents on applications as aforesaid, to issue all Letters Patent for said improvements to said ASSIGNEE, its successors, legal representatives and assigns, in accordance with the terms of this instrument.

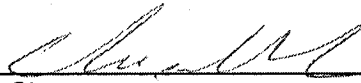
AND WE DO HEREBY sell, assign, transfer, and convey to ASSIGNEE, its successors, legal representatives, and assigns all claims for damages and all remedies arising out of any violation of the rights assigned hereby that may have accrued prior to the date of assignment to ASSIGNEE, or may accrue hereafter, including, but not limited to, the right to sue for, collect, and retain damages for past infringements of said Letters Patent before or after issuance.

AND WE HEREBY covenant and agree that we will communicate to said ASSIGNEE, its successors, legal representatives and assigns, any facts known to us respecting said improvements, and testify in any legal proceeding, sign all lawful papers, execute all divisional, continuing and reissue applications, make all rightful oaths and generally do everything possible to aid said ASSIGNEE, its successors, legal representatives and assigns, to obtain and enforce proper patent protection for said improvements in all countries.

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 2

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 24th day of October, 2008.


Chi-She Chen

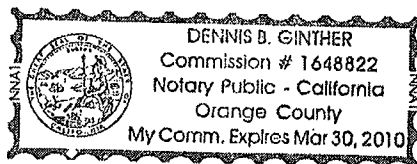
STATE OF California }
COUNTY OF Orange } ss.

On October, 24, 2008, before me, Dennis B. Ginter, notary public, personally appeared Chi-She Chen who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]

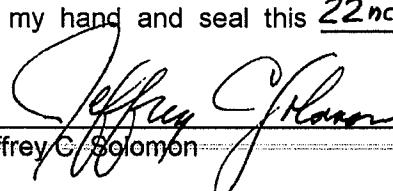



Notary Signature

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 3

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 22nd day of October, 2008.



Jeffrey C. Solomon

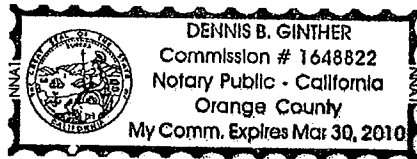
STATE OF California }
COUNTY OF Orange } ss.

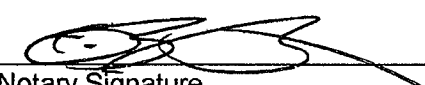
On October 22, 2008, before me, Dennis B. Ginter, notary public, personally appeared Jeffrey C. Solomon who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]





Notary Signature

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 4

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 24 day of October, 2008.

Scott H. Milton
Scott Milton

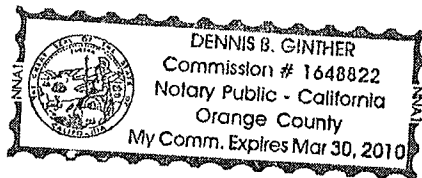
STATE OF California }
COUNTY OF Orange } ss.

On October 24, 2008, before me Dennis B. Ginther, notary public, personally appeared Scott Milton who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]



[Signature]
Notary Signature

Application No.: 12/131,873
Filing Date: June 2, 2008

PATENT
Client Code: NETL.040A
Page 5

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 22 day of October, 2008.

Jayesh Bhakta
Jayesh Bhakta

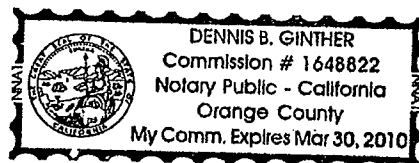
STATE OF California }
COUNTY OF Orange } ss.

On October 22, 2008, before me, Dennis B. Ginther, notary public, personally appeared Jayesh Bhakta who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]



[Signature]
Notary Signature

Electronic Acknowledgement Receipt

EFS ID:	4192851
Application Number:	12131873
International Application Number:	
Confirmation Number:	6645
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Quyen Nguyen
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.040A
Receipt Date:	28-OCT-2008
Filing Date:	02-JUN-2008
Time Stamp:	19:15:49
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Assignee showing of ownership per 37 CFR 3.73(b).	Statement37.pdf	42749 96d55263057fe72dc8c581d248d9001771b19f3a	no	1

Warnings:

Information:

2	Power of Attorney	POA_NETL.pdf	48001 ce2334a8d558e7ea3aec4aaadce6f8b122147048	no	1
Warnings:					
Information:					
3	Assignee showing of ownership per 37 CFR 3.73(b).	Assignment.pdf	276431 bfa3be860ca99c9defab2d76a9234359dc6dc008	no	5
Warnings:					
Information:					
Total Files Size (in bytes):			367181		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

STATEMENT UNDER 37 CFR § 3.73(b)
and
CHANGE OF CORRESPONDENCE ADDRESS

Applicant : Chi-She Chen et al.
App. No. : 12/131,873
Filed : June 2, 2008
For : NON-VOLATILE MEMORY MODULE
Examiner : Unknown
Group Art Unit : 2189

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

This document is being filed with a copy of a Revocation and General Power of Attorney, which is signed by the Assignee, Netlist, Inc.

Netlist, Inc. is the Assignee of the entire right, title, and interest of the above-referenced application by virtue of the attached copy of an Assignment that is being forwarded to the Recordation Branch concurrently under separate cover.

The undersigned is an agent of Customer Number 20,995 and is authorized to act on behalf of the Assignee as provided in the attached copy of the Revocation and General Power of Attorney. Please recognize or change the correspondence address for the above-identified application to **Customer No. 20,995**. All correspondence is to be directed to **Customer No. 20,995**.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: _____

10/28/08

By: _____

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

**PETITION FOR EXPRESS
ABANDONMENT TO AVOID
PUBLICATION UNDER 37 CFR 1.138(c)**File the petition electronically using EFS-Web
Or Mail the petition to:**Mail Stop Express Abandonment**

Commissioner for Patents

P.O. Box 1450, Alexandria, VA 22313-1450

Application Number	12/131,873
Filing Date	June 2, 2008
First Named Inventor	Chi-She Chen et al.
Art Unit	2189
Examiner Name	Unknown
Attorney Docket Number	NETL.040A

Petition for Express Abandonment to Avoid Publication under 37 CFR 1.138(c)

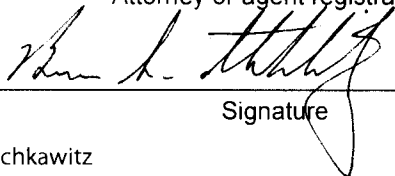
I hereby petition to expressly abandon the above-identified application to avoid publication.

Petition Fee – must be filed with petition to avoid delays in recognizing the petition.

- a. ☒ The Director is hereby authorized to charge the petition fee under 37 CFR 1.17(h) to Deposit Account No. 11-1410.
- b. ☐ Check in the amount of \$ _____ is enclosed.
- c. ☐ Payment by credit card (Form PTO-2038 is enclosed).

NOTE: A paper requesting express abandonment of an application is not effective unless and until an appropriate USPTO official recognizes and acts on the paper. See the Manual of Patent Examining Procedure (MPEP), section 711.01. In addition, the paper will not stop publication of the application unless a petition under 37 CFR 1.138(c) is recognized and acted on by the Pre-Grant Publication Division in sufficient time to avoid publication (e.g., more than four (4) weeks prior to the projected publication date).

**TO REQUEST A REFUND OF SEARCH FEE AND EXCESS CLAIMS FEE
(IF ELIGIBLE), PLEASE ALSO INCLUDE FORM PTO/SB/24B WITH THIS FORM.**I am the: ☐ applicant.
☐ assignee of record of the entire interest. See 37 CFR 3.71.
Statement under 37 CFR 3.73(b) is enclosed. (Form PTO/SB/96)
☒ attorney or agent of record. Attorney or agent registration number is 47,677
☐ attorney or agent acting under 37 CFR 1.34, who is authorized under 37 CFR 1.138(b)
because the application is expressly abandoned in favor of a continuing application.
Attorney or agent registration number is _____



Signature

October 30, 2008

Date

Bruce S. Itchkawitz

949-760-0404

Typed or printed name

Telephone Number

Note: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.138(c). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Mail Stop Express Abandonment, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO 9199 and select option 2.

Electronic Patent Application Fee Transmittal

Application Number:	12131873			
Filing Date:	02-Jun-2008			
Title of Invention:	NON-VOLATILE MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Bruce S. Itchkawitz/Vicki Morrison			
Attorney Docket Number:	NETL.040A			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Petition fee- 37 CFR 1.17(h) (Group III)	1464	1	130	130
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				130

Electronic Acknowledgement Receipt

EFS ID:	4207846
Application Number:	12131873
International Application Number:	
Confirmation Number:	6645
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Valerie Jones
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.040A
Receipt Date:	30-OCT-2008
Filing Date:	02-JUN-2008
Time Stamp:	20:00:59
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 130
RAM confirmation Number	15068
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Petition for review by the Office of Petitions.	NETL040APETITION.pdf	65504	no	1
			11a171eb1f23c65f067e28c18f033d5d05ff66e8		

Warnings:**Information:**

2	Fee Worksheet (PTO-06)	fee-info.pdf	30378	no	2
			d0bbfdcaffe41d14c1dfa9de15b110ed7f183010		

Warnings:**Information:**

Total Files Size (in bytes):			95882
------------------------------	--	--	-------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/131,873	06/02/2008	Chi-She Chen	NETL.000GEN

CONFIRMATION NO. 6645

POA ACCEPTANCE LETTER

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614



Date Mailed: 11/12/2008

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 10/28/2008.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/atesfai/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/131,873	06/02/2008	Chi-She Chen	NETL.000GEN

CONFIRMATION NO. 6645

ABANDONMENT/TERMINATION
LETTER

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614



Date Mailed: 11/17/2008

NOTICE OF ABANDONMENT

Express Abandonment

The letter of Express Abandonment filed on 10/30/2008 is acknowledged.

A copy of this notice MUST be returned with the reply.

/kebuchanan/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

OCT 15 2012

PTO/SB/81 (01-09)

Approved for use through 11/30/2011, OMB 0551-0036

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	12/131,873
	Filing Date	2008-08-02
	First Named Inventor	Chi-She Chen
	Title	Non-Volatile Memory Module
	Art Unit	
	Examiner Name	
	Attorney Docket Number	062453-002cont

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71. Statement under 37 CFR 3.73(b) (Form PTO/SB/95) submitted herewith or filed on _____.

SIGNATURE of Applicant or Assignee of Record

Signature

Date

Name

Chi-She Chen

Telephone

909-5988899

Title and Company

Senior Design Engineer

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ Total of 4 forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

RECEIVED
CENTRAL FAX CENTER

OCT 15 2012

PTO/SB/51 (01-09)

Approved for use through 11/30/2011 OMB 0851-0033
U.S. Patent and Trademark Office, U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	12/131,873
	Filing Date	2008-08-02
	First Named Inventor	Chi-She Chen
	Title	Non-Volatile Memory Module
	Art Unit	
	Examiner Name	
	Attorney Docket Number	062453-002cont

I hereby revoke all previous powers of attorney given in the above-identified application

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71.

Statement under 37 CFR 3.73(b) (Form PTO/SB/95) submitted herewith or filed on _____

SIGNATURE of Applicant or Assignee of Record

Signature

Name

Title and Company

Date

Telephone

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ Total of 4 forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2

OCT 15 2012

PTO/SB/1 (01-09)

Approved for use through 11/30/2011. OMB 0551-0036

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	12/131,873
	Filing Date	2008-08-02
	First Named Inventor	Chi-Sha Chen
	Title	Non-Volatile Memory Module
	Art Unit	
	Examiner Name	
	Attorney Docket Number	082453-002cont

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71. Statement Under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____

SIGNATURE of Applicant or Assignee of Record

Signature Scott H. Milton Date 10/11/12

Name Scott Milton Telephone 714-731-4542

Title and Company Vice President, System Engineering

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ Total of 4 forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2

OCT 15 2012

PTO/SB/01 (01-09)

Approved for use through 11/30/2011. OMB 0551-0035

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	12/131,873
	Filing Date	2008-06-02
	First Named Inventor	Chi-She Chen
	Title	Non-Volatile Memory Module
	Art Unit	
	Examiner Name	
	Attorney Docket Number	082453-002cont

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City State Zip

Country

Telephone Email

I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71.

Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____

SIGNATURE of Applicant or Assignee of Record

Signature	<i>Jayesh Bhakta</i>	Date	10/11/2012
Name	Jayesh Bhakta	Telephone	749-439-0205
Title and Company	Chief, module Arch. Netlist Inc.		

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ *Total of 4 forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to its (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete. Including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

OCT 15 2012

PTO/SB/97 (07-09)

Approved for use through 07/31/2012. OMB 0851-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Certificate of Transmission under 37 CFR 1.8

I hereby certify that this correspondence is being facsimile transmitted to the United States Patent and Trademark Office

on October 15, 2012
Date



Signature

Pamela Wilson

Typed or printed name of person signing Certificate

n/a

650-320-7700

Registration Number, if applicable

Telephone Number

Note: Each paper must have its own certificate of transmission, or this certificate must identify each submitted paper.

Power of Attorney (sb-81) - 4 pages (12/131,873)

This collection of information is required by 37 CFR 1.8. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1.8 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
 United States Patent and Trademark Office
 Address: COMMISSIONER FOR PATENTS
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
 www.uspto.gov

APPLICATION NUMBER	FILING DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/131,873	06/02/2008	Chi-She Chen	NETL.000GEN

CONFIRMATION NO. 6645

20995
 KNOBBE MARTENS OLSON & BEAR LLP
 2040 MAIN STREET
 FOURTEENTH FLOOR
 IRVINE, CA 92614



Cc: NIXON PEABODY LLP
 P.O. BOX 60610
 PALO ALTO, CA 94306

Date Mailed: 10/31/12

DENIAL OF REQUEST FOR POWER OF ATTORNEY

The request for Power of Attorney filed 10/15/12 is acknowledged. However, the request cannot be granted at this time for the reason stated below.

- ☐ The Power of Attorney you provided did not comply with the new Power of Attorney rules that became effective on June 25, 2004. See 37 CFR 1.32.
- ☐ The revocation is not signed by the applicant, the assignee of the entire interest, or one particular principal attorney having the authority to revoke.
- ☐ The Power of Attorney is from an assignee and the Certificate required by 37 CFR 3.73(b) has not been received.
- ☐ The person signing for the assignee has omitted their empowerment to sign on behalf of the assignee.
- ☒ The inventor(s) is without authority to appoint attorneys since the assignee has intervened as provided by 37 CFR 3.71.
- ☐ The signature(s) of _____, a co-inventor in this application, has been omitted. The Power of Attorney will be entered upon receipt of confirmation signed by said co-inventor(s).
- ☐ The person(s) appointed in the Power of Attorney is not registered to practice before the U.S. Patent and Trademark Office.

Questions relating to this Notice should be directed to the Application Assistance Unit.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/131,873	06/02/2008	Chi-She Chen	NETL.000GEN

CONFIRMATION NO. 6645

MISCELLANEOUS NOTICE



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 11/01/2012

A communication which cannot be delivered in electronic form has been mailed to the applicant.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.101PR
		Application Number	
Title of Invention	A METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117		☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name	Suffix
	TBD		TBD	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City		State/Province	Country of Residence	
Citizenship under 37 CFR 1.41(b)				
Mailing Address of Applicant:				
Address 1				
Address 2				
City		State/Province		
Postal Code		Country		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	20995		
Email Address	efiling@kmob.com	Add Email	Remove Email

Application Information:

Title of the Invention	A METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY		
Attorney Docket Number	NETL.101PR	Small Entity Status Claimed ☐	
Application Type	Provisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	6	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.101PR
		Application Number	
Title of Invention	A METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY		

Publication Information:
☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One: ☒ Customer Number ☐ US Patent Practitioner ☐ Limited Recognition (37 CFR 11.9)

Customer Number 20995

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

			Remove
Application Number	Country ¹	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☐ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

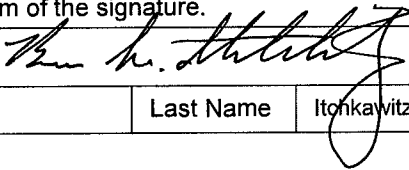
Assignee 1

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.101PR
		Application Number	
Title of Invention	A METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY		

If the Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information:				
Address 1				
Address 2				
City		State/Province		
Country		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee Data may be generated within this form by selecting the Add button.				

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature				Date (YYYY-MM-DD)	2011-09-23
First Name	Bruce	Last Name	Itonkavitz	Registration Number	47677

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

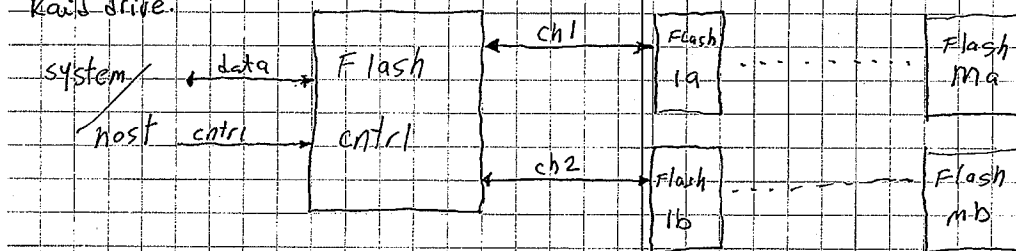
1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Appendix

11967745
[092211

A method of using standard Flash controllers to implement RAID storage on Flash products.

Description: using standard dual or quad channel Flash controller to emulate a RAID drive.



The flash or non-volatile or non-volatile memory controller will have 2 or more channels with each having access to 2 or more Flash devices. Flash controller will have multiple copy of its firmware within flash controller under control of internal ucontroller (embedded into flash ctrl) and a scheduler or mini kernel to manage each firmware once the system provided enough data to flash ctrl internal buffer for device / flash programming function, the scheduler or mini kernel will assign 1 or more channels in a pair setting to store two or more copies of data into the flash devices.

Ex. Host sends data = "Netlist" to the Flash controller, once the internal Flash controller buffer filled up or incoming data were completed, Flash controller scheduler will send one copy to channel 1, and one replica (exact copy) to channel 2, in case of dual channel simple RAID configuration. This will help controller to have multiple copy and increase storage endurance (endurance) capabilities. The same concept can be applied to multiple paired channel to improve endurance such as 4-channel or 8-channel RAID storage. one way is for controller to hold multiple pair and perform Enhanced ECC during read to correct any issues from multiple copies and send the correct results to host. Currently SLC flash devices have 100K program/erase cycles. Assuming a card of 1GB density build with SLC technology and file size of approximately card size or 1GB, the standard controllers (non-RAID) will be able to have an endurance of 100K under best case using dual channel or multiple paired channel one can increase this by at least 1 magnitude factor. The probability of the same cell (1-bit) failure location, on the same block, on the same page, on the same plane between two channels are very low, This flash controller firmware will pass the uncorrectable bit stream to mini kernel and by performing Enhanced ECC kernel will be able to correct the error. Need to define proper math to calculate this probability.

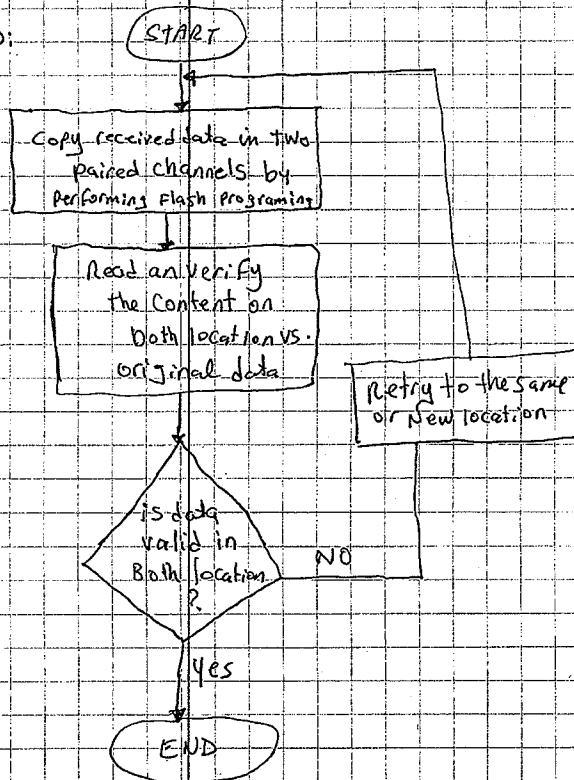
Under above defined configuration Controller will read the same page address (logical add) from both paired channels and will transmit the one that has No ECC error, or the one with correctable ECC Error to the host.

To improve performance one can use more than two channels. Ex. 4ch or 8ch to use interleaving capabilities to increase program/erase speed with larger page sizes.

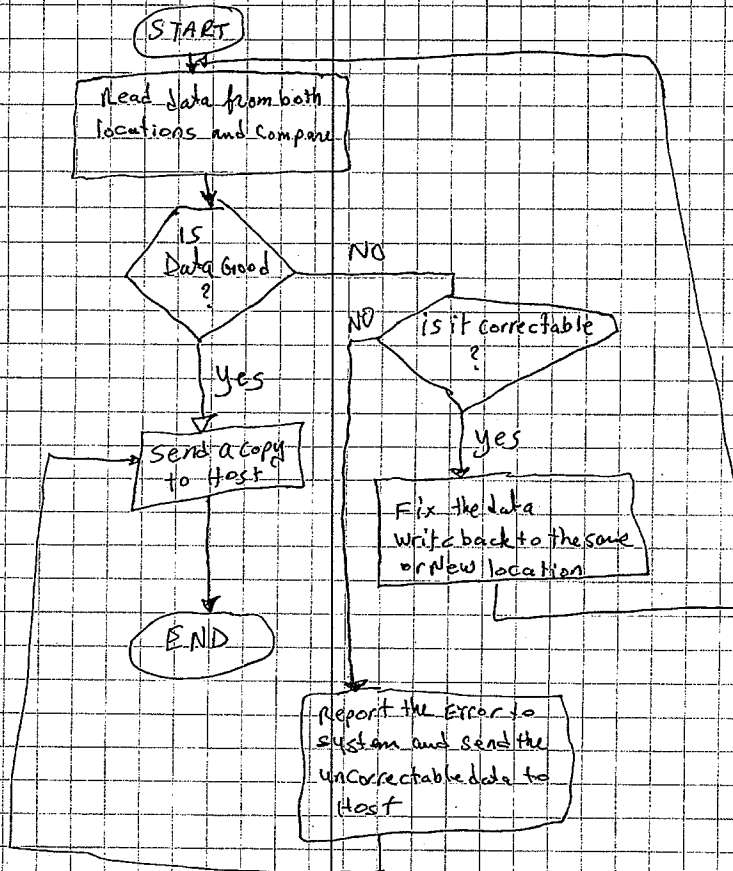
04-22-2011 controller with dedicated pair channels will have less firmware complexity, as their logical address of paired blocks would be across the designated two paired channels. Ex. ch0-1, or ch2-3, or ch4-5, or ch6-7. Controllers without dedicated pair channels firmware might put a copy on a non-paired channel. Ex ch0 or ch1-2, etc. and this kernel needs to be a bit more complex to keep track of all physical paired channels vs. logical paired channels as desired. The RAID storage Flash will improve redundancy, endurance, lower ECC, longer Flash life, better wear level and one can have fault-tolerant Flash storage systems.

Flowchart of RAID Storage on Flash:

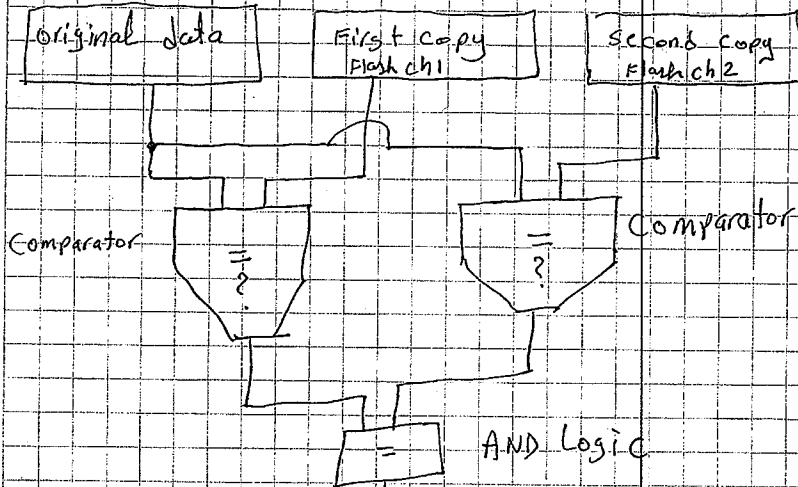
Write Sequence (RAID):



read sequence (RAID)



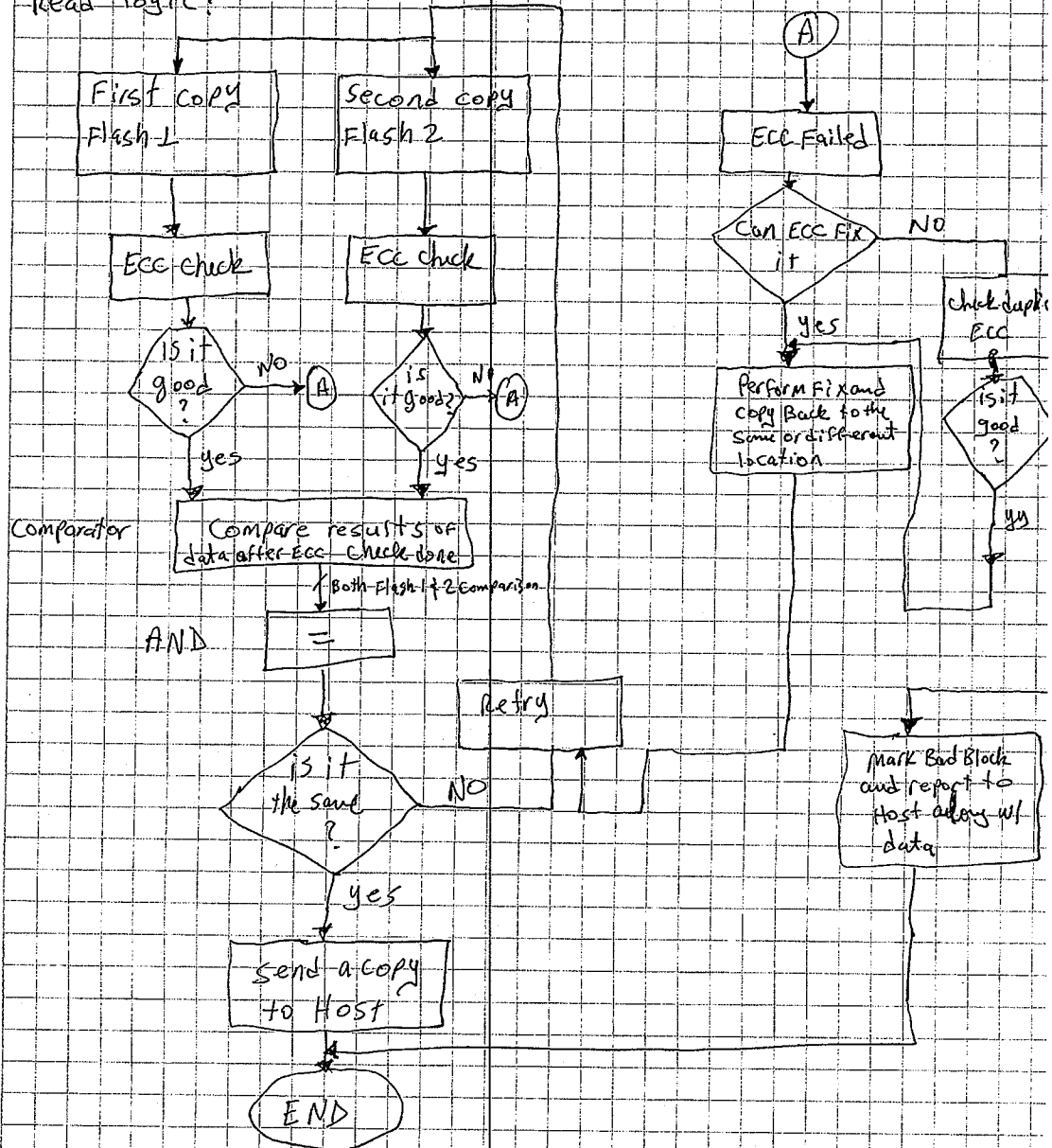
Write logic:



IF AND Result equal to "True" Write were Successful.

IF AND Result were "FALSE" Need to repeat Dual channel write till a successful write accomplished. This can be done to the same location or a new location.

Read logic:



Probability of getting the same uncorrectable error associated with one Read trial within two channel is defined as:

$P(A) = \frac{1}{100K}$, $P(B) = \frac{1}{100K}$ where 100K is the Program/erase endurance of SLC Flash products. As flash page program/erase approaches 100K cycle the probability of getting uncorrectable error will increase for that location.

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{100K} \cdot \frac{1}{100K} = \frac{1}{10^8} = \\ = 0.0001 \cdot 0.0001 = 0.00000001 = 1 \times 10^{-10}$$

Assuming Flash controllers will follow MFG suggested Ecc method to maximize Flash 100K program/erase cycles, then using RAID Flash Storage one can improve Flash life due to uncorrectable errors to 1×10^{-10} .

Endurance of Flash, means after a number of program/erase cycles the Floating gates start to "wear out", and becomes defective. A certain amount of defective gates can be compensated by ECC (Error Checking and Correction), However once the amount of defective gates (bit error rate) within a block is larger than ECC the block is mapped as bad block.

(1) $\text{bit error rate} > \text{ECC Capability}$ = bad block of a block

the endurance of typical SLC Flash is equal 100k program/erase cycle with an ECC of 1-bit engine.

Example of endurance calculation of 2GB SLC Flash product:

1) assuming size of Data is 256KB (File size), what is the Endurance of 2GB SLC card

$$\text{Page size} = 4\text{KB} (4096)$$

$$\text{Block size} = 64 \text{ Pages} = 256\text{KB}$$

$$\text{Device size} = 8192 \text{ Block} = 8192 \times 256\text{KB} = 2\text{GB}$$

Since the File size is equal block size of 256KB, then the product Endurance is:

$$8192 \text{ block} \times 100\text{k} = 781.25 \text{ Million block program/erase cycle.}$$

2) What is the lifetime of this product, assuming an application of 1 cycle every minute?

$$\text{lifetime} = \text{Endurance} / \text{Frequency in years}$$

$$\begin{aligned} \text{lifetime} &= \frac{\text{Approximately } 800.00\text{M}}{(1 \text{ cycle} \times 60 \text{ min} \times 24 \text{ hrs} \times 365 \text{ days})} \\ &= \frac{800.00\text{M}}{525600} \\ &= 1522 \text{ years} \end{aligned}$$

3) How about lifetime of the same product, assuming an application of 1 cycle every second?

$$\begin{aligned} \text{lifetime} &= \frac{\approx 800\text{M}}{(1 \text{ cycle} \times 60 \text{ sec} \times 60 \text{ min} \times 24 \text{ hrs} \times 365 \text{ days})} \\ &= \frac{800\text{M}}{31536000} \\ &= 25.36 \text{ years} \end{aligned}$$

4) Now let's assume the file size can be larger, what would the results be if the file is 2MB, 200MB, 2GB (matching card size)? (1-second program/erase)

a) for 2 MB File we will consume 8 Blocks as such:

$$2048 \text{ Byte (2MB)} : 256 \text{ KB (Block Size)} = 8 \text{ Blocks}$$

So $800 \text{M cycle} / 8 = 100 \text{M cycle then}$

$$\begin{aligned} \text{life time} &= 100 \text{M} / 31536000 \\ &= 3.17 \text{ years} \end{aligned}$$

b) 200MB File size, Program/erase every second?

$$\text{So } 200 \text{MB} : 256 \text{KB} = 800 \text{ Blocks}$$

$$\begin{aligned} \text{life time} &= 1 \text{M} / 31536000 \\ &= 0.0317 \text{ years} \end{aligned}$$

what if we write/erase every hours instead?

$$\begin{aligned} \text{then life time} &= 1 \text{M} / 8760 \\ &= 114 \text{ years instead} \end{aligned}$$

or what if we write/erase every minute instead?

$$\begin{aligned} \text{then life time} &= 1 \text{M} / 525600 \\ &= 1.9 \text{ years} \end{aligned}$$

c) 2GB file size, Program/erase every hours & minutes?

$$\text{So } 2048 \text{MB} : 256 \text{KB} = 8192 \text{ Blocks (entire Product)}$$

$$\begin{aligned} \text{then life time} &= 100 \text{K} / 8760 \quad (\text{once every hour}) \\ &= 11.4 \text{ years} \end{aligned}$$

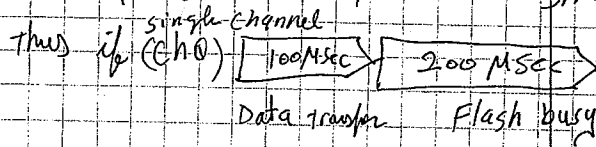
$$\begin{aligned} \text{or life time} &= 100 \text{K} / 525600 \quad (\text{once every minute}) \\ &= 0.19 \text{ year} \end{aligned}$$

The lifetime of each Flash product is determined by File size and Frequency of program/erase cycles.

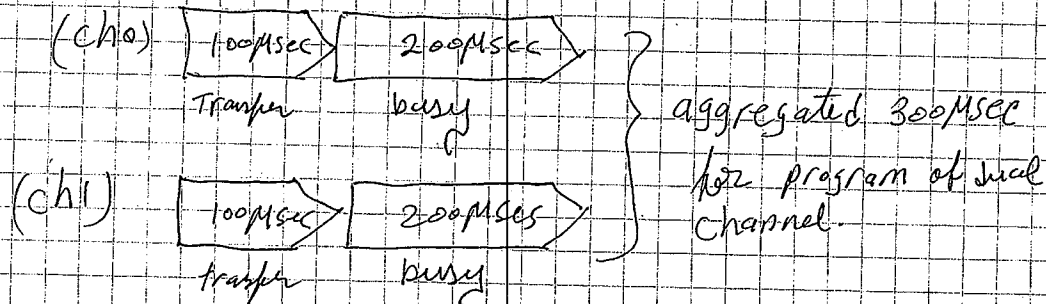
$$(2) \text{ lifetime} = \text{Endurance} (\# \text{Block} \times 100 \text{K}^{\text{SLC}}) / \text{Frequency of Prog/erase in year}$$

Using a Flash RAID storage, the lifetime of product would not improve, as both devices will wear out at the same time, thus the endurance is the same under RAID setting.

The transfer time + program (Flash busy time) of 1 channel & 2 channel would be the same as 2-channel Flash controllers have dedicated buffers, Flash controllers, and ECC engines per each channel.



the dual channel



there would be no additional penalty for RAID Implementation, during First write attempt, if the controller perform Read-Verify then based on controller clock speed & Architecture there would be additional time penalty for each successful verification and/or each non-successive verification + re-write operation.

The Read page is 35Msec Max in this example, thus when we perform read-verify it will take the controller: max 35Msec to read a page into data buffer and

then transfer it into internal controller buffer in either Asynchronous or Synchronous fashion which can be 200nT/sec (Synch) or 20ns (Asynch).

The verification portion will be internal to Flash architecture and it changes from vendor to vendor.

Assuming we read-verify one block then we can take:

$$(\text{Read}) 35\text{Msec} + (\text{Verify}) \approx 20\text{Msec} = 55\text{Msec}.$$

if we need to do worst case re-write and re-verify then we would have:

$$\text{1st Program } 300\text{Msec} + \text{Read-Verify } \overset{\text{1st}}{55\text{Msec}} + \text{2nd Program } 300\text{Msec} + \text{2nd Read-Verify } 55\text{Msec} = 710\text{Msec}.$$

The time needed to read a block, and correct the error, and re-write would be:

$$1 \text{ Block read } 35\text{Msec} + (\text{Verify}) \approx 20\text{Msec} + \text{write new location } 300\text{Msec} +$$

Using this probability one can re-calculate the lifetime of an application writing a 2GB file into a 2GB std vs. Raid flash storage with a frequency of 1 wr/each second and see the benefit.

$$\text{Storage size} = 2 \text{ GB}$$

$$\text{File size} = 2 \text{ GB}$$

$$\text{Page size} = 4 \text{ KB}$$

$$\text{Block size} = 64 \text{ pages} = 256 \text{ KB}$$

$$\text{Device size} = 8192 \text{ blocks} = 2 \text{ GB}$$

$$\text{Endurance} = \# \text{ of blocks} \times 100 \text{K program/erase (SLC)}$$

$$= 1 \times 100 \text{K}$$

$$= 100 \text{K}$$

So the standard (non-RAID) configuration will have an Endurance of 100K program/erase cycle.

$$\text{Lifetime} = \text{Endurance} / \text{Frequency (year)}$$

$$= 100 \text{K} / (1 \times 60 \text{sec} \times 60 \text{min} \times 24 \text{hrs} \times 365)$$

$$= 100 \text{K} / 31536000$$

$$= 0.00317 \text{ year}$$

Now assuming dual channel RAID storage and same condition we have:

$$\text{Endurance} = \text{ch1}(100 \text{K}) \times \text{ch2}(100 \text{K}) = 10 \text{ Billion}$$

$$\text{Lifetime} = 10 \text{B} / 31536000 = 317 \text{ years.}$$

This clearly shows that the raid storage product lifetime is superior by more than two factors as earlier estimated.

Figure 1:

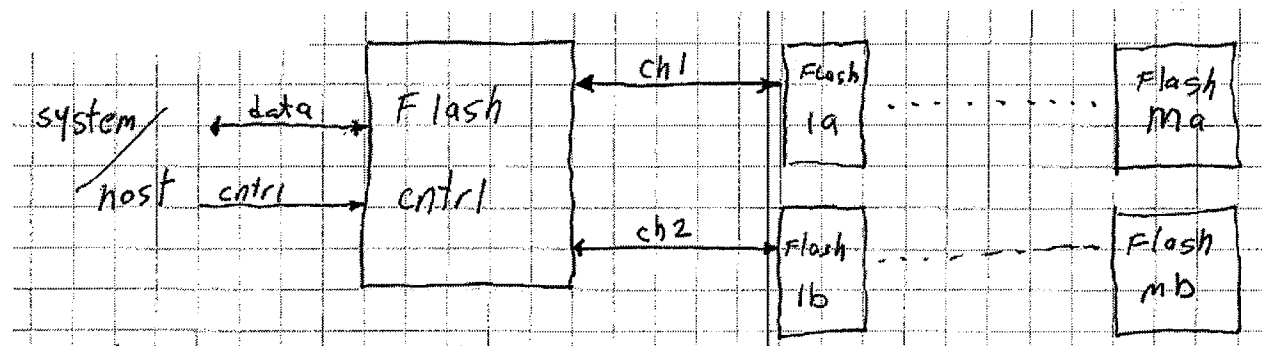


Figure 2:

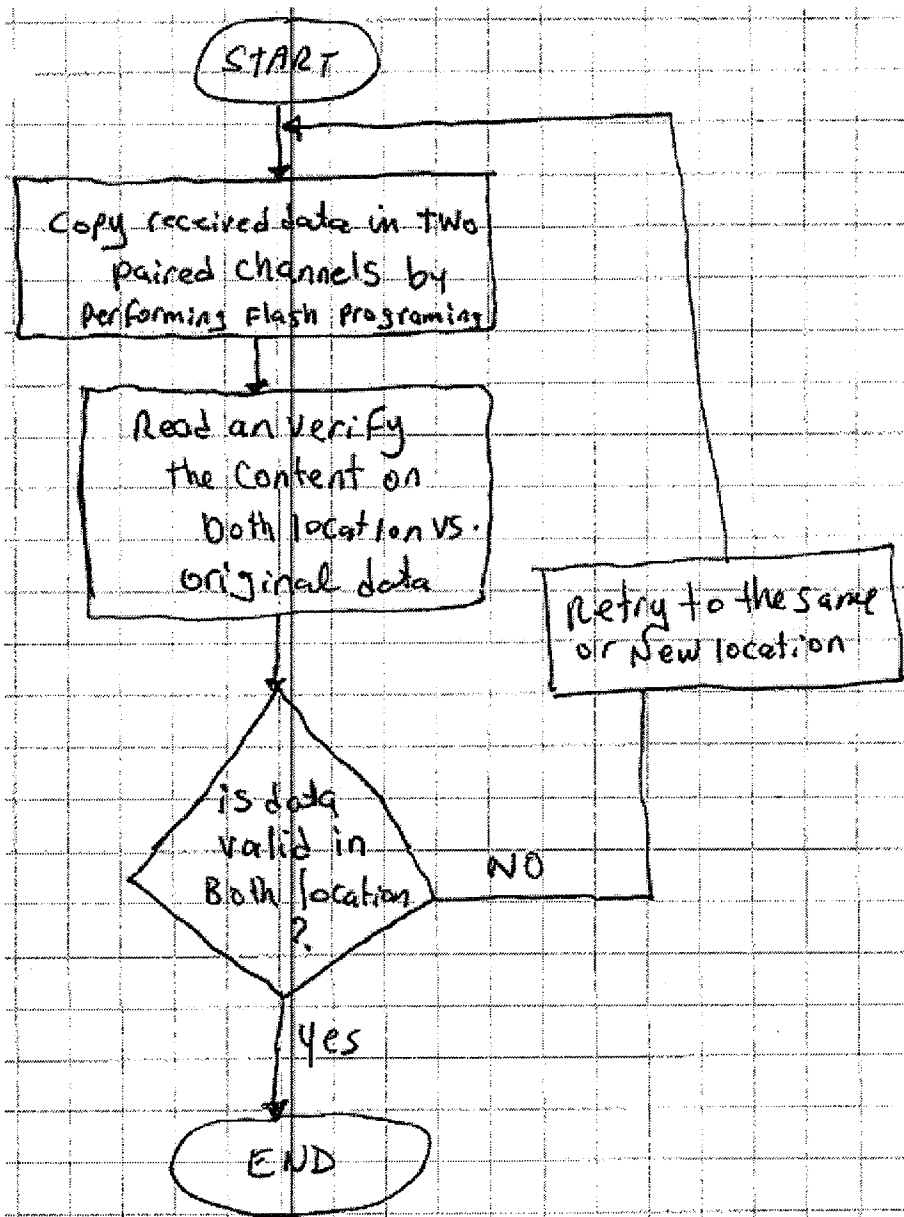


Figure 3:

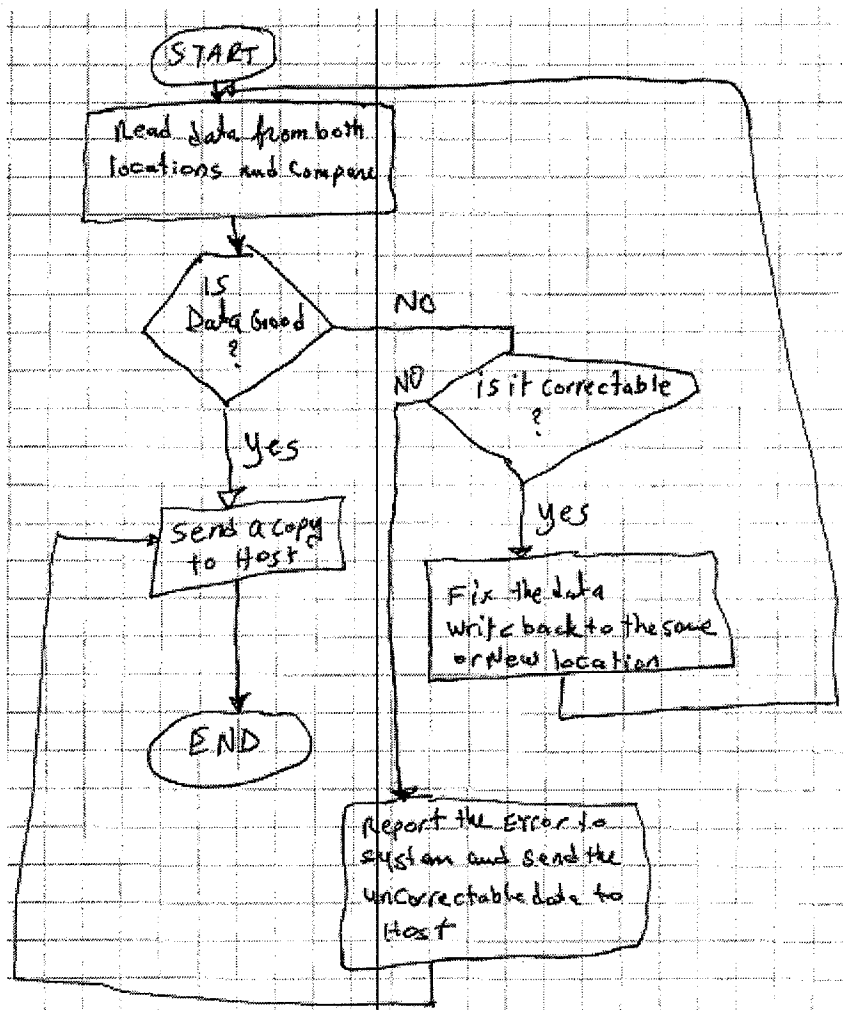


Figure 4:

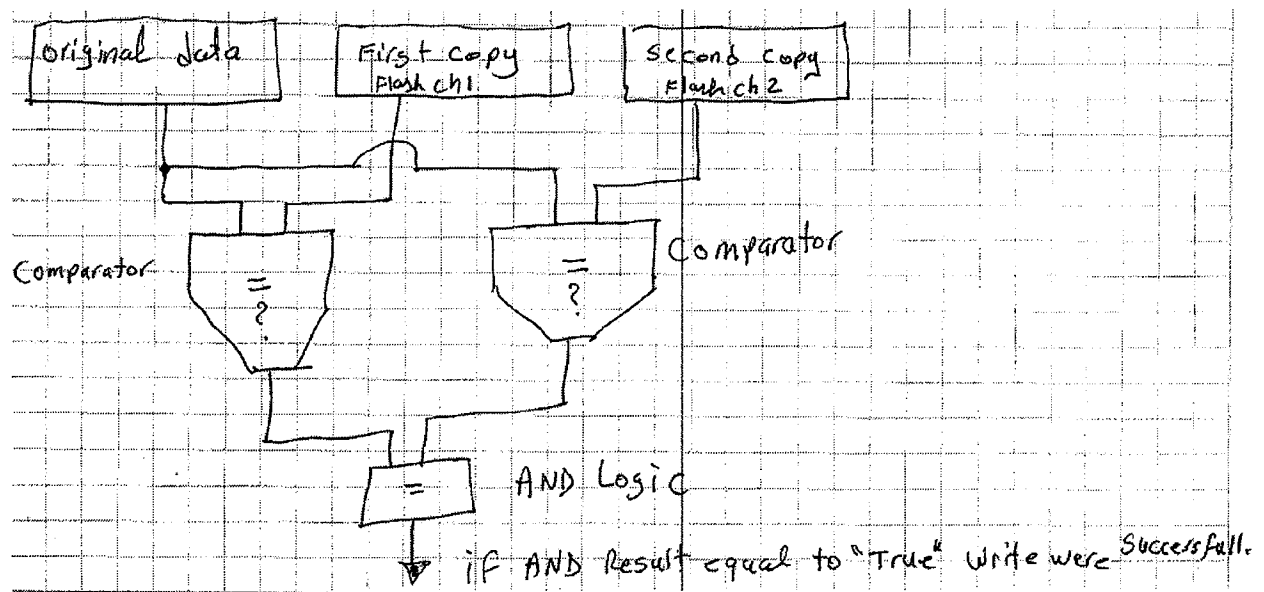


Figure 5:

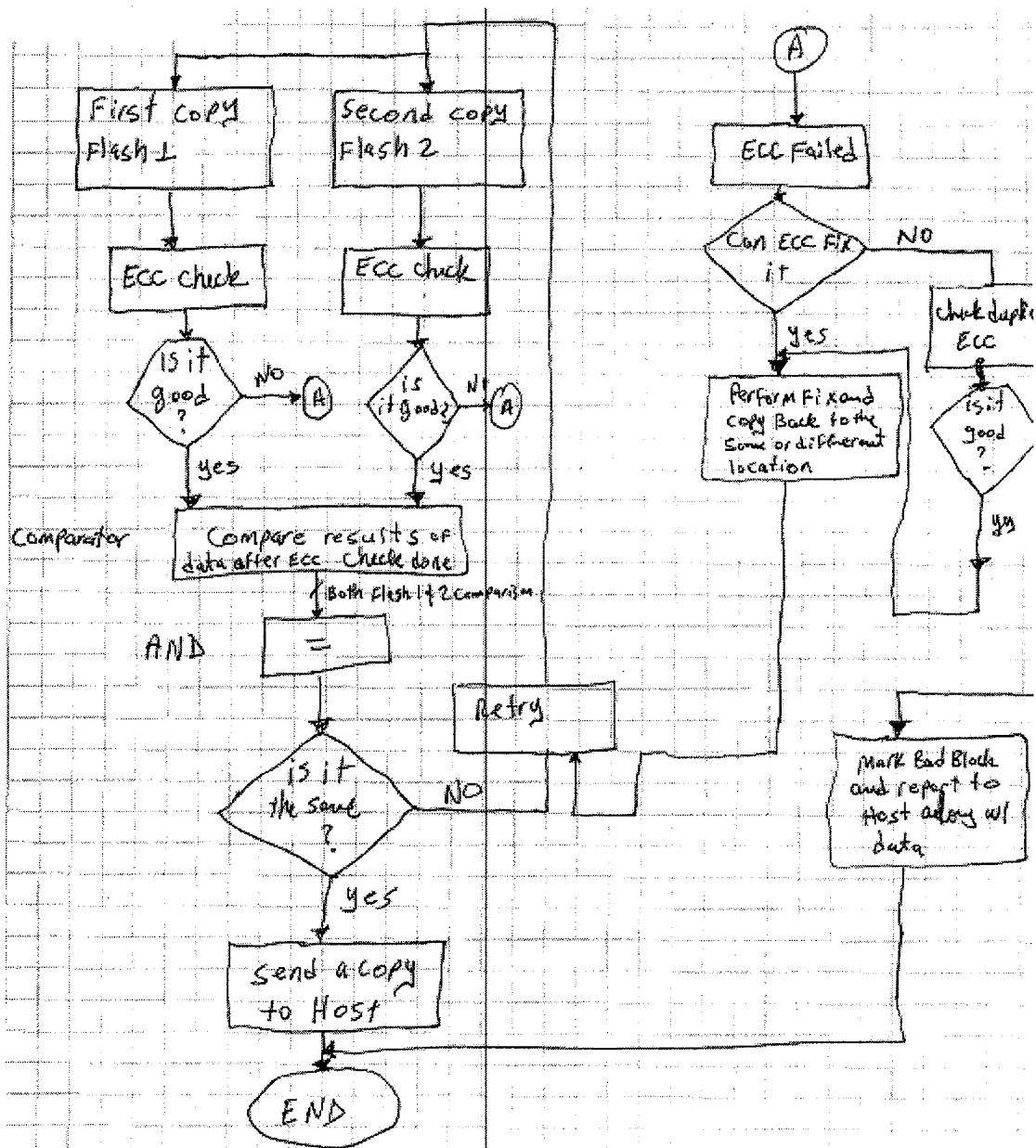
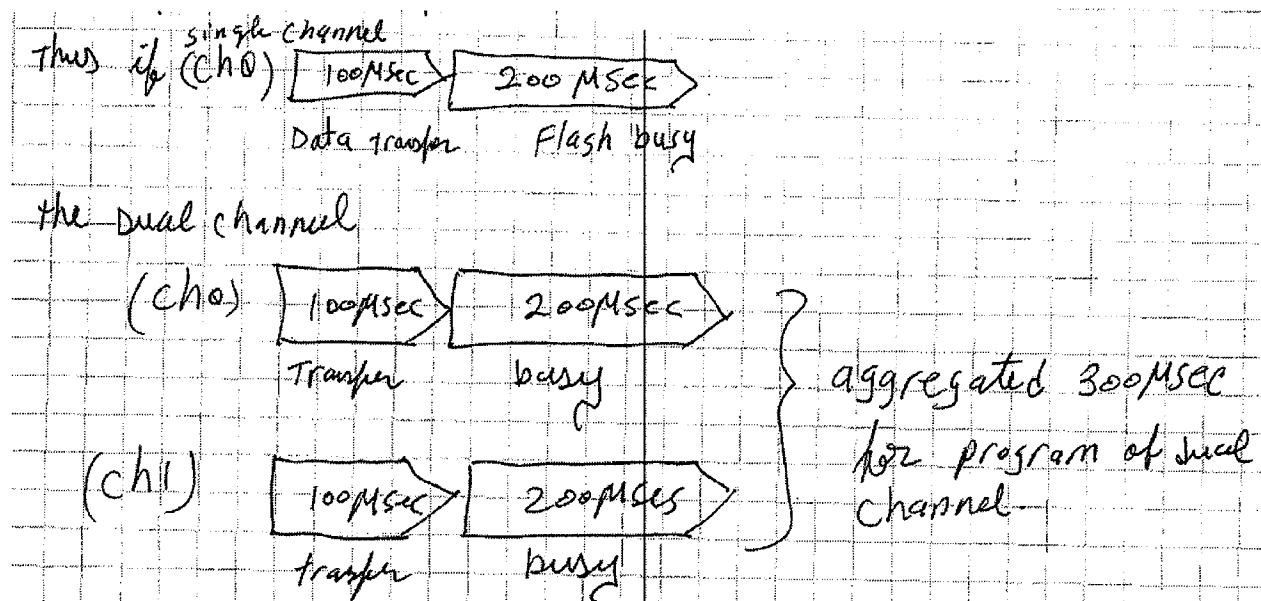


Figure 6:



**A METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT
FLASH RAID STORAGE FUNCTIONALITY**

[0001] A method of using standard flash controllers to implement RAID storage on flash products.

DESCRIPTION

[0002] Using a standard dual or quad channel flash controller to emulate a RAID drive.

[0003] Referring to Figure 1, the flash or none volatile or non-volatile memory controller can have two or more channels with each having access to two or more flash devices. The flash controller can have multiple copies of its firmware within the flash controller under control of an internal microcontroller (e.g., embedded into the flash controller) and a scheduler or mini-kernel to manage each firmware. Once the system has provided enough data to the flash controller's internal buffer for device/flash programming function, the scheduler or mini-kernel can assign one or more channels in a pair setting to store two or more copies of data into the flash devices.

[0004] For example, the host sends data = "Netlist" to the flash controller. Once the internal flash controller buffer field up or incoming data were completed, the flash controller scheduler will send one copy to channel 1, and one replica (exact copy) to channel 2, in case of dual channel simple RAID configuration. This will help the controller to have multiple copy and increase storage endurance (endurance) capabilities. The same concept can be applied to multiple paired channels to improve endurance such as 4-channel or 8-channel RAID storage. One way is for the controller to hold multiple pairs and perform enhanced ECC during read to correct any issues from multiple copies and send the correct results to the host. Currently, SLC flash devices have 100K program/erase cycles. Assuming a card of 1GB density built with SLC technology and file size of approximately card size or 1 GB, the standard controllers (non-RAID) will be able to have an endurance of 100K under a best case. Using dual channel, or multiple paired-channel, one can increase this by at least one magnitude factor. The probability of the same cell (1-bit) failure location, on the same block, on the same page, on the same plane, between two channels is very low. This flash controller firmware will pass the uncorrectable bit

stream to mini-kernel and by performing enhanced ECC kernel will be able to correct the error. The proper math to calculate this probability can be defined.

[0005] Under the above-defined configuration, the controller will read the same page address (logical add) from both paired channels and will transmit the one that has no ECC error, or the one with correctable ECC error to the host.

[0006] To improve performance, one can use more than two channels, for example 4-channel or 8-channel, to use interleaving capabilities to increase the program/erase speed with larger page sizes.

[0007] The controller with dedicated pair channels can have less firmware complexity, as their logical address of paired blocks would be across the designated two paired channels. For example, channels 0-1 or channels 2-3, or channels 4-5, or channels 6-7. Controllers without dedicated pair channels firmware might put a copy on a nonpaired channel. For example, channels 0-1 or channels 1-2, etc., and this kernel can be a bit more complex to keep track of the physical paired channels vs. logical paired channels, as desired. The RAID storage on flash can improve redundancy, endurance, lower ECC, longer flash life, better wear out, and one can have fault-tolerant flash storage systems.

[0008] Figure 2 shows a flow chart of example RAID storage on flash for a write sequence (RAID). Figure 3 shows a flow chart of example RAID storage on flash for a read sequence (RAID). Figure 4 shows example write logic.

[0009] If AND result equals "true," the write was successful. If AND result was "false," can repeat dual channel write until a successful write is accomplished. This can be done to the same location or a new location.

[0010] Figure 5 shows example read logic. The probability of getting the same uncorrectable error associated with one read trial within two channels is defined as:

$$P(A) = \frac{1}{100K}, \quad P(B) = \frac{1}{100K}$$

where 100K is the program/erase endurance of SLC flash products. As flash page program/erase approaches 100K cycle, the probability of getting uncorrectable error will increase for that location.

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{100K} \cdot \frac{1}{100K} = \frac{1}{10^8} = 0.00001 \cdot 0.00001 = 0.0000000001 = 1 \times 10^{-10}$$

[0011] Assuming flash controllers will follow MFG suggested ECC method to maximize flash 100K program/erase cycles, then using RAID flash storage one can improve flash life due to uncorrectable errors to 1×10^{-10} .

[0012] Endurance of flash, means after a number of program/erase cycles the floating gates start to “wear out,” and becomes defective. A certain amount of defective gates can be compensated by ECC (error checking and correction). However, once the amount of defective gates (bit error rate) within a block is larger than ECC the block can be mapped as a bad block.

[0013] (1) bit error rate of a block > ECC capability = bad block

[0014] The endurance of typical SLC flash is equal 100K program/erase cycle with an ECC of 1-bit engine.

[0015] The following is an example of an endurance calculation of 2GB SLC flash product:

[0016] (1) Assuming size of data is 256 KB (file size), what is the endurance of 2 GB SLC card?

Page size = 4 KB (4096)

Block size = 64 pages = 256 KB

Device size = 8192 Block = $8192 \times 256 \text{ KB} = 2 \text{ GB}$

Since the file size is equal block size of 256 KB, then the produce endurance is:

$8192 \text{ block} \times \text{look} = 781.25 \text{ million block program/erase cycle}$

[0017] (2) What is the lifetime of this product, assuming an application of 1 cycle every minute?

lifetime = endurance/frequency in years

lifetime = approximately $800.00 \text{ M} / (1 \text{ cycle} \times 60 \text{ min} \times 24 \text{ hrs} \times 365 \text{ days})$

$= 800.00 \text{ M} / 525600 = 1522 \text{ years}$

[0018] 3) How about lifetime of the same product, assuming an application of 1 cycle every second?

lifetime = $\approx 800 \text{ M} / (1 \text{ cycle} \times 60 \text{ sec} \times 60 \text{ min} \times 24 \text{ hrs} \times 365 \text{ days})$

$= 800 \text{ M} / 31536000 = 25.36 \text{ years}$

[0019] 4) Now let's assume the file size can be larger, what would the results be if the file is 2 MB, 200 MB, 2 GB (matching card size)? (1 second program/erase)

a) for 2 MB file, we will consume 8 blocks as such:

$$2048 \text{ Kbyte (2 MB)} \div 256 \text{ KB (block size)} = 8 \text{ blocks}$$

$$\text{So } 800 \text{ M cycle} / 8 = 100 \text{ M cycle; then}$$

$$\text{lifetime} = 100 \text{ M} / 31536000 = 3.17 \text{ years}$$

b) 200 MB file size, program/erase every second?

$$\text{so } 200 \text{ MB} \div 256 \text{ KB} = 800 \text{ blocks}$$

$$\text{lifetime} = 1 \text{ M} / 31536000 = 0.0317 \text{ years}$$

What if we write/erase every hour instead?

$$\text{then lifetime} = 1 \text{ M} / 8760 = 114 \text{ years instead.}$$

Or what if we write/erase every minute instead?

$$\text{then lifetime} = 1 \text{ M} / 525600 = 1.9 \text{ years}$$

c) 2 GB file size, program/erase every hours and minutes?

$$\text{so } 2048 \text{ MB} \div 256 \text{ KB} = 8192 \text{ blocks (entire product)}$$

$$\text{then lifetime} = 100 \text{ K} / 8760 = 11.4 \text{ years} \quad (\text{once every hour})$$

$$\text{or lifetime} = 100 \text{ K} / 525600 = 0.19 \text{ years} \quad (\text{once every minute})$$

The lifetime of each flash product is determined by file size and frequency of program/erase cycles.

[0020] (2) lifetime = endurance (#block x SLC 100 K) / Frequency of program/erase in year. Using a flash RAID storage, the lifetime of product would not improve, as both devices will wear out at the same time, thus the endurance is the same under RAID setting.

[0021] The transfer time and program (flash busy time) of 1-channel and 2-channel would be the same as 2-channel flash controllers have dedicated buffers, flash controllers, and ECC engines per each channel. See Figure 6.

[0022] There would be no additional penalty for RAID implementation, during first write attempt, if the controller performs read-verify then based on controller clock speed and architecture there would be additional time penalty for each successful verification and/or each non-successive verification and re-write operation.

[0023] The read page is 35 μsec max in this example, thus when we perform read-verify it will take the controller: max 35 μsec to read a page into data buffer and then transfer it

into internal controller buffer in either asynchronous or synchronous fashion which can be 200 MT/sec (synch) or 20 ns (asynch).

[0024] The verification portion will be internal to flash architecture and it changes from vendor to vendor.

[0025] Assuming we read-verify one block then we can take:

(Read) 35 μ sec + (verify) $\simeq$ 20 μ sec = 55 μ sec.

If we need to do worst case re-write and re-verify then we would have

1st program 300 μ sec + 1st read-verify 55 μ sec + 2nd program 300 μ sec + 2nd read-verify 55 μ sec = 710 μ sec

The time needed to read a block and correct the error and re-write would be:

1 block read 35 μ sec + (verify) $\simeq$ 20 μ sec + write new location 300 μ sec + read-verify 35 μ sec + $\simeq$ 20 μ sec = 410 μ sec

[0026] However, the system can have the data by 55 μ sec, the rest is internal to fix the issue. Reading dual channel verification in the first trial were bad:

Read 35 μ sec + verify $\simeq$ 20 μ sec + read (2nd) 35 μ sec + verify (2nd) 20 μ sec = 110 μ sec.

[0027] Read-verify and we get one channel bad (discrepancies) and can fix and re-write, then check again for goodness of data in both channels:

Read 35 μ sec + verify 20 μ sec + perform ECC fix $\simeq$ 30 μ sec + write new location 300 μ sec + read (2nd) 35 μ sec + verify (2nd) 20 μ sec = 440 μ sec.

[0028] Read-verify has discrepancies, retry not matched, ECC can't fix, checking duplicate ECC is good, perform fix, retry, re-verify.

[0029] Read 35 μ sec + verify 20 μ sec + retry read (2) 35 μ sec + verify (2) 20 μ sec + ECC check $\simeq$ 30 μ sec + check ECC results $\simeq$ 20 μ sec + checking duplicate ECC $\simeq$ 30 μ sec + fix re-write 300 μ sec + read 35 μ sec + verify 20 μ sec = 545 μ sec.

[0030] So in this example with assumption that write block will take 300 μ sec, and read will take 35 μ sec, our worst write would be = 710 μ sec, and our worst read would be = 545 μ sec.

[0031] This analysis clearly shows that even using this dual redundancy method, the worst case assumption still below 1 msec, and roughly equal to one block erase time of 700 μ sec.

[0032] Under flash RAID storage with dual channel configuration, if we assume we are using a 2 GB flash product using SLC type devices with 100K program/erase cycles once we

use the suggested ECC (for example, 8-bit / 512B) then if we use an application that writes a file of 2 GB size and perform repetitive write overtime to this flash card, under normal configuration, the card bit error rate will be larger than ECC capabilities once the write cycles approaches 100K, and beyond. Then the card will have bad blocks which impede further write and will generate uncorrectable errors. Using a RAID (for example, dual channel/redundancy) we want to find the probability of getting the same cell, within the same page, within the same block within both channels.

[0033] Using two independent events (mutually independent event) the probability will be:

$$Pr\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n Pr(A_i)$$

$$Pr(A \cap B) = Pr(A) \cdot Pr(B) = 100K \times 100K$$

$$Pr(A \cap B) = 10000000000 = 10 \text{ Billion}$$

$A(1)$ = Probability of uncorrectable error = 100K
 $A(2)$ = " " " " = 100K

[0034] Using this probability one can re-calculate the lifetime of an application writing a 2 GB file into a 2 GB std. vs. RAID flash storage with a frequency of 1 wr/each second and see the benefit

storage size = 2 GB

file size = 2 GB

page size = 4 KB

block size = 64 pages = 256 KB

device size = 8192 blocks = 2 GB

Endurance = # of blocks * 100 K program/erase (SLC) = 1 * 100 K = 100 K

[0035] So the standard (non-RAID) configuration will have an endurance of 100 K program/erase cycle. Thus

lifetime = endurance / frequency (year)

= 100 K / (1 * 60 sec * 60 min * 24 hrs * 365)

= 100 K / 31536000 = 0.00317 year

[0036] Now assuming dual channel RAID storage and save condition we have:

Endurance = ch 1 (100 K) x ch 2 (100 K) = 10 Billion

lifetime = 10 B / 31536000 = 317 years

[0037] This clearly shows that the RAID storage product lifetime is superior by more than two factors as earlier estimated.

[0038] Various embodiments of the present invention have been described above. Although this invention has been described with reference to these specific embodiments, the descriptions are intended to be illustrative of the invention and are not intended to be limiting. Various modifications and applications may occur to those skilled in the art without departing from the true spirit and scope of the invention.

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

A METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT
FLASH RAID STORAGE FUNCTIONALITY

First Named Inventor/Applicant Name:

TBD TBD

Filer:

Bruce S. Itchkawitz/Nicole Jevsevar

Attorney Docket Number:

NETL.101PR

Filed as Large Entity

Provisional Filing Fees

Description

Fee Code

Quantity

Amount

**Sub-Total in
USD(\$)**

Basic Filing:

Provisional application filing

1005

1

220

220

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Post-Allowance-and-Post-Issuance:

Extension-of-Time:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				220

Electronic Acknowledgement Receipt

EFS ID:	11039140
Application Number:	61538775
International Application Number:	
Confirmation Number:	1170
Title of Invention:	A METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY
First Named Inventor/Applicant Name:	TBD TBD
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Robin Leu
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.101PR
Receipt Date:	23-SEP-2011
Filing Date:	
Time Stamp:	21:08:57
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 220
RAM confirmation Number	14733
Deposit Account	111410
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	ADS.pdf	246945	no	4
			882a332b22abb43955d06393914bda385c921cd8		
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
2	Appendix to the Specification	Appendix.pdf	1360494	no	11
			d04b761872a6bbdb5b80e8db1ca2876a072571d4		
Warnings:					
Information:					
3	Drawings-only black and white line drawings	Drawings.pdf	268965	no	6
			a882b5f809caf43b97bf387d0522c9afd5cf9b20		
Warnings:					
Information:					
4	Specification	Specification.pdf	322701	no	7
			83e2d49df99701ec28b056504114f9d37cb6d797		
Warnings:					
Information:					
5	Fee Worksheet (SB06)	fee-info.pdf	29566	no	2
			7f8a8997e97893e56eabcd0f54b9544da7d1e2f2		
Warnings:					
Information:					
Total Files Size (in bytes):			2228671		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/538,775	09/23/2011		NETL.101PR

CONFIRMATION NO. 1170

FORMALITIES LETTER



OC000000050122155

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 09/30/2011

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- A surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of \$50 for a non-small entity, must be submitted.

SUMMARY OF FEES DUE:

Total fee(s) required within **TWO MONTHS** from the date of this Notice is \$50 for a non-small entity

- \$50 Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/sduong/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/538,775	09/23/2011		220	NETL.101PR		

CONFIRMATION NO. 1170

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000050122154

Date Mailed: 09/30/2011

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Power of Attorney:

Bruce Itchkawitz--47677

If Required, Foreign Filing License Granted: 09/29/2011

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/538,775**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID
STORAGE FUNCTIONALITY

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international

patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and

Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

Please Direct All Correspondence to Customer Number 20995

TRANSMITTAL LETTER
PROVISIONAL APPLICATION MISSING PARTS

Applicant	:	Mike Hossein Amidi
App. No.	:	61/538,775
Filed	:	September 23, 2011
For	:	A METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY

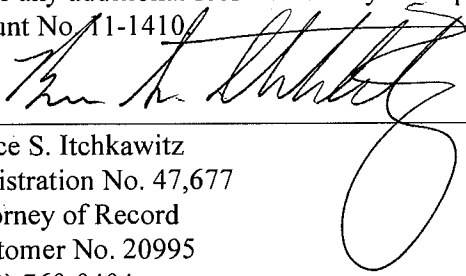
Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

In response to the Notice to File Missing Parts of Provisional Application, which was mailed by the Office on September 30, 2011, enclosed are:

- (X) Supplemental ADS in 3 pages.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.



Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

SUPPLEMENTAL APPLICATION DATA SHEET**Application Information**

Application Number:: 61/538,775
Filing Date:: September 23, 2011
Application Type:: Provisional
Title:: A METHOD OF USING STANDARD FLASH
CONTROLLERS TO IMPLEMENT FLASH RAID
STORAGE FUNCTIONALITY
Attorney Docket Number:: NETL.101PR
Small Entity?:: NO

Inventor Information

Applicant Authority Type:: Inventor
Primary Citizenship Country:: US
Status:: Full Capacity
Given Name:: Mike
Middle Name:: Hossein
Family Name:: Amidi
City of Residence:: Lake Forest
State:: California
Country of Residence:: US
Street:: 25162 Paseo Equestre
City:: Lake Forest
State:: California
Country:: US
Postal or Zip Code:: 92630

Correspondence Information

Correspondence Customer Number:: 20995

Phone Number:: (949) 760-0404

Fax Number:: (949) 760-9502

E-Mail Address:: efiling@kmob.com

Representative Information

Representative Customer Number:: 20995

Domestic Priority Information

Application::	Continuity Type::	Parent Application::	Parent Filing Date::
This Application	*	*	*
*			

Foreign Priority Information

Country::	Application Number::	Filing Date::	Priority Claimed::
*	*	*	*

Assignment Information

Assignee Name:: *

Street:: *

City:: *

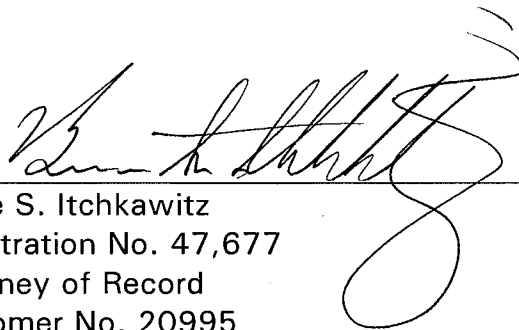
State or Province:: *

Country:: *

Postal or Zip Code:: *

Dated: November 29, 2011

By:

A handwritten signature in black ink, appearing to read "Bruce S. Itchkawitz", written over a horizontal line.

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

12343438/112911

Electronic Patent Application Fee Transmittal

Application Number:	61538775			
Filing Date:	23-Sep-2011			
Title of Invention:	METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz			
Attorney Docket Number:	NETL.101PR			
Filed as Large Entity				
Provisional Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late provisional filing fee/cover sheet	1052	1	50	50
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				50

Electronic Acknowledgement Receipt

EFS ID:	11502954
Application Number:	61538775
International Application Number:	
Confirmation Number:	1170
Title of Invention:	METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Erica Van Sciver
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.101PR
Receipt Date:	29-NOV-2011
Filing Date:	23-SEP-2011
Time Stamp:	19:51:06
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$50
RAM confirmation Number	7564
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	RESPONSE.pdf	32255 fad35a758589ee366b09da86953015350e073177	no	1
Warnings:					
Information:					
2	Application Data Sheet	SUPP_ADS.pdf	52927 a61e2b8c71ac28dcf57a7db2fdb422a2f34c963	no	3
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
3	Fee Worksheet (SB06)	fee-info.pdf	30147 4c5e3504bb9b541a67ea146992b543e5a1cf7495	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			115329		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/538,775	09/23/2011		270	NETL.101PR		

CONFIRMATION NO. 1170

UPDATED FILING RECEIPT



OC000000051212445

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 12/05/2011

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Mike Hossein Amidi, Lake Forest, CA;

Power of Attorney:

Bruce Itchkawitz--47677

If Required, Foreign Filing License Granted: 09/29/2011

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/538,775**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID
STORAGE FUNCTIONALITY

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international

patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and

Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage, facilitate, and accelerate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY TO PROSECUTE APPLICATIONS BEFORE THE USPTO

I hereby revoke all previous powers of attorney given in the application identified in the attached statement under 37 CFR 3.73(b).

I hereby appoint:



Practitioners associated with the Customer Number:

151145

OR



Practitioner(s) named below (if more than ten patent practitioners are to be named, then a customer number must be used):

Name	Registration Number	Name	Registration Number

as attorney(s) or agent(s) to represent the undersigned before the United States Patent and Trademark Office (USPTO) in connection with any and all patent applications assigned only to the undersigned according to the USPTO assignment records or assignment documents attached to this form in accordance with 37 CFR 3.73(b).

Please change the correspondence address for the application identified in the attached statement under 37 CFR 3.73(b) to:



The address associated with Customer Number:

151145

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

Assignee Name and Address:

Netlist, Inc.
175 Technology Drive, Suite 150
Irvine, CA 92618

A copy of this form, together with a statement under 37 CFR 3.73(b) (Form PTO/SB/96 or equivalent) is required to be filed in each application in which this form is used. The statement under 37 CFR 3.73(b) may be completed by one of the practitioners appointed in this form if the appointed practitioner is authorized to act on behalf of the assignee, and must identify the application in which this Power of Attorney is to be filed.

SIGNATURE of Assignee of Record

The individual whose signature and title is supplied below is authorized to act on behalf of the assignee

Signature		Date	12/5/17
Name	Noel B. Whitley	Telephone	949-679-0115
Title	VP, IP & Licensing		

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

STATEMENT UNDER 37 CFR 3.73(b)

Applicant/Patent Owner: Mike Hossein Amidi

Application No./Patent No.: 61/538,775 Filed/Issue Date: 09-23-2011

Titled: **METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE
FUNCTIONALITY**

NETLIST, INC., a **Corporation**

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)

the patent application/patent identified above, by virtue of either:

A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 027472, Frame 0827, or for which a copy therefore is attached.

OR

B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☐ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Khaled Shami/

Signature

Date

Khaled Shami, Reg. No. 38,745

Attorney for Assignee

Printed or Typed Name

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Acknowledgement Receipt

EFS ID:	31160852
Application Number:	61538775
International Application Number:	
Confirmation Number:	1170
Title of Invention:	METHOD OF USING STANDARD FLASH CONTROLLERS TO IMPLEMENT FLASH RAID STORAGE FUNCTIONALITY
First Named Inventor/Applicant Name:	Mike Hossein Amidi
Customer Number:	20995
Filer:	Khaled Shami/Casey Berger
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	NETL.101PR
Receipt Date:	07-DEC-2017
Filing Date:	23-SEP-2011
Time Stamp:	15:30:05
Application Type:	Provisional

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00160010000_POA373.pdf	1154671	yes	2
			d6e17819a8774ab785b94f6ff55984ed720e4a37		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Power of Attorney	1	1
	Assignee showing of ownership per 37 CFR 3.73	2	2

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

Total Files Size (in bytes):	1154671
-------------------------------------	---------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/538,775	09/23/2011	Mike Hossein Amidi	NETL.101PR

151145
Shami Messinger PLLC
1000 Potomac Street NW
Fifth Floor
Washington, DC 20007

CONFIRMATION NO. 1170
POA ACCEPTANCE LETTER



Date Mailed: 12/12/2017

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 12/07/2017.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/qtran/

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.096PR
		Application Number	
Title of Invention	HIGH DENSITY DIMMS		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

- ☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117
				☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name	Suffix
	Hyun		Lee	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City	Ladera Ranch	State/Province	CA	Country of Residence US
Citizenship under 37 CFR 1.41(b)		US		
Mailing Address of Applicant:				
Address 1	21 Thalia Street			
Address 2				
City	Ladera Ranch	State/Province	CA	
Postal Code	92694	Country	US	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	20995		
Email Address	efiling@kmob.com	Add Email	Remove Email

Application Information:

Title of the Invention	HIGH DENSITY DIMMS		
Attorney Docket Number	NETL.096PR	Small Entity Status Claimed ☐	
Application Type	Provisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	10	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.096PR
		Application Number	
Title of Invention	HIGH DENSITY DIMMS		

Publication Information:
☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.

☐ C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.
Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One: ☒ Customer Number ☐ US Patent Practitioner ☐ Limited Recognition (37 CFR 11.9)

Customer Number 20995

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

			Remove
Application Number	Country ¹	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☐ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Assignee Information:

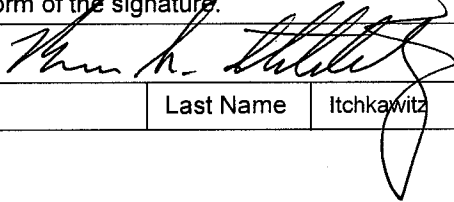
Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

Assignee 1

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.096PR
		Application Number	
Title of Invention	HIGH DENSITY DIMMS		

If the Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information:				
Address 1				
Address 2				
City		State/Province		
Country		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee Data may be generated within this form by selecting the Add button.				

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature				Date (YYYY-MM-DD)	2011-07-28
First Name	Bruce	Last Name	Itchkawitz	Registration Number	47677

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

HIGH DENSITY DIMMS**BACKGROUND**

[0001] As technology advances and the usage of tablet notebook computer increases, more data needs to be transferred among the data centers and to/from the end users. In many cases, the data centers are built by clustering many servers that are networked to increase the performance.

[0002] Although there are many types of networking that are based on the applications, the basic concept is to increase the performance by dynamically allocating the computing resources. In recent years, the server technology has been evolved to serve many specific applications such as 'finance transaction' (e.g., POS, inter-bank transaction, stock market transaction), 'scientific computation' (e.g., fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), 'medical diagnostics' (e.g., diagnostics based on the fuzzy logic, medical data processing), 'simple information sharing and searching' (e.g., web search, retail store website, company home page), 'email' (information distribution and archive), 'security service', 'entertainment' (e.g., video-on-demand), etc. However, all these applications share the same information transfer bottleneck, which resides in the CPU efficiently being able to transfer the data. This is because all data transfers are going through the CPU I/O channels.

[0003] The data transfer limitation by the CPU also applies to the data transfer from/to either HD or SSD to/from the memory subsystem that is connected to the FSB (Front Side Bus).

[0004] There have been various configurations to increase the data transfer from/to the main storage, such as HD or DDS, to the local DRAMs for the faster data throughput rate. Especially, the EMM that is supported with EcoRAM (Spansion) (see, Appendix B) presents an idea of constructing a fast SSD that assumes the physical form factor of a DIMM. Spansion populates the EcoRAM with a large quantity of Flash memories and a limited amount of DRAMs that serves as a data buffer. This arrangement delivers a much higher throughput rate than standard SSD since this EcoRAM is connected to the CPU via the HT interface while SSD is connected via SATA or USB. The read random access

throughput rate of EcoRAM is near 3GB/s compared with the standard PCIe-based NAND SSD which is about 400MB/s. This gives a 7.5X improvement. The write random access throughput rate of EcoRAM over the standard PCIe-based NAND SSD is less than 2X (197MB/s vs. 104MB/s) due to the fact that the write speed is predominated by the Flash write access time. The idea of EcoRAM is to build a SSD with the form factor of a standard DIMM such that it can be connected to the Front Side bus. However, due to the interface protocol difference between DRAM and Flash, Spansion developed a new interface device (Spansion EcoRAM Accelerator) that occupies one of the CPU sockets. Spansion EcoRAM is well suited for the "Read Access Intensive" applications, such as 'medical diagnostics, simple information sharing and searching, and entertainment. However, it does not fit to serve the applications that require extensive data manipulations, such as 'scientific computation', 'security service' and 'finance transactions'.

[0005] As it has been stated in the previous paragraphs, the two deficiencies of EcoRAM are: 1) it requires an interface device that occupies a CPU socket, thus less CPU, and in turn it reduces the overall computation efficiency; and 2) the high disparity in the throughput rate between read and write operations limits the utilization of this solution.

[0006] In addition to these deficiencies, the throughput rate of EcoRAM is much lower than DRAM, such that the read access throughput rate is only 25% of the DRAM read access rate; the write access throughput rate is less than 2% of the DRAM access rate.

SUMMARY

[0007] This application presents certain embodiments that overcome the shortcomings of the prior art: the significantly slower access time compare to the DRAM, and the high parity in the throughput rate between read and write throughput rate (15 to 1 ratio).

[0008] This application discloses certain embodiments of a Flash-DRAM-hybrid DIMM with integrated address spaces of Flash and DRAM. The application also presents a method of controlling such devices. In certain embodiments, the device is able to configure the actual memory density of the DIMM and/or the ratio of DRAM memory to Flash memory for optimal use with a particular application (e.g., POS, inter-bank transaction, stock market transaction, 'scientific computation,' such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, 'medical diagnostics,' such as diagnostics based

on the fuzzy logic, medical data processing, 'simple information sharing and searching,' such as web search, retail store website, company home page, 'email' or information distribution and archive, 'security service,' 'entertainment,' such as video-on-demand).

[0009] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, where DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 shows the data path in a conventional memory arrangement.

[0011] Figure 2 shows the general architecture of existing non-volatile memory module products. A part of SSD/HD is migrated into the Front Side Bus (FSB) to support the RAID (*Redundant Array of Independent Disks*) operation.

[0012] Figure 3 shows a currently available concept of connecting a Flash DIMM to the FSB.

[0013] Figure 4 shows the general architecture of a currently available product that uses the Flash DIMM with a DIMM controller.

[0014] Figure 5 shows an example memory module that contains DRAMs, Flash, and Central DIMM Controller (CDC), Data Manager (DMgr) in accordance with certain embodiments disclosed herein. The CDC provides two main functions: the standard RDIMM register function, and the data transfer management function between DRAM and Flash.

[0015] Figure 6 shows an example connection of the CDC to the register and DMgr, to MCH, DRAM, and Flash in accordance with certain embodiments described herein.

[0016] Figure 7 shows an example of the functional blocks in the CDC in accordance with certain embodiments described herein.

[0017] Figure 8A shows the general concept of an existing Flash-DRAM hybrid DIMM.

[0018] Figure 8B shows example functional blocks in a DRAM controller that is in the CDC in accordance with certain embodiments described herein.

[0019] Figure 9 shows an example state diagram of data transfer between the Flash memory and the DRAM memory in accordance with certain embodiments described herein.

[0020] Figure 10 shows an example of how the MCH addresses the Flash area in accordance with certain embodiments described herein.

[0021] Figure 11 is a table that shows example amounts of DRAMs that are used to support different applications in accordance with certain embodiments described herein.

[0022] Figures 12A-12C show a conventional NVDIMM, a conventional FDHDIMM, and an example FDHDIMM compatible with certain embodiments described herein, respectively, to illustrate some differences between the three modules.

BRIEF DESCRIPTION OF THE APPENDICES

[0023] Appendix A: “HyperVault: Multi-Terra Bytes of FSB Memory for HPC,” description regarding example device in accordance with certain embodiments described herein.

[0024] Appendix B: “Spansion® EcoRAM™ Solution: IOZone Solution Performance White Paper,” Version 0.4, description of currently available product.

DETAILED DESCRIPTION

[0025] A Flash and DRAM hybrid DIMM (FDHDIMM) that interfaces with MCH operates in two modes; as a standard DDR DIMM with SSD, or as a high density DIMM.

[0026] As a standard DDR DIMM with SSD, the FDHDIMM supports two views to the MCH. First, the MCH views the FDHDIMM as a combination of DRAM DIMM and SSD. In this case the MCH needs to manage two address spaces, one for the DRAMs and one for the Flash. The advantage of this mode is that the CPU does not need to be in the data path when data is moved from Flash to DRAM or vice versa. Second, the MCH views the FDHDIMM as an On-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space.

[0027] Thus, for the normal standard DDR DIMM with SSD operation, the MCH physically fetches data from the SSD to the DDR. This action will be much quicker than the

data going through the CPU, which happens when the current data move from SSD to DIMMs occurs, since all data movement occurs on a DIMM.

[0028] In the high density DIMM mode, the MCH recognizes the physical density of DIMM, which is the density of Flash alone. The DRAM serves as a data buffer that buffers read data, and serves as a temporary storage for the write data. The MCH has two options to execute a read operation.

[0029] The first read option is to send an active command (actually RAS) for the FDHDIMM to pre-fetch the read data from the Flash to the DRAMs, where the pre-fetch size is a page (1KB or 2KB, optional). This option will allow the data to transfer from the Flash to DRAMs, and provides the data latency and the throughput rate that is the same as any DRAM operation as long as the operations are done onto the pages that are opened with the pre-active command. However, this option requires much a longer separation period between the RAS and the first CAS command.

[0030] The other read option is executing the RAS and CAS (active and read/write) commands just as the standard DRAM. However, this option requires the initial Flash read time after the first CAS command (i.e. extensive CAS latency). This option supports the same throughput rate as the standard DRAM DIMM as long as the accesses occur from the open pages.

[0031] Certain embodiments described herein overcome the shortcomings of having the long separation period between the RAS and the first CAS command, and of having the extensive CAS latency.

[0032] In Figure 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to CPU via separated memory control ports. This figure shows the data flow path between the server main memory (SSD or HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which includes the IRQ (Interrupt) service that the CPU performs periodically. Furthermore, since all the data traffic goes through the CPU, the data move from/to SSD/HD to/from the DRAM DIMM occupies the CPU internal data bus, thus further hindering the overall CPU performance.

[0033] Figure 2 shows the general architecture of a non-volatile memory module that is currently being exercised. Non-Volatile DIMM (NVDIMM) is a Flash-DRAM hybrid DIMM (FDHDIMM), which uses a backup power (either battery or charged capacitor) to copy the data in the DRAM memory into the Flash memory when power loss occurs while the system is in operation. In this architecture, a part of SSD/HD is migrated into the FSB in the form of Flash. However, the density of the Flash is rather low since the Flash memory size does not have to be any larger than the DRAM memory size. The purpose of this type of architecture is to provide a non-volatile storage that is connected to the FSB to support the RAID (*Redundant Array of Independent Disks*) type of operation.

[0034] Figure 3 shows the system level architecture of a currently available product, which includes a Flash DIMM controller chip and Flash DIMM. In this figure, the Flash DIMM controller chip occupies a CPU socket and communicates with the CPU via HT (Hyper Transport). Although this product utilizes the HT interface, the actual interconnection between the controller chip and CPU is not limited to HT. Although this architecture does not address the shortcomings that associate with the traditional way (Figure 1) of connecting the SSD/HD and DRAM DIMMs, it allows increase of the density of non-volatile memory space that is connected to the virtual FSB since the CPU views the Flash DIMM controller chip as another processor with a large density of memory on its FSB.

[0035] There are two shortcomings with this architecture. First, it occupies one or two CPU sockets. Thus, it reduces the utilization of the server Mother Board (MB). Second, the Flash access speed is limited by 4 items: 1) the read/write speed of the Flash memory, 2) the FSB bus size, 3) the Flash DIMM controller chip latency, and 4) the HT interconnect efficiency, which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0036] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The parity in the access speed (15 to 1) between the read operation and write operation clearly shows that this architecture is limited by the read/write speed of the Flash memory. The discrepancy of the access speed between this type

of architecture and JEDEC standard DIMM grows wider as the memory technology advances, since as the memory technology advances, the Flash access rate stays relatively constant while the DRAM access speed increases.

[0037] Figure 4 shows the general architecture of the currently available FDHDIMM product, which receives two sets of control signals from the MCH/CPU while a single data bus interfaces with the MCH. In this architecture, since the MCH/CPU does not have a view to the data path between the DRAM memory and the Flash memory, it requires extra complication in designing the MCH.

[0038] Figure 5 shows the basic structure of an example memory module in accordance with certain embodiments described herein.

[0039] One major difference from the prior art is that the example memory module contains two on-DIMM controllers: the Central DIMM Controller (CDC) and the Data Manager (DMgr) to manage the interface between Flash, DRAM, and the MCH. This architecture addresses all the shortcomings that are associated with the prior art, such as the MCH views of the example memory module as a very high density DRAM DIMM.

[0040] The CDC controls the read/write access to/from the Flash memory to the DRAM memory, and DRAM memory to MCH/CPU. Since there is no direct access from the MCH/CPU to the Flash memory, the Flash access speed has very minimal impact to the overall access speed. The CDC performs the memory address translation/mapping and address domain conversion, Flash access control, error correction, data width modulation between the Flash memory and DRAM, etc. The CDC ensures that the example memory module operates with the industry standard DDR, DDR2, DDR3, DDR4 protocols.

[0041] The DMgr controls the data flow rate, the data transfer size, data buffer size, error monitoring and correction.

[0042] Figure 6 shows an example interoperability among the components of certain embodiments described herein. After the initialization, the CDC receives a command from CPU to prepare for a certain memory space, which is configurable or programmable by the CPU. The CPU can partition or parse the memory space in various ways that are optimum for the particular application that it needs to execute. The CDC maps the actual physical DRAM and Flash memory space to support the CPU view of the memory space. The CDC

also manages to support any booting operation by copying the necessary information from the Flash memory to the DRAM memory during system power up. This activity may utilize the CDC to initiate a data copying operation from the Flash memory to the DRAM memory just as the CPU copies the boot information from the SSD/HD to the DRAM memory. The DMgr controls the data traffic between the Flash memory and the DRAM memory. The DMgr receives the data transfer size, formatting information (540), direction of data flow (511, 512, 521, 522), and the starting time of the actual data transfer from CDC. The DMgr also receives the data flow path from the CDC to correctly establish the data transfer fabric. The DMgr also functions as a bi-directional data transfer fabric such that it has more than 2 sets of data ports facing the Flash and the DRAM memory. The multiplexors 511 and 512 provide the data paths from the DRAMs (DRAM-A and DRAM-B) to the MCH and the Flash memory, the multiplexors 521 and 522 provide the data paths from the MCH and the Flash memory to the DRAMs (e.g., DRAM-A and DRAM-B).

[0043] Two sets of multiplexors (511, 512, 521, 522) allow independent data transfer to the Flash memory from the DRAM memory. For example, the DMgr transfers data from DRAM-A to the MCH (via 511) at the same time as DRAM-B to the Flash memory (via 512), or it transfers data from DRAM-B to the MCH (via 511) at the same time as the Flash memory to DRAM-A (via 521). This special arrangement of data transfer connect also allows direct data transfer from the MCH to the Flash memory (e.g., via 521 and 512), and allows direct data transfer from the Flash memory to the MCH (e.g., via 521 and 511).

[0044] The Data Format (540) also contains a data buffer to support a wide data bus facing the Flash memory, and to match the data flow between the DRAM memory and the Flash memory.

[0045] Figure 7 shows example functional blocks of the CDC in accordance with certain embodiments described herein. The CDC has four major functional blocks: DRAM control block (DRAMCtrl), Flash control block (FlashCtrl), MCH/CPU command interpreter (CmdInt), DRAM-Flash interface scheduler (Scheduler), and the DMgr control block (DMgrCtrl).

[0046] DRAMCtrl (710) generates DRAM commands that are independent from the DRAM commands from the MCH. However, if the MCH initiates a read/write operation from/to the same DRAM that is currently executing a command from the DRAMCtrl (710), the DRAMCtrl (710) aborts the operation. The FlashCtrl (720) generates Flash commands for the proper read/write operations. The CmdInt (730) constantly intercepts the MCH command and provides the correct interpretation. For example, CmdInt issues an interrupt signal to the DRAMCtrl (710) if the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl initiated. The Scheduler (740) schedules the Flash –DRAM interface operation such that there is no resource conflict in the DMgr. The Scheduler assigns the time slots for the DRAMCtrl, FlashCtrl operation based on the current status and the command from the MCH. The DMgrCtrl (750) controls the data paths between the DRAM memory and the Flash memory, between the DRAM memory and the MCH, and between the MCH and the Flash memory.

[0047] The memory space mapping in Figure 8A for an existing Flash-DRAM hybrid DIMM shows two separate and independent address spaces, one for the Flash memory and the other for the DRAM memory. The CPU treats these two address spaces no differently than the conventional address spaces of DRAM and SSD/HD. The memory in each address space is controlled by the conventional method of controlling the DRAM and SSD/HD. However, the on-DIMM data path between Flash and DRAM allows direct data transfer to occur between the Flash memory and the DRAM memory. This method of transferring data bypasses the conventional method of transferring data from SSD/HD to the FSB memory via the CPU, thus reducing the data transfer time. Since this method also removes the CPU from the data path, it allows the over efficiency of the CPU usage, and thus improves the overall system performance. However, the shortcoming of this method are that: 1) the MCH needs to manage two address spaces, 2) the MCH needs to manage two different memory protocols simultaneously, 3) the MCH needs to map the DRAM memory space into the Flash memory space, 4) the data interface time cannot be controlled due to the difference in the data access time between the Flash memory and the DRAM memory, and 5) the MCH is completely blind about the data transfer activity between the Flash memory and the DRAM memory, and

as the result, controlling the data transfer between the Flash memory and the DRAM memory is practically impossible.

[0048] The memory space mapping in Figure 8B for a memory module in accordance with certain embodiments described herein shows an overlapping memory address space between the DRAM memory and the Flash memory. In this case, the MCH does not see any difference between the DRAM memory space and the Flash memory space. The MCH sees the large Flash memory space as the DRAM memory space, and accesses this memory space with the standard DDR protocol. The DRAM memory space holds the copy of data from the selected Flash memory space such that the MCH can access these data via FSB.

[0049] Figure 9 shows an example transfer of data from the Flash memory to the DRAM and vice versa in accordance with certain embodiments described herein. The data transfer from the Flash memory to the DRAM memory occurs when the CDC receives a command to move data from the Flash memory to the DRAM memory. The data transfer from the DRAM memory to the Flash memory occurs when all the pages in a particular block are closed. The CDC partitions the DRAM memory space into number of blocks for the purpose of supporting the optimum operation. The block size is configurable by the CPU, such that the number pages in a block can be optimized to support a particular application or a task. For example, when CPU concurrently executes more than one task, the CPU can configure a different block size for each task based on the DRAM space it needs for the particular task.

[0050] The MCH can access the memory module of certain embodiments described herein using the standard FSB DIMM access protocol (e.g., JEDEC). Since any page (or block) that does not reside in the DRAM memory has to be copied from the Flash memory, and the copying process takes time in the order of tens of microseconds, the MCH initiates the copying of the page ahead by writing into two specific RC registers along with the valid block address. When CDC receives these particular RC write commands from the MCH, it recognizes it as the block address that it needs to prepare, and it will copy the block to the DRAM memory. Once the block is in the DRAM memory space, the MCH can access the data with the standard DRAM DIMM operation protocol.

[0051] In Figure 9, after power up (910) and initialization (920), CPU/MCH initiates the data move from the Flash memory to the DRAM memory by writing to the control register in CDC (930), which copies a block of data from the Flash to the DRAM and erases the block in the Flash (840). Once the block of data is moved to the DRAM, the MCH can operate on this data block with the standard RDIMM access protocol (950). The CDC checks if any of the copied blocks are closed (960). If there are any closed blocks, the CDC writes the updated block from the DRAM memory to the Flash memory (970). If the CPU/MCH reopens (980) the closed block that is currently being written back into the Flash memory, the CDC stops the Flash write and erases the block being written (955). Otherwise, the CDC finishes the Flash write (990).

[0052] The dotted activity in Figure 9 is an independent/parallel activity of the CDC. When the CDC, at any time, receives the DRAM load command (935), it starts the Load DRAM w/RC (930), and initiates another branch of activity that includes 940 – 990. As it is shown in this figure, the CDC controls the DRAM memory to the Flash memory operation, such that the Flash is completely hidden from the MCH. The MCH accesses the memory module using the standard DDR protocol.

[0053] Figure 10 shows the definition of the block address field which includes the address bits that are not used by the JEDEC standard (A0 – A4) in accordance with certain embodiments described herein. In this figure, two sets of address bits AD6 to AD17 are allocated for the Block address.

[0054] For example, with the minimum block size of 256K bytes, 24 bit block address (two sets of AD6 to AD17) allows access to 4TB of Flash memory space. If the memory module has 1GB of DRAMs, it can hold 4K blocks of data in the DRAM memory.

[0055] The 4K blocks can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks can be configured. For example if the memory module is configured to be 2 ranks with 8 banks/rank, then each bank holds 250 blocks or 62K 1KB pages.

[0056] The time to copy a block of data from the Flash memory to the DRAM memory is around 400µs with 8 DMgr. Figure 10 shows the DMgr configuration to support this block of data transfer in accordance with certain embodiments described herein.

[0057] The time to copy a block of data from the DRAM memory to the Flash memory is about 22ms. The CDC manages the block write operation by queuing the blocks that are ready to be written back to the Flash memory. If any page in the block is reopened, then the CDC will stop the block write-back operation until all the pages in the block are closed, and then restart the write-back operation.

[0058] Since the write-back operation takes 22ms while any copying operation takes 400μs, to keep the read and write access time to be similar, the read to write ratio is 55 to 1. This means that, if the block usage time (i.e. the duration that block is being open) in the DRAM memory is average 22ms, then the block write-back operation is completely hidden from the MCH. If the block usage time is 11ms, then the CDC makes sure that, on average over 100ms, there are no more than 9 closed blocks in the queue to be written-back to the Flash memory. Otherwise, the memory module cannot support the standard DDR operation. Thus, the number of closed blocks in the queue to be written-back to the Flash memory varies with the average block usage time.

[0059] Max number of closed blocks to be written-back to Flash = ((#of blocks per bank)/(ratio of 'flash_block_write_time' to 'flash_read_time'))*((Block usage time)/('flash_block_write_time'))

[0060] This equation indicates that if the block usage time is the same as the 'flash_block_write_time', then each bank should not have more than 4 closed blocks, on average over 88ms (22ms * 4) that need to be written back to the Flash.

of block per bank = 250 (for 1GB DRAM memory)
The ratio of 'flash_block_write_time' to 'flash_read_time' = 55
Floor [250/55] = 4.

[0061] The above equation also indicates that the bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash.

[0062] Table 1 shows the maximum allowed closed blocks in the queue to be written back to the Flash memory.

[0063] Figure 12A schematically illustrates a conventional NVDIMM having a switch that connects to the MCH for normal operation. When power loss occurs, the switch

connects to the Flash for saving the DRAM contents to the Flash. When power restoration occurs, the switch connects to the Flash for retrieving the DRAM contents from the Flash.

[0064] Figure 12B schematically illustrates a conventional FDHDIMM. The MCH interfaces with DRAM. The interface is controlled by the FDHDIMM controller. Although the MCH interfaces with the DRAM, the DRAM access time is governed by the Flash access time. The DRAM is only used as a data buffer: data from the Flash memory is transferred to the DRAM at the Flash access speed, and buffered (collected) into the DRAM, which transfers the buffered data to the MCH at the rate of DRAM I/O speed. Since the Flash memory access speed is very slow (a few hundred microseconds), the average data throughput rate is limited by the Flash access speed.

[0065] Figure 12C schematically illustrates an example embodiment of a FDHDIMM in accordance with certain embodiments described herein. The CDC controller of this example embodiment receives the standard DDR commands from the MCH, and interprets, and produces commands to control the Flash memory and the DRAM memory. The CDC also produces a set of switch control signals for the data path router amongst DRAMs, Flash and MCH. The switch control signals are independently operated without any exclusivity.

[0066] Various embodiments of the present invention have been described above. Although this invention has been described with reference to these specific embodiments, the descriptions are intended to be illustrative of the invention and are not intended to be limiting. Various modifications and applications may occur to those skilled in the art without departing from the true spirit and scope of the invention.

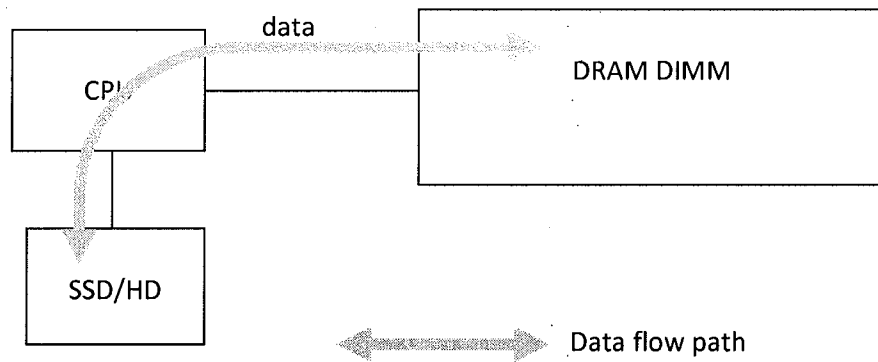


Figure 1: Conventional memory arrangement

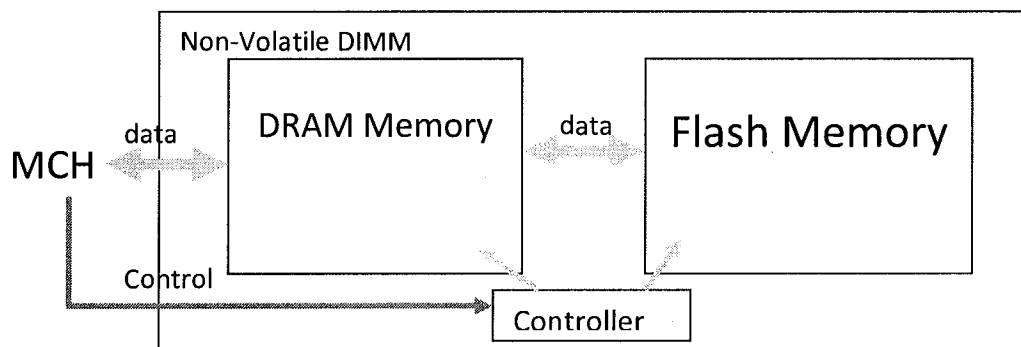
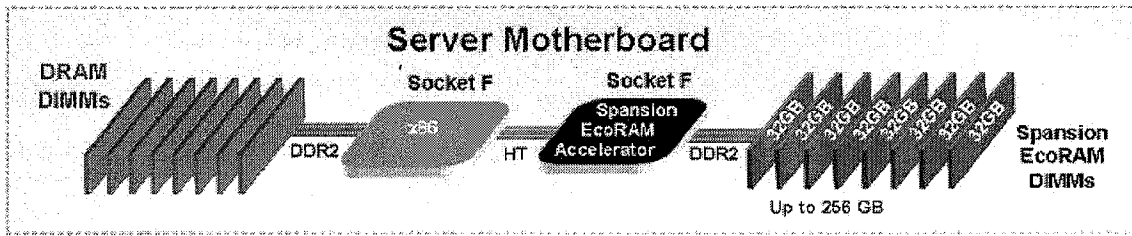


Figure 2: General Architecture of Non-Volatile DIMM (NVDIMM)
(density of DRAM memory = density of Flash memory)

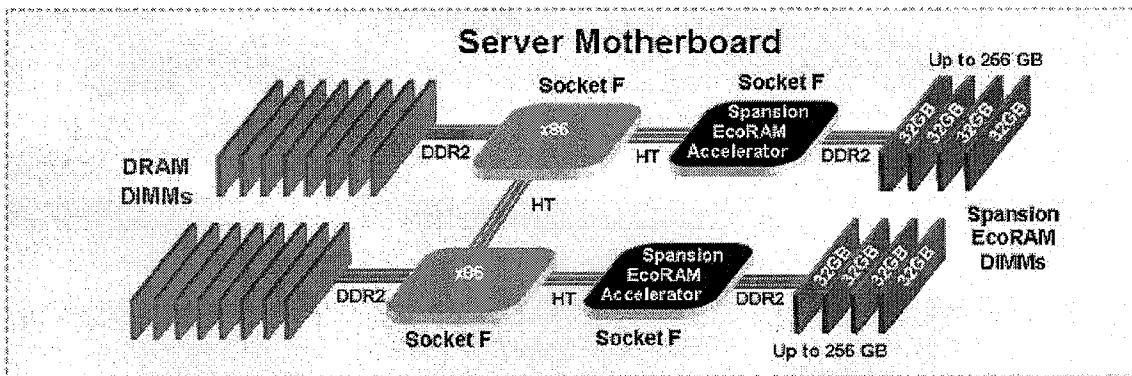
Spanion EcoRAM Configurations

256GB Spanion EcoRAM Solution – Single Accelerator



256GB Single Accelerator Spanion EcoRAM Solution

256 GB Spanion EcoRAM Solution – Dual Accelerator



256GB Dual Accelerator Spanion EcoRAM Solution

Figure 3: system level architecture of a currently available Flash DIMM product

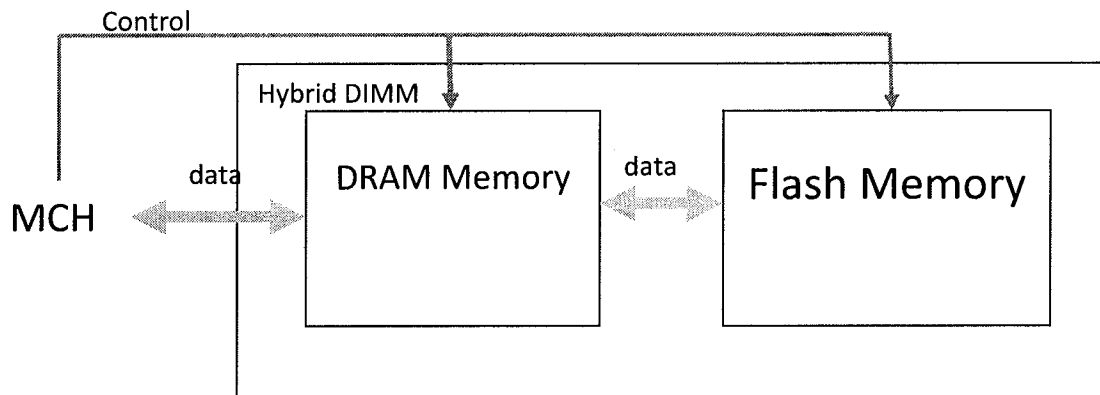


Figure 4: General Architecture of current Flash-DRAM Hybrid DIMM
(density of DRAM memory << density of Flash memory)

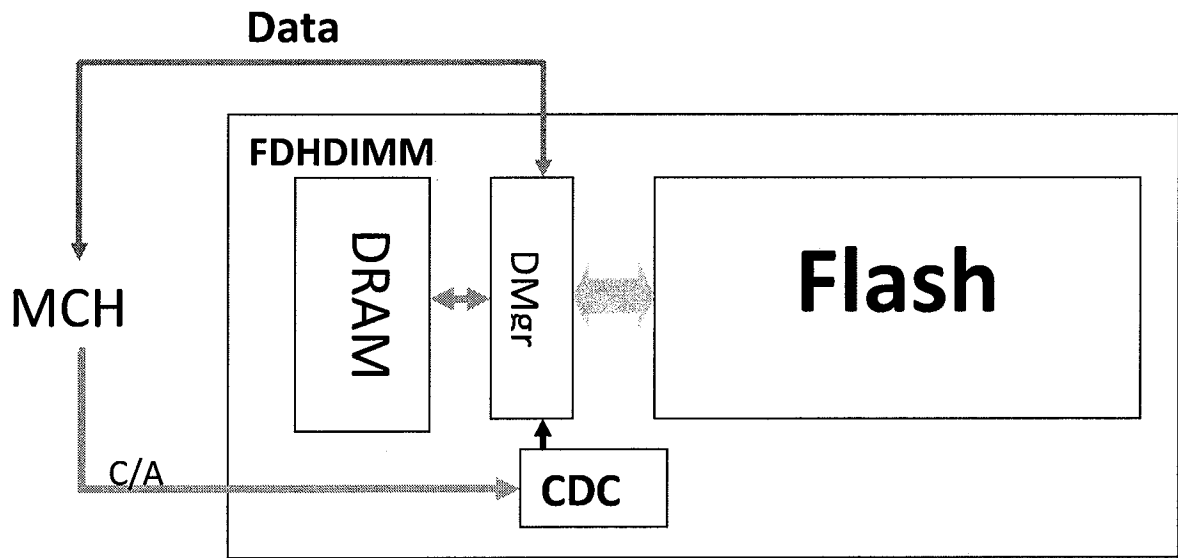


Figure 5: Embodiment that contains DRAMs, Flash, and Central DIMM Controller (CDC), Data Manager (DMgr)

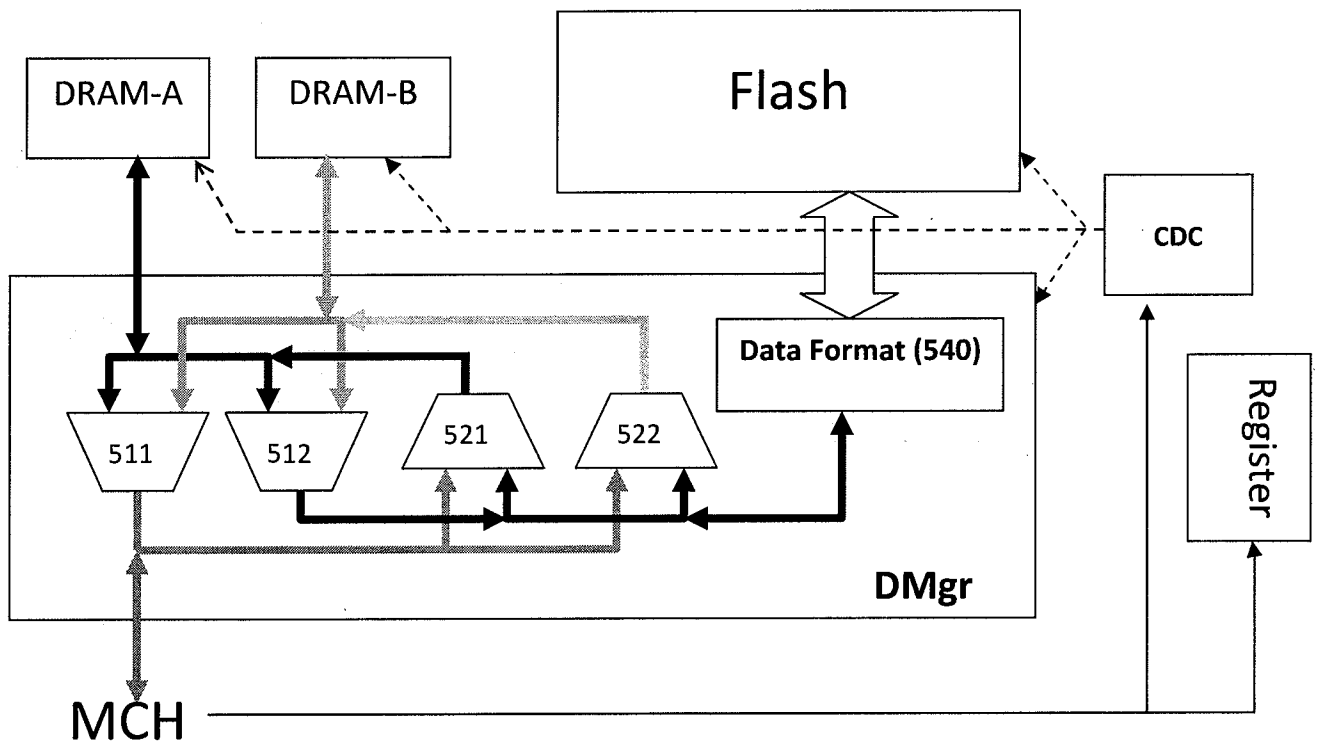


Figure 6: Connection of CDC to the register and DMgr, to MCH, DRAM, and Flash

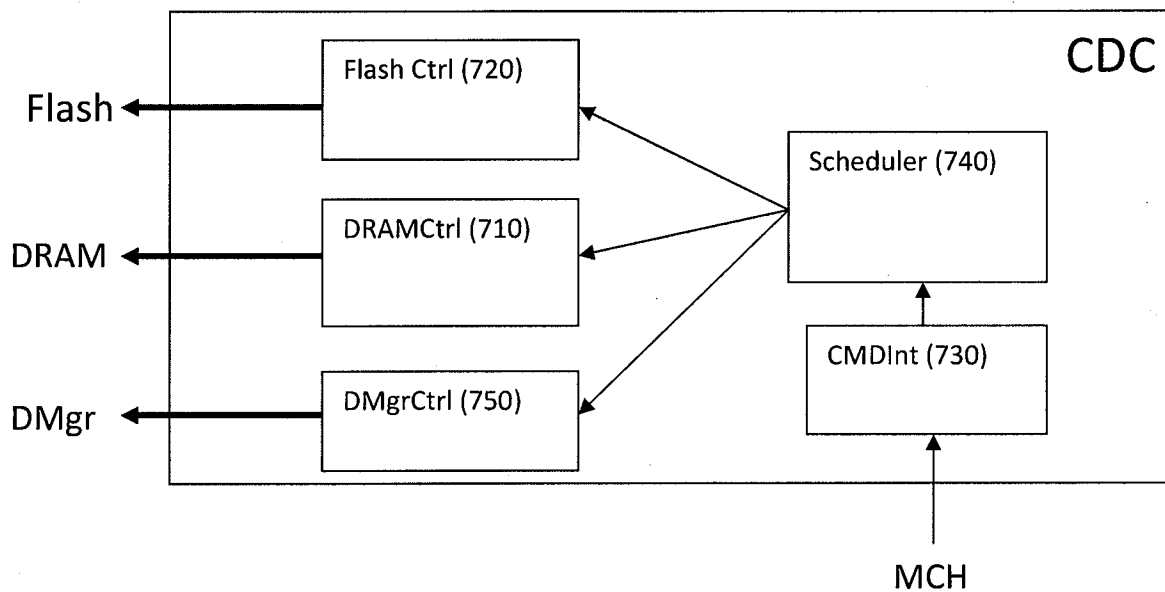


Figure 7: Functional blocks in CDC

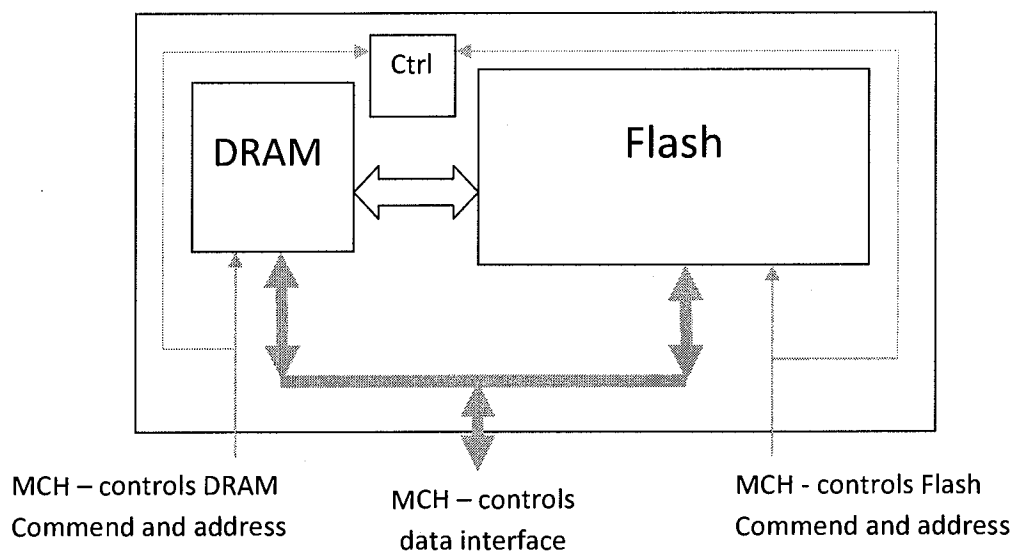


Figure 8A: General concept of the current Flash-DRAM Hybrid

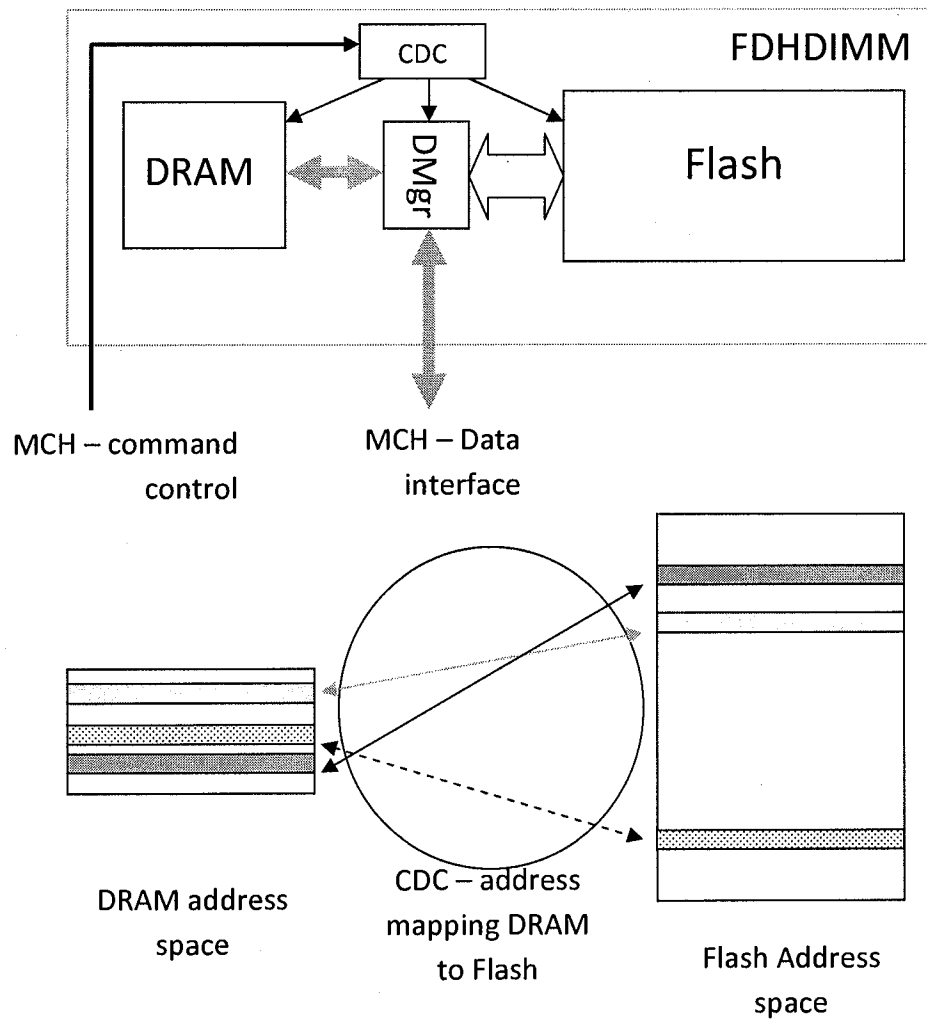


Figure 8B: Embodiment with single memory space with single control

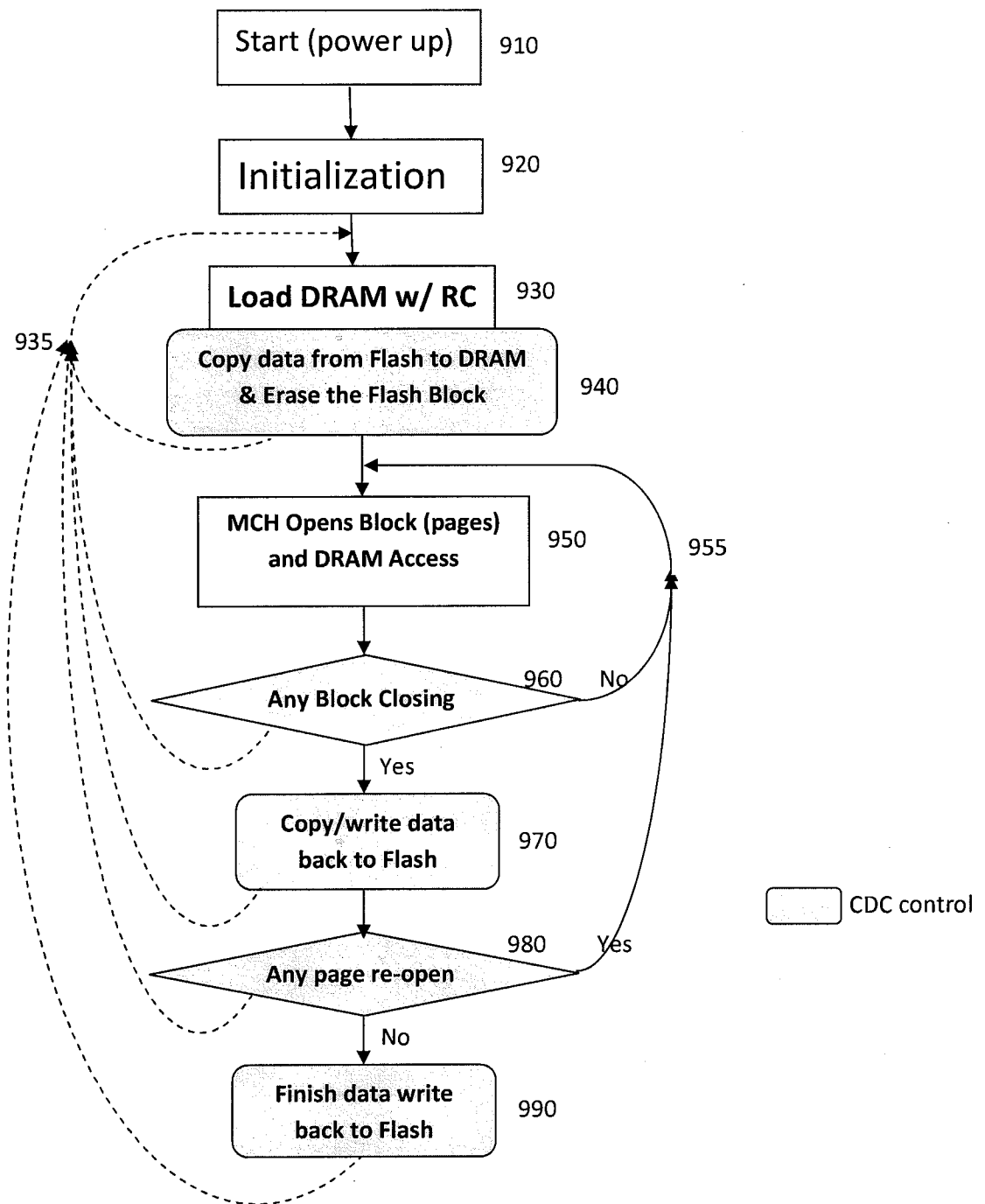


Figure 9: Operational state diagram of an embodiment

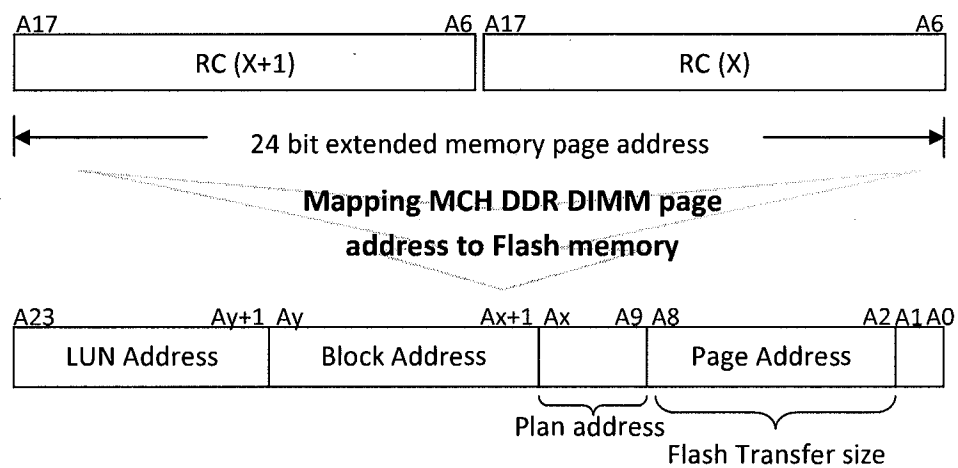


Figure 10: Mapping the DRAM page address to the Flash memory address field

DRAM density (GB)	# of blocks per bank	Flash wr-time to rd-time ratio	Avg block use time (sec)	Flash write time (sec)	Max allowed Closed Blk in queue to be written back to Flash
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

Figure 11: example maximum allowed closed blocks in the queue to be written into Flash memory

Figure 12A:

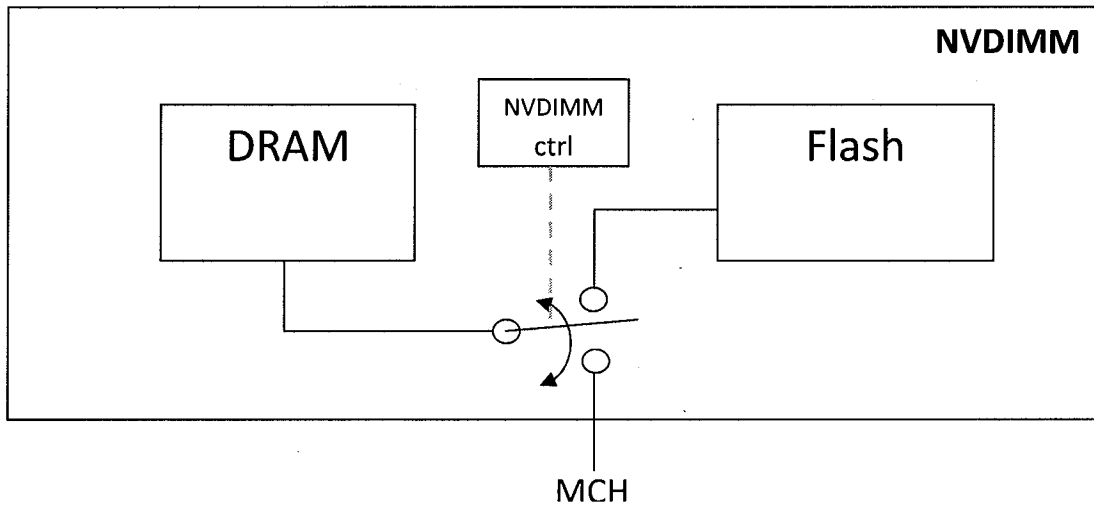


Figure 12B:

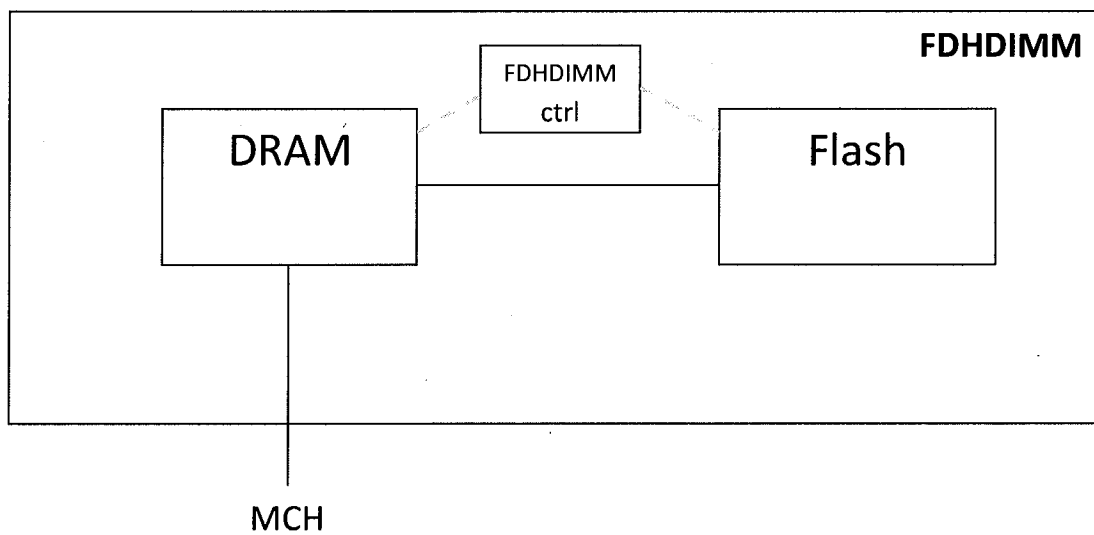
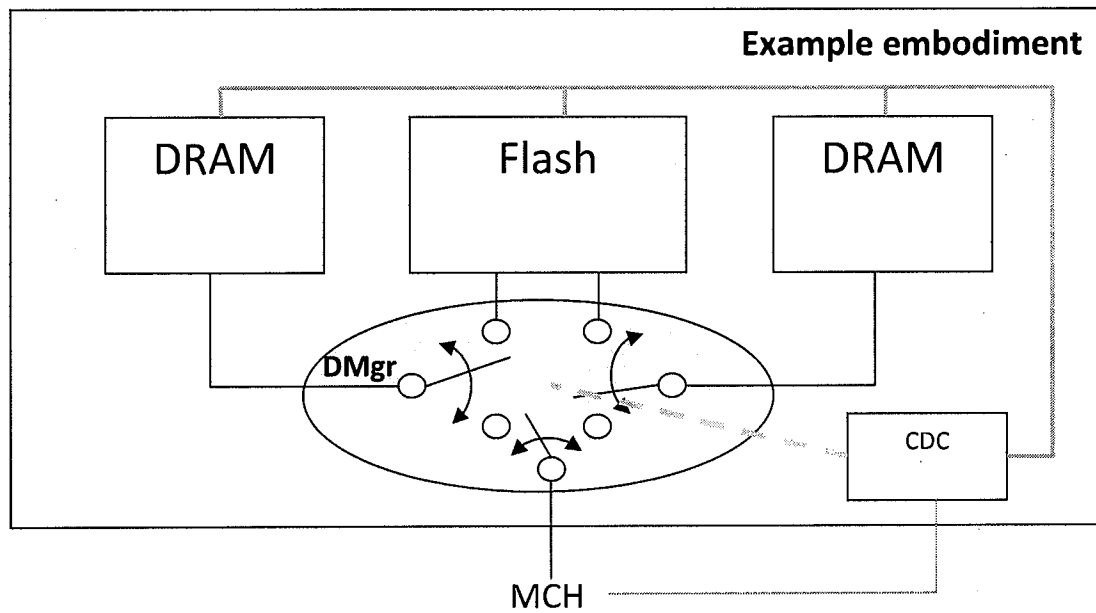


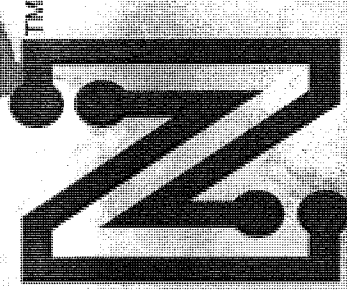
Figure 12C:



Appendix A

Hyper Vault

Multi-Terra Bytes of FSB Memory for HPC

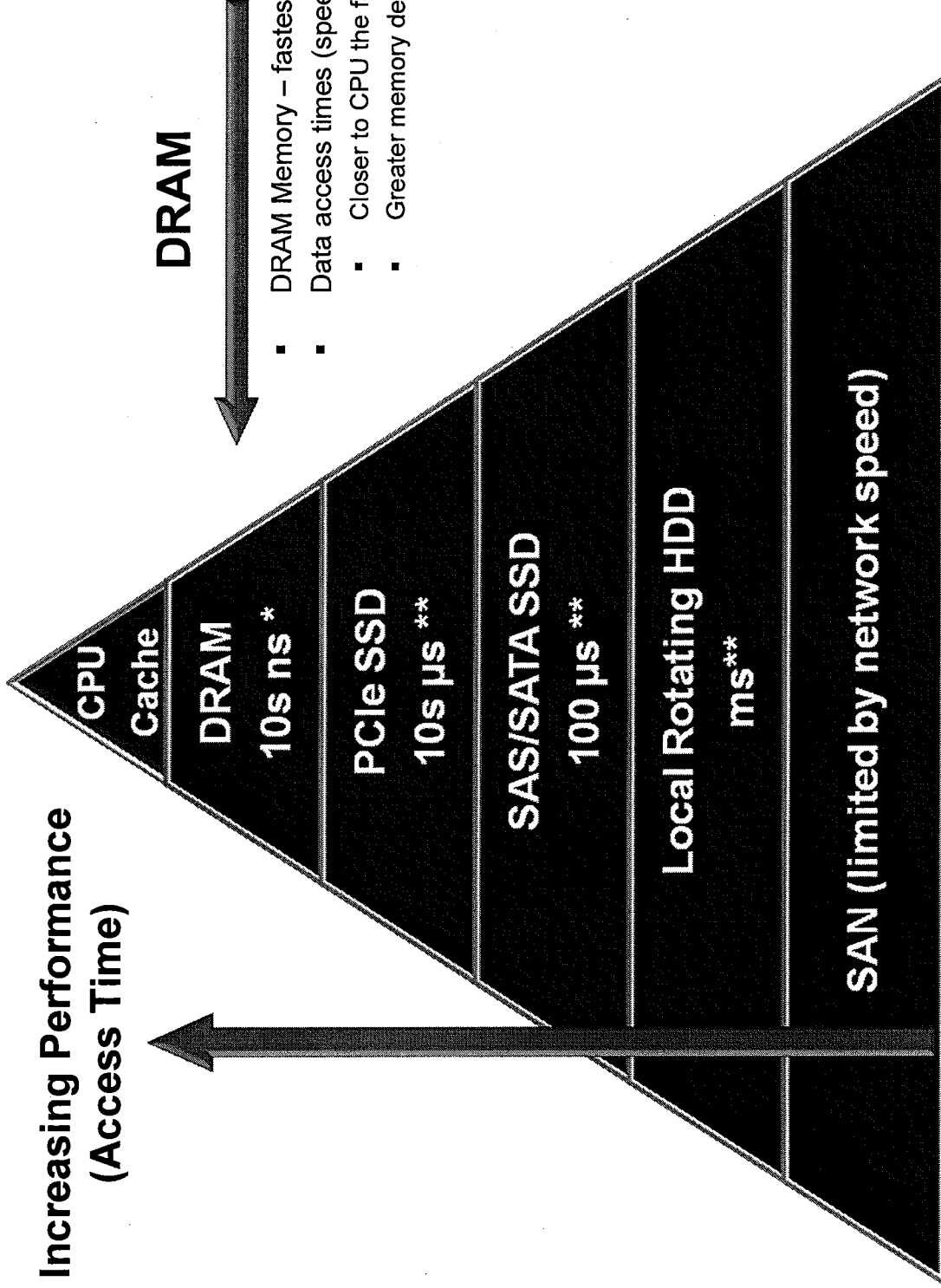


NETLIST

DRAM in Tiered Storage Hierarchy

SCALING THE CLOUD

Increasing Performance
(Access Time)



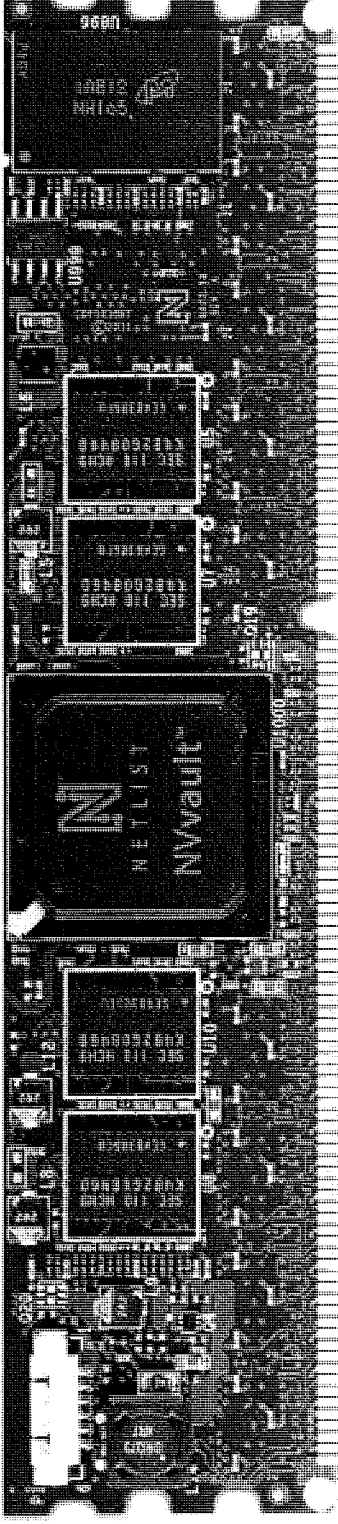
Notes:

[*] Based on Netlist calculations

[**] Based on product survey

NVvault DDR3 Product

SCALING THE CLOUD



Features

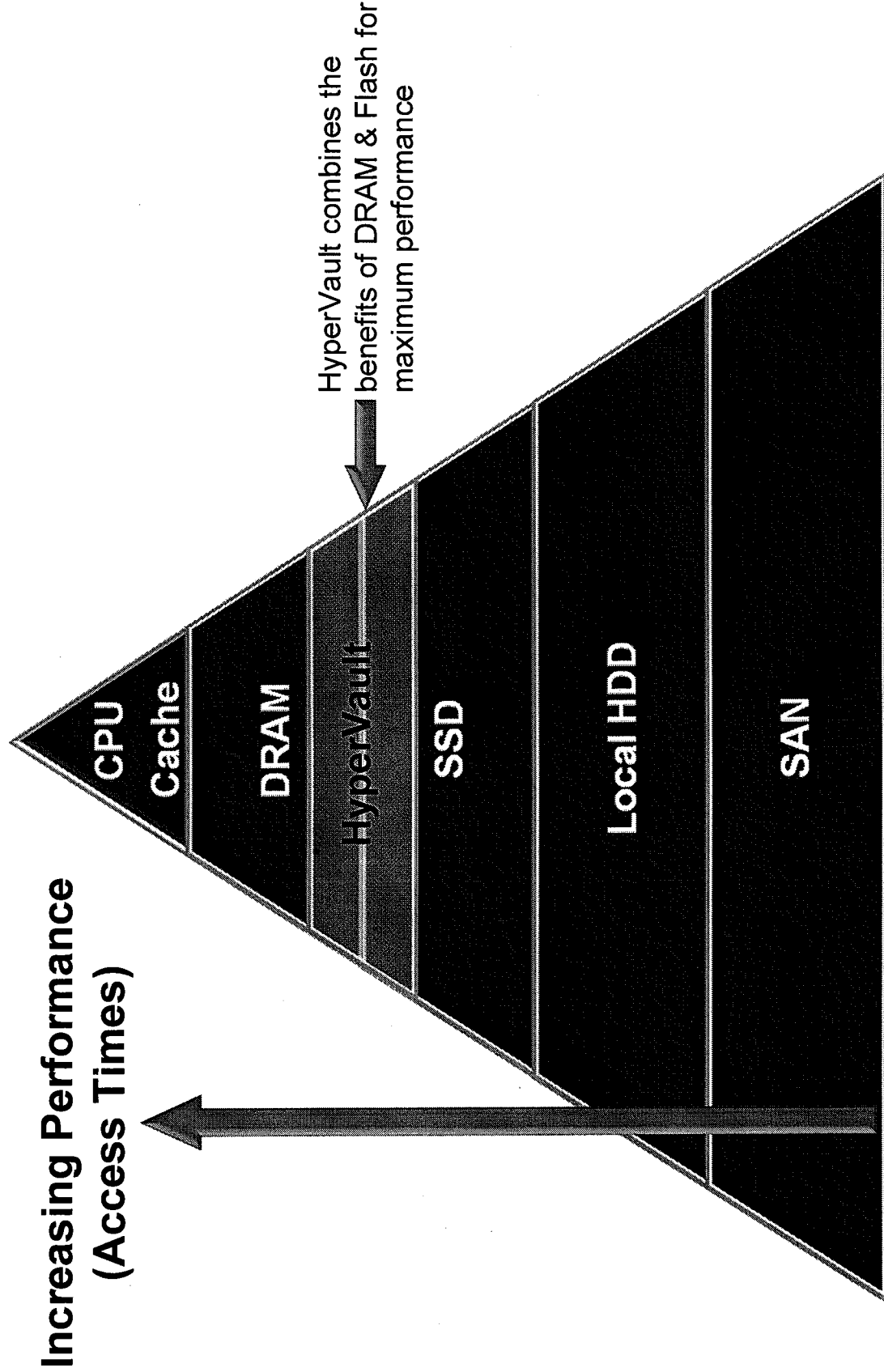
- 1333 MT/s throughput
- Fast R/W access time performance
- Fast backup/restore times (seconds)
- Standard DDR3 memory slot interface
- In-field programmability
- Programmable backup size
- Ultracapacitor for backup power
- 1GB/2GB capacity 7/11 (4GB/8GB Q4'11)

Benefits

- Accelerated write speeds for fast cache backup
- Low latency interface
- Plug 'n play hardware
- Eliminates batteries & reduces TCO
- No write life cycle limitations or performance degradation over time – writes to flash only upon power loss or system failure

HyperVault – Hybrid RDIMM Memory

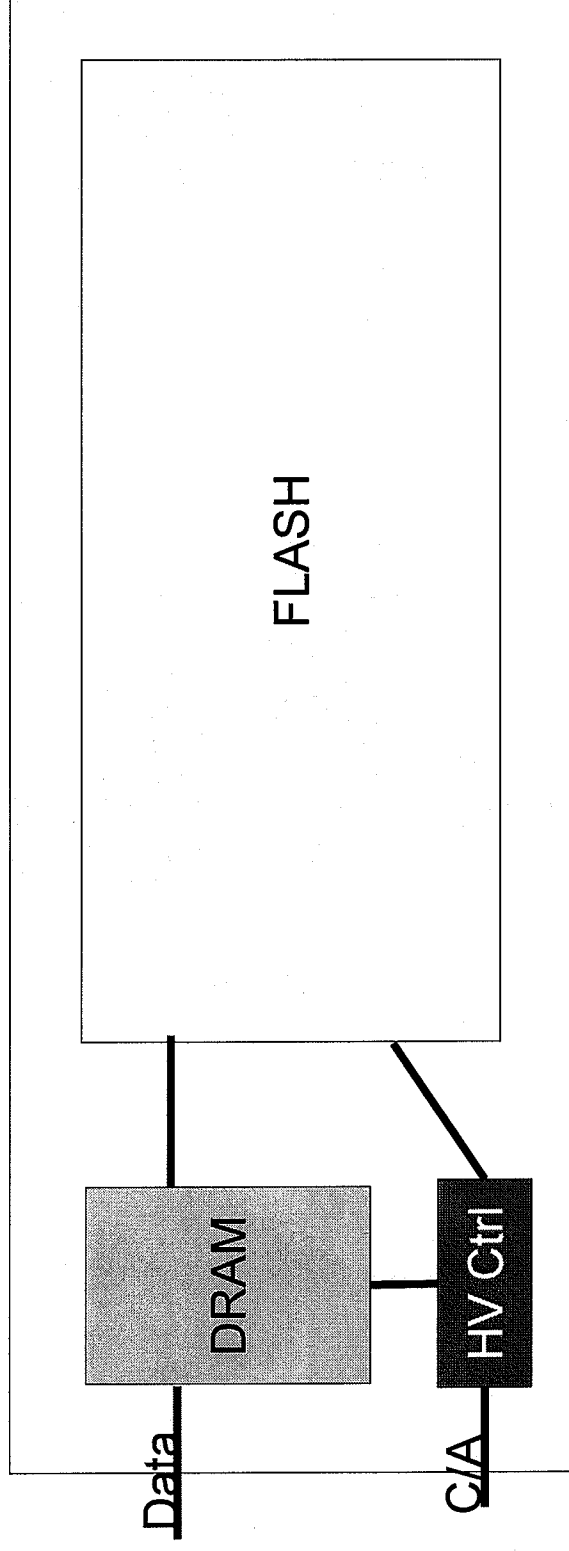
SCALING THE CLOUD



HyperVault – Hybrid RDIMM Memory

SCALING THE CLOUD

- What is HyperVault
 - Hybrid RDIMM with Flash and DRAM
 - Very High Density DIMM (1/4 to 1 TB)
 - Supports standard DDR DIMM interface



HyperVault – Hybrid RDIMM Memory

SCALING THE CLOUD

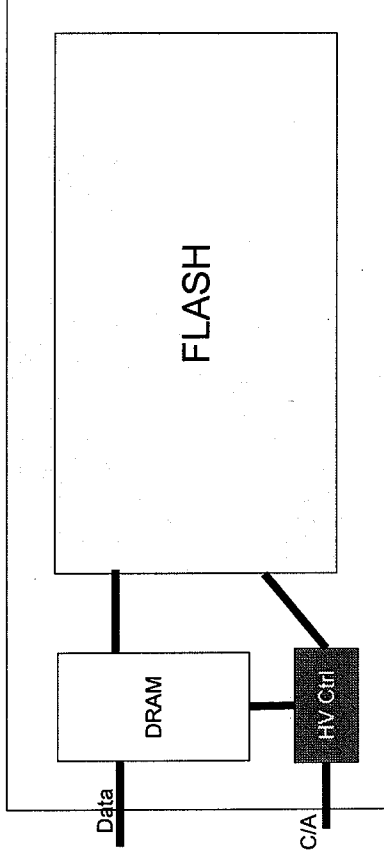
- What is HyperVault
 - No parity in Read Access vs. Write Access speed
 - Low power operation
 - Configurable page sizes
 - Configurable number of ranks per DIMM
 - Configurable number of banks per rank
 - Extended addressable memory space by
 - Concatenated RAS commands
 - Concatenated RAS and 1st CAS commands
 - Configurable NV memory space to support RAID operation
 - Automatic Data backup on the event of power loss
 - Automatic Data restoration as power comes back

HyperVault – Hybrid RDIMM Memory

SCALING THE CLOUD

- What is HV Ctrl

- HyperVault Controller
- Data path manager
 - Between MCH and DRAM
 - Between DRAM and FLASH
- Data format manipulator
 - Reformats the Data (by burst-size and by page-size)
 - Reformats the Data (by the number of banks per rank, by the number of ranks)

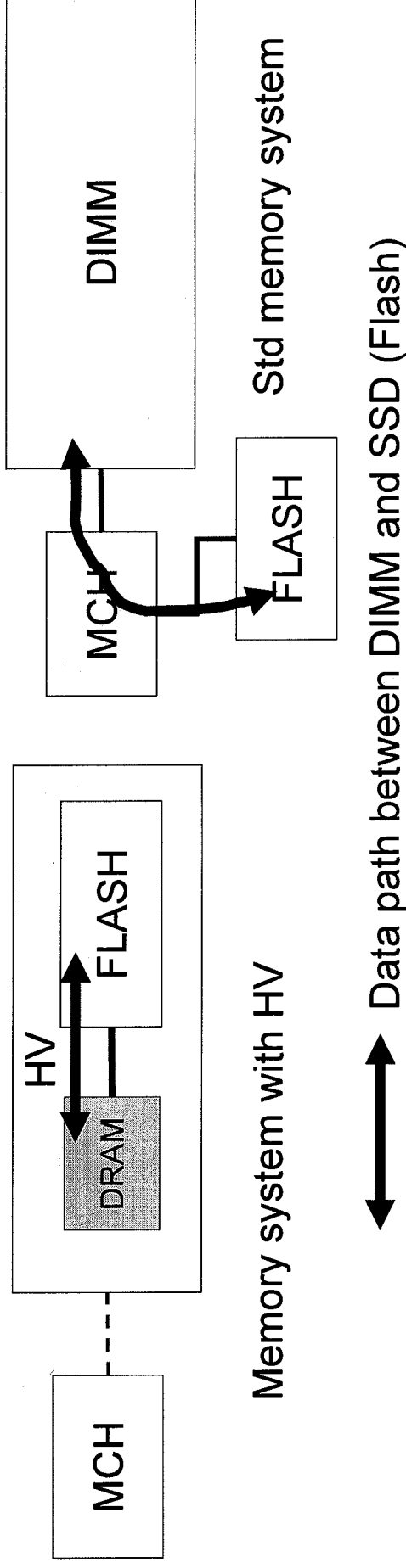


- Memory address manager
 - Manages the virtual FLASH memory address space, and the physical DRAM address space
 - Address Mapping between DRAM and FLASH
- Error manager (correction and recovery)
- Flow rate controller

HyperVault – Hybrid RDIMM Memory

SCALING THE CLOUD

- Why HyperVault
 - Integration of SSD into the FSB memory
 - Improves CPU usage by minimizing the data transfer through CPU between SSD and FSB memory
 - Improves the Data transfer time between FLASH and DRAM
 - Expandable Memory concept
 - Field / User configurable SSD density based on the target application
 - Adjustable SSD (Flash) over DRAM ratio for optimum performance
 - Automatic power loss protection



Appendix B



**SpanSion® EcoRAM™ Solution
IOZone Solution Performance White Paper**

Version 0.4

For

The SpanSion EcoRAM Solution Configurations

Copyright © 2009 SPANSION® INC.

This document contains information proprietary to SPANSION® INC.

Use or disclosure without written permission
of an officer of SPANSION® INC. is expressly forbidden.

Overview

The Spansion[®] EcoRAM[™] Solution consists of one or two Spansion EcoRAM Accelerators, 8 or 16 Spansion EcoRAM DIMMs (16GB or 32GB each) and associated driver software working together to provide a memory-expansion solution for AMD “Socket F” server motherboards.

Functionally, the Spansion EcoRAM solution initially appears as a POSIX-compliant device and operates like an ultra-fast SSD (Solid State Disk). In addition to the operating mode as an ultra-fast SSD, the Spansion EcoRAM solution also has a mode of operating as extremely high density main memory expansion.

In Expanded Memory Mode (EMM), Spansion EcoRAM memory supports read operations at cache-line granularity from any CPU in the system: data is returned in response to a CPU load instruction. The CPU accesses Spansion EcoRAM memory directly without first requiring a copy to a DRAM buffer and the initial access latency is a mere few hundred nanoseconds regardless of the size of data accessed. Like DRAM, Spansion EcoRAM memory supports ECC at cache-line granularity providing DRAM-class reliability levels.

Due to the extremely low read access latency of the Spansion EcoRAM's Expanded Memory Mode, there is no need to copy any portion of the data-set into DRAM to achieve higher performance. Low-latency, direct read accesses to Spansion EcoRAM occur as a cache line fill (64 Bytes). This capability is well suited to small, random read accesses on extremely large data sets where performance would traditionally require an equally large amount of DRAM to cache the IO accesses. Application workload throughput increases as much as two orders of magnitude with Spansion EcoRAM memory vs. Traditional IO, depending on the application workload's dataset size to DRAM ratio, memory access locality of reference and memory access granularity.

Testing Summary

Platform Configuration

- Server: Super Micro H8QME+2, 16GB DDR2 DRAM
- Operating System: CentOS 5.0 (2.6.18.3 kernel)

Spanion EcoRAM Configurations

256GB Spanion EcoRAM Solution – Single Accelerator

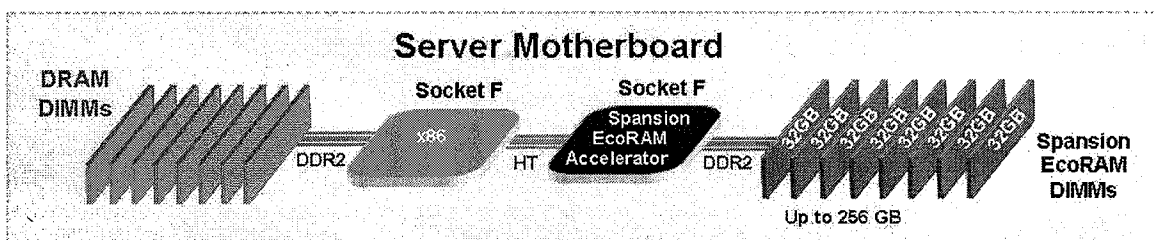


Figure 1: 256GB Single Accelerator Spanion EcoRAM Solution

256 GB Spanion EcoRAM Solution – Dual Accelerator

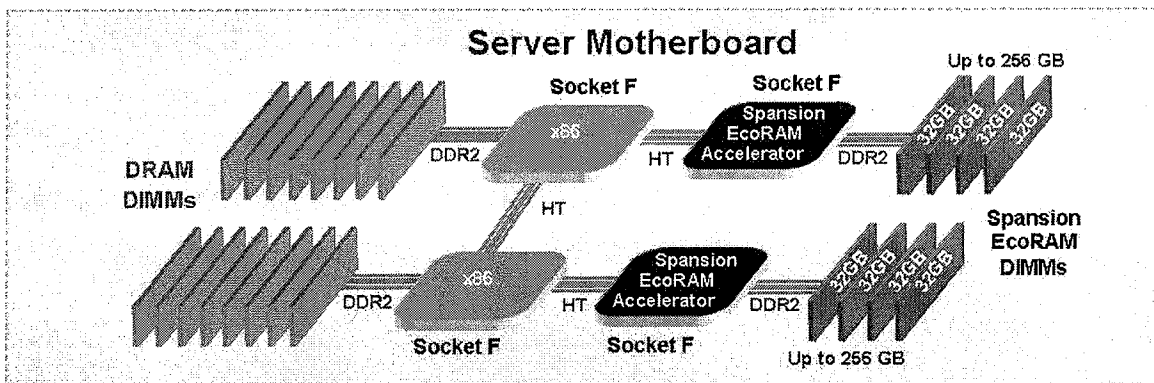


Figure 2: 256GB Dual Accelerator Spanion EcoRAM Solution

512 GB SpanSion EcoRAM Solution – Dual Accelerator

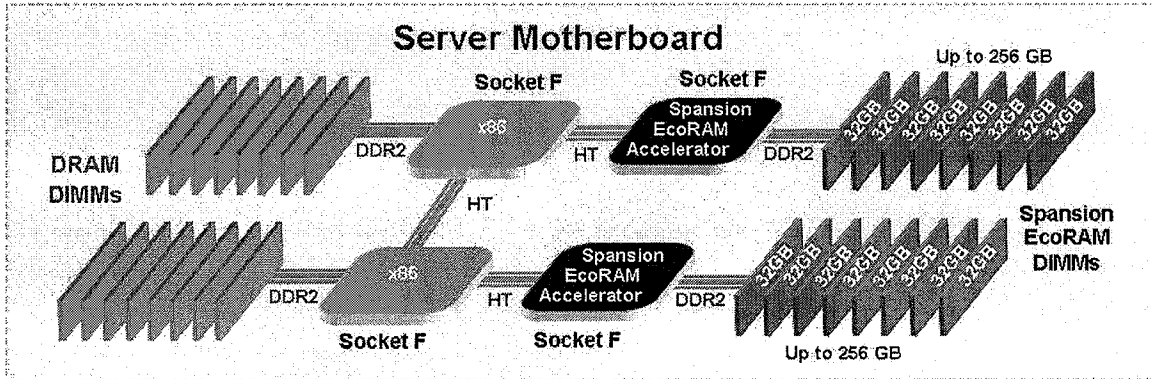


Figure 3: 512GB Dual Accelerator SpanSion EcoRAM Solution

Test Tools

The performance analysis tool used in this evaluation is the open-source file system benchmark, IOZone v3.291 compiled for 64bit. In all tests, the number of threads used equals the number of cores available.

For write testing, two instances of IOZone were loaded and NUMA controlled.

Multiple passes of each test were run and averaged maintaining a Standard Deviation of <2%. PCIe-Based SSD results are from obtained from published product literature and have not been independently verified.

Test Details

File Sizes are extra large at 4GB because smaller files may not provide realistic performance. All Tests run each thread on a unique 4GB File.

The 256GB SpanSion EcoRAM configurations use a single 2382 2.60GHz Opteron CPU, a single accelerator and 8 EcoRAM DIMMs. The 512GB configurations use dual 2382 2.60 GHz Opteron CPUs, dual accelerators and 16 EcoRAM DIMMs.

IOZone Command Line for Expanded Memory Mode Tests on Single SpanSion EcoRAM Accelerator

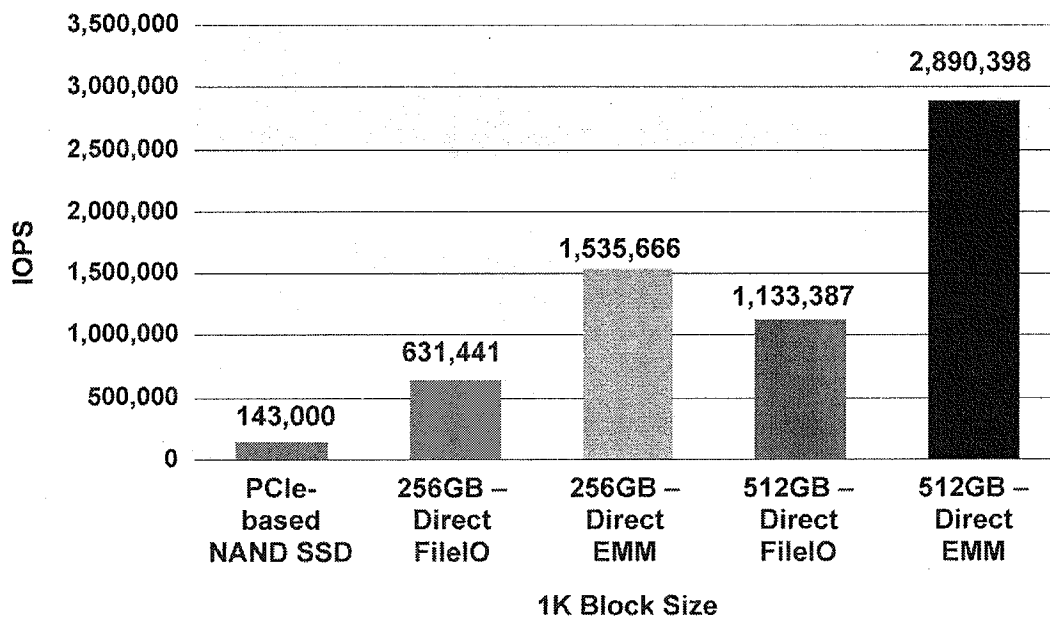
```
iozone -P 0 -L 64 -t 4 -F /uvfs1/file0 /uvfs1/file1 /uvfs1/file2 /uvfs1/file3 -R -i 1 -w -s 4g -r $BS -B
```

IOZone Command Line for FileIO Tests on Single SpanSion EcoRAM Accelerator

```
iozone -P 0 -L 64 -t 4 -F /uvfs1/file0 /uvfs1/file1 /uvfs1/file2 /uvfs1/file3 -R -i 1 -w -s 4g -r $BS
```

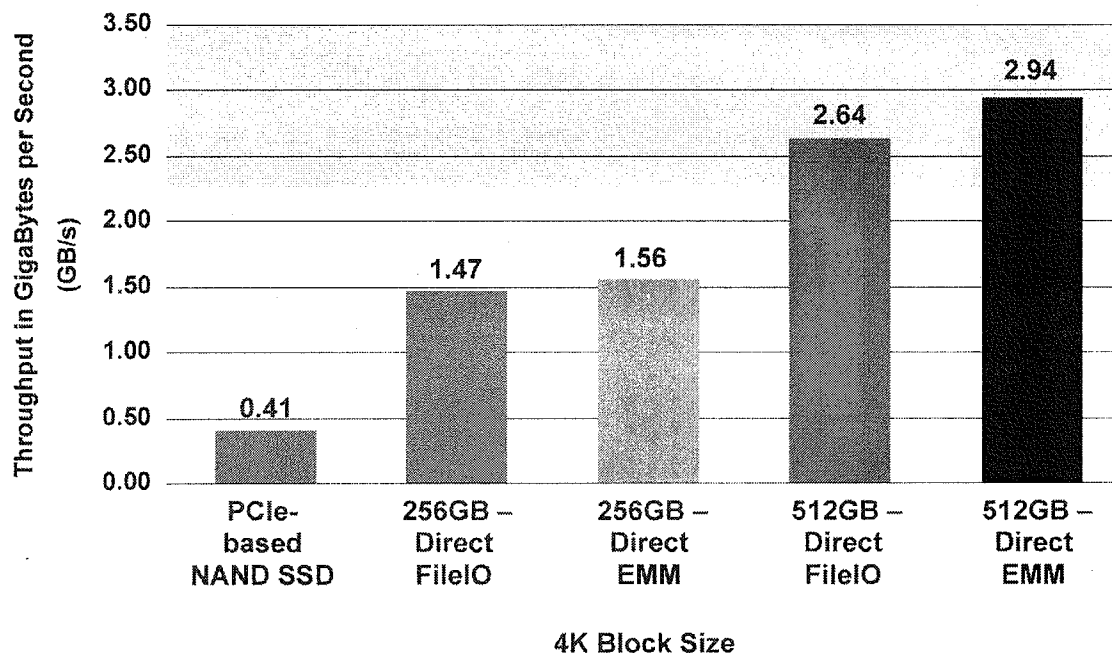
Results

IOZone Small Block Random Read IOPS – 1KB Block Size

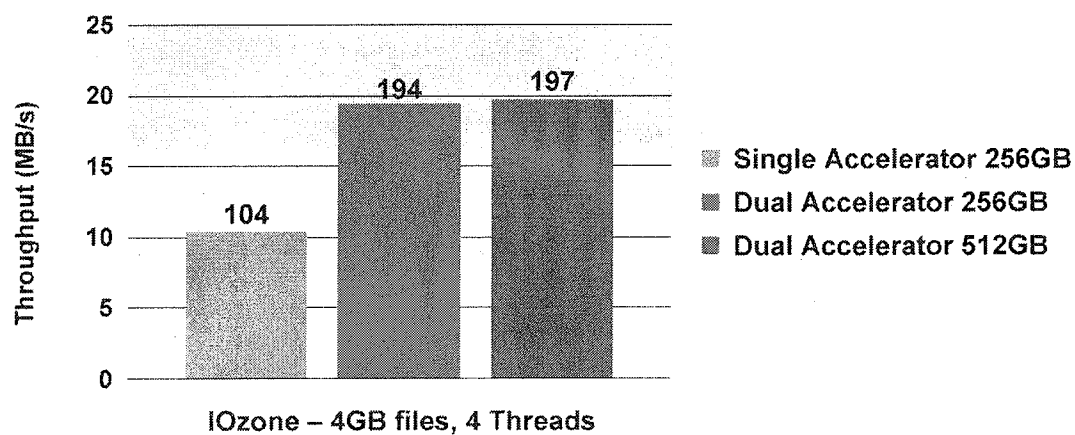


The extreme value of EcoRAM is in small block, random read accesses, particularly when using EcoRAM's EMM (Expanded Memory Mode).

IOZone Small Block Random Read Throughput Performance



IOZone Maximum Sustained Write Throughput – Direct Mode



Conclusions

This paper publishes the raw random read and sequential write throughput of a server equipped with Spansion EcoRAM using the IOZone Benchmarking Suite. Spansion EcoRAM's FileIO mode provides outstanding throughput for traditional IO access methods achieving significantly higher bandwidth and IOPS than the leading PCIe-Based NAND SSDs. For very small read accesses (as small as 64 bytes), the direct access capability of Spansion EcoRAM's Expanded Memory Mode maintains superior throughput at much smaller access sizes, due to the very low device latency (100-200ns.) and the elimination of buffer copies and system calls when accessing data. These performance capabilities, coupled with Spansion EcoRAM's ultra-high density and DRAM-class device reliability enables the expansion of a server's addressable memory space by incorporating Spansion EcoRAM for read-centric workloads.

Revision History

Revision	Date	Notes
.10	03/09/2009	First draft released
.20	07/15/2009	Results Update
.30	07/22/2009	Formatting
.40	10/08/2009	Results Update

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

HIGH DENSITY DIMMS

First Named Inventor/Applicant Name:

Hyun Lee

Filer:

Bruce S. Itchkawitz/Jessica Noda

Attorney Docket Number:

NETL.096PR

Filed as Large Entity

Provisional Filing Fees

Description

Fee Code

Quantity

Amount

**Sub-Total in
USD(\$)**

Basic Filing:

Provisional application filing

1005

1

220

220

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Post-Allowance-and-Post-Issuance:

Extension-of-Time:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				220

Electronic Acknowledgement Receipt

EFS ID:	10622596
Application Number:	61512871
International Application Number:	
Confirmation Number:	8113
Title of Invention:	HIGH DENSITY DIMMS
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Valerie Jones
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.096PR
Receipt Date:	28-JUL-2011
Filing Date:	
Time Stamp:	20:12:49
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 220
RAM confirmation Number	6325
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	ads.pdf	240305	no	4
			86196679691278421b532762c787e1a5f84c81db		
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
2	Specification	specification.pdf	722158	no	13
			840eae8ab61f3fb6ba1ccde8d448cc638b6525a8		
Warnings:					
Information:					
3	Drawings-only black and white line drawings	drawings.pdf	320706	no	10
			7bea41a4ca8883a2c23361de330a9b41674122d9		
Warnings:					
Information:					
4	Appendix to the Specification	appendix_a.pdf	2523426	no	9
			3474b0bf6e2acc29cee072df9471de4342ff9a0c		
Warnings:					
Information:					
5	Appendix to the Specification	appendix_b.pdf	570775	no	8
			807409f3e16244857ca06f9fbe0e18434ed6fdb5		
Warnings:					
Information:					
6	Fee Worksheet (SB06)	fee-info.pdf	29458	no	2
			f73572d8b63725e793e38bf764208dca3dd906ad		
Warnings:					
Information:					
Total Files Size (in bytes):			4406828		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/512,871	07/28/2011		220	NETL.096PR		

CONFIRMATION NO. 8113

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000049296462

Date Mailed: 08/17/2011

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;

Power of Attorney:

Bruce Itchkawitz--47677

If Required, Foreign Filing License Granted: 08/12/2011

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/512,871**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

HIGH DENSITY DIMMS

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international

patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and

Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

NON-VOLATILE MEMORY MODULE**BACKGROUND OF THE INVENTION**

[0001] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (*e.g.*, a server system or a personal computer) and are accessed by the processor of the computer system to provide volatile memory to the computer system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Figure 1 is a block diagram of an example 512MB DDR2 (PC2-3200) memory module in accordance with certain embodiments described herein.

[0003] Figure 2 is a block diagram of an example memory module with ECC (error-correcting code) having a first memory bank with nine volatile memory elements and a second memory bank with five non-volatile memory elements in accordance with certain embodiments described herein.

[0004] Figure 3 is a block diagram of an example memory module with ECC and having a logic element and microcontroller unit integrated into a single device.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0005] Figure 1 is a block diagram of an example memory module 10 compatible with certain embodiments described herein. The memory module 10 comprises a printed-circuit card (PCB) 20, a first memory bank 30 on the PCB 20, and a second memory bank 40 on the PCB 20. The first memory bank 30 comprises a first plurality of volatile memory elements 32 and the second memory bank 40 comprises a second plurality of non-volatile memory elements 42. Certain embodiments described herein advantageously provide non-volatile storage via the second memory bank 40 in addition to high-performance (*e.g.*, high speed) storage via the first memory bank 30. Certain embodiments described herein utilize the second memory bank 40 as a flash “mirror” to provide back-up of the volatile first memory bank 30 in the event of power interruptions.

[0006] In certain embodiments, the memory module 10 is configured to be in electrical communication with a host system. Examples of host systems include, but are not

limited to, blade servers, 1U servers, personal computers (PCs), and other applications in which space is constrained or limited.

[0007] In certain embodiments, the memory module 10 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 10 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.

[0008] In certain embodiments, the PCB 20 has an industry-standard form factor. For example, the PCB 10 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 10 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, VLP mini-DIMM, and micro-DIMM are also compatible with certain embodiments described herein.

[0009] The PCB 20 comprises an interface 22 that is configured to be in electrical communication with the host system (not shown). For example, the interface 22 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 22 provides a conduit for power voltage as well as data and address signals between the memory module 10 and the host system. For example, the interface 22 can comprise a standard 240-pin DDR2 edge connector.

[0010] In certain embodiments, the first plurality of volatile memory elements 32 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 32 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of Figure 1, the first memory bank 30 comprises eight 64Mx8 DDR2 SDRAM elements 32. In addition, volatile memory elements 32 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements 32 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-

grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0011] In certain embodiments, the second plurality of non-volatile memory elements 42 comprises two or more flash memory elements. Types of flash memory elements 42 compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, and multi-level cell (MLC). For example, in the block diagram of Figure 1, the second memory bank 40 comprises 512 MB of flash memory organized as four 128Mbx8 NAND flash memory elements 42. In addition, non-volatile memory elements 42 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Non-volatile memory elements 42 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0012] Figure 2 is a block diagram of an example memory module 10 with ECC (error-correcting code) having a first memory bank 30 with nine volatile memory elements 32 and a second memory bank 40 with five non-volatile memory elements 42 in accordance with certain embodiments described herein. The additional memory element 32 of the first memory bank 30 and the additional memory element 42 of the second memory bank 40 provide the ECC capability.

[0013] As schematically illustrated by Figure 1, in certain embodiments, the memory module 10 further comprises a voltage monitor circuit 50, a microcontroller unit (MCU) 60, a logic element 70, and a capacitor module 80. In certain embodiments, the MCU 60 provides memory management for the second memory bank 40 and controls data transfer between the first memory bank 30 and the second memory bank 40. The MCU 60 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. In certain embodiments, the capacitor module 80 comprises a step-up transformer 82, a step-down transformer 84, and a capacitor bank 86.

[0014] In certain embodiments, the logic element 70 comprises a field-programmable gate array (FPGA). Other types of logic elements 70 compatible with certain

embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 70 is a custom device. In certain embodiments, the logic element 70 comprises various discrete electrical elements, while in certain other embodiments, the logic element 70 comprises one or more integrated circuits. Figure 3 is a block diagram of an example memory module 10 having a microcontroller unit 60 and logic element 70 integrated into a single device.

[0015] Under normal operation in certain embodiments, the host system accesses the first memory bank 30, rather than the second memory bank 40, because the volatile memory elements 32 have superior read/write characteristics. The voltage monitor circuit 50 monitors the power supplied by the host system via the interface 22. Upon detecting a low power (*e.g.*, indicative of a powering down of the host system), the voltage monitor circuit 50 sends a signal to the MCU 60 which responds by executing a write cycle on the second memory bank 40. During this write cycle, data is read from the first memory bank 30 and is transferred to the second memory bank 40 via the MCU 60.

[0016] As schematically illustrated by Figure 1, the logic element 70 of certain embodiments is in electrical communication with the second memory bank 40 and the MCU 60. The logic element 70 can provide signal level translation between the volatile memory elements 32 (*e.g.*, 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 42 (*e.g.*, 3V TTL for NAND flash memory elements). In certain embodiments, the logic element 70 is also programmed to perform address decoding for the first memory bank 30 and the second memory bank 40.

[0017] During normal operation in certain embodiments, the step-up transformer 82 keeps the capacitor bank 86 charged at a peak value. In certain embodiments, the step-down transformer 84 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (*e.g.*, 1.8V to the volatile DRAM elements 32 and 3.0V to the non-volatile flash memory elements 42) during power down. In certain embodiments, the memory module 10 further comprises switch 90 (*e.g.*, FET switch) that switches between the charged capacitor bank 86 and system power received via the interface 22. The switch 90 of

certain embodiments advantageously ensures that the volatile memory elements 32 are powered long enough for the data to be stored in the non-volatile memory elements 42, and then switches over to the system power.

[0018] When normal operation resumes, data is transferred from the non-volatile memory bank 40 to the volatile memory bank 30 via the MCU 60. The host system can then resume accessing the volatile memory bank 30 of the memory module 10. In certain embodiments, the transfer of data from the volatile memory bank 30 to the non-volatile memory bank 40, or from the non-volatile memory bank 40 to the volatile memory bank 30, takes less than one minute per GB.

[0019] Various embodiments of the present invention have been described above. Although this invention has been described with reference to these specific embodiments, the descriptions are intended to be illustrative of the invention and are not intended to be limiting. Various modifications and applications may occur to those skilled in the art without departing from the true spirit and scope of the invention.

3051314
102506

Figure 1:

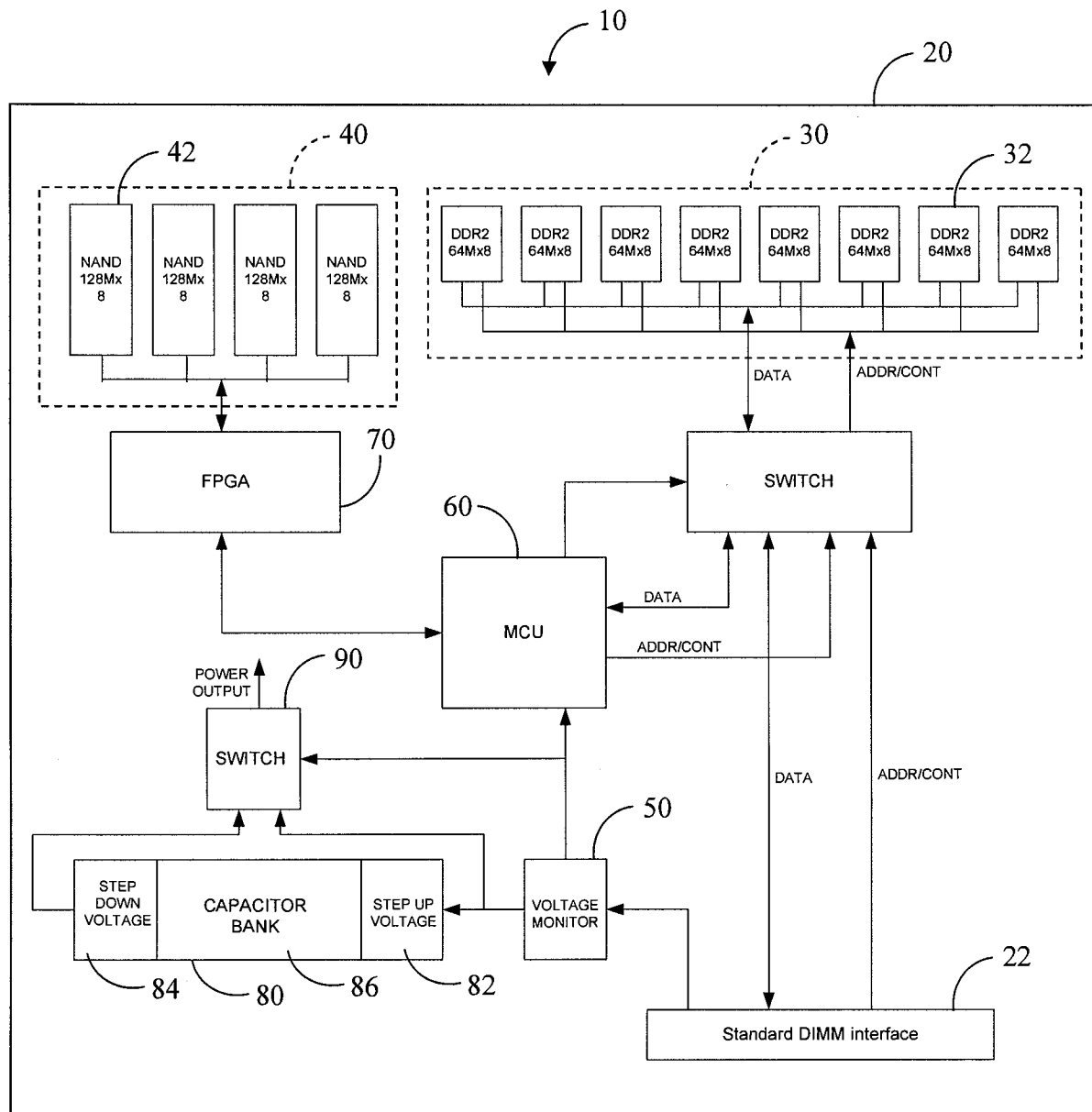


Figure 2:

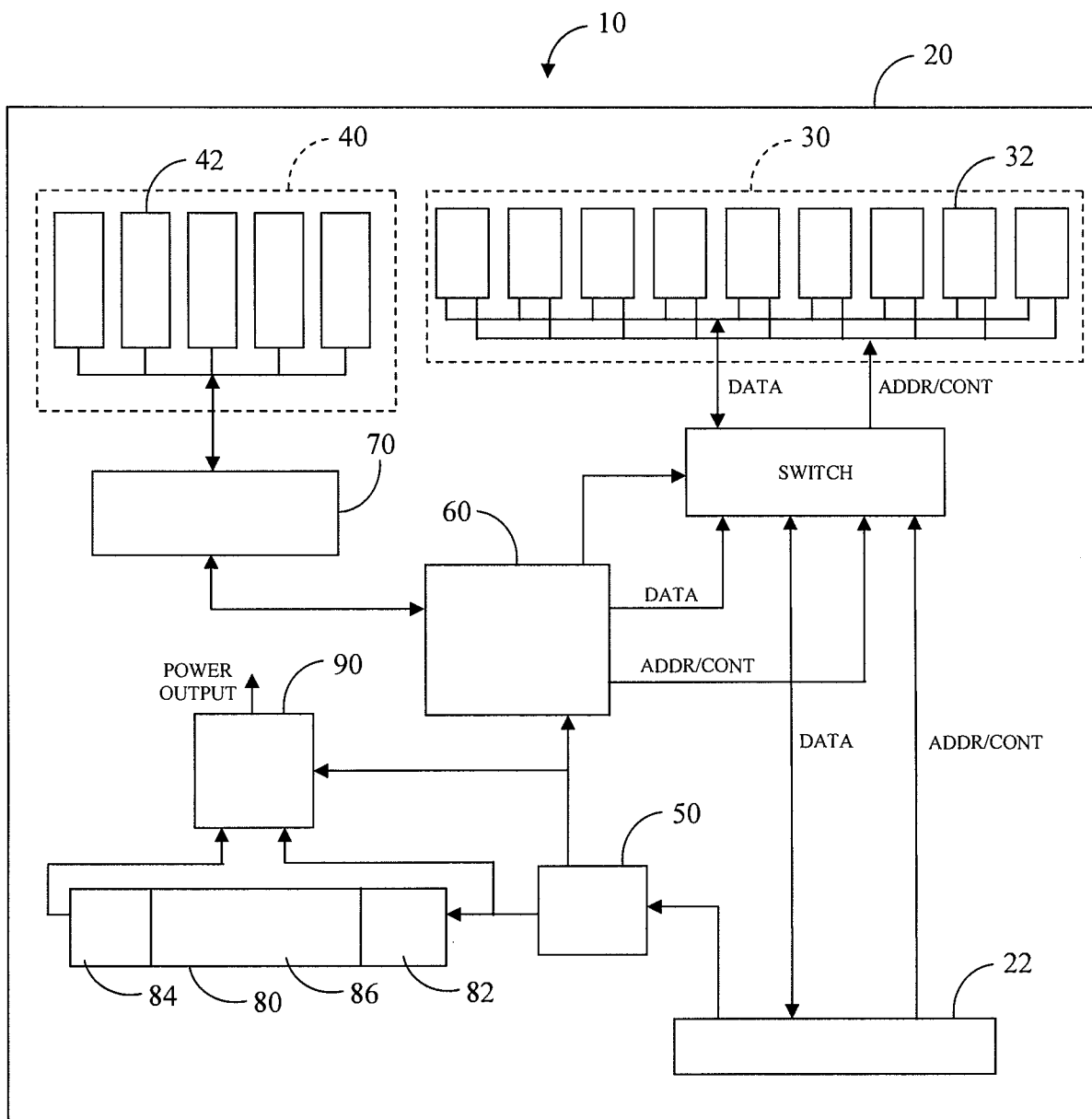
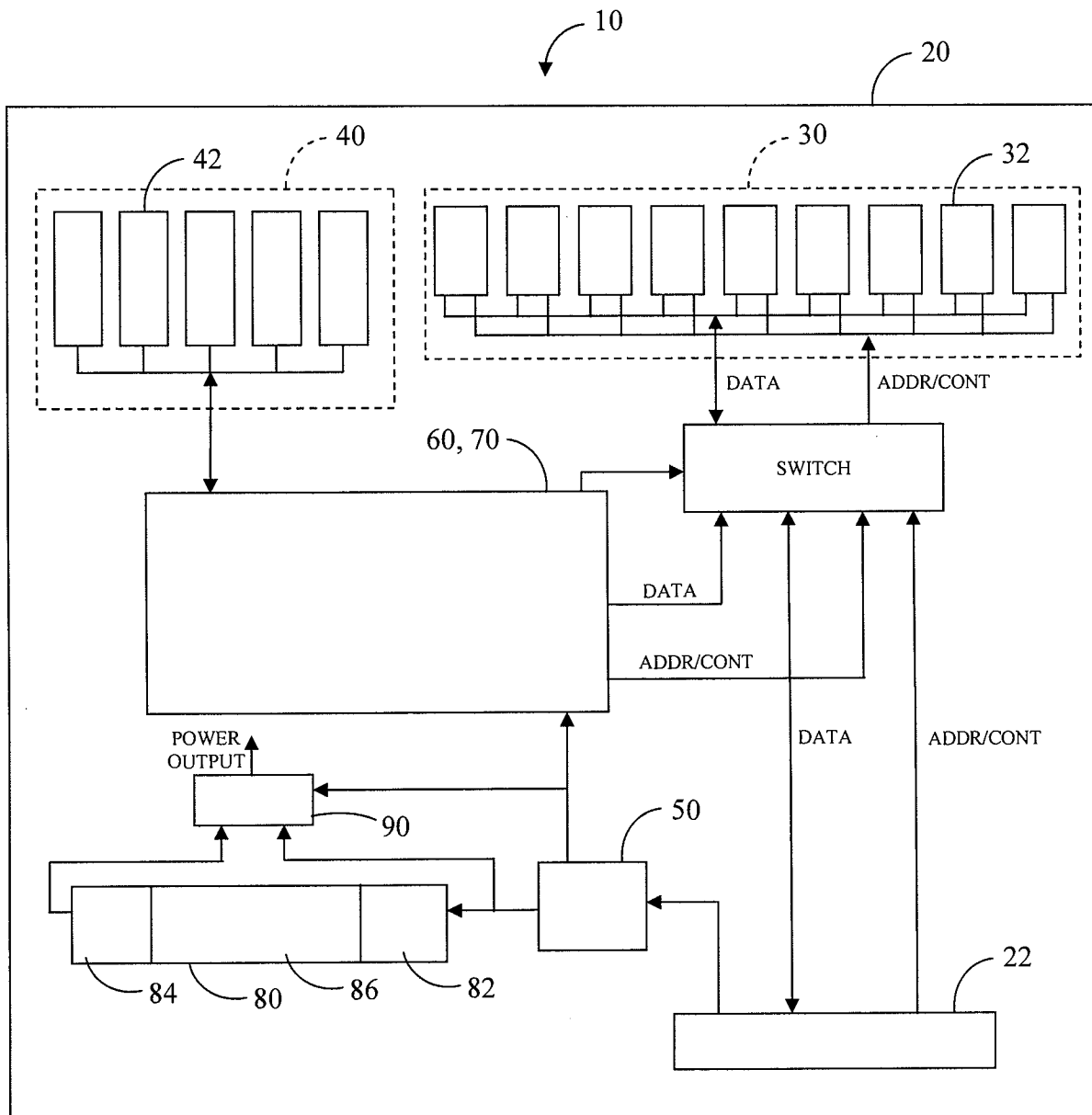


Figure 3:



**Provisional Application
COVER SHEET**

Attorney Docket No.: NETL.040PR
First Named Inventor: Jayesh R. Bhakta
Title: NON-VOLATILE MEMORY MODULE

Direct all correspondence to Customer No.: 20995

Date: June 1, 2007

Page 1 of 2

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

The following enclosures are transmitted herewith to be filed in the Provisional Patent Application of:

Inventors:

First Inventor : Jayesh R. Bhakta
Residence Address : 12220 Rose Street, Cerritos, CA 90703

Second Inventor : Chi She Chen
Residence Address : 944 Crystal Water Lane, Walnut, CA 91789

Third Inventor : Jeffrey C. Solomon
Residence Address : 16 Silver Fir, Irvine, CA 92604

APPLICATION ELEMENTS:

- (X) Specification in 5 pages.
(X) Drawings in 3 pages.

FILING FEES:

FEE CALCULATION				
FEE TYPE		LARGE FEE	CALCULATION	TOTAL
Basic Filing	37 CFR § 1.16(d)	1005 (\$200)		\$200
			TOTAL FEE DUE	\$200

This invention WAS NOT made by an agency of the United States Government or under a contract with an agency of the United States Government.

**Provisional Application
COVER SHEET**

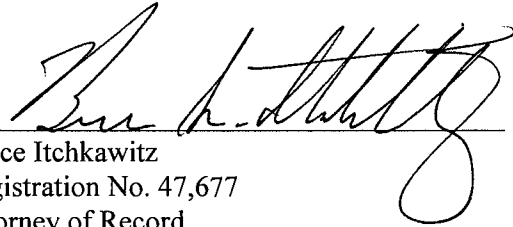
Attorney Docket No.: NETL.040PR
First Named Inventor: Jayesh R. Bhakta
Title: NON-VOLATILE MEMORY MODULE

Direct all correspondence to Customer No.: 20995

Date: June 1, 2007

Page 2 of 2

Commissioner is hereby authorized to charge any additional fees which may be required, or credit any overpayment to Account No. 11-1410.



Bruce Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

3832877
060107

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

NON-VOLATILE MEMORY MODULE

First Named Inventor/Applicant Name:

Jayesh R. Bhakta

Filer:

Bruce S. Itchkaeitz/Amy Perez

Attorney Docket Number:

NETL.040PR

Filed as Large Entity

Provisional Filing Fees

Description

Fee Code

Quantity

Amount

**Sub-Total in
USD(\$)**

Basic Filing:

Provisional application filing

1005

1

200

200

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Post-Allowance-and-Post-Issuance:

Extension-of-Time:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				200

Electronic Acknowledgement Receipt

EFS ID:	1832654
Application Number:	60941586
International Application Number:	
Confirmation Number:	2324
Title of Invention:	NON-VOLATILE MEMORY MODULE
First Named Inventor/Applicant Name:	Jayesh R. Bhakta
Customer Number:	20995
Filer:	Bruce S. Itchkaeitz/Brandon Hamada
Filer Authorized By:	Bruce S. Itchkaeitz
Attorney Docket Number:	NETL.040PR
Receipt Date:	01-JUN-2007
Filing Date:	
Time Stamp:	18:53:16
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment was successfully received in RAM	\$200
RAM confirmation Number	2593
Deposit Account	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)	Multi Part /.zip	Pages (if appl.)
-----------------	----------------------	-----------	------------------	------------------	------------------

1	Specification	prov.pdf	256635	no	5
Warnings:					
Information:					
2	Drawings	draw.pdf	46305	no	3
Warnings:					
Information:					
3	Transmittal of New Application	trans.pdf	42954	no	2
Warnings:					
Information:					
4	Fee Worksheet (PTO-06)	fee-info.pdf	8114	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			354008		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
60/941,586	06/01/2007		200	NETL.040PR		

CONFIRMATION NO. 2324

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT

Date Mailed: 06/15/2007

Receipt is acknowledged of this provisional Patent Application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Be sure to provide the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION when inquiring about this application. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please mail to the Commissioner for Patents P.O. Box 1450 Alexandria Va 22313-1450. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections (if appropriate).**

Applicant(s)

Jayesh R. Bhakta, Cerritos, CA;
Chi She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 06/14/2007

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US60/941,586**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

NON-VOLATILE MEMORY MODULE

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The

PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

OCT 15 2012

PTO/SB/81 (01-03)

Approved for use through 11/30/2011. OMB 0551-0055

U.S. Patent and Trademark Office, U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	807941586
	Filing Date	2007-06-01
	First Named Inventor	Chi-She Chen
	Title	Non-Volatile Memory Module
	Art Unit	
	Examiner Name	
	Attorney Docket Number	062453-002prov

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the:

☒ Applicant/inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71.

Statement under 37 CFR 3.73(b) (Form PTO/SB/99) submitted herewith or filed on _____

SIGNATURE of Applicant or Assignee of Record

Signature

Date

Name

Telephone

Title and Company

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ *Total of 4 forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

OCT 15 2012

PTO/EB/1 (01-09)

Approved for use through 11/30/2011 OMB 0851-0035
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	60/941,566
	Filing Date	2007-06-01
	First Named Inventor	Chi-She Chen
	Title	Non-Volatile Memory Module
	Art Unit	
	Examiner Name	
	Attorney Docket Number	062453-002prov

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71.

Statement under 37 CFR 3.73(b) (Form PTO/SB/95) submitted herewith or filed on: _____

SIGNATURE of Applicant or Assignee of Record	
Signature	Date
Name	Telephone
Title and Company	

Jeffrey O. Solomon
Oct. 11, 2012
949-679-0157
Senior Design Engineer, Netlist

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ *Total of 4 forms are submitted

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

RECEIVED
CENTRAL FAX CENTER

OCT 15 2012

PTO/SB/01 (01-03)

Approved for use through 11/30/2011. OMB 0551-0035

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	60/941,588
	Filing Date	2007-06-01
	First Named Inventor	Chi-She Chen
	Title	Non-Volatile Memory Module
	Art Unit	
	Examiner Name	
	Attorney Docket Number	082453-002prov

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71 Statement under 37 CFR 3.73(b) (Form PTO/SB/06) submitted herewith or filed on _____.

SIGNATURE of Applicant or Assignee of Record

Signature	<i>Scott H. Milton</i>	Date	10/11/12
Name	Scott Milton	Telephone	714-731-4542
Title and Company	Vice President, System Engineering		

NOTE: Signatures of all the inventors or assignees of record of the entire interest of their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ Total of 4 forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

OCT 15 2012

PTO/SB/81 (01-09)

Approved for use through 11/30/2011, OMB 0551-0035

U.S. Patent and Trademark Office: U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	80/941,586
	Filing Date	2007-08-01
	First Named Inventor	Chi-She Chen
	Title	Non-Volatile Memory Module
	Art Unit	
	Examiner Name	
	Attorney Docket Number	062453-002prov

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

46188

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71.

Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____.

SIGNATURE of Applicant or Assignee of Record

Signature	<i>Jayesh Bhakia</i>	Date	10/11/2012
Name	Jayesh Bhakia	Telephone	949-439-0205
Title and Company	Chief module Arch. McList Inc.		

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ Total of 4 forms are submitted.

This collection of information is required by 37 CFR 1.81, 1.92 and 1.93. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

OCT 15 2012

PTO/SB/97 (07-09)

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Certificate of Transmission under 37 CFR 1.8

I hereby certify that this correspondence is being facsimile transmitted to the United States Patent and Trademark Office

on October 15, 2012
Date



Signature

Pamela Wilson

Typed or printed name of person signing Certificate

n/a

650-320-7700

Registration Number, if applicable

Telephone Number

Note: Each paper must have its own certificate of transmission, or this certificate must identify each submitted paper.

Power of Attorney (sb-81) - 4 pages (60/941,586)

This collection of information is required by 37 CFR 1.8. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1.8 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
60/941,586	06/01/2007	Jayesh R. Bhakta	062453-002prov

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

CONFIRMATION NO. 2324
POA ACCEPTANCE LETTER



Date Mailed: 11/02/2012

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 10/15/2012.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/ddinh/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101